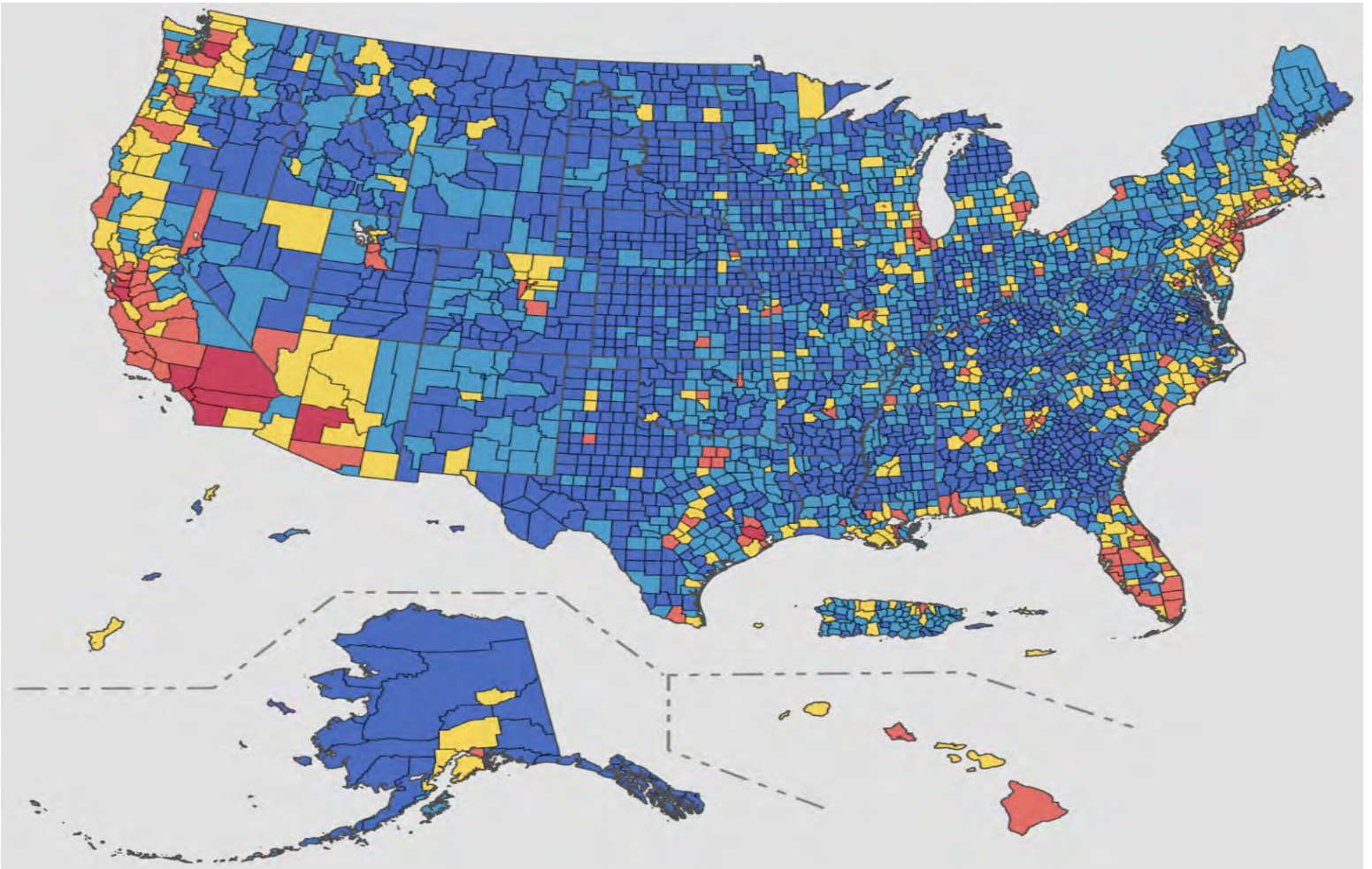




# National Risk Index Data

Technical Documentation

December 2025 - v1.20



FEMA

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## Revision History

| <b><i>Revision Number</i></b> | <b><i>Date Issued</i></b> | <b><i>Details</i></b> | <b><i>Federal Lead/Author</i></b>   |
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This document was created in accordance with guidance issued by the Secretary of Homeland Security that published documents should contain as few acronyms or abbreviations as possible. Compliance with this guidance may result in some concepts, scientific notation, mathematical methods, industry terminology, States, governmental bodies, nonprofit or private entities, products, databases, and working groups spelled out in a manner unfamiliar to the reader.

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Federal Emergency Management Agency. (2025). FEMA National Risk Index Data v 1.20.0 [Dataset]. Department of Homeland Security.

<https://www.fema.gov/flood-maps/products-tools/national-risk-index>

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## Acronym List

|         |                                                               |
|---------|---------------------------------------------------------------|
| EF-     | Enhanced Fujita Scale                                         |
| F-      | Fujita Scale                                                  |
| FSim    | Fire Simulation                                               |
| GVM     | Global Volcano Model                                          |
| LCMA    | Large Central Metropolitan Areas                              |
| NetCDF  | Network Common Data Form                                      |
| SHELDUS | Spatial Hazard Events & Losses Database for the United States |
| SLOSH   | Sea, Lake, and Overland Surges from Hurricanes                |



# 1. Introduction

The National Risk Index is a dataset developed to help illustrate the United States communities most at risk for 18 natural hazards. It was designed and built by the Federal Emergency Management Agency in close collaboration with various stakeholders and partners in academia; local, state, and federal government; and private industry. The National Risk Index leverages available source data for natural hazard and community risk factors to develop a baseline relative risk measurement for each county (or county-equivalent) and Census tract for all 50 states, the District of Columbia, American Samoa, Commonwealth of the Northern Mariana Islands, Guam, Commonwealth of Puerto Rico, and the U.S. Virgin Islands. The National Risk Index data are intended to help users better understand the natural hazard risk of their communities. Intended users include planners and emergency managers at the local, regional, state, and federal levels, as well as other decision makers and interested members of the general public. Specifically, it can support decision making to:

- Update emergency operations plans
- Enhance hazard mitigation plans
- Prioritize and allocate resources
- Identify the need for more refined risk assessments
- Encourage community-level risk communication and engagement
- Educate homeowners and renters
- Support enhanced codes and standards
- Inform long-term community recovery

This documentation provides a detailed overview of the National Risk Index, including its background, data sources, and processing methodologies. It describes the concepts used to develop the National Risk Index and calculate its components. The methodologies for computing each hazard type's Expected Annual Loss are also explained in depth in the sections for each hazard type (Sections 6 through 23).

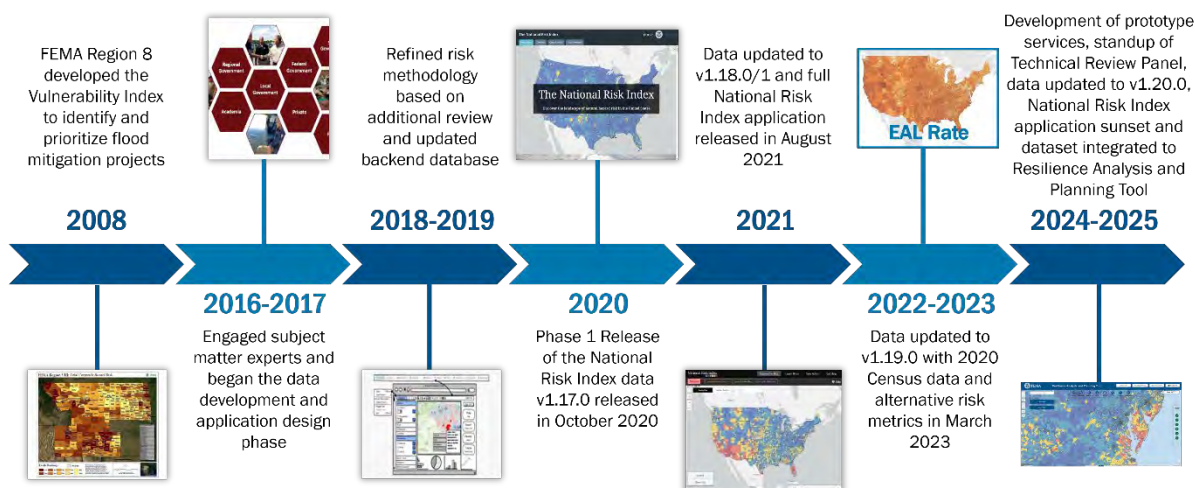
Note: This document is specific to the 2025 release (version 1.20.0).

## 2. Background

All communities in the United States experience natural hazards, and there is a wide range of environmental, social, and economic factors that influence each community's risk to natural hazards. The likelihood that a community may experience a natural hazard can vary drastically, as can the associated consequences. Additionally, a community's risk is influenced by many social, economic, and ecological factors. The Federal Emergency Management Agency, along with numerous federal, state, and local governments; academic institutions; nonprofit groups; and private industry (see [Figure 1](#)) collaborated to develop the National Risk Index as a baseline risk assessment dataset.

In 2016, the Federal Emergency Management Agency's Natural Hazards Risk Assessment Program started work on the National Risk Index by adopting an established vision for a multi-hazard view of risk that combines the likelihood and consequence of natural hazards with social factors and resilience capabilities. The goal was to take a broad, holistic view and create a nationwide baseline of natural hazard risk. Through various partnerships and working groups, the Federal Emergency Management Agency developed a methodology and procedure to create the dataset, which has been used in both a standalone application (since deprecated) and as part of other Federal Emergency Management Agency risk assessment tools.

### National Risk Index Timeline



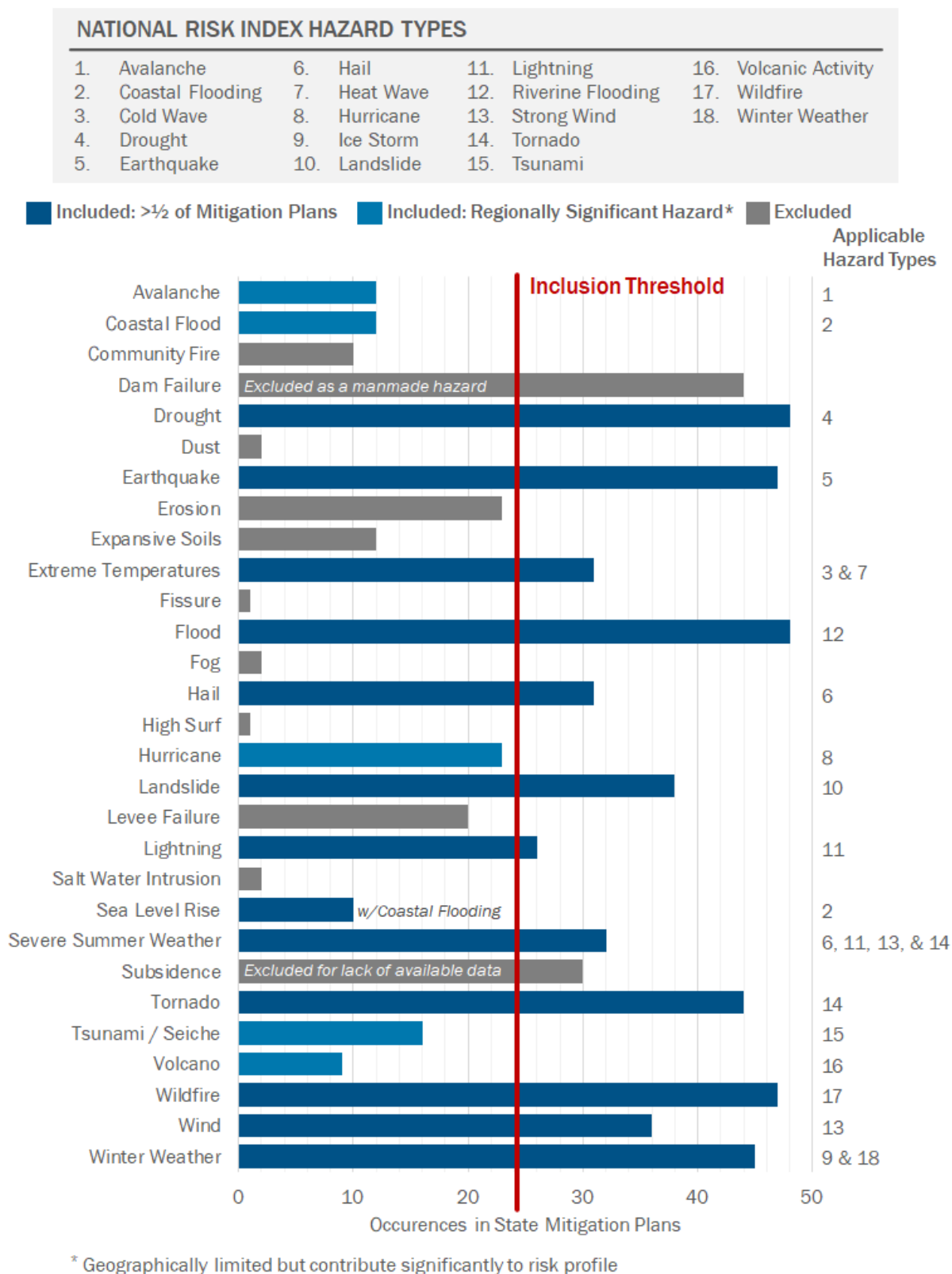
**Figure 1: Timeline of the Development of the National Risk Index**

The National Risk Index Team conducted multiple workshops and sessions to discuss and determine the methodologies for translating raw source data into natural hazard risk factors for input into the National Risk Index. The key objective of these exercises was to ensure that a vetted risk model or equation was leveraged throughout all methodological development and that certain factors were not being interpreted inconsistently across the 18 hazard types.

## 2.1. Natural Hazard Selection

A community's susceptibility to natural hazards varies from location to location. The 18 hazard types evaluated by the National Risk Index were chosen after reviewing Federal Emergency Management Agency-approved State Hazard Mitigation Plans for all 50 states in early 2016. Tribal hazard mitigation plans were not available at the time of the analysis, and island territories were excluded from the hazard selection process since data for most hazard types were not available at the time. Note that Washington, District of Columbia, was initially excluded from the hazard selection analysis process; however, it was added to the project scope in 2017, after the hazard selection process was completed.

Natural hazards that were included in at least half of the Federal Emergency Management Agency-approved state plans, or those that were deemed to be of regional significance, were selected (see [Figure 2](#)). A regionally significant hazard is defined as having the capacity to cause widespread, catastrophic damage, such as Hurricane, Tsunami, and Volcanic Activity, but otherwise affected fewer than 25 states. It should be noted that one natural hazard, Subsidence, fit these criteria, but could not be evaluated as there was no reliable, nationwide dataset cataloging this type of hazard occurrence.



**Figure 2: Determination of Hazard Inclusion Based on State Hazard Mitigation Plans from January 2016**

The State Hazard Mitigation Plan review revealed that both Dam Failure and Levee Failure hazards are profiled by many states. However, these were not included in the National Risk Index because they were either a non-natural hazard – in the case of Dam and Levee Failure – or lacked necessary national, public datasets from which to develop Risk Index Ratings in the National Risk Index methodology, such as with Subsidence.

## 2.2. Literature Review

Beginning in 2017, the project team started reviewing literature in the fields of hazard mitigation, emergency management, hazard risk science, and other related fields. Centering around a search for natural hazard and exposure variables, the literature review identified multiple datasets, risk indices, research reports, methodologies, indicator lists, and existing risk assessments at national and global scales.

The team identified important risk indicator categories and specific indicators during the review (see [Table 1](#)).

**Table 1: Literature Review Risk Indicators and Categories**

| <i>Risk Indicator Categories</i> | <i>Individual Risk Indicators</i> |                         |
|----------------------------------|-----------------------------------|-------------------------|
| ▪ Social                         | ▪ Income                          | ▪ Road Systems          |
| ▪ Economic                       | ▪ Age                             | ▪ Economic Productivity |
| ▪ Environmental                  | ▪ Illnesses                       | ▪ Housing               |
| ▪ Infrastructure                 | ▪ Hospitals                       | ▪ Community Revenue     |

After review, the team concluded the National Risk Index would involve three components: natural hazard risk (likelihoods and consequences), Social Vulnerability, and Community Resilience. The National Risk Index has continued to perform literature reviews in the intervening years to improve the National Risk Index Methodology and data used to develop risk profiles.

## 2.3. Working Groups

After the initial detailed literature review and hazard analysis, the National Risk Index Team convened three working groups comprised of intended users, subject matter experts, and interested stakeholders from all levels of government, private industry, nonprofits, and academia. Each working group was responsible for an aspect of the National Risk Index Methodology. Experts in each group helped guide the data and application development.

The Natural Hazards Working Group assessed and recommended datasets associated with the identified 18 hazard types selected (as well as Subsidence prior to its recommended removal) and determined the best ways to incorporate associated data.

The Social Vulnerability and Community Resilience Working Group reviewed and evaluated existing efforts to measure social vulnerability and community resilience to understand which components were most important (vulnerability, resilience, or both) and which indices should be used. As a result, both Social Vulnerability and Community Resilience are components of the National Risk Index.

The Data Analytics Working Group oversaw spatial processing, normalization, and aggregation of data to arrive at the National Risk Index Methodology and calculation procedure that integrated the datasets identified by the other two working groups. Together, the groups discussed and developed the initial version of National Risk Index, including the datasets and indices to incorporate, definitions of index components, data management strategies, metadata requirements, data processing and index creation methodologies, and the data visualization and interactive web mapping application requirements.

The National Risk Index has continued to engage experts in the intervening years to help improve the quality and accuracy of the risk profile. For example, in 2022, the National Risk Index team convened a new Social Vulnerability Working Group to obtain information to help determine the most suitable Social Vulnerability product for use in the risk equation. This working group included over 100 social vulnerability subject matter experts across public, private, non-profit, and academic sectors. Its efforts culminated in the decision to transition from the University of South Carolina's Hazards Vulnerability & Resilience Institute's Social Vulnerability Index to the Centers for Disease Control and Prevention's/Agency for Toxic Substances and Disease Registry's Social Vulnerability Index as the new basis for the consequence-enhancing community risk factor of the National Risk Index. This served as the basis for understanding available social vulnerability metrics that led to the incorporation of the Census Bureau's Community Resilience Estimates team's Social Vulnerability Measure in v1.20.0.

In 2024, the National Risk Index team established a Technical Review Panel comprised of Federal subject matter experts to help inform the development of the National Risk Index. The panel serves as a consultative body of stakeholders to assist in the continuous oversight and improvement of the National Risk Index data and methods.

## 2.4. Subject Matter Expert Review

Initial development of the National Risk Index began in 2017 and proceeded through the end of 2019. Over this period, the National Risk Index Team continually iterated on the data processing and risk calculation methodologies and engaged with subject matter experts throughout. See [Appendix A – Contributors](#) for the full list of organizations whose members contributed to the subject matter expert reviews and the development of the National Risk Index.

At major milestones, the team paused development to engage in broader, more comprehensive review periods by subject matter experts. The first major milestone arrived in January 2019, where teams of experts were tasked to evaluate two competing draft methodologies: "Methodology 1," which relied on unitless standardization of Expected Annual Loss, and "Methodology 2," which standardized Expected Annual Loss to a dollar value measurement. Over the course of two weeks

and many meetings, dozens of experts provided feedback to the National Risk Index Team, resulting in a clear consensus that, although both methodologies were valid, Methodology 2 created a more robust measurement of risk and a more valuable dataset for the hazard planning and mitigation communities.

With clear direction on the methodology, the National Risk Index Team continued iterating through improvements to data sourcing and processing. From July through September 2019, a final comprehensive subject matter expert review period was conducted to focus on the new Methodology's results. More than 40 experts participated in over 20 review sessions and helped the Team reach concurrence on the validity and value of the dataset. From these sessions, the National Risk Index Team was equipped to begin final iterations of the Methodology and source data processing for Version 1.18.0.

In the preparation of Version 1.19.0 (released March 2023), the National Risk Index Team continued to engage with dozens of subject matter experts on specific hazard types and risk factors to enhance the Methodology and source data. As a result of these reviews, the National Risk Index made a handful of significant structural changes to how Risk is calculated and communicated. This included revising the Risk equation to introduce the Community Risk Factor, a scaling factor unique to each Census tract that varies as a function of the community's Social Vulnerability and Community Resilience. The Community Risk Factor is discussed in [Section 4.3 Community Risk Factor](#). Additionally, Values were introduced as a third results type, and Scores and Ratings were redefined. These results types are discussed in further detail in [Section 3.2 Values, Scores and Ratings](#).

During the development of Version 1.20.0, the National Risk Index Team introduced a formalized subject matter expert review process to routinize in-depth reviews of specific hazards, via the Technical Review Panel. In addition, for components marked for significant methodological overhauls, the National Risk Index team established hazard-specific working groups to facilitate systematic collaboration with subject matter experts from professional organizations, state and local governments, and federal agencies. The Tsunami and Inland Flooding hazards were selected as the first hazards under this process, resulting in increased collaboration with subject matter experts to address data and methodology concerns.

## 2.5. Data and Methodologies

The National Risk Index is a first-of-its-kind, nationwide, holistic assessment of baseline risk to natural hazards. Although it is based on extensive research and best practices in the risk assessment fields, the Methodology is unique and carefully constructed to meet the specific needs of natural hazard risk assessment at both small and large geographic scales. A detailed overview of the risk calculation is available in [Section 3.1. Risk Calculation](#).

The National Risk Index's most important and principal component, Expected Annual Loss, is a robust measurement that quantifies in dollars the anticipated economic damage resulting from natural hazards each year. Details of its equation and analytical techniques are available in [Section 4.4](#). Expected Annual Loss consists of the best available datasets for 18 hazard types of national

and regional significance, with source data processed to match the unique nature of each hazard type. Full processing details for each hazard type are available in Sections 6 through 23. Per the direction established at initiation, the National Risk Index also includes measurements of Social Vulnerability and Community Resilience to quantify overall risk. These key components are detailed fully in [Section 4.1. Social Vulnerability](#) and [Section 4.2. Community Resilience](#), respectively.



### 3. Risk Analysis Overview

Natural hazard risk, in the most general terms, is often defined as the likelihood (or probability) of a natural hazard event happening multiplied by the expected consequence if a natural hazard event occurs. The generalized form of a risk equation is given in [Equation 1](#).

#### Equation 1: Generalized Risk Equation

$$\text{Risk} = \text{Likelihood} \times \text{Consequence}$$

#### 3.1. Risk Calculation

In the National Risk Index, risk is defined as the potential for negative impacts as a result of a natural hazard. The risk equation behind the National Risk Index includes three components: a natural hazards risk component, a consequence enhancing component, and a consequence reduction component. Expected Annual Loss is the natural hazards risk component, measuring the expected loss of building value, population, and/or agriculture value each year due to natural hazards. Social Vulnerability is the consequence-enhancing component and analyzes demographic characteristics to measure the susceptibility of social groups to the adverse impacts of natural hazards. Community Resilience is the consequence-reduction component and uses demographic characteristics to measure a community's ability to prepare for, adapt to, withstand, and recover from the effects of natural hazards. The Social Vulnerability and Community Resilience components are combined into one Community Risk Factor which is multiplied by the Expected Annual Loss component to calculate risk using [Equation 2](#).

#### Equation 2: Generalized National Risk Index Risk Equation

$$\text{Risk} = \text{Expected Annual Loss} \times \text{Community Risk Factor}$$

$$\text{where Community Risk Factor} = f\left(\frac{\text{Social Vulnerability}}{\text{Community Resilience}}\right)$$

The Risk Index Values form an absolute basis for measuring risk within the National Risk Index. They are used to generate all Risk Index Scores and Ratings. As described in [Equation 2](#) above, Risk Index Values are determined by multiplying Expected Annual Loss by the Community Risk Factor, which is a scaling factor unique to each community, that varies as a function of the community's Social Vulnerability and Community Resilience Values. All Risk Index Values are calculated at the Census tract level. County and county-equivalent (e.g., boroughs, municipios, parishes) Values are calculated by summing the Values from their tracts.

The function  $f(\cdot)$  is a transformation that maps the ratio between Social Vulnerability and Community Resilience to Community Risk Factor Values. This mapping is constructed so that higher social vulnerability and lower community resilience, relative to all other communities at the same level (county or Census tract), result in higher Risk Index Values for a given level of Expected Annual

Loss. For more details regarding the way the components are estimated before entering the risk equation, see [Sections 4.1 Social Vulnerability](#), [4.2 Community Resilience](#), [4.3 Community Risk Factor](#), and [4.4 Expected Annual Loss](#), respectively.

## 3.2. Values, Scores and Ratings

The National Risk Index provides three distinct types of results for Risk and each component used to derive Risk: Expected Annual Loss, Social Vulnerability, and Community Resilience:

- **Values.** Values for Risk and Expected Annual Loss are in units of dollars and represent the community's average economic loss from natural hazards each year. For Social Vulnerability and Community Resilience, Values are the index values for the community provided by the source data sets.
- **Scores.** Scores represent the national percentile ranking of the community's component Value compared to all other communities at the same level (county or Census tract).
- **Ratings.** Ratings are provided in one of five qualitative categories describing the community's component Value in comparison to all other communities at the same level. Rating categories range from "Very Low" to "Very High."

Risk and Expected Annual Loss are calculated first as monetary Values. Expected Annual Loss Values represent the economic losses that could feasibly occur due to natural hazards in a given year. Expected Annual Loss Values are calculated using the Expected Annual Loss equation (see [Section 4.4](#)), which accounts for the economic value of exposed population, buildings, and agriculture to a given hazard type (Exposure), the probability of that hazard type's occurrence each year (Annualized Frequency), and the expected proportion of the exposed Values that may be lost due to a hazard occurrence (Historic Loss Ratio).

Risk Index Values for each community are then determined by multiplying their Expected Annual Loss Value by the Community Risk Factor, as described in [Equation 2](#) and [Section 3.1](#). The Community Risk Factor scales the Expected Annual Loss Value such that communities with higher Social Vulnerability and/or lower Community Resilience will have higher representative Risk Index Values. This is intended to represent the added burden that natural hazards pose to communities with more vulnerable populations, or those that do not have the means to recover quickly from natural hazards.

The Risk Index and Expected Annual Loss are provided as both hazard-specific Values and composite Values from the summation of all 18 hazard types. More information on the calculation of Risk and EAL Values can be found in [Section 3.1 Risk Calculation](#), [Section 5 Expected Annual Loss Factors](#), and in the Expected Annual Loss section for each individual hazard.

From the Values, each component is represented by a Score that represents a community's national percentile ranking relative to all other communities at the same level (county or Census tract). All Scores are constrained to a range of 0 (lowest possible value) to 100 (highest possible value). To

achieve this range, Scores for each component are determined by the given Value's percentile ranking in the national distribution (rounded to the nearest hundredth).

All calculations are performed separately at two levels—county and Census tract—so Scores are relative only within their level. It must be stressed that Scores represent a community's relative ranking among all other communities for a given component and level. Scores are national percentiles, not absolute measurements, and should be expected to change over time either by their own changing measurements or changes in other communities.

The qualitative Rating categories describe the nature of a community's component Value in comparison to all other communities at the same level, ranging from "Very Low" to "Very High." Like Scores, Ratings are also based on their component's Value. Rating categories are relative and for Risk and Expected Annual Loss there are no specific numeric Values or Scores that determine the Rating. For example, a community's Risk Index Score for one hazard could be 8.9 with a Rating of "Relatively Low," while its Risk Index Score for another hazard may be 11.3 with a Rating of "Very Low." The Rating is intended to classify a community for a specific component in relation to all other communities at the same level.

To determine Ratings, an unsupervised machine learning technique known as k-means clustering is applied to each value. This approach divides all communities into groups or clusters such that the communities within each cluster are as similar as possible (minimized variance or inertia) while the clusters are as different as possible (maximized variance).

A cubed root transformation is applied to both Risk and Expected Annual Loss Values before k-means clustering. Without the transformation, these values are heavily skewed by an extreme range of population and building value densities between urban and rural communities. By applying a cubed root transformation, the National Risk Index controls for this characteristic and provides Ratings with greater differentiation and usefulness.

K-means clustering for Rating designation is performed in the National Risk Index processing database using the k-means clustering algorithm in the Python library scikit-learn.<sup>1</sup> The algorithm is initialized with the following parameters:

- Number of clusters (n\_clusters): 5
- Maximum iterations (max\_iter): 500
- Number of times the algorithm is run with different centroid seeds to arrive at the output with the least inertia (n\_init): 20
- Relative tolerance in the cluster centers of consecutive iterations to declare convergence (tol):  $1 \times 10^{-15}$

---

<sup>1</sup> See scikit-learn clustering documentation retrieved from <https://scikit-learn.org/stable/modules/clustering.html#k-means>.

- Random number generation seed for centroid initialization (random\_state): 42

All other parameters are defaults. The algorithm works by selecting five random initial Values, one for each cluster centroid, and then the rest of the Values are iteratively assigned to a cluster based on proximity to the centroid. Cluster centroids are updated in each iteration as the mean value of the Values within each cluster. The algorithm stops when it completes the maximum number of iterations or the centroid calculations converge within the established tolerance, whichever occurs first. The finalized cluster of the lowest Values is assigned the Rating “Very Low,” the next lowest cluster receives a Rating of “Relatively Low,” and so on.

As with Scores, Ratings are relative measures that are calculated at both the county and Census tract levels. As such, a Census tract with a “Very High” Rating may reside within a county with a “Low” or even “Very Low” Rating, and vice versa. A community’s Rating can also be expected to change over time either by their own changing measurements or changes in other communities.

Ratings for Social Vulnerability and Community Resilience have specific numerical boundaries based on national percentiles. The Ratings are divided into quintiles as described below:

- **Very High:** 80<sup>th</sup> to 100<sup>th</sup> percentiles
- **Relatively High:** 60<sup>th</sup> to 80<sup>th</sup> percentiles
- **Relatively Moderate:** 40<sup>th</sup> to 60<sup>th</sup> percentiles
- **Relatively Low:** 20<sup>th</sup> to 40<sup>th</sup> percentiles
- **Very Low:** 0<sup>th</sup> to 20<sup>th</sup> percentiles



**Figure 3: Calculating Values, Scores, and Ratings for Risk and Expected Annual Loss**

The National Risk Index provides Risk Index Values, Scores, and Ratings for counties (and county-equivalents) and Census tracts for all 50 states and the District of Columbia. Expected Annual Loss Scores and Ratings are available for all 50 states, the District of Columbia, American Samoa, Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and U.S. Virgin Islands. Social Vulnerability Scores and Ratings are available for all 50 states, the District of Columbia, and Puerto Rico. Community Resilience Scores and Ratings are available for all 50 states and the District of Columbia.

Component Values of 0 (zero) or missing Expected Annual Loss Values (“nulls”) receive Ratings that reflect the logic behind the score. A community where the Expected Annual Loss is zero either has no building, population, or agriculture value exposed to the hazard type, or has a calculated annualized frequency of zero for the hazard type. These communities are displayed as having “No Expected Annual Losses” for the designated hazard type, and they will have “No Rating” for their Risk Index for that hazard type.

In collaboration with subject matter experts most familiar with a hazard type and the source data, a priori definitions of hazard type applicability have also been applied to help distinguish between where no risk exists for the hazard type and where the hazard type is deemed not able to occur. For example, Avalanche Expected Annual Loss is not computed for areas with insufficient slope and snow depth. These communities are displayed as “Not Applicable” for Expected Annual Loss computation for the designated hazard type.

Finally, if a factor used to calculate the Expected Annual Loss of a Census tract or county for a hazard type has a null value, the community is rated as “Insufficient Data.” For example, certain hazard types only have source data used to determine annualized frequency or exposure available for the conterminous United States of America, meaning that both Alaska and Hawai‘i are rated as “Insufficient Data” to compute the Expected Annual Loss for those hazard types. Census tracts and counties without Social Vulnerability or Community Resilience data are given a Rating of “Data Unavailable.” If a community has “Insufficient Data” to compute Expected Annual Loss or “Data Unavailable” for Social Vulnerability or Community Resilience, the community is given a Risk Index Rating of “Insufficient Data.” When a hazard type is not applicable or there are insufficient data for a community, Expected Annual Loss for that hazard type is simply not included in the community’s final summation and scoring. A summary of non-numerical Ratings is provided in [Table 2](#).

**Table 2: Definitions of Ratings without Numerical Scores**

| <i><b>Rating</b></i> | <i><b>Expected Annual Loss</b></i>                   | <i><b>Social Vulnerability</b></i> | <i><b>Community Resilience</b></i> | <i><b>Risk Index</b></i>                             |
|----------------------|------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------------------------|
| Not Applicable       | Community is not considered at risk for hazard type. | n/a                                | n/a                                | Community is not considered at risk for hazard type. |

| <b>Rating</b>     | <b>Expected Annual Loss</b>                           | <b>Social Vulnerability</b>                  | <b>Community Resilience</b>                  | <b>Risk Index</b>                                                                    |
|-------------------|-------------------------------------------------------|----------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------|
| Data Unavailable  | n/a                                                   | Social Vulnerability data are not available. | Community Resilience data are not available. | n/a                                                                                  |
| Insufficient Data | Hazard source data are not available.                 | n/a                                          | n/a                                          | Social Vulnerability, Community Resilience, or hazard source data are not available. |
| No EAL            | Hazard type exposure or annualized frequency is zero. | n/a                                          | n/a                                          | n/a                                                                                  |
| No Rating         | n/a                                                   | n/a                                          | n/a                                          | Expected Annual Loss is zero.                                                        |

### 3.3. Assumptions and Limitations

The National Risk Index dataset is meant for planning purposes only and are intended for use as a tool for broad, nationwide comparisons. Nationwide datasets used as inputs are in many cases not as accurate as locally available data. Users with access to local data for each risk component should consider substituting those data to calculate a more accurate Expected Annual Loss Value at the local level.

The National Risk Index team recognizes that modeling natural hazard risk for the entire country has inherent uncertainty and inaccuracy, despite using the best available data and methods. This is due to statistical uncertainty in the methods used, the difficulties in modeling complex natural phenomena, the availability of a consistent and national-scale dataset, and sometimes ongoing debates in the scientific communities about how to model a natural hazard or its impacts. A discussion of hazard-specific modeling uncertainties, assumptions, limitations, or future considerations are provided in each of the hazard-specific sections of this document (see Sections 6 through 23).

The National Risk Index does not consider the intricate economic and physical interdependencies that exist across geographic regions. The user should be mindful that hazard impacts in surrounding counties or Census tracts can cause indirect losses in a location regardless of the location's risk profile.

The periods of record vary across hazard types and risk components with the most recent source datasets including a period of record up to 2024. It should be noted that the Expected Annual Loss Values represent an extrapolation based on a snapshot in time. Extending source data collection beyond that time may result in varying Census tract or county Expected Annual Loss Values due to changes in recorded hazard type severity and annualized frequency, as well as fluctuations in local economic value and/or population density.

Most of the hazard types use an annualized frequency model to determine Expected Annual Loss. This makes it difficult to accurately estimate Expected Annual Loss for high consequence, low frequency events. Certain rare hazard types (such as Earthquake, Hurricane, Tsunami, and Volcanic Activity) benefit from using a probabilistic model that estimates the likelihood of a hazard occurrence over an extended period of time, which can then be annualized. Of these, only Earthquake has probabilistic source data that are sufficient for accurately estimating Expected Annual Loss.<sup>2</sup> More details on the general methods used to calculate Expected Annual Loss are described in Section 5 and the specific approaches for each hazard type are detailed in Sections 6 through 23.

Best available nationwide data for some risk factors are rudimentary. More sophisticated risk analysis methodologies are available but require more temporally and spatially granular data for hazard exposure, annualized frequency, and Historic Loss Ratio measurements.

The National Risk Index makes various efforts to account for uncertainties present in its source data to calculate the best possible baseline measurement of natural hazard risk across communities nationwide. For instance, the use of buffered areas and fishnet intersections allows the annualized frequency calculations to account for the influence of chance determining the precise location at which a hazard event was recorded. Additionally, the Bayesian Credibility applied to the Historic Loss Ratio calculation attenuates some of the uncertainties and biases often associated with reported loss data (see [Section 5.4 Historic Loss Ratio](#) for more information).

The Methodology also makes various efforts to control for possible discrepancies in source data but cannot correct for all accuracy problems present in that data. The National Risk Index's processing database is a complex system, and localized inaccuracies in source data have the potential to propagate. Therefore, the National Risk Index and its components should be considered a baseline measurement and a guideline for determining natural hazard risk but should not be used as an absolute measurement of risk.

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<sup>2</sup> Federal Emergency Management Administration (FEMA). (2017). Hazus Estimated AELs for the United States: FEMA Publication 366. Retrieved from [https://www.fema.gov/sites/default/files/2020-07/fema\\_earthquakes\\_hazus-estimated-annualized-earthquake-losses-for-the-united-states\\_20170401.pdf](https://www.fema.gov/sites/default/files/2020-07/fema_earthquakes_hazus-estimated-annualized-earthquake-losses-for-the-united-states_20170401.pdf).



## 4. Risk Components Overview

The Risk Index is based on three components: Social Vulnerability, Community Resilience, and Expected Annual Loss, with Expected Annual Loss in turn comprised of Exposure, Annualized Frequency, and Historic Loss Ratio factors, for a total of five risk factors. Each risk factor contributes to either the likelihood or consequence aspect of risk and are classified as one of two risk types: risk based on geographic location, represented by the Community Risk Factor, or risk based on the nature and historical occurrences of natural hazards, represented by Expected Annual Loss. The five risk factors are summarized in [Table 3](#) and further described in this section.

**Table 3: Risk Components and Factors**

| <i>Risk Component</i> | <i>Risk Factors</i>  | <i>Risk Factor Description</i> | <i>Risk Contribution</i> | <i>Risk Type Assignment</i> |
|-----------------------|----------------------|--------------------------------|--------------------------|-----------------------------|
| Social Vulnerability  | Social Vulnerability | Consequence Enhancer           | Consequence              | Geographic Risk             |
| Community Resilience  | Community Resilience | Consequence Reducer            | Consequence              | Geographic Risk             |
| Expected Annual Loss  | Exposure             | Expected Consequence           | Consequence              | Natural Hazard Risk         |
| Expected Annual Loss  | Annualized Frequency | Probability of Occurrence      | Likelihood               | Natural Hazard Risk         |
| Expected Annual Loss  | Historic Loss Ratio  | Expected Consequence           | Consequence              | Natural Hazard Risk         |

### 4.1. Social Vulnerability

Social Vulnerability is broadly defined as the susceptibility of social groups to the adverse impacts of natural hazards, including disproportionate death, injury, loss, or disruption of livelihood. Social Vulnerability considers the social, economic, demographic, and housing characteristics of a community that influence its ability to prepare for, respond to, cope with, recover from, and adapt to environmental hazards.

As a consequence-enhancing risk factor, the Social Vulnerability Score represents the national percentile ranking of social vulnerability for a given county or Census tract in comparison to all other communities at the same level. The higher a county's or Census tract's Social Vulnerability is, the higher the risk. Because social vulnerability is unique to a geographic location—specifically, a county or Census tract—it is a geographic risk factor.



#### 4.1.1. SOCIAL VULNERABILITY SOURCE DATA

**Social Vulnerability source data provider:** [Census](#) Community Resilience Estimates Social Vulnerability Measure

The [Census](#) Community Resilience Estimates Social Vulnerability Measure is a location-specific assessment of social vulnerability that utilizes 10 socioeconomic variables (listed below) deemed to contribute to a community's reduced ability to prepare for, respond to, and recover from hazards.

- |                                      |                                                |
|--------------------------------------|------------------------------------------------|
| ▪ Income-to-Poverty Ratio            | ▪ Disability                                   |
| ▪ Single or zero caregiver household | ▪ No health insurance coverage                 |
| ▪ Unit-level crowding                | ▪ Being aged 65 years or older                 |
| ▪ Communication barrier              | ▪ No vehicle access                            |
| ▪ Lack of Employment                 | ▪ Households without broadband internet access |

The dataset was acquired from the Census Community Resilience Estimates team, and users looking for more information should consult the [Census Community Resilience Estimates web page](#) for more details.

#### 4.1.2. PROCESSING SOCIAL VULNERABILITY SOURCE DATA

The Social Vulnerability Scores provided in the data are derived from the raw national percentiles and 90% confidence intervals found in the source data from Census. For a detailed explanation of our methods, see Appendix D, we multiply this proportion by 100 to obtain a percentile ranking that accounts for the uncertainty reported by Census

County and Census tract Social Vulnerability Scores were classified into five qualitative categories, from "Very Low" to "Very High," by dividing Scores into quintiles (see [Section 3.2 Values, Scores and Ratings](#).) Social Vulnerability percentile rankings are available for all counties and Census tracts.

### 4.2. Community Resilience

Community Resilience is defined by National Institute of Standards and Technology as the ability of a community to prepare for anticipated natural hazards, adapt to changing conditions, and withstand and recover rapidly from disruptions.<sup>3</sup>

There are multiple, well-established ways to define community resilience at the local level, and key drivers of resilience vary between locations. Because there are no nationally available, bottom-up community resilience indices available, the Social Vulnerability and Community Resilience Working Group chose to utilize a top-down approach. The National Risk Index relies on using broad factors to define resilience at a national level and create a comparative metric to use as a risk factor.

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<sup>3</sup> NIST. (2020). *Community Resilience*. Retrieved from: <https://www.nist.gov/topics/community-resilience>.

In the National Risk Index data, Community Resilience is the consequence reduction risk factor and represents the relative level of community resilience in comparison to all other communities at the same level. A higher Community Resilience score results in lower Risk Index Values. Because Community Resilience is unique to a geographic location—specifically, a county—it is a geographic risk factor.

#### 4.2.1. COMMUNITY RESILIENCE SOURCE DATA

**Community Resilience source data:** [University of South Carolina's Hazards and Vulnerability Research Institute Baseline Resilience Indicators for Communities](#)

Community resilience data are obtained from the University of South Carolina's Hazards and Vulnerability Research Institute Baseline Resilience Indicators for Communities. These obtained data were used to create a FEMA modified version of the Baseline Resilience Indicators for Communities for use in the National Risk Index dataset. This index provides a sound methodology for quantifying community resilience by identifying the ability of a community to prepare and plan for, absorb, recover from, and more successfully adapt to the impacts of natural hazards. The dataset includes a set of 49 indicators that represent six types of resilience: social, economic, community capital, institutional capacity, housing/infrastructure, and environmental. It uses a local scale within a nationwide scope, and the national dataset serves as a baseline for measuring relative resilience. The data are used to compare one place to another and determine specific drivers of resilience, and a higher score indicates a stronger and more resilient community.

To better align the National Risk Index data with administration priorities, several inputs associated with gender, race, and ethnicity were removed from the Community Resilience data. The remaining inputs were used to calculate Community Resilience Scores, Values and Ratings for every applicable county. For details regarding the retrieval of Census data and calculation of variable inputs, see Appendix D. For more information regarding the methods used to calculate the resilience measure, see the University of South Carolina's Hazards and Vulnerability Research Institute website.<sup>4, 5</sup>

#### 4.2.2. PROCESSING COMMUNITY RESILIENCE SOURCE DATA

The FEMA modified Baseline Resilience Indicators for Communities index is available for all 50 states and the District of Columbia but it is only available at the county level. Community Resilience Scores were inferred from counties to Census tracts by assigning each Census tract the Score of its parent county. The Community Resilience Scores presented in the data download represent national percentiles of the values derived from the FEMA modified version of the index.

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<sup>4</sup> The methods and formulas used to calculate Community Resilience values for the National Risk Index dataset are identical to those of the Baseline Resilience Indicators for Communities. The only difference being that Social and Economic pillars for resilience were recalculated with input variables for food insecurity, racial/ethnic income inequality, and gender income inequality removed.

<sup>5</sup> [BRIC - McCausland College of Arts and Sciences | University of South Carolina](#)

Community Resilience Scores were classified into five qualitative categories, from “Very Low” to “Very High,” by dividing Scores into quintiles (see [Section 3.2 Values, Scores and Ratings](#)).<sup>6, 7</sup> For more information regarding how Community Resilience values were derived from the FEMA modified version of the index data, see Appendix E.

### 4.3. Community Risk Factor

The Community Risk Factor is a scaling factor that incorporates Social Vulnerability and Community Resilience in the National Risk Index to arrive at a distribution of Risk Index Values that better reflects the impacts communities experience from natural hazards. By design, the Community Risk Factor ensures that higher Social Vulnerability and lower Community Resilience, relative to all other communities at the same level (county or Census tract), result in higher Risk Index Values for a given level of Expected Annual Loss.

To generate a Community Risk Factor Value for a community, its Social Vulnerability Value is divided by its Community Resilience Value to obtain a Community Risk Ratio. Community Risk Ratio values are mapped to a triangular distribution with minimum 0.5, mode 1, and maximum 2 using national percentiles, as in [Equation 3](#).

#### Equation 3: Community Risk Factor Equation

$$\text{Community Risk Ratio}_{\text{value}} = \frac{\text{Social Vulnerability}}{\text{Community Resilience}}$$

$$\text{Community Risk Ratio}_{\text{NPCTL}} = \text{National Percentile}(\text{Community Risk Ratio}_{\text{value}})$$

$$\text{Community Risk Factor} = \text{PPF}(\text{Community Risk Ratio}_{\text{NPCTL}})$$

where:

*Community Risk Ratio<sub>value</sub>* is the ratio between the Social Vulnerability and Community Resilience Values for a particular community.

*Social Vulnerability* is the Social Vulnerability Score for a particular community.

*Community Resilience* is the value derived from the FEMA modified version of the Baseline Resilience Indicators for Communities Value for a particular community.

<sup>6</sup> Cutter, S.L., Ash, K.D., & Emrich, C.T. (2014). The geographies of community disaster resilience. *Global Environmental Change*, 29, 65-77. <https://doi.org/10.1016/j.gloenvcha.2014.08.005>.

<sup>7</sup> See also Mitigation Framework Leadership Group (MitFLG), FEMA. (2016). Draft Interagency Concept for Community Resilience Indicators and National-Level Measures. Washington, DC: Department of Homeland Security (DHS). Retrieved from <https://www.govinfo.gov/content/pkg/GOVPUB-HS-PURL-gpo131289/pdf/GOVPUB-HS-PURL-gpo131289.pdf>.

*Community Risk Ratio*<sub>NPCTL</sub> is the national percentile ranking of the Community Risk Ratio for a particular community.

*Community Risk Factor* is the scaling factor used to achieve a distribution of Risk Index Values that better reflects the impacts communities experience from natural hazards.

*PPF*(·) is the probability point function of a triangular distribution with minimum 0.5, mode 1, and maximum 2.

Several methods were evaluated to develop a Community Risk Factor function that maps Expected Annual Loss, Social Vulnerability, and Community Resilience to a distribution of Risk Index Values that provides a better representation of experienced impacts from natural hazards. Given that academic research on quantifying actual versus experienced impacts is still in its relative infancy, consensus knowledge regarding community risk indices and their relationship with experienced impacts from natural hazards is limited. For this reason, three-point estimation was selected to provide a framework for potential Community Risk Factor distributions.

Three-point estimation is a risk analysis technique that is used to model the distribution of a random variable when information is scarce. It involves the use of existing information to establish a range, shape, and mode to generate an approximate distribution for the variable in question.

A variety of shapes, ranges, and modes were evaluated for the Community Risk Factor function, and evaluation criteria for potential candidates were based upon the potential distribution's compatibility with the National Risk Index's underlying principles and its consistency with analogous methods used to calculate the Risk Index in previous versions of the data. Distribution shapes that were considered include the lognormal distribution, the triangular distribution, the program evaluation and review technique (PERT) distribution, and the Wedge distribution. Ranges that were considered include [0,2], [0,5], [0.5,2], [0.2,5], and [1,5], and potential values for the mode varied across ranges and distribution shapes. A triangular distribution with a range of 0.5 to 2 and a mode of 1 was ultimately selected for several reasons. First, the range allows for absolute Values of risk greater than Expected Annual Loss for communities with fewer resources and absolute Values of risk less than Expected Annual Loss for communities with more resources. Second, the bounds are generally consistent with those used to calculate risk in previous versions of the National Risk Index data. Third, its shape highlights communities near both tails of the distribution without assigning extreme Values to a small number of communities at the ends of the distribution. Fourth, since Expected Annual Loss is the primary driver of risk, a value of 1 was selected as the most appropriate mode for the Community Risk Factor.

#### 4.4. Expected Annual Loss

The Expected Annual Loss for each Census tract or county is the average economic loss in dollars resulting from natural hazards each year. Expected Annual Loss is computed for each hazard type and only quantifies loss for relevant consequence types (i.e., buildings, population, or agriculture).

For example, many hazard types only significantly impact buildings and population, so the loss to agriculture is not included in the computation. However, the Expected Annual Loss for Drought only quantifies the damage to agriculture in its computation. Agriculture is considered a relevant consequence type for hazard types where it has historically contributed greater than 1% of the total reported losses. Those are: Cold Wave, Hail, Heat Wave, Hurricane, Inland Flooding, Strong Wind, Tornado, Wildfire, and Winter Weather.

All losses are quantified as a dollar amount. While building and agriculture (crops and livestock) losses are quantified in dollars in the source data, population loss is quantified as the number of fatalities and injuries and must be converted to ensure all Expected Annual Loss Values use a common unit of measurement. Population loss is monetized into a population equivalence value using a Value of Statistical Life approach in which each fatality or ten injuries is treated as \$13.7 million of economic loss, an inflation-adjusted Value of Statistical Life used by the Federal Emergency Management Agency.<sup>8</sup> To adjust for inflation, all historical losses are converted to December 2024 dollars.

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<sup>8</sup> FEMA. (2022). Benefit-cost sustainment and enhancements: Draft Standard Economic Value Methodology Report. Retrieved from [https://www.fema.gov/sites/default/files/documents/fema\\_standard-economic-values-methodology-report\\_092022.pdf](https://www.fema.gov/sites/default/files/documents/fema_standard-economic-values-methodology-report_092022.pdf).

## 5. Expected Annual Loss Factors

The National Risk Index represents natural hazards in terms of Expected Annual Loss, which incorporates data for natural hazard exposure, annualized frequency, and Historic Loss Ratio. A single “mental model” was leveraged throughout all methodological processes in calculating these Expected Annual Loss factors to ensure consistency across the 18 hazard types.

### 5.1. Natural Hazards

The National Risk Index team defines natural hazards as environmental phenomena that have the potential to impact societies and the human environment. These should not be confused with other types of hazards, such as human-caused hazards. For example, a flood resulting from changes in river flows is a natural hazard, while a cyber-attack is considered a human-caused hazard, therefore annualized losses from cyber-attacks are excluded from the National Risk Index.

Natural hazards can also cause secondary natural hazard events that create additional hazards. For example, Volcanic Activity can create other hazards, such as ash and lava spread. The National Risk Index only considers main natural hazard events and not their results or after-effects.

Natural hazards and disasters are different. A natural hazard is the threat of an event that will likely have a negative impact. Natural hazards only become disasters when they significantly impact communities negatively. For example, communities that have high social vulnerability or low community resilience are more likely to experience disasters. The populations in these communities may be disproportionately affected due to social characteristics and capacity restraints which leave them more at-risk to natural hazards. The National Risk Index is designed to help communities understand their relative natural hazard risk and the impacts they could expect during or after a disaster.

The National Risk Index considers 18 hazard types. These hazard types are listed below and described in more detail in the hazard type-specific sections of this report (Sections 6 through 23).

[Avalanche](#)  
[Coastal Flooding](#)  
[Cold Wave](#)  
[Drought](#)  
[Earthquake](#)  
[Hail](#)

[Heat Wave](#)  
[Hurricane](#)  
[Ice Storm](#)  
[Inland Flooding](#)  
[Landslide](#)  
[Lightning](#)

[Strong Wind](#)  
[Tornado](#)  
[Tsunami](#)  
[Volcanic Activity](#)  
[Wildfire](#)  
[Winter Weather](#)

Source data necessary to calculate Expected Annual Loss is not available for all geographic areas for all hazard types. [Table 4](#) identifies where there is insufficient data (ID) for specific states, where there is sufficient data (Y), and territories or where the given hazard type is not applicable (N/A) (e.g., Avalanche in Puerto Rico). This is not to be conflated with instances where there is no calculated Expected Annual Loss.

**Table 4: Insufficiency of Source Data by Hazard Type and State/Territory**

| <i>Hazard Type</i> | <i>Continental<br/>U.S. States<br/>and DC</i> | <i>Alaska</i> | <i>Hawai'i</i> | <i>American<br/>Samoa</i> | <i>Northern<br/>Mariana<br/>Islands</i> | <i>Guam</i> | <i>Puerto<br/>Rico</i> | <i>U.S.<br/>Virgin<br/>Islands</i> |
|--------------------|-----------------------------------------------|---------------|----------------|---------------------------|-----------------------------------------|-------------|------------------------|------------------------------------|
| Avalanche          | Y                                             | Y             | N/A            | N/A                       | N/A                                     | N/A         | N/A                    | N/A                                |
| Coastal Flooding   | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Cold Wave          | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Drought            | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | Y                                  |
| Earthquake         | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Hail               | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | Y                                  |
| Heat Wave          | Y                                             | ID            | ID             | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Hurricane          | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Ice Storm          | Y                                             | Y             | Y              | N/A                       | N/A                                     | N/A         | N/A                    | N/A                                |
| Landslide          | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | ID                                 |
| Lightning          | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | ID                                 |
| Inland Flooding    | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Strong Wind        | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | Y                                  |
| Tornado            | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | Y                      | Y                                  |
| Tsunami            | ID<br>(for Eastern<br>U.S.)                   | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |
| Volcanic Activity  | Y                                             | Y             | Y              | Y                         | Y                                       | N/A         | Y                      | Y                                  |
| Wildfire           | Y                                             | Y             | Y              | ID                        | ID                                      | ID          | ID                     | ID                                 |
| Winter Weather     | Y                                             | Y             | Y              | Y                         | Y                                       | Y           | Y                      | Y                                  |

## 5.2. Natural Hazard Annualized Frequency

The natural hazard annualized frequency is defined as the expected frequency or probability of a hazard occurrence per year. Annualized frequency is derived either from the number of recorded hazard occurrences each year over a given period or the modeled probability of a hazard occurrence each year. The National Risk Index considers that natural hazards can occur in places where they may have not yet been recorded to-date and that hazards may have occurred in locations without being recorded. Therefore, the National Risk Index has built-in minimum representative annualized

frequency values for certain geographical areas and hazard types, such as Hurricane, Ice Storm, Landslide, Tornado, and Tsunami.

### 5.2.1. SELECTING SOURCE DATA

Annualized frequency data were derived from multiple sources and depend on the hazard types. Data sources were identified through public knowledge, guidance by subject matter experts, and research. Examples of selected data sources include the National Weather Service, the National Oceanographic and Atmospheric Administration, the U.S. Geological Survey, the United States Army Corps of Engineers, the Smithsonian databases, and the United States Department of Agriculture. See the hazard type-specific sections ([Sections 6 through 23](#)) for more information on spatial data sources.

### 5.2.2. ANNUALIZED FREQUENCY METHODOLOGY

The annualized frequency is the expected frequency for a given hazard type and measures the actual or expected number of hazard occurrences each year in events or event-days. Not all hazard occurrences are considered relevant for the annualized frequency calculation. Subject matter experts established that some hazard occurrences meet certain criteria to be included as a hazard occurrence capable of causing damage (e.g., Hail size of diameter greater than 0.75 in). See the hazard type-specific sections for more information on these criteria. Annualized frequency is defined as the number of historical occurrences of a hazard type within a known period of record per geographic area, as seen below in [Equation 4](#).

#### Equation 4: Annualized Frequency Equation

$$\text{Annualized Frequency} = \frac{\text{Number of Recorded Hazard Occurrences}}{\text{Period of Record}}$$

In some cases, as with Wildfire and Earthquake, the best available source data consists of probabilistic statistics contained in raster files that are used to compute an annualized frequency. In these cases, the annualized frequency value represents the probability of a hazard occurrence in a given year.

For hazard types that track actual hazard occurrences, the historical hazard occurrence count quantifies either the number of distinct hazard events that have occurred (e.g., Hurricanes to hit the area) or the count of days on which a hazard has occurred (e.g., on how many days a Heat Wave event was reported). The determination of whether hazard occurrence was defined by distinct event or event-days was based on subject matter expert review of the source data. This determination depended on how hazard occurrence was recorded as well as how losses were reported. [Table 5](#) gives the hazard occurrence basis (event or event-day) for each hazard type.



**Table 5: Geographic Level of Historical Hazard Occurrence Count Determination and Hazard Occurrence Basis**

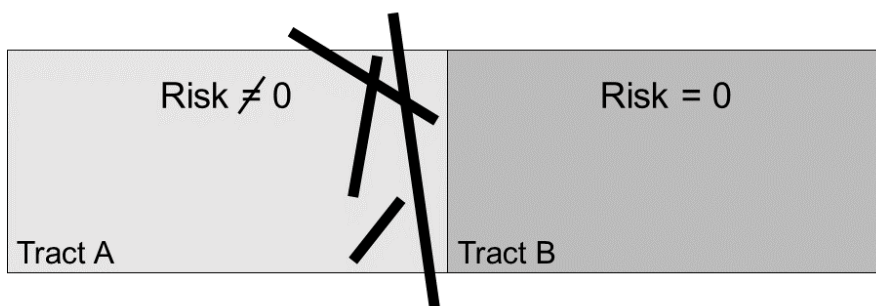
| <i>Hazard Type</i> | <i>Hazard Occurrence Basis</i> | <i>Geographic Level of Historical Hazard Occurrence Count Determination</i> | <i>Annualized Occurrence Measure</i> |
|--------------------|--------------------------------|-----------------------------------------------------------------------------|--------------------------------------|
| Avalanche          | Distinct events                | County                                                                      | Average annualized frequency         |
| Coastal Flooding   | No event count                 | No event count                                                              | Average annualized frequency         |
| Cold Wave          | Event-days                     | Census block                                                                | Annualized probability               |
| Drought            | Event-days                     | Census tract                                                                | Average annualized frequency         |
| Earthquake         | No event count                 | No event count                                                              | Annualized probability               |
| Hail               | Distinct events                | 49-km Fishnet                                                               | Average annualized frequency         |
| Heat Wave          | Event-days                     | Census block                                                                | Average annualized frequency         |
| Hurricane          | Distinct events                | 49-km Fishnet                                                               | Average annualized frequency         |
| Ice Storm          | Event-days                     | 49-km Fishnet                                                               | Average annualized frequency         |
| Landslide          | Distinct events                | Census tract                                                                | Average annualized frequency         |
| Lightning          | Event-days                     | Census block                                                                | Annualized probability               |
| Inland Flooding    | Event-days                     | County                                                                      | Average annualized frequency         |
| Strong Wind        | Distinct events                | 49-km Fishnet                                                               | Average annualized frequency         |
| Tornado            | Distinct events                | 49-km Fishnet (by sub-type)                                                 | Average annualized frequency         |
| Tsunami            | Distinct events                | Census tract                                                                | Annualized probability               |
| Volcanic Activity  | Distinct events                | Census block                                                                | Annualized probability               |
| Wildfire           | No event count                 | No event count                                                              | Annualized probability               |
| Winter Weather     | Event-days                     | Census block                                                                | Average annualized frequency         |

While the data generally presents information at the Census tract and county level, often the data used to determine this information are captured at either a lower or higher level. Predominantly, Expected Annual Loss factors are assessed at the Census block level, so the number of hazard occurrences (events or event-days) is determined for each Census block.

Depending on the nature of the hazard type and its source data, the hazard occurrence count used to calculate annualized frequency are initially captured at the Census block, Census tract, county, or 49-by-49 km fishnet grid cell level. See each hazard type's annualized frequency section (e.g., [Section 6.5](#), [Section 7.4](#), etc.) for the specific hazard occurrence count methodology. [Table 5](#) provides the geographic level at which hazard occurrence count information is determined for use in annualized frequency calculations for each hazard type.

For large geographic areas and areas with a statistically significant number of hazard occurrences recorded, [Equation 4](#) is used as one approach for calculating annualized frequency for some hazard types. However, for hazard types with few hazard occurrences historically recorded, due to urban bias and varying demographics across the United States, this equation is not always accurate or representative. Additionally, as geographic boundaries are partitioned into much smaller regions (counties, Census tracts, and Census blocks), further challenges are uncovered resulting from the fact that geographic areas that have not been historically impacted by a hazard type and/or recorded hazard occurrences are being calculated as having no risk from that hazard type. Since the Expected Annual Loss equation is multiplicative, any individual factor of zero results in a Risk Index Score of 0.

Consider an example ([Figure 4](#)) where four Tornadoes hit a single Census tract ("Tract A") near its geographic border. Using [Equation 4](#), the annualized frequency for "Tract A" would be calculated using a 4 in the numerator. However, given the Tornado event locations (specifically, their proximity to the neighboring tracts), these four events could easily have occurred within, say, "Tract B." Therefore, "Tract B" should not be represented as having no (zero) risk, and yet, it would be zero if the annualized frequency was deemed to be zero based on the fact that no Tornado has historically occurred in "Tract B." Natural hazard events cannot be expected to respect arbitrarily drawn political boundaries. Thus, in evaluating risk, hazard occurrence definition should account for events in nearby Census blocks or Census tracts that easily could have impacted a given community.



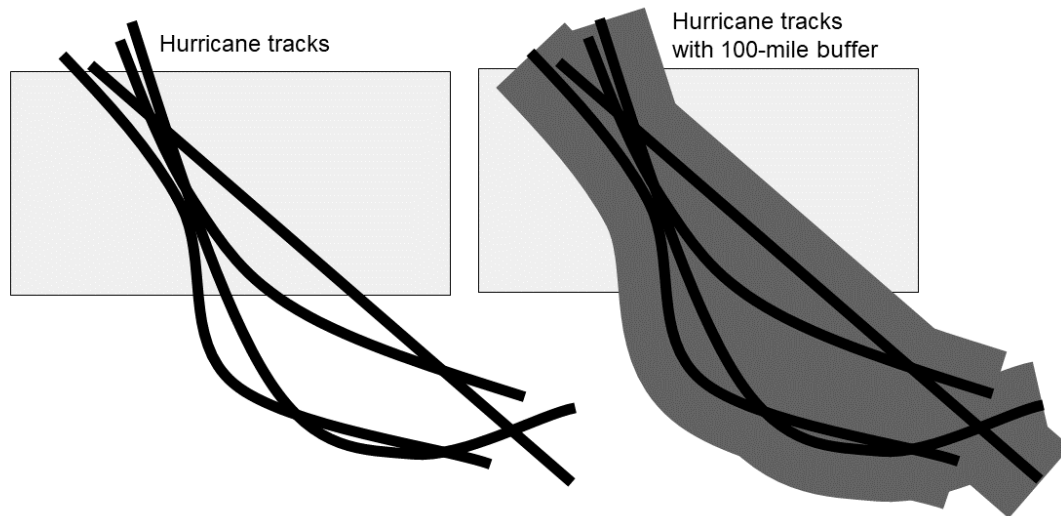
**Figure 4: Example of the Issues with a Simplistic Annualized Frequency Methodology**

Three main solutions were incorporated to spread the area of hazard influence used to calculate annualized frequency and/or exposure. Hazard type-specific annualized frequency methodologies may use some or all of these approaches:

1. **Hazard Occurrence Counting Using a 49-by-49 km<sup>9</sup> Fishnet Grid:** This approach involves creating a fishnet grid covering the United States and counting the number of hazard occurrences (event or event-days) within each cell. Communities within the cell inherit the hazard occurrence count (or receive an area-weighted hazard occurrence count when intersecting multiple cells; see [Section 5.2.3 Data Aggregation](#)) and annualized frequency is then calculated according to [Equation 4](#). Hazard types using this approach include Hail, Hurricane, Ice Storm, Strong Wind, and Tornado.
2. **Minimum Annual Frequency:** A minimum annual frequency is assigned to communities that have not experienced a hazard occurrence recorded by the source data but are determined to be at some risk due to their location (see the [Determining County-Level Possibility of Hazard Occurrence section](#)). Appropriate minimum annual frequency values for most hazards were identified by hazard type subject matter experts. The estimated values are low given the fact that historical occurrences had never been recorded over the period of record, which sometimes dated back multiple centuries. Minimum annual frequency values were typically defined in the format of “once in the period of record,” or similar. Hazard types using this approach include Avalanche, Hurricane, Ice Storm, Landslide, Inland Flooding, Tornado, and Tsunami.
3. **Hazard Occurrence Shape Buffering:** Hazard types with widespread and/or unpredictable locations are buffered using expert-determined distances to create more representative areas with potential exposure to hazard types. Buffering also allows occurrences with relatively small surface areas to be smoothed together into general representative shapes to eliminate gaps that may exist between historically recorded hazard occurrences (see [Figure 5](#)). Hazard types using this approach include Hail, Hurricane, Strong Wind, Tornado, Tsunami, and Volcanic Activity.

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<sup>9</sup> The 49-by-49 km fishnet cell size was chosen to approximate the average area of a county.



**Figure 5: Example of Buffering Hazard Occurrences to Determine Areas Applicable to Minimum Annual Frequency Values**

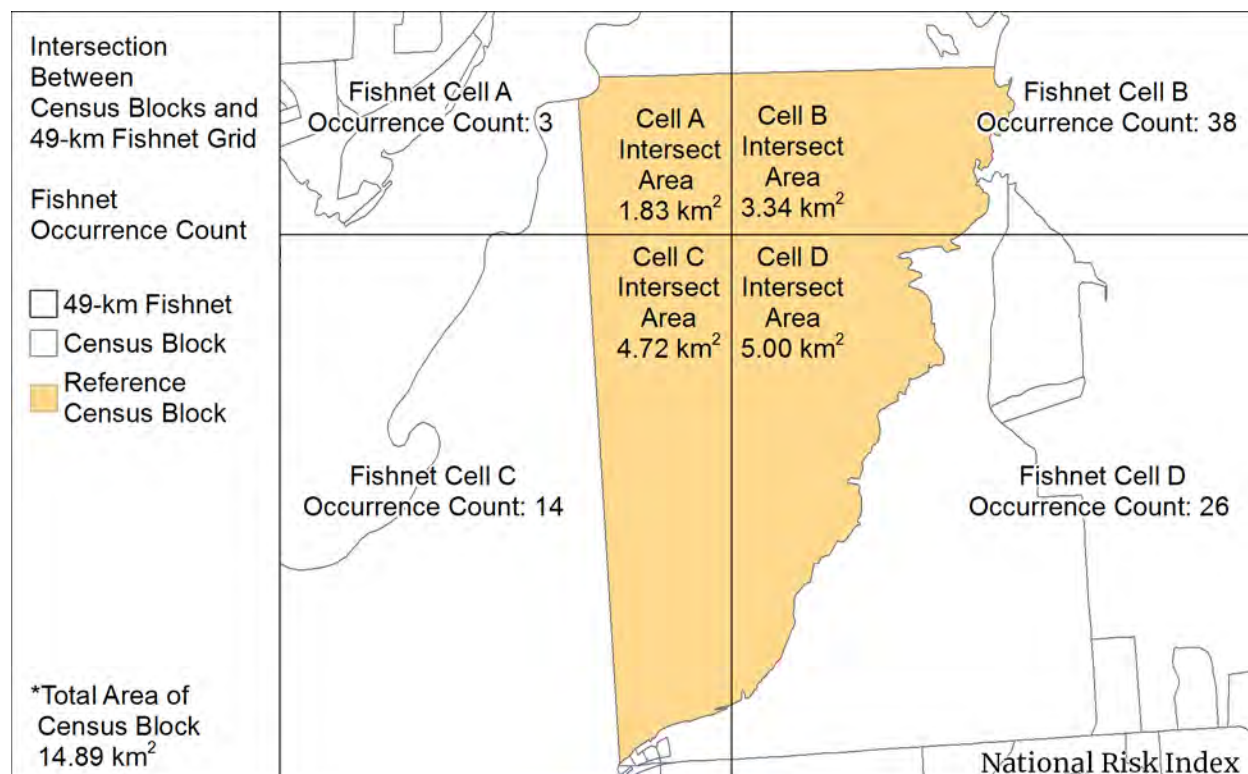
Some hazard types do not require any of these solutions due to the nature of the source data or the widespread prevalence of the hazard type. For example, the spatial data for Cold Wave, Heat Wave, and Winter Weather occurrences cover areas the size and shape of National Weather Service Forecast Zones and counties. These hazard occurrences happen across the entire United States, so it is not necessary to spread the hazard types' area of influence any further.

### 5.2.3. DATA AGGREGATION

In most instances, annualized frequency is calculated first at the Census block level. In cases where the hazard occurrence count is evaluated at the fishnet level (see [Table 5](#)), the Census block inherits the hazard occurrence count from the fishnet cell that encompasses it, or, if a Census block intersects multiple fishnet cells, an area-weighted count is calculated as computed in [Equation 5](#). Applying this equation to the example in [Figure 6](#) results in a Census block hazard occurrence count of about 22. This fishnet-aggregated count is used to calculate the Census block annualized frequency.

#### **Equation 5: Census Block Area-Weighted Fishnet Hazard Occurrence Count Calculation**

$$\text{Census Block Hazard Occurrence Count} = \frac{\sum(\text{Fishnet Hazard Occurrence} \times \text{Area of Fishnet Intersection})}{\text{Area of Census Block}}$$



**Figure 6: Aggregation from Fishnet Cell to Census Block Example**

The National Risk Index aggregates data from the Census block to the Census tract and county level, usually by leveraging area-weighted aggregation as computed in [Equation 6](#). These Census tract- and county-level annualized frequency values may not exactly match that of dividing the number of historical hazard occurrences at the Census tract and county level by the period of record, as they are based on an area-weighted aggregation of Census block hazard occurrence values.

**Equation 6: Census Tract and County Annualized Frequency Aggregations**

*Census Tract Annualized Frequency*

$$= \frac{\sum(\text{Census Block Annualized Frequency} \times \text{Area of Census Block})}{\text{Area of Census Tract}}$$

*County Annualized Frequency*

$$= \frac{\sum(\text{Census Block Annualized Frequency} \times \text{Area of Census Block})}{\text{Area of County}}$$

For a few hazard types, annualized frequency is calculated at the Census tract level, after which the Census block simply inherits the value of its parent Census tract (see [Table 5](#)). Avalanche and Inland Flooding are the only hazard types for which annualized frequency is calculated at the county level directly, and the Census tracts and Census blocks then inherit the value of their parent county.

## 5.3. Exposure

Exposure is defined as the representative value of buildings, population, or agriculture potentially exposed to a natural hazard occurrence. Data sources with the best available national-level data for each hazard type were selected to perform a spatial analysis and compute areas of exposure.

### 5.3.1. SELECTING SOURCE DATA

The initial spatial processing of the source data for each hazard type is used to identify areas of exposure. Data sources were selected for their accuracy, length of period of record, and spatial component, based on the best available, national-level data per hazard type. Sources were identified through public knowledge, subject matter expert recommendations, and research. Providers of exposure data include:

- [National Oceanic and Atmospheric Administration](#)
- [Spatial Hazard Events & Losses Database for the United States \(SHELDUS\)](#)
- [United States Army Corps of Engineers](#)
- [U.S. Geological Survey](#)
- [United States Department of Agriculture](#)
- [Federal Emergency Management Agency](#)

### 5.3.2. CONSEQUENCE TYPES

A consequence is defined in the National Risk Index as economic loss or bodily harm to individuals that is directly caused by a hazard occurrence. Consequences of hazard occurrences are categorized into three distinct types: buildings, population, and agriculture. A consequence type is included for a hazard type when the consequence type losses contribute greater than 1% of the total reported losses for the hazard type.

#### Buildings

Building exposure value is defined as the dollar value of the buildings determined to be exposed to a hazard according to a hazard type-specific methodology. The maximum possible building exposure of an area (Census block, Census tract, or county) is its building value as recorded in Hazus 6.0,<sup>10</sup> which are derived from the 2020 Census<sup>11</sup> and then inflation adjusted to December 2024.

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<sup>10</sup> FEMA. (2022). *Hazus 6.0 Release*. Retrieved from <https://msc.fema.gov/portal/resources/hazus>.

<sup>11</sup> United States Census Bureau. (2020). *2020 Census*. Retrieved from <https://www.census.gov/programs-surveys/decennial-census/decade.2020.html>.

## Population

Population exposure is defined as the estimated number of people determined to be exposed to a hazard according to a hazard type-specific methodology. The maximum possible population exposure of an area (Census block, Census tract, or county) is its population as recorded in Hazus 6.0. Population loss is monetized into a population equivalence value using a Value of Statistical Life approach in which each fatality or ten injuries is treated as \$13.7 million of economic loss (2024 dollars).

## Agriculture

Agriculture exposure value is defined as the estimated dollar value of the crops and livestock determined to be exposed to a hazard according to a hazard type-specific methodology. This is derived from the United States Department of Agriculture 2017 Census of Agriculture<sup>12</sup> county-level value of crop and pastureland with 2018 values for the United States Territories. All dollar values are inflation-adjusted to 2024 dollars.

### 5.3.3. EXPOSURE METHODOLOGY

Exposure is typically calculated at the Census block level and then aggregated to the Census tract and county levels by summing the Census block exposure values within the parent Census tract or parent county. See the hazard type-specific exposure sections ([Sections 6 through 23](#)) for more information.

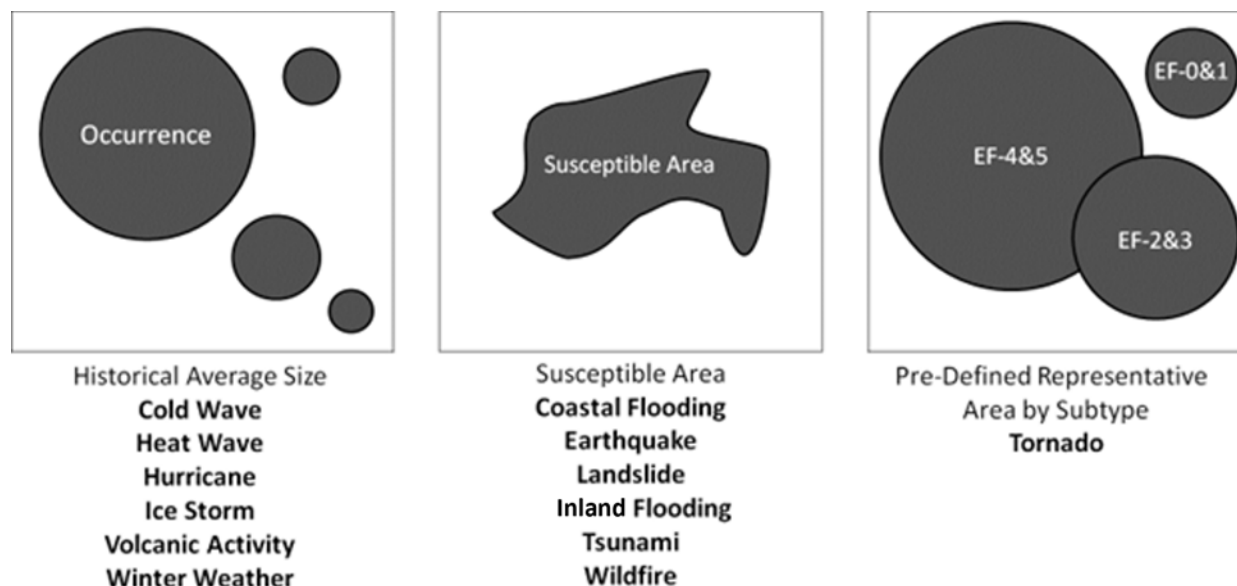
Some hazard type exposure areas are represented as polygons in the source data, while others are represented as points, lines, or raster cells. Exposure is based on either historical hazard occurrence locations or areas of identifiable risk (e.g., Tsunami inundation zones). Eventually, every relevant record in the source data is processed into a polygon via a hazard type-specific methodology. This polygon represents an area of exposure to the hazard type.

To calculate the hazard type's representative size for a given area, the National Risk Index leverages a few techniques, such as using subject matter experts to define a single representative hazard type size, calculating historical average hazard occurrence sizes, or defining the size of probabilistic/susceptible zones for hazard types within the area of interest using existing source data ([Figure 7](#)).

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<sup>12</sup> USDA. (2017). *2017 Census of Agriculture*. Retrieved from <https://www.nass.usda.gov/Publications/AgCensus/2017/index.php>.





**Figure 7: Examples of Representative Hazard Type Size**

To estimate exposure, the hazard occurrence or susceptible zone polygons are intersected with the appropriate administrative layer polygons, and the resulting intersect shape defines the area of exposure. Once the area of exposure is defined, one of three generalized approaches is executed within the processing database to estimate the exposure values within the administrative area. The approach used for a hazard type was determined by the hazard type's recorded historical hazard occurrences, hazard susceptibility maps, and subject matter experts.

[Appendix B – Hazard Data Characteristics Comparison](#) describes the type of exposure method used for each of the 18 hazard types. The general approaches to modeling exposure include:

1. **Widespread Hazard Occurrence Exposure.** The entire administrative area is considered to be exposed. This approach is leveraged for hazard types where the extent likely spans the entire administrative area, and the boundaries are indefinable.
2. **Developed Area/Agriculture Area Density Concentrated Exposure.** The determined area of exposure intersected with the developed area is multiplied by the density of either the population or building value within the developed land of the administrative area to calculate the worst-case concentration of consequence for the hazard type. To estimate agriculture exposure, this method uses the density of crop and livestock values within the crop or pastureland area. For some hazards, the area of exposure is intersected with the entire administrative area instead of the developed or agriculture area.
3. **Pre-Defined Representative Exposure.** Subject matter experts define a default, representative exposure value, or area.



### Approach 1. Widespread Hazard Occurrence Exposure

For certain hazard types where extent is widespread with indefinable boundaries, the entire community is considered exposed. For these hazard types, exposure values are defined as the entire community's building value, agriculture value, or population as recorded by Hazus 6.0 or the 2017 Census of Agriculture.

### Approach 2. Developed Area/Agriculture Area Density Concentrated Exposure

Exposure is calculated for most of the hazard types using the developed area density approach. This approach uses the area of the hazard occurrence exposure shape (intersection of hazard occurrence shape with the Census block) multiplied by the developed area density of the administrative area to generate the worst-case representative building value or population that could be exposed to a future hazard occurrence within the area. This can result in exposure values exceeding the total values of the Census block. In these cases, exposure is capped at the total Census block value.

The Hazus 6.0 data provides building value and population estimates at each administrative reference layer (Census block, Census tract, and county). For certain hazard types, a density estimate was needed for the hazard type's exposure calculation. Rather than only calculating an average density value for each administrative layer (i.e., by dividing the population of a Census block by the area of the Census block), the developed area density is calculated based on the developed area of the Census block and the crop and livestock area densities are based on the crop or pastureland area of the Census block defined by the process described in [Dasymetric Developed Area, and Agricultural Area Census Blocks](#).

With an estimate of the developed area and crop and pastureland area for each record of the administrative reference layers, densities were then calculated. Using Hazus data's Building Stock Value and Population estimates for each administrative layer, the ratio of developed areas within an administrative reference over its whole area was used to calculate the developed area building density and developed area population density.

To compute the developed area building and population densities, the building and population values of the administrative layer (Census block, Census tract, or county) are divided by the total developed area of the administrative layer, as in [Equation 7](#).

#### Equation 7: Census Block Developed Area Building and Population Density

$$BldgValueDen_{CB} = \frac{BldgValue_{CB}}{DevArea_{CB}}$$

$$PopDen_{CB} = \frac{Pop_{CB}}{DevArea_{CB}}$$

where:

|                     |                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| $BldgValueDen_{CB}$ | is the developed area building value density calculated at the Census block level (in dollars per square kilometer). |
| $BldgValue_{CB}$    | is the total building value of the Census block as recorded in Hazus 6.0 (in dollars).                               |
| $DevArea_{CB}$      | is the total developed area of the Census block (in square kilometers).                                              |
| $PopDen_{CB}$       | is the developed area population density calculated at the Census block level (in people per square kilometer).      |
| $Pop_{CB}$          | is the total population of the Census block as recorded in Hazus 6.0.                                                |

For agriculture, the United States Department of Agriculture 2017 Census of Agriculture provides an estimated dollar value of crop and livestock within each county. The county crop value is divided by the total crop area of the county to find its crop value density, and the county livestock value is divided by the total pastureland area of the county to find its pastureland value density. The resulting rates are then applied to the Census blocks within the county respectively for the Census block's crop and pastureland area (see [Equation 8](#)).

#### Equation 8: County Agriculture Value Density

$$AgValueDen_{Co} = \frac{AgValue_{Co}}{AgArea_{Co}}$$

where:

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| $AgValueDen_{Co}$ | is the crop or livestock value density calculated at the county level (in dollars per square kilometer).       |
| $AgValue_{Co}$    | is the total crop or livestock value of the county as reported in the 2017 Census of Agriculture (in dollars). |
| $AgArea_{Co}$     | is the total crop or pastureland area of the county (in square kilometers).                                    |

### Approach 3. Pre-defined Representative Exposure

Avalanche and Tornado each have unique methods of calculating exposure. For Avalanche, a single exposure value for population, defined by subject matter experts, is pre-determined and assigned to all areas deemed at risk of Avalanche occurrences. For Tornado, an average historical damage area is calculated for each EF-scale grouping subtype. For each sub-type, the representative footprint area is multiplied by the average building and population densities of the Census block to find exposure.

#### **5.3.4. DATA AGGREGATION**

Exposure is calculated at the Census block level and then is aggregated to the Census tract and county level by summing the Census block exposure values within the parent Census tract or parent county (with the exception of Earthquake, which is initially calculated at the Census tract level). Detailed methodologies per hazard type are explained in the hazard type-specific sections of this report ([Sections 6 through 23](#)).

### **5.4. Historic Loss Ratio**

The Historic Loss Ratio is the representative percentage of a location's hazard type exposure that experiences loss due to a hazard occurrence or the average rate of loss associated with the hazard occurrence. The Historic Loss Ratio is an area-specific estimate of the percentage of the exposed consequence type (building value, population, or agriculture value) expected to be lost due to a single hazard occurrence. In concept, it is the average of the loss ratios associated with past hazard occurrences and is used to estimate the potential impact of a future hazard occurrence.

To begin the determination of this value, a Loss Ratio per Basis (event or event-day) is calculated for each historical loss-causing hazard occurrence as the value of the loss divided by the exposed value for each relevant consequence type. A Bayesian credibility analysis is then performed with sets of individual Loss Ratios per Basis at multiple geographic levels (e.g., county, surrounding area, hazard specific regional area, etc.) and community characteristic groupings (e.g., topography, land cover, proximity to target county, urban-rural designations) to better balance Historic Loss Ratio accuracy with geographic precision and community characteristics. The resulting Historic Loss Ratio (by hazard type-consequence type combination) is a Bayesian-adjusted ratio that is the summed weighted average of various geospatial and community characteristic groupings of the consequence Loss Ratio per Basis sets for the hazard type. This resulting Bayesian-adjusted Historic Loss Ratio value—computed for each county-hazard type-consequence type combination—serves as a foundation for estimating the ratio of loss to exposed consequence type value that is expected from a single hazard occurrence.

Computation of the Historic Loss Ratio also considers zero-loss hazard occurrences for some hazard types prior to performing the Bayesian credibility spatial modeling analysis. This ensures that Historic Loss Ratio is multiplied by annualized frequency within the Expected Annual Loss equation without overinflating the Expected Annual Loss value. Additionally, the Bayesian-adjusted Historic Loss Ratio values are evaluated against historic loss totals to avoid underestimating losses in areas with credible historic data and may be subject to additional smoothing techniques to improve estimates in neighboring counties.

#### **5.4.1. SOURCE DATA**

The National Risk Index uses multiple sources for its data to calculate Historic Loss Ratios. For documentation of historical hazard event losses, the National Risk Index primarily relies on the Spatial Hazard Event and Loss Database for the United States (SHELDUS) data set. The National Center for Environmental Information Storm Events Database is used to support Historic Loss Ratio

work with the Tornado and Inland Flood hazard. The National Center for Health Statistics’ Urban-Rural Classification Scheme for Counties is used to determine the time-dependent classification of a county at the date of a hazard event.

Source Data: SHELDUS

Historical Losses source data provider: [Arizona State University \(ASU\), Spatial Hazard Event and Loss Database for the United States \(SHELDUS\)](#) <sup>13</sup>

Arizona State University’s SHELDUS loss data are used to calculate Historic Loss Ratio for most hazard types. SHELDUS provides county-level data that corresponds to nearly all of the hazard types. It also offers a further degree of description by identifying hazard occurrences by peril type as well as the hazard event itself. SHELDUS represents the best available national dataset on building, population, and agriculture losses. However, it is to be understood that the losses reported in SHELDUS often do not capture all losses of an event or underreport impacts from a peril type overall. That is to say, SHELDUS losses should generally be considered a baseline lower bound of historical impacts and calculated values in excess of SHELDUS figures do not inherently imply overestimations of risk.

Through its website, Arizona State University provides summary SHELDUS data that aggregates property damage, crop losses, injuries, and fatalities due to a peril or hazard by month, year, and county since 1960. However, Arizona State University allowed unaggregated data collected at the peril occurrence level to be shared with the Federal Emergency Management Agency for the development of the National Risk Index. Much of this data was originally collected by the National Oceanographic and Atmospheric Administration and published in the monthly Storm Data and Unusual Weather Phenomena report, though information may have also been extracted from additional resources. The records have been processed by Arizona State University to enable appropriate spatial aggregation by distributing losses among multiple counties for events with losses reported at the forecast zone or even the state level. For example, in [Table 6](#), a Winter Weather injury is recorded in SHELDUS records as 0.5 for two neighboring counties. Both occurred in the same date range and have the same level of property damage. This implies that the specific county where the injury occurred could not be determined because the reporting covered two counties, so Arizona State University split the injury evenly between them.

Table 6: Sample SHELDUS Peril Occurrence Data

| SHELDUS ID | Hazard Begin Date | Hazard End Date | County FIPS | Fatalities | Injuries | Property Damage | Crop Damage | Peril |
|------------|-------------------|-----------------|-------------|------------|----------|-----------------|-------------|-------|
| 25773      | Jan. 22, 1999     | Jan. 22, 1999   | 01033       | 0          | 0        | 5000            | 0           | Hail  |

<sup>13</sup> Center for Emergency Management and Homeland Security, Arizona State University. (2024). SHELDUS, Version 22.0. [online database]. Retrieved from <https://cemhs.asu.edu/sheldus>.

| <b>SHELDUS ID</b> | <b>Hazard Begin Date</b> | <b>Hazard End Date</b> | <b>County FIPS</b> | <b>Fatalities</b> | <b>Injuries</b> | <b>Property Damage</b> | <b>Crop Damage</b> | <b>Peril</b> |
|-------------------|--------------------------|------------------------|--------------------|-------------------|-----------------|------------------------|--------------------|--------------|
| 26427             | Sept. 14, 1999           | Sept. 14, 1999         | 04013              | 0                 | 2               | 7,000,000              | 0                  | Wind         |
| 27491             | Sept. 18, 1999           | Sept. 18, 1999         | 12099              | 0                 | 0               | 1,000                  | 0                  | Hail, Wind   |
| 9884227           | Dec. 17, 2010            | Dec. 20, 2010          | 06003              | 0                 | 0.5             | 100,000                | 0                  | Storm-Winter |
| 9884228           | Dec. 17, 2010            | Dec. 20, 2010          | 06017              | 0                 | 0.5             | 100,000                | 0                  | Storm-Winter |

### Source Data: National Centers for Environmental Information Storm Events Database

**Historical Losses source data provider:** [National Centers for Environmental Information Storm Events Database](#)<sup>14</sup>

Much of the data originates from the Storm Events Database, which records loss data for building damage, agriculture damage, and population death and injuries due to hazard and peril occurrences. Unlike SHELDUS, the Storm Events Database includes peril occurrences regardless of whether there was any reported loss. Some records in the Storm Events Database are designated with a forecast zone rather than a county, so each must be joined to a county via a time dependent county-forecast zone correlation table.

### Source Data: National Center for Health Statistics

**Historical Losses source data provider:** [2023 National Center for Health Statistics Urban-Rural Classification Scheme for Counties](#),<sup>15</sup>

The Bayesian credibility process used to generate Historic Loss Ratios for most hazard types blends a combination of county, surrounding area, hazard-specific regional area, and national loss ratio sets to balance regional variation and reduce statistical noise. However, problems can arise when blending loss ratios across communities with vastly different exposures. For instance, a credible regional/surrounding area average loss ratio could be driven by small outlier loss events in rural areas with low exposure. These outlier events can then influence the Historic Loss Ratio of a neighboring urban area with very high exposure leading to significant overestimation of Expected Annual Loss. To help address this, counties are temporally divided into two groups: (1) large central metropolitan areas, as defined by the National Center for Health Statistics Urban-Rural Classification Scheme where NCHSURCode = 1 and (2) all other counties (non-Large Central Metropolitan Areas, where NCHSURCode <> 1). This means that Loss Ratio per Basis loss ratios from non-Large Central

<sup>14</sup> NCEI. (2025). *Storm Events Database, Version 3.1*. [online database]. Retrieved from <https://www.ncdc.noaa.gov/stormevents/>.

<sup>15</sup> NCHS. (2023). *2023 NCHS Urban-Rural Classification Scheme for Counties*. Retrieved from [https://www.cdc.gov/nchs/data/series/sr\\_02/sr02\\_166.pdf](https://www.cdc.gov/nchs/data/series/sr_02/sr02_166.pdf)

Metropolitan Areas cannot influence the Historic Loss Ratio estimates of Large Central Metropolitan Areas and vice versa. This approach is applied to specific hazard-consequence types utilizing the Historic Loss Ratio Filter function.

The National Center for Health Statistics Urban-Rural Classification Scheme provides a classification code value (NCHSURCode) for each county and county equivalent. The NCHSURCode values are 1- Large Central metro, 2-Large fringe metro, 3-Medium metro, 4-Small metro, 5-Micropolitan, 6- Noncore (see footnote reference for expanded definition of each code). There have been four versions of the National Center for Health Statistics Urban-Rural Classification Scheme: 2001, 2006, 2013, and 2023 releases. As the population and characteristics of a county can change over time, a specific county can have multiple urban-rural code designations over time. When used in Historic Loss Ratio generation, each hazard-county loss event uses the event date to determine which vintage of the County National Center for Health Statistics Urban-Rural Code will be associated with the record, see [Table 7](#).

**Table 7: Sample of County Urban-Rural Code Designations with Historic Loss Ratio usage Range**

| <i>County FIPS</i> | <i>County Name</i>        | <i>Vintage</i>        | <i>NCHS Urban-Rural Code (NCHSURCode)</i> | <i>Date Range for Application to Hazard Events in HLR</i> |
|--------------------|---------------------------|-----------------------|-------------------------------------------|-----------------------------------------------------------|
| 01073              | Jefferson County, Alabama | 2001; 1990 Based Code | 3                                         | 1960 – 1999                                               |
| 01073              | Jefferson County, Alabama | 2006 Code             | 1                                         | 2000 – 2009                                               |
| 01073              | Jefferson County, Alabama | 2013 Code             | 1                                         | 2010 – 2019                                               |
| 01073              | Jefferson County, Alabama | 2023 Code             | 2                                         | 2020 – 2029                                               |
| 48085              | Collin County, Texas      | 2001; 1990 Based Code | 2                                         | 1960 – 1999                                               |
| 48085              | Collin County, Texas      | 2006 Code             | 2                                         | 2000 – 2009                                               |
| 48085              | Collin County, Texas      | 2013 Code             | 1                                         | 2010 – 2019                                               |
| 48085              | Collin County, Texas      | 2023 Code             | 1                                         | 2020 – 2029                                               |

#### 5.4.2. PREPARING HAZARD LOSS DATA FOR USE IN HISTORIC LOSS RATIO

Both SHELDUS and National Centers for Environmental Information sources document historical events via a peril designation. Sets of perils are associated with a hazard definition. The National Risk Index's hazard definitions are similar to those of SHELDUS; however, they are not identical. For

example, SHELDUS classifies all flooding perils under the hazard Flood, while the National Risk Index explores two flooding hazard types (Coastal and Inland) and classifies the different flooding perils accordingly. Peril-level data are mapped via a control table in the processing database to the appropriate hazard types (see [Table 8](#)).

**Table 8: Source Data Peril to National Risk Index Hazard Mapping**

| <i>National Risk Index Hazard Type</i> | <i>Perils in SHELDUS</i>                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Avalanche                              | Avalanche, Avalanche-Snow, Snow-Slide                                                                                                                                                                                                                                                                                                                                                            |
| Coastal Flooding                       | Coastal, Coastal Storm, Flood-Coastal, Flood-Tidal                                                                                                                                                                                                                                                                                                                                               |
| Cold Wave                              | Cold Wave, Freeze, Frost, Wind Chill                                                                                                                                                                                                                                                                                                                                                             |
| Drought                                | Drought                                                                                                                                                                                                                                                                                                                                                                                          |
| Earthquake                             | Earthquake, Fire-following Earthquake, Landslide following Earthquake, Liquefaction                                                                                                                                                                                                                                                                                                              |
| Hail                                   | Hail                                                                                                                                                                                                                                                                                                                                                                                             |
| Heat Wave                              | Heat, Heat Wave                                                                                                                                                                                                                                                                                                                                                                                  |
| Hurricane                              | Cyclone-Extratropical, Cyclone-Subtropical, Cyclone-Unspecified, Hurricane/Tropical Storm, Nor'easter, Storm Surge, Tropical Depression, Tropical Storm                                                                                                                                                                                                                                          |
| Ice                                    | Ice Storm                                                                                                                                                                                                                                                                                                                                                                                        |
| Inland Flooding                        | Flood-Flash, Flood-Ice Jam, Flooding, Flood-Lakeshore, Flood-Lowland, Flood-Inland, Flood-Small Stream, Flood-Snowmelt in SHLEDUS used for Inland Flooding Historic Loss Ratio<br><br>Hail Flooding, Lakeshore Flood, Thunderstorm Winds/Flood, Thunderstorm Winds/Flash Flood, Thunderstorm Winds/Flooding in National Centers for Environmental Information used for Inland Flooding Frequency |
| Landslide                              | Debris Flow, Landslide, Landslide-Slump, Mass Movement, Mud Flow, Mudslide, Rockslide, Rockfall                                                                                                                                                                                                                                                                                                  |
| Lightning                              | Fire-St Elmo's, Lightning                                                                                                                                                                                                                                                                                                                                                                        |
| Strong Wind                            | Derecho, Wind, Wind-Straight Line                                                                                                                                                                                                                                                                                                                                                                |
| Tornado                                | Fire-Tornado, Tornado, Waterspout, Wind-Tornadic, Wind-Vortex                                                                                                                                                                                                                                                                                                                                    |
| Tsunami                                | Tsunami, Tsunami/Seiche                                                                                                                                                                                                                                                                                                                                                                          |
| Volcanic Activity                      | Ashfall, Lahar, Lava Flow, Pyroclastic Flow, Vog, Volcano                                                                                                                                                                                                                                                                                                                                        |
| Wildfire                               | Fire-Brush, Fire-Bush, Fire-Forest, Fire-Grass, Wildfire                                                                                                                                                                                                                                                                                                                                         |
| Winter Weather                         | Blizzard, Snow Squall, Storm-Winter, Winter Weather                                                                                                                                                                                                                                                                                                                                              |

SHELDUS loss records were acquired for all perils and all counties in the United States. Loss types include property damage, injuries, fatalities, and crop damage. Property damage and crop damage are quantified in nominal dollars as they were reported at the time the loss occurred. The loss records utilized for the Historic Loss Ratio computation of most hazard types range from January 1996 through December 2023 as loss data captured during and after 1996 were deemed to be the most accurately and uniformly collected due to the standardization of collection practices that began in 1995. However, data from January 1960 to December 2023 are used to compute Historic Loss Ratios for two hazard types: Tornado (EF-2 and above) and Volcanic Activity (see [Section 19.6](#) and [Section 21.5](#)). Older event record data are also used to identify which counties had ever experienced losses for a specific hazard type, ensuring that these were set in the processing database as counties where there was some possibility of the hazard type occurring (see [Determining County-Level Possibility of Hazard Occurrence section](#)).

Not all perils to which loss is attributed in SHELDUS and National Centers for Environmental Information are included as National Risk Index hazard types. However, all SHELDUS records that attribute loss to at least one National Risk Index hazard type is extracted. Loss records in SHELDUS can attribute losses for a single loss-causing event to multiple perils. For example, losses from a single storm can be attributed to Wind, Hail, Tornado, and Severe Storm/Thunderstorm. In the processing database, these multi-peril occurrence records are expanded to multiple records, each attributing a portion of the total loss to a single hazard type. The loss reallocation in these cases does not estimate what degree of loss may be due to perils not included in the hazard types, like Severe Storm/Thunderstorm. Instead, a conservative approach is taken that assumes that all economic loss is due to National Risk Index hazard types.

Loss reallocation for each relevant consequence type is based on comparisons between the typical loss caused by each hazard type (see [Table 9](#)). To arrive at these percentages, loss attributed to a single hazard type was aggregated across the 1996-2023 period of record for each consequence type, and this loss was compared to the aggregated loss of the other hazard types within the combination as a portion of the combined loss of all hazards in the combination.

**Table 9: Loss Allocation by Hazard Type Combination**

| <i>Hazard Type Combination</i> | <i>Building</i> | <i>Population</i> | <i>Agriculture</i>     |
|--------------------------------|-----------------|-------------------|------------------------|
| Hail, Strong Wind              | 90%/10%         | 50%/50%           | 50%/50%                |
| Hail, Strong Wind, Tornado     | 35%/5%/60%      | 20%/20%/60%       | 40%/40%/20%            |
| Hail, Tornado                  | 35%/65%         | 25%/75%           | 70%/30%                |
| Lightning, Strong Wind         | 45%/55%         | 50%/50%           | 10% <sup>16</sup> /90% |
| Strong Wind, Tornado           | 5%/95%          | 20%/80%           | 60%/40%                |

<sup>16</sup> Agriculture is not used as a consequence type for Lightning.



Past losses occurring in counties that have been dissolved are included in SHELDUS data as well. These counties are flagged in the SHELDUS data as all records are assigned the name of the county (County FIPS code) at the time the loss was reported. SHELDUS provided a county table that includes the date range when each county definition was applicable. Most changes are due to renaming, a change in the county FIPS code, or the absorption of one county by another. More complex boundary changes necessitate additional processing. The Historic Loss Ratio Methodology attempts to reapportion loss from these dissolved counties to their present-day equivalents if the loss occurred during the period of record for a particular hazard type.

The United States Census Bureau maintains a list of Substantial Changes to Counties and County Equivalent Entities<sup>17</sup> that was used to map the dissolved counties to their present-day equivalents. Most of these counties were completely absorbed by new or existing counties, and the economic loss of the dissolved county could be 100% reallocated to its present-day equivalent (see [Table 10](#)). If a county was dissolved into two or more new or existing counties, the population of the county at the time it was dissolved was compared to the population of the present-day counties to estimate the proportion of loss that should be attributed to each present-day county. The exception to this rule is the reapportionment of the Yellowstone National Park county-equivalent. Loss allocation was divided roughly according to land area reapportionment because the permanent population of the national park is so low.

**Table 10: Dissolved County Allocation of SHELDUS Loss**

| <i>Dissolved County</i>            | <i>Year Dissolved</i> | <i>Present-day Counties</i>     | <i>Loss Allocation</i>                        |
|------------------------------------|-----------------------|---------------------------------|-----------------------------------------------|
| Skagway-Yakutat, Alaska            | 1980                  | Skagway, Yakutat                | 67%/33%                                       |
| Aleutians Islands, Alaska          | 1987                  | Aleutians East, Aleutians West  | 20%/80%                                       |
| Skagway-Yakutat-Angoon, Alaska     | 1992                  | Yakutat, Skagway, Hoonah-Angoon | 20%/20%/60%                                   |
| Yellowstone National Park, Montana | 1997                  | Gallatin, Park                  | 60%/40%                                       |
| Skagway-Hoonah-Angoon, Alaska      | 2007                  | Skagway, Hoonah-Angoon          | Bldg.: 25%/75%<br>Pop: 25%/75%<br>Ag: 50%/50% |

<sup>17</sup> United States Census Bureau. (2020). *Substantial Changes to Counties and County Equivalent Entities: 1970-Present*. Retrieved from <https://www.census.gov/programs-surveys/geography/technical-documentation/county-changes.2020.html>

| <i>Dissolved County</i>                   | <i>Year Dissolved</i> | <i>Present-day Counties</i>                        | <i>Loss Allocation</i>                                   |
|-------------------------------------------|-----------------------|----------------------------------------------------|----------------------------------------------------------|
| Prince of Wales – Outer Ketchikan, Alaska | 2008                  | Ketchikan Gateway, Prince of Wales-Hyder, Wrangell | Bldg: 10%/80%/10%<br>Pop: 10%/80%/10%<br>Ag: 33%/33%/34% |
| Wrangell-Petersburg, Alaska               | 2008                  | Petersburg, Wrangell                               | 64%/36%                                                  |
| Bedford City, Virginia                    | 2013                  | Bedford                                            | 100%                                                     |
| Shannon, South Dakota                     | 2015                  | Oglala Lakota                                      | 100%                                                     |
| Wade Hampton, Alaska                      | 2015                  | Kusilvak                                           | 100%                                                     |
| Valdez-Cordova, Alaska                    | 2019                  | Chugach, Copper River                              | Bldg: 54%/46%<br>Pop: 73%/27%<br>Ag: 0%/100%             |

## Consequence Types

The consequence types in the loss data sources are treated as direct corollaries to consequence types measured for natural hazard exposure.

### Building

Building loss is defined as the SHELDUS or National Centers for Environmental Information reported damage to property caused by the hazard occurrence in dollars (inflation-adjusted to 2022 dollars). In the calculation of Historic Loss Ratios, property loss is treated as analogous to the building values recorded in Hazus 6.0. However, SHELDUS property damage can include other types of property, like vehicles or infrastructure, which would not be reported in the Census data used by Hazus to estimate building value. This is a caveat to consider when working with the data. SHELDUS and Hazus data remain the best available estimates of loss and value that could be utilized.

### Population

Population loss is defined as the SHELDUS or National Centers for Environmental Information reported number of fatalities and injuries caused by the hazard occurrence. To combine fatalities and injuries for the computation of population loss value, an injury is counted as one-tenth (1/10) of a fatality. Note that for the Cold Wave and Heat Wave hazards, additional Historic Loss Ratio values are independently generated for both fatality and injury losses.

## Agriculture

Agriculture loss is defined as the SHELDUS or National Centers for Environmental Information reported damage to crops and livestock caused by the hazard occurrence in dollars (inflation-adjusted to 2022 dollars). SHELDUS also tracks crop indemnity payments for United States Department of Agriculture-insured crop loss; however, the total crop/livestock damage value was considered to be more inclusive, and the crop indemnity data are not used.

### 5.4.3. LOSS RECORD FORMATTING

The nature and structure of the source loss data requires that some pre-processing and re-formatting of the data within the database to ensure that all historical hazard occurrences are structured appropriately for inclusion in the Historic Loss Ratio calculations, including (1) per-basis record expansion; (2) single-day, timeframe, or consecutive-day aggregation of the SHELDUS and National Centers for Environmental Information loss records; and (3) the insertion of records representing zero-loss hazard occurrences. The processes are described in more detail below.

#### Loss Record Expansion to per Basis Records

A series of manipulations of the SHELDUS and National Centers for Environmental Information hazard occurrence records are performed to adapt the data for use. For hazards in which the occurrence basis is event-day, records of hazard events that span multiple days have their loss split evenly into a single record per day. For example, the January 2009 Ice Storm event (peril “Ice”) in [Table 11](#) lasted three days. The basis of Ice Storm occurrences is the event-day as this definition better captures the variability in duration for Ice Storm events. Without the resolution of knowing which event-day the damage occurred on, the loss is divided among the days so that each event-day record has an equal portion of the total loss (see [Table 12](#)).

**Table 11: SHELDUS Loss Records**

| <i>SHELDUS ID</i> | <i>Hazard Begin Date</i> | <i>Hazard End Date</i> | <i>County FIPS</i> | <i>Fatalities</i> | <i>Injuries</i> | <i>Property Damage</i> | <i>Crop Damage</i> | <i>Peril</i>   |
|-------------------|--------------------------|------------------------|--------------------|-------------------|-----------------|------------------------|--------------------|----------------|
| 10043726          | Jul. 01, 2002            | Jul. 17, 2002          | 08067              | 0                 | 0               | 8,000,000              | 0                  | Wildfire       |
| 10044246          | Jul. 01, 2002            | Jul. 17, 2002          | 08067              | 0                 | 0               | 2,500,000              | 0                  | Wildfire       |
| 10053354          | May 02, 2003             | May 02, 2003           | 01047              | 0                 | 0               | 5,000                  | 0                  | Hail           |
| 10053765          | May 02, 2003             | May 02, 2003           | 01047              | 0                 | 0               | 45,000                 | 0                  | Hail           |
| 10090870          | June 12, 2006            | June 14, 2006          | 12129              | 0                 | 0               | 20,000                 | 0                  | Tropical Storm |
| 10090997          | June 12, 2006            | June 13, 2006          | 12129              | 0                 | 0               | 5,000                  | 0                  | Storm Surge    |
| 10139562          | Jan. 26, 2009            | Jan. 28, 2009          | 05007              | 0                 | 0               | 30,000,000             | 0                  | Ice            |

**Table 12: SHELUDS Loss Allocation Date Expansion Records**

| County<br>FIPS | Utilize Start<br>Date | Utilize End<br>Date | Hazard<br>Type | Basis     | Property<br>Damage | Injuries | Fatalities | Crop<br>Damage |
|----------------|-----------------------|---------------------|----------------|-----------|--------------------|----------|------------|----------------|
| 05007          | Jan. 26, 2009         | Jan. 26, 2009       | Ice Storm      | Event-Day | 10,000,000         | 0        | 0          | 0              |
| 05007          | Jan. 27, 2009         | Jan. 27, 2009       | Ice Storm      | Event-Day | 10,000,000         | 0        | 0          | 0              |
| 05007          | Jan. 28, 2009         | Jan. 28, 2009       | Ice Storm      | Event-Day | 10,000,000         | 0        | 0          | 0              |

This record count expansion process is performed because loss ratios will ultimately be computed for each event or event-day record. Having a record for each hazard occurrence per basis unit better supports the process of determining loss ratio averages and variance. For some hazard types, a cap on the number of days to which a single occurrence could be expanded was set (see [Table 14](#)) to prevent certain errors in the date fields from propagating. If the date range for a loss record extends beyond this cap, dates from the start date to the cap are included in the date expansion and have losses allocated to them. The rest of the days over the cap value are discarded.

### Loss Record Aggregation of per Basis Records

The Historic Loss Ratio Methodology assumes that multiple reports of loss that occur in the same county during the same date range and are due to the same hazard type are classified as part of the same hazard occurrence. For event-day based hazards, following the date expansions process described above, multiple loss-causing records occurring on the same day are replaced by a single record with the summed losses for each consequence type. For example, the two Hail event records from [Table 11](#) (peril “Hail”) that both occurred on 5/2/2003 are aggregated into a single record in [Table 13](#). This single-day timeframe aggregation ensures that a single-day recorded loss occurring within the date range of a multiple-day recorded loss is treated as the same event-day as one of the days within the multiple-day event. Some event-based hazard types use timeframe aggregation to replace multiple loss-causing records (occurring in the same county with the same Start and End Date combination) with a single loss record with the summed losses for each consequence type. For example, the two event records from [Table 11](#) where the peril is Wildfire have the same Start and End Date combination. These events are aggregated into a single record in [Table 13](#). This addresses instances where SHELUDS reports damages impacting different areas of the United States for the same multi-day event.

For a few event-based hazard types, a consecutive-day aggregation takes place in which loss records that occur in the same county on the same or consecutive days are combined into a single loss record with the summed loss. For example, the two multi-day Hurricane event records from [Table 11](#) (peril “Tropical Storm” and “Storm Surge”) that occurred over consecutive days from June 12, 2006 to June 14, 2006 are aggregated into a single record in [Table 13](#). This aggregation allows loss records that are due to the same loss-causing events to be logically combined so that each occurrence’s loss ratio is accurately computed. Treating each loss record as a separate occurrence

with a lower loss value could potentially dilute and underestimate the Historic Loss Ratio of the county.

**Table 13: SHELATUS Loss Aggregated Records**

| <i>County FIPS</i> | <i>Utilize Start Date</i> | <i>Utilize End Date</i> | <i>Hazard Type</i> | <i>Number of Records Aggregated</i> | <i>Property Damage</i> | <i>Injuries</i> | <i>Fatalities</i> | <i>Crop Damage</i> |
|--------------------|---------------------------|-------------------------|--------------------|-------------------------------------|------------------------|-----------------|-------------------|--------------------|
| 08067              | July 1, 2002              | July 17, 2002           | Wildfire           | 2                                   | 10,500,000             | 0               | 0                 | 0                  |
| 01047              | May 02, 2003              | May 02, 2003            | Hail               | 2                                   | 50,000                 | 0               | 0                 | 0                  |
| 12129              | June 12, 2006             | June 14, 2006           | Hurricane          | 2                                   | 25,000                 | 0               | 0                 | 0                  |

Hazard types are processed using one or more of the methods previously described. The nature of the hazard and its loss reporting inform which processes are utilized. [Table 14](#) describes which processes are used for each hazard type.

**Table 14: Loss Record Processing by Hazard Type**

| <i>Hazard Type</i> | <i>Day Expansion Performed?</i> | <i>Maximum Expansion Days</i> | <i>Consecutive Day Aggregation Performed?</i> | <i>Timeframe Aggregation Performed?</i> |
|--------------------|---------------------------------|-------------------------------|-----------------------------------------------|-----------------------------------------|
| Avalanche          | No                              | N/A                           | No                                            | No                                      |
| Coastal Flooding   | No                              | N/A                           | Yes                                           | No                                      |
| Cold Wave          | Yes                             | 31                            | No                                            | Yes                                     |
| Drought            | Yes                             | 365                           | No                                            | Yes                                     |
| Earthquake         | No                              | N/A                           | No                                            | Yes                                     |
| Hail               | No                              | (blank)                       | No                                            | Yes                                     |
| Heat Wave          | Yes                             | 31                            | No                                            | Yes                                     |
| Hurricane          | No                              | N/A                           | Yes                                           | No                                      |
| Ice Storm          | Yes                             | 31                            | No                                            | Yes                                     |
| Inland Flooding    | Yes                             | 31                            | No                                            | Yes                                     |
| Landslide          | No                              | N/A                           | No                                            | Yes                                     |
| Lightning          | No                              | 1                             | No                                            | Yes                                     |

| <i>Hazard Type</i> | <i>Day Expansion Performed?</i> | <i>Maximum Expansion Days</i> | <i>Consecutive Day Aggregation Performed?</i> | <i>Timeframe Aggregation Performed?</i> |
|--------------------|---------------------------------|-------------------------------|-----------------------------------------------|-----------------------------------------|
| Strong Wind        | No                              | (blank)                       | No                                            | Yes                                     |
| Tornado            | No                              | (blank)                       | No                                            | No                                      |
| Tsunami            | No                              | N/A                           | Yes                                           | No                                      |
| Volcanic Activity  | No                              | N/A                           | No                                            | No                                      |
| Wildfire           | No                              | N/A                           | No                                            | Yes                                     |
| Winter Weather     | Yes                             | 31                            | No                                            | Yes                                     |

Once this reallocation and aggregation of loss records has been completed, each building and agriculture loss value is inflation-adjusted to 2024 using the Bureau of Labor Statistics Consumer Price Index<sup>18</sup> as seen in [Equation 9](#).

#### Equation 9: Conversion to 2022 Dollars

$$V_{MO_{2022}} = V_{orig} \times \frac{CPI_{MO_{2022}}}{CPI_{MO_{Year}}}$$

where:

$V_{MO_{2022}}$  is the dollar value in 2022 dollars.

$V_{orig}$  is the original dollar value (assumed dollar value at the time of the loss event).

$CPI_{MO_{2022}}$  is the Consumer Price Index for the month of the loss event in 2022 dollars.

$CPI_{MO_{Year}}$  is the Consumer Price Index for the month/year of the loss event.

#### Loss Ratio Per Basis Calculation

After all pre-processing is complete, the Loss Ratio per Basis is calculated for each event or event-day occurrence for each consequence type (building, population, or agriculture) according to [Equation 10](#).

<sup>18</sup> Bureau of Labor Statistics. (2024). *Consumer Price Index for all urban consumers* [online dataset]. Retrieved from <https://www.bls.gov/data/>.

**Equation 10: Loss Ratio per Basis Calculation**

$$LRB_{HazCoCnsqType} = \frac{Loss_{HazCoCnsqType}}{HLRExposure_{HazCoCnsqType}}$$

where:

$LRB_{HazCoCnsqType}$  is the Loss Ratio per Basis (event or event-day) representing the ratio of loss to exposure for a specific hazard occurrence experienced by a specific county. Calculation is performed for each relevant consequence type (building, population, and agriculture).

$Loss_{HazCoCnsqType}$  is the loss (by consequence type) experienced from the hazard event or event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{HazCoCnsqType}$  is the total value (by consequence type) estimated to have been exposed to the event or event-day hazard occurrence (in dollars or people).

The definition of the Historic Loss Ratio exposure variable in the loss ratio per basis formula does not always match the definition of the exposure factor utilized in the Expected Annual Loss formula. For hazard types that can occur almost anywhere or affect large geographic areas, the Historic Loss Ratio exposure is the entire county's building value, population, or agriculture value. Hazard types that only occur in certain susceptible zones, such as floodplains and tsunami inundation zones, use the Historic Loss Ratio exposure value associated with those susceptible zones. Tornado Historic Loss Ratio exposure is defined by the area footprint of specific historical Tornado paths. Avalanche is a unique case that requires the use of default exposure population values (described in [Section 6.3](#)). Specific methods of determining Historic Loss Ratio exposure in the Loss Ratio per Basis calculation are found in the Historic Loss Ratio section for each hazard type. [Table 15](#) lists the exposure types used in each hazard type's Loss Ratio per Basis calculation.

**Table 15: Historic Loss Ratio Exposure Types Used in Loss Ratio per Basis Calculation**

| <b>Hazard Type</b> | <b>Historic Loss Ratio Exposure Type</b>                                     |
|--------------------|------------------------------------------------------------------------------|
| Avalanche          | Default Value for Population; Value Defined by Hazard Intersect for Building |
| Coastal Flooding   | Value Defined by Hazard Intersect                                            |
| Cold Wave          | Total County Value                                                           |
| Drought            | Total County Value                                                           |
| Earthquake         | Total County Value                                                           |
| Hail               | Total County Value                                                           |

| <b>Hazard Type</b> | <b>Historic Loss Ratio Exposure Type</b>                                                                     |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| Heat Wave          | Total County Value                                                                                           |
| Hurricane          | Total County Value                                                                                           |
| Ice Storm          | Total County Value                                                                                           |
| Inland Flooding    | Value Defined by Hazard Intersect                                                                            |
| Landslide          | Value Defined by Hazard Intersect                                                                            |
| Lightning          | Total County Value                                                                                           |
| Strong Wind        | Total County Value                                                                                           |
| Tornado            | Historical Footprint Matched to Specific SHELDUS Loss                                                        |
| Tsunami            | Exposure Value provided by Estimated Average Annualized Tsunami Losses for the United States Report (P-2426) |
| Volcanic Activity  | Value Defined by Hazard Intersect                                                                            |
| Wildfire           | Value Defined by Hazard Intersect                                                                            |
| Winter Weather     | Total County Value                                                                                           |

### Zero-Loss Hazard Occurrences

Hazards may occur without resulting in recorded loss to buildings, population, or agriculture. For example, Lightning may strike with a high frequency but have few loss-causing occurrences. SHELDUS does not record events in which no loss was reported. In an effort to capture zero-loss hazard occurrences, a count of historical occurrences is estimated from hazard source data and compared to a count of loss-causing events from SHELDUS and the National Centers for Environmental Information Storm Events Database. A county-level annual rate from the hazard source data is calculated as the count of total hazard occurrences divided by the hazard's period of record. This rate is then multiplied by the number of years in the SHELDUS period of record to estimate an expected hazard occurrence count.

When more occurrences are estimated by the hazard historical occurrence source than SHELDUS or the National Centers for Environmental Information Storm Events Database, a number of zero-loss records are inserted into the set of Loss Ratio per Basis to make up the difference between historical occurrences and loss-causing events from SHELDUS so that the counts for both metrics are equal.

Computing loss ratio averages and variances without including the zero-loss records produces markedly different results than when they are included. For example, a county with 100 historical Lightning strikes may only have two loss-causing events, one causing \$40,000 in damage to buildings and the other causing \$60,000. If the building exposure value is \$10 million, the loss



ratios for each loss-causing event would be 0.004 and 0.006, respectively. If only the Loss Ratio per Basis for two loss-causing occurrences were considered, the average would be 0.005. Including the 98 Lightning strikes that did not result in loss lowers the average to 0.0001, a more accurate approximation of the average Lightning strike's impact on the county as not every Lightning strike is a loss-causing occurrence.

The outputs of the Loss Ratio per Basis calculation (see [Equation 10](#)) and all corrective record insertion are stored in the Loss Ratio per Basis table within the processing database, and are then used to compute Bayesian metrics and calculate the weighting factors that are applied to find the hazard type Bayesian-adjusted Historic Loss Ratio for each consequence type for the county. [Table 16](#) illustrates the content of the Loss Ratio per Basis database table after the corrective record insertions. Notice the loss ratios for three Ice Storm event-days in one county in January 2009. These have been expanded from a single SHELDS record based on duration days and consequence types. Also, one zero-loss record for each relevant consequence type has been inserted to recognize an Ice Storm event-day that occurred within the county (based on the historical occurrence source data) but resulted in no economic loss. These records can then be used to calculate loss ratio averages and variance.

**Table 16: Sample Data from the Loss Ratio per Basis Table**

| <i>Hazard Type</i> | <i>Peril</i>              | <i>Date</i>   | <i>Conseq. Type</i> | <i>Conseq. Exposure</i> | <i>Conseq. Loss per Basis</i> | <i>Conseq. Ratio per Basis Unit</i> | <i>Record Type</i>        |
|--------------------|---------------------------|---------------|---------------------|-------------------------|-------------------------------|-------------------------------------|---------------------------|
| Ice Storm          | Ice                       | Jan. 26, 2009 | Population          | 221339                  | 0.01666667                    | 7.53E-08                            | Peril Basis Expansion     |
| Ice Storm          | Ice                       | Jan. 27, 2009 | Population          | 221339                  | 0.01666667                    | 7.53E-08                            | Peril Basis Expansion     |
| Ice Storm          | Ice                       | Jan. 26, 2009 | Building            | 2.3138E+10              | 5881140.47                    | 0.00025                             | Peril Basis Expansion     |
| Ice Storm          | Ice                       | Jan. 27, 2009 | Building            | 2.3138E+10              | 5881140.47                    | 0.00025                             | Peril Basis Expansion     |
| Ice Storm          | Ice                       | Nov. 02, 1998 | Population          | 221339                  | 0                             | 0                                   | SHELDUS Native Record     |
| Ice Storm          | Ice                       | Nov. 02, 1998 | Building            | 2.3138E+10              | 310468.525                    | 0.0000134                           | SHELDUS Native Record     |
| Ice Storm          | Inserted Zero-Loss Record |               | Population          | 221339                  | 0                             | 0                                   | Inserted Zero-Loss Record |

| <i>Hazard Type</i> | <i>Peril</i>              | <i>Date</i> | <i>Conseq. Type</i> | <i>Conseq. Exposure</i> | <i>Conseq. Loss per Basis</i> | <i>Conseq. Ratio per Basis Unit</i> | <i>Record Type</i>        |
|--------------------|---------------------------|-------------|---------------------|-------------------------|-------------------------------|-------------------------------------|---------------------------|
| Ice Storm          | Inserted Zero-Loss Record |             | Building            | 2.3138E+10              | 0                             | 0                                   | Inserted Zero-Loss Record |

#### 5.4.4. HISTORIC LOSS RATIO CALCULATION METHODOLOGY

A county Historic Loss Ratio could be computed as the average of the individual hazard occurrence loss rates (referred to here as Loss Ratio per Basis) for a given hazard-consequence type combination. However, a county Historic Loss Ratio cannot be calculated in these simple terms and be considered accurate. Many counties that have not experienced a loss-causing hazard occurrence during the time period captured from SHEL DUS may be in close proximity to counties that share similar characteristics and have experienced loss to the hazard type. For example, it may be inaccurate to say that a county's likely loss ratio to Hurricane is zero just because it has not experienced a loss-causing Hurricane occurrence during the 28-year window, especially if it borders counties that have experienced loss to Hurricanes. A better approximation of the Historic Loss Ratio is achieved by applying a Bayesian spatial weighting matrix to smooth the loss ratio data spatially and ensure that the Historic Loss Ratio is represented in a rational way without allowing anomalous hazard occurrences to distort the data.

To implement Bayesian credibility weighting, loss ratio averages and variances need to be computed for various sets of Loss Ratio per Basis. These Loss Ratio per Basis sets can be defined by both a geographic and a county characteristic component. The goal in the creation of a Loss Ratio per Basis set is to construct a group of loss ratios that have elements of commonality/similarity at several levels: county, surrounding areas,<sup>19</sup> hazard specific regional, and national. These geographic levels define which spatial grouping (or set) of Loss Ratios per Basis are used to calculate the average and variance values. The county-level grouping includes all Loss Ratios per Basis for the county, the surrounding area groupings include Loss Ratios per Basis for all counties that intersect a target county's buffered area, the regional grouping includes Loss Ratios per Basis for all counties within the defined region, and national includes all Loss Ratios per Basis.

#### Loss Ratio per Basis Sets

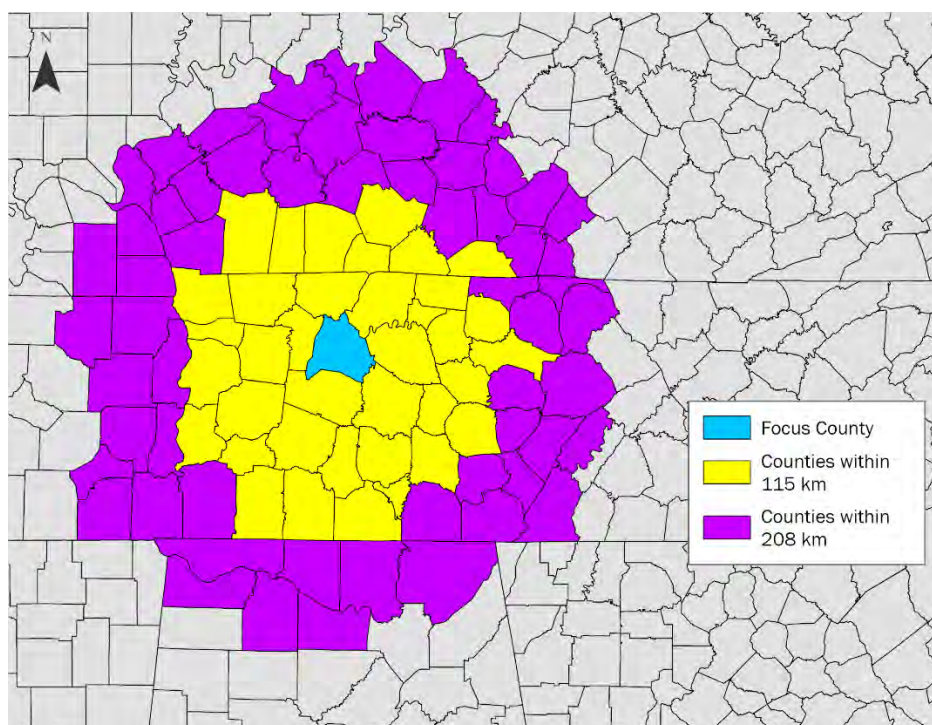
In the National Risk Index, these groups of similarly characterized counties are referred to as Loss Ratio per Basis Sets. An unlimited number of Loss Ratio Basis Sets can be created by choosing different geographic and desired area characteristics to reference. For example, Loss Ratio per Basis Sets can be created by utilizing urban/rural designation, topography or land cover, proximity, and

<sup>19</sup> Two surrounding area definitions are implemented for each county using a buffered area around their centroid with a radius of 115km and 208km, respectively. See the [Intersection](#) section for more information on the use of buffered areas to represent a county average.

other characteristics, in order to better understand what a “similar” county is. Loss Ratio per Basis Set definitions are made up of Bayesian Level and National Center for Health Statistics Urban-Rural Code Filter settings and applied to hazard-consequence type combination. Each of these settings can be thought of as a condition for the inclusion of a specific loss ratio record into the set.

### Bayesian Levels

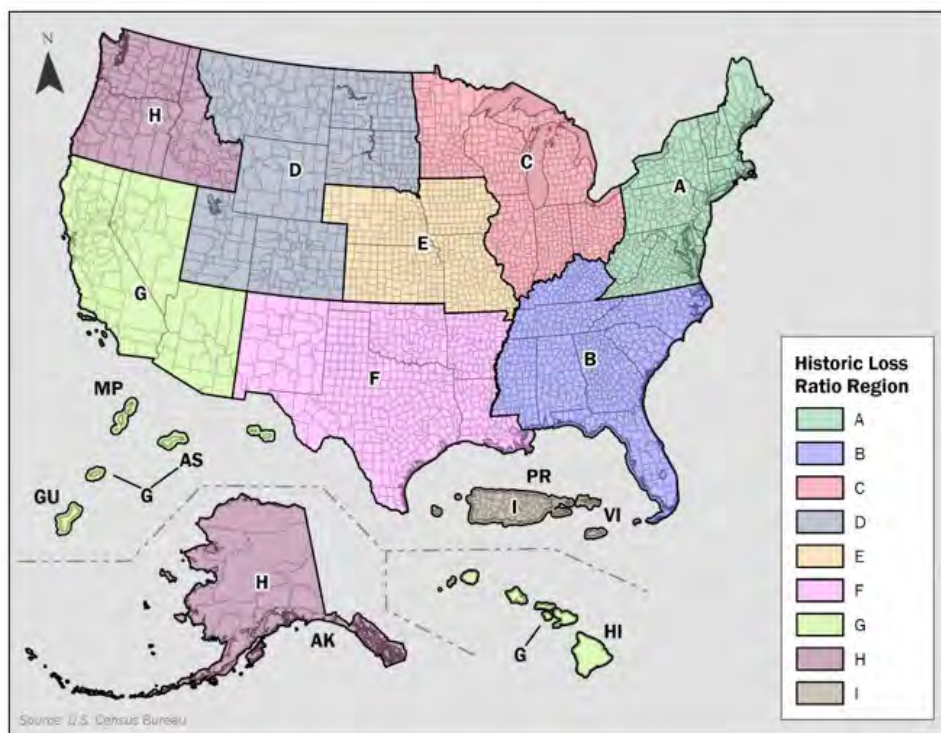
Bayesian levels are geographic definitions that represent a shape or boundary, such as a state or custom-defined area, and serve as a selector for the counties to be associated with the Bayesian level. Bayesian levels can be generally characterized as representing the county, surrounding areas, regional, and national. For the surrounding area category, two different “Neighbor Counties” Bayesian Levels were utilized; one with the focus county buffered by 115km and the other by 208km. Note that each county in the country has its own Neighbor Counties shape, see [Figure 8](#) for an example.



**Figure 8: Example of Neighbor County Based on Sets**

For the regional category, several hazard-specific Bayesian Levels were utilized, each representing county grouping relevant to the hazard. For more information on the hazard-specific Bayesian levels and their associated maps, please see the Historic Loss Record subsections for each hazard ([Sections 6 through 23](#)). Additionally in the regional category it was desired to have a regional definition for geographical groupings larger than states but smaller than the nationwide grouping was required. Because the Federal Emergency Management Agency has a pre-existing definition of regions that is logical and groups states by similar geographical and climatological characteristics, a decision was made to modify the existing region definition rather than create new region definitions.

Thus, Historic Loss Ratio region definitions for specific hazard types are derived from the Federal Emergency Management Agency administrative region definitions, with the only difference being that Federal Emergency Management Agency Regions 1, 2, and 3 are merged to form a region that is closer in size to that of the other regions, and the Caribbean Territories are given their own region (see [Figure 9](#)).



**Figure 9: Historic Loss Ratio Region Definitions.**

### Filter (County Attribute Filters)

Filters are applied to further segment the sets defined by Bayesian levels. Filters use additional county attributes to subdivide the counties within a Bayesian level. The National Risk Index utilized a National Center for Health Statistics Urban-Rural Classification Scheme Urban-Rural Filter definition, see [Section 5.4.1](#).

Conceptually for a hazard-consequence type-Bayesian level combination, the Loss Record per Basis Set will contain all loss ratio records associated with the Bayesian level. If a Filter is applied to the Bayesian level, then those Loss Ratios per Basis are further refined to only include those that meet that filter's specifications – reducing the number of Loss Ratios per B that are used in Historic Loss Ratio calculations.

## Set Usage Threshold

Usage thresholds are established for determining eligibility of a specific Loss Ratio per Basis set to be included in a specific county's Historic Loss Ratio calculation. These "Set Usage Thresholds" are established as either a record count or percentage of overall number of Loss Records per Basis that is converted to a record count at the time of evaluation and help to ensure that a minimum number of Loss Records per Basis records are available to be mathematically valid. Set Usage Thresholds can be used in three different modes:

1. **Total Record Count Mode** – All Loss Record per Basis records in the set are counted regardless of loss ratio value.
2. **Non-Zero Record Count Mode** – Only Loss Ratio per Basis records with a non-zero loss ratio value are counted.
3. **Non-Zero-Padded Record Count Mode** – Excludes Loss Ratios per Basis created by the zero-padding process, which simulates losses for non-reporting events (e.g., lightning strikes with no recorded damage).

Each Loss Ratio per Basis Set definition can be configured with a Set Usage threshold value (e.g., 25 records) and a mode, determining its eligibility for inclusion of the set in the Bayesian credibility calculation for the specific county hazard-consequence Historic Loss Ratio. This means that each Loss Ratio per Basis set definition is unique to each county given the thresholds and which counties are considered similar to the target county.

For more information on the hazard-consequence specific Loss Ratio per Basis Set definitions (Bayesian Level, Filter, and Usage Threshold and mode), please see the Historic Loss Record subsections for each hazard ([Sections 6 through 23](#)).

## Bayesian Credibility

To apply Bayesian credibility weighting factors and balance Historic Loss Ratio accuracy with characteristic and geographic precision in areas where small sample sizes result in volatile Historic Loss Ratio estimates, Loss Ratio per Basis averages, variance, and other statistics may be calculated at Loss Ratio per Basis sets. The formulas in [Equation 11](#) illustrate the computation of the loss ratio average, variance, and standard error.

**Equation 11: Hazard-Consequence Loss Ratio per Basis Set Ratio Average, Variance, and Standard Error Computations**

$$avgLRB_{HazSetCnsqType} = \frac{\sum LRB_{HazSetCnsqType}}{CountOccurrences_{HazSetCnsqType}}$$

$$varLRB_{HazSetCnsqType} = \frac{\sum \left( LRB_{HazSetCnsqType} - avgLRB_{HazSetCnsqType} \right)^2}{CountOccurrences_{HazSetCnsqType} - 1}$$

$$setLRB_{HazSetCnsqType} = \frac{\sqrt{varLRB_{HazSetCnsqType}}}{\sqrt{\frac{CountOccurrences_{HazSetCnsqType}}{N_{Set}}}}$$

where:

$avgLRB_{HazSetCnsqType}$  is the average value of all Loss Ratio per Basis (event or event-day) records of the consequence type for the Loss Ratio per Basis set due to the hazard type.

$LRB_{HazSetCnsqType}$  is a single Loss Ratio per Basis (event or event-day) of the consequence type within the Loss Ratio per Basis set due to the hazard type occurrence basis.

$CountOccurrences_{HazSetCnsqType}$  is the total number of records of hazard occurrences (events or event-days) in the Loss Ratio per Basis set by consequence type (includes any zero-loss occurrences).

$varLRB_{HazSetCnsqType}$  is the Loss Ratio per Basis variance for the hazard and consequence type.

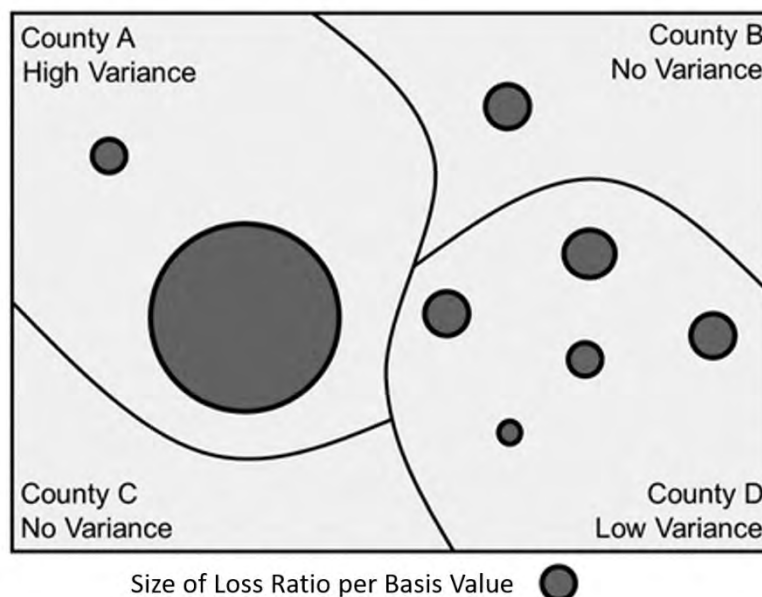
$setLRB_{HazLevelCnsqType}$  is the Loss Ratio per Basis variance of the Loss Ratio per Basis set for the hazard and consequence type.

$LRB_{HazLevelCnsqType}$  is the Loss Ratio per Basis standard error of the Loss Ratio per Basis set for the hazard and consequence type.



$N_{Set}$  is the number of counties that reside within a particular county's Bayesian level

Credibility is determined by sample size, Loss Ratio per Basis variance, and the number of SHEL DUS loss records per county contained within a Loss Ratio per Basis Set for the hazard and consequence type.<sup>20</sup> To ensure that each Loss Ratio per Basis Set contains a sufficient number of observed loss events, a minimum number of records from the SHEL DUS loss database is required for inclusion in the target county's Historic Loss Ratio calculation. If a given Loss Ratio per Basis set exceeds this threshold, then its credibility (i.e., its contribution to the county's calculated Historic Loss Ratio value) increases as a function of sample size and decreased Loss Ratio per Basis variance. In other words, the higher the credibility at a given geographic level (Loss Ratio per Basis Set), the higher the contribution to the county's calculated Historic Loss Ratio value.<sup>21</sup> Figure 10 illustrates possible Loss Ratio per Basis variance and SHEL DUS record counts in neighboring counties. Weighting factors in the Bayesian credibility calculation determines the contribution of each Loss Ratio per Basis set to the final, Bayesian-adjusted Historic Loss Ratio value.



**Figure 10: Example of Variance in County Loss Ratio per Basis Values**

Weighting factors are derived from the standard error values (calculated using [Equation 11](#)) and a Boolean indicator variable at each geographic level according to [Equation 12](#). The Boolean indicator

<sup>20</sup> Since Loss Ratio per Basis Sets at higher Bayesian levels will always have (weakly) more occurrences than lower levels, Loss Ratio per Basis standard errors are calculated on a per-county basis within the set. This attenuates the influence of Loss Ratio per Basis sets at the national and region level by assigning greater credibility at lower levels containing counties with a relatively large number of historic loss events.

<sup>21</sup> However, if the minimum number of records required for inclusion into the county's Historic Loss Ratio calculation is not satisfied, then that geographic level's credibility is set equal to zero.

variable is set equal to 1 if the minimum loss record threshold is satisfied and 0 otherwise. Levels not used for a specific hazard type are removed from the computation.

**Equation 12: Historic Loss Ratio Bayesian Weighting Factor Calculation**

$$\begin{aligned}
 Wt_{Denom} &= \frac{1}{setLRB_{HazNtlCnsqType}} + \delta_{CountRegCnsqType} * \frac{1}{setLRB_{HazRegCnsqType}} + \delta_{CountSur208CnsqType} \\
 &\quad * \frac{1}{setLRB_{HazSur208CnsqType}} + \delta_{CountSur115CnsqType} * \frac{1}{setLRB_{HazSur115CnsqType}} \\
 &\quad + \delta_{CountCoCnsqType} * \frac{1}{setLRB_{HazCoCnsqType}} \\
 Wt_{HazNtlCnsqType} &= \frac{1/setLRB_{HazNtlCnsqType}}{Wt_{Denom}} \\
 Wt_{HazRegCnsqType} &= \frac{1/setLRB_{HazRegCnsqType}}{Wt_{Denom}} * \delta_{CountRegCnsqType} \\
 Wt_{HazSur208CnsqType} &= \frac{1/setLRB_{HazSur208CnsqType}}{Wt_{Denom}} * \delta_{CountSur208CnsqType} \\
 Wt_{HazSur115CnsqType} &= \frac{1/setLRB_{HazSur115CnsqType}}{Wt_{Denom}} * \delta_{CountSur115CnsqType} \\
 Wt_{HazCoCnsqType} &= \frac{1/setLRB_{HazCoCnsqType}}{Wt_{Denom}} * \delta_{CountCoCnsqType}
 \end{aligned}$$

where:

$Wt_{Denom}$  is the sum of the inverted variances calculated at each geographic level and is used as a denominator for the level weighting factors.

$Wt_{HazX CnsqType}$  is the weighting factor to be applied to the average consequence type Loss Ratio per Basis for the hazard type at X level (national, regional, surrounding, county).

$setLRB_{HazX CnsqType}$  is the consequence type Loss Ratio per Basis variance for the hazard type at X level (national, regional, surrounding, county).



$\delta_{CountX_{CnsqType}}$  is the Boolean variable equal to 1 if the minimum loss record threshold is satisfied

For several hazard types, regional Bayesian Historic Loss Ratio weighting supplies a more accurate estimation of Historic Loss Ratio for areas that have not experienced losses due to hazard occurrences during the period of record. This is especially true for areas where hazard type annualized frequency and severity are dependent on their geographic location and climate. For example, Winter Weather will have a quite different degree of impact on the Northeast than on the Southwest. For this reason, the Bayesian spatial weighting incorporates regional weighting rather than national for select hazard types.

To use this regional weighting, a regional definition for geographical groupings larger than states, but smaller than the nationwide grouping, was required. Because the Federal Emergency Management Agency has a pre-existing definition of regions that groups states by similar geographical and climatological characteristics, a modified version of these regions are used as a default region definition for most hazards. Region definitions for some hazards are defined using existing categorizations defined by authoritative government agencies where available) or spatially constrained geographic clustering.<sup>22</sup>

The Historic Loss Ratio for each relevant consequence type is calculated as the sum of its weighted average county, surrounding area, regional, and national average Loss Ratios per Basis (see [Equation 13](#)). Geographic levels not used for a specific hazard type are removed from the computation.

### Equation 13: County Bayesian-Adjusted Historic Loss Ratio Calculation

$$HLR_{HazCo_{CnsqType}} = \left( avgLRB_{HazNtl_{CnsqType}} \times Wt_{HazNtl_{Cnsqt}} \right) + \left( avgLRB_{HazReg_{CnsqType}} \times Wt_{HazReg_{Cnsqt}} \right) + \left( avgLRB_{HazSur208_{CnsqType}} \times Wt_{HazSur208_{Cnsqt}} \right) + \left( avgLRB_{HazSur115_{CnsqType}} \times Wt_{HazSur115_{Cnsqt}} \right) + \left( avgLRB_{HazCO_{CnsqType}} \times Wt_{HazCO_{Cnsqt}} \right)$$

where:

$HLR_{HazCo_{CnsqType}}$  is the Bayesian-adjusted Historic Loss Ratio for the hazard type at the county level by consequence type.

$avgLRB_{HazX_{CnsqType}}$  is the average Loss Ratio per Basis by consequence type for the hazard type at X level (national, regional, surrounding, county).

$Wt_{HazX_{CnsqType}}$  is the weighting factor applied to the Loss Ratio per Basis by consequence type for the hazard type at X level (national, regional, surrounding, county).

<sup>22</sup> For more details regarding the methods used to defined regions for specific hazards, see the Historic Loss Ratio portion of their hazard subsection.

This resulting Bayesian-adjusted Historic Loss Ratio value, computed for each county-hazard type-consequence type combination, serves as a prediction of the ratio of loss to exposed consequence type value that is expected from a single hazard occurrence. When multiplied by the annualized frequency of hazard occurrence in an area and the consequence type value exposed to the hazard type, the Historic Loss Ratio enables the estimation of a community's Expected Annual Loss for that consequence type and hazard type.

### Ex-Post Back Testing and Geographic Smoothing Refinements

When Bayesian-adjusted Historic Loss Ratio values are computed for a given hazard and consequence type, an initial set of county level Expected Annual Loss values are calculated and compared with their corresponding annualized losses reported in the SHELDUS period of record. If a given county's Expected Annual Loss is less than its corresponding annualized SHELDUS losses, then a new Historic Loss Ratio value is computed by solving for the combination of Bayesian weights that minimizes the absolute difference between the two. This back testing refinements was implemented to mitigate instance where Bayesian credibility weighting may underestimate losses in comparison to our historic loss data.

If back testing results in an increase in one or more Historic Loss Ratio values, then a smoothing refinement is applied to neighboring counties if the back-test adjusted Historic Loss Ratio is greater than the national median. This refinement uses an exponential decay function over a 312km radii of influence around the back-test adjusted focus county to smooth out nearby county Historic Loss Ratio values by mean-adjusting them upward subject to predefined constraints.<sup>23</sup> To prevent double counting, counties that lie within the radii of influence of more than one back-tested focus county are assigned a weighted average adjustment based upon their relative distance from each source.

### Hurricane Regression

Historic Loss Ratios for Hurricane are calculated using multiple regression. The model regresses county level sociogeographic information on county level mean loss ratios and fitted values are assigned to each county for which a hurricane hazard event is possible. For more information on how Historic Loss Ratio estimates are determined for Hurricane see [Section 13.6 Hurricane Historic Loss Ratio](#).

### Historic Loss Ratio Inheritance

The county Bayesian-adjusted Historic Loss Ratio is inherited by the Census blocks and Census tracts within the parent county when used in the Expected Annual Loss calculations, as in [Equation 14](#).

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<sup>23</sup> To prevent adverse impacts of smoothing adjustments on Expected Annual Loss, an Expected Annual Loss influence cap was implemented smoothing adjustments which set the maximum allowable increase in Expected Annual Loss to a county as a result of Historic Loss Ratio smoothing equal to the ex-ante national mean Expected Annual Loss for the hazard and consequence type combination. Additionally, National Center for Health Statistics urban-rural county classifications were used to prevent focus counties from influencing nearby counties more urban than themselves.

**Equation 14: Census Tract and Census Block Historic Loss Ratio Inheritance**

$$HLR_{HazCoCnsqType} = HLR_{HazCTCnsqType} = HLR_{HazCBCnsqType}$$

where:

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| $HLR_{HazCoCnsqType}$ | is the Bayesian-adjusted Historic Loss Ratio, a hazard type-county-consequence type specific value. |
| $HLR_{HazCTCnsqType}$ | is the inherited Historic Loss Ratio for the hazard type at the Census tract level.                 |
| $HLR_{HazCBCnsqType}$ | is the inherited Historic Loss Ratio for the hazard type at the Census block level.                 |

**5.4.5. LIMITATIONS AND ASSUMPTIONS IN HISTORIC LOSS RATIO METHODOLOGY**

Several factors are not entirely accounted for in the calculation of Historic Loss Ratio. Certain processes, such as Bayesian credibility adjustments, attempt to correct some of these limitations. This section addresses some of the assumptions that are intrinsic within the current methodology and how these can limit the accuracy of the calculation.

Evaluating historical economic loss from SHELUDS over a relatively brief period of time and comparing it to a static Historic Loss Ratio exposure value does not account for changes in development patterns over these years. For example, a hazard occurrence in 1995 may have a low Historic Loss Ratio when its loss is compared to its 2022 Hazus-derived exposure value; however, because of increased development and population influx over the years, its Historic Loss Ratio would be much higher if the same loss were compared to the actual 1995 exposure value. There is an inherent assumption in the methodology that all buildings, population, and agriculture exposed to the hazard are static in economic value and quantity over the data period. Additionally, the SHELUDS loss values are inflation-adjusted to 2022 dollars, and Hazus-derived exposure values are also expressed in 2022 valuations.

Since the Historic Loss Ratio calculation is based on historical occurrences, it does not project reductions due to enhanced mitigation efforts and improved building standards that have changed over time (i.e., a seawall being built after a destructive flooding occurrence may reduce the damage caused by subsequent flooding occurrences).

Characterizing agriculture losses from occurrences is complex and can vary based on a number of factors, including supply and demand, substitution effects, crop rotation, and seasonality. The simplified Historic Loss Ratio calculations use crop and livestock distribution and values based on agriculture data from CropScape and the Census of Agriculture.

There are many cases where the geographic precision of the recorded loss is imperfectly captured in hazard occurrence reports from National Weather Service and other sources of SHELDUS data. The regional reporting data used to compile SHELDUS may mention multiple counties for a loss-causing occurrence. In these cases, the loss is spread equally over the counties where the hazard occurred, though the loss may have occurred only in one county. Also, loss may only occur in a portion of the county, yet the Historic Loss Ratio will apply to the entire county due to loss not being recorded with any granularity below the county level.

## 5.5. Validating Expected Annual Loss Estimates to Historical Losses

The variety of hazard types and source data presents a significant challenge to provide accurate and meaningful results for the variety of potential lenses through which the results may be viewed, such as:

- Hazard type Expected Annual Loss rankings within a county
- County Expected Annual Loss rankings within a hazard type
- County Expected Annual Loss rankings across all hazard types
- Hazard type Expected Annual Loss rankings all counties

In an attempt to validate the Expected Annual Loss, historical losses from SHELDUS and the National Centers for Environmental Information Storm Events Database for the period from 1996 to 2023 were aggregated for the United States for each hazard type and divided by the period of record (28 years) to give a rough nationwide annualized loss estimate.<sup>24</sup> This value was compared to the aggregated Expected Annual Loss estimate for its corresponding hazard type. All but 4 (Earthquake, Inland Flooding, Tsunami, and Volcanic Activity) of the hazard type Expected Annual Losses are within a factor of 2 of the experienced historical losses.

When evaluating the historical record, losses for some hazard types from the recent past are not representative of their potential future impacts. For example, Earthquakes and Volcanic Activity have the potential to impact high-value urban areas in ways not observed in the SHELDUS period of record. The data and methods used to calculate Expected Annual Losses for these hazard types account for the (albeit small) probability that such outlier events may occur. For this reason, Earthquake and Volcanic Activity Expected Annual Losses are significantly higher than their historical losses provided by SHELDUS.

These exceptions aside, a relatively high level of agreement between the calculated Expected Annual Loss and the historical loss records serves as an indication that the estimated annual hazard loss is fairly aligned with actual recorded historical loss.

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<sup>24</sup> For Cold Wave, the historic loss data were aggregated from the National Centers for Environmental Information Storm Events Database for 1996 to 2019 and divided by the 27-year period of record.

## 5.6. Expected Annual Loss Rate

The three primary types of results provided by the National Risk Index dataset are Expected Annual Loss, Social Vulnerability, and Community Resilience (see [Section 3.2 Values, Scores and Ratings](#)). In addition to those, the National Risk Index provides a set of secondary Expected Annual Loss-related results, called Expected Annual Loss rates, which are calculated from the basic risk components and Expected Annual Loss factors. These Expected Annual Loss rates are designed to reflect the average expected annual percentage loss for the building value, population, and agriculture value within a community. They provide relative natural hazard intensities regardless of the community's exposure value enabling the comparison of expected losses that are controlled for community size. [Equation 15](#) introduces the basic equation to calculate Expected Annual Loss Rate metrics for each hazard type.

### Equation 15. Hazard Type Expected Annual Loss Rate

$$EAL\ Rate_{Hazard\ Consequence\ Type} = \frac{EAL_{Hazard\ Consequence\ Type}}{Full\ Community\ Exposure_{Consequence\ Type}}$$

where:

$EAL\ Rate_{Hazard\ Consequence\ Type}$  is the Expected Annual Loss Rate for the community from a specific hazard type for a specific consequence type.

$EAL_{Hazard\ Consequence\ Type}$  is the Expected Annual Loss value for the community from a specific hazard type for a specific consequence type.

$Full\ Community\ Exposure_{Consequence\ Type}$  is the full community exposure for a specific consequence type. For hazard types with widespread occurrence, exposure and full community exposure may be equivalent; but for hazard types that are limited to susceptible areas such as floodplains, the exposure value can be significantly smaller than the full community exposure.

Expected Annual Loss Rates are calculated for each relevant hazard and consequence type using the equation above. [Equation 16](#) introduces the Composite Expected Annual Loss Rates calculation, which consider all 18 hazard types, for each relevant consequence type.

### Equation 16. Composite Expected Annual Loss Rate

$$EAL\ Rate_{Composite\ Consequence\ Type} = \frac{EAL_{Composite\ Consequence\ Type}}{Full\ Community\ Exposure_{Consequence\ Type}}$$

where:

$EAL\ Rate_{Composite\ Consequence\ Type}$  is the Expected Annual Loss Rate for the community from all relevant hazard types for a specific consequence type.

$EAL_{CompositeConsequenceType}$  is the Expected Annual Loss value for the community from all relevant hazard types for a specific consequence type.

$FullCommunityExposure_{ConsequenceType}$  is the full community exposure for a specific consequence type.

Total Expected Annual Loss Rates, which consider all three consequence types, are not calculated. These rates are difficult to interpret and vary dramatically depending on each community's building, population, and agriculture profile.<sup>25</sup> While a Total Expected Annual Loss Rate value can be calculated, the results are meaningless for supporting decisions. The National Risk Index team cautions against attempting to calculate or use any "all consequence" variations of the Expected Annual Loss Rate as described in this document. However, an approach for normalizing these variations across consequence types to develop a Total Expected Annual Loss Rate National Percentile metric is discussed in the next section.

### 5.6.1. EXPECTED ANNUAL LOSS RATE NATIONAL PERCENTILE

Although it is not meaningful to calculate an all-consequence type total Expected Annual Loss Rate, it is feasible to calculate an Expected Annual Loss Rate National Percentile as a means of ranking communities by Expected Annual Loss Rate across all consequence types.

The first step is to calculate Expected Annual Loss Rate National Percentiles for each consequence type ([Equation 17](#)). This is applicable to composite and hazard type metrics.

#### Equation 17. Expected Annual Loss (EAL) Rate National Percentile (NPCTL)

$$EAL\ Rate\ NPCTL_{ConsequenceType} = National\ Percentile(EAL\ Rate_{ConsequenceType})$$

National percentiles are calculated by comparing each community to all other communities at the same level where a selected hazard type is applicable. For example, when calculating national percentiles for the Coastal Flooding hazard type, only communities where Coastal Flooding is deemed possible are included in the set. So, the county with the lowest Expected Annual Loss Rate for Coastal Flooding, among counties where Coastal Flooding is possible, will appear close to the 0<sup>th</sup> percentile rather than the 95<sup>th</sup> percentile, because Coastal Flooding is not applicable to 95% of counties.

<sup>25</sup> There are two primary reasons for this. First, Agriculture Expected Annual Loss Rates tend to be significantly higher than Population or Building Expected Annual Loss Rates because of relatively high Historic Loss Ratios for agriculture compared to the other consequence types. This creates dramatic fluctuation when two counties with similar hazards have significantly different levels of agriculture development. Second, the Value of Statistical Life rightly implies that the lives within a community are far more important than the buildings or agriculture in that community. However, the dollar value equivalent of a community's population drastically outweighs the other consequence types when summing the full community exposure across consequence types.

Next, the Interim Total Expected Annual Loss Rate is calculated for each hazard type and composite using [Equation 18](#).

**Equation 18. Interim Total Expected Annual Loss (EAL) Rate**

$$\begin{aligned} & \text{Interim Total EAL Rate} \\ &= \text{EAL Rate NPCTL}_{Bldg} \times \frac{EAL_{Bldg}}{EAL_{Total}} \\ &+ \text{EAL Rate NPCTL}_{Pop} \times \frac{EAL_{Pop}}{EAL_{Total}} \\ &+ \text{EAL Rate NPCTL}_{Ag} \times \frac{EAL_{Ag}}{EAL_{Total}} \end{aligned}$$

This equation applies an Expected Annual Loss-weighted average to ensure that the consequence type contributing the most Expected Annual Loss for the community is given the most weight in determining the Expected Annual Loss Rate National Percentile (NPCTL).

The Expected Annual Loss Rate National Percentile (NPCTL) is then calculated using [Equation 19](#).

**Equation 19. Expected Annual Loss (EAL) Rate National Percentile (NPCTL)**

$$EAL \text{ Rate NPCTL} = \text{National Percentile}(\text{Interim Total EAL Rate})$$

Taking the national percentile of the Interim Expected Annual Loss Rate National Percentile ensures that the result follows a typical uniform statistical distribution of a percentile calculation.

Within the data download, the National Risk Index provides national percentiles for the total Expected Annual Loss metric for composite and each hazard type. They are not provided for individual consequence types.

### 5.6.2. SOCIAL VULNERABILITY AND COMMUNITY RESILIENCE ADJUSTED EXPECTED ANNUAL LOSS RATE NATIONAL PERCENTILES

A Social Vulnerability and Community Resilience Adjusted Expected Annual Loss Rate National Percentile metric is also provided in the data download to address the community risk factors. [Equation 20](#) is used to calculate this rate:

**Equation 20. Adjusted Expected Annual Loss (EAL) Rate**

$$\text{Adjusted EAL Rate} = \text{EAL Rate} \times \text{Community Risk Factor}$$

$$\text{where Community Risk Factor} = f\left(\frac{\text{Social Vulnerability}}{\text{Community Resilience}}\right)$$

Social Vulnerability and Community Resilience Adjusted Expected Annual Loss Rate National Percentile is only available as a composite metric, not for individual hazard types.



### 5.6.3. UNDERSTANDING EXPECTED ANNUAL LOSS RATES

Expected Annual Loss Rate metrics are calculated as decimals but can be effectively understood and communicated as the fraction of a value (whether buildings, people, or agriculture) that could be expected to be lost annually on average. In the National Risk Index, Expected Annual Loss Rates are expressed as decimals and can be interpreted in the following manner:

- A building Expected Annual Loss Rate of 0.00013 is equivalent to an expectation of \$1 out of every \$7.7K of building value lost to natural hazards annually on average.
- A population Expected Annual Loss Rate of 0.000043 is equivalent to an expectation of 1 out of 23K people dying from natural hazards annually on average.
- An agriculture Expected Annual Loss Rate of 0.008 is equivalent to an expectation of \$1 out of every \$125 of agriculture value lost to natural hazards annually on average.

In each of the examples above, it is critical to understand the basis upon which the values are calculated. First, they are expected rates, meaning that they are based on models that cannot predict future losses. Second, they are average rates, meaning that actual annual experience may deviate significantly from the projected rates (e.g., some years may experience 10 times the rate, while others may have no losses). Finally, they are annual rates, meaning that they apply to only a 1-year period. A 10-year period would have 10 times the average expected loss.

### 5.6.4. CALCULATING EXPECTED ANNUAL LOSS

Expected Annual Loss is calculated using a multiplicative equation that considers the consequence risk factors of natural hazard exposure, Historic Loss Ratio, and the likelihood risk factor of natural hazard annualized frequency for 18 hazard types. The Expected Annual Loss value for each consequence type is calculated by multiplying the exposure value of an area by the estimated annualized frequency and the Historic Loss Ratio (see [Equation 21](#)). See [Section 5 Expected Annual Loss Factors](#) for further explanation of these Expected Annual Loss factors and how they are computed. Expected Annual Loss Values are computed at the Census block level (or the Census tract level for some hazard types) for each relevant consequence type and summed up to produce a total Expected Annual Loss for each hazard type. A composite Expected Annual Loss is also summed from all hazard type Expected Annual Loss Values for the community (see [Equation 21](#)).



**Equation 21: Expected Annual Loss Values**

$$\begin{aligned}
 \text{Expected Annual Loss}_{\text{Hazard Type Consequence Type}} &= \text{Exposure}_{\text{Hazard Type Consequence Type}} \times \text{Annualized Frequency}_{\text{Hazard Type}} \\
 &\times \text{Historic Loss Ratio}_{\text{Hazard Type Consequence Type}} \\
 \text{Expected Annual Loss}_{\text{Hazard Type Total}} &= \text{Expected Annual Loss}_{\text{Hazard Type Building Value}} \\
 &+ \text{Expected Annual Loss}_{\text{Hazard Type Population Value}} \\
 &+ \text{Expected Annual Loss}_{\text{Hazard Type Agriculture Value}} \\
 \text{Expected Annual Loss}_{\text{Composite}} &= \sum_{i=1}^{18} \text{Expected Annual Loss}_{\text{Hazard Type Total } i}
 \end{aligned}$$

A cube root transformation is applied to the hazard type Expected Annual Loss Value for each community to address skew. The resulting transformed Values are then min-max normalized (0.00 – 100.00 scale) to produce an Expected Annual Loss score for each hazard type (see [Equation 22](#)). The composite Expected Annual Loss Score is calculated using the same cube root transformation and min-max normalization process shown in [Equation 22](#). County and Census tract Expected Annual Loss Scores were classified into five qualitative categories, from “Very Low” to “Very High,” using k-means clustering (see [Section 3.2 Values, Scores and Ratings](#)).

**Equation 22: Expected Annual Loss Scores**

$$\begin{aligned}
 \text{EAL Cube Root}_{\text{Hazard Type Total}} &= \left( \sqrt[3]{\text{Expected Annual Loss}_{\text{Hazard Type Total}}} \right) \\
 \text{Expected Annual Loss Score}_{\text{Hazard Type Total}} &= \frac{\text{EAL Cube Root}_{\text{Hazard Type Total}} - \text{Min}(\text{EAL Cube Root}_{\text{Hazard Type Total}})}{\text{Max}(\text{EAL Cube Root}_{\text{Hazard Type Total}}) - \text{Min}(\text{EAL Cube Root}_{\text{Hazard Type Total}})}
 \end{aligned}$$

While each hazard type uses the same factors to calculate Expected Annual Loss, these computations require different approaches due to the varying nature of the hazard types and the differences in source data format. A set of common analytical techniques (see [Section 5.6.5 Analytical Techniques](#)) are leveraged to achieve the best possible consistency between all hazard types for accurate calculation. The process for computing the Expected Annual Loss and its factors for each hazard type are described in the specific sections for each hazard type ([Sections 6 through 23](#)).

See [Table 17](#) for a simplified example of a county-level Expected Annual Loss calculation for the Hail hazard type. All three consequence types are included in the calculation of the Hail Expected Annual

Loss. By multiplying the county's consequence exposure, annualized frequency, and specific Historic Loss Ratio for each consequence type, an Expected Annual Loss Value for that consequence type is determined. The Values for each consequence are summed up to produce total Expected Annual Loss Value for Hail for the county. This total Expected Annual Loss Value is used to derive the hazard type's Expected Annual Loss Score for that county. This computation includes a min-max normalization using the total Expected Annual Loss Values of all counties in the United States for each hazard type. The total Expected Annual Loss for Hail is summed with the total Expected Annual Loss Values for the 17 other hazard types to calculate the composite Expected Annual Loss, which is scored in the same way.

**Table 17: Example of a County-Level Expected Annual Loss Calculation for Hail**

| <i>EAL Factor</i>    | <i>Building Value</i> | <i>Population &amp; Population Equivalence</i> | <i>Agriculture Value</i> |
|----------------------|-----------------------|------------------------------------------------|--------------------------|
| Exposure             | \$28.21 B             | 310,235 people or \$3.6 T                      | \$77.03 M                |
| Annualized Frequency | 3.9 events/year       | 3.9 events/year                                | 3.9 events/year          |
| HLR                  | 3.1e-5                | 1.3e-8                                         | 3.2e-6                   |
| EAL                  | \$3.47 M              | 0.016 people or \$185,600                      | \$968                    |

### 5.6.5. ANALYTICAL TECHNIQUES

Arriving at a dollar value representing the Expected Annual Loss due to each of the 18 hazard types for every county and Census tract in the United States requires multiple analytical techniques utilized across all hazard types to ensure the most accurate and consistent representation of Expected Annual Loss.

### Processing Database

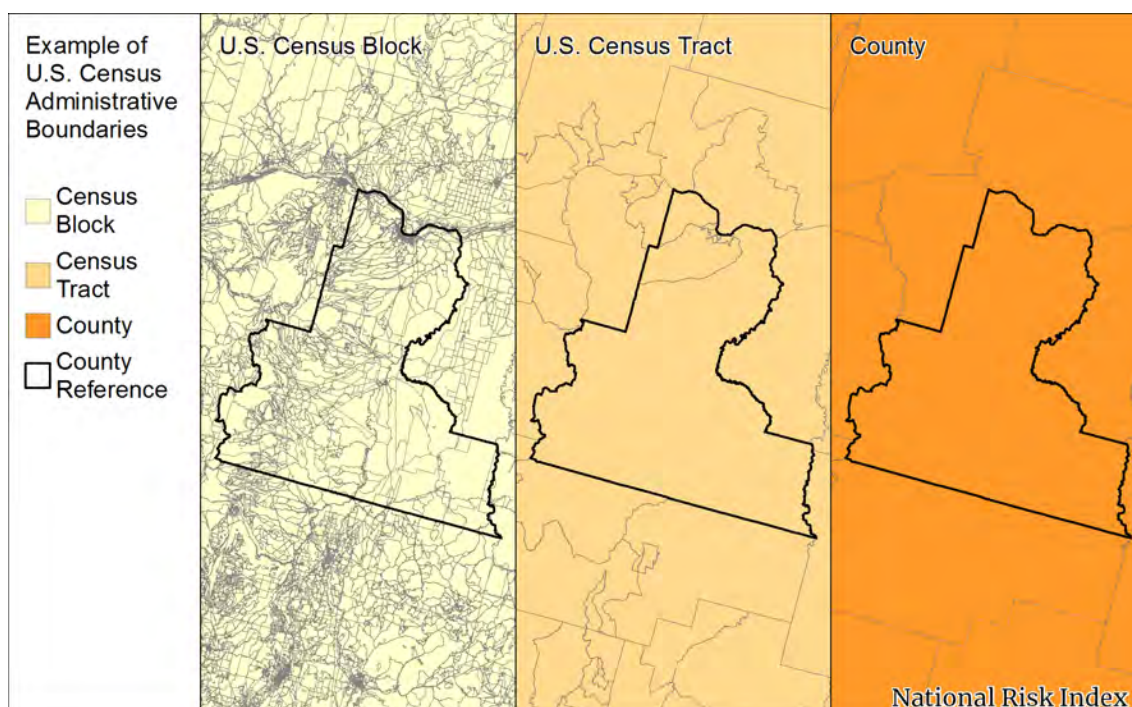
To support the processing of the National Risk Index, dedicated SQL Server and PostgreSQL database environments were established. Using a relational database with spatial capabilities to store and analyze each dataset used to compute the National Risk Index's Values and Scores provides a variety of benefits. The database allows for computational efficiencies when calculating the factors of the Expected Annual Loss for more than 11 million Census blocks in the United States. Grouping and aggregation functions are used to easily aggregate these Values into the Census tract- and county-level Values. Implementation of methodologies in stored procedures and functions allows for the application and adaptation of complex business logic and spatial analysis. The processing database also makes quality control easier by allowing complex calculations to be processed in steps, with the output for each step accessible in its own table. Records for each Census block are checked to identify outliers and any potential problems with the methodology or algorithms. Additionally, repeatable processes are modified and run in smaller portions, cutting down on processing time as methodology is adapted. For example, a change in source data for a hazard type only requires the replacement of source data tables for that hazard type and for the reprocessing of

a single hazard type. The processing database also supports version control and allows backups of each version to be stored securely.

Most spatial functions, such as buffering and intersection, are performed within the processing database. However, some processes necessitate the additional use of ArcGIS tools and functions. The outputs of these external processes are transferred and stored within the processing database, where they are used to compute the components of the Expected Annual Loss.

### Geographic/Administrative Layers

Expected Annual Loss factors may be calculated at three different administrative layers: Census block, Census tract, and county. The most granular level is the Census block. Where possible, Values are calculated at this level and then aggregated. The source of the boundaries for these layers is the United States Census Bureau's 2021 TIGER/Line shapefiles<sup>26</sup>, except for Connecticut which uses the TIGER 2024 boundaries. The shapefiles include United States Territories and some large bodies of water. All spatial layers are transformed into the NAD83 coordinate reference system. [Figure 11](#) provides examples of Census block, Census tract, and county boundaries.



**Figure 11: Example of Census Block, Census Tract, and County Shapes**

The National Risk Index also supplies a relational dataset mapping Census tract and county data to American Indian Areas. Federal Emergency Management Agency utilizes two authoritative sources for the locations of tribal lands, including reservations, statistical areas, and trust lands. The first is the Homeland Infrastructure Foundation-Level Data American Indian/Alaska Native/Native Hawai'ian

<sup>26</sup> United States Census Bureau. (2021). *TIGER/Line Shapefiles and Geodatabases* [cartographic dataset]. Retrieved from [https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List\\_230945507.html](https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List_230945507.html).

Areas<sup>27</sup> shapefile that is adapted from the United States Census Bureau's Topologically Integrated Geographic Encoding and Referencing/Line American Indian Area Geography shapefile.<sup>28</sup> The shapefile includes federally recognized American Indian reservations and off-reservation trust land areas, state-recognized American Indian reservations, and Hawai'ian homelands. The second shapefile, used internally by Federal Emergency Management Agency and referred to as the Federal Emergency Management Agency Mitigation Planning Jurisdiction Layer,<sup>29</sup> is adapted from the Bureau of Indian Affairs geospatial data.<sup>30</sup> This data includes Land Area Representations and Tribal Statistical Area, among other types of tribal areas. While the Homeland Infrastructure Foundation-Level Data and Federal Emergency Management Agency datasets have most areas in common, they are not identical.

To build the relational dataset, each shapefile was intersected against the Census block layer. If a tribal area covered at least 5% of at least one Census block within a Census tract, a relationship was established between the area and the Census tract. Census tracts that intersect a tribal area, but with less than 5% coverage in a Census block, were visually inspected to ensure that areas of intersection arose out of natural imprecision in the boundaries rather than valid cases of small tribal areas within the Census tract. Following this process, 11 relationship records at the Census tract level and 9 records at the county level were manually included in the final dataset.

The work of the National Risk Index to evaluate risk at the county and Census tract level has not been duplicated to give tribal entity risk. Instead, the National Risk Index provides data to identify the spatial relationships between tribal areas and communities; so, that associated county and Census tract profiles are applied to understand tribal entities' risk and Expected Annual Loss data.

### Determining County-Level Possibility of Hazard Occurrence

Not all hazard types are able to occur in all areas. For example, Coastal Flooding is not able to occur in Kansas, and Avalanches are not able to occur on flat terrain. The National Risk Index logically differentiates areas where a given hazard type is unlikely or has never occurred from areas where that hazard type is not able to occur using a control table in the database that designates where each hazard type is able to occur. This designation is based on counties that intersect past hazard occurrence polygons generated through spatial processing, have some possibility of occurrence as identified by probabilistic or susceptibility source data, or have a recorded loss due to hazard occurrence. Hazard type Expected Annual Loss is only calculated for communities where it is possible for the hazard type to occur.

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<sup>27</sup> Department of Homeland Security (DHS). (2025). *AIANNH Areas*. [cartographic dataset]

<sup>28</sup> United States Census Bureau. (2024). *American Indian Area Geography*. [cartographic dataset] Retrieved from <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>.

<sup>29</sup> FEMA. (2020). *FEMA Mitigation Planning Jurisdiction Layer*. [cartographic dataset] Supplied by FEMA.

<sup>30</sup> BIA Branch of Geospatial Support, U.S. Department of the Interior. (2018). *American Indian and Alaskan Native LARs*. [cartographic dataset] In 2025, data migrated to <https://onemap-bia-geospatial.hub.arcgis.com/>.

## Base Calculation and Aggregation

One of the National Risk Index's strengths is that it determines the Expected Annual Loss for an area at the lowest geographical level deemed appropriate, predominantly the Census block level. Expected Annual Loss is determined by assessing the combination of a specific community's annualized frequency and associated consequence if it were to occur (for example, how often Inland Flooding occurs in the area and what buildings, population, and agriculture are potentially affected). For many hazard types, annualized frequency and exposure are highly localized. Modeling the annualized frequency in coordination with its exposure provides the best assessment.

The Census block is currently the lowest administrative level at which population and building value data are nationally, consistently, and publicly available. By performing the Expected Annual Loss calculation at the Census block level, the National Risk Index is more accurately assessing Expected Annual Loss by looking at specific annualized frequency and exposure combinations at the lowest possible resolution. The National Risk Index provides the most relevant aggregations to its users, namely Expected Annual Loss Values at the Census tract and county levels. For all hazard types, Census tract- and county-level exposure and annualized frequency are calculated by aggregating Values from the Census block level. The processing database ensures that the aggregated Expected Annual Loss Values do not exceed the aggregated level's actual exposure values.

Data are also aggregated at the state level. Exposure and Expected Annual Loss Values of all Census blocks within each state are summed to give that state's Values for each hazard type by consequence types. State-level exposure and Expected Annual Loss aggregation for all hazard types except Earthquake is performed in this way. Earthquake exposure and Expected Annual Loss Values are extracted from the Federal Emergency Management Agency Hazus Earthquake Analysis at the Census tract level and aggregated to state level estimations.

Expected Annual Loss Values are provided at the state level for each hazard-consequence type combination (i.e., all 18 hazard types for building, population, and agriculture; including Composite Risk). Total state building values, population, and agriculture values are set as ceilings on values by consequence type. The sum of the Expected Annual Losses for each hazard type for each relevant consequence type is used to calculate the state's national Expected Annual Loss percentile using a cumulative distribution function to determine the state's relative Expected Annual Loss. Predictably, this statistic places more populated states with higher building values in the highest percentiles while small and sparsely populated states are in the lower percentiles.

State level Risk Index Scores and Ratings are not currently supplied in the dataset. The Social Vulnerability measure from the Census and the FEMA modified version of the Baseline Resilience Indicators for Communities index, the base for Social Vulnerability and Community Resilience components, respectively, are only provided at the Census tract and/or county level. It was determined that deriving meaningful state level representations of these values was out of the current scope of the National Risk Index; so, state-level Risk Index Scores are not calculated within the dataset. Hazard type annualized frequency and Historic Loss Ratios are also not calculated at the state level.



## Representation of Hazard Types as Spatial Polygons

Expected Annual Loss factors for each hazard type are derived from one or more sources of spatial hazard information. This can include identified hazard-susceptible zones, spatiotemporal records of past hazard occurrences, and countywide records of economic loss due to a hazard occurrence. The format of spatial source data varies by hazard type. Annualized frequency and exposure calculations typically require spatiotemporal records of past hazard occurrences or probabilistic modeling.

Necessary conversions to prepare for intersection with reference layers are performed with PostGIS spatial operations in the PostgreSQL database environment. The most common method of hazard conversion used for calculations is the buffering of points and lines to form polygons.

Point and line representations of hazard occurrences or hazard-susceptible zones are buffered by different distances depending on the hazard type. Point buffers allow for better representation of hazard occurrence coverage or area of possible impact. Path representations, such as those for Tornado and Hurricane, are included in the source data as a series of points with a common identifier (e.g., StormID). These are connected by a line or multi-segmented line. The line is then buffered by a distance depending on the severity of the Tornado, using the Enhanced Fujita (EF-) scale, or Hurricane, using the Saffir-Simpson scale, hazard occurrence. See the spatial processing discussion in the sections for each hazard type ([Sections 6 through 23](#)) for more detail on the buffering techniques used.

Conversion from raster to polygon vector format is performed by using ArcGIS Pro's Raster to Polygon conversion tool. In vector format, attributes from the source raster data are used to filter or select the data needed for methodology calculations for a specific hazard type.

## Dasymetric Developed Area, and Agricultural Area Census Blocks

For more refined estimations of hazard exposure and frequency at the Census block level, the spatial representations of most hazards were intersected with a dasymetric subset of the Census blocks. To create the dasymetric layer, several spatial datasets are utilized as inputs to custom code to isolate developed area within the 2021 TIGER Census block boundaries<sup>31</sup>, except for Connecticut which uses the TIGER 2024 boundaries. For the 50 states and the District of Columbia, three input datasets identifying buildings are pixelated and are each spatially intersected with the Census block polygons to establish a building-Census block relationship. These datasets are the Oak Ridge National Lab Building Footprint dataset<sup>32</sup>, the Microsoft Bing Building Footprint dataset<sup>33</sup>, and the United States Army Corps of Engineers' National Structure Inventory dataset<sup>34</sup>.

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<sup>31</sup> United States Census Bureau. (2021). *TIGER Census Blocks National Geodatabase* [cartographic dataset]. Retrieved from [https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List\\_230945507.html](https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List_230945507.html).

<sup>32</sup> FEMA, ORNL (2021). *USA Structures* [online dataset]. Retrieved from <https://gis-fema.hub.arcgis.com/pages/usa-structures>.

<sup>33</sup> Microsoft (2018). *Microsoft Building Footprints* [online dataset]. Retrieved from <https://www.microsoft.com/en-us/maps/building-footprints>.

<sup>34</sup> USACE (2022). *NSI* [online dataset]. Retrieved from <https://nsi.sec.usace.army.mil/downloads/>.

These same three building datasets are also each spatially intersected with the water bodies in the 2021 TIGER Areal Hydrography dataset<sup>35</sup> to establish a building-water body relationship. Any building footprint polygons or National Structure Inventory points that intersect a water body are filtered out of the input dataset. Census blocks which contain only water are removed from input into the dasymetric processing of the TIGER Census block boundaries. Additional processing ensures that building footprints in Census blocks intersecting 2019 National Land Cover Database raster pixels with the classification Open Water are not included in the dasymetric output layer.

For the entire national dataset, all pixels marked for dasymetric inclusion for a Census block are integrated as a union, and then the Census block boundary is used to clip the unioned shape to generate the final product: the Census block dasymetric geometry. The tract-level dasymetric geometry is an aggregation of the Census block boundaries and the tract area is the summation of Census block dasymetric area within the tract. The area of the dasymetric Census block is considered the developed area of the block.

For most hazards, polygons intersect with the dasymetric Census block or Census tract layer to calculate hazard frequency and building and population exposure. Other hazards use the 2019 National Land Cover Database raster files<sup>36</sup> to determine developed area within the Census block. There are five classes of developed areas (Developed, Developed Open Space, and Developed Low, Medium, and High Intensity) and TIGER Census block intersection with pixels with these classifications is used to create the developed area layer. The same process is used with the classifications Pasture/Hay and Cultivated Crops to create an agricultural area layer. The dasymetric, developed area, and agricultural area vector layers are created using a custom raster-vector intersect tool. Note that, when developed area is referred to in this document, the dasymetric definition of developed area is what is being described.

## Intersection

Determining areas of spatial intersection between hazard occurrences or hazard-susceptible zones and the various levels of reference layers is an essential function used in calculating Expected Annual Loss. The results of these intersections are stored in the processing database and used for multiple purposes. For many hazard types, the quantification of a hazard type's exposure is done at the Census block level. This requires the computation of intersecting areas of exposure. [Figure 12](#) provides an example of a hazard occurrence shape intersecting a Census block.

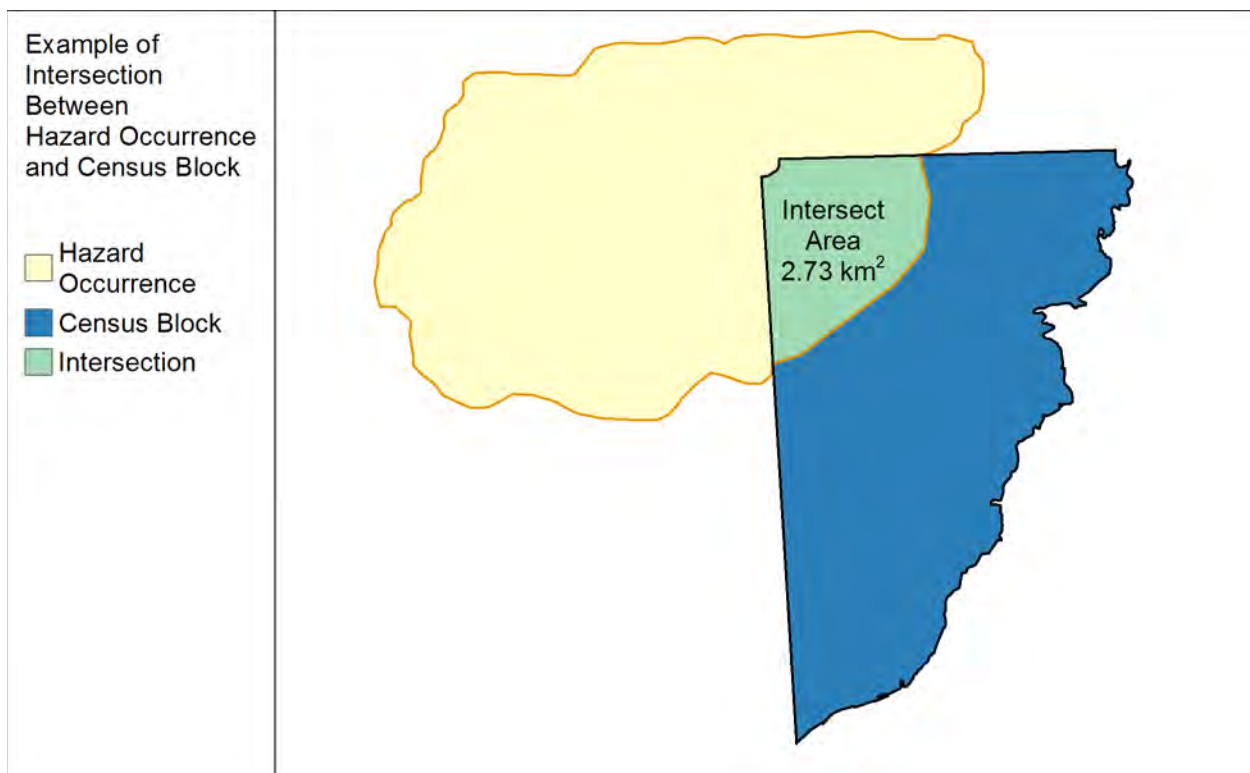
A custom intersect tool was developed to perform intersections more rapidly between various layers used to calculate hazard exposure and frequency. This tool stores the unique identifiers of the intersecting polygons, the area of intersection, and, if the intersection is a partial overlap of the reference polygon (typically the dasymetric Census block), the resulting intersect polygon in a results table. If the input polygon (typically a hazard representation polygon) completely contains the

<sup>35</sup> United States Census Bureau. (2021). *TIGER Areal Hydrography National Geodatabase* [cartographic dataset]. Retrieved from [https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List\\_230945507.html](https://www.census.gov/programs-surveys/geography/geographies/mapping-files.2021.List_230945507.html).

<sup>36</sup> Multi-Resolution Land Characteristics Consortium. (2019). *NLCD* [online dataset]. Retrieved from <https://www.mrlc.gov/data>.

reference polygon, only the area of the reference polygon and not the polygon itself are stored. This tool was used to perform all intersections except the intersections for creating dasymetric, developed, and agricultural area and the intersection of the Wildfire raster with the Census blocks. These were performed using the raster-vector intersect tool.

Annualized frequency computations typically involve counting the number of hazard occurrence polygons that intersect the Census block. Widespread hazards, like Hurricanes, often require a larger aggregation layer to represent the annualized frequency of hazard occurrences more accurately. For these hazard types, the intersection is performed with a 49-by-49 km fishnet grid, and the count of the fishnet grid cell is inherited by the Census blocks it encompasses using an area-weighted value when a Census block intersects more than one cell.



**Figure 12: Example of Intersection Between Hazard Occurrence and Census Block**

The 49-by-49 km grid cell size was used because of analysis that roughly estimated the average Census tract size to be 4,900 m<sup>2</sup> (or 70-by-70 m) and the average county size to be 2,500 km<sup>2</sup> (or 50-by-50 km), which was reduced slightly to 49-by-49 km to ensure the county size was a multiple of the Census tract size. Though the use of a grid at the average Census tract resolution was discarded, the use of the 49-by-49 km fishnet grid was maintained for the calculation of annualized frequency for widespread hazard types.



## 6. Avalanche

An Avalanche is a mass of snow in swift motion traveling down a slope.

This section provides an overview of the data and methodology used to estimate risk from Avalanche events. For the Avalanche hazard, an avalanche susceptibility layer was developed using slope and snow depth data to identify areas across the United States that could feasibly experience avalanche events. Historical loss and frequency data for the building and population consequence types were sourced from the Colorado Avalanche Information Center, SHELDUS, and National Oceanic and Atmospheric Administration's National Climatic Data Center Storm Events Database. Agricultural losses are not considered in the Avalanche Expected Annual Loss calculations. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 6.1. Avalanche Spatial Source Data

**Susceptible Area Source:** [1/3 arc-second Digital Elevation Models – U.S. Geological Survey National Map 3D Elevation Program \(3DEP\)](#) <sup>37</sup>

The 1/3 arc-second digital elevation models are a series of tiled elevations layers with a resolution of approximately 10-meters. These elevation models were produced under the 3D Elevation Program (3DEP). The data represent a seamless topographic bare-earth surface for the entirety of the United States, including Alaska, Hawai'i, and the United States Territories.

**Susceptible Area Source:** [Snow Data Assimilation System \(SNODAS\) Data Products at NSIDC, Version 1, Historical Snow Depth Data](#) <sup>38</sup>

The National Oceanic and Atmospheric Administration National Weather Service's National Operational Hydrologic Remote Sensing Center SNOW Data Assimilation System (SNODAS) contains snowpack properties, such as depth and snow water equivalent. SNODAS is a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center to provide the best possible estimates of snow cover and associated parameters to support hydrologic modeling and analysis. SNODAS data output is a gridded data set at 1-km resolution providing daily estimated snow depths from 2003 through 2024 for the contiguous United States.

**Susceptible Area Source:** National Water and Climate Center, Snow Survey and Water Supply Forecasting Program, [Snow Telemetry \(SNOTEL\) Network, Snow Depth Data](#) <sup>39</sup>

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<sup>37</sup> U.S. Geological Survey. (2023). *1/3rd arc-second Digital Elevation Models (DEMs) - USGS National Map 3DEP*. Downloadable Data Collection: U.S. Geological Survey.

<sup>38</sup> National Operational Hydrologic Remote Sensing Center. (2004). *Snow Data Assimilation System (SNODAS) Data Products at NSIDC*. (G02158, Version 1). [Data Set]. Boulder, Colorado USA. National Snow and Ice Data Center. <https://doi.org/10.7265/N5TB14TC>. [Historical Snow Depth Data]. Date Accessed 8-27-2024.

<sup>39</sup> United States, US Department of Agriculture, Natural Resource Conservation Service, National Water and Climate Center. Water and Climate Information System, Report Generator. (2024). *Air and Water Database*. <https://nwcc-apps.sc.egov.usda.gov/>. Retrieved on February 29, 2024.

The Natural Resources Conservation Service manages a comprehensive network of manually measured snow courses and automated Snow Telemetry (SNOTEL) monitoring sites throughout the western United States and Alaska. The SNOTEL Network is comprised of 900 automated data collection sites across remote, high elevation mountain watershed areas in the western United States with an additional 90 stations across Alaska. The records for each station are unique to that station, so some stations have daily snow depth records covering more than 50 years, but most of the stations with daily snow depth data have only been active since the mid-1990s.

**Susceptible Area Source:** [Northern Hemisphere 0.25 Degree Resolution Machine Learning Snow Depth Data](#)<sup>40</sup>

The Northern Hemisphere 0.25 degree resolution machine learning snow depth data is a data fusion framework based on Random Forest Regression of snow products from Advanced Microwave Scanning Radiometer for the Earth Observing System, Advanced Microwave Scanning Radiometer-2, Global Snow Monitoring for Climate Research, the Northern Hemisphere Snow Depth, ERA-Interim, and Modern-Era Retrospective Analysis for Research and Applications, version 2. It incorporates geolocation (latitude and longitude), and topographic data (elevation), as input independent variables. More than 30,000 ground observation sites were used as the dependent variable to train and validate the model in different time periods. This fusion framework resulted in a series of continuous daily snow depth data from 1980 through 2019 over the Northern Hemisphere with a spatial resolution of 0.25° (approximately 27.75 km).

**Historical Occurrence Source:** [Colorado Avalanche Information Center \(CAIC\) Accident Data](#)<sup>41</sup>

The Colorado Avalanche Information Center maintains a database of all reported avalanche fatalities going back to the 1950s. The data are most robust starting in 1994, which was chosen as the cutoff point for fatality frequency determination. The database contains the information for just under 1,000 avalanche events, with ~1/3 of the entries containing coordinate data. The remaining entries contain a physical description of the event locality, allowing the approximate placement of the event occurrence.

**Historical Occurrence Source:** [Arizona State University. Spatial Hazard Event and Loss Database for the United States \(SHELDUS\)](#)<sup>42</sup>

The best alternative to the Colorado Avalanche Information Center database for non-fatality events is SHELDUS, which contains event details from 1960 through 2023. The SHELDUS dataset contains over 1,600 avalanche events, ~300 of which produced fatalities, overlapping with the Colorado

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<sup>40</sup> Hu, Y., Che, T., Dai, L., Zhu, Y., Xiao, L., Deng, J., & Li, X. (2023). A long-term daily gridded snow depth dataset for the Northern Hemisphere from 1980 to 2019 based on machine learning. *Big Earth Data*, 8(2), 274–301. <https://doi.org/10.1080/20964471.2023.2177435>

<sup>41</sup> Colorado Avalanche Information Center (CAIC). (2023). *US Avalanche Accident Data* [online database]. Retrieved from <https://avalanche.state.co.us/accidents/statistics-and-reporting>. Dated 9/13/2024.

<sup>42</sup> CEMHS. (2025). *Spatial Hazard Events and Losses Database for the United States, Version 23.0*. [Online Database]. Phoenix, Arizona: Center for Emergency Management and Homeland Security, Arizona State University.

Avalanche Information Center database; however, most of the identified events were property and non-fatality injury events, supplying a better source for frequency calculations. The SHELDUS data are only identified down to the county level, limiting more precise location identification. For more information on SHELDUS, see [Section 5.4.1 Source Data](#).

**Historical Occurrence Source:** [National Oceanic and Atmospheric Administration National Climatic Data Center Storm Events Database](#)<sup>43</sup>

The National Oceanic and Atmospheric Administration National Climatic Data Center Storm Events Database, containing ~450 avalanche events, was used to supplement additional events not seen in either the Colorado Avalanche Information Center or SHELDUS databases. Storm Data are provided by the National Weather Service and contains statistics on personal injuries and damage estimates covering the United States. The data date back to as early as 1950 through to the present, updated monthly with up to a 120 day delay possible. The data contains a chronological listing, by state, of hurricanes, tornadoes, thunderstorms, hail, floods, drought conditions, lightning, high winds, snow, temperature extremes, and other weather phenomena.

## 6.2. Avalanche Spatial Data Processing

### 6.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Studies of avalanche hazards have indicated that there are two primary factors that lead to potential avalanche events: slope angle and snow depth.<sup>44</sup> While other factors such as land cover and slope aspect play a part in the frequency and size of avalanche events, they are generally considered to be minor factors.

According to subject matter experts, Avalanche events are most likely to occur in regions where the terrain slope is between 30 and 50° and the snow depth is 1 meter or more.<sup>45</sup> Slope and snow depth data were overlaid to identify areas which satisfied those two requirements, creating a national susceptibility map for all areas with the potential for an avalanche event, including insignificant events. With the focus being on avalanche events that would have a significant impact on lives and property, the broad scope of the susceptibility map was limited, using the identified locational qualifiers in [Section 6.2.5](#).

### 6.2.2. SLOPE PROCESSING

Historical avalanche release events frequently occur where slope angles are between 30 and 50 degrees; slopes less than 30 degrees are not steep enough to initiate an avalanche release, while those greater than 50 degrees are often too steep for enough snow to accumulate for a large avalanche event. Slope data was obtained using the U.S. Geological Survey nationwide 1/3 arc-

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43 NOAA, National Weather Service. (2024). NCDC Storm Events Database. [Online Database] <https://www.ncdc.noaa.gov/stormevents/>. Data obtained 8/29/2024.

44 Maggioni, M., & Gruber, U. (2003). The influence of topographic parameters on avalanche release dimension and frequency. *Cold Regions Science and Technology*, 37.3: 407-419.

45 Per communication with avalanche experts at the Colorado Avalanche Information Center (October 1, 2024).

second (~10-meter) digital elevation model. The 10-meter digital elevation model was upscaled to produce a 100-meter digital elevation model, from which the slope was calculated using the ArcGIS Pro Surface Parameters tool across the continental United States and Alaska. While a finer-grained analysis was possible, it was determined that the slopes identified on smaller scales included smaller stream valleys and road cuts that were not aerially large enough to produce significant avalanche events.

### **6.2.3. SNOW DEPTH PROCESSING: CONTINENTAL UNITED STATES**

Snowfall and snow depth are the other main factors for an avalanche event to occur. While avalanches are primarily known from areas of deep snow coverage, avalanche events can occur any time the snow depth exceeds 1 meter. The SNODAS data was analyzed to obtain the maximum daily snow depth at every grid cell for the entire 18-year history of the dataset. This maximum snow depth data was used to create a maximum snow depth triangular irregular network layer, from which a maximum snow depth raster was created. To minimize snow depth extrapolations amongst distant snow depth points, a 5-km buffer was added to all of the snow depth grid points to create a mask from which to extract the snow depth raster. All raster cells with a snow depth of 1-meter or greater were then extracted from the layer.

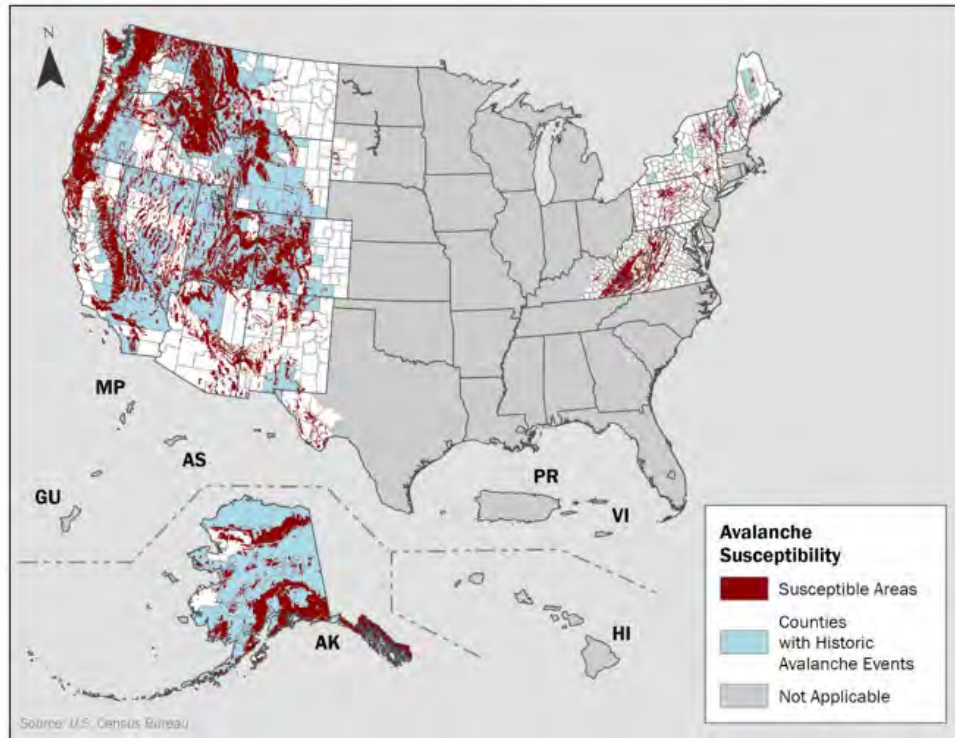
### **6.2.4. SNOW DEPTH PROCESSING: ALASKA**

SNODAS data were not available for Alaska, so a combination of SNOTEL and Hu et al.'s (2023) 0.25° machine learning-produced daily snow depth data were combined to identify areas with the potential to obtain greater than 1-meter snow depths across Alaska. For each SNOTEL station, a maximum daily snow depth was identified across the entire history of each station. The SNOTEL data points were supplemented with the 0.25° machine learning-produced daily snow depth data. The machine learning snow depth data is partially produced using passive microwave brightness to determine a semi-global map of daily snow depth. This method is known to underestimate snow depths when actual depths are greater than 1 meter, however since the avalanche threshold is 1 meter, the data can produce depth limits within the boundaries needed for this analysis. The 0.25° machine learning-produced daily snow depth data was contoured at 0.2 meters. The contours were used as “soft” lines with the SNOTEL data being used as “hard” points to create a triangular irregular network surface. This triangular irregular network was translated to a raster and, as with the conterminous United States, the snow depth was then selected for all areas that exceeded 1 meter in depth.

### **6.2.5. SUSCEPTIBILITY AREA REFINEMENT**

The historical record was used to identify locations where avalanche events have been known to occur. Outside of these areas, states, counties, and urban areas without previous avalanche hazards were examined for the possibility of future avalanche events. In that regard, midwestern counties and states east of the Rocky Mountains and the Badlands and areas south of Virginia/Kentucky were assumed to have a near-zero chance of an impactful avalanche event. In the Northeastern United States, areas outside of the Appalachian Mountains were considered to have low likelihood of

avalanche events. Susceptible zones across the continental United States were also analyzed where they intercepted urban areas as identified by the Census<sup>46</sup>. After careful analysis, it was decided that select urban areas should be excluded from the susceptibility layer if they were located within steep river valleys. This was likely a relic of spatial resolution of the data sources versus actual avalanche hazard and primarily impacted urban areas in the Northeastern United States. Urban areas in the western states that are located near mountain ranges were deemed potentially at risk. All states and counties identified as unlikely for future impactful avalanche events were removed from the analyses. These states and counties are illustrated in [Figure 13](#).



**Figure 13: Avalanche Susceptibility Zones**

## 6.3. Avalanche Non-Spatial Data Processing

### 6.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Avalanche, in events, is computed as the number of unique SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration-recorded Avalanche events that occurred within a county from January 1994 to December 2023. Because the exact location of the event within the county cannot be determined

<sup>46</sup> United States Census Bureau. (2024). 2020 Census Urban Areas. TIGER/Line Shapefiles [Data set]. <https://catalog.data.gov/dataset/tiger-line-shapefile-2020-nation-u-s-2020-census-urban-area>

from the SHELDUS or Colorado Avalanche Information Center record, historical event counts are not supplied at the Census tract level.

### 6.3.2. PERIOD OF RECORD

Combined, the SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration databases provide robust avalanche records dating from Jan 01, 1994 through Dec. 31, 2023, resulting in a period of record of 30 years. While the SHELDUS data do extend back to 1960, an analysis of the data shows extremely limited, and sometimes incorrect, event identifications prior to 1994, so a cutoff of 1994 was used for frequency determinations.

### 6.3.3. HISTORICAL EVENT FREQUENCY

Events from the National Climatic Data Center Storm Events Database, Colorado Avalanche Information Center Accident Data, and SHELDUS database were cross checked to ensure only unique events were included. All three of the databases were checked for quality and accuracy of the locations, as spatial locations occasionally did not match the physical description (i.e., a data point in Washington described an event in Utah). These historical events were then used to determine the frequency of events at the county level based on the number of historical events within that county across the period of record, 30 years.

## 6.4. Avalanche Exposure

### 6.4.1. BUILDING

Avalanches occur primarily in sparsely populated mountainous areas, so exposure values tend to be low. Avalanche exposure is determined by identifying the Hazus dasymetric shapes that overlap with the Avalanche Susceptibility Zones. The Hazus dasymetric shapes represent the highest likelihood of buildings within the susceptibility zones.

### 6.4.2. POPULATION

While building exposure estimates are straightforward, population exposure is difficult since people are typically exposed to avalanches while enjoying extracurricular activities in mountainous backcountries. For population exposure, default county values were identified based on historical fatality and injury records for each physiographic region ([Figure 15](#)). These default values are 3, 5, or 10 people per avalanche event based on how frequently people are injured or killed in each region ([Table 18](#)).

**Table 18: Avalanche Population Exposure Per Region**

| <i>Region</i>   | <i>Population Exposure</i> |
|-----------------|----------------------------|
| APPALACHIAN     | 5                          |
| ARCTIC LOWLANDS | 3                          |



| <i>Region</i>     | <i>Population Exposure</i> |
|-------------------|----------------------------|
| BASIN AND RANGE   | 10                         |
| COLORADO PLATEAUS | 5                          |
| COLUMBIA PLATEAU  | 5                          |
| PACIFIC BORDER    | 10                         |
| ROCKY MOUNTAINS   | 10                         |

In counties where there were no historical injuries or fatalities, a susceptibility weighting value was derived by dividing the total susceptible area within the county by the county with the largest amount of susceptible area within the region. This ratio was then multiplied by the regional population exposure to obtain the county-specific population exposure.

#### 6.4.3. AGRICULTURE

Avalanches occur during the winter months at times when temperatures are below freezing so agriculture is typically minimally impacted, and while avalanches have been known to knock over trees, the historical record of avalanche events impacting agricultural orchards is unknown.

#### 6.4.4. COUNTY-LEVEL EXPOSURE ESTIMATION

At the county level, the building exposure value is the summation of the building exposure value of all the Census tracts within the county, which is the summation of all exposure values within the Census blocks. The Census tract inherits its population exposure value from the parent county. For example, if a county has a default population exposure value set to 5 people, each tract within that county is also assigned a population exposure value of 5 people. This assumes that those 5 exposed people within that county may at any moment be located in any given Census tract where avalanches are possible.

### 6.5. Annualized Frequency

The annualized frequency value represents the number of Avalanche loss-causing occurrences, in events, each year over the period of record (30 years). This annualized frequency is calculated at the county level. The Census tract inherits the parent county-level value, and the Census tract value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the SHEL DUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration Avalanche event counts for the county and divide by the period of record using [Equation 23](#).



**Equation 23: County Avalanche Annualized Frequency**

$$Freq_{AVLN_{Co}} = \frac{EventCount_{AVLN_{Co}}}{PeriodRecord_{AVLN}}$$

where:

$Freq_{AVLN_{Co}}$  is the annualized frequency of Avalanche events determined for a specific county (events per year).

$EventCount_{AVLN_{Co}}$  is the total number of unique SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration Avalanche events that have impacted the county within the period of record.

$PeriodRecord_{AVLN}$  is the period of record for Avalanche (30 years).

**6.5.1. MINIMUM ANNUAL FREQUENCY**

If a county intersects an Avalanche susceptibility zone but has not experienced a loss-causing Avalanche event, it is assigned a minimum annual frequency of 0.01 or once in 100 years. This was determined by subject matter experts to be an acceptable assumption.

**6.5.2. ANNUALIZED FREQUENCY INHERITANCE**

The Census tract inherits its annualized frequency value from the parent county as in [Equation 24](#).

**Equation 24: Census Tract Avalanche Annualized Frequency Inheritance**

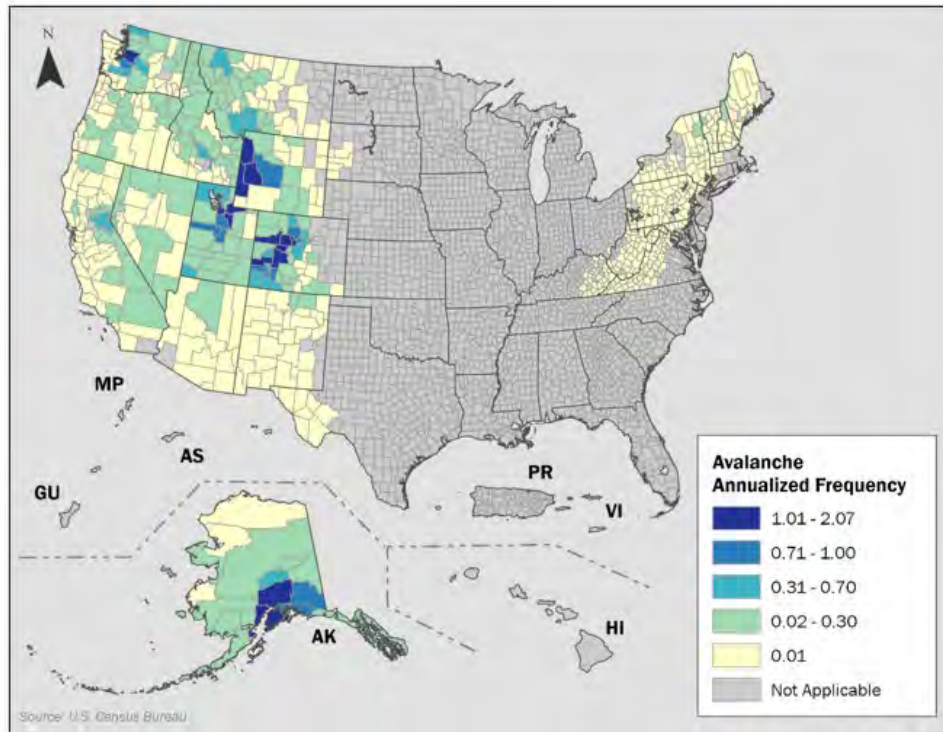
$$Freq_{AVLN_{CT}} = Freq_{AVLN_{Co}}$$

where:

$Freq_{AVLN_{CT}}$  is the annualized frequency of Avalanche events determined for a specific Census tract (events per year).

$Freq_{AVLN_{Co}}$  is the annualized frequency of Avalanche events determined for a specific county (events per year).

[Figure 14](#) displays Avalanche annualized frequency at the county level.



**Figure 14: Avalanche Annualized Frequency by County**

## 6.6. Avalanche Historic Loss Ratio

The Avalanche Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to an Avalanche occurrence, or the average rate of loss associated with an Avalanche occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Avalanche hazard type.

Loss data are provided by SHELDUS,<sup>47</sup> Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. For building value losses, SHELDUS data from 1996 through 2023 are included in the Historic Loss Ratio for Building calculations. For population losses, SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration events from 1994 through 2023 are included in the Historic Loss Ratio for Population calculations. Three peril types are mapped to the hazard Avalanche (see [Table 19](#)). These native records are aggregated on a timeframe basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

<sup>47</sup> For Avalanche loss information, SHELDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

**Table 19: Avalanche Peril Types and Recorded Events from 1994-2023**

| Loss Source                                                       | Peril Type     | Total Loss Records | Total Records per Event Basis                                                        |
|-------------------------------------------------------------------|----------------|--------------------|--------------------------------------------------------------------------------------|
| SHELDUS                                                           | Avalanche      | 1,418              | 1,067 (unique events not in Colorado Avalanche Information Center database)          |
|                                                                   | Avalanche-Snow | 0                  | 0                                                                                    |
|                                                                   | Snow-Slide     | 0                  | 0                                                                                    |
| Colorado Avalanche Information Center Fatality data <sup>48</sup> | N/A            | 663                | 663                                                                                  |
| National Oceanic and Atmospheric Administration Injury data       | N/A            | 453                | 32 (unique events not in Colorado Avalanche Information Center or SHELDUS databases) |

The Historic Loss Ratio Building exposure value used in the Loss Ratio per Basis calculation is determined by summing the developed area exposure values of the Census tracts that intersect the Avalanche susceptibility layer times the dasymetric rate for the county. For building Loss Ratio per Basis calculations, the Loss Ratio per Basis for each SHELDUS -documented event is calculated using [Equation 25](#).

**Equation 25: Loss Ratio per Basis Building Calculation for a Single Avalanche**

$$LRB_{AVLN Co CnsqType} = \frac{LOSS_{AVLN Co CnsqType}}{HLR_{Exposure_{AVLN Co CnsqType}}}$$

where:

$LRB_{AVLN Co CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Avalanche event.

$LOSS_{AVLN Co CnsqType}$  is the loss experienced from the Avalanche event documented to have occurred in the county (in 2022 dollars).

<sup>48</sup> The Colorado Avalanche Information Center data is not from the SHELDUS database and represents a separate accumulation of avalanche induced fatalities in the United States.

$HLRExposure_{AVLN_{CoCnsqType}}$  is the maximum default value of all the Census tracts within the county estimated to have been exposed to the Avalanche event occurrence (in 2022 dollars).

For Historic Loss Ratio Population calculations, the Loss Ratio per Basis uses a combination of the SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration databases for a total of 815 fatality and/or injury events. The Colorado Avalanche Information Center database contains all recorded fatalities due to avalanche events, therefore all fatality events in the SHELDUS and National Oceanic and Atmospheric Administration databases are excluded from the calculations. The Historic Loss Ratio population exposure is set using [Table 18](#). For population Loss Ratio per Basis calculations, the Loss Ratio per Basis for each documented event is calculated using [Equation 26](#).

**Equation 26: Loss Ratio per Basis Population Calculation for a Single Avalanche**

$$LRB_{AVLN_{CoCnsqType}} = \frac{LOSS_{AVLN_{CoCnsqType}}}{HLRExposure_{AVLN_{CoCnsqType}}}$$

where:

$LRB_{AVLN_{CoCnsqType}}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Avalanche event.

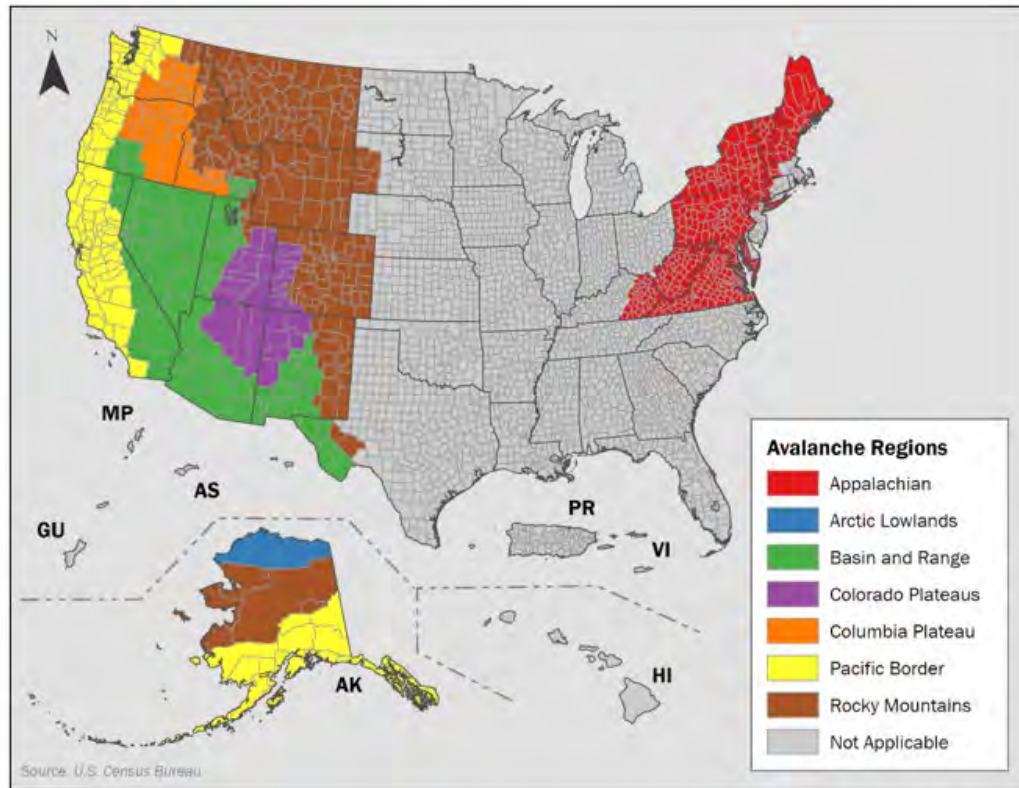
$LOSS_{AVLN_{CoCnsqType}}$  is the loss experienced from the Avalanche event documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{AVLN_{CoCnsqType}}$  is the number of people assumed to have been involved with each avalanche event.

Using all three databases allowed for the calculation and determination of zero-loss events over the course of the period of record. For building Loss Ratio per Basis calculations, average Loss Ratio per Basis values are calculated across Avalanche regions ([Figure 15](#)). Avalanche physiographic regions are based on the U.S. Geological Survey's physiographic provinces for the contiguous United States<sup>49</sup>. Information from Muhs and Budahn, 2019<sup>50</sup> was used to supplement the U.S. Geological Survey map for continuation into Alaska. Each county was then assigned a physiographic province based on where the majority of the county lay. The smaller provinces were then merged together in order to provide continuous provinces across large areas of the United States.

49 U.S. Geological Survey. (1946). Physiographic divisions of the conterminous U.S.: U.S. Geological Survey data release, <https://doi.org/10.5066/P9B1S3K8>.

50 Muhs, D. R., & Budahn, J. R. (2019). New geochemical evidence for the origin of North America's largest dune field, the Nebraska Sand Hills, central Great Plains, USA. *Geomorphology*, 332, 188-212



**Figure 15: Avalanche Physiographic Regions**

The average building Loss Ratio per Basis values across regions were calculated using only data from the SHELDES database from 1996 through 2023, since this database was the only one that specifically had building losses. Zero loss events within the SHELDES database were included within the average calculations. The total average Loss Ratio per Basis for each region was used as the Historic Loss Ratio for Building values for avalanche ([Table 20](#)).

**Table 20: Historic Loss Ratio Values for Avalanche**

| <i>Region</i>     | <i>HLRB Value</i> | <i>HLRP Value</i> |
|-------------------|-------------------|-------------------|
| APPALACHIAN*      | 0.000030054       | 0.032800000       |
| ARCTIC LOWLANDS*^ | 0.000030054       | 0.002974689       |
| BASIN AND RANGE   | 0.000071771       | 0.003505438       |
| COLORADO PLATEAUS | 0.000050349       | 0.004013063       |

| <i>Region</i>    | <i>HLRB Value</i> | <i>HLRP Value</i> |
|------------------|-------------------|-------------------|
| COLUMBIA PLATEAU | 0.003982173       | 0.055517241       |
| PACIFIC BORDER   | 0.000104972       | 0.004728560       |
| ROCKY MOUNTAINS  | 0.000030054       | 0.002974689       |

\*No building losses were available therefore the minimum Historic Loss Ratio for Buildings value was used, the Rocky Mountains region.

^No population losses were available therefore the minimum Historic Loss Ratios for Population value was used, the Rocky Mountains region.

Individual loss ratios were then calculated out for individual counties that experienced loss events. Counties where no historical building losses occurred would receive the regional Historic Loss Ratio for Building values, while counties that had experienced historical losses were assigned Historic Loss Ratio for Building values (regional, county, or combined weighted) based on the calculated Expected Annual Loss for Building compared to the annual total historical loss values.

For Historic Loss Ratio for Population calculations, population losses were calculated by giving each fatality a weight of 1, while each injury was given a weight of 0.1. These values were then added together. The population loss value was calculated out for all fatality and injury events within the SHELDUS, Colorado Avalanche Information Center, and National Oceanic and Atmospheric Administration databases, which was then divided by the population regional default exposure ([Table 18](#)). This ratio was added up at the county level and then divided by the estimated number of avalanches over the period of record (30 years). This provided the zero-loss event padding for an accurate loss rate assessment. Historic Loss Ratio for Population values were then calculated out for all counties with a historical population loss event. A Historic Loss Ratio for Population regional value was calculated by summing the regional loss ratios and dividing that by the total number of estimated Avalanches over the period of record. A Historic Loss Ratio for Population background value was chosen from the lowest calculated county-wide Historic Loss Ratio for Population value and attributed to all counties with no historical population loss events. Counties were then assigned Historic Loss Ratio for Population (regional, county, county minimal, or combined weighted) based on the calculated Expected Annual Loss for Population compared to the annual total historical loss values.

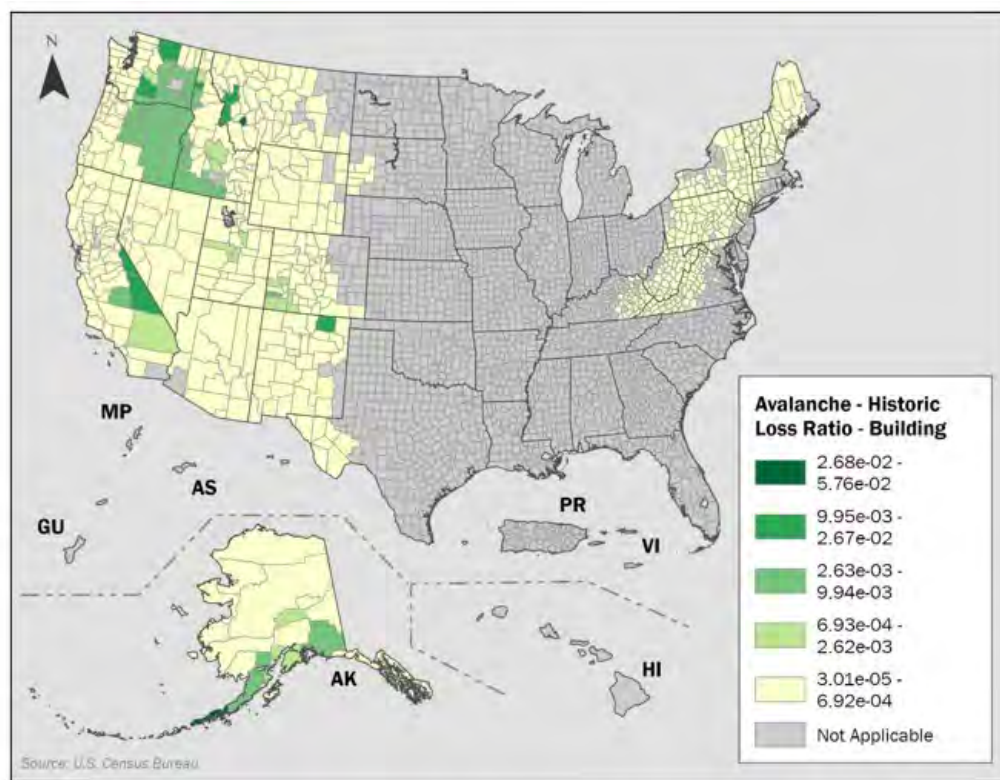
After the Loss Ratios per Basis are calculated for all Landslide events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Landslide.



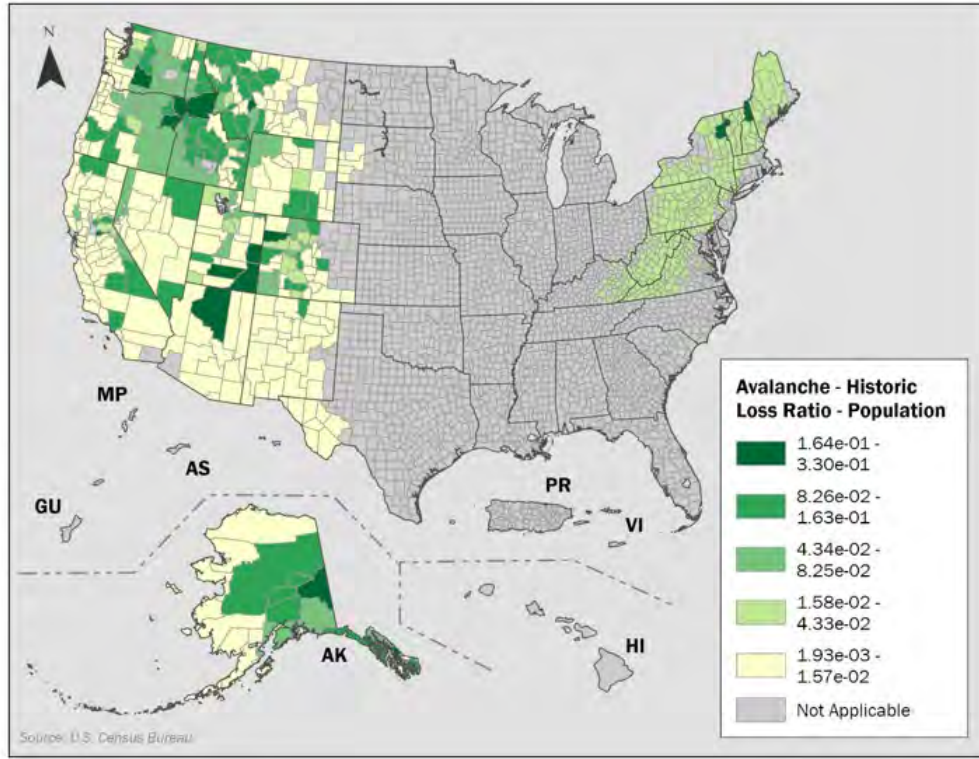
**Table 21: Avalanche Historic Loss Ratio Bayesian Weighting Level Sets**

| Consequence Type | Set ID | Bayesian Level                  | Set Refinement Filter | Set Utilization Threshold (Set Must Have) |
|------------------|--------|---------------------------------|-----------------------|-------------------------------------------|
| Population       | P1     | County                          | N/A                   | N/A                                       |
|                  | P2     | Avalanche Physiographic Regions | N/A                   | N/A                                       |
| Building         | B1     | County                          | N/A                   | N/A                                       |
|                  | B2     | Avalanche Physiographic Regions | N/A                   | N/A                                       |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values (see [Section 5.4.4 Historic Loss Ratio](#) Calculation Methodology). [Figure 16](#) and [Figure 17](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Avalanche. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

**Figure 16: Avalanche Bayesian-Adjusted Historic Loss Ratio – Building Value**





**Figure 17: Avalanche Bayesian-Adjusted Historic Loss Ratio – Population**

## 6.7. Avalanche Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census tract level using [Equation 27](#).

### Equation 27: Census Tract Expected Annual Loss to Avalanche

$$EAL_{AVLN_{CT_{Bldg}}} = Exposure_{AVLN_{CT_{Bldg}}} \times Freq_{AVLN_{CT}} \times HLR_{AVLN_{CT_{Bldg}}}$$

$$EAL_{AVLN_{CT_{Pop}}} = Exposure_{AVLN_{CT_{Pop}}} \times Freq_{AVLN_{CT}} \times HLR_{AVLN_{CT_{Pop}}}$$

where:

$EAL_{AVLN_{CT_{Bldg}}}$  is the building Expected Annual Loss due to Avalanche occurrences for a specific Census tract (in dollars).

$Exposure_{AVLN_{CT_{Bldg}}}$  is the building value exposed to Avalanche occurrences in the Census tract (in dollars).

$Freq_{AVLN_{CT}}$  is the Avalanche annualized frequency for the Census tract (events per year).

|                              |                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $HLR_{AVLN_{CT_{Bldg}}}$     | is the Bayesian-adjusted building Historic Loss Ratio for Avalanche for the Census tract.                                 |
| $EAL_{AVLN_{CT_{Pop}}}$      | is the population equivalence Expected Annual Loss due to Avalanche occurrences for a specific Census tract (in dollars). |
| $Exposure_{AVLN_{CT_{Pop}}}$ | is the population equivalence value exposed to Avalanche occurrences in the Census tract (in dollars).                    |
| $HLR_{AVLN_{CT_{Pop}}}$      | is the Bayesian-adjusted population Historic Loss Ratio for Avalanche for the Census tract.                               |

The total Expected Annual Loss value at the county level is the sum of the aggregated building and population equivalence Expected Annual Loss values at the Census tract level as in [Equation 28](#).

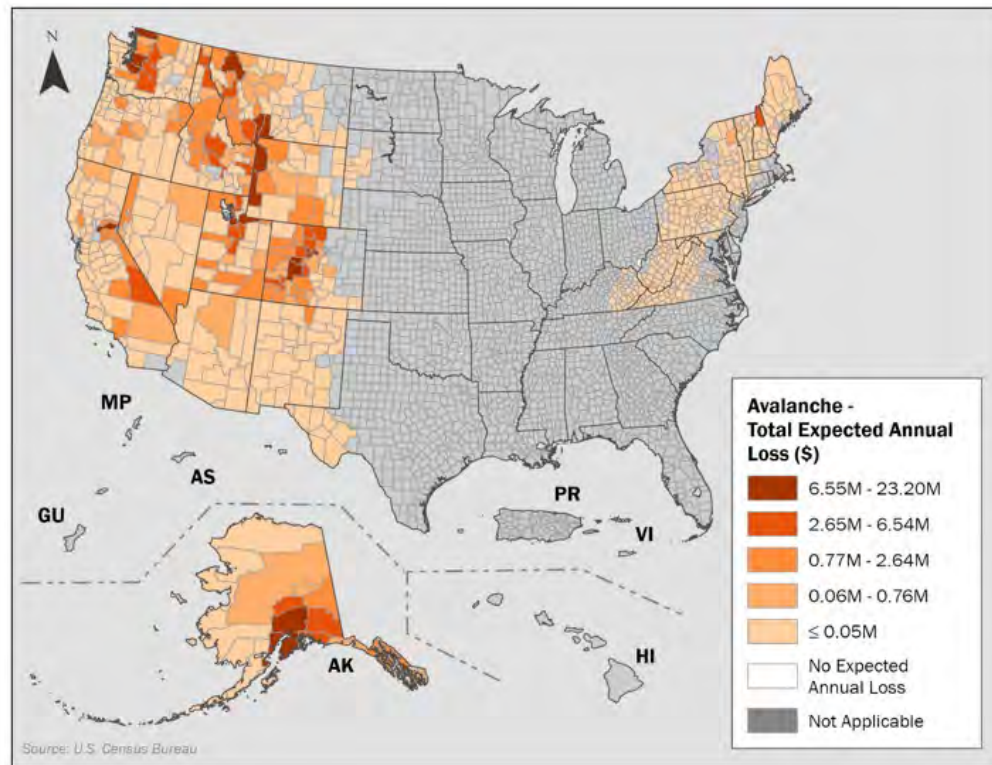
**Equation 28: County Expected Annual Loss to Avalanche**

$$EAL_{AVLN_{Co}} = \max(EAL_{AVLN_{CT_{Bldg}}}) + \max(EAL_{AVLN_{CT_{Pop}}})$$

where:

|                                |                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{AVLN_{Co}}$              | is the total Expected Annual Loss due to Avalanche occurrences for a specific county (in dollars).                                       |
| $\max(EAL_{AVLN_{CT_{Bldg}}})$ | is the maximum building Expected Annual Loss due to Avalanche occurrences of all Census tracts in the county (in dollars).               |
| $\max(EAL_{AVLN_{CT_{Pop}}})$  | is the maximum population equivalence Expected Annual Loss due to Avalanche occurrences of all Census tracts in the county (in dollars). |

[Figure 18](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Avalanche occurrences.



**Figure 18: Total Expected Annual Loss by County to Avalanche**

With the Avalanche total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Avalanche Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Avalanche Expected Annual Loss for population by the total population of the community.

## 6.8. Avalanche Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- This process is assumed to identify all areas that are potentially prone to Avalanche events. However, this is just a static snapshot of a very dynamic hazard – the triggering conditions that could cause an Avalanche event are variable year to year, month to month, and even day to day.

- This process is known to overestimate some areas where there are steep slopes but minimal potential for impactful Avalanche events, such as along river valleys, and areas in the southern states where no Avalanches are anticipated. While these areas were limited by excluding counties and urban areas known to not have historical Avalanche events, in other areas they are likely to still be prevalent. Further refinement of the identification process is needed to better differentiate these areas.
- The current potential Avalanche hazard map does not contain a gradational scale, which would identify areas within this susceptible region that are more or less likely to have Avalanche events. Features such as solar aspect angle and prevailing wind direction are known to have an impact on Avalanche potential. This may also be informed by previous inventories of North American Public Avalanche Danger Scale (NAPADS), which is a system used by Avalanche forecasters that rates avalanche danger based on the likelihood, size, and distribution of expected avalanches.<sup>51</sup>
- Even though the current source for historical Avalanche events is more comprehensive than was previously available, the number of Avalanche events is still vastly underreported since the historical data only record impactful events. The most comprehensive historical record from the Colorado Avalanche Information Center only contains records with fatalities and does not include records for injuries or population impact. The location of all the historical events is sometimes not fully accurate, with many records only being able to be identified down to the county level. There are also no indications in the current data of the cause of the Avalanche events as natural versus human-induced. With a greater historical record of Avalanche events, including natural events that are known to not have caused a significant impact on people or property, a more nuanced Avalanche hazard identification procedure can be implemented. These additional historical event records may be available from regional and local sources.

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<sup>51</sup> <https://avalanche.org/avalanche-encyclopedia/#north-american-public-avalanche-danger-scale>

## 7. Coastal Flooding

Coastal Flooding is when water inundates or covers normally dry coastal land as a result of high or rising tides or storm surges.

This section provides an overview of the data and methodology used to estimate risk from Coastal Flooding. The Coastal Flooding hazard utilizes the Federal Emergency Management Agency's National Flood Hazard Layer, the National Oceanic and Atmospheric Administration High Tide Flooding probability data and Sea, Lake, and Overland Surges from Hurricanes (SLOSH) maximum-of-maximum raster data to estimate annualized losses to coastal flooding and storm surge along the Pacific, Atlantic, and Gulf coastlines. For the Great Lakes region, a distinct methodology using the Federal Emergency Management Agency's Geospatial Flood Risk Assessment product was employed. The Coastal Flooding hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 7.1. Coastal Flooding Spatial Source Data

**Susceptible Area Source:** [National Flood Insurance Program, National Flood Hazard Layer \(NFHL\)](#)<sup>52</sup>

The National Flood Hazard Layer contains several layers depicting flood information, including levee locations, Flood Insurance Rate Map boundaries, and Special Flood Hazard Areas or floodplain polygons. The Special Flood Hazard Area polygons for 1% annual chance and 0.2% annual chance were downloaded in shapefile format for use in the calculation of Coastal Flooding exposure and annualized frequency.

**Susceptible Area Source:** CoreLogic Special Flood Hazard Area Layer

The CoreLogic digitized floodplain boundaries supplement the Federal Emergency Management Agency's official digital National Flood Hazard Layer data in areas where only paper Flood Insurance Rate Maps exist. These boundaries have been compiled by CoreLogic through the digitization of existing paper flood maps and the use of legacy paper the Federal Emergency Management Agency products. the Federal Emergency Management Agency has licensed this data from CoreLogic to supplement its National Flood Hazard Layer data while the Federal Emergency Management Agency engages with communities where digital data-coverage gaps exist in the Federal Emergency Management Agency's National Flood Hazard Layer.

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<sup>52</sup> National Flood Insurance Program, FEMA. (2025). NFHL [online dataset]. Retrieved from <https://www.fema.gov/national-flood-hazard-layer-nfhl>.

**Susceptible Area Source:** [National Oceanic and Atmospheric Administration Office for Coastal Management, High Tide Flooding \(HTF\) Probability](#)<sup>53</sup>

The High Tide Flooding Probability data set provides estimates of the probability of Coastal Flooding events due to high tides for United States coastal regions. The dataset categorizes flooding events across three severity levels: Minor flooding events (1–2-year floods), Moderate flooding events (10–50 floods), and Major flooding events (100-year floods). For each severity level, there is a GeoTIFF raster layer where each raster cell contains information about the probability of High Tide Flooding at a given location, represented by a number between 0 and 100.

**Susceptible Area Source:** The Federal Emergency Management Agency, Geospatial Flood Risk Assessment

The Geospatial Flood Risk Assessment contains Hazus dasymetric developed area polygon features. Each feature has an attribute value that describes an average annual loss rate due to flood hazard associated with a one-story single-family home with a slab-on-grade foundation. The model and framework are built on years of investment in flood hazard information and analysis through the National Flood Insurance Program and Risk Mapping, Assessment, and Planning program. The Geospatial Flood Risk Assessment depicts inland flood losses derived from geographic factors such as elevation, distance to rivers, and relative elevation from a river, combining catastrophe modeling, National Flood Hazard Insurance Program historical data, and actuarial pricing approach. This information is used to inform potential annualized loss rates for different building types at the dasymetric Census block level. This dataset is only incorporated for the Great Lakes region.

**Hazard Possible Area Source:** National Oceanic and Atmospheric Administration [National Hurricane Center \(NHC\), Sea, Lake, and Overland Surges from Hurricane \(SLOSH\)](#)<sup>54, 55</sup>

Sea, Lake, and Overland Surges from Hurricane Maximum of the Maximum raster files are modeled based on hurricane categories 1–5 along the Gulf and Atlantic coastline from Texas to Maine. These areas represent near-worst case scenarios and were derived from the storm surge inundation maps created by the National Hurricane Center. Cell values represent the storm surge level above ground in one-foot increments. The hurricane category 4 and 5 raster files were downloaded for use in the determination of hazard possibility.

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<sup>53</sup> Sweet, W.V., B.D. Hamlington, R.E. Kopp, C.P. Weaver, P.L. Barnard, D. Bekaert, W. Brooks, M. Craghan, G. Dusek, T. Frederikse, G. Garner, A.S. Genz, J.P. Krasting, E. Larour, D. Marcy, J.J. Marra, J. Obeysekera, M. Osler, M. Pendleton, D. Roman, L. Schmied, W. Veatch, K.D. White, and C. Zuzak, 2022: Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines. NOAA Technical Report NOS 01. National Oceanic and Atmospheric Administration, National Ocean Service, Silver Spring, MD, 111 pp. <https://oceanservice.noaa.gov/hazards/sealevelrise/noaa-nos-techrpt01-global-regional-SLR-scenarios-US.pdf>.

<sup>54</sup> NHC, NOAA. (2018). National Storm Surge Hazard Maps, Version 2 [online dataset]. Retrieved from <https://www.nhc.noaa.gov/nationalsurge/#data>.

<sup>55</sup> NHC, NOAA. (2022). National Storm Surge Hazard Maps, Version 3 [online dataset]. Retrieved from <https://www.nhc.noaa.gov/nationalsurge/#data>.



## 7.2. Coastal Flooding Spatial Data Processing

Coastal spatial processing included numerous complex steps in order to complete Expected Annual Loss calculations. The process uniquely modeled Coastal Flooding exposure by the sub-type of flooding and calculated corresponding annualized frequencies for each flooding sub-type. The sub-types of flooding included in the Coastal Flooding hazard type are:

- National Oceanic and Atmospheric Administration Minor High Tide Flooding Event Area
- National Oceanic and Atmospheric Administration Moderate High Tide Flooding Event Area
- National Oceanic and Atmospheric Administration Major High Tide Flooding Event Area
- Special Flood Hazard Area 100-year flood area
- Special Flood Hazard Area 500-year flood area
- The Federal Emergency Management Agency Geospatial Flood Risk Assessment

All National Oceanic and Atmospheric Administration High Tide Flooding Coastal Flood Frequency rasters were converted to polygons and cell values were assigned into 8 event probability/frequency categories.

All spatial datasets were first extracted and, if necessary, converted to polygon vector format. National Flood Hazard Layer data were extracted for 1% annual chance coastal floodplains (100-year) and 0.2% annual chance floodplains (500-year) according to their flood-zone sub-type. “V” Zones and “A” Zones with the word “coastal” in the sub-type signify coastal 1% annual chance floodplains and were extracted. Additionally, where National Flood Hazard Layer data were unavailable, flood hazard areas in which the flood zone category begins with “V” were extracted from the CoreLogic data. The National Flood Hazard Layer and CoreLogic selections were combined into a single layer representing Special Flood Hazard Area 100-year flood area. The 0.2% annual chance extract was combined with an extract of “X” Zone with sub-type “AREA WITH REDUCED FLOOD RISK DUE TO LEVEE” to create a Special Flood Hazard Area 500-year flood area. For the Great Lakes Region, a Great Lakes flood area is determined by the extent of all polygons within the Federal Emergency Management Agency Geospatial Flood Risk Assessment with a “Great Lakes Flooding Premium without Fees” greater than 0.

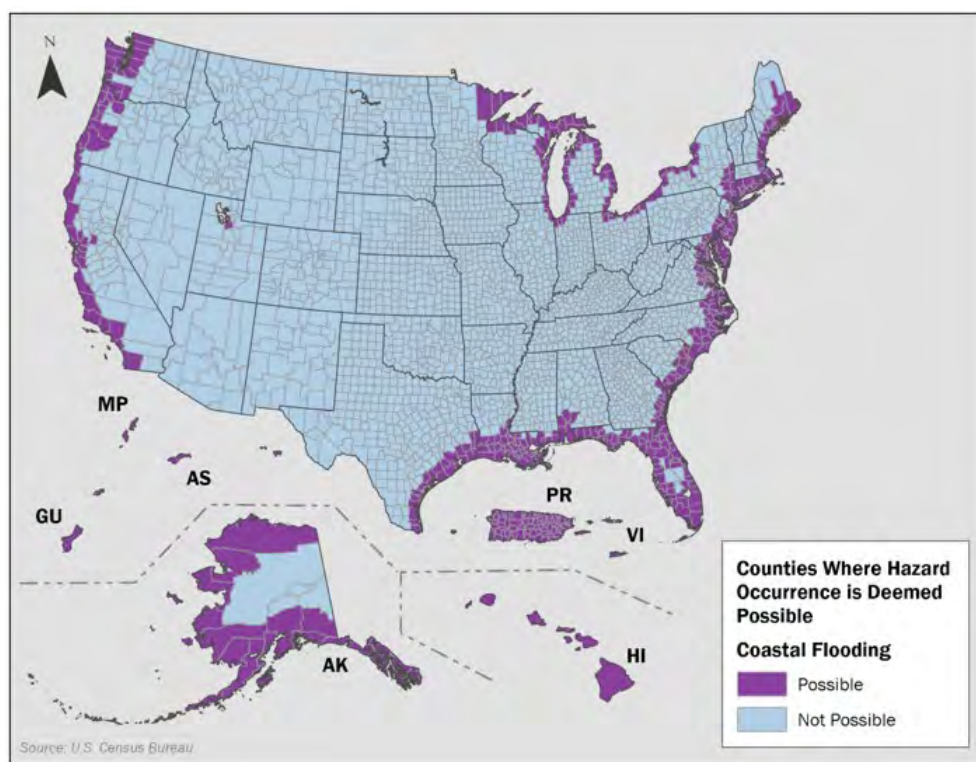
By region, the highest available SLOSH raster file was converted into a polygon vector format and intersected with Census block to support the designation as possible for Coastal Flooding (see [Section 7.2.1 Determination of Possibility of Hazard Occurrence](#)).

### 7.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

To distinguish between areas with no Coastal Flooding occurrences and those where such occurrences are not deemed possible, a control table was generated to designate which Census blocks have some probability of being impacted by a Coastal Flooding occurrence. Coastal Flooding



was deemed possible in different regions based on different criteria. For the Gulf Coast and southeast Atlantic coast from Texas to North Carolina and Puerto Rico and the U.S. Virgin Islands, Census blocks that intersected the SLOSH Maximum of the Maximum raster layer for Category 5 storms were deemed possible for Coastal Flooding. For the northeast Atlantic coast from Virginia to Maine, Census blocks that intersected the SLOSH Maximum of the Maximum layer for Category 4 storms were deemed possible for Coastal Flooding. The SLOSH Maximum of the Maximum layer for Category 2 storms was used to determine Coastal Flooding possibility in Southern California, and SLOSH Maximum of the Maximum layer for Category 4 storms was used for Hawai'i. For American Samoa and Guam, all Census blocks along the coastline were deemed possible. All of Northern Mariana Islands were included as possible. For the Great Lakes, any Census block in a Census tract along the coastline was deemed possible for Coastal Flooding, as well as any Census blocks which intersected a coastal 1% or 0.2% annual chance floodplain in the National Flood Hazard Layer as defined in 7.2. The National Flood Hazard Layer with these criteria were also used to designate a portion of the shoreline of the Great Salt Lake in Utah possible for Coastal Flooding. On the coasts of Alaska, Washington, Oregon, and most of California, Hazus coastal area designations were used to determine Coastal Flooding possibility. (see [Figure 19](#)).



**Figure 19: Map of Counties Deemed Possible for Coastal Flooding Occurrence**

### 7.3. Coastal Flooding Exposure

To identify areas of exposure, each event sub-type layer (National Oceanic and Atmospheric Administration Minor, Moderate, and Major High Tide Flooding, Special Flood Hazard Area 100-year flood area, Special Flood Hazard Area 500-year flood area, and the Federal Emergency Management

Agency Geospatial Flood Risk Assessment Great Lakes flood area) is independently intersected with the Census block developed area polygons within the processing database. The Special Flood Hazard Areas 100-year flood area and Special Flood Hazard Area 500-year flood area intersection results are further refined by erasing those developed area intersects that are within the National Oceanic and Atmospheric Administration Minor, Moderate, or Major High Tide Flooding area. The Federal Emergency Management Agency Geospatial Flood Risk Assessment Great Lakes flood area is overlaid with Census block developed area polygons, and any developed area polygon that intersects the Great Lakes susceptible zone is considered exposed to the Great Lakes flooding hazard-subtype. The resulting tables contain the layer polygon's unique identifier, Census block number, and the developed area of intersection (see [Table 22](#)). All area values are square kilometers.

**Table 22: Sample Data from the Census Block Intersection Table**

| <i>Coastal Flood Zone<br/>100yr ID</i> | <i>Census Block</i> | <i>Area Developed km2</i> |
|----------------------------------------|---------------------|---------------------------|
| 391                                    | 150030099021008     | 0.001979101               |
| 445                                    | 150030098012011     | 0                         |
| 2112                                   | 480079501002007     | 1.59E-06                  |

To determine exposure value for buildings and population, the intersected areas from each Coastal Flooding sub-type are multiplied by the developed area building value density and the developed area population density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 29](#)). These Census block developed area building and population value densities have been calculated by dividing the total exposure values (as recorded in Hazus 6.0) by the developed land area (in square kilometers; see [Section 5.3.3 Exposure Methodology](#)). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

**Equation 29: Census Block Coastal Flooding Sub-Type Building and Population Exposure**

$$Exposure_{CFLDSubCB_{Bldg}} = IntsctDevArea_{CFLDSubCB} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{CFLDSubCB_{Pop}} = IntsctDevArea_{CFLDSubCB} \times DevAreaDen_{CB_{Pop}} \times VSL$$

where:

$Exposure_{CFLDSubCB_{Bldg}}$  is the estimated building value exposed to the Coastal Flooding sub-type in a specific Census block (in dollars).

|                                 |                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| $IntsctDevArea_{CFLDSub_{CB}}$  | is the intersected developed area of the Coastal Flooding sub-type with the Census block (in square kilometers). |
| $DevAreaDen_{CB_{Bldg}}$        | is the developed area building value density of the Census block (in dollars per square kilometer).              |
| $Exposure_{CFLDSub_{CB_{Pop}}}$ | is the population exposed to the Coastal Flooding sub-type in a specific Census block (in people).               |
| $DevAreaDen_{CB_{Pop}}$         | is the developed area population density of the Census block (in people per square kilometer).                   |
| $VSL$                           | is the Value of Statistical Life (\$13.7 per person).                                                            |

These calculations are performed for each of the eight layers so that exposure values for each sub-type of flood zone and consequence type are calculated.

## 7.4. Coastal Flooding Annualized Frequency

The annualized frequency value represents the modeled frequency of a Coastal Flooding occurrence, in events, per year. Coastal Flooding annualized frequency is calculated at the Census block level by Coastal Flooding sub-type. The separate intersection of the Census block with the 1% annual chance floodplain delta layer, 0.2% annual chance floodplain delta layer, and each High Tide Flooding layer subtype layer governs which sub-type frequencies are applicable to each Census block.

Each sub-type of Coastal Flooding has a different annualized frequency, as listed below:

- Special Flood Hazard Area 1% annual chance: 0.01
- Special Flood Hazard Area 0.2% annual chance: 0.002
- National Oceanic and Atmospheric Administration High Tide Flooding Minor Subtype
- National Oceanic and Atmospheric Administration High Tide Flooding Moderate Subtype
- National Oceanic and Atmospheric Administration High Tide Flooding Major Subtype

For the Special Flood Hazard Area floodplain sub-types, a constant value of is applied to the floodplain. Floodplains for 100-year event receive a frequency value of 0.01, and floodplains for 500-year events receive a frequency value of 0.002.<sup>56</sup> Then area-weighted frequencies are

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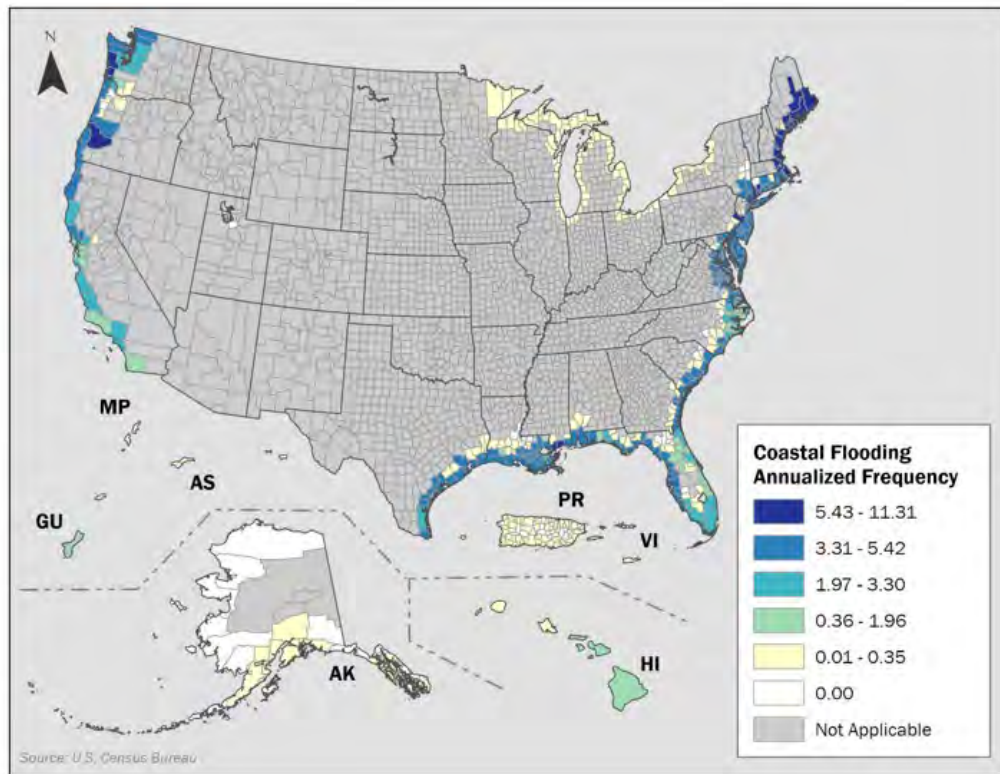
<sup>56</sup> For the Great Lakes Region, areas outside the Special Flood Hazard Area 100 and 500-year flood that intersect the Federal Emergency Management Agency Geospatial Flood Risk Assessment Great lakes layer receive a minimum annualized frequency of 0.0001.

calculated for each subtype by intersecting their respective floodplain layer with Census block developed area shapes.

Frequencies for each National Oceanic and Atmospheric Administration High Tide Flooding subtype are calculated by intersecting their respective high tide flood layer with the Census block developed area polygons. Then area-weighted frequencies for each subtype are calculated using the midpoint of the probability classification categories specified by National Oceanic and Atmospheric Administration. To adjust frequencies for areas protected by levees, these areas are clipped out before sub-type area weighted frequencies are computed (for a detailed list, see [Table 23](#)).

**Table 23: High Tide Flooding Probability Classification**

| <i>Category</i> | <i>High Tide Flooding<br/>Probability Classification</i> | <i>Midpoint Probability</i> |
|-----------------|----------------------------------------------------------|-----------------------------|
| <b>1</b>        | >10 events per year                                      | 15                          |
| <b>2</b>        | 5-10 events per year                                     | 7.5                         |
| <b>3</b>        | 1-5 events per year                                      | 3                           |
| <b>4</b>        | 99%-20% annual chance                                    | 0.6                         |
| <b>5</b>        | 20%-10% annual chance                                    | 0.15                        |
| <b>6</b>        | 10%-5% annual chance                                     | 0.075                       |
| <b>7</b>        | 5%-2% annual chance                                      | 0.035                       |
| <b>8</b>        | <2% annual chance                                        | 0.015                       |



**Figure 20: Coastal Flooding Annualized Frequency by County**

## 7.5. Coastal Flooding Historic Loss Ratio

The Coastal Flooding Historic Loss Ratio is the representative percentage of a location's hazard exposure area that experiences loss due to a Coastal Flooding occurrence, or the average rate of loss associated with a Coastal Flooding occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Coastal Flooding hazard type.

Loss data are provided by SHELUDS<sup>57</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDS events from 1996 to 2023 are included in the Historic Loss Ratio calculation. Four peril types are mapped to the hazard Coastal Flooding (see [Table 24](#)). These native records are aggregated on a consecutive day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 24: Coastal Flooding Peril Types and Recorded Events from 1996-2023**

| Peril Type in SHELUDS | Total SHELUDS Loss Records | Total Records per Event Basis |
|-----------------------|----------------------------|-------------------------------|
| Coastal               | 1                          | 1                             |

<sup>57</sup> For Coastal Flooding loss information, SHELUDS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

| <i>Peril Type in SHELUS</i> | <i>Total SHELUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-----------------------------|----------------------------------|--------------------------------------|
| Coastal Storm               | 22                               | 22                                   |
| Flood-Coastal               | 921                              | 783                                  |
| Flood-Tidal                 | 1                                | 1                                    |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is determined by summing the developed area exposure values of the Census tracts that intersect the National Oceanic and Atmospheric Administration High Tide Flooding Moderate High Tide Flooding sub-type event layer. The Loss Ratio per Basis for each SHELUS-documented event and each consequence type (building and population) is calculated using [Equation 30](#).

**Equation 30: Loss Ratio per Basis Calculation for a Single Coastal Flooding Event**

$$LRB_{CFLD\ Co\ CnsqType} = \frac{LOSS_{CFLD\ Co\ CnsqType}}{HLRExposure_{CFLD\ Co\ CnsqType}}$$

where:

$LRB_{CFLD\ Co\ CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Coastal Flooding event. Calculation is performed for each consequence type (building and population).

$LOSS_{CFLD\ Co\ CnsqType}$  is the loss (by consequence type) experienced from the specific Coastal Flooding event documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{CFLD\ Co\ CnsqType}$  is the value (by consequence type) of the susceptible area estimated to have been exposed to the Coastal Flooding event (in dollars or people).

Since Coastal Flooding frequency is based on flooding probabilities, no zero-loss occurrences are inserted into the Loss Ratio table. Additionally, both the Loss and Historic Loss Ratio exposure values of each Coastal Flooding Loss Data Loss Ratio per Basis are measured in 2022 dollars and Historic Loss Ratio exposure values are derived using population and building values from the 2020 Census.

After the Loss Ratio per Basis data are calculated for all Coastal Flooding events, various sets of Loss Ratio per Basis data are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific



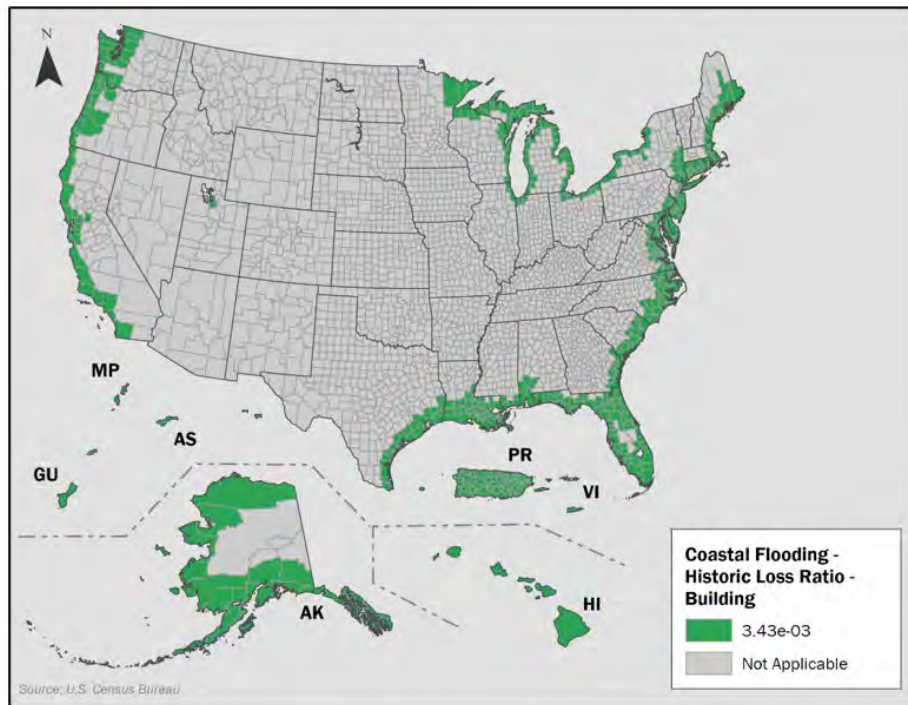
county. The following table displays the Loss Ratio per Basis sets used in creating the Historic Loss Ratios for Coastal Flooding.

**Table 25: Coastal Flooding Historic Loss Ratio Bayesian Weighting Level Sets**

| Consequence Type | Set ID | Bayesian Level                         | Set Refinement Filter | Set Utilization Threshold (Set Must Have) |
|------------------|--------|----------------------------------------|-----------------------|-------------------------------------------|
| Population       | P1     | County                                 | N/A                   | N/A                                       |
|                  | P2     | Coastal Flooding Physiographic Regions | N/A                   | N/A                                       |
|                  | P3     | National                               | N/A                   | N/A                                       |
| Building         | B1     | County                                 | N/A                   | N/A                                       |
|                  | B2     | Coastal Flooding Physiographic Regions | N/A                   | N/A                                       |
|                  | B3     | National                               | N/A                   | N/A                                       |

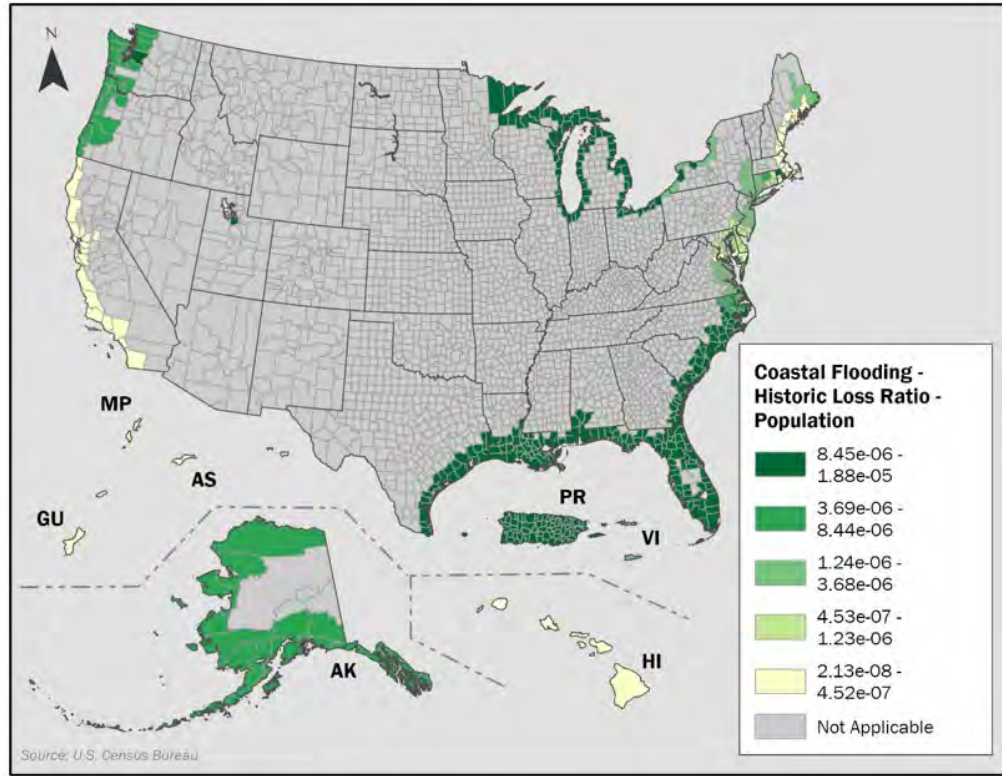
The regional definition for Coastal Flooding is derived from the default Historic Loss Ratio Regions (see [Section 5.4 Historic Loss Ratio](#)). For the building consequence type, all counties are assigned the national average loss ratio value.

[Figure 21](#) and [Figure 22](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Coastal Flooding. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 21: Coastal Flooding Bayesian-Adjusted Historic Loss Ratio – Building Value**





**Figure 22: Coastal Flooding Bayesian-Adjusted Historic Loss Ratio – Population**

## 7.6. Coastal Flooding Expected Annual Loss

Expected Annual Loss is calculated for each Coastal Flooding sub-type exposure at a Census block level and summed to get an Expected Annual Loss value for the Census block as in [Equation 31](#).

### Equation 31: Census Block Coastal Flooding Expected Annual Loss Calculation

$$EAL_{CFLD_{Subtype} \text{ } CB_{CnsqType}} = Freq_{CFLD_{Subtype} \text{ } CB} \times Exp_{CFLD_{Subtype} \text{ } CB_{CnsqType}} \times HLR_{CFLD_{CB} \text{ } CnsqType}$$

where:

$EAL_{CFLD_{Subtype} \text{ } CB_{CnsqType}}$

is the Coastal Flooding Expected Annual Loss by sub-type for the consequence type for a specific Census block.

$Freq_{CFLD_{Subtype} \text{ } CB}$

is the area weighted average annualized frequency of the Coastal Flooding subtype.

$C$

is the consequence type exposure value associated with the Coastal Flooding subtype in the Census block (in dollars).

$HLR_{CFLD_{CB} CnsqType}$  is the Coastal Flooding historic loss rate by subtype for the consequence type from the Census Block's parent county.

Using these values, the Coastal Flooding Expected Annual Loss is computed at the Census block level using [Equation 32](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 32: Census Block Expected Annual Loss to Coastal Flooding

$$EAL_{CFLD_{CB} CnsqType} = EAL_{CFLD_{NHTMinor} CB CnsqType} + EAL_{CFLD_{NHTMo derate} CB CnsqType} + EAL_{CFLD_{NHTMajor} CB CnsqType} + EAL_{CFLD_{100Yr} CB CnsqType} + EAL_{CFLD_{500Yr} CB CnsqType}$$

where:

$EAL_{CFLD_{CB} CnsqType}$  is the Coastal Flooding Expected Annual Loss for the consequence type for a specific Census block.

$EAL_{CFLD_{NHTMinor} CB CnsqType}$  is the Expected Annual Loss derived from the Minor National Oceanic and Atmospheric Administration High Tide Flooding events layer for the consequence type for a specific Census block.

$EAL_{CFLD_{NHTMo derate} CB CnsqType}$  is the Expected Annual Loss derived from the Moderate National Oceanic and Atmospheric Administration High Tide Flooding events layer for the consequence type for a specific Census block.

$EAL_{CFLD_{NHTMajor} CB CnsqType}$  is the Expected Annual Loss derived from the Major National Oceanic and Atmospheric Administration High Tide Flooding events layer for the consequence type for a specific Census block.

$EAL_{CFLD_{100Yr} CB CnsqType}$  is the Expected Annual Loss derived from the 100-year Special Flood Hazard Area Federal Emergency Management Agency floodplain layer for the consequence type for a specific Census block.

$EAL_{CFLD_{500Yr} CB CnsqType}$  is the Expected Annual Loss derived from the 500-year Special Flood Hazard Area Federal Emergency Management Agency floodplain layer for the consequence type for a specific Census block.

The total Expected Annual Loss values at the Census tract and county level are sums of the aggregated building and population equivalence Expected Annual Loss values at the Census block level as in [Equation 33](#).

**Equation 33: Census Tract and County Expected Annual Loss to Coastal Flooding**

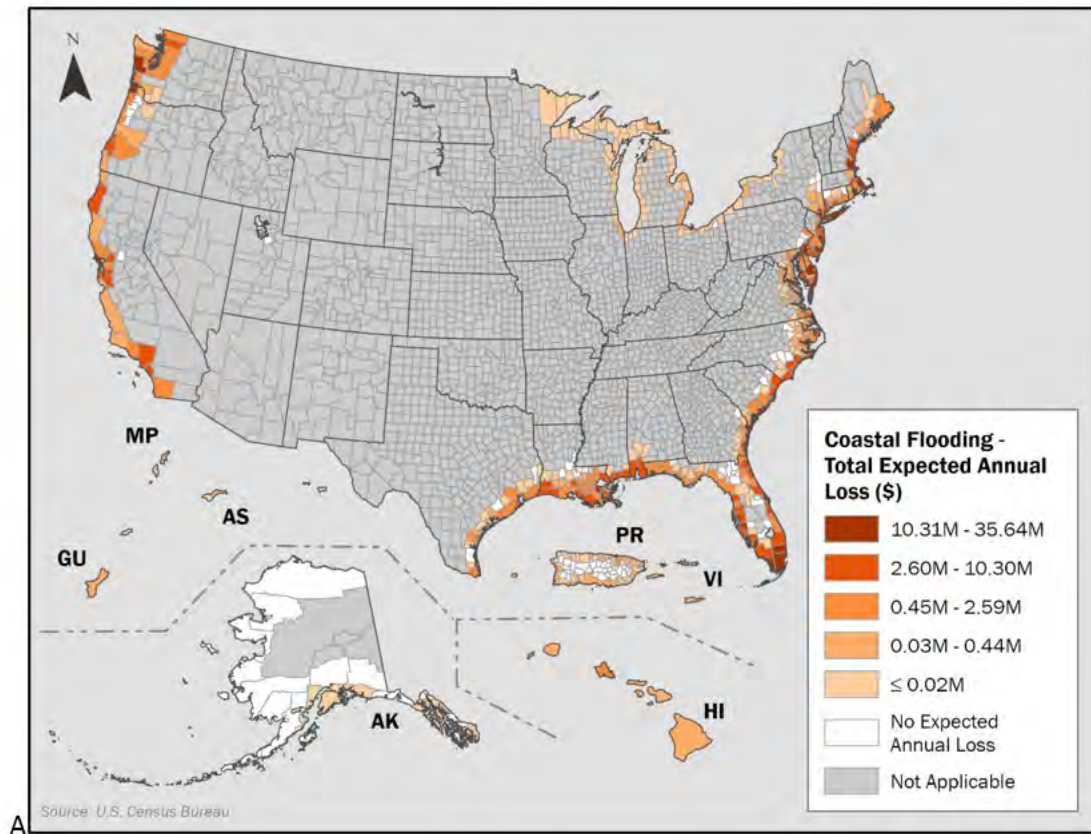
$$EAL_{CFLD_{CT}} = \sum_{CB}^{CT} EAL_{CFLD_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{CFLD_{CB_{Pop}}}$$

$$EAL_{CFLD_{Co}} = \sum_{CB}^{Co} EAL_{CFLD_{CB_{Bldg}}} + \sum_{CB}^{Co} EAL_{CFLD_{CB_{Pop}}}$$

where:

|                                         |                                                                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{CFLD_{CT}}$                       | is the total Expected Annual Loss due to Coastal Flooding occurrences for a specific Census tract (in dollars).                                       |
| $\sum_{CB}^{CT} EAL_{CFLD_{CB_{Bldg}}}$ | is the summed building Expected Annual Loss value due to Coastal Flooding occurrences for all Census blocks in the Census tract (in dollars).         |
| $\sum_{CB}^{CT} EAL_{CFLD_{CB_{Pop}}}$  | is the summed population equivalence Expected Annual Loss due to Coastal Flooding occurrences for all Census blocks in the Census tract (in dollars). |
| $EAL_{CFLD_{Co}}$                       | is the total Expected Annual Loss due to Coastal Flooding occurrences for a specific county (in dollars).                                             |
| $\sum_{CB}^{Co} EAL_{CFLD_{CB_{Bldg}}}$ | is the summed building Expected Annual Loss value due to Coastal Flooding occurrences for all Census blocks in the county (in dollars).               |
| $\sum_{CB}^{Co} EAL_{CFLD_{CB_{Pop}}}$  | is the summed population equivalence Expected Annual Loss due to Coastal Flooding occurrences for all Census blocks in the county (in dollars).       |

[Figure 23](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Coastal Flooding occurrences.



**Figure 23: Total Expected Annual Loss by County to Coastal Flooding**

With the Coastal Flooding total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Coastal Flooding Expected Annual Loss for buildings by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Coastal Flooding Expected Annual Loss for population by the total population of the community.

## 7.7. Coastal Flooding Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- There are several assumptions in the Geospatial Flood Risk Assessment loss data that result in limitations to these modeled impacts. The first is that Geospatial Flood Risk Assessment assumes all buildings are single-story, single-family homes with a slab-on-grade foundation

that has a replacement cost value of \$250,000. Any adjustments made for different occupancy types are estimations and might not fully capture specific vulnerabilities, and foundation types other than slab-on-grade are estimated using generalized damage functions, which may not reflect accurate damage levels. As such, there is potential for underestimation or overestimation of losses due to reliance on generalized damage functions rather than specific empirical data.

## 8. Cold Wave

A Cold Wave is a rapid fall in temperature within 24 hours and extreme low temperatures for an extended period. The temperatures classified as a Cold Wave are dependent on the location and defined by the local National Weather Service weather forecast office.

The Cold Wave hazard modeling uses Mesonet data to provide impact areas for this hazard that are derived from the National Weather Service's warnings and advisories. SHELDUS data were used for historical losses for Cold Wave events and generate the Historic Loss Ratios, with data from the Centers for Disease Control and Prevention Compressed Mortality Database used to supplement the SHELDUS mortality data. These are overlaid with the exposure values provided in Hazus 6.0 for Building, Population, and Agriculture for the final Expected Annual Loss. The Cold Wave hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 8.1. Cold Wave Spatial Source Data

**Historical Occurrence Compiling Source:** [Iowa State University, Iowa Environmental Mesonet](#)<sup>58</sup>

The National Weather Service is continuously issuing weather alerts based on current weather conditions.<sup>59</sup> Each alert is coded by type of phenomenon and level of significance, (e.g., Watch, Warning, or Advisory) and conceptually can indicate the potential for weather event activities in a specific area. Archived National Weather Service alerts are aggregated, continuously updated, and made available for download in shapefile format by Iowa State University's Iowa Environmental Mesonet. Data include geometry for each alert's issued area and attributes related to each alert's severity and phenomena type. Weather alerts are also timestamped with the time of issuance and the time of expiration. A table describing this dataset's attributes can be found in [Appendix C – Mesonet-National Weather Service Weather Event Attribute Description](#).

The geometry shape for each alert record represents the geographic area for which the National Weather Service alert applied. However, the Mesonet shapes are simplified versions of the more detailed National Weather Service Public Forecast Zone shape originally associated with the alert record. Because the Mesonet tabular data still retain the reference IDs for the National Weather Service Public Forecast Zones, these reference IDs can be used to relate to the more detailed zone associated with each alert record.

The National Weather Service Public Forecast Zones can be downloaded in shapefile format<sup>60</sup> and represent the codified areas for which weather alerts are issued by National Weather Service. The

<sup>58</sup> Department of Agronomy, Iowa State University. (2025). *Iowa Environmental Mesonet* [online database]. Retrieved from <https://mesonet.agron.iastate.edu/request/gis/watchwarn.phtml>.

<sup>59</sup> NWS, NOAA. (2018). *Active Alerts* [online dataset]. Retrieved from <https://www.weather.gov/alerts>.

<sup>60</sup> NWS, NOAA. (2022). *NWS Public Forecast Zones* [online dataset]. Retrieved from <https://www.weather.gov/gis/PublicZones>.



Public Forecast Zones shape definitions are predominantly derived from county boundaries. While the Public Forecast Zone boundaries are more refined than those substituted into the Mesonet data, they are not at the same resolution as the current county boundaries derived from Census blocks.

Utilizing the Public Forecast Zone shapefile in conjunction with the Public Forecast Zone – County Correlation file,<sup>61</sup> a determination was made as to which Public Forecast Zones have single-county coverage and which are either sub-county zones or made of portions of multiple counties. For perspective, the following approximate distributions of forecast zone composition were found:

- 70% of the zones are single-county coverage.
- 20% are cases where a single county is subdivided into multiple zones.
- 10% are zones that breach parts of multiple contiguous counties.

For those Forecast Zones covering a single county, the United States Census Topologically Integrated Geographic Encoding and Referencing 2021 county boundaries (Topologically Integrated Geographic Encoding and Referencing 2024 boundaries for Connecticut) are substituted.

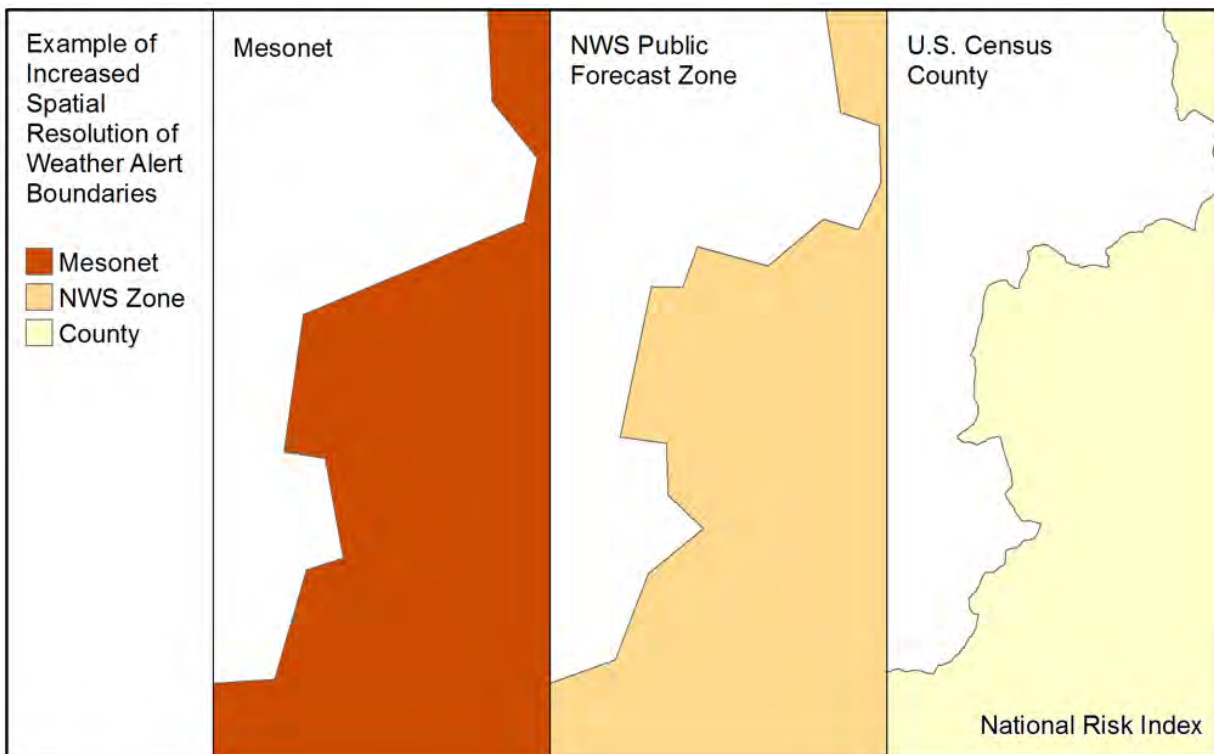
Another aspect of the National Weather Service Public Forecast Zones is that they can and have changed over time. In the Mesonet data (2005 through 2024), there are many Forecast Zones referenced that do not exist in the current National Weather Service Public Forecast Zone shapefile. This occurs when a National Weather Service Public Forecast Zone has been modified in shape, renamed, and/or “retired” from use.

Further research found that the National Weather Service maintains a downloadable Change History log of the various changes in Forecast Zone areas since 1997. This text file does not contain the pre- or post-shape of the altered forecast zone. Archived versions of these changes are likely available via contact with National Weather Service, but the effort to match the National Weather Service issued alert record to the version-controlled shape representation of the forecast zone at the time of alert issue is beyond the scope of the processing effort, though a Mesonet representative was contacted to see if Forecast Zone shapes associated with each year of alert data had been archived. Unfortunately, no such archival information is currently available. For cases where the more refined National Weather Service Forecast Zone shape is unavailable, the simplified Mesonet boundary version shape is used. See [Figure 24](#) for an example of the differences in the spatial resolution of weather alert boundaries.

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<sup>61</sup> NWS, NOAA. (2022). *Zone-County Correlation File* [online dataset]. Retrieved from <https://www.weather.gov/gis/ZoneCounty>.





**Figure 24: Three Boundary Definitions: Mesonet, Forecast Zone, and United States Census County**

## 8.2. Cold Wave Spatial Data Processing

Cold Wave weather event alerts were extracted from the dataset based on the Valid Time Extent Code significance code (SIG field) and the phenomena code (PHENOM or TYPE field) values. Only Warning alerts (SIG = 'W') and Advisory alerts (SIG = 'Y') of the Phenomena type "Extreme Cold" (EC) or "Wind Chill" (WC) were considered Cold Wave occurrences (see [Table 26](#)).

To remove unintended errors in spatial results due to the use of the simplified event area shapes contained in the Mesonet data, event areas with a higher resolution version were substituted. This substitution uses the National Weather Service Public Forecast Zone shape associated with the alert record or, in cases where the forecast zone is for a single county, a better resolution version of the county boundary area.

**Table 26: Original Mesonet Cold Wave Records**

| WFO | ISSUED                | EXPIRED               | PHENOM | SIG | NWS_UGC | AREA_KM2   |
|-----|-----------------------|-----------------------|--------|-----|---------|------------|
| DLH | Dec. 25, 2017 6:00 AM | Dec. 26, 2017 6:00 PM | WC     | W   | MNZ018  | 4648.70996 |
| BIS | Jan. 03, 2017 9:06 PM | Jan. 04, 2017 6:07 PM | WC     | A   | NDZ020  | 1888.72131 |
| MSO | Feb. 06, 2014 2:33 PM | Feb. 06, 2014 5:25 PM | EC     | W   | MTZ043  | 5891.24316 |

Cold Wave occurrences are measured in event-days as this more accurately represents the variability of Cold Wave event duration.<sup>62</sup> To capture this, each native alert record with a duration greater than a single day was replaced with multiple records, one for each day of the original record's duration.

If a Cold Wave event's duration on any given day is less than 6 hours, then the event is assigned to the day having the greatest duration of the event. This accounts for cases where the event occurs in the late evening and endures for a greater length of time on the next calendar day than on the day the alert was issued.

For cases where the event duration is longer, the following logic is used: If a weather event's duration is greater than 6 hours, assign the event to all days on which 6 or more hours occur. For example, if a 14-hour weather event was issued for 2 AM until 6 PM on January 1, then the event would be assigned to January 1. If the alert was issued from 11 PM on January 1 to 1 PM on January 2, then the event would be assigned to only January 2. If the alert was issued from 6 PM on January 1 to 9 AM on January 2, then the event would be assigned to both January 1 and January 2. To illustrate this concept, the Cold Wave event in [Table 27](#) is expanded to create the Cold Wave event-day records in [Table 28](#).

Additionally, there are some data quality issues with the Mesonet data. For example, some warnings have an expiration date that is prior to the issue date. In these cases, a single record is used and assigned the issue date.

**Table 27: Sample Cold Wave Data after Zone Shape Re-Sourcing**

| <i>ColdwaveID</i> | <i>WFO</i> | <i>Issued</i>             | <i>Expired</i>           | <i>PHENOM</i> | <i>SIG</i> | <i>NWS_UGC</i> | <i>AreaKm2</i> | <i>NewShapeSource</i> |
|-------------------|------------|---------------------------|--------------------------|---------------|------------|----------------|----------------|-----------------------|
| 1189968           | DLH        | Jan. 05, 2014<br>12:00 AM | Jan. 07, 2014<br>5:07 PM | WC            | W          | WIZ002         | 3917.1735      | Census County         |

**Table 28: Sample Data from the Cold Wave Date Expansion Table**

| <i>ExpansionID</i> | <i>ColdwaveID</i> | <i>Issued</i>             | <i>Expired</i>            | <i>DateType</i>               | <i>ColdwaveHours</i> |
|--------------------|-------------------|---------------------------|---------------------------|-------------------------------|----------------------|
| 10771              | 1189968           | Jan. 05, 2014<br>12:00 AM | Jan. 06, 2014<br>12:00 AM | Expanded Dates<br>- Issued    | 24                   |
| 10772              | 1189968           | Jan. 06, 2014<br>12:00 AM | Jan. 07, 2014<br>12:00 AM | Expanded Dates<br>- New Dates | 24                   |
| 10773              | 1189968           | Jan. 07, 2014<br>12:00 AM | Jan. 07, 2014<br>5:07 PM  | Expanded Dates<br>- Expired   | 17.11666             |

To avoid overestimating the area of influence a “single” distinct weather event has due to multiple National Weather Service alerts being issued for that same weather event, a process to combine all

<sup>62</sup> To avoid double counting, Warnings and Advisories issued on the same day are mapped to a single Cold Wave event-day.

Cold Wave event areas occurring on the same day (Year, Month, Day specific) into one representative event shape was performed. This process results in a single event impact area shape for each day on which a Cold Wave event occurred. These Cold Wave event-day polygons can then be intersected with the Census block polygons to determine Cold Wave exposure and annualized frequency.

#### 8.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Cold Waves can occur almost anywhere in the United States as the definition of a Cold Wave is locally defined by the area's Weather Forecast Office. For example, a forecast office in Texas may define a Cold Wave differently than a forecast office in New York. Therefore, all counties are deemed possible for Cold Wave occurrence.

### 8.3. Cold Wave Non-Spatial Data Processing

#### 8.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Cold Wave, in event-days, is computed as the area-weighted sum of distinct Cold Wave event-day polygons that intersect a Census block.

Historical event-day counts are also supplied at the Census tract and county levels as the number of distinct Cold Wave event-day polygons that intersect the Census tract and county, respectively.

#### 8.3.2. PERIOD OF RECORD

The National Weather Service's system of recording Watch, Warnings, and Advisories in the 1990s and early 2000s was too difficult for automated processing. In 2005 the Valid Time Extent Code system was implemented, which allowed for the easy automated parsing of alert data. Therefore, National Weather Service weather events data were downloaded for 2005 through 2024. The date range is Nov. 12, 2005 to Nov. 12, 2024, so the period of record for which Cold Wave data are utilized is 19.0 years.

### 8.4. Cold Wave Exposure

To identify areas of exposure, the Cold Wave event-day polygons (also referred to as Cold Wave Date Expansions to acknowledge the spatiotemporal processing described in [Section 8.2 Cold Wave Spatial Data Processing](#)) are intersected with the Census block polygons within the processing database. The resulting table contains the Cold Wave event-day's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 29](#)).

**Table 29: Sample Data from the Cold Wave Expansion Census Block Intersection Table**

| <i>ColdwaveDateExpansionID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|--------------------------------|--------------------|---------------------------|
| 2025                           | 120830011043089    | 0.0331315054931641        |

| <b>ColdwaveDateExpansionID</b> | <b>CensusBlock</b> | <b>IntersectedAreaKm2</b> |
|--------------------------------|--------------------|---------------------------|
| 2025                           | 120830011043090    | 0.00229587890625          |
| 2025                           | 120830011043091    | 0.00324445764160156       |

To determine exposure value, the average coverage of a Cold Wave event-day is found by summing the intersected areas for all Cold Wave event-day polygons that intersect the Census block and dividing this sum by the number of intersecting event-day polygons. This is multiplied by the developed area building value density, the developed area population density, and the agriculture area value density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 34](#)). These Census block densities have been calculated by dividing the Census block total exposure values (as recorded in Hazus 6.0) by the developed or agriculture land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.<sup>63</sup>

#### Equation 34: Census Block Cold Wave Exposure

$$Exposure_{CWAV_{CB}Bldg} = \frac{\sum IntsctArea_{CWAV_{CB}}}{EventDayCount_{CWAV_{CB}}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{CWAV_{CB}Pop} = \left( \frac{\sum IntsctArea_{CWAV_{CB}}}{EventDayCount_{CWAV_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

$$Exposure_{CWAV_{CB}Ag} = \frac{\sum IntsctArea_{CWAV_{CB}}}{EventDayCount_{CWAV_{CB}}} \times AgValueDen_{CB}$$

where:

$Exposure_{CWAV_{CB}Bldg}$  is the building value exposed to Cold Wave event-days in a specific Census block (in dollars).

$\sum IntsctArea_{CWAV_{CB}}$  is the sum of the intersected areas of past Cold Wave event-days with the Census block (in square kilometers).

$EventDayCount_{CWAV_{CB}}$  is the total number of Cold Wave event-day polygons that intersect the Census block.

$DevAreaDen_{CB_{Bldg}}$  is the developed area building value density of the Census block (in dollars per square kilometer).

<sup>63</sup> The exposure values associated with population will apply to both Injury and Fatality consequence subtypes. More details are provided in [Section 5.4.4](#).

$Exposure_{CWAV_{CBPop}}$  is the population equivalence value exposed to Cold Wave event-days in a specific Census block (in dollars).

$DevAreaDen_{CBPop}$  is the developed area population density of the Census block (in people per square kilometer).

$VSL$  is the Value of Statistical Life (\$13.7 million per person).

$Exposure_{CWAV_{CBAg}}$  is the agriculture value exposed to Cold Wave event-days in a specific Census block (in dollars).

$AgValueDen_{CB}$  is the agriculture value density of the Census block (in dollars per square kilometer).

It should be noted that, for a Cold Wave event-day polygon's intersection with a Census block to be included, the area of the intersection must cover at least 5% of the Census block. This is a spatial modeling technique to correct for the small intersect "slivers" generated by differing versions of county boundary geometry being used.

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value and the Census of Agriculture-reported crop and livestock value for the Census block are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

#### 8.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 35](#)).

**Equation 35: Census Tract and County Cold Wave Exposure Aggregation**

$$Exposure_{CWAV CT Bldg} = \sum_{CB}^{CT} Exposure_{CWAV CB Bldg}$$

$$Exposure_{CWAV Co Bldg} = \sum_{CB}^{Co} Exposure_{CWAV CB Bldg}$$

$$Exposure_{CWAV CT Pop} = \sum_{CB}^{CT} Exposure_{CWAV CB Pop}$$

$$Exposure_{CWAV Co Pop} = \sum_{CB}^{Co} Exposure_{CWAV CB Pop}$$

$$Exposure_{CWAV CT Ag} = \sum_{CB}^{CT} Exposure_{CWAV CB Ag}$$

$$Exposure_{CWAV Co Ag} = \sum_{CB}^{Co} Exposure_{CWAV CB Ag}$$

where:

$Exposure_{CWAV CT Bldg}$  is the building value exposed to Cold Wave event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{CWAV CB Bldg}$  is the summed value of all buildings exposed to Cold Wave for each Census block within the Census tract (in dollars).

$Exposure_{CWAV Co Bldg}$  is the building value exposed to Cold Wave event-days in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{CWAV CB Bldg}$  is the summed value of all buildings exposed to Cold Wave for each Census block within the county (in dollars).

$Exposure_{CWAV CT Pop}$  is the population equivalence value exposed to Cold Wave event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{CWAV CB Pop}$  is the summed value of all population equivalence exposed to Cold Wave for each Census block within the Census tract (in dollars).

$Exposure_{CWAV Co Pop}$  is the population equivalence value exposed to Cold Wave event-days in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{CWAV_{CBPop}}$  is the summed value of all population equivalence exposed to Cold Wave for each Census block within the county (in dollars).

$Exposure_{CWAV_{CTAg}}$  is the agriculture value exposed to Cold Wave event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{CWAV_{CBAg}}$  is the summed value of all agriculture value exposed to Cold Wave for each Census block within the Census tract (in dollars).

$Exposure_{CWAV_{CoAg}}$  is the agriculture value exposed to Cold Wave event-days in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{CWAV_{CBAg}}$  is the summed value of all agriculture value exposed to Cold Wave for each Census block within the county (in dollars).

## 8.5. Cold Wave Annualized Frequency

The number of recorded Cold Wave occurrences, in event-days, each year over the period of record (19.0 years) is used to estimate the annualized frequency of Cold Waves in an area. Because a Cold Wave event can occur over several days or a single day, an event-day basis was used to estimate annualized frequency as this method better captures the variability in duration between occurrences. The annualized frequency is calculated at the Census block level, and this Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the same intersection between Cold Wave event-days (or Cold Wave Date Expansion) polygons and Census block polygons that were used to calculate exposure. The count of distinct Cold Wave event-day polygons intersecting each Census block is recorded and used to calculate the annualized frequency of Cold Wave event-days as in [Equation 36](#).

### Equation 36: Census Block Cold Wave Annualized Frequency

$$Freq_{CWAV_{CB}} = \frac{EventDayCount_{CWAV_{CB}}}{PeriodRecord_{CWAV}}$$

where:

$Freq_{CWAV_{CB}}$  is the annualized frequency of Cold Wave event-days determined for a specific Census block (event-days per year).

$EventDayCount_{CWAV_{CB}}$  is the number of Cold Wave event-days that intersect the Census block.



$PeriodRecord_{CWAV}$  is the period of record for Cold Wave (19.0 years).

### 8.5.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. To achieve this, the annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 37](#). Given this, it is possible that the annualized frequency value reported by the dataset does not exactly match that achieved by dividing the number of Cold Wave events at the Census tract and county level by the period of record.

#### Equation 37: Census Tract and County Area-Weighted Cold Wave Annualized Frequency Aggregation

$$Freq_{CWAV_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{CWAV_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{CWAV_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{CWAV_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{CWAV_{CT}}$  is the area-weighted Cold Wave annualized frequency for a specific Census tract (event-days per year).

$Freq_{CWAV_{CB}}$  is the annualized frequency of Cold Wave event-days determined for a specific Census block (event-days per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

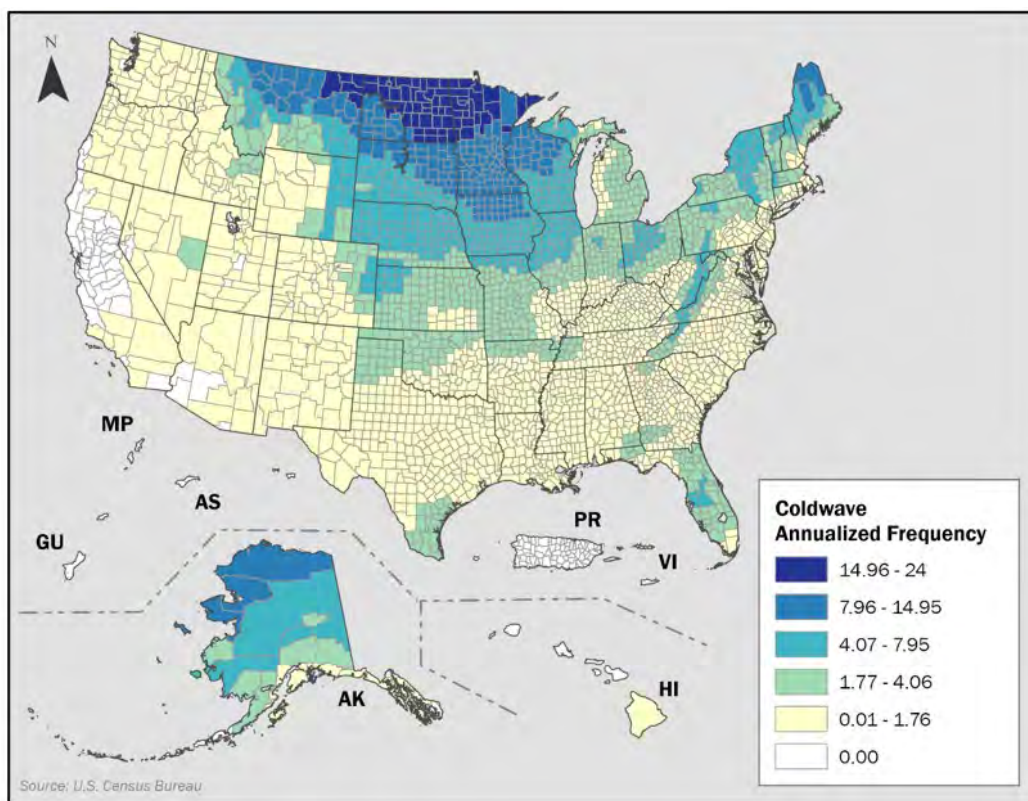
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{CWAV_{Co}}$  is the area-weighted Cold Wave annualized frequency for a specific county (event-days per year).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 25](#) displays Cold Wave annualized frequency at the county level.



**Figure 25: Cold Wave Annualized Frequency by County**

## 8.6. Cold Wave Historic Loss Ratio

The Cold Wave Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Cold Wave event-day, or the average rate of loss associated with the occurrence of a Cold Wave event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Cold Wave hazard type.

Loss data are provided by SHELDUS<sup>64</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. Four peril types are mapped to the hazard Cold Wave (see [Table 30](#)). These native records are expanded on an event-day basis (to a maximum of 31 event-days) and aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

<sup>64</sup> For Heat Wave loss information, SHELDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Association's National Centers for Environmental Information.

**Table 30: SHELDUS Event Types and Recorded Events from 1996-2023**

| Event Type | Total SHELDUS Records with Loss | Total SHELDUS Loss Records per Event Basis |
|------------|---------------------------------|--------------------------------------------|
| Cold Wave  | 0                               | 0                                          |
| Freeze     | 1,336                           | 2,331                                      |
| Frost      | 1,265                           | 2,238                                      |
| Wind Chill | 1,348                           | 3,846                                      |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is a county-level value that represents the dollar value of the total building value, the entire population of a county as recorded in the Hazus 6.0 data,<sup>65</sup> or the total Census of Agriculture-reported crop and livestock value. The Loss Ratio per Basis for each SHELDUS-documented event-day and each consequence type (building, injury, fatality and agriculture) is calculated using [Equation 38](#).

**Equation 38: Loss Ratio per Basis Calculation for a Single Cold Wave Event-Day**

$$LRB_{CWAV Co CnsqType} = \frac{Loss_{CWAV Co CnsqType}}{HLRExposure_{Co CnsqType}}$$

where:

$LRB_{CWAV Co CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Cold Wave event-day. Calculation is performed for each consequence type (building, injury, fatality, and agriculture).

$Loss_{CWAV Co CnsqType}$  is the loss (by consequence type) experienced from the Cold Wave event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{Co CnsqType}$  is the total value (by consequence type) of the county estimated to have been exposed to the Cold Wave event-day (in dollars or people).

Cold Wave event-days can occur with a high frequency in areas but often result in no recorded loss to building value, population, or agriculture value. SHELDUS does not record events in which no loss occurred, so zero-loss event-days are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELDUS period of record (1996 through 2023). For Cold Wave, the historical event-day count is extracted using the intersection between the Cold Wave event-day polygons from the years 2005-2024 and the Census block polygons used to calculate exposure and annualized frequency (see [Table 29](#)). An annual rate

<sup>65</sup> These values define Historic Loss Ratio exposure for both the Injury and Fatality consequence subtypes for population.

is calculated as the event-day count divided by the period of record of 19.0 years, and this rate is multiplied by the SHELATUS period of record of 28 years to estimate a historical event-day count for the appropriate time range.

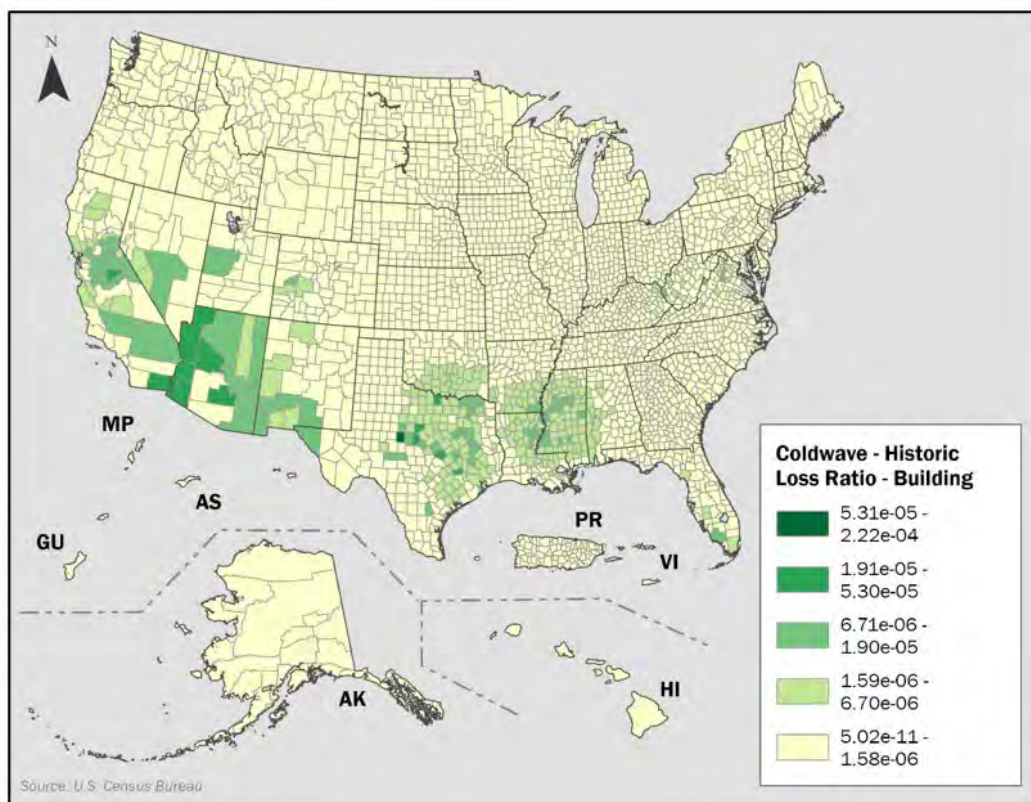
If the number of loss-causing Cold Wave event-day records from the SHELATUS database is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for all Cold Wave events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Record per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Cold Wave.

**Table 31: Cold Wave Historic Loss Ratio Bayesian Weighting Level Sets**

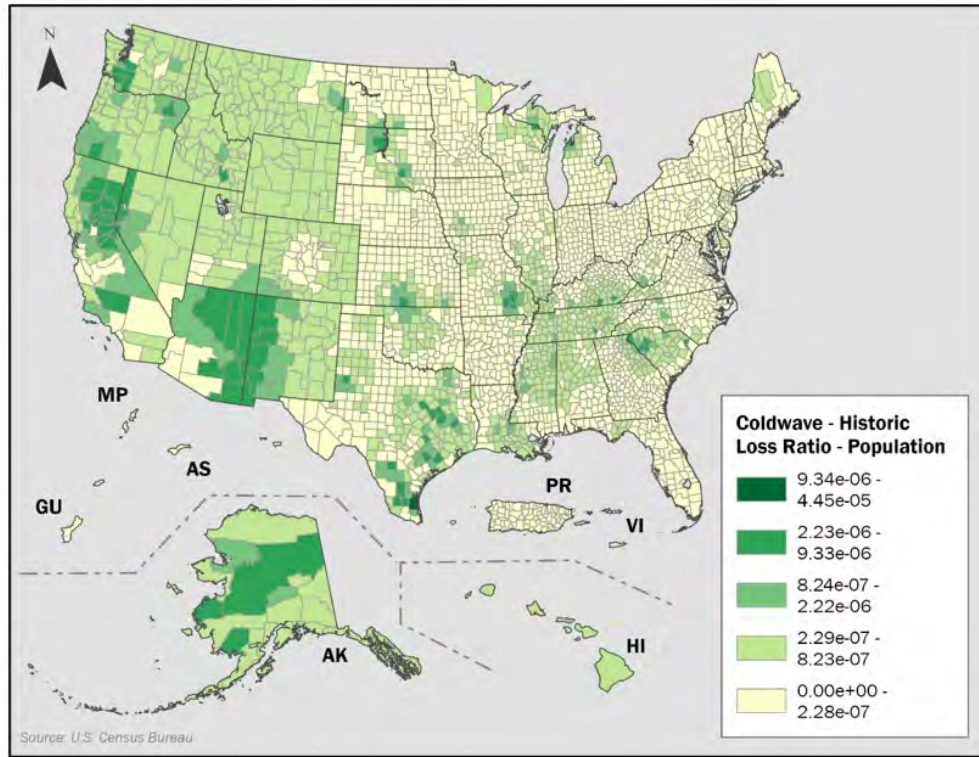
| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>             | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-----------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A2            | Neighbor Counties (Touch & 115km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A3            | Neighbor Counties (Touch & 208km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A4            | Census Regions                    | N/A                                                 | N/A                                              |
|                         | A5            | National                          | N/A                                                 | N/A                                              |
| Population              | P1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P4            | Census Regions                    | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P5            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
| Building                | B1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | B2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B4            | Census Regions                    | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | B5            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Cold Wave (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 26](#), [Figure 27](#), and [Figure 28](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Cold Wave. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

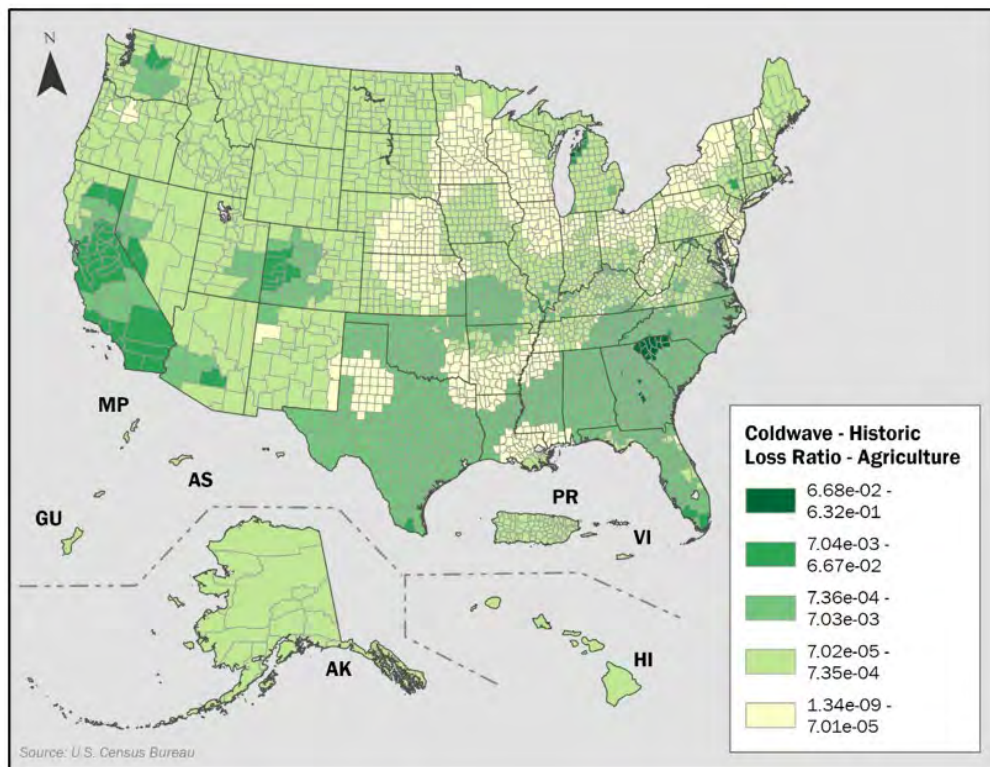


**Figure 26: Cold Wave Bayesian-Adjusted Historic Loss Ratio – Building Value**





**Figure 27: Cold Wave Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 28: Cold Wave Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

## 8.7. Cold Wave Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level. Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels. For the building and agriculture consequence types, Expected Annual Loss is calculated as in [Equation 39](#).

### Equation 39: Census Block Expected Annual Loss to Cold Wave for Building and Agriculture

$$EAL_{CWAV\ CB\ Bldg} = Exposure_{CWAV\ CB\ Bldg} \times Freq_{CWAV\ CB} \times HLR_{CWAV\ CB\ Bldg}$$

$$EAL_{CWAV\ CB\ Ag} = Exposure_{CWAV\ CB\ Ag} \times Freq_{CWAV\ CB} \times HLR_{CWAV\ CB\ Ag}$$

where:

$EAL_{CWAV\ CB\ Bldg}$  is the building Expected Annual Loss due to Cold Wave occurrences for a specific Census block (in dollars).

$Exposure_{CWAV\ CB\ Bldg}$  is the building value exposed to Cold Wave occurrences in the Census block (in dollars).

$Freq_{CWAV\ CB}$  is the Cold Wave annualized frequency for the Census block (event-days per year).

$HLR_{CWAV\ CB\ Bldg}$  is the Bayesian-adjusted building Historic Loss Ratio for Cold Wave for the Census block.

$EAL_{CWAV\ CB\ Ag}$  is the agriculture Expected Annual Loss due to Cold Wave occurrences for a specific Census block (in dollars).

$Exposure_{CWAV\ CB\ Ag}$  is the agriculture value exposed to Cold Wave occurrences in the Census block (in dollars).

$HLR_{CWAV\ CB\ Ag}$  is the Bayesian-adjusted agriculture Historic Loss Ratio for Cold Wave for the Census block.

For the population consequence type, initial or “baseline” Expected Annual Loss for Population values are calculated like other hazards on an impacted people basis (i.e., Fatalities + 0.1\*Injuries). To scale loss estimates up to match the annualized fatality rate reported in the National Center for Health Statistics Compressed Mortality data, baseline Expected Annual Loss for Population values are divided into consequence-subtypes for fatalities and injuries using the proportion of impacted people reported in SHEL DUS associated with fatalities and injuries, respectively. Census block level Expected Annual Loss for the fatality consequence subtype is calculated by scaling up the baseline



population Expected Annual Loss values attributed to fatalities so that the national total matches the annualized fatality count reported in the National Center for Health Statistics Compressed Mortality data, as in [Equation 40](#).

**Equation 40: Census Block Expected Annual Loss to Cold Wave for Fatalities**

$$\begin{aligned}
 EAL_{CWAV_{CBPopBaseline}} &= Exposure_{CWAV_{CBPop}} \times Freq_{CWAV_{CB}} \times HLR_{CWAV_{CBPop}} \\
 SHELDUSRatio_{CWAV_{Fatality}} &= \frac{SHELDUSTotal_{CWAV_{Fatality}}}{SHELDUSTotal_{CWAV_{Fatality}} + 0.1 * SHELDUSTotal_{CWAV_{Injury}}} \\
 EAL_{CWAV_{CBPopBaselineFatality}} &= EAL_{CWAV_{CBPopBaseline}} \times SHELDUSRatio_{CWAV_{Fatality}} \\
 CMScaler_{CWAV} &= \frac{CMAnnualizedTotal_{CWAV}}{\sum_{CB}^{Nat} EAL_{CWAV_{CBPopBaselineFatality}}} \times VSL_{2024} \\
 EAL_{CWAV_{CBPopFatality}} &= EAL_{CWAV_{CBPopBaselineFatality}} \times CMScaler_{CWAV}
 \end{aligned}$$

where:

|                                      |                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $EAL_{CWAV_{CBPopBaselineFatality}}$ | is the baseline Expected Annual Loss fatality value due to Cold Wave for a specific Census block.                  |
| $EAL_{CWAV_{CBPopBaseline}}$         | is the baseline population Expected Annual Loss value due to Cold Wave for a specific Census block.                |
| $Exposure_{CWAV_{CBPop}}$            | is the population value exposed to Cold Wave occurrences for a specific Census block                               |
| $Freq_{CWAV_{CB}}$                   | is the Cold Wave annualized frequency for the Census block (event-days per year).                                  |
| $HLR_{CWAV_{CBPopFatality}}$         | is the Bayesian-adjusted fatality Historic Loss Ratio for Cold Wave for the Census block.                          |
| $SHELDUSRatio_{CWAV_{Fatality}}$     | is the proportion of the national total impacted people due to Cold Wave reported in the SHELDUS period of record. |
| $SHELDUSTotal_{CWAV_{Fatality}}$     | is the national total number of fatalities due to Cold Wave reported in the SHELDUS period of record.              |

$SHELDUSTotal_{CWAVInjury}$ 

is the national total number of injuries due to Cold Wave reported in the SHELDUS period of record

 $EAL_{CWAV_{CBPopBaselineFatality}}$ 

is the baseline population Expected Annual Loss value attributed to the fatality consequence subtype due to Cold Wave for a specific Census block.

 $CMAnnualizedTotal_{CWAV}$ 

is the annualized national total fatalities due to Cold Wave reported in the National Center for Health Statistics Compressed Mortality data.

 $\sum_{CB}^{Nat} EAL_{CWAV_{CBPopBaselineFatality}}$ 

is the sum of baseline Cold Wave fatality Expected Annual Loss values across all Census blocks in the nation.

 $CMScaler_{CWAV}$ 

is the ratio between annualized national fatality totals from the National Center for Health Statistics and the baseline Cold Wave Expected Annual Loss fatality calculation.

 $EAL_{CWAV_{CBPopFatality}}$ 

is the scaled population equivalence Expected Annual Loss fatality value due to Cold Wave for a specific Census block.

 $VSL_{2024}$ 

is the inflation adjusted value of a statistical life as measured in 2024 USD.

For the Cold Wave injury consequence subtype, Expected Annual Loss is calculated using the Bayesian-adjusted population Historic Loss Ratio and the proportion of impacted people reported in SHELDUS associated with injuries as in [Equation 41](#).

**Equation 41: Census Block Expected Annual Loss to Cold Wave for Injuries**

$$EAL_{CWAV_{CBPopBaseline}} = Exposure_{CWAV_{CBPop}} \times Freq_{CWAV_{CB}} \times HLR_{CWAV_{CBPop}}$$

$$EAL_{CWAV_{CBPopInjury}} = EAL_{CWAV_{CBPopBaseline}} \times \left(1 - SHELDUSRatio_{PopFatality}\right) \times VSL_{2024}$$

where:

 $EAL_{CWAV_{CBPopBaseline}}$ 

is the baseline population Expected Annual Loss value due to Cold Wave for a specific Census block

 $Exposure_{CWAV_{CBPop}}$ 

is the population value exposed to Cold Wave occurrences for a specific Census block

|                              |                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| $Freq_{CWAVCB}$              | is the Cold Wave annualized frequency for the Census block (event-days per year).                                   |
| $HLR_{CWAVCBPop}$            | is the Bayesian-adjusted population Historic Loss Ratio for Cold Wave for the Census block.                         |
| $EAL_{CWAVCBPopInjury}$      | is the scaled population equivalence Expected Annual Loss injury value due to Cold Wave for a specific Census block |
| $SHELDUSRatio_{PopFatality}$ | is the proportion of the national total impacted people due to Cold Wave reported in the SHELDUS period of record.  |
| $VSL_{2024}$                 | is the inflation adjusted value of a statistical life as measured in 2024 USD.                                      |

To calculate Cold Wave Expected Annual Loss for the population consequence type, fatality and injury Expected Annual Loss values are summed within Census blocks as in [Equation 42](#).

**Equation 42: Census Block Expected Annual Loss to Cold Wave for Population**

$$EAL_{CWAVCBPop} = EAL_{CWAVCBPopFatality} + EAL_{CWAVCBPopInjury}$$

where:

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{CWAVCBPop}$         | is the population equivalence Expected Annual Loss value due to Cold Wave occurrences for a specific Census block (in dollars). |
| $EAL_{CWAVCBPopFatality}$ | is the scaled population equivalence Expected Annual Loss fatality value due to Cold Wave for a specific Census block.          |
| $EAL_{CWAVCBPopInjury}$   | is the population equivalence Expected Annual Loss injury value due to Cold Wave for a specific Census block.                   |
| $EAL_{CWAVCBPop}$         | is the population equivalence Expected Annual Loss value due to Cold Wave occurrences for a specific Census block (in dollars). |
| $Exposure_{CWAVCBPop}$    | is the population equivalence value exposed to Cold Wave occurrences in the Census block (in dollars).                          |
| $HLR_{CWAVCBPop}$         | is the Bayesian-adjusted population Historic Loss Ratio for Cold Wave for the Census block.                                     |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated values building, population, and agriculture Expected Annual Loss values at the Census block level (see [Equation 43](#)).

**Equation 43: Census Tract and County Expected Annual Loss to Cold Wave**

$$EAL_{CWAV_{CT}} = \sum_{CB}^{CT} EAL_{CWAV_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{CWAV_{CB_{Pop}}} + \sum_{CB}^{CT} EAL_{CWAV_{CB_{Ag}}}$$

$$EAL_{CWAV_{Co}} = \sum_{CB}^{Co} EAL_{CWAV_{CB_{Bldg}}} + \sum_{CB}^{Co} EAL_{CWAV_{CB_{Pop}}} + \sum_{CB}^{Co} EAL_{CWAV_{CB_{Ag}}}$$

where:

$EAL_{CWAV_{CT}}$  is the total Expected Annual Loss due to Cold Wave occurrences for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{CWAV_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{CWAV_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{CWAV_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the Census tract (in dollars).

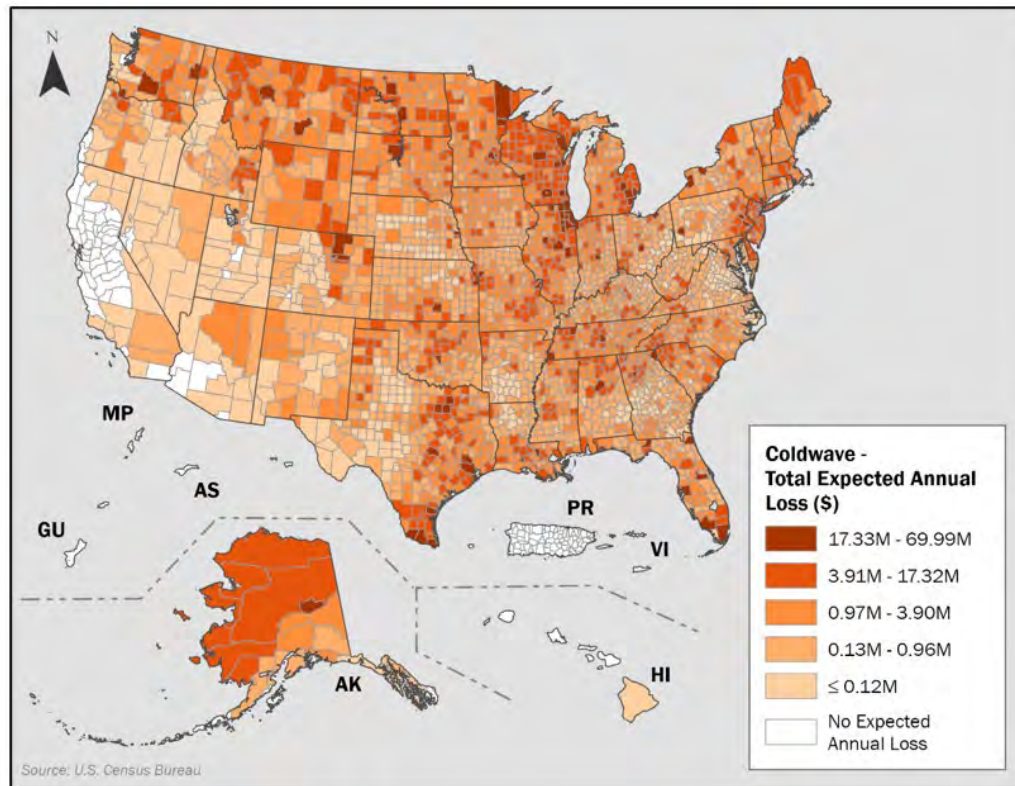
$EAL_{CWAV_{Co}}$  is the total Expected Annual Loss due to Cold Wave occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{CWAV_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{CWAV_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{CWAV_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Cold Wave occurrences for all Census blocks in the county (in dollars).

[Figure 29](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Cold Wave occurrences.



**Figure 29: Total Expected Annual Loss by County to Cold Wave**

With the Cold Wave total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Cold Wave Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Cold Wave Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Cold Wave Expected Annual Loss for agriculture by the total agriculture value of the community.

## 8.8. Cold Wave Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The Cold Wave hazard relies on Centers for Disease Control and Prevention data for fatality counts to estimate Population consequence type losses. These Compressed Mortality data are suppressed at the county level or at any geographic level where there are fewer than 10

reported fatalities. Additionally, mortality statistics are reported by the location of the decedent's residence, not necessarily by the location where the fatality occurred.

- There is currently no adequate injury information in SHELDER to have an injury Historic Loss Ratio distinct from the fatality Historic Loss Ratio.
- The ICD-10 Codes in the Centers for Disease Control and Prevention WONDER do not consider contributing causes of death based on the Cold hazard type, only underlying causes of death.
- National Weather Service Warnings and Advisories are issued at the discretion of each individual National Weather Service Forecast Office, based on regionally determined thresholds. Though this accounts for varying cold-related capacities across the country (Alaska, for example, is more accustomed to lower temperatures than Florida), it results in an inconsistent record of dangerous cold events.

## 9. Drought

A Drought is a deficiency of precipitation over an extended period of time resulting in a water shortage.

Drought hazard modeling uses datasets from the National Drought Mitigation Center and SHELUS as the basis for calculating historical event impact areas and associated losses. The Drought hazard only calculates risk for the Agriculture consequence type. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 9.1. Drought Spatial Source Data

**Historical Occurrence Source:** [University of Nebraska-Lincoln National Drought Mitigation Center \(NDMC\), United States Drought Monitor](#)<sup>66</sup>

The National Drought Mitigation Center provides shapefiles representing areas experiencing Drought on a weekly basis since 2000 (see [Figure 30](#)). Each Drought polygon is categorized by intensity from Abnormally Dry to Exceptional Drought. The Drought Monitor uses multiple indices and indicators to classify Drought severity, and they rely on local condition reports from expert observers (see [Table 32](#)).



**Figure 30: Sample Drought Shape**

<sup>66</sup> NDMC, University of Nebraska-Lincoln & NOAA. (2025). U.S. Drought Monitor [online database]. Retrieved from <https://droughtmonitor.unl.edu/DmData/GISData.aspx>.



**Table 32: Drought Category Descriptions**

| <i>Category Type</i> | <i>Value</i> | <i>Description</i>                                                                                            |
|----------------------|--------------|---------------------------------------------------------------------------------------------------------------|
| Dryness              | D0           | Abnormally Dry – used for areas showing dryness but not yet in drought, or for areas recovering from drought. |
| Drought Intensity    | D1           | Moderate Drought                                                                                              |
| Drought Intensity    | D2           | Severe Drought                                                                                                |
| Drought Intensity    | D3           | Extreme Drought                                                                                               |
| Drought Intensity    | D4           | Exceptional Drought                                                                                           |

## 9.2. Drought Spatial Data Processing

The drought shapefiles associated with each week from January 2000 through March 2025 were extracted and loaded into the processing database. The data initially consisted of 11,885 drought-week records. Only the most severe Drought events were analyzed, so only Drought Intensity categories D3 (Extreme Drought) and D4 (Exceptional Drought) were utilized. Drought-week polygons were then intersected with the Census tract polygons to calculate exposure and annualized frequency.

### 9.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Drought can occur almost anywhere under the right conditions, so all counties were deemed possible for Drought occurrence.

## 9.3. Drought Non-Spatial Data Processing

### 9.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Drought in event-days is computed as the number of distinct Drought event-week polygons that intersect a Census tract multiplied by seven. This count uses the same Drought Census tract intersection table used to find exposure at the Census tract level and are used to compute annualized frequency at the Census tract level.

A historical event-day count is also supplied at the county level as the number of distinct Drought event-week polygons that intersect the county multiplied by seven.

### 9.3.2. PERIOD OF RECORD

The United States Drought Monitor data include Droughts from Jan. 01, 2000 to Mar. 31, 2025, so the period of record for which Drought data are utilized is 25.25 years.

## 9.4. Drought Exposure

Only the Agriculture consequence type is considered for the Drought hazard. However, according to the United States Department of Agriculture Farm Service Agency Guidelines for the Livestock Indemnity Program, drought is not considered a primary cause of livestock death in the United States.<sup>67</sup> So, for the Drought hazard, only the crop portion of the Agriculture consequence type is considered in the exposure. To identify areas of exposure, the Drought-week polygons are intersected with the Census tract polygons within the processing database. The resulting table contains the Drought-week polygon's unique identifier, Census tract number, the intersected area, and the area of intersection containing crops (see [Table 33](#)). All areas are square kilometers.

**Table 33: Sample Data from the Drought Census Tract Intersection Table**

| <i>DroughtID</i> | <i>CensusTract</i> | <i>IntersectedAreaKm2</i> | <i>AreaCropPastureKm2</i> |
|------------------|--------------------|---------------------------|---------------------------|
| 4146             | 47065011001        | 10.5401941730042          | 0                         |
| 4146             | 47073050602        | 16.8104900265808          | 0                         |
| 4146             | 47089070900        | 169.275131709686          | 169.275131709686          |

To determine exposure value, the average coverage of a Drought event-week is found by summing the intersected areas for all Drought event-weeks that intersected the Census tract and dividing this sum by the number of intersecting event-weeks. This is multiplied by the total crop value density of the Census tract (see [Equation 44](#)). The Census tract crop value density has been calculated by dividing the total crop value of the Census tract by its crop land area (in square kilometers).

**Equation 44: Census Tract Drought Exposure**

$$Exposure_{DRT_{CT}_{Ag}} = \frac{\sum IntsctArea_{DRT_{CT}_{Ag}}}{EventWeekCount_{DRT_{CT}}} \times AgValueDen_{CT}$$

where:

$Exposure_{DRT_{CT}_{Ag}}$  is the agriculture (crop only) value exposed to Drought for a specific Census tract (in dollars).

$\sum IntsctArea_{DRT_{CT}_{Ag}}$  is the sum of the intersected areas of past Drought event-weeks with the Census tract (in square kilometers).

$EventWeekCount_{DRT_{CT}}$  is the total number of Drought event-week polygons that intersect the Census tract.

<sup>67</sup> USDA, FSA. (2022). Livestock Indemnity Program Fact Sheet. Retrieved from <https://www.fsa.usda.gov/tools/informational/fact-sheets/livestock-indemnity-program-lip>

$AgValueDen_{CT}$  is the agriculture (crop only) value density of the Census tract (in dollars per square kilometer).

The crop value derived from USDA 2017 Census of Agriculture<sup>68</sup> and CropScape<sup>69</sup> data for the Census tract is considered a ceiling on exposure. If the calculated exposed crop value exceeds the CropScape-derived value, then the CropScape value is used as the crop exposure value for the Census tract.

#### 9.4.1. EXPOSURE AGGREGATION

To calculate exposure at the county level, the exposure values for each Census tract within the county are summed as in [Equation 45](#).

##### Equation 45: County Drought Exposure Aggregation

$$Exposure_{DRGT Co Ag} = \sum_{CT}^{Co} Exposure_{DRGT CT Ag}$$

where:

$Exposure_{DRGT Co Ag}$  is the agriculture (crop only) value exposed to Drought for a specific county (in dollars).

$\sum_{CT}^{Co} Exposure_{DRGT CT Ag}$  is the summed value of all agriculture (crop only) areas exposed to Drought for each Census tract within the county (in dollars).

### 9.5. Drought Annualized Frequency

The annualized frequency value represents the number of recorded Drought occurrences, in event-days, each year over the period of record (25.25 years). The annualized frequency is calculated at the Census tract level, and the Census tract-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the same intersection between Drought event-week polygons and Census tract polygons that were used to calculate exposure. The historical event-day count described above is used to calculate the annualized frequency of Drought event-days as in [Equation 46](#).

<sup>68</sup> USDA. (2017). 2017 Census of Agriculture. Retrieved from <https://www.nass.usda.gov/Publications/AgCensus/2017/index.php>.

<sup>69</sup> George Mason University. (2025). CropScape – Crop Land Data Layer. Retrieved from <https://idia.gmu.edu/cropscape-cropland-data-layer/>.

**Equation 46: Census Tract Drought Annualized Frequency**

$$Freq_{DRGT\ CT} = \frac{EventDayCount_{DRGT\ CT}}{PeriodRecord_{DRGT}}$$

where:

$Freq_{DRGT\ CT}$  is the annualized frequency of Drought determined for a specific Census tract (event-days per year).

$EventDayCount_{DRGT\ CT}$  is the number of Drought event-days (event-weeks multiplied by seven) that intersect the Census tract.

$PeriodRecord_{DRGT}$  is the period of record for Drought (25.25 years).

**9.5.1. ANNUALIZED FREQUENCY AGGREGATION**

The dataset presents area-weighted average annualized frequency values at the county level, so these values may not exactly match that of dividing the number of recorded Drought event-days at the county level by the period of record. The annualized frequency values at the Census tract level are aggregated to the county level using area-weighted functions as in [Equation 47](#).

**Equation 47: County Area-Weighted Drought Annualized Frequency**

$$Freq_{DRGT\ Co} = \frac{\sum_{CT}^{Co} (Freq_{DRGT\ CT} \times Area_{CT})}{Area_{Co}}$$

where:

$Freq_{DRGT\ Co}$  is the Drought annualized frequency calculated for a specific county (event-days per year).

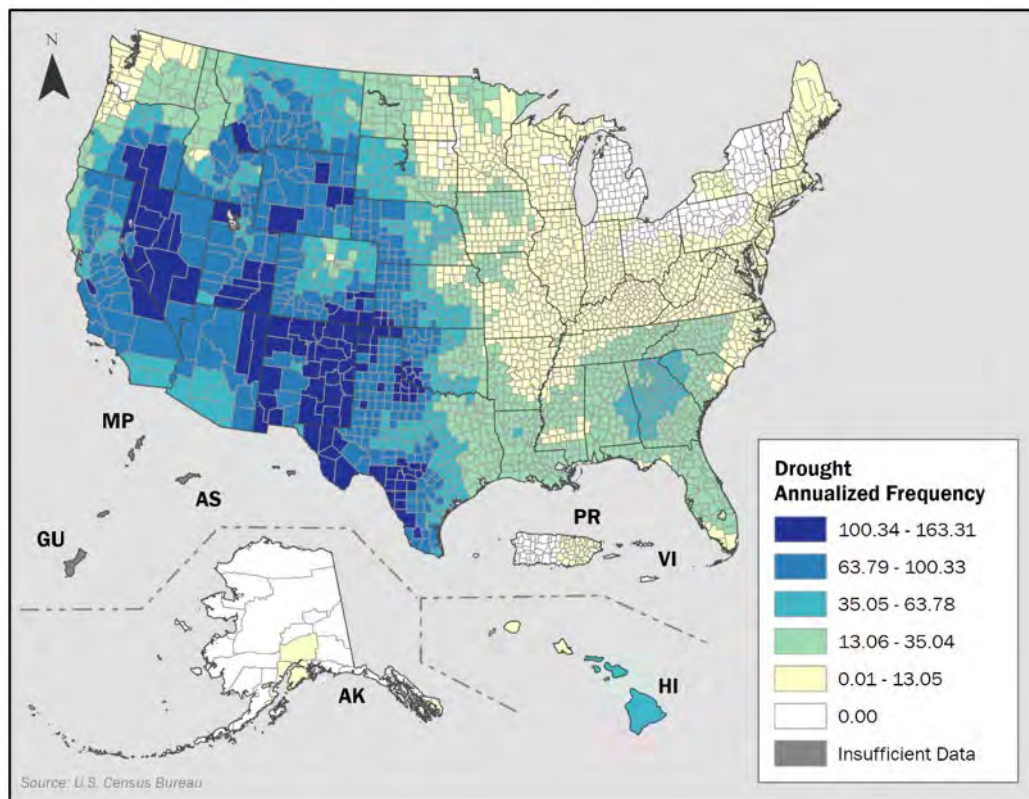
$Freq_{DRGT\ CT}$  is the annualized frequency of Drought determined for a specific Census tract (event-days per year).

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$\sum_{CT}^{Co}$  is the sum for all Census tracts in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 31](#) displays Drought annualized frequency at the county level.



**Figure 31: Drought Annualized Frequency by County**

## 9.6. Drought Historic Loss Ratio

The Drought Historic Loss Ratio is the representative percentage of a location's Drought exposure that experiences loss due to a Drought event-day, or the average rate of loss associated with the occurrence of a Drought event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Drought hazard type.

Loss data are provided by SHELUDUS<sup>70</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. Data from the SHELUDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. One peril type is mapped to the hazard Drought (see [Table 34](#)). These native loss records are expanded on an event-day basis (to a maximum of 365 days) and are aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

<sup>70</sup> For Drought loss information, SHELUDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

**Table 34: Drought Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Drought                       | 5,872                              | 160,852                              |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is a county-level value that represents the dollar value of the total crop value of the county as estimated in the United States Department of Agriculture 2017 Census of Agriculture data. The Loss Ratio per Basis for each SHEL DUS-documented event-day is calculated using [Equation 48](#).

**Equation 48: Loss Ratio per Basis Calculation for a Single Drought Event-Day**

$$LRB_{DRGT Co Ag} = \frac{LOSS_{DRGT Co Ag}}{HLRExposure_{Co Ag}}$$

where:

$LRB_{DRGT Co Ag}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Drought event-day. Calculation is performed for agriculture (crop only).

$LOSS_{DRGT Co Ag}$  is the agriculture loss experienced from the Drought event-day documented to have occurred in the county (in dollars).

$HLRExposure_{Co Ag}$  is the total agriculture (crop only) value of the county estimated to have been exposed to the Drought event-day (in dollars).

Drought event-days can occur with a high frequency in areas but often result in no recorded loss to agriculture. SHEL DUS does not record events in which no loss occurred, so a number of zero-loss event-days are inserted into the data to align the event-day count in the Historic Loss Ratio calculation to the historical event-day count experienced within the SHEL DUS period of record (1996 through 2023). For Drought, the historical event-day count is extracted using the intersection between the Drought event-week polygons and the Census tract used to calculate annualized frequency and multiplying by 7 to convert weeks into days. An annual rate is calculated as the event-day count divided by the period of record of 25.25 years, and this rate is multiplied by the SHEL DUS period of record of 28 years to estimate a historical event-day count for the appropriate time range.

If the number of loss-causing Drought event-day records from SHEL DUS is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for all Drought events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis

Sets are defined by consequence type, Bayesian level, the occasional use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Drought.<sup>71</sup>

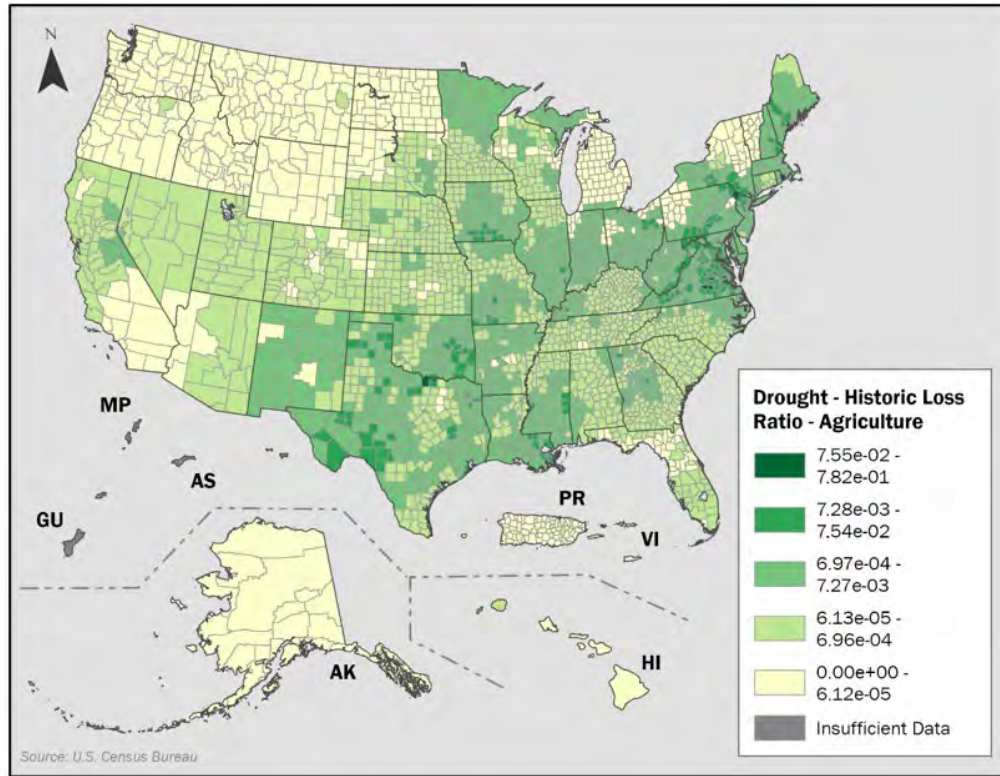
**Table 35: Drought Loss Ratio Bayesian Weighting Level Sets**

| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>                           | <i>Set Refinement Filter</i> | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-------------------------------------------------|------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                                          | N/A                          | 25 Loss Ratio per Basis records                  |
| Agriculture             | A2            | Neighbor Counties (Touch & 208km)               | N/A                          | 25 Loss Ratio per Basis records                  |
| Agriculture             | A3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
| Agriculture             | A4            | National                                        | N/A                          | N/A                                              |

Loss Ratio per Basis set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Drought (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 32](#) represents the final, Bayesian-adjusted county-level Historic Loss Ratio values for Drought. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

<sup>71</sup> Due to data limitations, a disproportionate number of counties deemed hazard possible for Drought had a county and/or surrounding area level variance in loss ratios that was either 0 or undefined. To prevent overestimation of losses, historic loss ratios for these counties were determined by a fixed weighting scheme of 50% for County, 25% for surrounding area, 15% for regional, and 10% for national.





**Figure 32: Drought Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

## 9.7. Drought Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census tract level as in [Equation 49](#). Performing the base calculations once at the Census tract level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 49: Census Tract Expected Annual Loss to Drought

$$EAL_{DRGT_{CT_{Ag}}} = Exposure_{DRGT_{CT_{Ag}}} \times Freq_{DRGT_{CT}} \times HLR_{DRGT_{CT_{Ag}}}$$

where:

$EAL_{DRGT_{CT_{Ag}}}$  is the agriculture Expected Annual Loss due to Drought occurrences for a specific Census tract (in dollars).

$Exposure_{DRGT_{CT_{Ag}}}$  is the agriculture (crop only) value exposed to Drought occurrences in the Census tract (in dollars).

$Freq_{DRGT_{CT}}$  is the Drought annualized frequency for the Census tract (event-days per year).

$HLR_{DRGT CT Ag}$  is the Bayesian-adjusted agriculture Historic Loss Ratio for Drought for the Census tract.

The total Expected Annual Loss values at the county level are the aggregated agriculture Expected Annual Loss values at the Census tract level as in [Equation 50](#).

**Equation 50: County Expected Annual Loss to Drought**

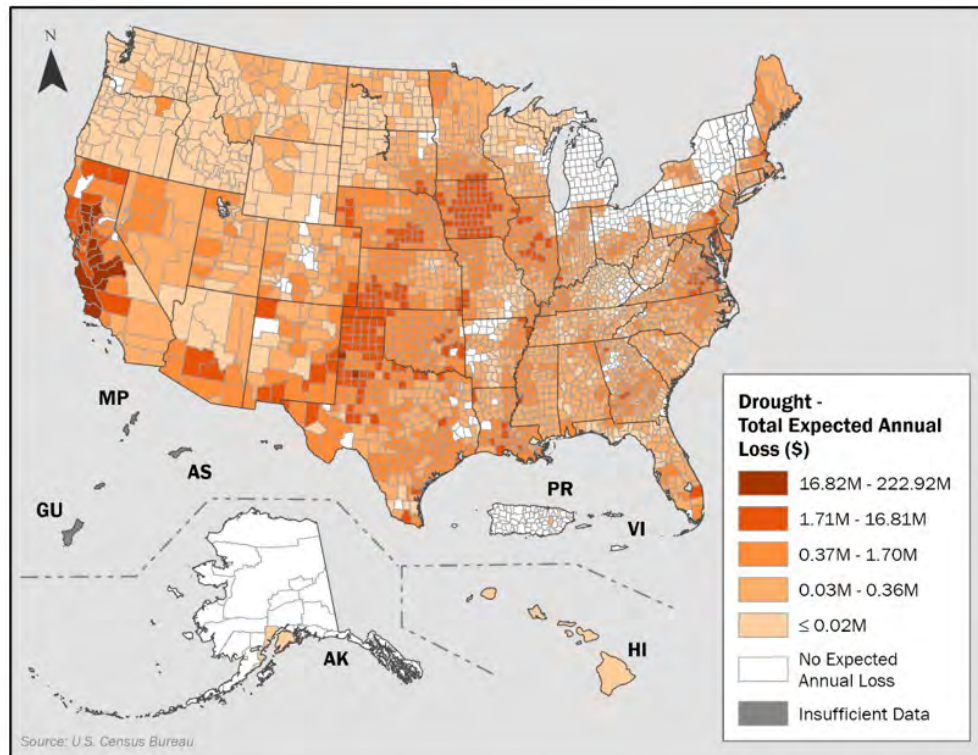
$$EAL_{DRGT Co Ag} = \sum_{CT}^{Co} EAL_{DRGT CT Ag}$$

where:

$EAL_{DRGT Co Ag}$  is the total Expected Annual Loss due to Drought for a specific county (in dollars).

$\sum_{CT}^{Co} EAL_{DRGT CT Ag}$  is the summed agriculture Expected Annual Loss to agriculture value due to Drought occurrences for all Census tracts in the county (in dollars).

[Figure 33](#) shows the total Expected Annual Loss (agriculture only) to Drought occurrences.



**Figure 33: Total Expected Annual Loss by County to Drought**

With the Drought total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)).

The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Agriculture Expected Annual Loss Rate is calculated by dividing the Drought Expected Annual Loss for agriculture by the total crop value of the community.

## 9.8. Drought Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- There is often a disconnect between where a drought happens and where its impacts are felt, making direct economic losses a difficult component to measure. Drought risk is also geographically variable – some regions depend on snowpack for its water supply, while others are more dependent on liquid precipitation. Additionally, mitigative efforts and legislative policies on water use can significantly alter the impacts of a drought occurrence. As such, modeling and measuring the impacts of Drought is an incredibly complex undertaking.
- This only measures the direct impacts of Drought on crops; livestock is not considered because it is assumed that other mitigative activities would be implemented to protect livestock.
- Time is a critical component for measuring Drought impacts. The timing and seasonality of a drought occurrence as well as individual crop viability impacts potential losses. Additionally, Drought is not a discretely occurring hazard and can last anywhere from weeks to years at a time. Information regarding flash droughts, which are rapid-onset drought conditions brought about by a combination of low precipitation and elevated temperatures, high winds, and/or changes in radiation<sup>72</sup> are also still limited.

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<sup>72</sup> <https://www.drought.gov/what-is-drought/flash-drought>

## 10. Earthquake

An Earthquake is a shaking of the earth's surface by energy waves emitted by slowly moving tectonic plates overcoming friction with one another underneath the earth's surface.

The Earthquake hazard is modeled by combining seismic frequency data from the U.S. Geological Survey to provide the hazard-susceptible area, and the Expected Annual Loss data provided by the Federal Emergency Management Agency's Hazus Earthquake Analysis. It is important to note that underlying hazard susceptibility data for Earthquake does not rely on observed historical events, but on probabilistic hazard data, meaning that the annualized frequency and historic loss ratio values are surrogate values and not used in the Expected Annual Loss calculations. In addition, the Exposure and Expected Annual Loss values are sourced directly from the Federal Emergency Management Agency's Hazus Earthquake Analysis for both Census tract and county level values. The Earthquake hazard calculates risk for only the Building and Population consequence types. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

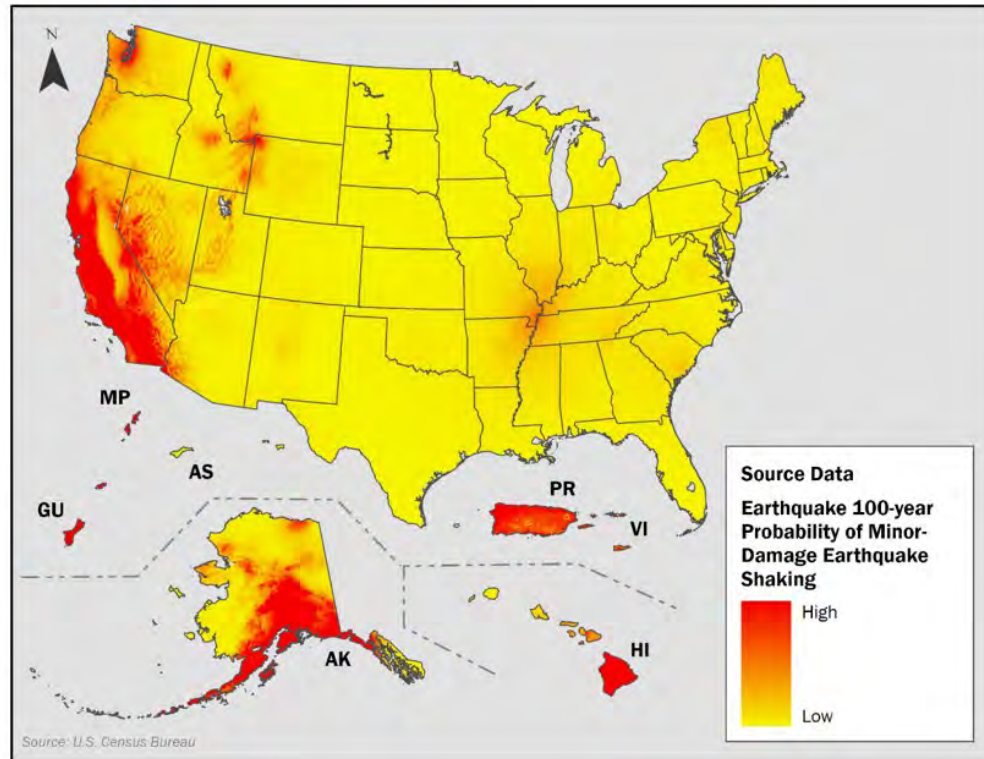
### 10.1. Earthquake Spatial Source Data

**Susceptible Area Source:** [U.S. Geological Survey 2023 National Seismic Hazard Model](#), and [2025 Puerto Rico and U.S. Virgin Islands Long-term National Seismic Hazard Model](#), Kishor Jaiswal

The 2023 50-State Update of the United States National Seismic Hazard Model and the 2025 Puerto Rico and U.S. Virgin Islands Long-term National Seismic Hazard Model defines the potential for earthquake ground shaking for various probability levels across the conterminous United States, Alaska, Hawai'i and Puerto Rico and the U.S. Virgin Islands and is applied in seismic provisions of building codes, insurance rate structures, risk assessments, and other public policy. The updated model and data, provided by Kishor Jaiswal of the U.S. Geological Survey, represent an assessment of the best available science in earthquake hazards and update the previous National Seismic Hazard Models for the conterminous United States (2018), Alaska (2007), Puerto Rico and the U.S. Virgin Islands (2003), Hawai'i (2001), and the Pacific Territories. The U.S. Geological Survey supplied a geodatabase of gridded point locations representing the 100-year probability of Minor-Damage Earthquake Shaking (see [Figure 34](#)). Cell values range from 0 to 100. These files are derived from the hazard model used to create U.S. Geological Survey National Seismic Hazard Maps.<sup>73</sup>

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<sup>73</sup> USGS. (2024). 2023 50-State Long-term National Seismic Hazard Model Retrieved from <https://www.usgs.gov/programs/earthquake-hazards/science/2023-50-state-long-term-national-seismic-hazard-model-0>



**Figure 34: Map of Earthquake Probability**

**Loss Quantification Source:** The Federal Emergency Management Agency’s Hazus Earthquake Analysis

The Federal Emergency Management Agency’s Hazus software uses a uniform engineering-based approach to measure damages, casualties, and economic losses from earthquakes nationwide. The Federal Emergency Management Agency uses Hazus to determine Earthquake risk throughout the United States at both the Census tract and county levels (see [Table 36](#) for sample data). The analysis also calculates an Annualized Earthquake Loss value that is used as the Expected Annual Loss value for buildings at the Census tract and county levels. A separate measure of annualized population loss is provided by the Federal Emergency Management Agency as the estimation of Level 4 severity injuries (instantaneous deaths or mortal injuries) expected annually due to Earthquakes. These loss values are combined to find the Expected Annual Loss.

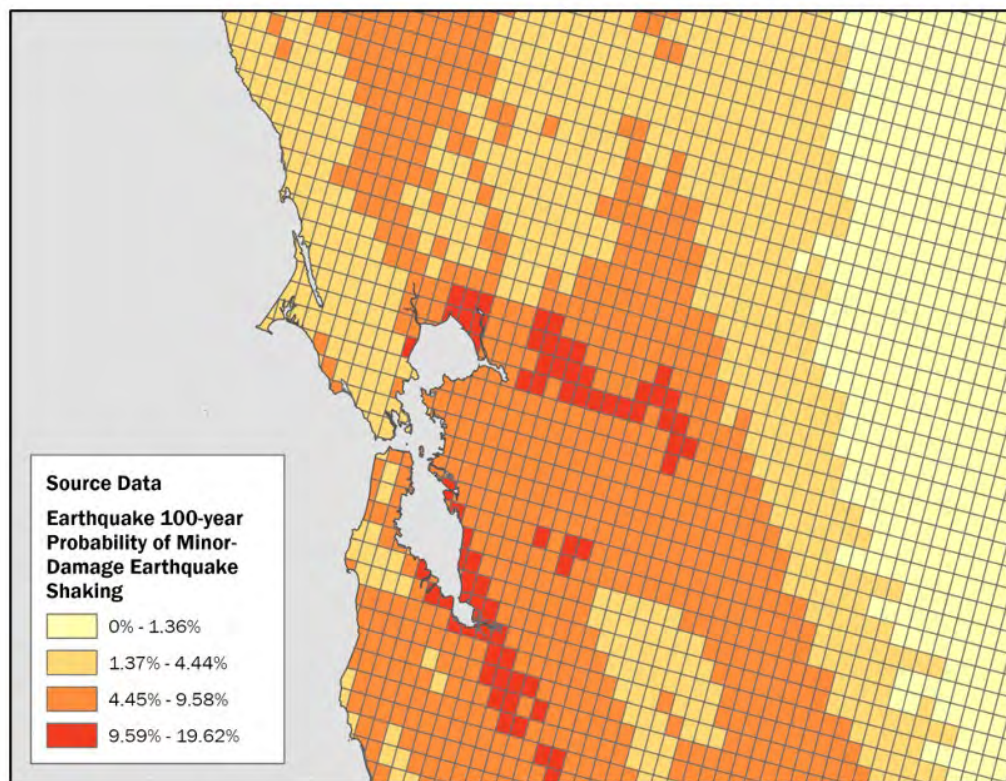


**Table 36: Sample Census Tract-Level Data from the Federal Emergency Management Agency Hazus Earthquake Analysis**

| <i>Census Tract</i> | <i>County Name</i> | <i>State</i> | <i>TotalExp_Bldg_1k</i> | <i>Pop_2020</i> | <i>AEL_1mil</i> |
|---------------------|--------------------|--------------|-------------------------|-----------------|-----------------|
| 15003009902         | Honolulu           | Hawai'i      | 369,069                 | 3,998           | 0.06929         |
| 06019004212         | Fresno             | California   | 1,783,600               | 12,943          | 1.047864        |
| 02013000100         | Aleutians East     | Alaska       | 457,812                 | 3,420           | 0.431396        |

## 10.2. Earthquake Spatial Data Processing

Raw frequency data was received from the U.S. Geological Survey as a point feature class, with each point corresponding to the expected annualized number of loss-causing Earthquake hazard occurrences in that location. Then, Thiessen polygons (Voronoi diagram) were generated from the point data to represent areas closest to each point, creating a grid where the point values were applied to the grid cell. This grid provides the annualized frequency data for the United States and the territories of Puerto Rico, U.S. Virgin Islands, Guam, Northern Mariana Islands, and American Samoa integrated with the necessary spatial attributes. See [Section 10.4](#) for how these data were used.

**Figure 35: Map of Earthquake Fishnet**

### 10.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

In the Federal Emergency Management Agency Hazus Earthquake Analysis, every county has some degree of Earthquake risk; therefore, all counties were deemed possible for Earthquake occurrence.

## 10.3. Earthquake Exposure

Like the other exposure values produced, Hazus bases its exposure values on the Hazus 6.1 building values and population data. Exposure values are extracted from the Federal Emergency Management Agency Hazus Earthquake Analysis data at the Census tract and county levels.

A small subset of exposure values from the Federal Emergency Management Agency Hazus Earthquake Analysis exceed the Hazus-recorded building values or populations for the Census tract or county. These values were left as is rather than being lowered to the Hazus values.

## 10.4. Earthquake Annualized Frequency

The annualized frequency value represents the area-weighted probability of Earthquake occurrences (at least minor-damage shaking) in events impacting a location in a given year. The annualized frequency is calculated at the Census block level.

Earthquake-probability fishnet polygons are intersected with the Census block polygons within the processing database. The resulting table contains the fishnet polygon's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 37](#)).

**Table 37: Sample Data from the Earthquake Fishnet Census Block Intersection Table**

| <i>EarthquakeFishnetID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|----------------------------|--------------------|---------------------------|
| 422655                     | 191930036004217    | 0.003866                  |
| 422655                     | 191930036004221    | 0.010595                  |
| 422655                     | 191930036004225    | 0.019825                  |

This intersection between Earthquake-probability fishnet polygons and Census block polygons is used to calculate annualized frequency at the Census block level as in [Equation 51](#).

**Equation 51: Census Block Area-Weighted Fishnet Earthquake Annualized Frequency**

$$Freq_{ERQK_{CB}} = \frac{\sum_{Fish}^{CB} (IntsctArea_{ERQK_{FishCB}} \times Prob_{ERQK_{FishCB}})}{Area_{CB}}$$

where:



|                                 |                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Freq_{ERQK_{CB}}$              | is the area-weighted annualized frequency of Earthquake determined for a specific Census block (probability per year).                                                    |
| $IntsctArea_{ERQK_{Fish_{CB}}}$ | is the intersected area of the Earthquake probability fishnet grid cell where the Earthquake probability was greater than 0 with the Census block (in square kilometers). |
| $Prob_{ERQK_{Fish_{CB}}}$       | is the probability of Earthquake event for the intersecting fishnet grid cell.                                                                                            |
| $\sum_{Fish}^{CB}$              | is the sum for all fishnet grid cells that intersect the Census block.                                                                                                    |
| $Area_{CB}$                     | is the total area of the Census block (in square kilometers).                                                                                                             |

#### 10.4.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level as aggregates of the Census block values. These values are surrogates as the final Expected Annual Loss values are extracted from the Federal Emergency Management Agency Hazus Earthquake Analysis, and it was not possible to derive an annualized frequency component from the Federal Emergency Management Agency Hazus Earthquake Analysis. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 52](#).

##### Equation 52: Census Tract and County Area-Weighted Earthquake Annualized Frequency Aggregation

$$Freq_{ERQK_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{ERQK_{CB}} \times Area_{CB})}{Area_{CT}}$$

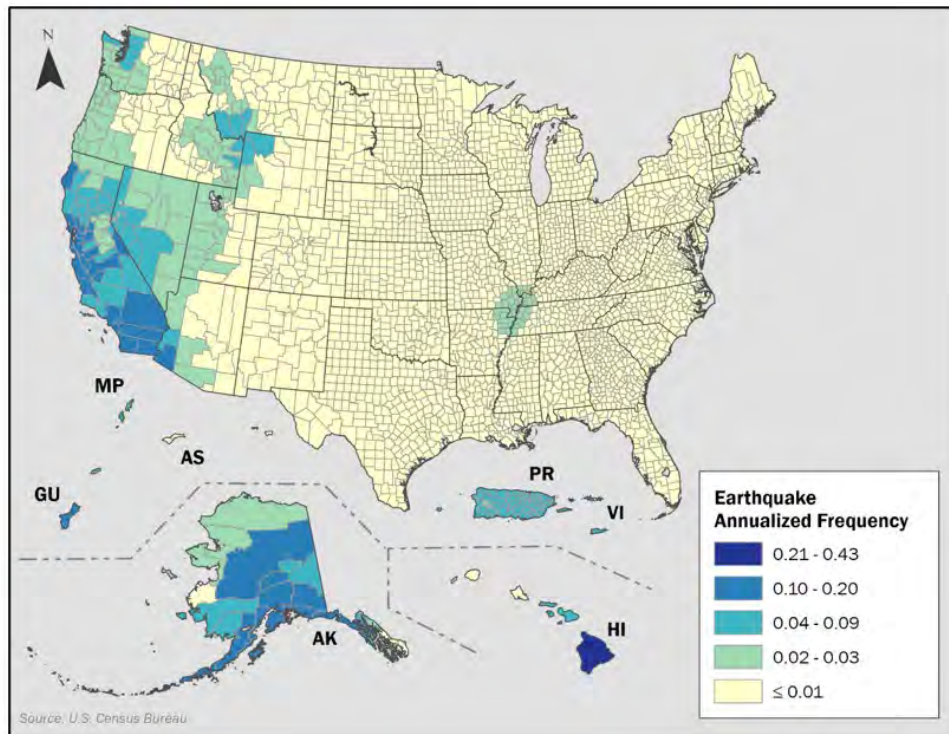
$$Freq_{ERQK_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{ERQK_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

|                    |                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| $Freq_{ERQK_{CT}}$ | is the area-weighted Earthquake annualized frequency calculated for a specific Census tract (probability per year).    |
| $Freq_{ERQK_{CB}}$ | is the area-weighted annualized frequency of Earthquake determined for a specific Census block (probability per year). |
| $Area_{CB}$        | is the total area of the Census block (in square kilometers).                                                          |
| $\sum_{CB}^{CT}$   | is the sum for all Census blocks in the Census tract.                                                                  |

|                    |                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| $Area_{CT}$        | is the total area of the Census tract (in square kilometers).                                                 |
| $Freq_{ERQK_{Co}}$ | is the area-weighted Earthquake annualized frequency calculated for a specific county (probability per year). |
| $\sum_{CB}^{Co}$   | is the sum for all Census blocks in the county.                                                               |
| $Area_{Co}$        | is the total area of the county (in square kilometers).                                                       |

[Figure 36](#) displays Earthquake annualized frequency at the county level.



**Figure 36: Earthquake Annualized Frequency by County**

## 10.5. Earthquake Historic Loss Ratio

The Earthquake Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to an Earthquake occurrence, or the average rate of loss associated with an Earthquake occurrence. Historic Loss Ratio values displayed are surrogate values as the final Expected Annual Loss values at the Census tract and county level are extracted from the Federal Emergency Management Agency Hazus Earthquake Analysis. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Earthquake hazard type.

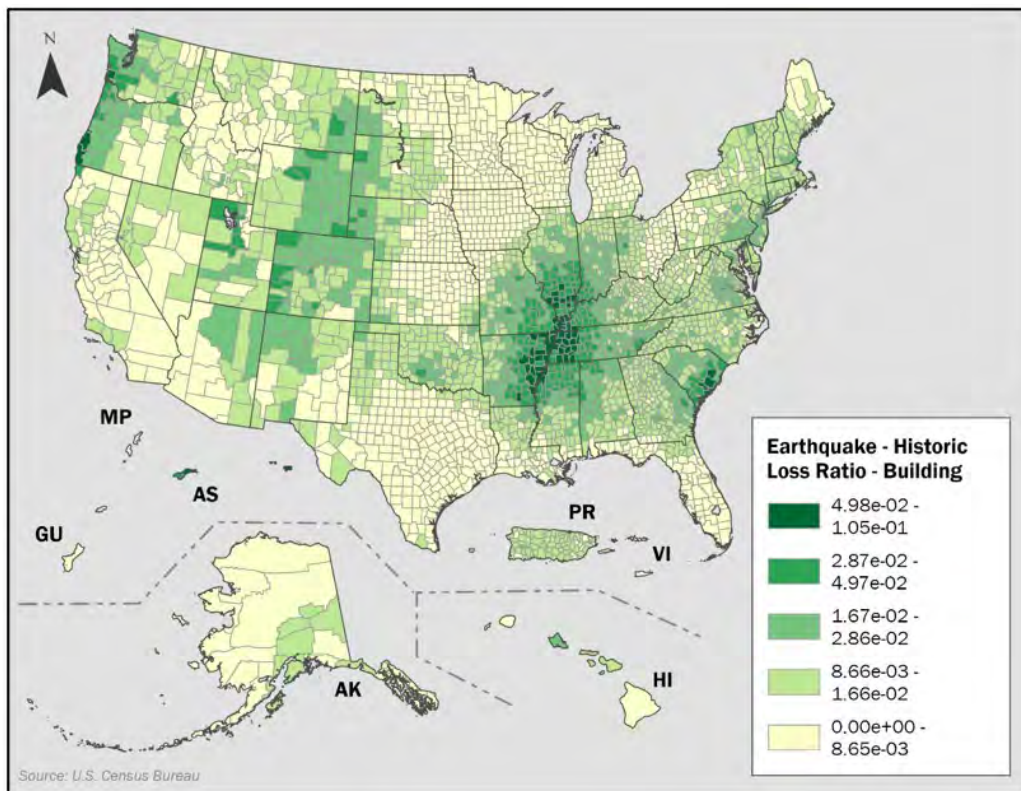
**Equation 53: Historic Loss Ratio Calculation for Earthquake**

$$HLR_{ERQK_{CoCnsqType}} = \frac{EAL_{ERQK_{CoCnsqType}}}{AFREQ_{ERQK_{Co}} \times EXP_{ERQK_{CoCnsqType}}}$$

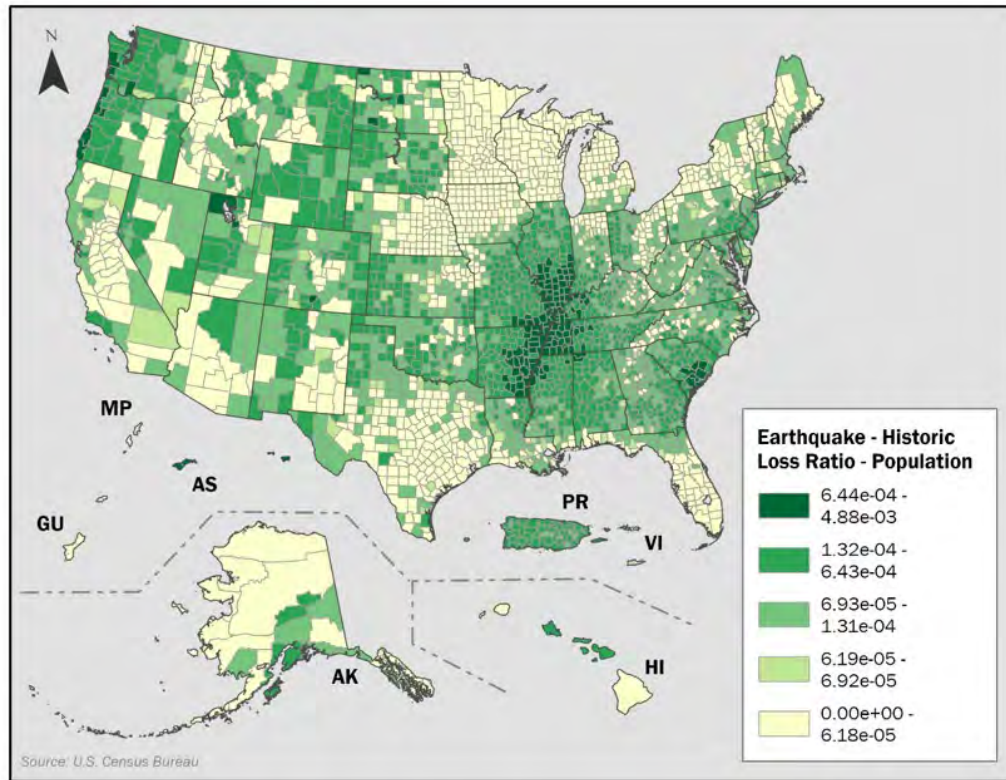
where:

|                           |                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $HLR_{ERQK_{CoCnsqType}}$ | is the Historic Loss Ratio (by consequence type) of a county for Earthquake.                                                                                      |
| $EAL_{ERQK_{CoCnsqType}}$ | is the Expected Annual Loss (by consequence type) of a county for Earthquake.                                                                                     |
| $AFREQ_{ERQK_{Co}}$       | is the annualized frequency of a county for Earthquake.                                                                                                           |
| $EXP_{ERQK_{CoCnsqType}}$ | is the value (by consequence type) of the sum of Census block exposure values obtained from the Federal Emergency Management Agency study (in dollars or people). |

[Figure 37](#) and [Figure 38](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Earthquake.



**Figure 37: Earthquake Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 38: Earthquake Bayesian-Adjusted Historic Loss Ratio – Population**

The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census tracts within the parent county.

## 10.6. Earthquake Expected Annual Loss

Expected Annual Loss values are extracted from the Federal Emergency Management Agency Hazus Earthquake Analysis data at the Census tract and county levels. Exposure, annualized frequency, and Historic Loss Ratio are provided at the Census tract and county level as surrogate values but are not used to compute the Expected Annual Loss values.

The Federal Emergency Management Agency Hazus Earthquake Analysis data estimate the Annualized Earthquake Losses, the estimated long-term value of earthquake losses to the general building stock in any single year in a specified geographic area, as well as an annualized population loss value. The Annualized Earthquake Loss is estimated using the Riemann Sums Method. This method averages the annual losses associated with various return periods, averages the annual losses associated with each of the eight return periods, and then multiplies the resulting losses by their differential probability to develop one loss value for each Census tract.<sup>74</sup> The population loss estimation is based on the correlation between building damage and the number and severity of

<sup>74</sup> FEMA. (2023). FEMA P-366: Hazus Estimated Annualized Earthquake Losses for the United States. [https://www.fema.gov/sites/default/files/documents/fema\\_p-366-hazus-estimated-annualized-earthquake-losses-united-states.pdf](https://www.fema.gov/sites/default/files/documents/fema_p-366-hazus-estimated-annualized-earthquake-losses-united-states.pdf).

casualties. The summed Federal Emergency Management Agency Hazus Earthquake Analysis loss values are used as the total Expected Annual Loss at the Census tract and county levels as in [Equation 54](#).

**Equation 54: Census Tract and County Expected Annual Loss to Earthquake**

$$EAL_{ERQK_{CT}} = AEL_{CT} + (PopLoss_{CT} \times VSL)$$

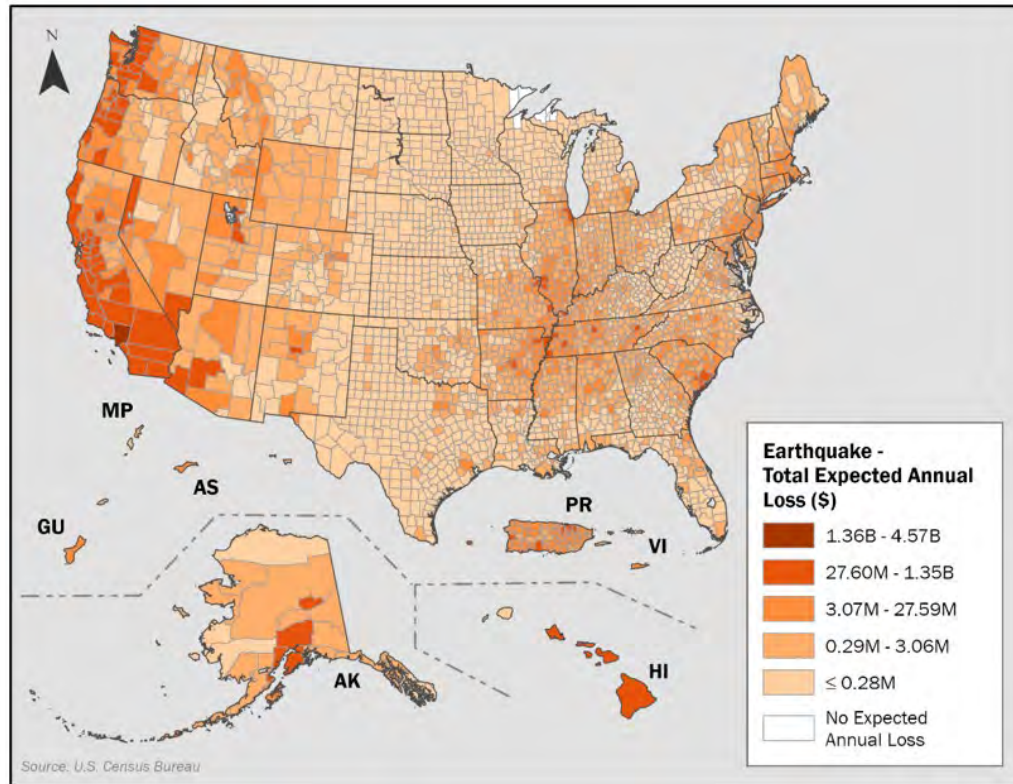
$$EAL_{ERQK_{Co}} = AEL_{Co} + (PopLoss_{Co} \times VSL)$$

where:

|                   |                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{ERQK_{CT}}$ | is the total Expected Annual Loss due to Earthquake occurrences for a specific Census tract (in dollars).                                                       |
| $AEL_{CT}$        | is the annual Earthquake loss to buildings for a specific Census tract by the Federal Emergency Management Agency Hazus Earthquake Analysis study (in dollars). |
| $PopLoss_{CT}$    | is the population loss estimation for a specific Census tract by the Federal Emergency Management Agency Hazus Earthquake Analysis study (in people).           |
| $VSL$             | is the Value of Statistical Life (\$13.7 million per person).                                                                                                   |
| $EAL_{ERQK_{Co}}$ | is the total Expected Annual Loss due to Earthquake occurrences for a specific county (in dollars).                                                             |
| $AEL_{Co}$        | is the annual Earthquake loss to buildings for a specific county by the Federal Emergency Management Agency Hazus Earthquake Analysis study (in dollars).       |
| $PopLoss_{Co}$    | is the population loss estimation for a specific county by the Federal Emergency Management Agency Hazus Earthquake Analysis study (in people).                 |

[Figure 39](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Earthquake occurrences.





**Figure 39: Total Expected Annual Loss by County to Earthquake**

With the Earthquake total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Earthquake Expected Annual Loss for buildings by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Earthquake Expected Annual Loss for population by the total population of the community.

## 10.7. Earthquake Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- Earthquake risk is based on probabilistic data for earthquake events and not on historical earthquake events. This means that the loss amounts and Ratings may differ from expected figures when looking solely at historical records.

- The surrogate frequency for Earthquake measures the true annual frequency of a *damaging* earthquake (MMI6) occurring. This differs from the way frequency is measured in other hazards, where the frequency measures hazard occurrence, regardless of whether an event results in damages.



## 11. Hail

Hail is a form of precipitation that occurs during thunderstorms when raindrops, in extremely cold areas of the atmosphere, freeze into balls of ice before falling towards the earth's surface.

The primary dataset available from the Storm Prediction Center includes all hail event records and basic latitude-longitude data which are interpolated into segments, then buffered to create hail impact areas for loss estimation. These are combined with calculated annualized frequency and SHELUS-derived historic loss ratios to create the Expected Annual Loss. The Hail hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 11.1. Hail Spatial Source Data

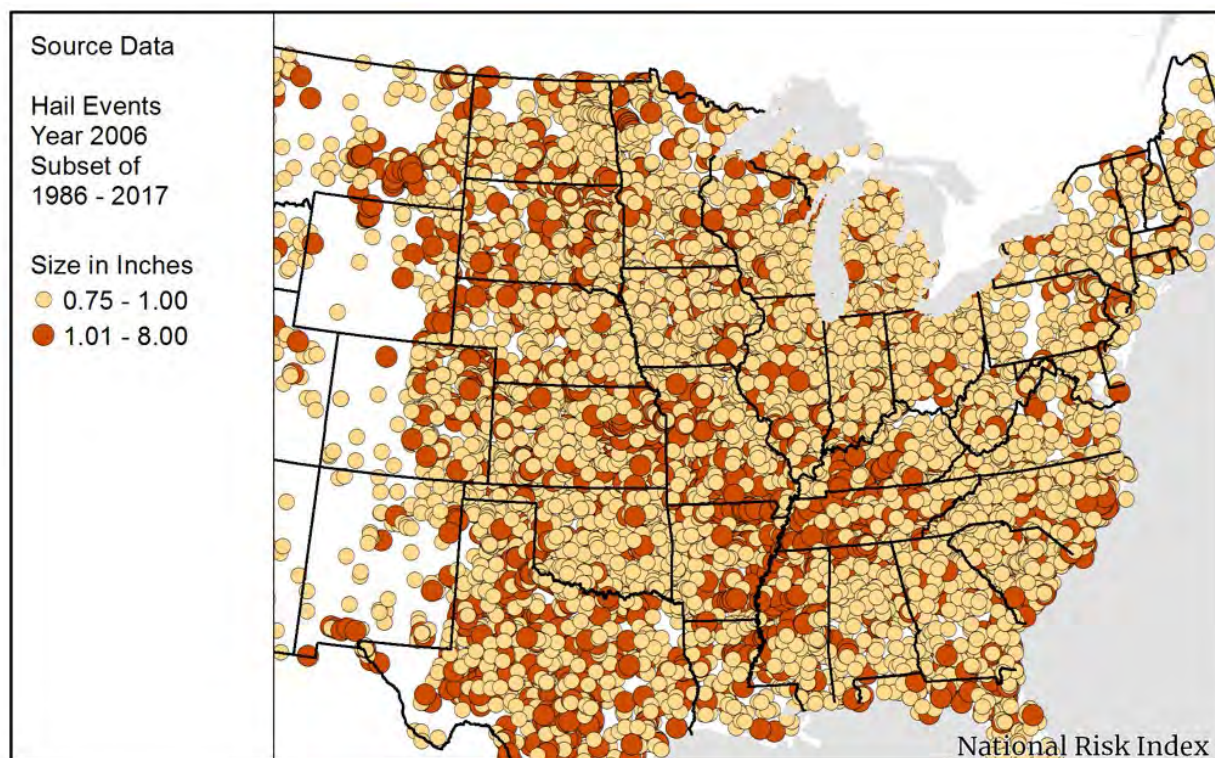
**Historical Occurrence Source:** [National Weather Service, Storm Prediction Center, Severe Weather Database Files](#) <sup>75</sup>

The Storm Prediction Center compiles all records of Hail events from the National Weather Service's monthly Storm Data publication and makes them available in CSV format on the Warning Coordination Meteorologist's website. These files record spatiotemporal information (start and end coordinates, date, time) as well as economic loss and hail size in inches (see [Table 38](#) and [Figure 40](#)).

**Table 38: Sample Hail Data from the Storm Prediction Center**

| Om (Hail ID) | Date                   | State        | Mag (Hail Size in inches) | Injuries | Fatalities | Loss (Property Loss in \$) | Closs (Crop Loss in \$) | Slon (Start Longitude) | Slat (Start Latitude) |
|--------------|------------------------|--------------|---------------------------|----------|------------|----------------------------|-------------------------|------------------------|-----------------------|
| 4095         | May 23, 2010 11:20 PM  | Alaska       | 0.75                      | 0        | 0          | 0                          | 0                       | -150.22                | 65                    |
| 317151       | Jul. 19, 2011 12:50 PM | Oregon       | 1.00                      | 1        | 0          | 0                          | 0                       | 45.3                   | -118.14               |
| 2016-06082   | Jun. 22, 2016 2:19 AM  | North Dakota | 1.75                      | 0        | 0          | 25000                      | 50000                   | -104.04                | 47.94                 |

<sup>75</sup> NWS, SPC. (2024). Severe Weather Database files, Hail, 1955-2023 [online dataset]. Retrieved from <http://www.spc.noaa.gov/wcm/>.

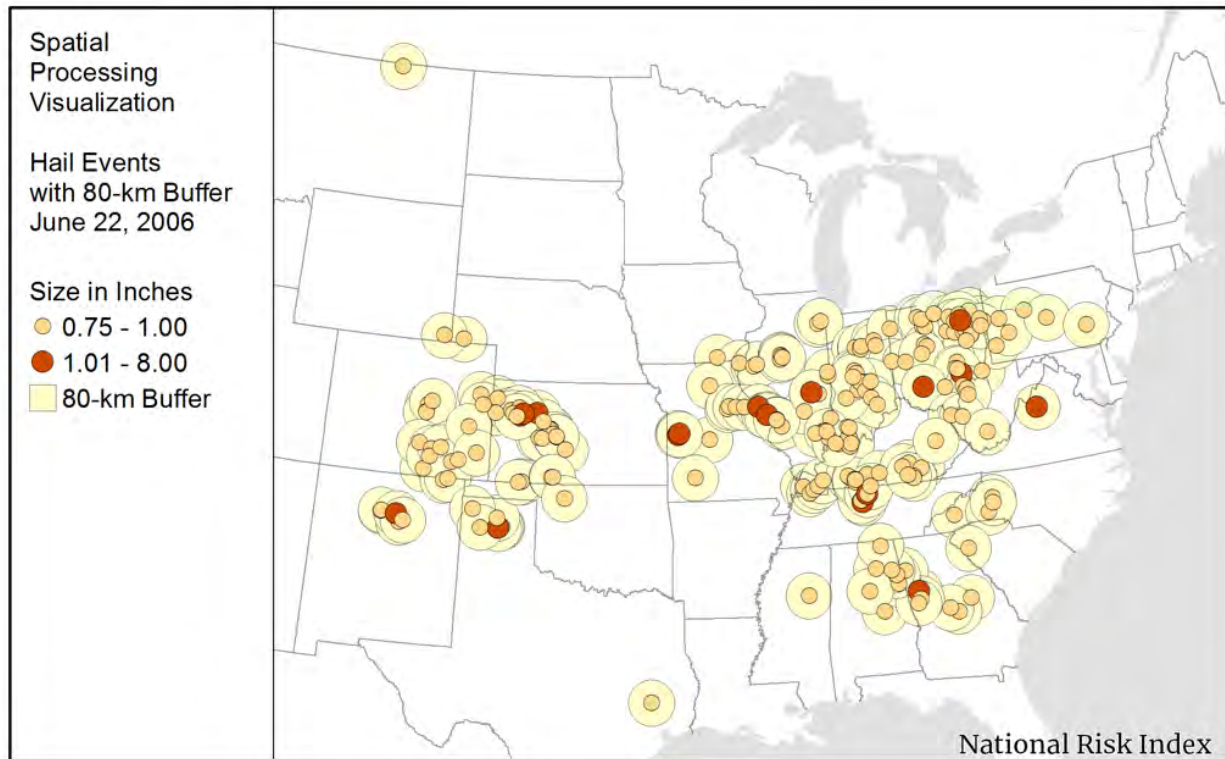


**Figure 40: Map of Hail Source Data Points**

## 11.2. Hail Spatial Data Processing

The source data includes fields for two sets of coordinates: a start and an end. This is mainly because the data share their format with the data for tornadoes. Most Hail events only have start coordinates (or the end coordinates match the start coordinates), so the points are projected from these coordinates. Any events outside of the period of record were filtered out. Additionally, smaller Hail size events were filtered out. Due to changes in National Weather Service standards for reportable Hail, events before 2010 were required to meet a Hail size threshold of 0.75-in, and those after 2010 must meet a 1.0-in size threshold. Anything below the threshold was not used in the analysis of Hail Expected Annual Loss. An 80-km buffer was created from the remaining points. The resulting Hail event polygons were used to estimate annualized frequency at the Census block level.

The buffer is not an attempt to represent the area of impact by a Hail event, but rather an effort to estimate the area where Hail may have been present. Hail reporting can be influenced by urban bias, meaning that a Hail event in a populated area is more likely to be reported than if the same event had occurred in a rural area. Additionally, the position of the Hail event reported in the source data is not guaranteed to be the actual location of the occurrence but may be the location of a nearby weather station or reporting center. The use of the 80-km buffering allows the reported location to be spread across a broader area (see [Figure 41](#)).



**Figure 41: Map of Buffered Hail Points**

#### 11.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Hail can occur almost anywhere under the right conditions, so all counties were deemed possible for Hail occurrence.

### 11.3. Hail Non-Spatial Data Processing

#### 11.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Hail in events is initially computed as the number of distinct Hail event polygons that intersect a 49-by-49-km fishnet grid cell. Buffering the Hail points and using the fishnet grid to count historical Hail events serves to spatially spread the influence of past Hail events to nearby areas that may also be susceptible to Hail but have not experienced Hail as frequently. However, using these methods can overestimate Hail frequency. To adjust for this, a national scaling factor is calculated (see [Equation 55](#)).

**Equation 55: National Scaling Factor for Hail Event Count**

$$NatlScalingFactor_{HAIL} = \frac{EventCount_{HAIL_{Ntl}}}{\sum FishnetIntsctCount_{HAIL_{Ntl}}}$$

where:

$NatlScalingFactor_{HAIL}$  is the Hail scaling factor applied to the fishnet grid cell event count.

$EventCount_{HAIL_{Ntl}}$  is the count of distinct Hail events that have occurred in the United States

$\Sigma FishnetIntsctCount_{HAIL_{Ntl}}$  is the summed total of all Hail event polygon-fishnet grid cell intersections in the United States

The scaling factor is then applied to the fishnet grid Hail event count (see [Equation 56](#)).

#### Equation 56: Scaled Hail Event Fishnet Count

$$ScaledEventCount_{HAIL_{Fish}} = EventCount_{HAIL_{Fish}} \times NatlScalingFactor_{HAIL}$$

where:

$ScaledEventCount_{HAIL_{Fish}}$  is the scaled count of Hail events within a fishnet grid cell (in events per year).

$EventCount_{HAIL_{Fish}}$  is the count of Hail event polygons that intersect a 49-by-49-km fishnet grid cell.

$NatlScalingFactor_{HAIL}$  is the Hail scaling factor to be applied to the fishnet grid cell event count.

The Census block Hail event count is then computed as the scaled event count of the fishnet grid cell that encompasses the Census block or, if the Census block intersects multiple fishnet grid cells, an area-weighted count of the cells that intersect the Census block (see [Appendix D – Fishnet Occurrence Count](#)). This scaled count is then used to compute Hail event annualized frequency.

Historical event counts are also supplied at the Census tract and county levels as the scaled, area-weighted count of Hail events intersecting fishnet grid cells that intersect the Census tract and county, respectively.

### 11.3.2. PERIOD OF RECORD

Hail data between Jan. 01, 1986 and Dec. 31, 2023 are analyzed, so the period of record for which Hail data are utilized is 38.0 years.

## 11.4. Hail Exposure

Because Hail can occur anywhere, the entire building, population, and agriculture value of a Census block, Census tract, and county are considered exposed to Hail. Population equivalence, which is used in select Expected Annual Loss calculations, is calculated by multiplying population by the Value of Statistical Life (\$13.7 million per person).

## 11.5. Hail Annualized Frequency

The number of recorded Hail occurrences in events each year over the period of record (38.0 years) is used to estimate the annualized frequency of Hail events in an area. This annualized frequency is calculated at the Census block level, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the Hail event polygons created from the source data (as described in [Section 11.2 Hail Spatial Data Processing](#)), as well as their corresponding computed duration days from the pre-processing of the data. The Census block Hail event count computed using the scaled event counts of the fishnet grid cells intersecting the Census block is divided by the period of record to compute frequency as in [Equation 57](#).

### Equation 57: Census Block Hail Annualized Frequency

$$Freq_{HAIL_{CB}} = \frac{ScaledEventCount_{HAIL_{Fish}}}{PeriodRecord_{HAIL}}$$

where:

$Freq_{HAIL_{CB}}$  is the annualized frequency of Hail events determined for a specific Census block (events per year).

$ScaledEventCount_{HAIL_{Fish}}$  is the scaled count of Hail events calculated for the Census block.

$PeriodRecord_{HAIL}$  is the period of record for Hail (38.0 years).

#### 11.5.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Hail events at the Census tract and county level by the period of record, as the event count for annualized frequency is a fishnet area-weighted event count including Hail events that may have impacted the surrounding area but not the county or Census tract itself. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 58](#).

### Equation 58: Census Tract and County Area-Weighted Hail Annualized Frequency Aggregation

$$Freq_{HAIL_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{HAIL_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{HAIL_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{HAIL_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:



$Freq_{HAIL_{CT}}$  is the area-weighted Hail annualized frequency calculated for a specific Census tract (events per year).

$Freq_{HAIL_{CB}}$  is the annualized frequency of Hail events determined for a specific Census block (events per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

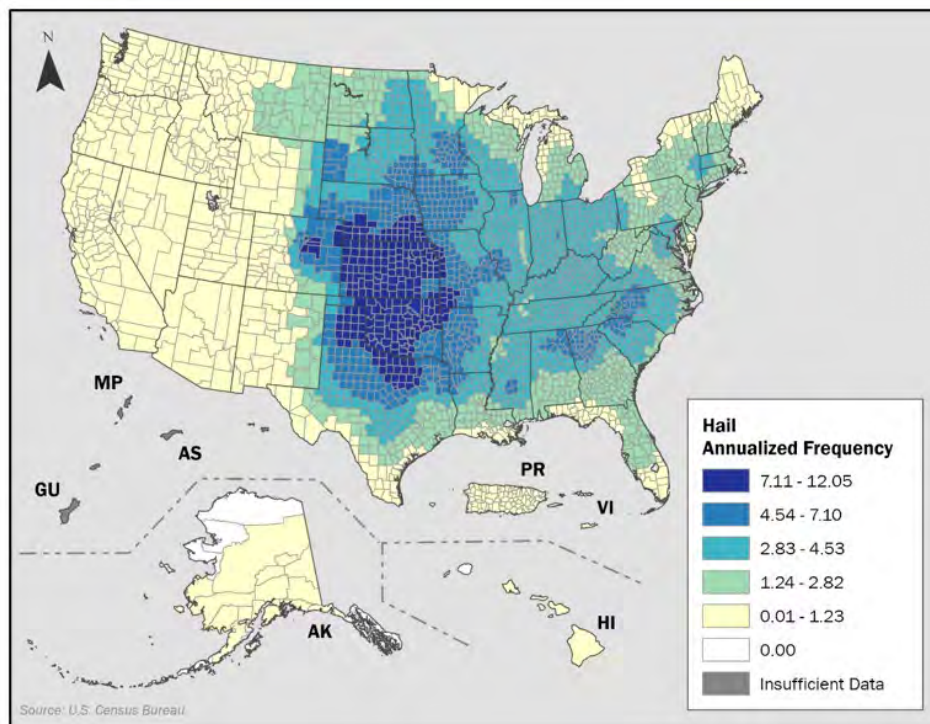
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{HAIL_{Co}}$  is the area-weighted Hail annualized frequency calculated for a specific county (events per year).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 42](#) displays Hail annualized frequency at the county level.



**Figure 42: Hail Annualized Frequency by County**

## 11.6. Hail Historic Loss Ratio

The Hail Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Hail event, or the average rate of loss associated with the occurrence of a Hail event. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Hail hazard type.

Loss data are provided by SHEL DUS<sup>76</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHEL DUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. One peril type is mapped to the hazard Hail (see [Table 39](#)). Native records of Hail storms that caused loss over more than one day (such as those that occurred overnight) have their loss assigned to the first day, and all records are aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 39: Hail Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Hail                          | 29,605                             | 20,134                               |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is a county-level value that represents the dollar value of the total building value or the entire population of a county as recorded in Hazus 6.0, or the total crop and livestock value of the county as estimated in the United States Department of Agriculture 2017 Census of Agriculture data. The Loss Ratio per Basis for each SHEL DUS-documented event and each consequence type (building, population, and agriculture) is calculated using [Equation 59](#).

**Equation 59: Loss Ratio per Basis Calculation for a Single Hail Event**

$$LRB_{HAILCoCnsqType} = \frac{LOSS_{HAILCoCnsqType}}{HLRExposure_{CoCnsqType}}$$

where:

$LRB_{HAILCoCnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Hail event-day. Calculation is performed for each consequence type (building, population, and agriculture).

$LOSS_{HAILCoCnsqType}$  is the loss (by consequence type) experienced from the Hail event documented to have occurred in the county (in dollars).

<sup>76</sup> For Hail loss information, SHEL DUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.



$HLRExposure_{CoCnsqType}$  is the total value (by consequence type) of the county estimated to have been exposed to the Hail event (in dollars).

Hail events can occur with a high frequency in areas, but often result in no recorded loss to buildings, population, or agriculture. SHELDTUS does not record events in which no loss occurred, so a number of zero-loss events are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELDTUS period of record (1996 through 2023). For Hail, the historical event count is extracted using an intersection between the Census blocks and the Hail event polygons for the years 1986-2023. An annual rate is calculated as the event count divided by the event polygon period of record of 38.0 years, and this rate is multiplied by the SHELDTUS period of record of 28 years to estimate a historical event count for the appropriate time range.

If the number of Hail event records from SHELDTUS is less than the scaled event count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

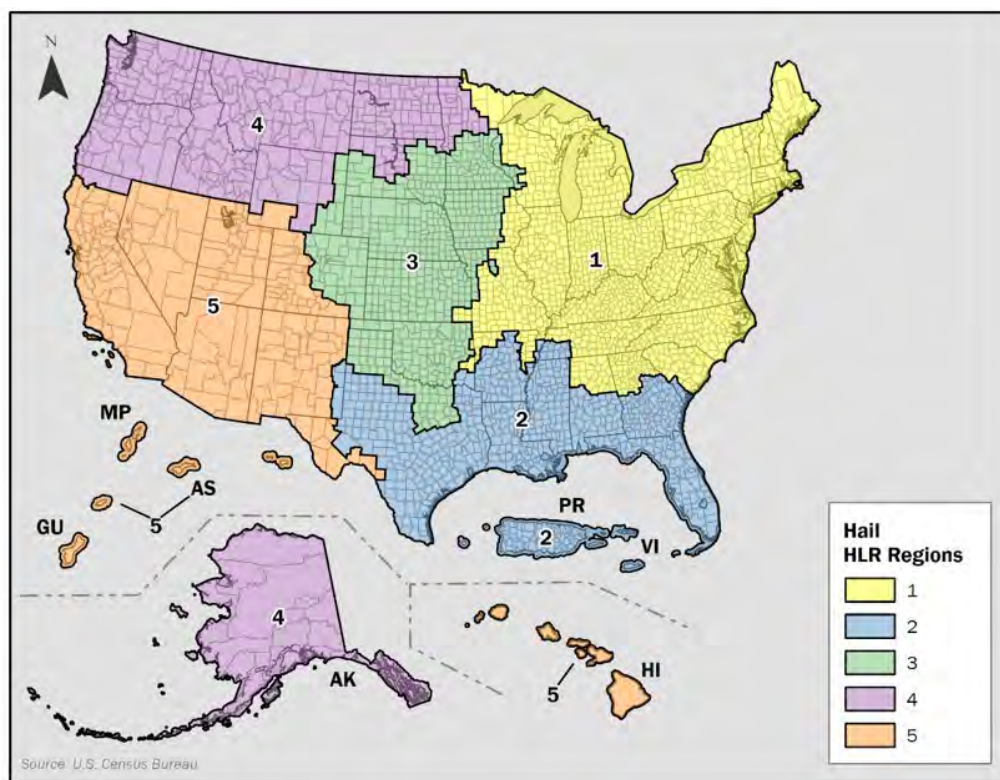
After the Loss Ratios per Basis are calculated for all Hail events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Hail.

**Table 40: Hail Historic Loss Ratio Bayesian Weighting Level Sets**

| <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>             | <b>Set Refinement Filter</b>                        | <b>Set Utilization Threshold (Set Must Have)</b> |
|-------------------------|---------------|-----------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A2            | Neighbor Counties (Touch & 115km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A3            | Neighbor Counties (Touch & 208km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A4            | Hail Regions                      | N/A                                                 | N/A                                              |
|                         | A5            | National                          | N/A                                                 | N/A                                              |
| Population              | P1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P4            | Hail Regions                      | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P5            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

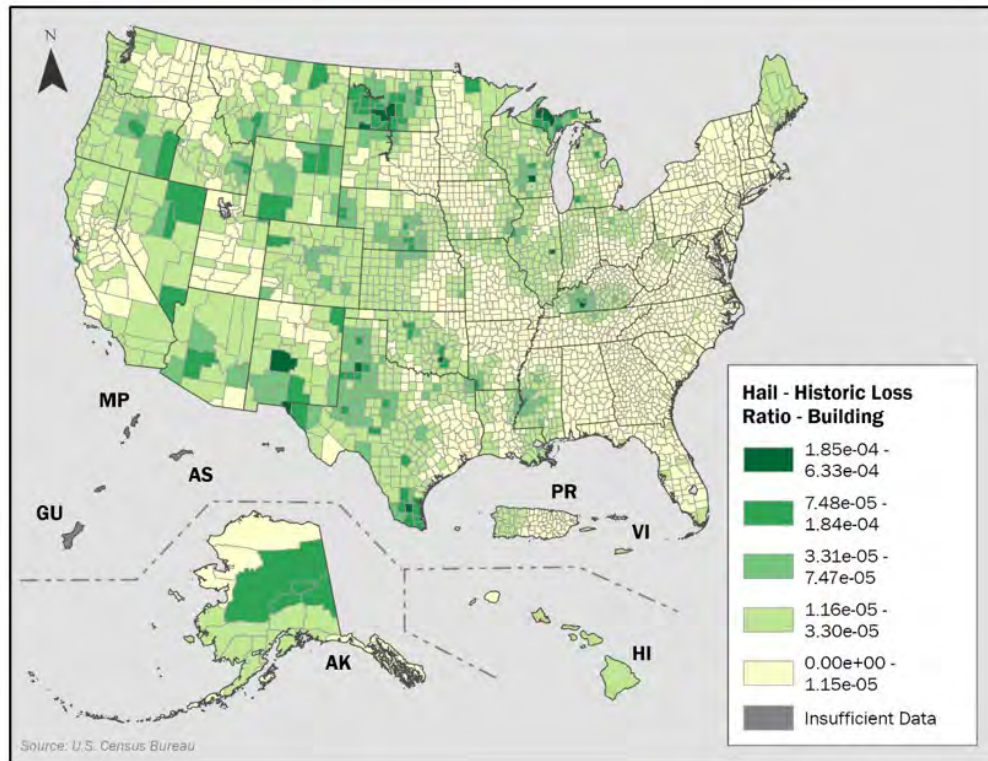
| Consequence Type | Set ID | Bayesian Level                    | Set Refinement Filter                               | Set Utilization Threshold (Set Must Have) |
|------------------|--------|-----------------------------------|-----------------------------------------------------|-------------------------------------------|
| Building         | B1     | County                            | N/A                                                 | 25 Loss Ratio per Basis records           |
| Building         | B2     | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records           |
| Building         | B3     | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records           |
| Building         | B4     | Hail Regions                      | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                       |
| Building         | B5     | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                       |

[Figure 43](#) displays the hazard-specific regions used in Loss Ratio per Basis sets for Historic Loss Ratio calculations.

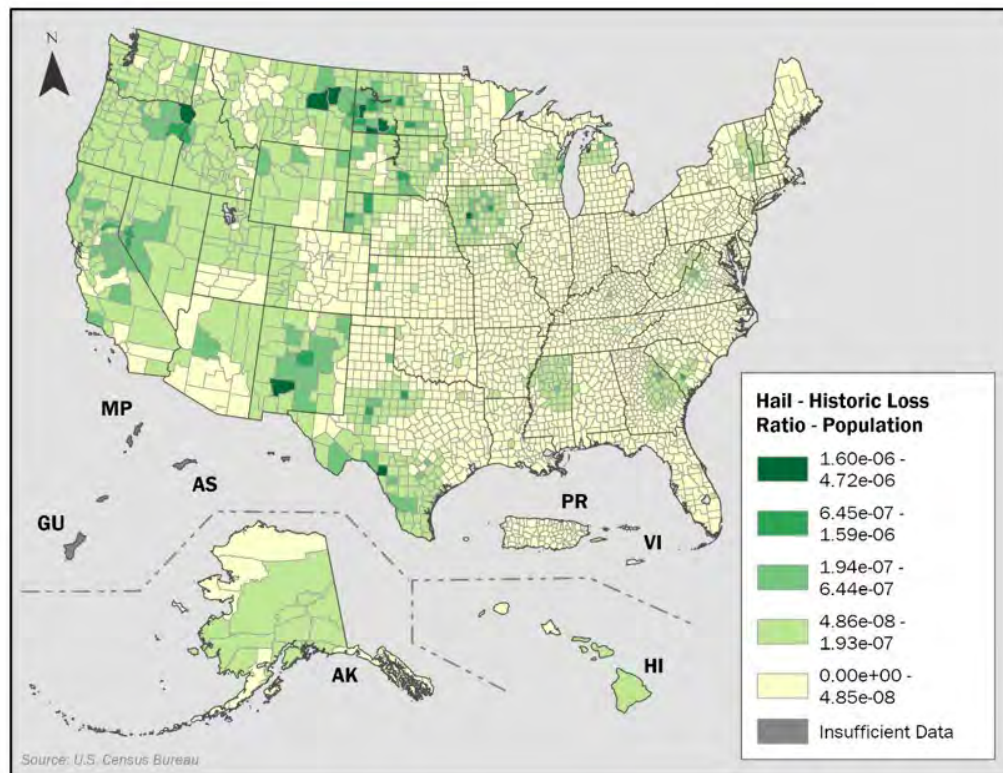


**Figure 43: Hail-specific Historic Loss Ratio Region**

Loss Ratio per Basis set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Hail (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 44](#), [Figure 45](#), and [Figure 46](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Hail. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

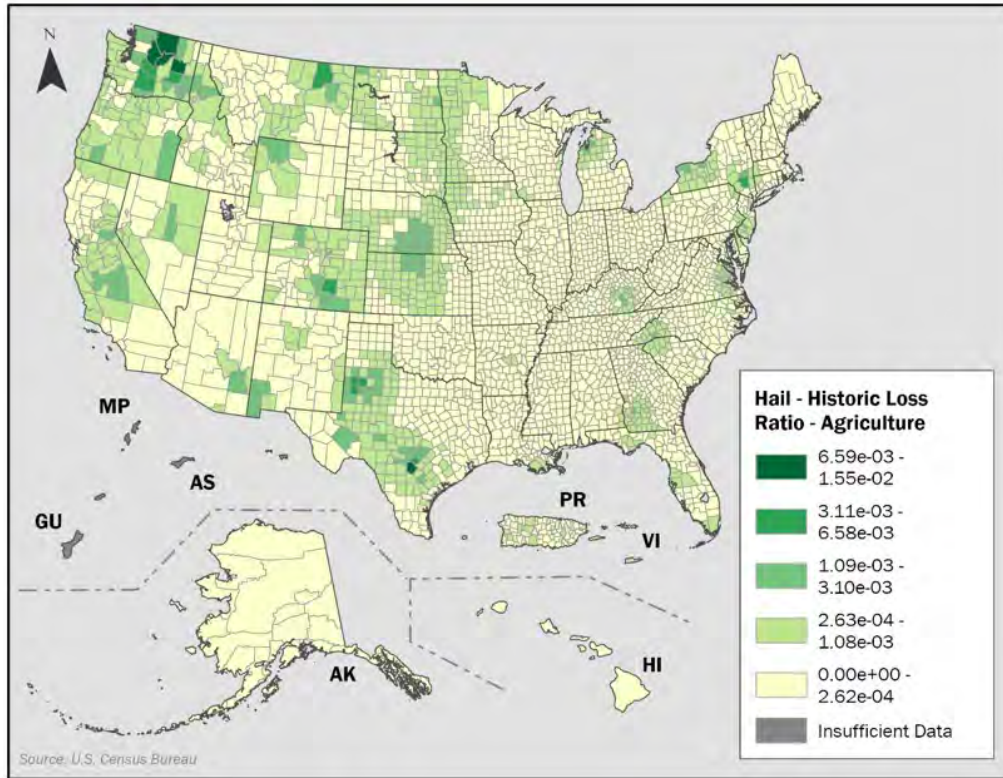


**Figure 44: Hail Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 45: Hail Bayesian-Adjusted Historic Loss Ratio – Population**





**Figure 46: Hail Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

### 11.7. Hail Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level (see [Equation 60](#)). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

**Equation 60: Census Block Expected Annual Loss to Hail**

$$EAL_{HAILCB_{Bldg}} = Exposure_{HAILCB_{Bldg}} \times Freq_{HAILCB} \times HLR_{HAILCB_{Bldg}}$$

$$EAL_{HAILCB_{Pop}} = Exposure_{HAILCB_{Pop}} \times Freq_{HAILCB} \times HLR_{HAILCB_{Pop}}$$

$$EAL_{HAILCB_{Ag}} = Exposure_{HAILCB_{Ag}} \times Freq_{HAILCB} \times HLR_{HAILCB_{Ag}}$$

where:

$EAL_{HAILCB_{Bldg}}$  is the building Expected Annual Loss due to Hail occurrences for a specific Census block (in dollars).

|                           |                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| $Exposure_{HAILCB\ Bldg}$ | is the building value exposed to Hail occurrences in the Census block (in dollars).                                  |
| $Freq_{HAILCB}$           | is the Hail annualized frequency calculated for the Census block (events per year).                                  |
| $HLR_{HAILCB\ Bldg}$      | is the Bayesian-adjusted building Historic Loss Ratio for Hail for the Census block.                                 |
| $EAL_{HAILCB\ Pop}$       | is the population equivalence Expected Annual Loss due to Hail occurrences for a specific Census block (in dollars). |
| $Exposure_{HAILCB\ Pop}$  | is the population equivalence value exposed to Hail occurrences in the Census block (in dollars).                    |
| $HLR_{HAILCB\ Pop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Hail for the Census block.                               |
| $EAL_{HAILCB\ Ag}$        | is the agriculture Expected Annual Loss due to Hail occurrences for a specific Census block (in dollars).            |
| $Exposure_{HAILCB\ Ag}$   | is the agriculture value exposed to Hail occurrences in the Census block (in dollars).                               |
| $HLR_{HAILCB\ Ag}$        | is the Bayesian-adjusted agriculture Historic Loss Ratio for Hail for the Census block.                              |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 61](#).

**Equation 61: Census Tract and County Expected Annual Loss to Hail**

$$EAL_{HAILCT} = \sum_{CB}^{CT} EAL_{HAILCB\ Bldg} + \sum_{CB}^{CT} EAL_{HAILCB\ Pop} + \sum_{CB}^{CT} EAL_{HAILCB\ Ag}$$

$$EAL_{HAILCo} = \sum_{CB}^{Co} EAL_{HAILCB\ Bldg} + \sum_{CB}^{Co} EAL_{HAILCB\ Pop} + \sum_{CB}^{Co} EAL_{HAILCB\ Ag}$$

where:

|                                     |                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $EAL_{HAILCT}$                      | is the total Expected Annual Loss due to Hail occurrences for a specific Census tract (in dollars).                         |
| $\sum_{CB}^{CT} EAL_{HAILCB\ Bldg}$ | is the summed building Expected Annual Loss due to Hail occurrences for all Census blocks in the Census tract (in dollars). |

$\sum_{CB}^{CT} EAL_{HAILCBPop}$  is the summed population equivalence Expected Annual Loss due to Hail occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{HAILCBAg}$  is the summed agriculture Expected Annual Loss due to Hail occurrences for all Census blocks in the Census tract (in dollars).

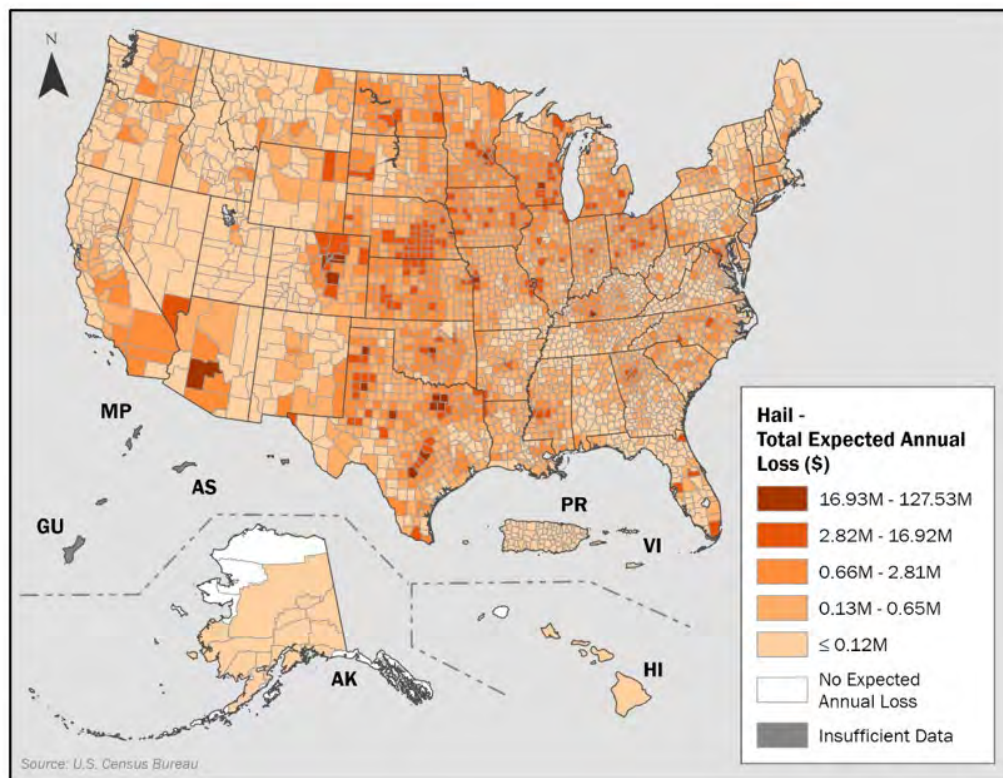
$EAL_{HAILCo}$  is the total Expected Annual Loss due to Hail occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{HAILCB Bldg}$  is the summed building Expected Annual Loss due to Hail occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{HAILCBPop}$  is the summed population equivalence Expected Annual Loss due to Hail events occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{HAILCB Ag}$  is the summed agriculture Expected Annual Loss due to Hail occurrences for all Census blocks in the county (in dollars).

[Figure 47](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Hail occurrences.



**Figure 47: Total Expected Annual Loss by County to Hail**

With the Hail total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Hail Expected Annual Loss for buildings by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Hail Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Hail Expected Annual Loss for agriculture by the total agriculture value of the community.

## **11.8. Hail Methodology Discussion**

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The Hail hazard currently does not incorporate the impacts of different diameter sizes. Hazard subtypes for varying diameter sizes would allow for more specificity in Historic Loss Ratio determination.



## 12. Heat Wave

A Heat Wave is a period of abnormally and uncomfortably hot and unusually humid weather typically lasting two or more days with temperatures outside the historical averages for a given area. The temperatures classified as Heat Wave are dependent on the location and defined by the local National Weather Service weather forecast office.

Similar to Cold Wave, the Heat Wave hazard modeling uses Mesonet data that are derived from the National Weather Service's warnings and advisories to provide impact areas for this hazard. SHELDUS data were used for historical losses for Heat Wave events and to generate the Historic Loss Ratios, with data from the Centers for Disease Control and Prevention's Compressed Mortality Database used to supplement the SHELDUS mortality data. These are overlaid with the exposure values provided in Hazus 6.0 for Building, Population, and Agriculture for the final Expected Annual Loss calculations. The Heat Wave hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 12.1. Heat Wave Spatial Source Data

**Historical Occurrence Compiling Source:** [Iowa State University, Iowa Environmental Mesonet](#)<sup>77</sup>

The National Weather Service is continuously issuing weather alerts based on current weather conditions.<sup>78</sup> Each alert is coded by type and significance and can conceptually serve as documentation of the potential for weather event activities in a specific area. Archived National Weather Service alerts are aggregated, continuously updated, and made available for download in shapefile format by Iowa State University's Iowa Environmental Mesonet. Data include geometry for each alert's issued area and attributes related to each alert's severity and phenomena type. Weather alerts are also timestamped with the time of issuance and the time of expiration. A table describing this dataset's attributes can be found in [Appendix C – Mesonet-National Weather Service Weather Event Attribute Description](#).

The geometry shape for each alert record represents the geographic area for which the National Weather Service alert applied. However, the Mesonet shapes are simplified versions of the more detailed National Weather Service Public Forecast Zone shape originally associated with the alert record. Because the Mesonet tabular data still retains the reference ID for the National Weather Service Public Forecast Zone, it is used to relate to the zone associated with each alert record.

The National Weather Service Public Forecast Zones can be downloaded in shapefile format<sup>79</sup> and represent the codified areas for which weather alerts are issued by National Weather Service. The

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<sup>77</sup> Department of Agronomy, Iowa State University. (2025). *Iowa Environmental Mesonet* [online database]. Retrieved from <https://mesonet.agron.iastate.edu/request/gis/watchwarn.phtml>.

<sup>78</sup> NWS, NOAA. (2018). *Active Alerts* [online dataset]. Retrieved from <https://www.weather.gov/alerts>.

<sup>79</sup> NWS, NOAA. (2022). *NWS Public Forecast Zones* [online dataset]. Retrieved from <https://www.weather.gov/gis/PublicZones>.

Public Forecast Zones shape definitions are predominantly derived from county boundaries. While the Public Forecast Zone boundaries are more refined than those substituted into the Mesonet data, they are not at the same resolution as the current county boundaries derived from Census blocks.

Utilizing the Public Forecast Zone shapefile in conjunction with the Public Forecast Zone – County Correlation file,<sup>80</sup> a determination was made as to which Public Forecast Zones have single-county coverage and which are either sub-county zones or made of portions of multiple counties. For perspective, the following approximate distributions of forecast zone composition were found:

- 70% of the zones are single-county coverage.
- 20% are cases where a single county is subdivided into multiple zones.
- 10% are zones that breach parts of multiple contiguous counties.

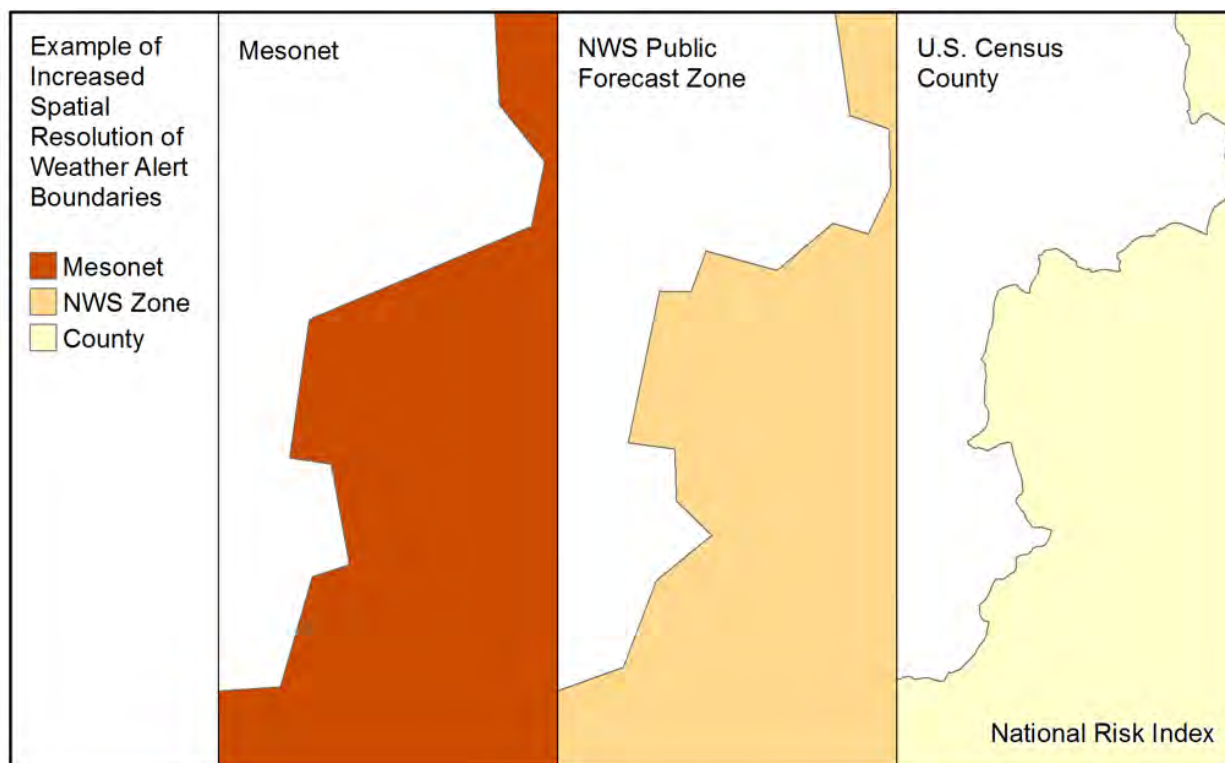
For those Forecast Zones covering a single county, the United States Census Topologically Integrated Geographic Encoding and Referencing 2021 county boundaries (Topologically Integrated Geographic Encoding and Referencing 2024 boundaries for Connecticut) are substituted.

Another aspect of the National Weather Service Public Forecast Zones is that they can and have changed over time. In the Mesonet data (2005 through 2024), there are many distinct Forecast Zones referenced that do not exist in the current National Weather Service Public Forecast Zone shapefile. This occurs when a National Weather Service Public Forecast Zone has been modified in shape, renamed, and/or “retired” from use.

The National Weather Service maintains a downloadable Change History log of the various changes in Forecast Zone areas since 1997. This text file does not contain the pre- nor post-shape of the altered forecast zone. Archived versions of these changes are likely available via contact with National Weather Service, but the effort to match the National Weather Service issued alert record to the version-controlled shape representation of the forecast zone at the time of alert issue is beyond the scope of the processing effort, though a Mesonet representative was contacted to see if Forecast Zone shapes associated with each year of alert data had been archived. Unfortunately, no such archival information is available. For cases where the more refined National Weather Service Forecast Zone shape is unavailable, the simplified Mesonet boundary version shape is used. See [Figure 48](#) for an example of the differences in the spatial resolution of weather alert boundaries.

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<sup>80</sup> NWS, NOAA. (2022). *Zone-County Correlation File* [online dataset]. Retrieved from <https://www.weather.gov/gis/ZoneCounty>.



**Figure 48: Three Boundary Definitions: Mesonet, Forecast Zone, and United States Census County**

## 12.2. Heat Wave Spatial Processing

Heat Wave exposure and annualized frequency values are determined by taking the spatial intersection of the National Weather Service Heat Wave event polygons and Census block geometries. Before this analysis could occur, the raw National Weather Service and Census data were prepared using the procedures outlined below.

Heat Wave weather event alerts were first extracted from the dataset based on the Valid Time Extent Code significance code (SIG field) and the phenomena code (PHENOM or TYPE field) values. Only Warning alerts (SIG = 'W') and Advisory alerts (SIG = 'Y') of the Phenomena type "Excessive Heat" (EH) or "Heat" (HT) were considered Heat Wave occurrences (see [Table 41](#)); Watch alerts (SIG = 'A') are excluded from this analysis.

To remove unintended error in spatial results due to the use of the simplified event area shapes contained in the Mesonet data, event areas with a higher resolution version were substituted. This substitution used the National Weather Service Public Forecast Zone shape associated with the alert record or, in cases where the forecast zone is for a single county, a better resolution version of the county boundary area.

**Table 41: Original Mesonet Heat Wave Records**

| <i>WFO</i> | <i>ISSUED</i>         | <i>EXPIRED</i>        | <i>PHENOM</i> | <i>SIG</i> | <i>NWS_UGC</i> | <i>AREA_KM2</i> |
|------------|-----------------------|-----------------------|---------------|------------|----------------|-----------------|
| PSR        | June 04, 2017 6:00 PM | June 08, 2017 3:00 AM | EH            | W          | AZZ554         | 583.5392        |
| MAF        | June 17, 2017 5:00 PM | June 18, 2017 2:00 AM | EH            | W          | TXZ045         | 3894.574        |
| VEF        | June 17, 2017 6:00 PM | June 27, 2017 4:00 AM | EH            | W          | NVZ017         | 7555.405        |

Heat Wave occurrences are measured in event-days as this more accurately represents the variability of Heat Wave event duration. To capture this, each native alert record with a duration greater than a single day was replaced with multiple records, one for each day of the original record's duration.

If a Heat Wave event's duration on any given day is less than 6 hours, then the event was assigned to the day having the greatest duration of the event. This handled cases where the event occurs in the late evening but endures for a greater length of time on the next calendar day than on the day the alert was issued.

For cases where the event duration is longer, the following business logic was used: If a weather event's duration is greater than 6 hours, assign the event to all days on which 6 or more hours occur. For example, if a 14-hour weather event was issued for 2 AM until 6 PM on July 1, then the event would be assigned to July 1. If the alert was issued from 11 PM on July 1 to 1 PM on July 2, then the event would be assigned to only July 2. If the alert was issued from 6 PM on July 1 to 9 AM on July 2, then the event would be assigned to both July 1 and July 2. To illustrate this concept, the Heat Wave events in [Table 42](#) are expanded to create the Heat Wave event-day records in [Table 43](#).

Additionally, there are some data quality issues with the Mesonet data. For example, some warnings have an expiration date that is prior to the issue date. In these cases, a single record was used and assigned the issue date.

**Table 42: Sample Heat Wave Data after Zone Shape Re-Sourcing**

| <i>HeatwaveID</i> | <i>WFO</i> | <i>Issued</i>         | <i>Expired</i>        | <i>PHENOM</i> | <i>SIG</i> | <i>NWS_UGC</i> | <i>AreaKm2</i> | <i>NewShapeSource</i> |
|-------------------|------------|-----------------------|-----------------------|---------------|------------|----------------|----------------|-----------------------|
| 47081             | PSR        | Jun. 04, 2017 6:00 PM | Jun. 08, 2017 3:00 AM | EH            | W          | AZZ554         | 577.2512       | NWS Forecast Zone     |
| 51174             | TWC        | Jun. 04, 2017 6:00 PM | Jun. 06, 2017 9:45 PM | EH            | W          | AZZ504         | 5763.599       | NWS Forecast Zone     |

**Table 43: Sample Data from the Heat Wave Date Expansion Table**

| <i>HeatwaveDate<br/>ExpansionID</i> | <i>HeatwaveID</i> | <i>Issued</i>                | <i>Expired</i>               | <i>DateType</i>            | <i>HeatwaveHours</i> |
|-------------------------------------|-------------------|------------------------------|------------------------------|----------------------------|----------------------|
| 2030                                | 47081             | Jun. 04,<br>2017 6:00<br>PM  | Jun. 05,<br>2017<br>12:00 AM | Expanded Dates - Issued    | 6                    |
| 2032                                | 47081             | Jun. 05,<br>2017<br>12:00 AM | Jun. 06,<br>2017<br>12:00 AM | Expanded Dates - New Dates | 24                   |
| 2031                                | 47081             | Jun. 06,<br>2017<br>12:00 AM | Jun. 07,<br>2017<br>12:00 AM | Expanded Dates - New Dates | 24                   |
| 2034                                | 47081             | Jun. 07,<br>2017<br>12:00 AM | Jun. 08,<br>2017 2:00<br>AM  | Expanded Dates - New Dates | 24                   |
| 1                                   | 51174             | Jun. 07,<br>2017 6:00<br>PM  | Jun. 07,<br>2017 6:00<br>PM  | Expired before Issued      | 0                    |

To avoid overestimating the area of influence a “single” distinct weather event has due to multiple National Weather Service alerts being issued for that same weather event, a process to combine all Heat Wave event areas occurring on the same day (Year, Month, Day specific) into one representative event shape was performed. This process results in a single event impact area shape for each day a Heat Wave event occurred. These Heat Wave event-day polygons could then be intersected with the Census block polygons to determine Heat Wave exposure and annualized frequency.

### 12.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Heat Waves can occur almost anywhere in the United States as the definition of a Heat Wave is locally defined by the area’s weather forecast office. For example, a forecast office in Texas may define a Heat Wave differently than a forecast office in New York. Therefore, all counties were deemed possible for Heat Wave occurrence.<sup>81</sup>

<sup>81</sup> On June 2, 2025, the Fairbanks and Juneau, AK Weather Forecast Offices replaced the use of Special Weather Statements with Heat Advisories ([https://www.weather.gov/media/notification/pdf\\_2025/scn25-39\\_heat\\_advisory\\_public\\_notice\\_AR.pdf](https://www.weather.gov/media/notification/pdf_2025/scn25-39_heat_advisory_public_notice_AR.pdf)). Prior to this date, Alaska had no Heat Advisory system in place and was therefore excluded from this analysis due to having insufficient data. Alaska’s first ever Heat Advisory was issued on June 15, 2025.

### 12.3. Heat Wave Non-Spatial Data Processing

#### 12.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Heat Wave, in event-days, is computed as the area-weighted sum of distinct Heat Wave event-day polygons that intersect a Census block. This count uses the same Heat Wave expansion Census block intersection table used to calculate exposure and is used to compute annualized frequency at the Census block level.

Historical event-day counts are also supplied at the Census tract and county levels as the number of distinct Heat Wave event-day polygons that intersect the Census tract and county, respectively.

#### 12.3.2. PERIOD OF RECORD

The National Weather Service’s system of recording watch, warnings, and advisories (WWA) in the 1990s and early 2000s was too difficult for automated processing. In 2005 the Valid Time Extent Code system was implemented, which allowed for the easy automated parsing of alert data. Therefore, National Weather Service weather events data were downloaded for 2005 through 2024. The date range is Nov. 12, 2005 to Nov. 12, 2024, so the period of record for which Heat Wave data are utilized is 19.0 years.

### 12.4. Heat Wave Exposure

To identify areas of exposure, the Heat Wave event-day polygons (also referred to as Heat Wave Date Expansions to acknowledge the spatiotemporal processing described in [Section 12.2 Heat Wave Spatial Processing](#)) are intersected with the Census block polygons within the processing database. The resulting table contains the Heat Wave event-day’s unique identifier, Census block number, and the intersected area in square kilometers (see [Table 44](#)).

**Table 44: Sample Data from the Heat Wave Expansion Census Block Intersection Table**

| <i>HeatwaveDateExpansionID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|--------------------------------|--------------------|---------------------------|
| 53297                          | 040131167132017    | 0.080384                  |
| 53297                          | 040131167133000    | 0.313492                  |
| 53297                          | 040131167133001    | 0.032176                  |

To determine exposure value, the average coverage of a Heat Wave event-day is found by summing the intersected areas for all Heat Wave event-day polygons that intersected the Census block and dividing this sum by the number of intersecting event-day polygons. This is multiplied by the developed area building value density, developed area population density, and the agriculture area value density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 62](#)). The densities for each consequence type in the Census block have been calculated by dividing the Census block total exposure values (building value and



population as recorded in Hazus 6.0) by the developed or agriculture land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

### Equation 62: Census Block Heat Wave Exposure

$$Exposure_{HWAV_{CB_{Bldg}}} = \frac{\sum IntsctArea_{HWAV_{CB}}}{EventDayCount_{HWAV_{CB}}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{HWAV_{CB_{Pop}}} = \left( \frac{\sum IntsctArea_{HWAV_{CB}}}{EventDayCount_{HWAV_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

$$Exposure_{HWAV_{CB_{Ag}}} = \frac{\sum IntsctArea_{HWAV_{CB}}}{EventDayCount_{HWAV_{CB}}} \times AgValueDen_{CB}$$

where:

$Exposure_{HWAV_{CB_{Bldg}}}$  is the building value exposed to Heat Wave event-days in a specific Census block (in dollars).

$\sum IntsctArea_{HWAV_{CB}}$  is the sum of the intersected areas of past Heat Wave event-days with the Census block (in square kilometers).

$EventDayCount_{HWAV_{CB}}$  is the total number of Heat Wave event-day polygons that intersect the Census block.

$DevAreaDen_{CB_{Bldg}}$  is the developed area building value density of the Census block (in dollars per square kilometer).

$Exposure_{HWAV_{CB_{Pop}}}$  is the population equivalence value exposed to Heat Wave event-days in a specific Census block (in dollars).

$DevAreaDen_{CB_{Pop}}$  is the developed area population density of the Census block (in people per square kilometer).

$VSL$  Is the Value of Statistical Life (\$13.7 per person).

$Exposure_{HWAV_{CB_{Ag}}}$  Is the agriculture value exposed to Heat Wave event-days in a specific Census block (in dollars).

$AgValueDen_{CB}$  is the agriculture value density of the Census block (in dollars per square kilometer).

It should be noted that, in order for a Heat Wave event-day polygon's intersection with a Census block to be included, the area of the intersection must cover at least 5% of the Census block. This is a spatial modeling technique to correct for the small intersect "slivers" generated by differing versions of county boundary geometry being used.

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value and the Census of Agriculture-reported crop and livestock value for the Census block are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

#### 12.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 63](#)).<sup>82</sup>

**Equation 63: Census Tract and County Heat Wave Exposure Aggregations**

$$\begin{aligned}
 Exposure_{HWAV_{CT}Bldg} &= \sum_{CB}^{CT} Exposure_{HWAV_{CB}Bldg} \\
 Exposure_{HWAV_{Co}Bldg} &= \sum_{CB}^{Co} Exposure_{HWAV_{CB}Bldg} \\
 Exposure_{HWAV_{CT}Pop} &= \sum_{CB}^{CT} Exposure_{HWAV_{CB}Pop} \\
 Exposure_{HWAV_{Co}Pop} &= \sum_{CB}^{Co} Exposure_{HWAV_{CB}Pop} \\
 Exposure_{HWAV_{CT}Ag} &= \sum_{CB}^{CT} Exposure_{HWAV_{CB}Ag} \\
 Exposure_{HWAV_{Co}Ag} &= \sum_{CB}^{Co} Exposure_{HWAV_{CB}Ag}
 \end{aligned}$$

where:

<sup>82</sup> The exposure values associated with population will apply to both Injury and Fatality consequence subtypes. More details are provided in Section 5.4.4.

|                                           |                                                                                                                                    |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{HWAV_{CT}Bldg}$                | is the building value exposed to Heat Wave event-days in a specific Census tract (in dollars).                                     |
| $\sum_{CB}^{CT} Exposure_{HWAV_{CB}Bldg}$ | is the summed value of all buildings exposed to Heat Wave for each Census block within the Census tract (in dollars).              |
| $Exposure_{HWAV_{Co}Bldg}$                | is the building value exposed to Heat Wave event-days in a specific county (in dollars).                                           |
| $\sum_{CB}^{Co} Exposure_{HWAV_{CB}Bldg}$ | is the summed value of all buildings exposed to Heat Wave for each Census block within the county (in dollars).                    |
| $Exposure_{HWAV_{CT}Pop}$                 | is the population equivalence value exposed to Heat Wave event-days in a specific Census tract (in dollars).                       |
| $\sum_{CB}^{CT} Exposure_{HWAV_{CB}Pop}$  | is the summed value of all population equivalence exposed to Heat Wave for each Census block within the Census tract (in dollars). |
| $Exposure_{HWAV_{Co}Pop}$                 | is the population equivalence value exposed to Heat Wave event-days in a specific county (in dollars).                             |
| $\sum_{CB}^{Co} Exposure_{HWAV_{CB}Pop}$  | is the summed value of all population equivalence exposed to Heat Wave for each Census block within the county (in dollars).       |
| $Exposure_{HWAV_{CT}Ag}$                  | is the agriculture value exposed to Heat Wave event-days in a specific Census tract (in dollars).                                  |
| $\sum_{CB}^{CT} Exposure_{HWAV_{CB}Ag}$   | is the summed value of all agriculture areas exposed to Heat Wave for each Census block within the Census tract (in dollars).      |
| $Exposure_{HWAV_{Co}Ag}$                  | is the agriculture value exposed to Heat Wave event-days in a specific county (in dollars).                                        |
| $\sum_{CB}^{Co} Exposure_{HWAV_{CB}Ag}$   | is the summed value of all agriculture areas exposed to Heat Wave for each Census block within the county (in dollars).            |

## 12.5. Heat Wave Annualized Frequency

The number of recorded Heat Wave occurrences, in event-days, each year over the period of record (19.0 years) is used to estimate the annualized frequency of Heat Waves in an area. Because a Heat Wave event can last over several days or a single day, an event-day basis was used to estimate annualized frequency as this method better captures the variability in duration between events. The annualized frequency is calculated at the Census block level, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the same intersection between Heat Wave event-days (or Heat Wave Date Expansion) polygons and Census block polygons that were used to calculate exposure. The count of distinct Heat Wave event-day polygons intersecting each Census block is recorded and used to calculate the annualized frequency of Heat Wave event-days as in [Equation 64](#).

**Equation 64: Census Block Heat Wave Annualized Frequency**

$$Freq_{HWAV_{CB}} = \frac{EventDayCount_{HWAV_{CB}}}{PeriodRecord_{HWAV}}$$

where:

$Freq_{HWAV_{CB}}$  is the Heat Wave annualized frequency calculated for a specific Census block (event-days per year).

$EventDayCount_{HWAV_{CB}}$  is the number of Heat Wave event-days that intersect the Census block.

$PeriodRecord_{HWAV}$  is the period of record for Heat Wave (19.0 years).

### 12.5.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Heat Wave event-days at the Census tract and county level by the period of record, as the annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 65](#).

**Equation 65: Census Tract and County Area-Weighted Heat Wave Annualized Frequency Aggregation**

$$Freq_{HWAV_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{HWAV_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{HWAV_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{HWAV_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{HWAV_{CT}}$  is the area-weighted Heat Wave annualized frequency calculated for a specific Census tract (event-days per year).

$Freq_{HWAV_{CB}}$  is the Heat Wave annualized frequency calculated for a specific Census block (event-days per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

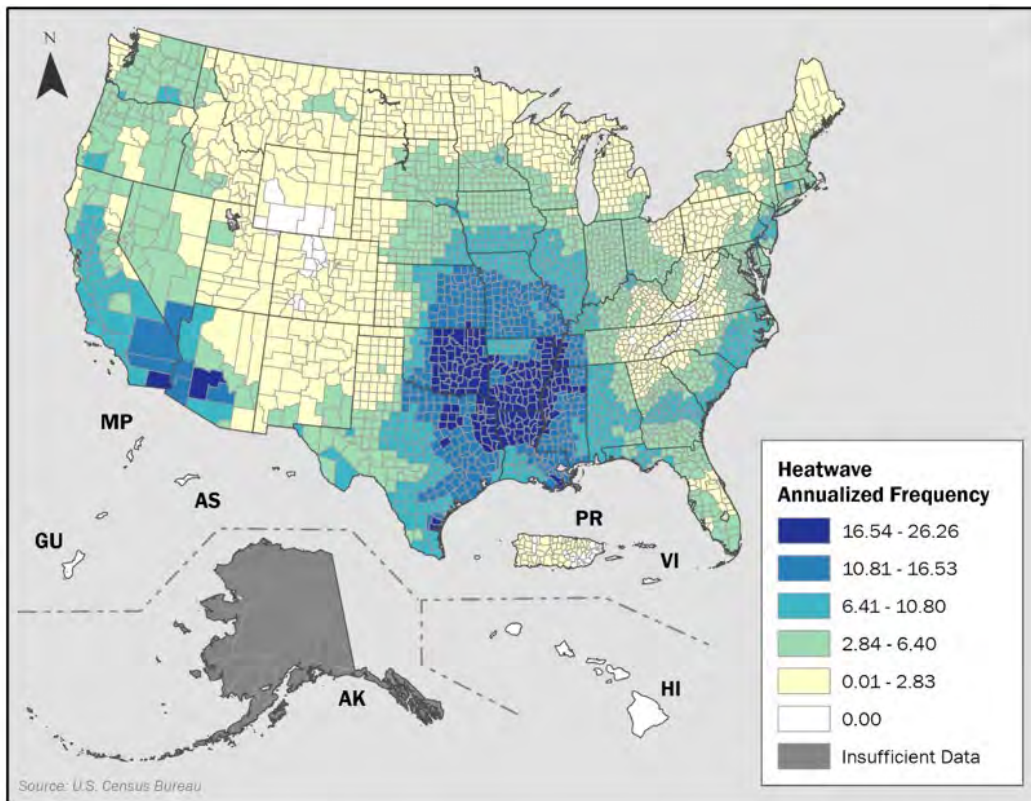
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{HWAV_{Co}}$  is the area-weighted Heat Wave annualized frequency calculated for a specific county (event-days per year).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 49](#) displays Heat Wave annualized frequency at the county level.



**Figure 49: Heat Wave Annualized Frequency by County**

## 12.6. Heat Wave Historic Loss Ratio

The Heat Wave Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Heat Wave event-day, or the average rate of loss associated with the occurrence of a Heat Wave event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Heat Wave hazard type.

Loss data are provided by SHELDDUS<sup>83</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELDDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. Two peril types are mapped to the hazard Heat Wave (see [Table 45](#)). These native records are expanded on an event-day basis (to a maximum of 31 event-days) and aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 45: Heat Wave Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELDDUS</i> | <i>Total SHELDDUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Heat                          | 3,445                              | 13,238                               |
| Heat Wave                     | 1                                  | 1                                    |

The Historic Loss Ratio exposure value used in the formula below is a county-level value, in dollars, which represents the total building value, the entire population, or the total agriculture value of a county. Population counts and building values are recorded in Hazus 6.0,<sup>84</sup> and agriculture value is sourced from the 2017 Census of Agriculture, with values adjusted to December 2024 United States Dollars. The Loss Ratio per Basis for each SHELDDUS-documented event-day and each consequence type (building, injury, fatality, and agriculture) is calculated using [Equation 66](#).

**Equation 66: Loss Ratio per Basis Calculation for a Single Heat Wave Event-Day**

$$LRB_{HWAV Co CnsqType} = \frac{Loss_{HWAV Co CnsqType}}{HLRExposure_{Co CnsqType}}$$

where:

$LRB_{HWAV Co CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Heat Wave event-day. Calculation is performed for each consequence type (building, injury, fatality, and agriculture).

<sup>83</sup> For Heat Wave loss information, SHELDDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

<sup>84</sup> These values define Historic Loss Ratio exposure for both the Injury and Fatality consequence subtypes for population.



$LOSS_{HWAVCoCnsqType}$  is the loss (by consequence type) experienced from the Heat Wave event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{CoCnsqType}$  is the total value (by consequence type) of the county estimated to have been exposed to the Heat Wave event-day (in dollars or people).

Heat Wave event-days can occur with a high frequency in areas but often result in no recorded loss to building value, population, or agriculture value. SHEL DUS does not record events in which no loss occurred, so zero-loss event-days are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHEL DUS period of record (1996 through 2023). For Heat Wave, the historical event-day count is extracted using the intersection between the Heat Wave event-day polygons from the years 2005-2024 and the Census block polygons used to calculate exposure and annualized frequency (see [Table 44](#)). An annual rate is calculated as the event-day count divided by the period of record of 19.0 years, and this rate is multiplied by the SHEL DUS period of record of 28 years to estimate a historical event-day count for the appropriate time range.

If the number of Heat Wave event-day records from SHEL DUS is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

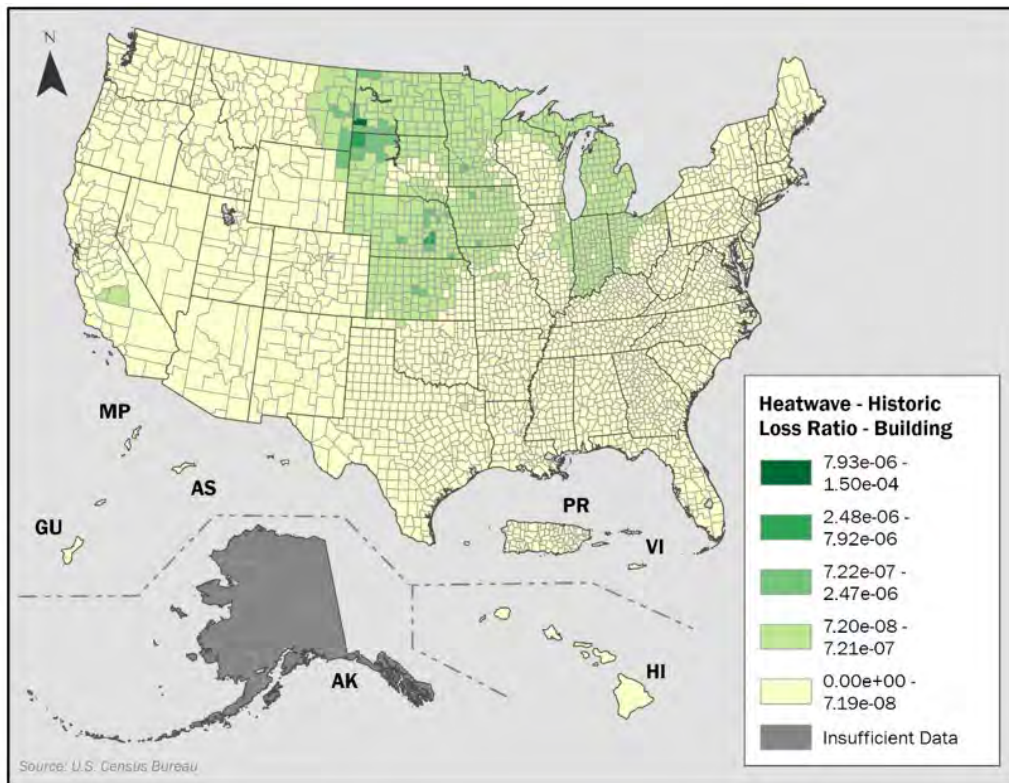
After the Loss Ratios per Basis are calculated for all Heat Wave events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Heat Wave.

**Table 46: Heat Wave Historic Loss Ratio Bayesian Weighting Level Sets**

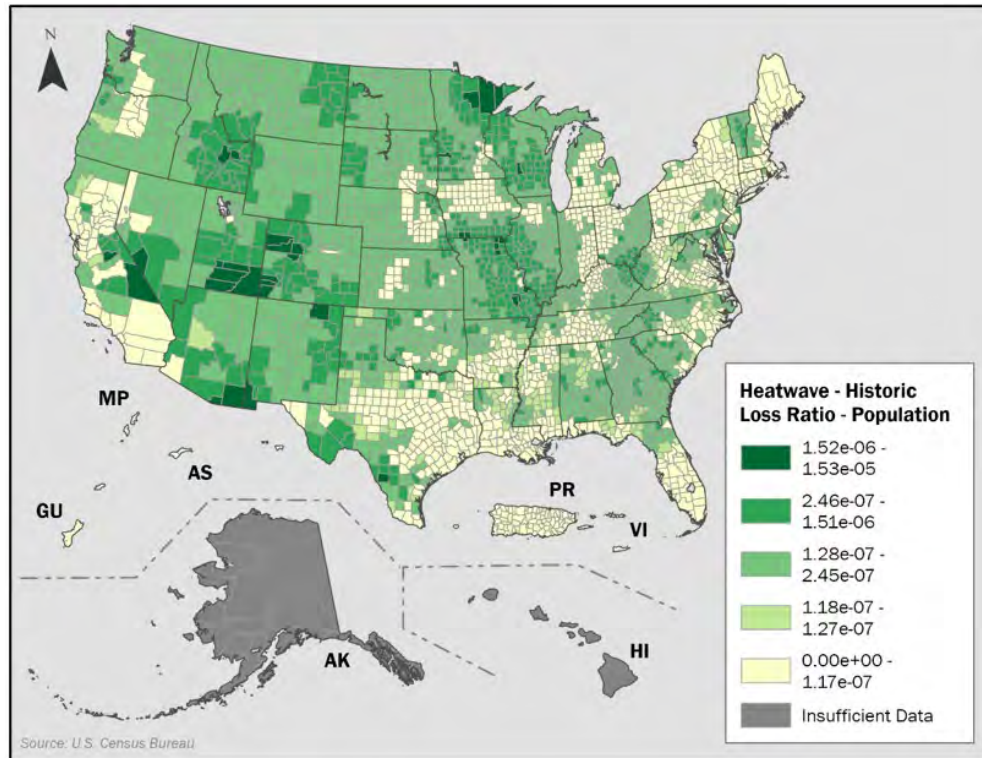
| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>             | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-----------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A2            | Neighbor Counties (Touch & 208km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A3            | Census Regions                    | N/A                                                 | N/A                                              |
|                         | A4            | National                          | N/A                                                 | N/A                                              |
| Population              | P1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Census Regions                    | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P4            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

| Consequence Type | Set ID | Bayesian Level                    | Set Refinement Filter                               | Set Utilization Threshold (Set Must Have) |
|------------------|--------|-----------------------------------|-----------------------------------------------------|-------------------------------------------|
| Building         | B1     | County                            | N/A                                                 | 25 Loss Ratio per Basis records           |
|                  | B2     | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records           |
|                  | B3     | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records           |
|                  | B4     | Census Regions                    | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                       |
|                  | B5     | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                       |

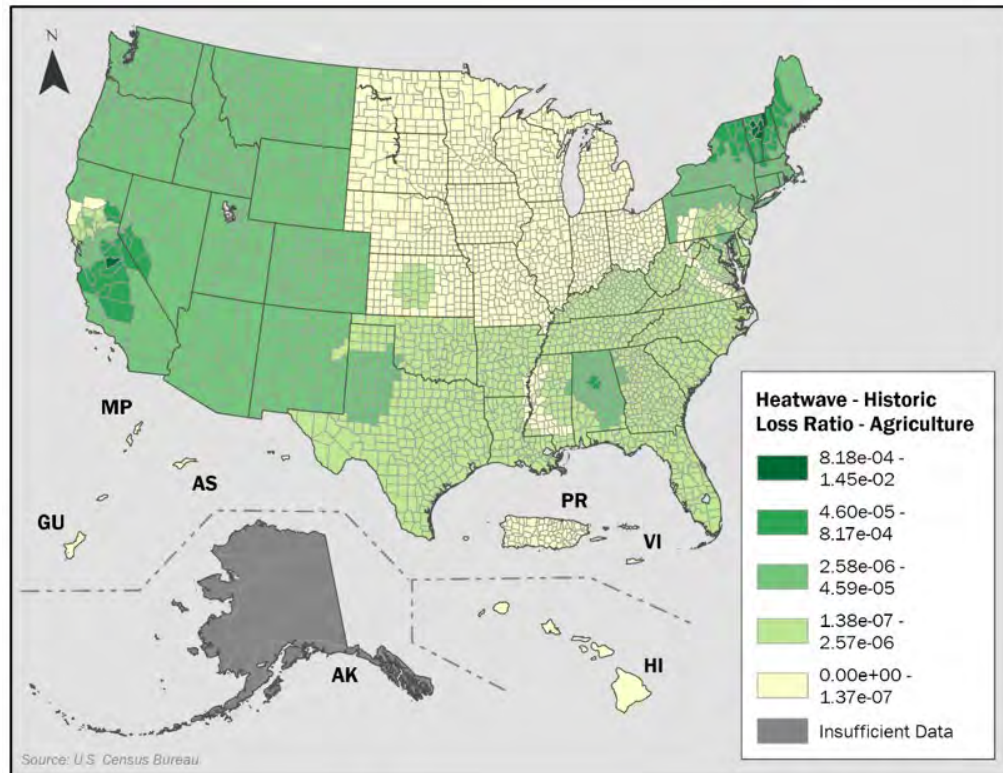
Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Heat Wave (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 50](#), [Figure 51](#), and [Figure 52](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Heat Wave. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 50: Heat Wave Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 51: Heat Wave Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 52: Heat Wave Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

## 12.7. Heat Wave Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level. Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels. For the building and agriculture consequence types, Expected Annual Loss is calculated as in [Equation 67](#).

### Equation 67: Census Block Expected Annual Loss to Heat Wave

$$EAL_{HWAV_{CB_{Bldg}}} = Exposure_{HWAV_{CB_{Bldg}}} \times Freq_{HWAV_{CB}} \times HLR_{HWAV_{CB_{Bldg}}}$$

$$EAL_{HWAV_{CB_{Ag}}} = Exposure_{HWAV_{CB_{Ag}}} \times Freq_{HWAV_{CB}} \times HLR_{HWAV_{CB_{Ag}}}$$

where:

$EAL_{HWAV_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Heat Wave occurrences for a specific Census block (in dollars).

$Exposure_{HWAV_{CB_{Bldg}}}$  is the building value exposed to Heat Wave occurrences in the Census block (in dollars).

$Freq_{HWAV_{CB}}$  is the Heat Wave annualized frequency calculated for the Census block (event-days per year).

$HLR_{HWAV_{CB_{Bldg}}}$  is the Bayesian-adjusted building Historic Loss Ratio for Heat Wave for the Census block.

$EAL_{HWAV_{CB_{Ag}}}$  is the agriculture Expected Annual Loss due to Heat Wave occurrences for a specific Census block (in dollars).

$Exposure_{HWAV_{CB_{Ag}}}$  is the agriculture value exposed to Heat Wave occurrences in the Census block (in dollars).

$HLR_{HWAV_{CB_{Ag}}}$  is the Bayesian-adjusted agriculture Historic Loss Ratio for Heat Wave for the Census block.

For the population consequence type, initial or “baseline” Expected Annual Loss for Population values are calculated like other hazards on an impacted people basis (i.e., Fatalities + 0.1\*Injuries). To scale loss estimates up to match the annualized fatality rate reported in the National Center for Health Statistics Compressed Mortality data, baseline Expected Annual Loss for Population values are divided into consequence subtypes for fatalities and injuries using the proportion of impacted people reported in SHELDUS associated with fatalities and injuries, respectively. Census block-level Expected Annual Loss for the fatality consequence subtype is calculated by scaling up the baseline population Expected Annual Loss values attributed to fatalities so that the national total matches the



annualized fatality count reported in the National Center for Health Statistics Compressed Mortality data, as in [Equation 68](#).

**Equation 68: Census Block Expected Annual Loss to Heat Wave for Fatalities**

$$EAL_{HWAV_{CBPopBaseline}} = Exposure_{HWAV_{CBPop}} \times Freq_{HWAV_{CB}} \times HLR_{HWAV_{CBPop}}$$

$$SHELDUSRatio_{HWAV_{Fatality}} = \frac{SHELDUSTotal_{HWAV_{Fatality}}}{SHELDUSTotal_{HWAV_{Fatality}} + 0.1 * SHELDUSTotal_{HWAV_{Injury}}}$$

$$EAL_{HWAV_{CBPopBaselineFatality}} = EAL_{HWAV_{CBPopBaseline}} \times SHELDUSRatio_{HWAV_{Fatality}}$$

$$CMScaler_{HWAV} = \frac{CMAnnualizedTotal_{HWAV}}{\sum_{CB}^{Nat} EAL_{HWAV_{CBPopBaselineFatality}}} \times VSL_{2024}$$

$$EAL_{HWAV_{CBPopFatality}} = EAL_{HWAV_{CBPopFatalityBaseline}} \times CMScaler_{HWAV}$$

where:

|                                      |                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $EAL_{HWAV_{CBPopBaselineFatality}}$ | is the baseline Expected Annual Loss fatality value due to Heat Wave for a specific Census block.                  |
| $EAL_{HWAV_{CBPopBaseline}}$         | is the baseline population Expected Annual Loss value due to Heat Wave for a specific Census block.                |
| $Exposure_{HWAV_{CBPop}}$            | is the population value exposed to Heat Wave occurrences for a specific Census block                               |
| $Freq_{HWAV_{CB}}$                   | is the Heat Wave annualized frequency for the Census block (event-days per year).                                  |
| $HLR_{HWAV_{CBPopFatality}}$         | is the Bayesian-adjusted fatality Historic Loss Ratio for Heat Wave for the Census block.                          |
| $SHELDUSRatio_{HWAV_{Fatality}}$     | is the proportion of the national total impacted people due to Heat Wave reported in the SHELDUS period of record. |
| $SHELDUSTotal_{HWAV_{Fatality}}$     | is the national total number of fatalities due to Heat Wave reported in the SHELDUS period of record.              |

|                                                           |                                                                                                                                                                               |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $SHELDUSTotal_{HWAV\ Injury}$                             | is the national total number of injuries due to Heat Wave reported in the SHELDUS period of record                                                                            |
| $CMAnnualizedTotal_{HWAV}$                                | is the annualized national total fatalities due to Heat Wave reported in the National Center for Health Statistics Compressed Mortality data.                                 |
| $\sum_{CB}^{Nat} EAL_{HWAV\ CB\ Pop\ Baseline\ Fatality}$ | is the sum of baseline Heat Wave fatality Expected Annual Loss values across all Census blocks in the nation.                                                                 |
| $CMScaler_{HWAV}$                                         | is the ratio between annualized national fatality totals from the National Center for Health Statistics and the baseline Heat Wave Expected Annual Loss fatality calculation. |
| $EAL_{HWAV\ CB\ Pop\ Fatality}$                           | is the scaled population equivalence Expected Annual Loss fatality value due to Heat Wave for a specific Census block.                                                        |
| $VSL_{2024}$                                              | is the inflation adjusted value of a statistical life as measured in 2024 USD.                                                                                                |

For the Heat Wave injury consequence subtype, Expected Annual Loss is calculated using the Bayesian-adjusted population Historic Loss Ratio and the proportion of impacted people reported in SHELDUS associated with injuries as in [Equation 69](#).

#### Equation 69: Census Block Expected Annual Loss to Heat Wave for Injuries

$$EAL_{HWAV\ CB\ Pop\ Baseline} = Exposure_{HWAV\ CB\ Pop} \times Freq_{HWAV\ CB} \times HLR_{HWAV\ CB\ Pop}$$

$$EAL_{HWAV\ CB\ Pop\ Injury} = EAL_{HWAV\ CB\ Pop\ Baseline} \times (1 - SHELDUSRatio_{Pop\ Fatality}) \times VSL_{2024}$$

where:

|                                 |                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| $EAL_{HWAV\ CB\ Pop\ Baseline}$ | is the baseline population Expected Annual Loss value due to Heat Wave for a specific Census block |
| $Exposure_{HWAV\ CB\ Pop}$      | is the population value exposed to Heat Wave occurrences for a specific Census block               |
| $Freq_{HWAV\ CB}$               | is the Heat Wave annualized frequency for the Census block (event-days per year).                  |
| $HLR_{HWAV\ CB\ Pop}$           | is the Bayesian-adjusted population Historic Loss Ratio for Heat Wave for the Census block.        |



$EAL_{HWAV_{CBPopInjury}}$  is the scaled population equivalence Expected Annual Loss injury value due to Heat Wave for a specific Census block

$SHELDUSRatio_{PopFatality}$  is the proportion of the national total impacted people due to Heat Wave reported in the SHELDUS period of record.

$VSL_{2024}$  is the inflation adjusted value of a statistical life as measured in 2024 United States Dollars.

To calculate Heat Wave Expected Annual Loss for the population consequence type, fatality and injury Expected Annual Loss values are summed within Census blocks as in [Equation 70](#).

**Equation 70: Census Block Expected Annual Loss to Heat Wave for Population**

$$EAL_{HWAV_{CBPop}} = EAL_{HWAV_{CBPopFatality}} + EAL_{HWAV_{CBPopInjury}}$$

where:

$EAL_{HWAV_{CBPop}}$  is the population equivalence Expected Annual Loss value due to Heat Wave occurrences for a specific Census block (in dollars).

$EAL_{HWAV_{CBPopFatality}}$  is the scaled population equivalence Expected Annual Loss fatality value due to Heat Wave for a specific Census block.

$EAL_{HWAV_{CBPopInjury}}$  is the population equivalence Expected Annual Loss injury value due to Heat Wave for a specific Census block.

The total Expected Annual Loss Values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 71](#).

**Equation 71: Census Tract and County Expected Annual Loss to Heat Wave**

$$EAL_{HWAV_{CT}} = \sum_{CB}^{CT} EAL_{HWAV_{CBBldg}} + \sum_{CB}^{CT} EAL_{HWAV_{CBPop}} + \sum_{CB}^{CT} EAL_{HWAV_{CBAg}}$$

$$EAL_{HWAV_{Co}} = \sum_{CB}^{Co} EAL_{HWAV_{CBBldg}} + \sum_{CB}^{Co} EAL_{HWAV_{CBPop}} + \sum_{CB}^{Co} EAL_{HWAV_{CBAg}}$$

where:

$EAL_{HWAV_{CT}}$  is the total Expected Annual Loss due to Heat Wave occurrences for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{HWAV CB Bldg}$  is the summed building Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{HWAV CB Pop}$  is the summed population equivalence Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{HWAV CB Ag}$  is the summed agriculture Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the Census tract (in dollars).

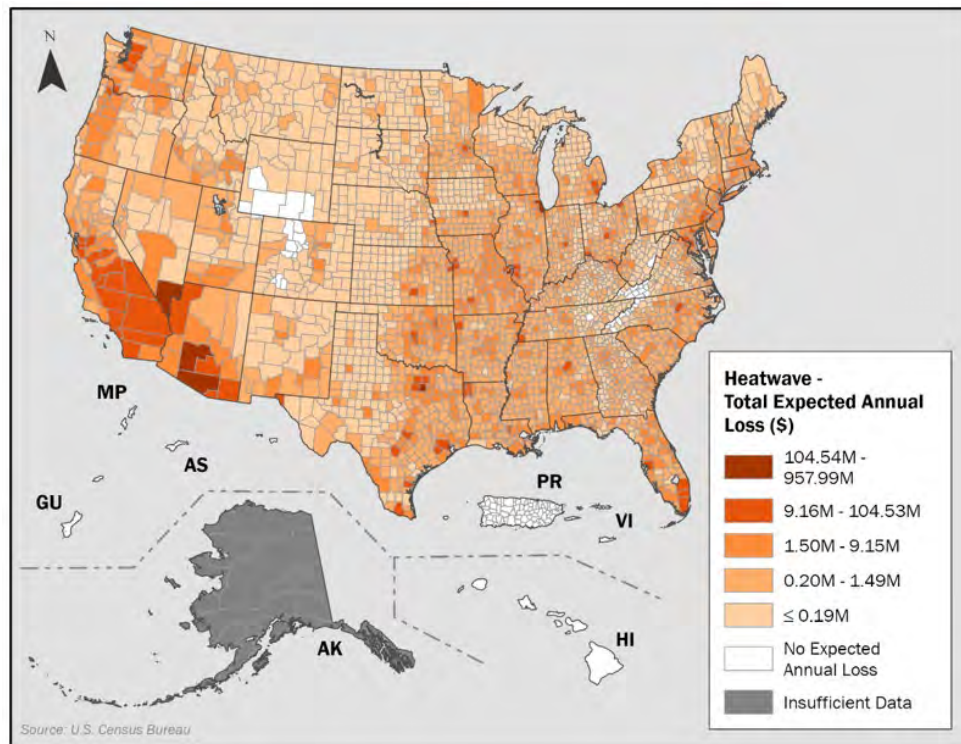
$EAL_{HWAV Co}$  is the total Expected Annual Loss due to Heat Wave occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{HWAV CB Bldg}$  is the summed building Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{HWAV CB Pop}$  is the summed population equivalence Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{HWAV CB Ag}$  is the summed agriculture Expected Annual Loss due to Heat Wave occurrences for all Census blocks in the county (in dollars).

[Figure 53](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Heat Wave occurrences.



**Figure 53: Total Expected Annual Loss by County to Heat Wave**

With the Heat Wave total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Heat Wave Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Heat Wave Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Heat Wave Expected Annual Loss for agriculture by the total agriculture value of the community.

## 12.8. Heat Wave Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- National Weather Service Warnings and Advisories are issued at the discretion of each individual National Weather Service Forecast Office, based on regionally determined thresholds. Though this accounts for varying heat-related capacities across the country (Florida, for example, is more accustomed to higher temperatures than Alaska), it results in an inconsistent record of dangerous heat events.
- The Heat Wave hazard relies on Centers for Disease Control and Prevention data for fatality counts to estimate Population consequence type losses. These Compressed Mortality data are suppressed at the county level or at any geographic level where there are fewer than 10 reported fatalities. Additionally, mortality statistics are reported by the location of the decedent's residence, and not necessarily by the location where the fatality occurred.
- There is currently no adequate injury information in SHELDER to have an injury Historic Loss Ratio distinct from the fatality Historic Loss Ratio.
- The ICD-10 Codes in CDC WONDER do not consider contributing causes of death based on the Heat hazard type, only underlying causes of death.

## 13. Hurricane

A Hurricane is a tropical cyclone or localized, low-pressure weather system that has organized thunderstorms but no front (a boundary separating two air masses of different densities) and maximum sustained winds of at least 74 miles per hour (mph). The Hurricane data also includes tropical storms for which wind speeds range from 39 to 74 mph.

The model for the Hurricane hazard is intended to reflect only the wind-related risks of hurricane events, not the flooding risks as those are incorporated into the Inland Flooding and Coastal Flooding hazards. As such, the methodology and data used to model hurricane frequency, historical occurrences, and exposure areas are based on hurricane storm tracks and windfields. Using data from the National Hurricane Center for storm tracks and windfield extents and historical loss data from SHELUDUS, this methodology identifies Census blocks where hurricane wind impacts are possible. The hurricane hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 13.1. Hurricane Spatial Source Data

**Historical Occurrence Source:** [National Oceanic and Atmospheric Administration, NHC, HURDAT2 Best Track Data](#)<sup>85</sup>

The NHC, a component of the National Oceanic and Atmospheric Administration’s National Centers for Environmental Prediction, maintains several databases, including the HURDAT2 Best Track Data Archive. The dataset is the most comprehensive source of information on Atlantic tropical and subtropical cyclones.<sup>86</sup> It contains a series of storm observation records at six-hour intervals with location, maximum wind speed, central pressure, and (beginning in 2004) cyclone size. The observation records are organized by storm with a unique identifier and include temporal data (date and time; see [Table 47](#) and [Figure 54](#)). The dataset is the result of a post-storm analysis and contains the official assessment of a storm’s path and characteristics. It also can include storm observations that were not available in real-time during the storm.

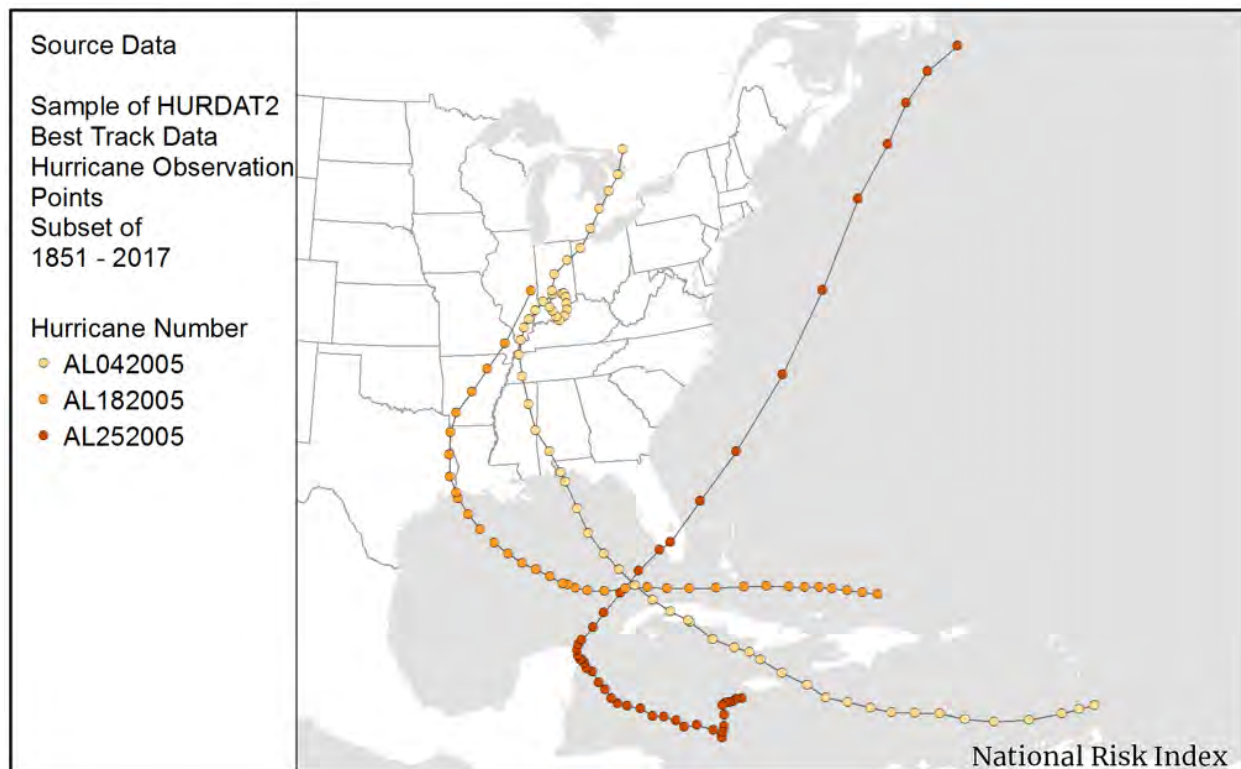
**Table 47: Sample Data from HURDAT2**

| DateObs                     | Basin | HurricaneNumber | HurricaneName | SystemStatus | Latitude | Longitude | MaxWindKts |
|-----------------------------|-------|-----------------|---------------|--------------|----------|-----------|------------|
| Oct. 01,<br>2016 6:00<br>AM | AL    | AL142016        | Matthew       | HU           | 13.4     | -72.5     | 140        |

<sup>85</sup> NHC, NOAA. (2025). HURDAT2 Best Track Data Archive [online dataset]. Retrieved from <https://www.nhc.noaa.gov/data/>.

<sup>86</sup> Landsea, C. W. & Franklin, J.L. (2013). Atlantic hurricane database uncertainty and presentation of a new database format. Monthly Weather Review, 141, 3576-3592.

| DateObs                      | Basin | HurricaneNumber | HurricaneName | SystemStatus | Latitude | Longitude | MaxWindKts |
|------------------------------|-------|-----------------|---------------|--------------|----------|-----------|------------|
| Oct. 01,<br>2016<br>12:00 PM | AL    | AL142016        | Matthew       | HU           | 13.4     | -73.1     | 135        |
| Oct. 01,<br>2016 6:00<br>PM  | AL    | AL142016        | Matthew       | HU           | 13.4     | -73.3     | 130        |



**Figure 54: Map of HURDAT2 Points**

**Historical Occurrence Source:** [National Oceanic and Atmospheric Administration, NHC, Tropical Cyclone Reports](#)<sup>87</sup>

The National Hurricane Center's Tropical Cyclone Reports contain comprehensive information on each tropical cyclone, including synoptic history, meteorological statistics, casualties and damages, and the post-analysis best track (six-hour positions and intensities). Tropical cyclones include depressions, storms, and hurricanes. These data are stored as shapefiles before being converted to text-only when they are transferred to the HURDAT2 Best Track Data.

<sup>87</sup> NHC, NOAA. (2025). Tropical Cyclone Reports [online dataset]. Retrieved from <https://www.nhc.noaa.gov/data/tcr/> on April 4, 2025)

This dataset provided the tracks for the 7 tropical storms and 11 hurricanes of the 2024 Atlantic Hurricane Season that had not yet been included in the available HURDAT2 dataset.

**Historical Occurrence Source:** [National Oceanic and Atmospheric Administration, Historical Hurricane Tracks](#)<sup>88</sup>

HURDAT2 does not include data on storm tracks in the Western or Southern Pacific Ocean, effectively excluding most storms that have historically impacted the Pacific territories. Data from the Historical Hurricane Tracks were therefore used to establish annualized frequency estimates for Guam, American Samoa, and Northern Mariana Islands. The Historical Hurricane Tracks is a free web-based tool developed by the National Oceanic and Atmospheric Administration's Office for Coastal Management with the National Hurricane Center and National Centers for Environmental Information. The tool contains global hurricane data for over 13,000 historical storm tracks dating as far back as 1851, with detailed reports for United States tropical cyclones since 1958. Storm track information displayed in the web-based app is sourced from HURDAT2 and National Centers for Environmental Information's [International Best Track Archive for Climate Stewardship](#) datasets.

## 13.2. Hurricane Spatial Processing

HURDAT2 data for both the Pacific and Atlantic basin were downloaded and loaded into the processing database. Upon loading, a record for every storm observation was created and attributed with the StormID to which it is associated.

Each storm observation record (point location) was categorized by its associated wind speeds (based on the Saffir-Simpson Hurricane Wind Scale).<sup>89</sup> The storm category was then used to assign a buffer radius (representing the average distance at which storm force winds are found) to each observation location. [Table 48](#) presents the storm category wind speed definitions and their associated average radius distance of storm force winds. These radii are derived through a process based on research conducted by Bell and Ray (2004).<sup>90</sup>

**Table 48: Hurricane Categorization**

| <i>Storm Category</i> | <i>Minimum Wind Speed (mph)</i> | <i>Maximum Wind Speed (mph)</i> | <i>Minimum Wind Speed (kts)</i> | <i>Maximum Wind Speed (kts)</i> | <i>Average Radius of Hurricane/Tropical Storm Force Winds (miles)</i> |
|-----------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-----------------------------------------------------------------------|
| Other                 | 0                               | 38.9                            | 0                               | 32.9                            | 0                                                                     |
| Tropical Storm        | 39                              | 73.9                            | 33                              | 63.9                            | 15                                                                    |
| Category 1            | 74                              | 95.9                            | 64                              | 82.9                            | 26.45                                                                 |

<sup>88</sup> <https://oceanservice.noaa.gov/news/historical-hurricanes/>

<sup>89</sup> Schott, T., Landsea, C., Hafele, G., Lorens, J., Taylor, A., Thurm, H., Ward, B., Willis, M., & Zaleski, W. (2019). The Saffir-Simpson Hurricane Wind Scale [PDF file]. Retrieved from <https://www.nhc.noaa.gov/pdf/sshws.pdf>.

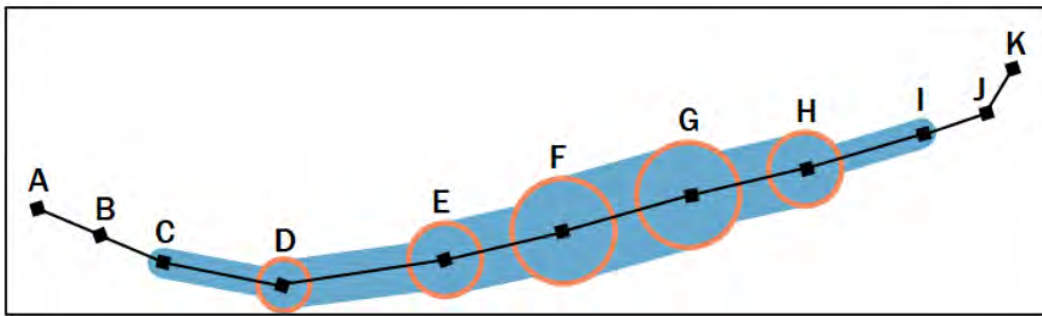
<sup>90</sup> Bell, K., & Ray, P.S. (2004). North Atlantic hurricanes 1977-99: Surface hurricane-force wind radii. *Monthly Weather Review*, 132(5), 1167-1189. DOI: [https://doi.org/10.1175/1520-0493\(2004\)132%3C1167:NAHSHW%3E2.0.CO;2](https://doi.org/10.1175/1520-0493(2004)132%3C1167:NAHSHW%3E2.0.CO;2)



| <b>Storm Category</b> | <b>Minimum Wind Speed (mph)</b> | <b>Maximum Wind Speed (mph)</b> | <b>Minimum Wind Speed (kts)</b> | <b>Maximum Wind Speed (kts)</b> | <b>Average Radius of Hurricane/Tropical Storm Force Winds (miles)</b> |
|-----------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-----------------------------------------------------------------------|
| <b>Category 2</b>     | 96                              | 110.9                           | 83                              | 95.9                            | 39.1                                                                  |
| <b>Category 3</b>     | 111                             | 129.9                           | 96                              | 112.9                           | 43.7                                                                  |
| <b>Category 4</b>     | 130                             | 156.9                           | 113                             | 136.9                           | 50.03                                                                 |
| <b>Category 5</b>     | 157                             | 9999                            | 137                             | 9999                            | 54.04                                                                 |

After the storms in the HURDAT2 data were processed, the resulting storm data and the Tropical Cyclone Report storms' observation points were connected to create multi-segment lines that represent the path of each storm. Each line segment between two consecutive storm observation points was attributed with the lowest storm category value of its endpoint observations (based on the assumption that this would be the minimum expected category along the path segment).

Each storm observation location and line segment were independently buffered by the average radius distance (of storm force winds) associated with its assigned storm category. All the buffered shapes associated with a given storm were then union-dissolved into a single polygon shape representing the area for which hurricane force winds were modeled for that particular storm to create historical Hurricane event path polygons (see [Figure 55](#), [Table 49](#), and [Figure 56](#)).

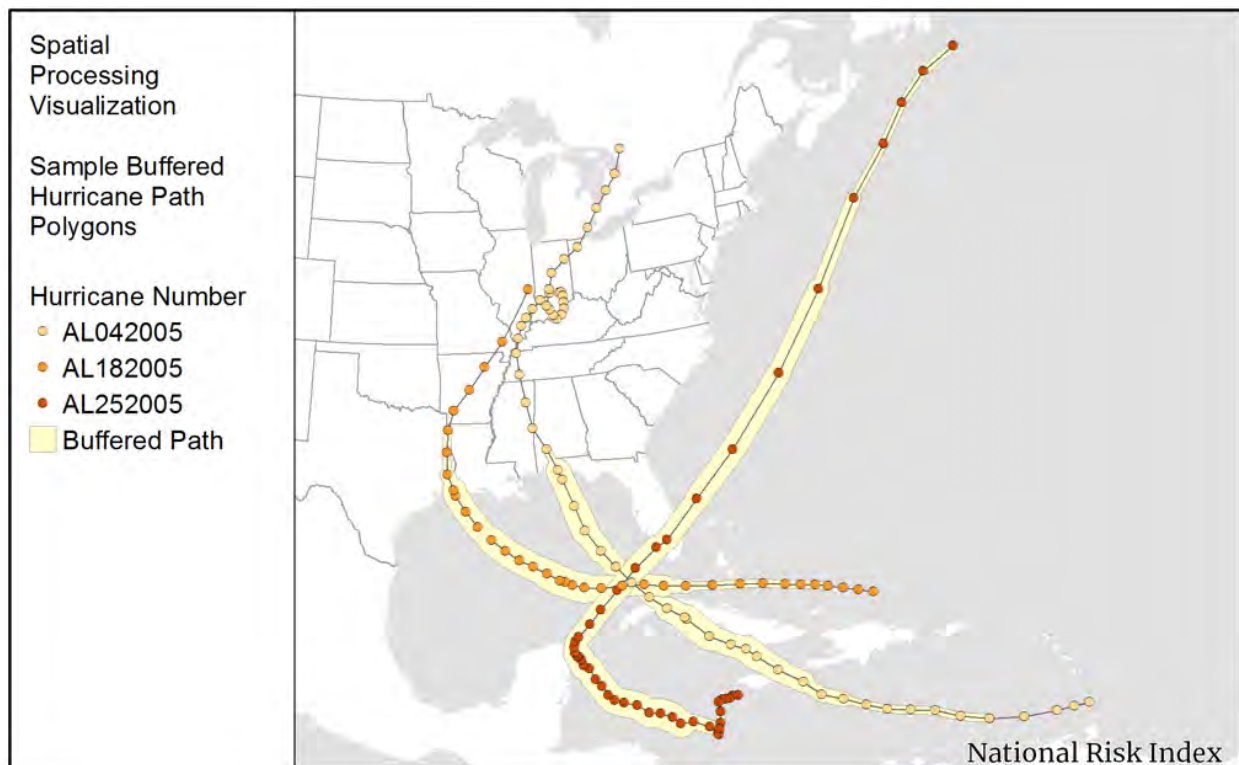


**Figure 55: Notional Example Hurricane Event Path Polygon**

**Table 49: Notional Example Hurricane Storm Observations**

| <b>Observation Point</b> | <b>Wind Speed (kt)</b> | <b>Storm Category</b> |
|--------------------------|------------------------|-----------------------|
| A                        | 15                     | Other                 |
| B                        | 25                     | Other                 |
| C                        | 50                     | Tropical Storm        |
| D                        | 65                     | Cat 1                 |

| <i>Observation Point</i> | <i>Wind Speed (kt)</i> | <i>Storm Category</i> |
|--------------------------|------------------------|-----------------------|
| E                        | 85                     | Cat 2                 |
| F                        | 100                    | Cat 3                 |
| G                        | 110                    | Cat 3                 |
| H                        | 90                     | Cat 2                 |
| I                        | 60                     | Tropical Storm        |
| J                        | 30                     | Other                 |
| K                        | 15                     | Other                 |

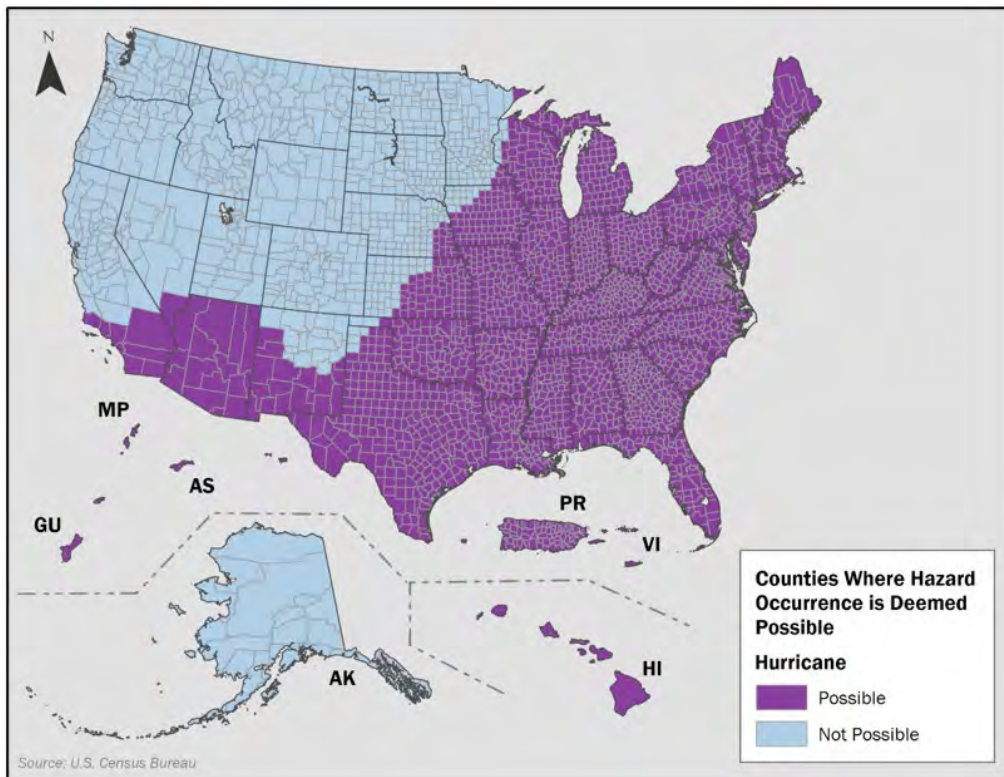


**Figure 56: Sample Buffered Hurricane Event Path Polygons**

For the Pacific Territories (Guam, American Samoa, and Northern Mariana Islands), the Historical Hurricane Tracks viewer was used to determine the number of hurricanes from 1949-2024 which passed within 1 mile of each territory. Storm tracks for Tropical Storms and all Hurricane categories were included in the count.

### 13.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

To distinguish between areas where no Hurricane events have occurred and those where such events are not deemed possible, a control table was generated to designate which counties have some probability of Hurricane occurrence. The Hurricane event path polygons processed to represent historical storms as described in [Section 13.2 Hurricane Spatial Processing](#) were buffered to an additional 100 miles, and for any Census blocks that intersected at least one buffered Hurricane event path polygon, their counties were included with some probability of event occurrence. Additionally, a subset of inland counties near the Atlantic basin that had sustained historical economic loss to Hurricanes according to SHELUS was also included (see [Figure 57](#)).



**Figure 57: Map of Counties Deemed Possible for Hurricane Occurrence**

## 13.3. Hurricane Non-Spatial Data Processing

### 13.3.1. HISTORICAL OCCURRENCE COUNT

Historical occurrence counts of Hurricanes, in events, are supplied at the Census tract and county levels as the number of distinct Hurricane event path polygons (see [Section 13.2 Hurricane Spatial Processing](#)) that intersect the Census tract and county, respectively. This count uses the same Hurricane Census block intersection table used to calculate exposure.

For the Pacific Territories, the historical occurrence counts are made by buffering the geographic area of each Territory by 1 mile and counting the number of Hurricane event path polygons that

intersected each of these buffered areas. These counts were then inherited by all Census tracts within each Territory.

### 13.3.2. PERIOD OF RECORD

The hurricane track datasets are organized by ocean basins: Atlantic and Pacific. The storms in the Atlantic datasets date from Jan. 01, 1851 to Sept. 30, 2024 (173.86 years), while those in the Pacific date from Jan. 01, 1949 to Nov. 30, 2024 (76.0 years).

## 13.4. Hurricane Exposure

To identify areas of exposure, the Hurricane event path polygons are intersected with the Census block polygons within the processing database. The resulting table contains the storm's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 50](#)).

**Table 50: Sample Data from the Hurricane Census Block Intersection Table**

| <i>HurricaneProcessedID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|-----------------------------|--------------------|---------------------------|
| 1561                        | 280870009003030    | 0.00563004156494141       |
| 1561                        | 280870009003031    | 0.000665592071533203      |
| 1561                        | 280870009003032    | 0.00911474768066406       |

To determine exposure value, the average coverage of a Hurricane occurrence is found by summing the intersected areas for all buffered Hurricane paths that intersected the Census block and dividing this sum by the number of intersecting Hurricane paths. This is multiplied by the developed area building value density, the developed area population density, and the agriculture area value density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 72](#)). These densities in the Census block have been calculated by dividing the total exposure values (building value, population, and agriculture as recorded in Hazus 6.0) by the developed or agriculture land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

### Equation 72: Census Block Hurricane Exposure

$$Exposure_{HRCN_{CB}Bldg} = \frac{\sum IntsctArea_{HRCN_{CB}}}{EventCount_{HRCN_{CB}}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{HRCN_{CB}Pop} = \left( \frac{\sum IntsctArea_{HRCN_{CB}}}{EventCount_{HRCN_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

$$Exposure_{HRCN_{CB}Ag} = \frac{\sum IntsctArea_{HRCN_{CB}}}{EventCount_{HRCN_{CB}}} \times AgValueDen_{CB}$$

where:

|                               |                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| $Exposure_{HRCN_{CB}Bldg}$    | is the building value exposed to Hurricanes in a specific Census block (in dollars).                 |
| $\sum IntsctArea_{HRCN_{CB}}$ | is the sum of the intersected areas of past Hurricanes with the Census block (in square kilometers). |
| $EventCount_{HRCN_{CB}}$      | is the total number of Hurricanes that intersect the Census block.                                   |
| $DevAreaDen_{CB}Bldg$         | is the developed area building value density of the Census block (in dollars per square kilometer).  |
| $Exposure_{HRCN_{CB}Pop}$     | is the population equivalence value exposed to Hurricanes in a specific Census block (in dollars).   |
| $DevAreaDen_{CB}Pop$          | is the developed area population density of the Census block (in people per square kilometer).       |
| $VSL$                         | is the Value of Statistical Life (\$13.7 million per person).                                        |
| $Exposure_{HRCN_{CB}Ag}$      | is the agriculture value exposed to Hurricanes in a specific Census block (in dollars).              |
| $AgValueDen_{CB}$             | is the agriculture value density of the Census block (in dollars per square kilometer).              |

In cases where a Census block is deemed potentially at risk for Hurricane damage due to its proximity to areas that have been hit by one or more Hurricanes in the past but has not experienced any historical Hurricane occurrences itself, the exposure value is estimated to be the full Census block building value and population value. These areas will likely have a low Historic Loss Ratio and/or annualized frequency, which will diminish the effect of using full Census block exposure values in the final Expected Annual Loss calculation.

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value and the Census of Agriculture-reported crop and livestock value for the Census block are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

### 13.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 73](#)).

**Equation 73: Census Tract and County Hurricane Exposure Aggregation**

$$Exposure_{HRCN_{CT}Bldg} = \sum_{CB}^{CT} Exposure_{HRCN_{CB}Bldg}$$

$$Exposure_{HRCN_{Co}Bldg} = \sum_{CB}^{Co} Exposure_{HRCN_{CB}Bldg}$$

$$Exposure_{HRCN_{CT}Pop} = \sum_{CB}^{CT} Exposure_{HRCN_{CB}Pop}$$

$$Exposure_{HRCN_{Co}Pop} = \sum_{CB}^{Co} Exposure_{HRCN_{CB}Pop}$$

$$Exposure_{HRCN_{CT}Ag} = \sum_{CB}^{CT} Exposure_{HRCN_{CB}Ag}$$

$$Exposure_{HRCN_{Co}Ag} = \sum_{CB}^{Co} Exposure_{HRCN_{CB}Ag}$$

where:

$Exposure_{HRCN_{CT}Bldg}$  is the building value exposed to Hurricanes in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{HRCN_{CB}Bldg}$  is the summed value of all buildings exposed to Hurricane for each Census block within the Census tract (in dollars).

$Exposure_{HRCN_{Co}Bldg}$  is the building value exposed to Hurricanes in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{HRCN_{CB}Bldg}$  is the summed value of all buildings exposed to Hurricane for each Census block within the county (in dollars).

$Exposure_{HRCN_{CT}Pop}$  is the population equivalence value exposed to Hurricanes in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{HRCN_{CB}Pop}$  is the summed value of all population equivalence exposed to Hurricane for each Census block within the Census tract (in dollars).



|                                          |                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{HRCN_{CoPop}}$                | is the population equivalence value exposed to Hurricanes in a specific county (in dollars).                                  |
| $\sum_{CB}^{Co} Exposure_{HRCN_{CBPop}}$ | is the summed value of all population equivalence exposed to Hurricane for each Census block within the county (in dollars).  |
| $Exposure_{HRCN_{CTAg}}$                 | is the agriculture value exposed to Hurricanes in a specific Census tract (in dollars).                                       |
| $\sum_{CB}^{CT} Exposure_{HRCN_{CBAg}}$  | is the summed value of all agriculture areas exposed to Hurricane for each Census block within the Census tract (in dollars). |
| $Exposure_{HRCN_{CoAg}}$                 | is the agriculture value exposed to Hurricanes in a specific county (in dollars).                                             |
| $\sum_{CB}^{Co} Exposure_{HRCN_{CBAg}}$  | is the summed value of all agriculture areas exposed to Hurricane for each Census block within the county (in dollars).       |

### 13.5. Hurricane Annualized Frequency

The annualized frequency value represents the estimated number of recorded Hurricane occurrences, in events, each year for a specific area. This annualized frequency is utilized at the Census block level, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations are determined by intersecting the same buffered Hurricane event path polygons that are used to calculate exposure with a 49-by-49-km fishnet grid. The count of distinct Hurricane event path polygons intersecting each grid cell is recorded, and each Census block inherits this fishnet-aggregated count from the grid cell that encompasses it. If the Census block intersects multiple fishnet grid cells, an area-weighted average count is calculated (see [Appendix D – Fishnet Occurrence Count](#)).

The Hurricane event count (determined from the fishnet-aggregated count) is then divided by the period of record (depending on the ocean basin of the location) as in [Equation 74](#).

#### Equation 74: Census Block Hurricane Annualized Frequency

$$Freq_{HRCN_{CB}} = \frac{EventCount_{HRCN_{CB}}}{PeriodRecord_{HRCN}}$$

where:

$Freq_{HRCN_{CB}}$  is the annualized frequency of Hurricane events determined for a specific Census block (events per year).

$EventCount_{HRCN_{CB}}$  is the number of historical Hurricane events calculated for the Census block.

$PeriodRecord_{HRCN}$  is the period of record for Hurricane events, either 173.86 for Atlantic storms or 76.0 for Pacific storms (in years).

### 13.5.1. MINIMUM ANNUAL FREQUENCY

If a Census block's historical Hurricane event count (inherited from the fishnet count) is 0, but the Census block is part of a county that was designated as one in which Hurricanes are possible, the Census block is assigned the minimum annual Hurricane frequency. This Minimum Annual Frequency is set at 0.01 (1 in 100 years). This was determined by subject matter experts to be an acceptable assumption.

### 13.5.2. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Hurricane events at the Census tract and county level by the period of record, as the event count for annualized frequency is a fishnet area-weighted event count including Hurricanes that may have impacted the surrounding area but not the county or Census tract itself. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 75](#).

#### Equation 75: Census Tract and County Area-Weighted Hurricane Annualized Frequency

$$Freq_{HRCN_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{HRCN_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{HRCN_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{HRCN_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{HRCN_{CT}}$  is the area-weighted Hurricane annualized frequency calculated for a specific Census tract (events per year).

$Freq_{HRCN_{CB}}$  is the annualized frequency of Hurricane events determined for a specific Census block (events per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

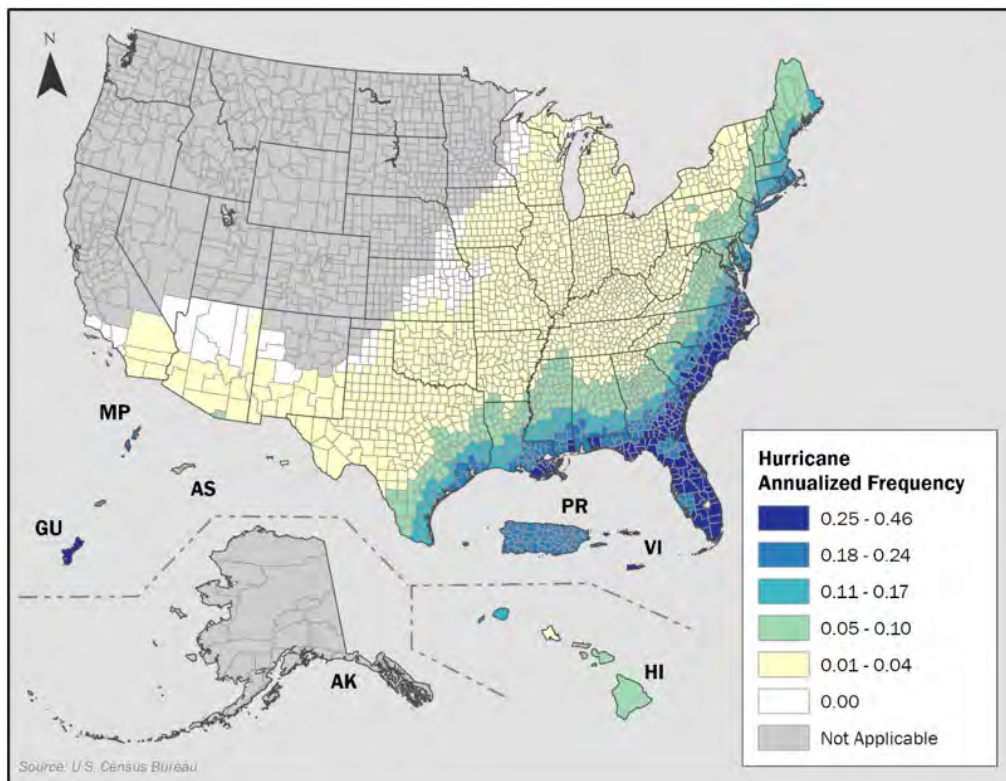
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{HRCN_{Co}}$  is the area-weighted Hurricane annualized frequency calculated for a specific county (events per year).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 58](#) displays Hurricane annualized frequency at the county level.



**Figure 58: Hurricane Annualized Frequency by County**

### 13.6. Hurricane Historic Loss Ratio

The Hurricane Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Hurricane occurrence, or the average rate of loss associated with the Hurricane occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Hurricane hazard type.

Loss data are provided by SHEL DUS<sup>91</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHEL DUS events from 1996 through 2023 are included in the Historic Loss

<sup>91</sup> For Hurricane loss information, SHEL DUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

Ratio calculation. Eight peril types are mapped to the hazard Hurricane (see [Table 51](#)). These native records are aggregated on a consecutive day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). Note that recorded Hurricane events only include those that made landfall as a Tropical Storm or Hurricane.

**Table 51: Hurricane Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Cyclone-Extratropical         | 0                                  | 0                                    |
| Cyclone-Subtropical           | 0                                  | 0                                    |
| Cyclone-Unspecified           | 1                                  | 1                                    |
| Hurricane/Tropical Storm      | 1,542                              | 1,348                                |
| Nor'easter                    | 0                                  | 0                                    |
| Storm Surge                   | 645                                | 538                                  |
| Tropical Depression           | 200                                | 190                                  |
| Tropical Storm                | 3,086                              | 3,850                                |

The Historic Loss Ratio exposure value used in the formula below is a county-level value that represents the dollar value of the total building value, the entire population of a county, or the dollar value of the total agriculture value as recorded in Hazus 6.0. The Loss Ratio per Basis for each SHEL DUS-documented event and each consequence type (building, population, and agriculture) is calculated using [Equation 76](#).

**Equation 76: Loss Ratio per Basis Calculation for a Single Hurricane Event**

$$LRB_{HRCNCoCnsqType} = \frac{LOSS_{HRCNCoCnsqType}}{HLRExposure_{CoCnsqType}}$$

where:

$LRB_{HRCNCoCnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Hurricane event. Calculation is performed for each consequence type (building, population, and agriculture).

$LOSS_{HRCNCoCnsqType}$  is the loss (by consequence type) experienced from the Hurricane event documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{CoCnsqType}$  is the total value (by consequence type) of the county estimated to have been exposed to the Hurricane event (in dollars or people).

Hurricane events (particularly tropical storms) may occur in areas without resulting in recorded loss to buildings, population, or agriculture. SHELDUS does not record events in which no loss occurred, so a number of zero-loss event records are inserted into the loss data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELDUS period of record (1996 through 2023). For Hurricane, the historical event count is extracted using the intersection between the Hurricane event path polygons and the Census block polygons used to calculate exposure (see [Table 50](#)). Unlike the count used for annualized frequency, this count is simply the number of distinct Hurricane event path polygons that have intersected Census blocks in the county. An annual rate is calculated as the event count divided by the period of record of 173.86 years for the Atlantic basin or 76.0 years for the Pacific basin. This rate is multiplied by the SHELDUS period of record of 28 years to estimate a historical event-day count for the appropriate time range. Because SHELDUS does not include any Hurricane loss records for the Pacific territories, the highest Historic Loss Ratio from the Gulf Coast was applied.

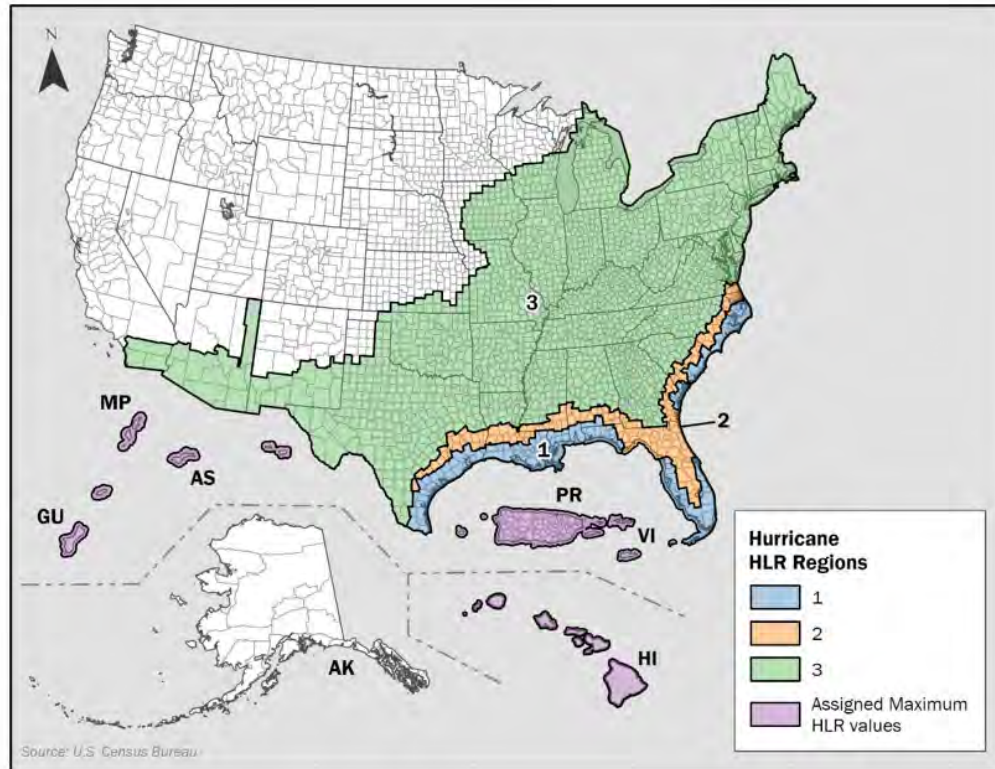
If the number of loss-causing Hurricane event records from SHELDUS is less than the scaled event count for the county, a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for each county, county-level mean loss ratios and event counts are computed and used as inputs into a frequency weighted Generalized Linear Model with a logit link function. The Generalized Linear Model generates Historic Loss Ratio estimates for each county, conditional on their distance from the Gulf/Atlantic coast, maximum sustained probabilistic wind speeds from the Hazus Hurricane inventory, National Center for Health Statistics Urban-Rural Classification, and Hurricane Historic Loss Ratio region definition.

The regional definition for Hurricane is derived from county-level boundaries that approximate hurricane prone wind speed regions identified in the American Society of Civil Engineers 7-05, Minimum Design Loads for Buildings and Other Structures.<sup>92</sup> This region definition was introduced specifically for Hurricane due to the exaggerated Expected Annual Loss values in certain large inland cities with high exposure value (large population and high building values), low hazard occurrence, and use of national weighting, which can skew the Historic Loss Ratio.<sup>93</sup>

<sup>92</sup> ASCE. (2005). Minimum design loads for buildings and other structures (ASCE/SEI 7-05). Reston, VA: ASCE.

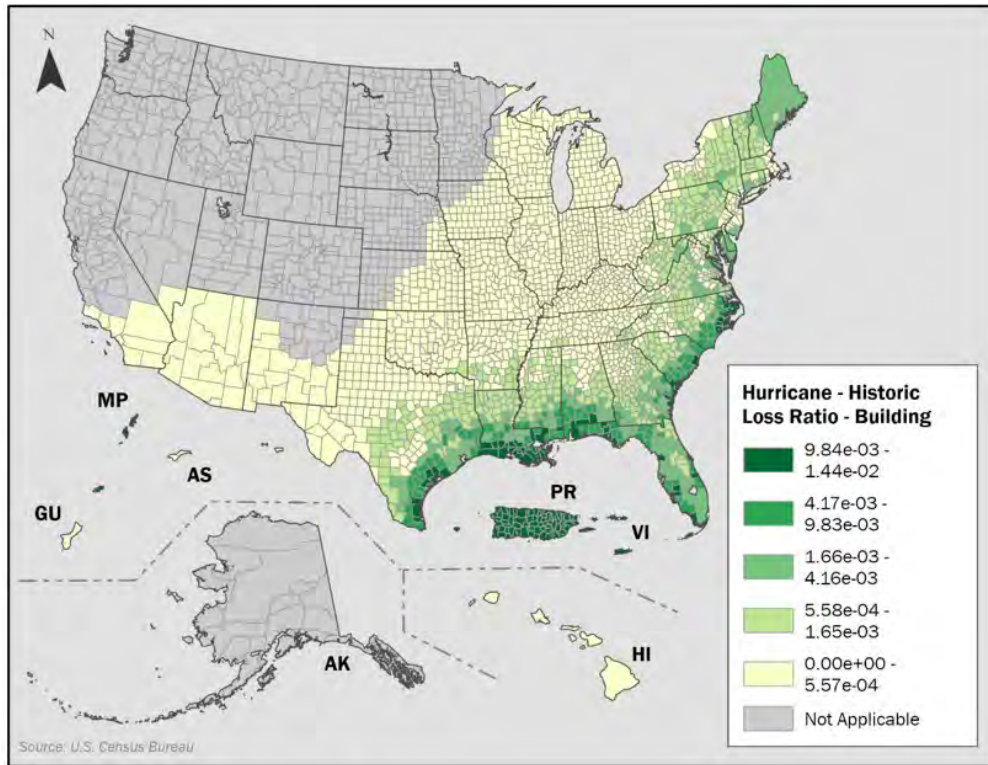
<sup>93</sup> Hawai'i, American Samoa, Guam, and Puerto Rico are not included in a Hurricane regional definition. These areas were assigned maximum Historic Loss Ratio values from the regression.



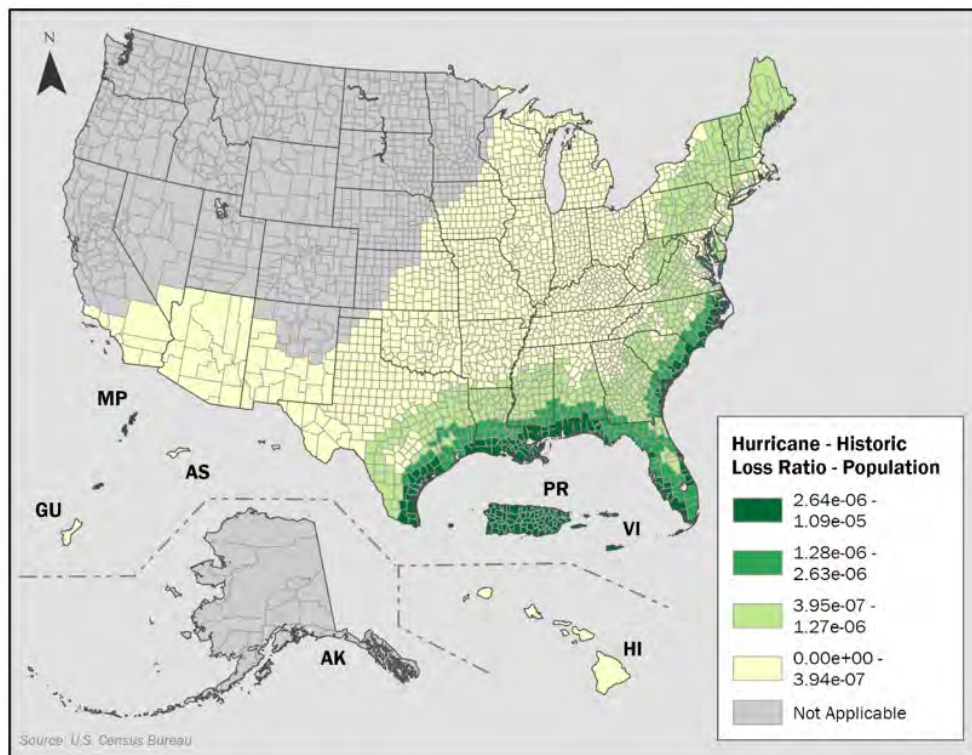
**Figure 59: Hurricane-specific Historic Loss Ratio Region**

[Figure 60](#), [Figure 61](#), and [Figure 62](#) represent the final county-level Historic Loss Ratio values for Hurricane, for building value, population, and agriculture value, respectively. The resulting Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

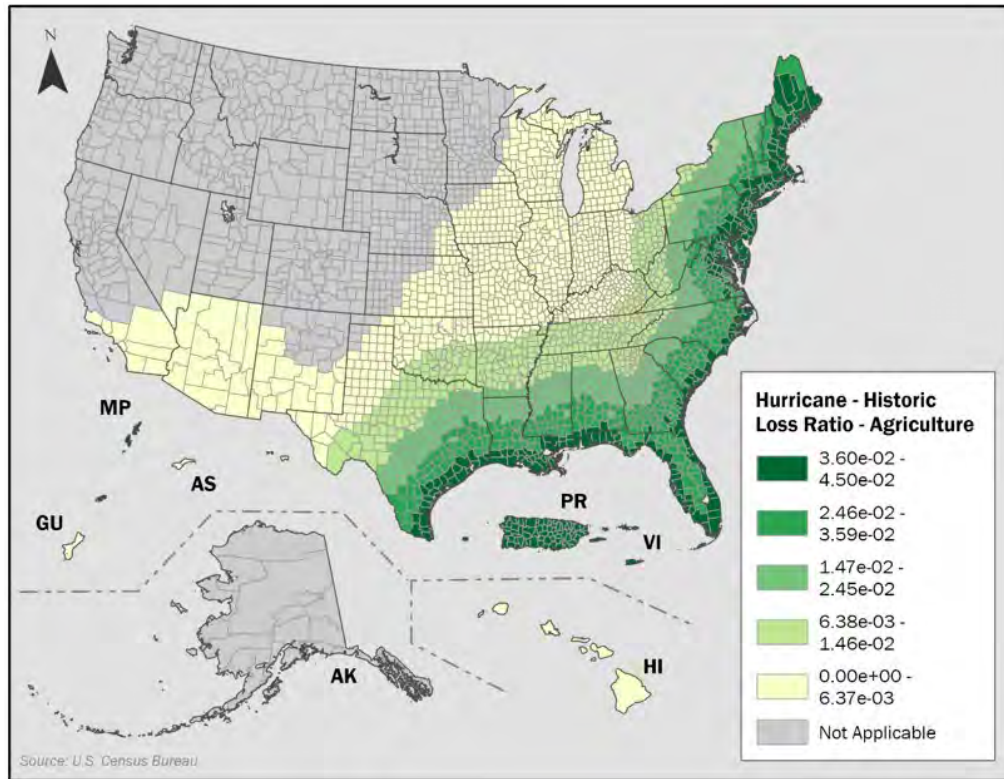




**Figure 60: Hurricane Historic Loss Ratio – Building Value**



**Figure 61: Hurricane Historic Loss Ratio – Population**



**Figure 62: Hurricane Historic Loss Ratio – Agriculture Value**

### 13.7. Hurricane Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 77](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

**Equation 77: Census Block Expected Annual Loss to Hurricanes**

$$EAL_{HRCN_{Bldg}} = Exposure_{HRCN_{CB_{Bldg}}} \times Freq_{HRCN_{CB}} \times HLR_{HRCN_{CB_{Bldg}}}$$

$$EAL_{HRCN_{Pop}} = Exposure_{HRCN_{CB_{Pop}}} \times Freq_{HRCN_{CB}} \times HLR_{HRCN_{CB_{Pop}}}$$

$$EAL_{HRCN_{Ag}} = Exposure_{HRCN_{CB_{Ag}}} \times Freq_{HRCN_{CB}} \times HLR_{HRCN_{CB_{Ag}}}$$

where:

$EAL_{HRCN_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Hurricane occurrences for a specific Census block (in dollars).

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{HRCN_{CB}Bldg}$ | is the building value exposed to Hurricane occurrences in the Census block (in dollars).                                  |
| $Freq_{HRCN_{CB}}$         | is the Hurricane annualized frequency for the Census block (events per year).                                             |
| $HLR_{HRCN_{CB}Bldg}$      | is the Bayesian-adjusted building Historic Loss Ratio for Hurricane for the Census block.                                 |
| $EAL_{HRCN_{CB}Pop}$       | is the population equivalence Expected Annual Loss due to Hurricane occurrences for a specific Census block (in dollars). |
| $Exposure_{HRCN_{CB}Pop}$  | is the population equivalence value exposed to Hurricane occurrences in the Census block (in dollars).                    |
| $HLR_{HRCN_{CB}Pop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Hurricane for the Census block.                               |
| $EAL_{HRCN_{CB}Ag}$        | is the agriculture Expected Annual Loss due to Hurricane occurrences for a specific Census block (in dollars).            |
| $Exposure_{HRCN_{CB}Ag}$   | is the agriculture value exposed to Hurricane occurrences in the Census block (in dollars).                               |
| $HLR_{HRCN_{CB}Ag}$        | is the Bayesian-adjusted agriculture Historic Loss Ratio for Hurricane for the Census block.                              |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 78](#).

**Equation 78: Census Tract and County Expected Annual Loss to Hurricanes**

$$EAL_{HRCN_{CT}} = \sum_{CB}^{CT} EAL_{HRCN_{CB}Bldg} + \sum_{CB}^{CT} EAL_{HRCN_{CB}Pop} + \sum_{CB}^{CT} EAL_{HRCN_{CB}Ag}$$

$$EAL_{HRCN_{Co}} = \sum_{CB}^{Co} EAL_{HRCN_{CB}Bldg} + \sum_{CB}^{Co} EAL_{HRCN_{CB}Pop} + \sum_{CB}^{Co} EAL_{HRCN_{CB}Ag}$$

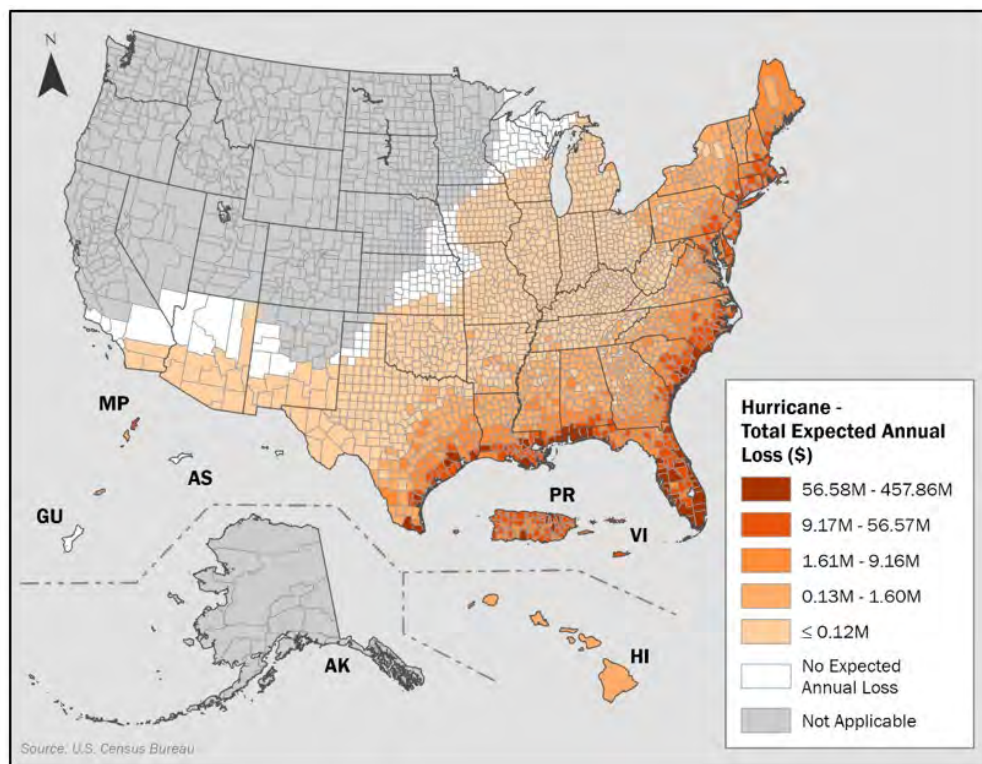
where:

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| $EAL_{HRCN_{CT}}$ | is the total Expected Annual Loss due to Hurricane occurrences for a specific Census tract (in dollars). |
|-------------------|----------------------------------------------------------------------------------------------------------|

|                                       |                                                                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sum_{CB}^{CT} EAL_{HRCN_{CB} Bldg}$ | is the summed building Expected Annual Loss due to Hurricane occurrences for all Census blocks in the Census tract (in dollars).               |
| $\sum_{CB}^{CT} EAL_{HRCN_{CB} Pop}$  | is the summed population equivalence Expected Annual Loss due to Hurricane occurrences for all Census blocks in the Census tract (in dollars). |
| $\sum_{CB}^{CT} EAL_{HRCN_{CB} Ag}$   | is the summed agriculture Expected Annual Loss due to Hurricane occurrences for all Census blocks in the Census tract (in dollars).            |
| $EAL_{HRCN_{Co}}$                     | is the total Expected Annual Loss due to Hurricane occurrences for a specific county (in dollars).                                             |
| $\sum_{CB}^{Co} EAL_{HRCN_{CB} Bldg}$ | is the summed building Expected Annual Loss due to Hurricane occurrences for all Census blocks in the county (in dollars).                     |
| $\sum_{CB}^{Co} EAL_{HRCN_{CB} Pop}$  | is the summed population equivalence Expected Annual Loss due to Hurricane occurrences for all Census blocks in the county (in dollars).       |
| $\sum_{CB}^{Co} EAL_{HRCN_{CB} Ag}$   | is the summed agriculture Expected Annual Loss due to Hurricane occurrences for all Census blocks in the county (in dollars).                  |

[Figure 63](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture combined) to Hurricane occurrences.





**Figure 63: Total Expected Annual Loss by County to Hurricane**

With the Hurricane total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Hurricane Expected Annual Loss value for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Hurricane Expected Annual Loss value for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Hurricane Expected Annual Loss value for agriculture by the total agriculture value of the community.

### 13.8. Hurricane Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The Hurricane hazard endeavors to reflect only the wind-related risks, not the associated inland flooding or coastal flooding risks. This is partially to avoid over-estimating risk, since both water-related hazards are already considered on their own. However, due to the inexact differentiation of water vs. wind-caused damages for historical hurricane events, the data

development team recognizes that the underlying data may combine losses from different hurricane impacts. Therefore, the hazard cannot at this time be accurately labeled as “hurricane wind,” though it is largely focused only on hurricane wind.

- Alaska was designated as “Hazard Not Possible” for hurricane wind. Despite a single historical event impacting the area, the data development team determined that tropical storm impacts at the current time are insufficient to allow for comparison to the rest of the country.



## 14. Ice Storm

An Ice Storm is a freezing rain situation (rain that freezes on surface contact) with significant ice accumulations of 0.25 inches or greater.

The Ice Storm hazard is based on a static dataset that has not been updated since 2014. While lacking data updates on the geographic extent of ice storm occurrences in the past ten years, the data were still sufficient for hazard susceptibility determinations. The historical loss data in SHELUS provided sufficient loss records for creating Historic Loss Ratios and valid Expected Annual Loss values. The Ice Storm hazard only calculates risk for the Building and Population consequence types. For a summary of the Hazard Data Characteristics, please see [Appendix B](#) – Hazard Data Characteristics Comparison.

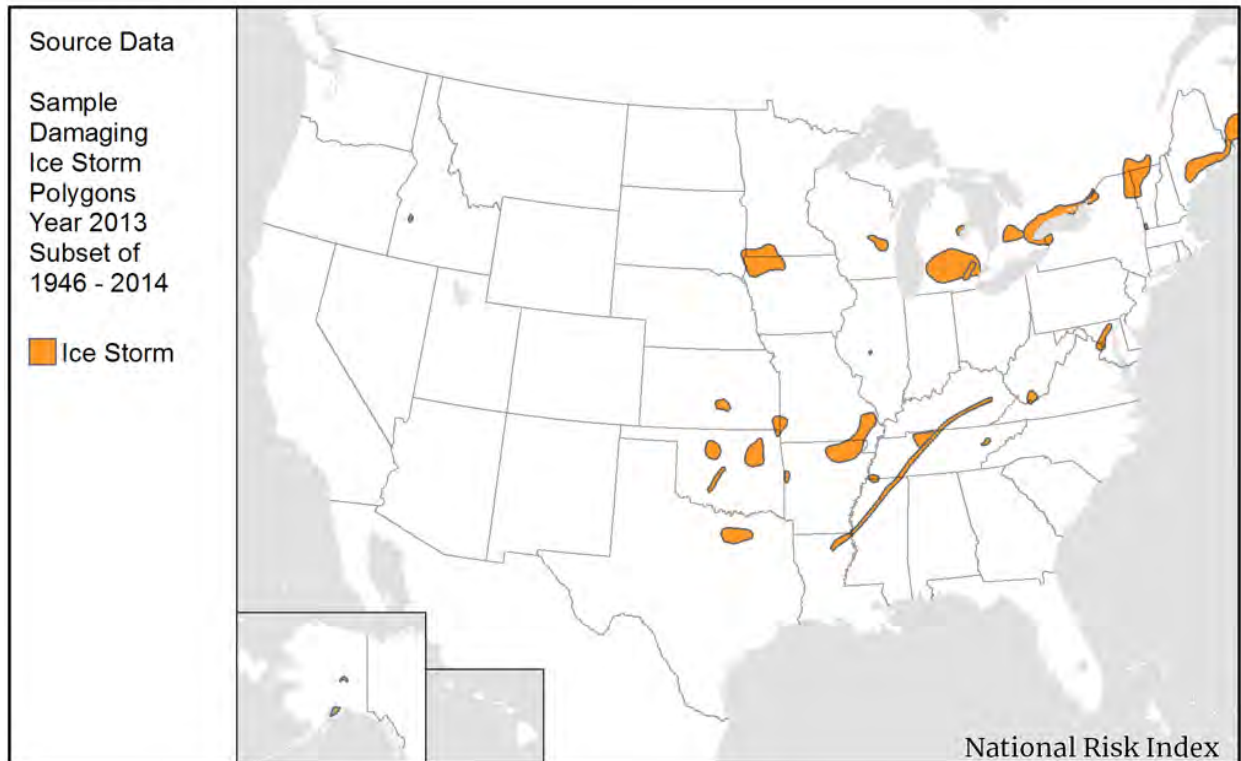
### 14.1. Ice Storm Spatial Source Data

**Historical Event Source:** USACE Cold Regions Research and Engineering Laboratory, Damaging Ice Storm GIS<sup>94</sup>

The Cold Regions Research and Engineering Laboratory Damaging Ice Storm GIS database includes footprint polygons representing the area where ice-sensitive structures (i.e., overhead power, phone and cable television lines, communication towers, and trees) were damaged by freezing rain storms in a subset of storms between 1946 and the spring of 2014, with modeled ice thicknesses designated as significant based on an established 50-year mean recurrence interval (see [Figure 64](#)). Start and end dates for Ice Storm occurrences are also included in the data. Ice Storms that cause only slippery roads are not included. This data source is not complete for all years in the period of record, as many weather stations did not begin storing electronic records until the early 1970s.

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<sup>94</sup> CRREL, USACE. (2014). Damaging Ice Storm GIS [online dataset]. Retrieved from <https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/490684/damaging-ice-storm-gis/>.



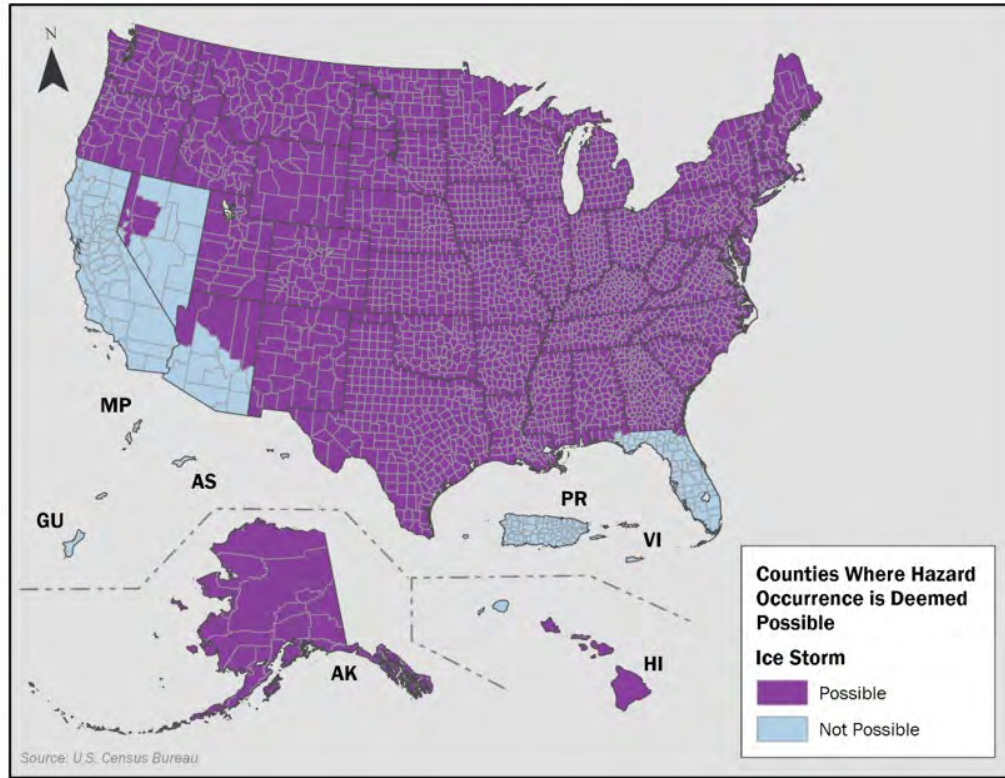
**Figure 64: Map of Sample Damaging Ice Storm Polygons**

## 14.2. Ice Storm Spatial Data Processing

Because the source data provides Ice Storm footprint polygons that work well, no spatial pre-processing was necessary beyond projecting the data to the North America Albers Equal Area Conic projection. However, some inaccuracies could be found in the storm event start and end dates, such as end dates that preceded their start dates or exceptionally long storms that were deemed suspect. Once Ice Storm durations were calculated, any negative or zero-day durations were set to 1, while any storms longer than 30 days were capped at 30. These durations were used to estimate exposure and annualized frequency.

### 14.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

To distinguish between areas with no Ice Storm occurrences and those where such occurrences are not deemed possible, a control table was generated to designate which counties have some probability of Ice Storm occurrence. This was initially determined by selecting only counties that intersected a past Ice Storm footprint polygon. However, this selection was widened to include all counties in states that intersected a past Ice Storm footprint polygon, except Florida and Hawai'i. Counties in Florida that intersected past Ice Storm footprint polygons were included as possible; however, the southern parts of the state that had not experienced an Ice Storm were not included. Any county that had sustained economic loss due to an Ice Storm as reported in SHELUDS was also included as one in which Ice Storm occurrence is possible. (See [Figure 65](#)).



**Figure 65: Map of Counties Deemed Possible for Ice Storm Occurrence**

### 14.3. Ice Storm Non-Spatial Data Processing

#### 14.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Ice Storm, in event-days, is computed as the number of distinct Ice Storm event polygons that intersect a 49-by-49-km fishnet grid cell multiplied by the number of duration days associated with each Ice Storm occurrence (see [Equation 79](#)).

#### Equation 79: Fishnet Ice Storm Event-Day Count

$$EventDayCount_{ISTM_{Fish}} = EventCount_{ISTM_{Fish}} \times \sum_{ISTM}^{Fish} Days_{ISTM}$$

where:

$EventDayCount_{ISTM_{Fish}}$  is the count of Ice Storm event-days calculated for a specific fishnet grid cell (in days).

$EventCount_{ISTM_{Fish}}$  is the count of distinct Ice Storm event polygons that intersect the fishnet grid cell.

$\sum_{ISTM}^{Fish} Days_{ISTM}$  is the sum of the duration days for each Ice Storm event polygon that intersects the fishnet grid cell (in days).

Historical event-day counts are supplied at the Census tract and county levels as the area-weighted Ice Storm event-day count of the fishnet grid cells that intersect the Census tract and county, respectively.

### 14.3.2. PERIOD OF RECORD

To capture the largest extent of credible data, records from Dec. 31, 1946 to Feb. 12, 2014 are analyzed. The period of record for which Ice Storm data are utilized is 67.16 years.

## 14.4. Ice Storm Exposure

To identify areas of exposure, the Ice Storm event-day polygons are intersected with the Census block polygons within the processing database. The resulting table contains the storm's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 52](#)).

**Table 52: Sample Data from the Ice Storm Census Block Intersection Table**

| <i>IceStormID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|-------------------|--------------------|---------------------------|
| 568               | 511610308024005    | 0.087504                  |
| 568               | 511610308024006    | 0.035518                  |
| 568               | 511610308024007    | 0.287145                  |

Because an Ice Storm event can occur over several days or a single day, an event-day basis was used to estimate exposure and annualized frequency as this method better captures the variability in duration between occurrences. To determine exposure value, the average coverage of an Ice Storm event-day is found by taking the sum of the products of the intersected areas for all storms multiplied by their event-day durations and dividing this sum by the total number of Ice Storm event-days for the Census block. This is divided by the total area of the Census block to calculate the average Ice Storm event-day coverage percentage and multiplied by the developed area building value density and the developed area population density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 80](#)). These Census block densities have been calculated by dividing the total exposure values (as recorded in Hazus 6.1) by the developed land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

**Equation 80: Census Block Ice Storm Exposure**

$$Exposure_{ISTM_{CB}Bldg} = \frac{\sum_{ISTM}^{CB} (IntsctArea_{ISTM_{CB}} \times Days_{ISTM})}{\sum_{ISTM}^{CB} (Days_{ISTM})} \Bigg/ Area_{CB} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{ISTM_{CB}Pop} = \left( \frac{\sum_{ISTM}^{CB} (IntsctArea_{ISTM_{CB}} \times Days_{ISTM})}{\sum_{ISTM}^{CB} (Days_{ISTM})} \Bigg/ Area_{CB} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

where:

$Exposure_{ISTM_{CB}Bldg}$  is the building value exposed to Ice Storm event-days for a specific Census block (in dollars).

$IntsctArea_{ISTM_{CB}}$  is the intersected area of the Ice Storm event polygon with the Census block (in square kilometers).

$Days_{ISTM}$  is the event-day duration of the Ice Storm event (in days).

$\sum_{ISTM}^{CB}$  is the sum for all Ice Storm event polygons intersecting the Census block.

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$DevAreaDen_{CB_{Bldg}}$  is the developed area building value density of the Census block (in dollars per square kilometer).

$Exposure_{ISTM_{CB}Pop}$  is the population equivalence value exposed to Ice Storm events for a specific Census block (in dollars).

$DevAreaDen_{CB_{Pop}}$  is the developed area population density of the Census block (in people per square kilometer).

$VSL$  is the Value of Statistical Life (\$13.7 million per person).

In cases where a Census block is deemed potentially at risk for Ice Storm damage, but has had no historical Ice Storm events, the exposure value is estimated to be the full Census block building value and population value. A low Historic Loss Ratio and low frequency of Ice Storm event-days will diminish the effect of using full Census block values in the final Expected Annual Loss calculation.

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value are

considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

#### 14.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 81](#)).

**Equation 81: Census Tract and County Ice Storm Exposure**

$$Exposure_{ISTM_{CT}Bldg} = \sum_{CB}^{CT} Exposure_{ISTM_{CB}Bldg}$$

$$Exposure_{ISTM_{Co}Bldg} = \sum_{CB}^{Co} Exposure_{ISTM_{CB}Bldg}$$

$$Exposure_{ISMT_{CT}Pop} = \sum_{CB}^{CT} Exposure_{ISTM_{CB}Pop}$$

$$Exposure_{ISTM_{Co}Pop} = \sum_{CB}^{Co} Exposure_{ISTM_{CB}Pop}$$

where:

$Exposure_{ISTM_{CT}Bldg}$  is the building value exposed to Ice Storm event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{ISTM_{CB}Bldg}$  is the summed value of all buildings exposed to Ice Storms for each Census block within the Census tract (in dollars).

$Exposure_{ISTM_{Co}Bldg}$  is the building value exposed to Ice Storm event-days in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{ISTM_{CB}Bldg}$  is the summed value of all buildings exposed to Ice Storms for each Census block within the county (in dollars).

$Exposure_{ISMT_{CT}Pop}$  is the population equivalence value exposed to Ice Storm event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{ISTM_{CB}Pop}$  is the summed value of all population equivalence exposed to Ice Storms for each Census block within the Census tract (in dollars).



$Exposure_{ISTM_{CoPop}}$  is the population equivalence value exposed to Ice Storm event-days in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{ISTM_{CBPop}}$  is the summed value of all population equivalence exposed to Ice Storms for each Census block within the county (in dollars).

## 14.5. Ice Storm Annualized Frequency

The annualized frequency value represents the estimated number of recorded Ice Storm occurrences, in event-days, each year for a specific area. This annualized frequency is calculated at the Census block level, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the Ice Storm footprint polygons from the source data as well as their corresponding computed duration days from the pre-processing of the data. The footprint polygons are intersected with a 49-by-49-km fishnet grid. The sum of Ice Storm event-days for the polygons intersecting each grid cell is recorded, and the Census block inherits this aggregated event-day count from the grid cell that encompasses it (see [Equation 79](#)). If the Census block intersects multiple fishnet grid cells, an area-weighted average count is calculated (see [Appendix D – Fishnet Occurrence Count](#)). Using this count, the Census block annualized frequency is calculated as in [Equation 82](#).

### Equation 82: Census Block Ice Storm Annualized Frequency

$$Freq_{ISTM_{CB}} = \frac{EventCount_{ISTM_{CB}}}{PeriodRecord_{ISTM}}$$

where:

$Freq_{ISTM_{CB}}$  is the area-weighted annualized frequency of Ice Storm event-days determined for a specific Census block (event-days per year).

$EventCount_{ISTM_{CB}}$  is the number of historical Ice Storm event-days calculated for the Census block.

$PeriodRecord_{ISTM}$  is the period of record for Ice Storm (67.16 years).

#### 14.5.1. MINIMUM ANNUAL FREQUENCY

If a Census block's historical Ice Storm event-day count is 0, but the Census block is part of a county that was designated as one in which Ice Storms are possible, the Census block is assigned the minimum annual Ice Storm frequency. This Minimum Annual Frequency is set at 0.01489, or once in the period of record (1 in 67.16 years).

### 14.5.2. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Ice Storm event-days at the Census tract and county level by the period of record, as the event count for annualized frequency is a fishnet area-weighted event count including Ice Storms that may have impacted the surrounding area but not the county or Census tract itself. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 83](#).

**Equation 83: Census Tract and County Area-Weighted Ice Storm Annualized Frequency Aggregation**

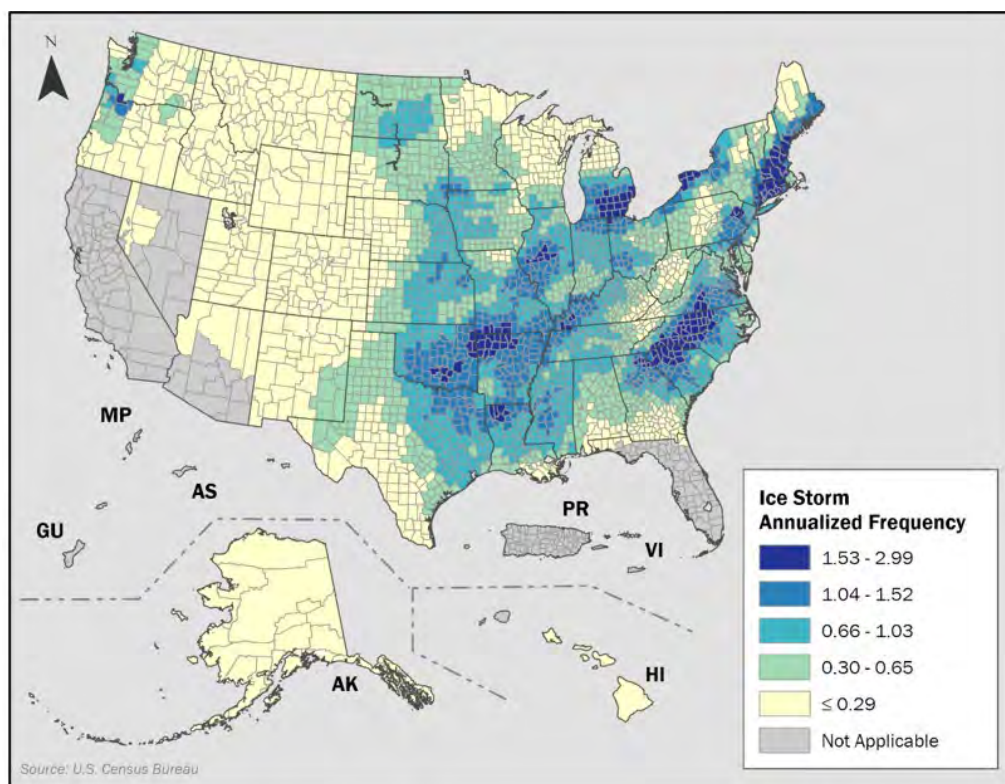
$$Freq_{ISTM_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{ISTM_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{ISTM_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{ISTM_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $Freq_{ISTM_{CT}}$ | is the area-weighted annualized frequency of Ice Storm event-days determined for a specific Census tract (event-days per year). |
| $Freq_{ISTM_{CB}}$ | is the area-weighted annualized frequency of Ice Storm event-days determined for a specific Census block (event-days per year). |
| $Area_{CB}$        | is the total area of the Census block (in square kilometers).                                                                   |
| $\sum_{CB}^{CT}$   | is the sum for all Census blocks in the Census tract.                                                                           |
| $Area_{CT}$        | is the total area of the Census tract (in square kilometers).                                                                   |
| $Freq_{ISTM_{Co}}$ | is the area-weighted Ice Storm annualized frequency determined for a specific county (event-days per year).                     |
| $\sum_{CB}^{Co}$   | is the sum for all Census blocks in the county.                                                                                 |
| $Area_{Co}$        | is the total area of the county (in square kilometers).                                                                         |

[Figure 66](#) displays Ice Storm annualized frequency at the county level.



**Figure 66: Ice Storm Annualized Frequency by County**

## 14.6. Ice Storm Historic Loss Ratio

The Ice Storm Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to an Ice Storm event-day, or the average rate of loss associated with the occurrence of an Ice Storm event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Ice Storm hazard type.

Loss data are provided by SHELUDUS<sup>95</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. One peril type is mapped to the hazard Ice Storm (see [Table 53](#)). These native records are expanded on an event-day basis (to a maximum of 31 event-days) and aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 53: Ice Storm Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELUDUS</i> | <i>Total SHELUDUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Ice                           | 4,053                              | 6,964                                |

<sup>95</sup> For Ice Storm loss information, SHELUDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

The Historic Loss Ratio exposure value used in the formula below is a county-level value that represents the dollar value of the total building value or the entire population of a county as recorded in Hazus 6.1. The Loss Ratio per Basis for each SHELdUS-documented event-day and each consequence type (building and population) is calculated using [Equation 84](#).

**Equation 84: Loss Ratio per Basis Calculation for a Single Ice Storm Event-Day**

$$LRB_{ISTM_{Co}CnsqType} = \frac{LOSS_{ISTM_{Co}CnsqType}}{HLRExposure_{Co}CnsqType}$$

where:

$LRB_{ISTM_{Co}CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Ice Storm event-day. Calculation is performed for each consequence type (building and population).

$LOSS_{ISTM_{Co}CnsqType}$  is the loss (by consequence type) experienced from the Ice Storm event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{Co}CnsqType$  is the total value (by consequence type) of the county estimated to have been exposed to the Ice Storm event-day (in dollars or people).

Ice Storm event-days may occur in areas without resulting in recorded loss to buildings or population. SHELdUS does not record events in which no loss occurred, so a number of zero-loss event-day records are inserted into the loss data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELdUS period of record (1996 through 2023). For Ice Storm, the historical event-day count is extracted using an intersection between the Ice Storm event-day polygons and the Census blocks. Unlike the count used for annualized frequency, this count is simply the summed duration days of distinct Ice Storm polygons that have intersected Census blocks in the county. An annual rate is calculated as the event-day count divided by the period of record of 67.16 years, and this rate is multiplied by the SHELdUS period of record of 28 years to estimate a historical event-day count for the appropriate time range.

If the number of loss-causing Ice Storm event-day records from SHELdUS is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for each county, Bayesian credibility weighting factors are computed and applied at several levels: county, 115km surrounding area buffer, 208km surrounding area buffer, regional, and national. The regional definition for Ice Storm is derived from the default National Risk Index Historic Loss Ratio Regions (see [Section 5.4](#)). For building and

population consequence types, Bayesian credibility weighting factors are computed and applied at each level for urban and rural counties separately (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

In an effort to correct for urban bias, a ceiling is applied to the Bayesian-adjusted population Historic Loss Ratio for Ice Storm. This is calculated as the average number of people (excluding zero population loss events) impacted by past Ice Storms per county divided by the county population. This affects a few highly populated counties where the Bayesian influence of injuries and fatalities in less populated surrounding counties may overinflate the Historic Loss Ratio of urban counties.

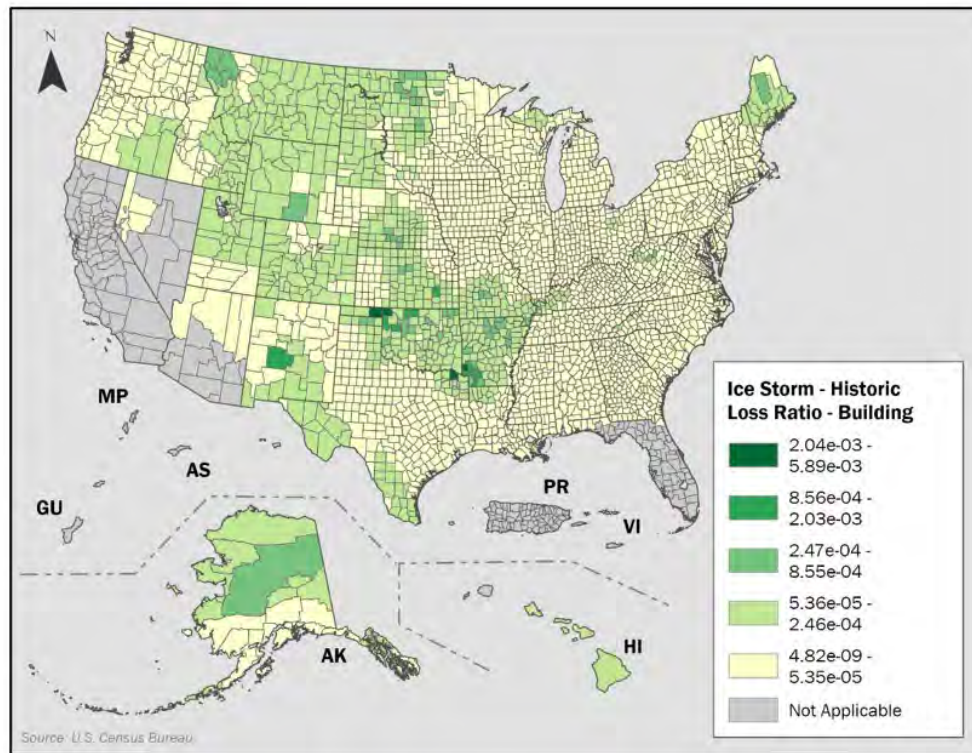
After the Loss Ratios per Basis are calculated for all Ice Storm events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Ice Storm .

**Table 54: Ice Storm Historic Loss Ratio Bayesian Weighting Level Sets**

| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>                           | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Population              | P1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 115km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Neighbor Counties (Touch & 208km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P4            | National Risk Index Historic Loss Ratio Regions | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P5            | National                                        | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
| Building                | B1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | B2            | Neighbor Counties (Touch & 115km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B3            | Neighbor Counties (Touch & 208km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B4            | National Risk Index Historic Loss Ratio Regions | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | B5            | National                                        | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

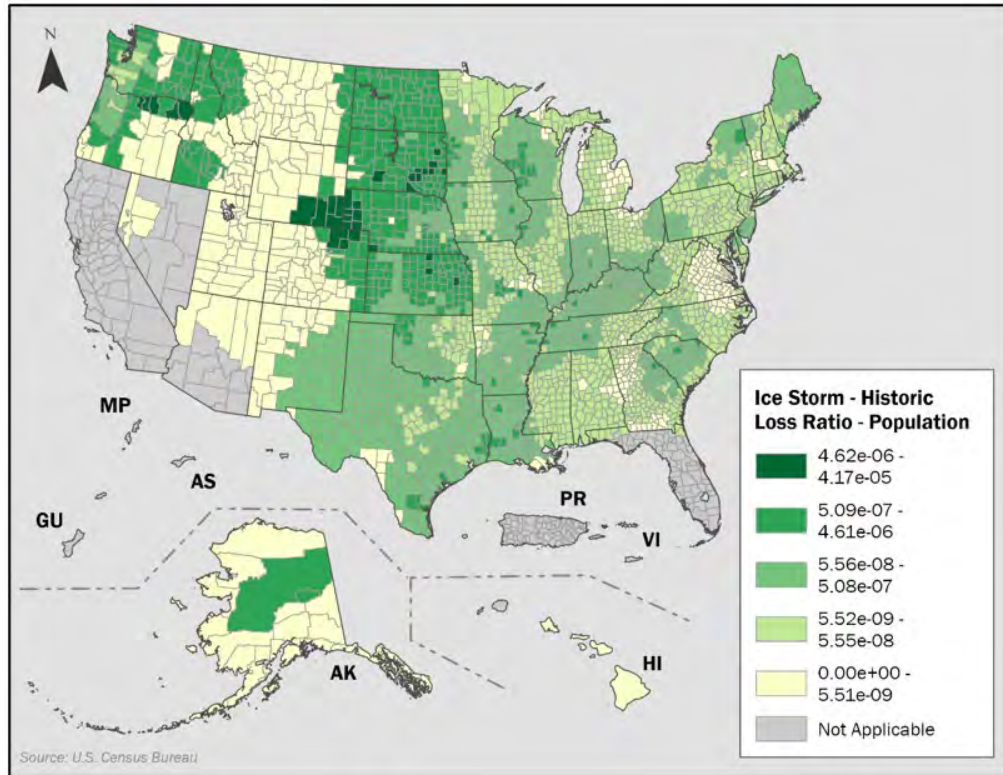
Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing techniques are then implemented to further refine the Historic Loss Ratio values for Ice Storm (see [Section 5.4.4](#)).

[Historic Loss Ratio Calculation Methodology](#)). [Figure 67](#) and [Figure 68](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Ice Storm. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 67: Ice Storm Bayesian-Adjusted Historic Loss Ratio – Building Value**





**Figure 68: Ice Storm Bayesian-Adjusted Historic Loss Ratio – Population**

## 14.7. Ice Storm Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 85](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 85: Census Block Expected Annual Loss to Ice Storms

$$EAL_{ISTM_{CB}Bldg} = Exposure_{ISTM_{CB}Bldg} \times Freq_{ISTM_{CB}} \times HLR_{ISTM_{CB}Bldg}$$

$$EAL_{ISTM_{CB}Pop} = Exposure_{ISTM_{CB}Pop} \times Freq_{ISTM_{CB}} \times HLR_{ISTM_{CB}Pop}$$

where:

$EAL_{ISTM_{CB}Bldg}$  is the building Expected Annual Loss due to Ice Storm occurrences for a specific Census block (in dollars).

$Exposure_{ISTM_{CB}Bldg}$  is the building value exposed to Ice Storm occurrences in the Census block (in dollars).

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $Freq_{ISTM_{CB}}$         | is the Ice Storm annualized frequency for the Census block (event-days per year).                                         |
| $HLR_{ISTM_{CB} Bldg}$     | is the Bayesian-adjusted building Historic Loss Ratio for Ice Storm for the Census block.                                 |
| $EAL_{ISTM_{CB} Pop}$      | is the population equivalence Expected Annual Loss due to Ice Storm occurrences for a specific Census block (in dollars). |
| $Exposure_{ISTM_{CB} Pop}$ | is the population equivalence value exposed to Ice Storm occurrences in the Census block (in dollars).                    |
| $HLR_{ISTM_{CB} Pop}$      | is the Bayesian-adjusted population Historic Loss Ratio for Ice Storm for the Census block.                               |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building and population equivalence Expected Annual Loss values at the Census block level as in [Equation 86](#).

**Equation 86: Census Tract and County Expected Annual Loss to Ice Storms**

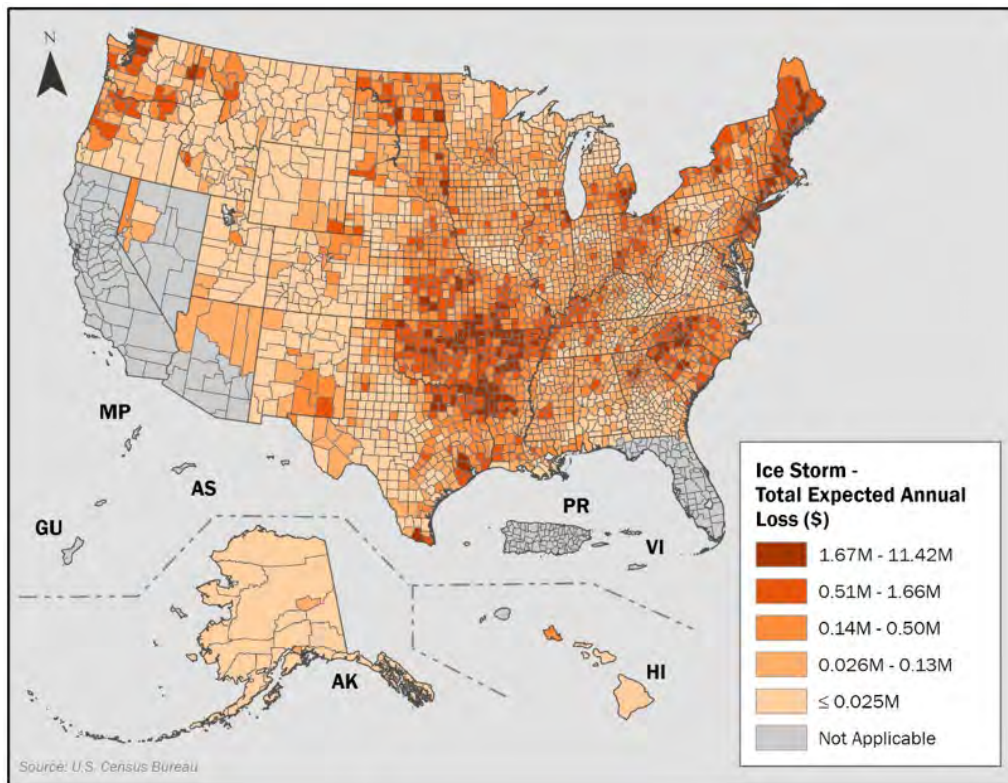
$$EAL_{ISTM_{CT}} = \sum_{CB}^{CT} EAL_{ISTM_{CB} Bldg} + \sum_{CB}^{CT} EAL_{ISTM_{CB} Pop}$$

$$EAL_{ISTM_{Co}} = \sum_{CB}^{Co} EAL_{ISTM_{CB} Bldg} + \sum_{CB}^{Co} EAL_{ISTM_{CB} Pop}$$

where:

|                                       |                                                                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{ISTM_{CT}}$                     | is the total Expected Annual Loss due to Ice Storm occurrences for a specific Census tract (in dollars).                                       |
| $\sum_{CB}^{CT} EAL_{ISTM_{CB} Bldg}$ | is the summed building Expected Annual Loss to Ice Storm occurrences for all Census blocks in the Census tract (in dollars).                   |
| $\sum_{CB}^{CT} EAL_{ISTM_{CB} Pop}$  | is the summed population equivalence Expected Annual Loss due to Ice Storm occurrences for all Census blocks in the Census tract (in dollars). |
| $EAL_{ISTM_{Co}}$                     | is the total Expected Annual Loss due to Ice Storm occurrences for a specific county (in dollars).                                             |
| $\sum_{CB}^{Co} EAL_{ISTM_{CB} Bldg}$ | is the summed building Expected Annual Loss due to Ice Storm occurrences for all Census blocks in the county (in dollars).                     |
| $\sum_{CB}^{Co} EAL_{ISTM_{CB} Pop}$  | is the summed population equivalence Expected Annual Loss due to Ice Storm events for all Census blocks in the county (in dollars).            |

[Figure 69](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Ice Storm occurrences.



**Figure 69: Total Expected Annual Loss by County to Ice Storm**

With the Ice Storm total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Ice Storm Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Ice Storm Expected Annual Loss for population by the total population of the community.

## 14.8. Ice Storm Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- There are many peril hazards within the SHELUS database that can be considered as either Ice Storm or Winter Weather. The definition used by the data development team means that

the losses have been incorporated into their respective hazards, but that these may differ from users' conceptualizations or expectations.

- The underlying data for Ice Storm occurrences, sourced from the United States Army Corps of Engineers' Cold Regions Research and Engineering Laboratory have not been updated since 2014. Any changes to trends in frequency and geographic extents of ice storm occurrences from 2014 through present day would therefore not be captured in loss estimations.

## 15. Inland Flooding

Inland Flooding occurs when streams and rivers exceed the capacity of their natural or constructed channels to accommodate water flow and water overflows the banks, spilling into adjacent low-lying, dry land (fluvial flooding), or when excessive rainfall overwhelms drainage systems leading to surface water accumulation (pluvial flooding). Inland flooding can occur anywhere it rains, even if there are no nearby waterbodies.

The Inland Flooding hazard calculates risk for all three consequence types: Building, Population, and Agriculture. Methods for the building consequence type received significant updates due to the incorporation of new source data. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 15.1. Inland Flooding Spatial Source Data

**Susceptible Area Source:** The Federal Emergency Management Agency, Geospatial Flood Risk Assessment

The Geospatial Flood Risk Assessment contains Hazus dasymetric developed area polygon features. Each feature has an attribute value that describes an average annual loss rate due to flood hazard associated with a one-story single-family home with a slab-on-grade foundation. The model and framework are built on years of investment in flood hazard information and analysis through the National Flood Insurance Program and Risk Mapping, Assessment, and Planning program. The Geospatial Flood Risk Assessment depicts inland flood losses derived from geographic factors such as elevation, distance to rivers, and relative elevation from a river, combining catastrophe modeling, National Flood Insurance Program historical data, and actuarial pricing approach. This information is used to inform potential annualized loss rates for different building types at the dasymetric Census block level.

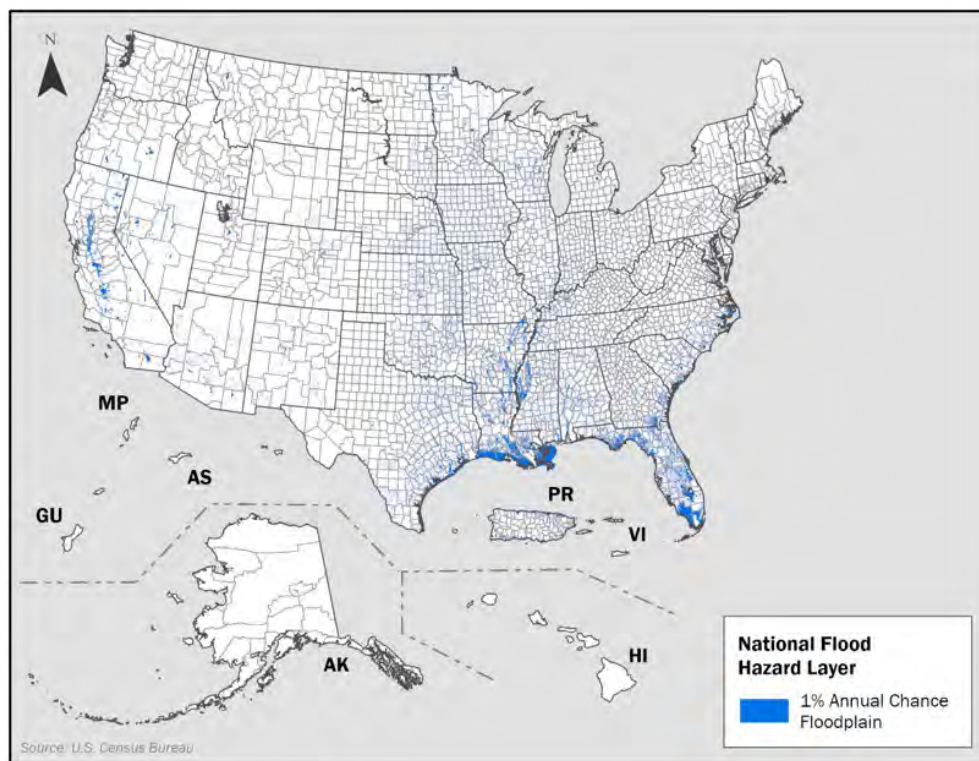
**Susceptible Area Source:** [The Federal Emergency Management Agency, National Flood Insurance Program, National Flood Hazard Layer](#)<sup>96</sup>

The National Flood Hazard Layer contains several layers depicting flood information, including levee locations, Flood Insurance Rate Map boundaries, and floodplain polygons. The polygons for the 1% annual chance floodplain were downloaded in shapefile format (see [Figure 70](#)) for use in the calculation of Inland Flooding exposure.

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<sup>96</sup> National Flood Insurance Program, FEMA. (2025). *NFHL* [online dataset]. Retrieved from <https://www.fema.gov/national-flood-hazard-layer-nfhl>.





**Figure 70: Map of 1% Annual Chance Floodplain**

**Susceptible Area Source:** CoreLogic SFHA Layer

The CoreLogic digitized floodplain boundaries supplement the Federal Emergency Management Agency's official digital National Flood Hazard Layer data in areas where only paper Flood Insurance Rate Maps exist. These boundaries have been compiled by CoreLogic through the digitization of existing paper flood maps and the use of legacy paper Federal Emergency Management Agency products. The Federal Emergency Management Agency has licensed this data from CoreLogic to supplement its National Flood Hazard Layer data while the Federal Emergency Management Agency engages with communities where digital data-coverage gaps exist in the Federal Emergency Management Agency National Flood Hazard Layer.

**Historical Event Source:** [National Centers for Environmental Information, Storm Events Database](#)<sup>97</sup>

The National Centers for Environmental Information Storm Events Database contains records of the occurrence of storms and other significant weather phenomena, including flooding events, since January 1950. Each flooding event record includes the affected counties, the dates of the event occurrence, and any reported loss. These records are used to calculate the annualized frequency for Inland Flooding.

<sup>97</sup> National Centers for Environmental Information (NCEI). (2020). *Storm Events Database, Version 3.1*. [online database]. Retrieved from <https://www.ncdc.noaa.gov/stormevents/>.



## 15.2. Inland Flooding Spatial Data Processing

The flood hazard areas in which the Flood Zone Category begins with “A” were extracted from the CoreLogic data and the National Flood Hazard Layer data. This selection criteria extracted the 1% annual chance flood hazard areas associated with possible Inland Flooding, as well as coastal hazard areas that experienced shallow flow or ponding with water depths of 1 to 3 feet (“AH”). These two selections from the source data were then combined to form a single layer of polygons for the 1% annual chance Inland Flood hazard type. The resulting layer was used in annualized frequency calculations as well as the calculation of exposure for population and agriculture.

For the building consequence type, the Geospatial Flood Risk Assessment provided baseline annualized loss rates at the Census block level and provided the foundation for calculating Expected Annual Loss for the building consequence type. These baseline loss rates describe the proportion of a Census block’s building value that would be lost due to flood hazard in a typical year if its entire building stock were made up of one-story single-family home with a slab-on-grade foundation.

To account for differences in physical flood susceptibility across building types, baseline loss rates were adjusted for each building category defined in the Hazus Flood model based upon their relative damage function values for the flood depth of 3 feet before being applied directly to their respective building values.

### 15.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

On examining the economic loss records in SHELDS, it was found that almost every county in the United States had sustained some form of loss due to Inland Flooding occurrences, so all counties were deemed possible for Inland Flooding occurrence.

## 15.3. Inland Flooding Non-Spatial Data Processing

### 15.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Inland Flooding, in event-days, is computed as the number of days in which Inland Flooding events (defined as having an Event Type of Flash Flood, Flood, Hail Flooding, Lakeshore Flood, Thunderstorm Winds/Flood, or Thunderstorm Winds/Flash Flood) were recorded in the National Centers for Environmental Information Storm Events Database within the county from January 1996 to December 2023. Multiple event records that occur on the same day in the same county are counted as a single Inland Flooding event as these recorded events are likely due to the same cause (heavy rain, for example) but occur in distinct parts of the county. This count is only performed for counties that intersect the 1% annual chance inland floodplain. Historical event-day counts are also supplied at the Census tract level. These values are inherited from the parent county as the exact location of the occurrence within the county cannot always be determined from the National Centers for Environmental Information Storm Events Database record.

### 15.3.2. PERIOD OF RECORD

The Inland Flooding annualized frequency calculation is based on the number of recorded Inland Flooding events in the National Centers for Environmental Information Storm Events Database from Jan. 01, 1996 to Dec. 31, 2023, so the period of record is 27 years.

### 15.4. Inland Flooding Exposure

To identify areas of exposure, the inland floodplain polygons were intersected with the Census block developed area polygons within the processing database. The resulting table contains the floodplain polygon's unique identifier, Census block number, the intersected area, the developed area of intersection, and the area of intersection containing crop or pastureland (see [Table 55](#)). All area values are square kilometers.

**Table 55: Sample Data from the Inland Flood Zone Census Block Intersection Table**

| <i>FloodZoneInland10<br/>OyrID</i> | <i>CensusBlock</i> | <i>AreaDevelopedKm2</i> | <i>IntersectedAreaKm2</i> | <i>AreaCropPastureKm2</i> |
|------------------------------------|--------------------|-------------------------|---------------------------|---------------------------|
| 413                                | 150010202021103    | 0.005357                | 0.005357                  | 0                         |
| 2805                               | 150010202021103    | 0.003001                | 0.003013                  | 0                         |
| 8069                               | 150010203001007    | 0.05579                 | 0.05579                   | 0.000463                  |

To determine exposure value for population, the sum of the developed areas of the inland floodplain polygons for each Census block is multiplied by the developed area population density of the Census block to model the conservative-case concentration of exposure within the Census block. To determine exposure value for agriculture, the sum of the agriculture area intersecting the inland floodplain polygons for each Census block is multiplied by the total agriculture area value density (see [Equation 87](#)). These densities have been calculated by dividing the total exposure values (population as recorded in Hazus 6.0) by the developed area or agriculture land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

#### Equation 87: Census Block Inland Flooding Exposure

$$\begin{aligned}
 Exposure_{IFLD_{CB_{Bldg}}} &= \sum IntsctArea_{IFLD_{Dev_{CB}}} \times DevAreaDen_{CB_{Bldg}} \\
 Exposure_{IFLD_{CB_{Pop}}} &= \left( \sum IntsctArea_{IFLD_{Dev_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL \\
 Exposure_{IFLD_{CB_{Ag}}} &= \sum IntsctArea_{IFLD_{Ag_{CB}}} \times AgValueDen_{CB}
 \end{aligned}$$

where:

|                                 |                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{IFLD\ CB\ Bldg}$     | is the population equivalence value exposed to Inland Flooding in a specific Census block (in dollars).                                          |
| $\sum IntsctArea_{IFLD\ DevCB}$ | is the sum of the intersected developed areas of inland floodplain polygons with the Census block (in square kilometers).                        |
| $DevAreaDen_{CB\ Pop}$          | is the developed area population density of the Census block (in people per square kilometer).                                                   |
| $VSL$                           | is the Value of Statistical Life (\$13.7 million per person).                                                                                    |
| $Exposure_{IFLD\ CB\ Pop}$      | is the agriculture value exposed to Inland Flooding in a specific Census block (in dollars).                                                     |
| $\sum IntsctArea_{IFLD\ AgCB}$  | is the sum of the intersected agriculture areas of inland floodplain polygons with the Census block (in square kilometers).                      |
| $AgValueDen_{CB}$               | is the agriculture value density of the Census block (in dollars per square kilometer).                                                          |
| $Exposure_{IFLD\ CB\ Bldg}$     | is the building value exposed to Inland Flooding in a specific Census block (in dollars).                                                        |
| $IntsctArea_{IFLD\ DevCB}$      | is the sum of the intersected developed areas of Geospatial Flood Risk Assessment premium polygons with the Census block (in square kilometers). |
| $DevAreaDen_{CB\ Bldg}$         | is the developed area building value density of the Census block (in dollars per square kilometer).                                              |

Because the exposure model uses a conservative-case concentration of exposure, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population value and the Census of Agriculture-reported crop and livestock value for the Census block are considered ceilings on exposure. For example, if the calculated exposed population value exceeds the Hazus-recorded population value, then the Hazus-recorded population value is used as the population exposure value for the Census block.

Exposure values for the building consequence type are derived from the mapping of GFRA building loss rates to the Hazus building inventory. If the Geospatial Flood Risk Assessment loss rate associated with a particular Census block is greater than 0, then the building exposure for the Census block is equal to its Hazus building value. Otherwise, if the Geospatial Flood Risk Assessment

loss rate associated with a particular Census block is equal to 0, then the building exposure for the Census block is also equal to zero.

#### 15.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 88](#)).

**Equation 88: Census Tract and County Inland Flooding Exposure Aggregation**

$$Exposure_{IFLDCTBldg} = \sum_{CB}^{CT} Exposure_{IFLDCB Bldg}$$

$$Exposure_{IFLDCOBldg} = \sum_{CB}^{CO} Exposure_{IFLDCB Bldg}$$

$$Exposure_{IFLDCTPop} = \sum_{CB}^{CT} Exposure_{IFLDCBPop}$$

$$Exposure_{IFLDCOPop} = \sum_{CB}^{CO} Exposure_{IFLDCBPop}$$

$$Exposure_{IFLDCTAg} = \sum_{CB}^{CT} Exposure_{IFLDCB Ag}$$

$$Exposure_{IFLDCOAg} = \sum_{CB}^{CO} Exposure_{IFLDCB Ag}$$

where:

$Exposure_{IFLDCTBldg}$  is the building value exposed to Inland Flooding in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{IFLDCB Bldg}$  is the summed value of all buildings exposed to Inland Flooding for each Census block within the Census tract (in dollars).

$Exposure_{IFLDCOBldg}$  is the building value exposed to Inland Flooding in a specific county (in dollars).

$\sum_{CB}^{CO} Exposure_{IFLDCB Bldg}$  is the summed value of all buildings exposed to Inland Flooding for each Census block within the county (in dollars).

$Exposure_{IFLD_{CTPop}}$  is the population equivalence value exposed to Inland Flooding in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{IFLD_{CBPop}}$  is the summed value of all population equivalence exposed to Inland Flooding for each Census block within the Census tract (in dollars).

$Exposure_{IFLD_{COPop}}$  is the population equivalence value exposed to Inland Flooding in a specific county (in dollars).

$\sum_{CB}^{CO} Exposure_{IFLD_{CBPop}}$  is the summed value of all population equivalence exposed to Inland Flooding for each Census block within the county (in dollars).

$Exposure_{IFLD_{CTAg}}$  is the agriculture value exposed to Inland Flooding in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{IFLD_{CBAg}}$  is the summed value of all agriculture value exposed to Inland Flooding for each Census block within the Census tract (in dollars).

$Exposure_{IFLD_{COAg}}$  is the agriculture value exposed to Inland Flooding in a specific county (in dollars).

$\sum_{CB}^{CO} Exposure_{IFLD_{CBAg}}$  is the summed value of all agriculture value exposed to Inland Flooding for each Census block within the county (in dollars).

## 15.5. Inland Flooding Annualized Frequency

The annualized frequency value represents the number of Inland Flooding occurrences, in event-days, each year over the period of record (27 years). Annualized frequency is initially calculated at the county level. The Census tracts and Census blocks inherit annualized frequency values from the counties that contain them, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the National Centers for Environmental Information Storm Events Database Inland Flooding events for the county (see [Section 15.5 Inland Flooding Annualized Frequency](#)) and divide by the period of record as in [Equation 89](#). Multiple event records that occur on the same day in the same county are counted as a single Inland Flooding occurrence.

**Equation 89: County Inland Flooding Annualized Frequency**

$$Freq_{IFLD_{CO}} = \frac{EventDayCount_{IFLD_{CO}}}{PeriodRecord_{IFLD}}$$

where:

$Freq_{IFLD_{CO}}$  is the annualized frequency of Inland Flooding events determined for a specific county (event-days per year).

$EventDayCount_{IFLD_{CO}}$  is the total number of Inland Flooding event-days (from the National Centers for Environmental Information Storm Events Database) that have impacted the county.

$PeriodRecord_{IFLD}$  is the period of record for Inland Flooding (27 years).

**15.5.1. MINIMUM ANNUAL FREQUENCY**

If a county intersects the 1% annual chance inland floodplain but has not experienced an Inland Flooding event-day, it is assigned a Minimum Annual Frequency of 0.01 or once in 100 years. If a county intersects the Geospatial Flood Risk Assessment layer but neither intersects the 1% annual chance inland floodplain nor has not experienced an Inland Flooding event-day, it is assigned a Minimum Annual Frequency of 0.005 (i.e., the mid-point between 0.01 and 0).

**15.5.2. ANNUALIZED FREQUENCY INHERITANCE**

The Census tracts and Census blocks inherit their annualized frequency values from the parent counties that contain them as in [Equation 90](#).

**Equation 90: Census Block and Tract Inland Flooding Annualized Frequency Inheritance**

$$Freq_{IFLD_{CB}} = Freq_{IFLD_{CT}} = Freq_{IFLD_{CO}}$$

where:

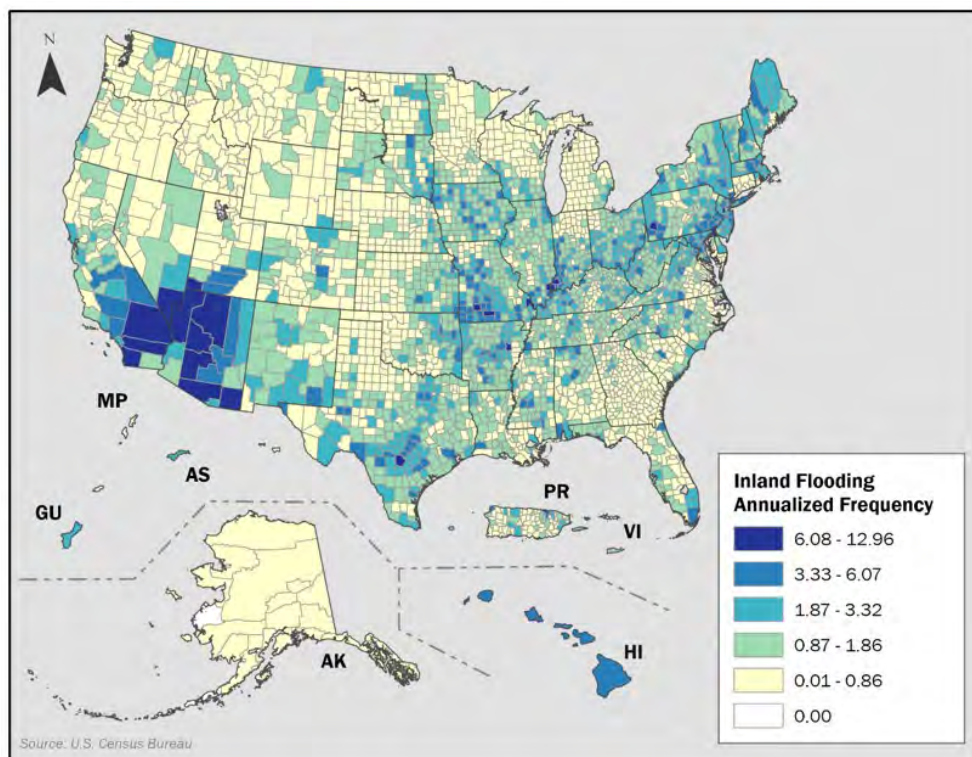
$Freq_{IFLD_{CB}}$  is the inherited annualized frequency of Inland Flooding event-days for a specific Census block within the parent county.

$Freq_{IFLD_{CT}}$  is the inherited annualized frequency of Inland Flooding event-days for a specific Census tract within the parent county.

$Freq_{IFLD_{CO}}$  is the annualized frequency of Inland Flooding event-days associated with a specific county.



[Figure 71](#) displays Inland Flooding annualized frequency at the county level.



**Figure 71: Inland Flooding Annualized Frequency by County**

## 15.6. Inland Flooding Historic Loss Ratio

The Inland Flooding Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to an Inland Flooding occurrence, or the average rate of loss associated with an Inland Flooding occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Inland Flooding hazard type for population and agriculture.<sup>98</sup>

For the Population and Agriculture consequence types, loss data are provided by SHELUDUS<sup>99</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDUS events from 1996 to 2023 are included in the Historic Loss Ratio calculation. Eight peril types are mapped to the hazard Inland Flooding (see [Table 56](#)). These native records are expanded on an event-day basis (to a maximum of 31 days) and aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

<sup>98</sup> Because losses for the Building consequence type are determined by loss rates obtained from the Geospatial Flood Risk Assessment methodology, no Inland Flooding Historic Loss Ratio values are computed for the Building consequence type.

<sup>99</sup> For Inland Flooding loss information, SHELUDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

**Table 56: Inland Flooding Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELUS</i> | <i>Total SHELUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-----------------------------|----------------------------------|--------------------------------------|
| Flood-Flash                 | 34,251                           | 37,286                               |
| Flood-Ice Jam               | 43                               | 136                                  |
| Flooding                    | 24,122                           | 108,504                              |
| Flood-Lakeshore             | 140                              | 668                                  |
| Flood-Lowland               | 0                                | 0                                    |
| Flood-Inland                | 60                               | 647                                  |
| Flood-Small Stream          | 256                              | 359                                  |
| Flood-Snowmelt              | 0                                | 0                                    |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is the value of the county's area that is susceptible to Inland Flooding. This value is determined by summing the developed area density or agriculture area density exposure values of the Census blocks that intersect the layer of the 1% annual chance floodplain (see [Section 15.4 Inland Flooding Exposure](#)). To prevent inflating the Loss Ratios per Basis of counties for which the areas of intersection with the floodplain were very small, counties with a calculated agriculture value exposure less than \$10,000 or a calculated population exposure less than one person were given an Loss Ratio per Basis of 0 for the consequence types that did not meet its respective threshold. The Loss Ratio per Basis for each SHELUS-documented event-day and each consequence type (population and agriculture) is calculated using [Equation 91](#).

**Equation 91: Loss Ratio per Basis Calculation for a Single Inland Flooding Event-Day**

$$LRB_{IFLD_{CoCnsqType}} = \frac{LOSS_{IFLD_{CoCnsqType}}}{HLRExposure_{IFLD_{CoCnsqType}}}$$

where:

$LRB_{IFLD_{CoCnsqType}}$

is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Inland Flooding event-day. Calculation is performed for each consequence type (population and agriculture).

$LOSS_{IFLD_{CoCnsqType}}$

is the loss (by consequence type) experienced from the Inland Flooding event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{IFLDcoCnsqType}$  is the value (by consequence type) of the susceptible area estimated to have been exposed to the Inland Flooding event-day (in dollars or people).

Inland Flooding event-days may occur in areas without resulting in recorded loss to buildings, population, or agriculture. SHELDUS does not record event-days in which no loss occurred, so a number of zero-loss event-days are inserted into the data to align the event-day count in the Historic Loss Ratio calculation to the historical event-day count experienced within the SHELDUS period of record (1996 to 2023). For Inland Flooding, the historical event-day count is computed as the number of Inland Flooding event-days recorded in the National Centers for Environmental Information Storm Events Database that have occurred within the county. (Multiple event records that occur on the same day in the same county are counted as a single event-day.) The period of record for both the SHELDUS and National Centers for Environmental Information data is the same 27 years, so the count does not need to be scaled by an annual rate.

If the number of loss-causing Inland Flooding event-day records from SHELDUS is less than the historical event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for each county, Bayesian credibility weighting factors are computed and applied at two levels: county and regional. The regional definition for Inland Flooding is derived from the Federal Emergency Management Agency regions with Regions 1, 2, and 3 merged (see [Section 5.4.4 Historic Loss Ratio](#) Calculation Methodology).

To address overestimation of population impacts, the Bayesian-adjusted population Historic Loss Ratios are compared to the ratio of the average number of people impacted (excluding zero population loss events) divided by the county population. The smaller of these two values is used as the county's population Historic Loss Ratio.

After the Loss Ratios per Basis are calculated for all Inland Flooding events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Agriculture and Population Historic Loss Ratios for Inland Flooding.

**Table 57: Inland Flooding Historic Loss Ratio Bayesian Weighting Level Sets**

| <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>                           | <b>Set Refinement Filter</b>                        | <b>Set Utilization Threshold (Set Must Have)</b> |
|-------------------------|---------------|-------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A2            | Neighbor Counties (Touch & 208km)               | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A3            | National Risk Index Historic Loss Ratio Regions | N/A                                                 | N/A                                              |
|                         | A4            | National                                        | N/A                                                 | N/A                                              |
| Population              | P1            | County                                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 208km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | National Risk Index Historic Loss Ratio Regions | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P4            | National                                        | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing techniques are then implemented to further refine the Historic Loss Ratio values for Inland Flooding (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 72](#), [Figure 73](#) and [Figure 74](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Inland Flooding. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

For the Building consequence type, surrogate Historic Loss Ratios are calculated from the resulting Expected Annual Loss building value (see [Equation 92](#)).

**Equation 92: Historic Loss Ratio Building Calculation for Inland Flooding**

$$HLR_{IFLD Co Bldg} = \frac{EAL_{IFLD Co Bldg}}{AFREQ_{IFLD Co} \times EXP_{IFLD Co Bldg}}$$

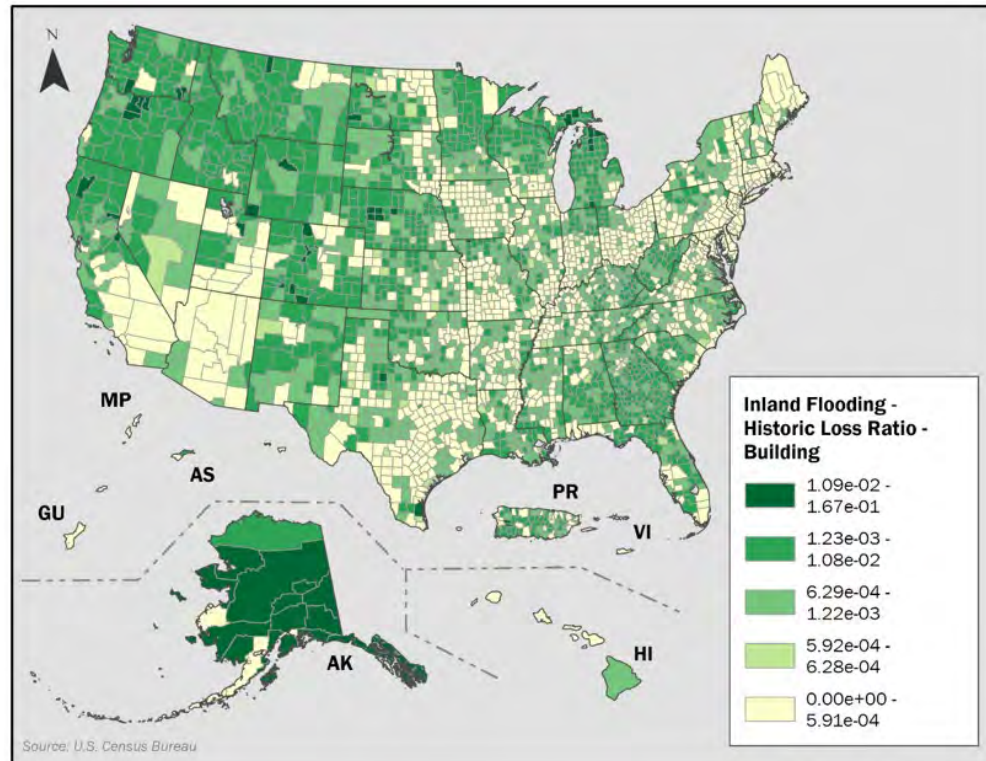
where:

$HLR_{IFLD Co Bldg}$  is the Building consequence type Historic Loss Ratio of a county for Inland Flooding.

$EAL_{IFLD_{Co}Bldg}$  is the Building Expected Annual Loss of a county for Inland Flooding.

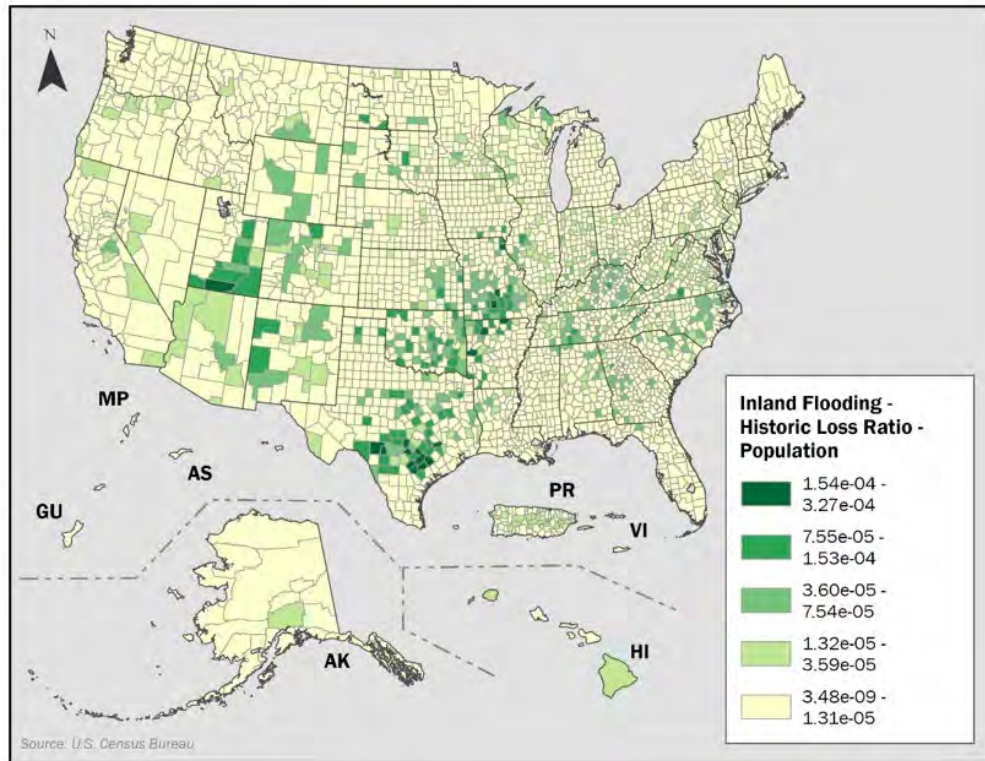
$AFREQ_{IFLD_{Co}}$  is the annualized frequency of a county for Inland Flooding.

$EXP_{IFLD_{Co}Bldg}$  is the sum of the Census block Building exposure values (in dollars).

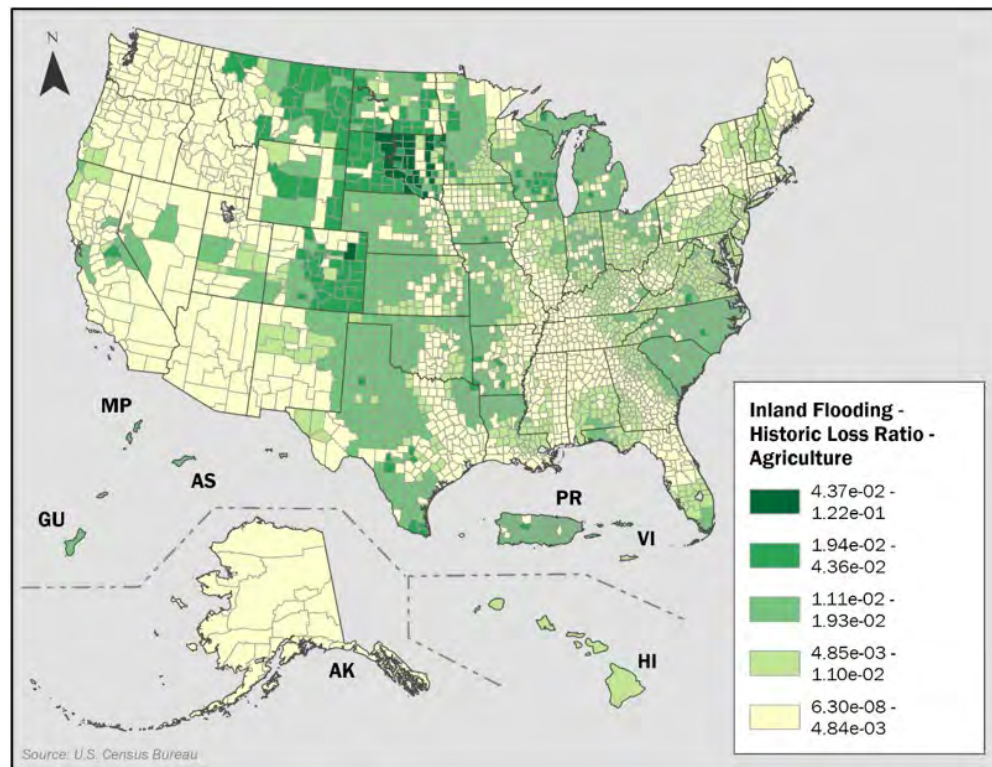


**Figure 72: Inland Flooding Bayesian-Adjusted Historic Loss Ratio – Building**





**Figure 73: Inland Flooding Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 74: Inland Flooding Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**



## 15.7. Inland Flooding Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss for population and agriculture is computed at the Census block level as in [Equation 93](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 93: Census Block Expected Annual Loss for Inland Flooding

$$EAL_{IFLDCB_{Pop}} = Exposure_{IFLDCB_{Pop}} \times Freq_{IFLDCB_{Pop}} \times HLR_{IFLDCB_{Pop}}$$

$$EAL_{IFLDCB_{Ag}} = Exposure_{IFLDCB_{Ag}} \times Freq_{IFLDCB_{Ag}} \times HLR_{IFLDCB_{Ag}}$$

where:

$Freq_{IFLDCB_{Pop}}$  is the Inland Flooding annualized frequency for the Census block.

$EAL_{IFLDCB_{Pop}}$  is the population equivalence Expected Annual Loss due to Inland Flooding occurrences for a specific Census block (in dollars).

$Exposure_{IFLDCB_{Pop}}$  is the population equivalence value exposed to Inland Flooding occurrences in the Census block (in dollars).

$HLR_{IFLDCB_{Pop}}$  is the Bayesian-adjusted population Historic Loss Ratio for Inland Flooding for the Census block.

$EAL_{IFLDCB_{Ag}}$  is the agriculture Expected Annual Loss due to Inland Flooding occurrences for a specific Census block (in dollars).

$Exposure_{IFLDCB_{Ag}}$  is the agriculture value exposed to Inland Flooding occurrences in the Census block (in dollars).

$HLR_{IFLDCB_{Ag}}$  is the Bayesian-adjusted agriculture Historic Loss Ratio for Inland Flooding for the Census block.

For building, Expected Annual Loss is calculated using baseline annual loss rates for each Census block from the Geospatial Flood Risk Assessment, build-type adjustment factors derived from damage functions in the Hazus flood model, and Census block building value distributions across building types from the Hazus 6.1 inventory (see [Equation 94](#)).

**Equation 94: Census Block Expected Annual Loss for Building**

$$EAL_{IFLD_{CB_{Bldg}}} = \sum_{BT}^{CB} (BuildValue_{CB_{BT}} \times BLR_{CB} \times Adj_{BT})$$

$$Adj_{BT} = \frac{DmgFun_{BT}}{DmgFun_{Baseline}}$$

where:

|                          |                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{IFLD_{CB_{Bldg}}}$ | is the building Expected Annual Loss due to Inland Flooding for a particular Census block.                                                                                                 |
| $\sum_{BT}^{CB}$         | is the sum of Expected Annual Loss due to Inland Flooding across building types within a particular Census block.                                                                          |
| $BuildValue_{CB_{BT}}$   | is the Hazus 6.0 building value for a particular Census block and building type.                                                                                                           |
| $BLR_{CB}$               | is the baseline building loss rate obtained from the Geospatial Flood Risk Assessment for a particular Census block.                                                                       |
| $Adj_{BT}$               | is the building loss rate adjustment factor for a particular building type.                                                                                                                |
| $DmgFun_{BT}$            | is the Hazus Flood structure damage function value associated with an average 100-yr event flood depth for a particular building type.                                                     |
| $DmgFun_{Baseline}$      | is the Hazus Flood structure damage function value associated with an average 100-yr event flood depth for the representative building type used in the Geospatial Flood Risk Assessment . |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 95](#).

**Equation 95: Census Tract and County Expected Annual Loss to Inland Flooding**

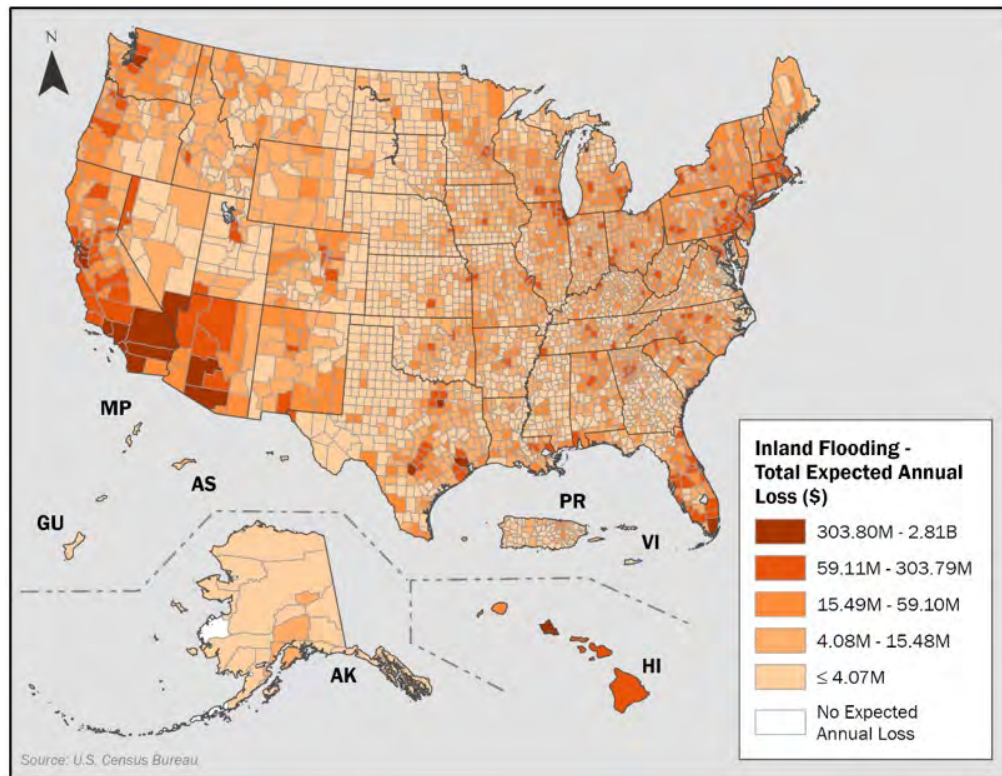
$$EAL_{IFLD_{CT}} = \sum_{CB}^{CT} EAL_{IFLD_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{IFLD_{CB_{Pop}}} + \sum_{CB}^{CT} EAL_{IFLD_{CB_{Ag}}}$$

$$EAL_{IFLD_{CO}} = \sum_{CB}^{CO} EAL_{IFLD_{CB_{Bldg}}} + \sum_{CB}^{CO} EAL_{IFLD_{CB_{Pop}}} + \sum_{CB}^{CO} EAL_{IFLD_{CB_{Ag}}}$$

where:

|                                       |                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{IFLD\ CT}$                      | is the total Expected Annual Loss due to Inland Flooding occurrences for a specific Census tract (in dollars).                           |
| $\sum_{CB}^{CT} EAL_{IFLD\ CB\ Bldg}$ | is the summed building Expected Annual Loss due to Inland Flooding for all Census blocks in the Census tract (in dollars).               |
| $\sum_{CB}^{CT} EAL_{IFLD\ CB\ Pop}$  | is the summed population equivalence Expected Annual Loss due to Inland Flooding for all Census blocks in the Census tract (in dollars). |
| $\sum_{CB}^{CT} EAL_{IFLD\ CB\ Ag}$   | is the summed agriculture Expected Annual Loss due to Inland Flooding for all Census blocks in the Census tract (in dollars).            |
| $EAL_{IFLD\ CO}$                      | is the total Expected Annual Loss due to Inland Flooding occurrences for a specific county (in dollars).                                 |
| $\sum_{CB}^{CO} EAL_{IFLD\ CB\ Bldg}$ | is the summed building Expected Annual Loss due to Inland Flooding for all Census blocks in the county (in dollars).                     |
| $\sum_{CB}^{CO} EAL_{IFLD\ CB\ Pop}$  | is the summed population equivalence Expected Annual Loss due to Inland Flooding for all Census blocks in the county (in dollars).       |
| $\sum_{CB}^{CO} EAL_{IFLD\ CB\ Ag}$   | is the summed agriculture Expected Annual Loss due to Inland Flooding for all Census blocks in the county (in dollars).                  |

[Figure 75](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Inland Flooding occurrences.



**Figure 75: Total Expected Annual Loss by County to Inland Flooding**

With the Inland Flooding total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Inland Flooding Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Inland Flooding Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Inland Flooding Expected Annual Loss for agriculture by the total agriculture value of the community.

## 15.8. Inland Flooding Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The Geospatial Flood Risk Assessment loss rates used to estimate inland flooding risks are based on an assumed building type of a slab-on-grade, single-family home, with a replacement cost value of \$250,000. This loss rate is useful as a figure for comparative risk

that exists across the nation, but the rate itself is unable to differentiate between different building types.

- Loss calculations use generalized damage curves that correspond to certain types of buildings and may not accurately reflect the variety of building types in a given area. There is the potential for underestimation or overestimation of losses due to reliance on generalized damage functions rather than specific empirical data.

## 16. Landslide

A Landslide is the movement of a mass of rock, debris, or earth down a slope.

The Landslide hazard incorporates the 2024 U.S. Geological Survey National Landslide Susceptibility Layer and the 2025 U.S. Geological Survey United States county-level landslide frequency data to determine Expected Annual Loss values for the Continental United States, Alaska, and Hawai'i. For landslide risk in Puerto Rico, the 2024 U.S. Geological Survey National Landslide Susceptibility Layer was used for susceptibility, but an independent method developed in partnership with subject matter experts was needed for landslide frequency. Puerto Rico landslide frequency was calculated using an inventory of Hurricane Maria-induced landslide locations to estimate landslide distribution and a post-1950's inventory of landslide inducing events to calculate an annualized frequency. The Landslide hazard only calculates risk for Building and Population consequence types; it does not include Agriculture consequences. For a summary of the Hazard Data Characteristics, please see [Appendix B](#) – Hazard Data Characteristics Comparison.

### 16.1. Landslide Spatial Source Data

**Susceptible Area Source:** [Slope-Relief Threshold Landslide Susceptibility Models for the United States and Puerto Rico](#)<sup>100</sup>

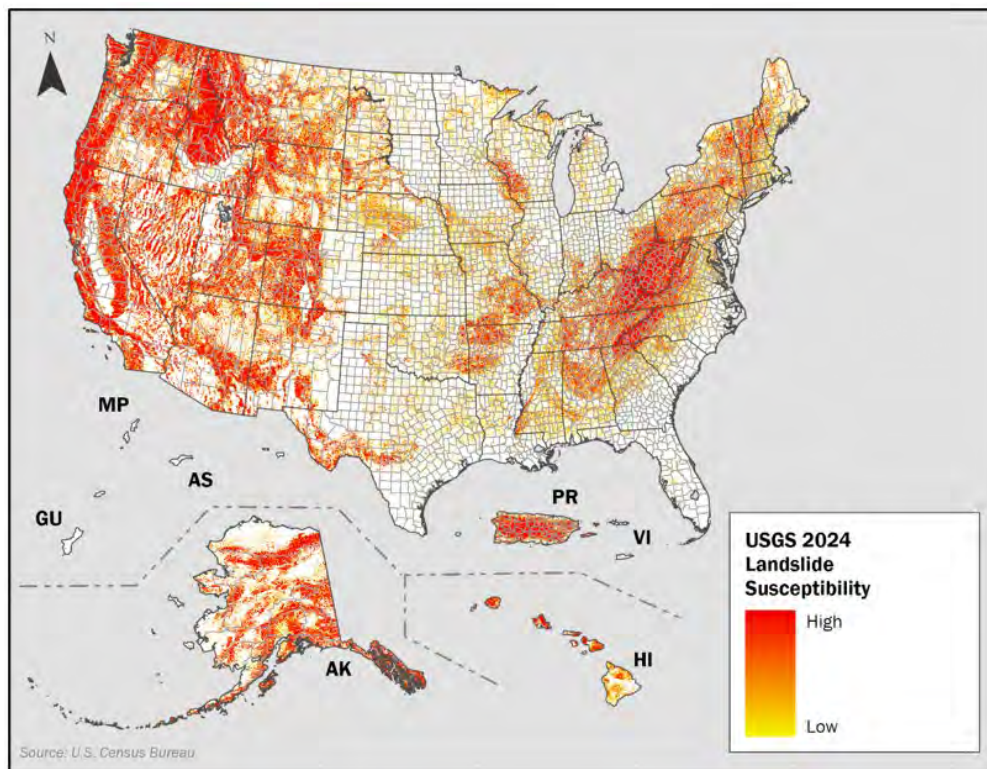
Per Mirus et al.'s 2024 study<sup>101</sup>, high-resolution 10-meter DEM data, in conjunction with a historical record of over 600,000 landslides<sup>102</sup> were used to identify landslide susceptible locations across the continental United States, Alaska, Hawai'i, and Puerto Rico ([Figure 76](#)) based on slope and relief variables. The results were then downsampled to a 90-meter grid to account for uncertainty in the DEM and historical landslide positions, counting all the susceptible 10-meter pixels within the 90-meter pixel to create a gradational susceptibility scale from 0 to 81. In the susceptibility scale 0 indicates areas not susceptible to a landslide, while 81 means that the entire grid pixel was susceptible to a landslide event.

<sup>100</sup> Belair, G.M., Jones, J.M., Martinez, S.N., Mirus, B.B., and Wood, N.J., 2024, Slope-Relief Threshold Landslide Susceptibility Models for the United States and Puerto Rico: U.S. Geological Survey data release, <https://doi.org/10.5066/P13KAGU3>.

<sup>101</sup> Mirus, B. B., Belair, G. M., Wood, N. J., Jones, J., & Martinez, S. N. (2024). Parsimonious high-resolution and slide susceptibility modeling at continental scales. *AGU Advances*, 5, e2024AV001214. <https://doi.org/10.1029/2024AV001214>

<sup>102</sup> Belair, G.M., Jones, E.S., Luna, L.V., and Mirus, B.B., 2019, Landslide Inventories across the United States (ver. 3.0, February 2025): U.S. Geological Survey data release, <https://doi.org/10.5066/P14AJF8I>





**Figure 76: Landslide Hazard Map Raster**

**Historical Occurrence Source:** [Map of slope-failure locations in Puerto Rico after Hurricane Maria](#)<sup>103</sup>

When Hurricane Maria made landfall in Puerto Rico on September 20, 2017, it triggered tens of thousands of slope failures across the island. Shortly after the storm, the U.S. Geological Survey undertook a massive and comprehensive landslide mapping effort, using publicly available high-resolution aerial and satellite imagery to record the locations of over 71,000 landslide headscarps. The resulting shapefile represents each landslide with a point located at the center of its headscarp, with attribute information on the municipio and barrio in which each landslide occurred, as well as the geographic coordinates of each point. While most landslides recorded in this dataset are assumed to have been triggered by Hurricane Maria, some may have occurred prior to the storm, either from Hurricane Irma which passed the island two weeks earlier, or other isolated events.

**Frequency Source:** [Constraining landslide frequency across the United States to inform county-level risk reduction](#) (Luna et al., 2025)<sup>104</sup>

Annualized landslide frequencies for the continental United States, Alaska, and Hawai'i were estimated by the U.S. Geological Survey using the [U.S. Geological Survey Landslide Inventories](#)

<sup>103</sup> Hughes, K.S., Bayouth García, D., Martínez Milian, G.O., Schulz, W.H., and Baum, R.L., 2019, Map of slope-failure locations in Puerto Rico after Hurricane María: U.S. Geological Survey data release, <https://doi.org/10.5066/P9BVM74>

<sup>104</sup> Luna, L. V., Woodard, J. B., Bytheway, J. L., Belair, G. M., and Mirus, B. B.: Constraining landslide frequency across the United States to inform county-level risk reduction, EGU sphere [preprint], <https://doi.org/10.5194/egusphere-2025-947>, 2025.

[across the United States \(Version 3, 2025 Update\)](#) to train a statistical model. The statistical model estimated county-level landslide frequencies as a function of standardized landslide susceptible area, standardized probability of potentially landslide triggering earthquakes, and ecoregion. For this model, landslide susceptible area was calculated using the 10-meter U.S. Geological Survey National Landslide Susceptibility Model. The average probability of an earthquake with a Modified Mercalli Intensity (MMI) greater than or equal to VI in 100 years from the 2023 United States National Seismic Hazard Model was used to indicate potentially landslide triggering earthquake probability. Level I ecoregion served as a proxy for other environmental factors that may influence landslide frequency but are not explicitly modeled, for example, climate, land cover, and geology.

## 16.2. Landslide Spatial Data Processing

### 16.2.1. EXPOSURE PROCESSING

Landslide exposure was determined by taking the intersection of the U.S. Geological Survey-identified landslide susceptible areas with each Census block in the United States and Puerto Rico. To do so, the raster formatted data were first converted to a vector format. Converting the raster dataset to vector format greatly improves the processing speed and repeatability of resource-intensive intersection functions performed within the processing database. The National Landslide Susceptibility Model identifies each pixel with a susceptibility grid weighting, where each 90-m pixel is given a weight of 0 to 81 depending on how many 10-m pixels within it are identified as susceptible to landslides. Mirus et al (2024) assigned numerical bins for grid values with the percentage of known historical landslides that occurred within each bin ([Table 58](#)). The percentage of total historical landslides was used to normalize the grid weights, providing a percentage of susceptibility within each pixel cell. Building value and population counts that fell within each grid bin were multiplied by the grid weight percent to identify exposed values. For instance, if a building fell within a susceptibility grid cell value of 48, then 16% of the building value would be used as susceptible.

**Table 58: Landslide Susceptibility bins and corresponding grid weights**

| <i>Grid Code Bin</i> | <i>Percent Historical Landslides</i> | <i>Grid Weight</i> |
|----------------------|--------------------------------------|--------------------|
| 64-81                | 75                                   | 1.000              |
| 48-63                | 12                                   | 0.160              |
| 32-47                | 6                                    | 0.080              |
| 16-31                | 4                                    | 0.053              |
| 1-15                 | 2                                    | 0.027              |
| 0                    | 1                                    | 0.000              |

### 16.2.2. FREQUENCY PROCESSING

The U.S. Geological Survey county-level annualized landslide frequencies for the continental United States, Alaska, and Hawai'i are presented as a rate of landslides per 1,000 square km per year for each county. These landslide frequency densities were multiplied by the county's area in order to get the number of landslides per year for each county. The county-level estimated landslide counts were then split into Census tracts using an area-weighted percentage based on the 2024 U.S. Geological Survey Landslide Susceptibility map and the bins used to assign grid exposure weights (Table 58), allocating a fraction of the counties' estimated landslides to each Census tract. The only deviation from this procedure within the continental United States, Alaska, and Hawai'i is for San Bernadino County, California (FIPS 06071). Due to the disproportionate tract sizes, building values, and susceptibility, the Census tract level distribution of landslide frequency was providing the largest tract in the western portion of the county with a disproportionately large fraction of the estimated yearly landslide losses, despite there being no recorded landslides occurrences in that tract. Therefore, it was decided that the frequencies would be adjusted for this county using a weighting factor of 50/50, where half the weighting used the standard methodology outlined above and the other half of the weighting used the percentage of the county's historical landslides in each tract as documented in the 2025 U.S. Geological Survey Landslide Inventories, Version 3.

For Puerto Rico, the national frequency methodology was not possible because the 2023 release of the National Seismic Hazard Model that is relied upon for MMI estimates does not include Puerto Rico, excluding it from analysis in the 2025 U.S. Geological Survey landslide frequency study. An alternative determination for Puerto Rico landslide frequencies was therefore needed. A vast majority of the historical count of landslide events within Puerto Rico (71,431 out of the total recorded 71,638 landslides) occurred during Hurricanes Irma and Maria. These data were collected as a part of the study of landslide susceptibility on the island of Puerto Rico (Hughes et al., 2019), and the number and spatial extent of these landslides are broadly agreed to be well documented and comprehensive by subject matter experts.

Historically, landslide counts and locations in Puerto Rico are available for an extremely limited number of triggering events. Hurricanes Irma and Maria produced an unprecedented number of landslides with almost complete coverage of the island. Studies of this, and other past events, help to elucidate the frequency of widespread triggering events as well as their potential number of landslides that are possible through similar future events.

The process of determining landslide frequency within Puerto Rico was done in two separate analyses. First, the landslide inventory compiled by Hughes et al. (2019) and leveraged in the Hughes and Shultz 2020 study<sup>105</sup> is regarded as having comprehensive and well-placed landslide points consistently across the ~71,000-point database. Using these landslide point locations, the relative percentage of landslides were produced within each Census tract and county. Second, an estimated yearly average number of landslides was determined using a 2019 study of historical

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<sup>105</sup> Hughes, K.S., and Schulz, W.H., 2020, Map depicting susceptibility to landslides triggered by intense rainfall, Puerto Rico: U.S. Geological Survey Open-File Report 2020–1022, 91 p., 1 plate, scale 1:150,000, <https://doi.org/10.3133/ofr20201022>.

weather-related landslide triggering events.<sup>106</sup> This study identified all major landslide triggering events from 1960 through 2017 (57 years) and categorized the landslides that occurred during these events as Unknown, None, Isolated, Many, and Widespread (see [Table 59](#)).

**Table 59: Estimated Number of Puerto Rican Landslides per Historical Event**

| <i>Landslide Occurrences</i> | <i>Number of Events</i> | <i>Est. Number of Landslides</i> | <i>Justification for the Est. Number of Landslides</i>                                                                                                                                                |
|------------------------------|-------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unknown                      | 1                       | 0                                |                                                                                                                                                                                                       |
| None                         | 21                      | 0                                |                                                                                                                                                                                                       |
| Isolated                     | 8                       | 10                               | Mentioned as being less than 10.                                                                                                                                                                      |
| Frequent (Many)              | 15                      | 40                               | These LSs are "isolated" and on the order of 10's, another factor of 10 less than a Widespread event was assumed.                                                                                     |
| Widespread                   | 26                      | 400                              | Hurricane Maria is mentioned as being 100x greater than normal. The paper reports the LS produced during Hurricane Maria as ~40,000 LSs, so a "typical" widespread event is estimated to have 400 LS. |

Using the numbers estimated from [Table 59](#), an estimated total number of landslides that occurred over the 57-year period was calculated and then converted into an annualized frequency for the whole of Puerto Rico. This estimated annual number of landslides (1,448.3 landslides per year) was split amongst the Census tracts and counties using the previously determined estimated frequency percentage.

### 16.2.3. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Any county with a Census block that intersected a Landslide-susceptibility polygon where the grid code was 1 or higher is assumed to have the potential for a Landslide hazard.

## 16.3. Landslide Non-Spatial Data Processing

### 16.3.1. HISTORICAL OCCURRENCE COUNT

The Landslide hazard uses a county-level landslide frequency estimated by a statistical model for the Continental United States, Alaska, and Hawai'i, so a historical occurrence count and period of record is not needed for frequency calculations and is therefore not calculated for those states. For Puerto

<sup>106</sup> Bessette-Kirton, Erin, Corina Cerovski-Darriau, William Schulz, Jeffrey A. Coe, Jason W. Kean, Jonathan W. Godt, Matthew A. Thomas, and K. Stephen Hughes. "Landslides triggered by Hurricane Maria: Assessment of an extreme event in Puerto Rico." *GSA Today* 29, no. 6 (2019): 4-10. <https://doi.org/10.1130/GSATG383A.1>.

Rico, the historical occurrence count is the number of landslides that occurred during Hurricanes Maria and Irma, as recorded by Hughes et al. (2019), and used in the calculations of the Puerto Rico frequency estimates.

## 16.4. Landslide Exposure

To identify areas of exposure, the Landslide-susceptibility polygons were intersected with the Census block polygons within the processing database. The resulting table contains the polygon's unique identifier, Census block number, the grid code exposure percent from [Table 58](#), and the intersected area in square kilometers (see [Table 60](#)).

**Table 60: Sample Data from the Landslide Census Block Intersection Table**

| <i>LandslideGridCellID</i> | <i>CensusBlock</i> | <i>Exposure Percent</i> | <i>IntersectedAreaKm2</i> |
|----------------------------|--------------------|-------------------------|---------------------------|
| 206991169                  | 210019706001006    | 0.160                   | 0.0000054990              |
| 207012504                  | 210019706001006    | 0.080                   | 0.0022838710              |
| 206885758                  | 210019706001006    | 0.080                   | 0.0015999071              |

To find exposure value, the sum of the intersection areas of the Landslide-susceptibility polygons for each Census block was multiplied by the average weighted grid percentage of the Census block and then multiplied by the developed area building value density and the developed area population density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 96](#)). These Census block densities have been calculated by dividing the total county values (as recorded in Hazus 6.0) by the developed land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

**Equation 96: Census Block Landslide Exposure**

$$Exposure_{LNDSCB Bldg} = \sum (IntsctArea_{LNDSCB} \times GridPcnt_{LNDSCB}) \times DevAreaDen_{CB Bldg}$$

$$Exposure_{LNDSCB Pop} = \left( \sum (IntsctArea_{LNDSCB} \times GridPcnt_{LNDSCB}) \times DevAreaDen_{CB Pop} \right) \times VSL$$

where:

$Exposure_{LNDSCB Bldg}$  is the building value exposed to Landslide susceptibility in a specific Census block (in dollars).

$GridPcnt_{LNDSCB}$  is the average weighted grid percentage of the Census block

$IntsctArea_{LNDSCB}$  is the intersected areas of Landslide-susceptibility polygons with the Census block (in square kilometers).

|                         |                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------|
| $DevAreaDen_{CB Bldg}$  | is the developed area building value density of the Census block (in dollars per square kilometer).              |
| $Exposure_{LNDSCB Pop}$ | is the population equivalence value exposed to Landslide susceptibility in a specific Census block (in dollars). |
| $DevAreaDen_{CB Pop}$   | is the developed area population density of the Census block (in people per square kilometer).                   |
| $VSL$                   | is the Value of Statistical Life (\$13.7 million per person).                                                    |

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

#### 16.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 97](#)).

##### Equation 97: Census Tract and County Landslide Aggregation

$$\begin{aligned}
 Exposure_{LNDSCTBldg} &= \sum_{CB}^{CT} Exposure_{LNDSCB Bldg} \\
 Exposure_{LNDSCo Bldg} &= \sum_{CB}^{Co} Exposure_{LNDSCB Bldg} \\
 Exposure_{LNDSCTPop} &= \sum_{CB}^{CT} Exposure_{LNDSCB Pop} \\
 Exposure_{LNDSCo Pop} &= \sum_{CB}^{Co} Exposure_{LNDSCB Pop}
 \end{aligned}$$

where:

|                                         |                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{LNDSCTBldg}$                 | is the building value exposed to Landslide susceptibility in a specific Census tract (in dollars).                                   |
| $\sum_{CB}^{CT} Exposure_{LNDSCB Bldg}$ | is the summed value of all buildings exposed to Landslide susceptibility for each Census block within the Census tract (in dollars). |



|                                        |                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{LNDSCoBldg}$                | is the building value exposed to Landslide susceptibility in a specific county (in dollars).                                                      |
| $\sum_{CB}^{Co} Exposure_{LNDSCoBldg}$ | is the summed value of all buildings exposed to Landslide susceptibility for each Census block within the county (in dollars).                    |
| $Exposure_{LNDSCoPop}$                 | is the population equivalence value exposed to Landslide susceptibility in a specific Census tract (in dollars).                                  |
| $\sum_{CB}^{CT} Exposure_{LNDSCoPop}$  | is the summed value of all population equivalence exposed to Landslide susceptibility for each Census block within the Census tract (in dollars). |
| $Exposure_{LNDSCoPop}$                 | is the population equivalence value exposed to Landslide susceptibility in a specific county (in dollars).                                        |
| $\sum_{CB}^{Co} Exposure_{LNDSCoPop}$  | is the summed value of all population equivalence exposed to Landslide susceptibility for each Census block within the county (in dollars).       |

## 16.5. Landslide Annualized Frequency

The annualized frequency value represents the estimated number of recorded Landslide occurrences, in events, each year for a specific area. The annualized frequency is calculated at the county level and downscaled to the Census tract level through an area-weighted proportional relationship of the susceptible regions (see [Equation 98](#)), and the Census block inherits this value. The Census block value is used in the Expected Annual Loss calculations.

### Equation 98: Census Tract Landslide Annualized Frequency

$$Freq_{LNDCT} = EstimatedYearlyEventCount_{LNDSCo} \times \frac{AreaWeightedSusceptible\%_{LNDSCo}}{AreaWeightedSusceptible\%_{LNDSCo}}$$

where:

|                                      |                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| $Freq_{LNDCT}$                       | is the annualized frequency of Landslide events determined for a specific Census tract (events per year).           |
| $EstimatedYearlyEventCount_{LNDSCo}$ | is the estimated number of Landslide events per county per year as determined by the U.S. Geological Survey.        |
| $AreaWeightedSusceptible\%_{LNDSCo}$ | is the area weighted average for the U.S. Geological Survey Susceptibility values over the area of the Census tract |
| $AreaWeightedSusceptible\%_{LNDSCo}$ | is the area weighted average for the U.S. Geological Survey Susceptibility values over the area of the county       |

### 16.5.1. MINIMUM ANNUAL FREQUENCY

The frequency rates, as determined within the U.S. Geological Survey's Constraining landslide frequency, identifies the potential landslide rates for the continental United States, Alaska, and Hawai'i at the county level. The frequency rate was developed using the susceptibility map, therefore there should be no census tract where the frequency rate is not identified while also being identified as susceptible to landslides.

For Puerto Rico, if a Census tract's historical Landslide event count is 0, but the Census tract is part of a county that was designated as one in which Landslides are possible according to the determination above, the Census tract is assigned the minimum annual Landslide frequency. This Minimum Annual Frequency is set at 0.01 (1 in 100 years).

### 16.5.2. ANNUALIZED FREQUENCY INHERITANCE AND AGGREGATION

The Census block inherits its annualized frequency value from the Census tract that contains it as in [Equation 99](#).

#### Equation 99: Census Block Landslide Inheritance

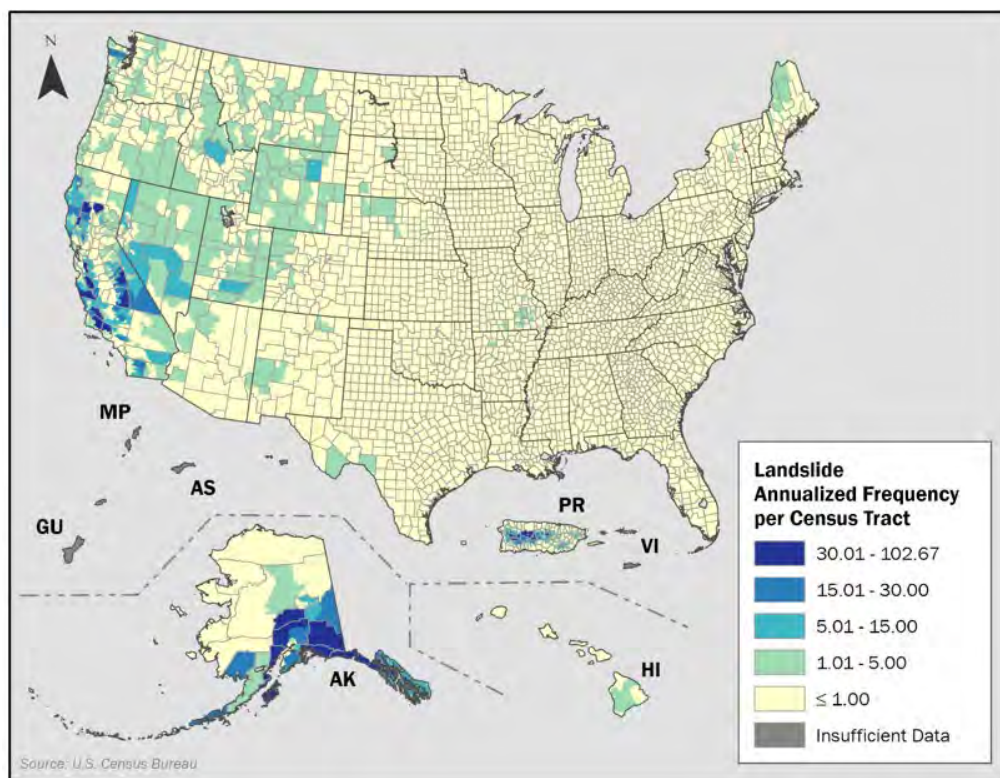
$$Freq_{LNDSCB} = Freq_{LNDSCCT}$$

where:

$Freq_{LNDSCB}$  is the annualized frequency of Landslide events determined for a specific Census block (events per year).

$Freq_{LNDSCCT}$  is the annualized frequency of Landslide events determined for a specific Census tract (events per year).

[Figure 77](#) displays Landslide annualized frequency at the Census tract level.



**Figure 77: Landslide Annualized Frequency by Census tract**

## 16.6. Landslide Historic Loss Ratio

The Landslide Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Landslide occurrence, or the average rate of loss associated with a Landslide occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Landslide hazard type.

Loss data are provided by SHELUDUS<sup>107</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. Seven peril types are mapped to the hazard Landslide (see [Table 61](#)).

**Table 61: Landslide Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELUDUS</i> | <i>Total SHELUDUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Debris Flow                   | 688                                | 686                                  |
| Landslide                     | 608                                | 585                                  |

<sup>107</sup> For Landslide loss information, SHELUDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information, U.S. Geological Survey Landslide News & Info, the USDA's Cost Estimating Guide for Road Construction, NASA's GLC, and the Oregon Department of Geology and Mineral Industries' Statewide Landslide Information Layer for Oregon

| <b>Peril Type in SHELdUS</b> | <b>Total SHELdUS Loss Records</b> | <b>Total Records per Event Basis</b> |
|------------------------------|-----------------------------------|--------------------------------------|
| Landslide-Slump              | 1                                 | 1                                    |
| Mud Flow                     | 0                                 | 0                                    |
| Mudslide                     | 171                               | 170                                  |
| Rock Slide                   | 53                                | 53                                   |
| Rock Fall                    | 32                                | 32                                   |
| Hurricane Maria added losses | 78                                | 78                                   |

Large value events within the SHELdUS database were evaluated if they provided a disproportionately large loss compared to other losses within the county. Through this process several high value SHELdUS landslide loss events were determined to be either duplicate events or not applicable to building losses. These non-applicable events were then evaluated based on historical accounts whether part of the losses were building losses or whether the entire event should be removed. For instance, the “the largest trash landslide in Ohio history” on March 9, 1996 within the Rumpke Sanitary Landfill in Hamilton County, Ohio, and the Bingham Pit Landslide on April 10, 2013 in Salt Lake City, Utah were both deemed not to be building losses.

Hurricane Maria landslide losses were not a part of the SHELdUS database and therefore were incorporated based on calculations from separate sources. Total losses due to Hurricane Maria were estimated to be approximately \$90 billion across Puerto Rico and the U.S. Virgin Islands.<sup>108</sup> However, that number encompasses all losses attributed to the hurricane, not just landslide losses. Within Puerto Rico alone, there was an estimated 300,000 homes lost from hurricane, flooding, and landslides.<sup>109</sup> To identify only potential building losses due to landslides, the Census sub-block dasymetric polygons that also contained landslide points from Hurricane Maria (Hughes et al., 2019) were identified. All known buildings within these sub-block dasymetric polygons were then identified and the building values summed to provide a potential loss of buildings by Hurricane Maria to be \$6.6B. This value was then distributed across the island using a 50-50 weighting of landslide frequency and total building value within the Census tracts.

The Historic Loss Ratio exposure value for Landslide is a county-level value that represents the dollar value of the total building value or the entire population of a county as recorded in Hazus 6.0. The Loss Ratio per Basis for each SHELdUS-documented event and each consequence type (building and population) is calculated using [Equation 100](#).

<sup>108</sup> <https://www.ncdc.noaa.gov/stormevents/eventdetails.jsp?id=721909>

<sup>109</sup> FEMA. 2018. Mitigation Assessment Team Report. Hurricanes Irma and Maria in Puerto Rico: Building Performance, Observations, Recommendations and Technical Guidance FEMA P-2020. Washington, DC.  
[https://www.fema.gov/sites/default/files/2020-07/mat-report\\_hurricane-irma-maria-puerto-rico\\_2.pdf](https://www.fema.gov/sites/default/files/2020-07/mat-report_hurricane-irma-maria-puerto-rico_2.pdf)

**Equation 100: Loss Ratio per Basis Calculation for a Single Landslide Event**

$$LRB_{LNDSCoCnsqType} = \frac{LOSS_{LNDSCoCnsqType}}{HLRExposure_{LNDSCoCnsqType}}$$

where:

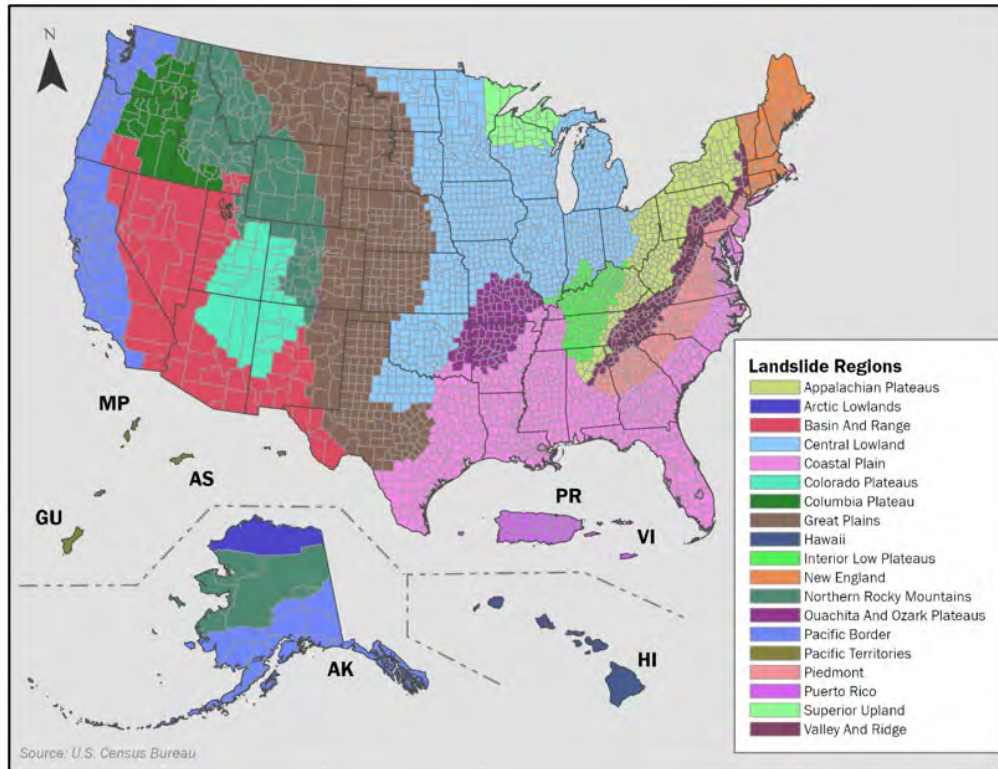
|                                |                                                                                                                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $LRB_{LNDSCoCnsqType}$         | is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Landslide event. Calculation is performed for each consequence type (building and population). |
| $LOSS_{LNDSCoCnsqType}$        | is the loss (by consequence type) experienced from the Landslide event documented to have occurred in the county (in dollars or impacted people).                                                                                          |
| $HLRExposure_{LNDSCoCnsqType}$ | is the total value (by consequence type) of the county estimated to have been exposed to the Landslide event (in dollars or people).                                                                                                       |

Landslide events can occur with a high frequency in areas but often result in no recorded loss to buildings or population. SHELDUS does not record events in which no loss occurred, so zero-loss events were inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELDUS period of record (1996 through 2023). For Landslide, the historical event count is the annual frequency times the period of record (28 years).

To reduce the influence of reporting gaps, the Historic Loss Ratio values were averaged across Physiographic Regions ([Figure 78](#)). Landslide Physiographic Regions were based on the U.S. Geological Survey physiographic provinces for the contiguous United States<sup>110</sup> and Alaska.<sup>111</sup> Each county was then assigned a physiographic province based on where the majority of the county lay. The smaller provinces were merged together in order to provide continuous provinces across large areas of the United States. Regional Historic Loss Ratio values were then calculated using an average Census tract exposure across the county for building losses and a countywide exposure for population losses ([Table 62](#)).

<sup>110</sup> U.S. Geological Survey, 1946, Physiographic divisions of the conterminous U.S.: U.S. Geological Survey data release, <https://doi.org/10.5066/P9B1S3K8>.

<sup>111</sup> Wahrhaftig, C., 1965. Physiographic divisions of Alaska (No. 482). US Govt. Print. Off., <https://doi.org/10.3133/pp482>.



**Figure 78: Landslide Physiographic Regions**

**Table 62: Regional Landslide Historic Loss Ratio Values**

| <i>Region</i>            | <i>Regional HLRB</i> | <i>Regional HLRP</i> |
|--------------------------|----------------------|----------------------|
| APPALACHIAN PLATEAUS     | 0.000085136          | 0.000002838          |
| ARCTIC LOWLANDS*^        | 0.000015941          | 0.000000300          |
| BASIN AND RANGE          | 0.000201879          | 0.000000300          |
| CENTRAL LOWLAND          | 0.000632909          | 0.000029325          |
| COASTAL PLAIN^           | 0.000770447          | 0.000000300          |
| COLORADO PLATEAUS        | 0.000096645          | 0.000013297          |
| COLUMBIA PLATEAU^        | 0.001534158          | 0.000000300          |
| GREAT PLAINS^            | 0.000026128          | 0.000000300          |
| HAWAI'I^                 | 0.000252652          | 0.000000300          |
| INTERIOR LOW PLATEAUS^   | 0.000015941          | 0.000000300          |
| NEW ENGLAND              | 0.000151945          | 0.000001685          |
| NORTHERN ROCKY MOUNTAINS | 0.000167025          | 0.000029455          |



| <i><b>Region</b></i>         | <i><b>Regional HLRB</b></i> | <i><b>Regional HLRP</b></i> |
|------------------------------|-----------------------------|-----------------------------|
| OUACHITA AND OZARK PLATEAUS^ | 0.000284085                 | 0.000000300                 |
| PACIFIC BORDER               | 0.000089825                 | 0.000001468                 |
| PIEDMONT^                    | 0.000084201                 | 0.000000300                 |
| PUERTO RICO                  | 0.000269864                 | 0.000000426                 |
| SUPERIOR UPLAND*^            | 0.000015941                 | 0.000000300                 |
| VALLEY AND RIDGE             | 0.000308041                 | 0.000003193                 |

\*No building losses were available therefore the minimum Building Historic Loss Ratio (HLRB) value was used, the Interior Low Plateaus region.

^Population losses did not meet the minimum threshold of 2 losses; therefore, the minimum Population Historic Loss Ratio (HLRP) value was used, the Basin and Range region.

If a county had any SHELDTUS loss events, a weighting between county level Historic Loss Ratio values and regional level Historic Loss Ratio values were used to obtain the annual historical losses at the county level, otherwise the regional level Historic Loss Ratio value was used.

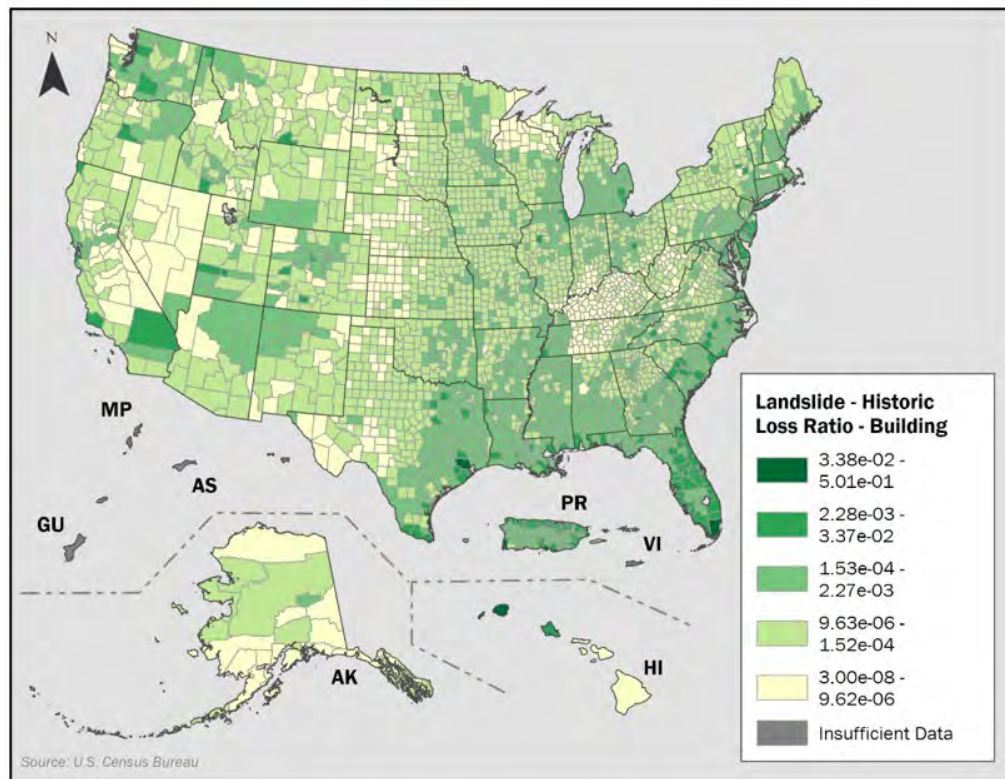
After the Loss Ratios per Basis are calculated for all Landslide events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Landslide.

**Table 63: Landslide Historic Loss Ratio Bayesian Weighting Level Sets**

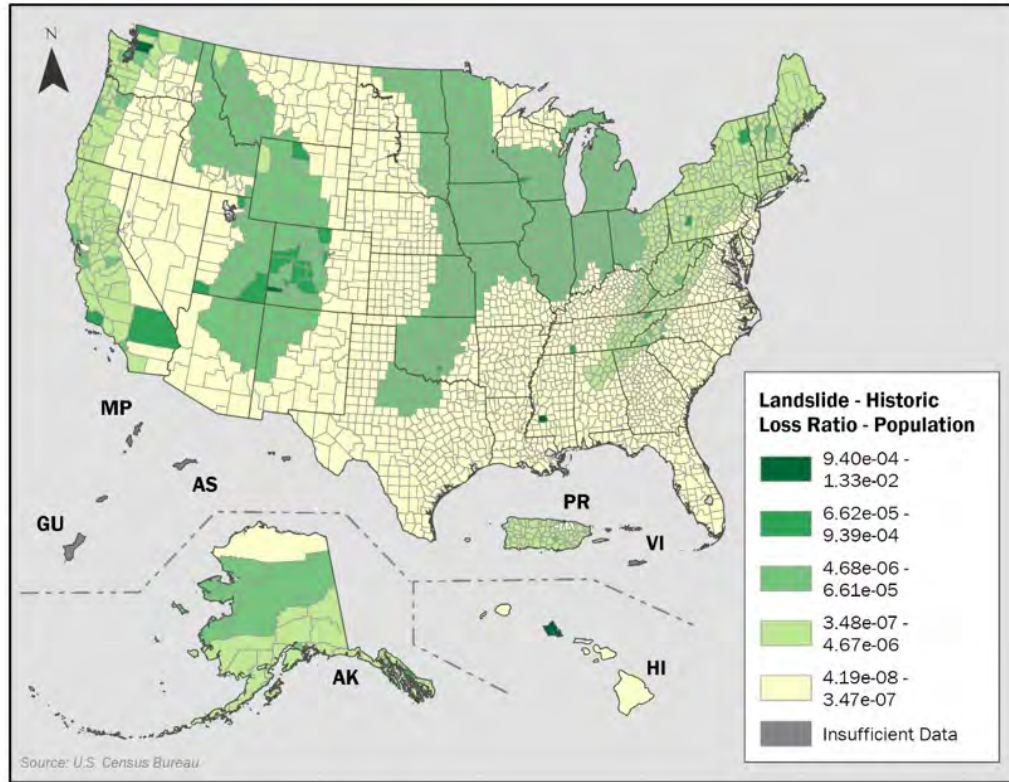
| <i><b>Consequence Type</b></i> | <i><b>Set ID</b></i> | <i><b>Bayesian Level</b></i>    | <i><b>Set Refinement Filter</b></i> | <i><b>Set Utilization Threshold (Set Must Have)</b></i> |
|--------------------------------|----------------------|---------------------------------|-------------------------------------|---------------------------------------------------------|
| Population                     | P1                   | County                          | N/A                                 | 1 Loss Ratio per Basis record                           |
|                                | P2                   | Landslide Physiographic Regions | N/A                                 | 2 Loss Ratio per Basis records                          |
| Building                       | B1                   | County                          | N/A                                 | 1 Loss Ratio per Basis record                           |
|                                | B2                   | Landslide Physiographic Regions | N/A                                 | 1 Loss Ratio per Basis record                           |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 79](#) and [Figure 80](#), represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Landslide. The resulting Bayesian-

adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 79: Landslide Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 80: Landslide Bayesian-Adjusted Historic Loss Ratio – Population**

## 16.7. Landslide Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 101](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 101: Census Block Expected Annual Loss to Landslide

$$EAL_{LNDSCB_{Bldg}} = Exposure_{LNDSCB_{Bldg}} \times Freq_{LNDSCB} \times HLR_{LNDSCB_{Bldg}}$$

$$EAL_{LNDSCB_{Pop}} = Exposure_{LNDSCB_{Pop}} \times Freq_{LNDSCB} \times HLR_{LNDSCB_{Pop}}$$

where:

$EAL_{LNDSCB_{Bldg}}$  is the building Expected Annual Loss due to Landslide occurrences for a specific Census block (in dollars).

$Exposure_{LNDSCB_{Bldg}}$  is the building value exposed to Landslide susceptibility in the Census block (in dollars).

$Freq_{LNDSCB}$  is the annualized frequency of Landslide events determined for a specific Census block (events per year).

$HLR_{LNDSCB Bldg}$  is the building Historic Loss Ratio for Landslide for the Census block.

$EAL_{LNDSCB Pop}$  is the population equivalence Expected Annual Loss due to Landslide occurrences for a specific Census block (in dollars).

$Exposure_{LNDSCB Pop}$  is the population equivalence value exposed to Landslide susceptibility in the Census block (in dollars).

$HLR_{LNDSCB Pop}$  is the population Historic Loss Ratio for Landslide for the Census block.

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building and population equivalence Expected Annual Loss values at the Census block level as in [Equation 102](#).

**Equation 102: Census Tract and County Expected Annual Loss to Landslide**

$$EAL_{LNDSC T} = \sum_{CB}^{CT} EAL_{LNDSCB Bldg} + \sum_{CB}^{CT} EAL_{LNDSCB Pop}$$

$$EAL_{LNDSCo} = \sum_{CB}^{Co} EAL_{LNDSCB Bldg} + \sum_{CB}^{Co} EAL_{LNDSCB Pop}$$

where:

$EAL_{LNDSC T}$  is the total Expected Annual Loss due to Landslide occurrences for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{LNDSCB Bldg}$  is the summed building Expected Annual Loss due to Landslide occurrences for all Census blocks in the Census tract (in dollars).

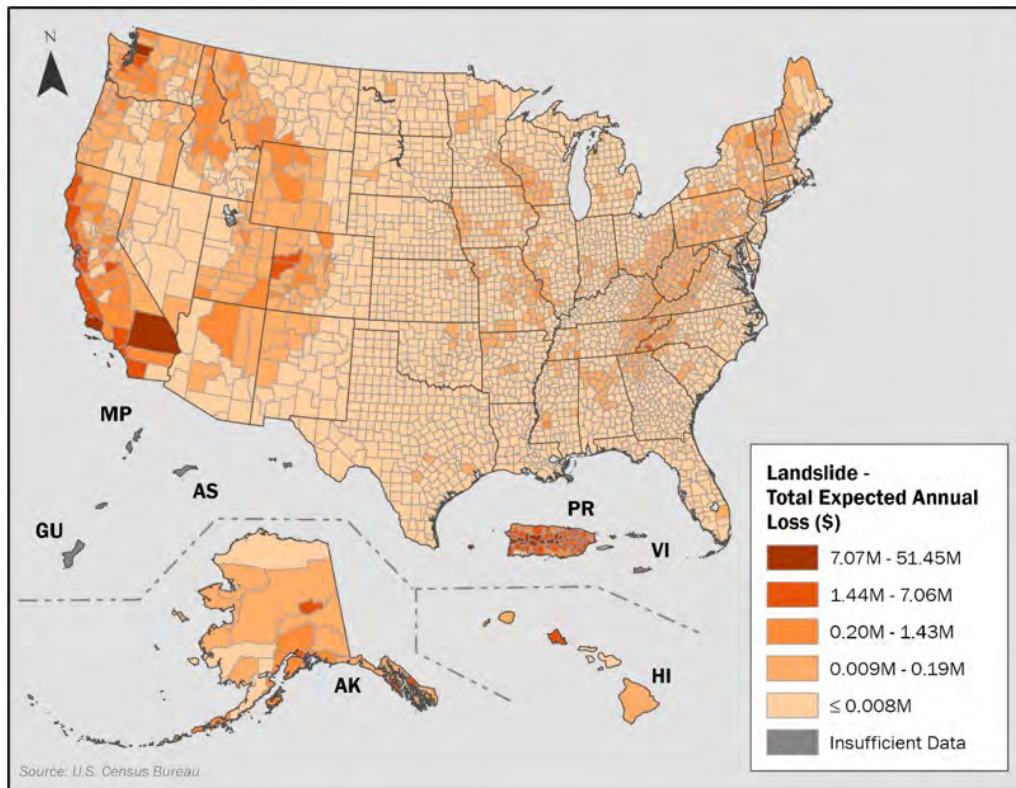
$\sum_{CB}^{CT} EAL_{LNDSCB Pop}$  is the summed population equivalence Expected Annual Loss due to Landslide occurrences for all Census blocks in the Census tract (in dollars).

$EAL_{LNDSCo}$  is the total Expected Annual Loss due to Landslide occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{LNDSCB Bldg}$  is the summed building Expected Annual Loss due to Landslide occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{LNDSCB Pop}$  is the summed population equivalence Expected Annual Loss due to Landslide occurrences for all Census blocks in the county (in dollars).

[Figure 81](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Landslide occurrences.



**Figure 81: Total Expected Annual Loss by County to Landslide**

With the Landslide total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Landslide Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Landslide Expected Annual Loss for population by the total population of the community.

## 16.8. Landslide Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The current (2024) U.S. Geological Survey landslide susceptibility map vastly improves resolution of landslide area identification from a 1-km grid to a 90-m grid, based on down-sampling of a 10-m binary identification system (either susceptible or not). A gradational approach to landslide susceptibility which incorporates additional landslide triggering factors



to help identify areas which are more or less susceptible to landslides would allow for more specific estimations of exposure and losses.

- The susceptible map data depicts areas of steep slopes where landslides may begin and do not represent where the landslide or debris flow may travel and deposit. While the locations are at least partially accounted for during the process of downsampling to a 90-meter resolution, explicit consideration of runout areas would improve accuracy of the loss and risk assessments.
- The landslide Expected Annual Loss calculations treat all types of landslides (rock falls, debris flows, shallow vs deep landslides, among others) the same. Additional information on the predicted landslide types and characteristics could help refine the geographic extent and severity of landslide occurrences.
- U.S. Geological Survey Landslide susceptibility data are currently limited to just the United States and Puerto Rico, leaving the other territories (American Samoa, Guam, U.S. Virgin Islands, Commonwealth of the Northern Mariana Islands) out of the Landslide risk assessment. Territorial inclusion is contingent on the availability of 10-m DEM data. Once the high-resolution DEM data is available for the other territories, it is hoped that the landslide susceptibility and frequency estimates could be extended to include them.
- The new landslide frequency identification is limited in scope to just the continental United States, Alaska, and Hawai'i. It is hoped for future releases to include Puerto Rico and the remaining territories (contingent on the inclusion of susceptibility data and seismic hazard models).
- The landslide frequency estimates consider spatially homogeneous triggering thresholds, rather than probabilities of landslide triggering that vary spatially and with increasing magnitude of the triggering event. Future iterations could consider that increasing storm severity and ground motion would have increasing likelihood of landslide occurrence with increased numbers of landslides, and that these relationships vary spatially.
- Landslide susceptibility and frequency identification in the Continental United States, Alaska, and Hawai'i primarily considers seismically induced landslides and does not explicitly consider rainfall, whereas the current Puerto Rico susceptibility and frequency calculations are based solely on weather-induced landslides. Ideally in future iterations, a consistent and comprehensive assessment of triggering events (earthquakes, rainfall, volcanic activity, among others) would be considered across the United States and its territories.



## 17. Lightning

Lightning is a visible electrical discharge or spark of electricity in the atmosphere between clouds, the air, and/or the ground often produced by a thunderstorm.

The Lightning hazard encompasses the Continental United States, Alaska, Hawai'i, and Puerto Rico, but does not include the remaining Territories due to insufficient data. Unlike other hazards, Lightning modeling has required greater attention to the treatment and interpretation of annualized frequency. The Lightning hazard only calculates risk for the Building and Population consequence types. For a summary of the Hazard Data Characteristics, please see [Appendix B](#) – Hazard Data Characteristics Comparison.

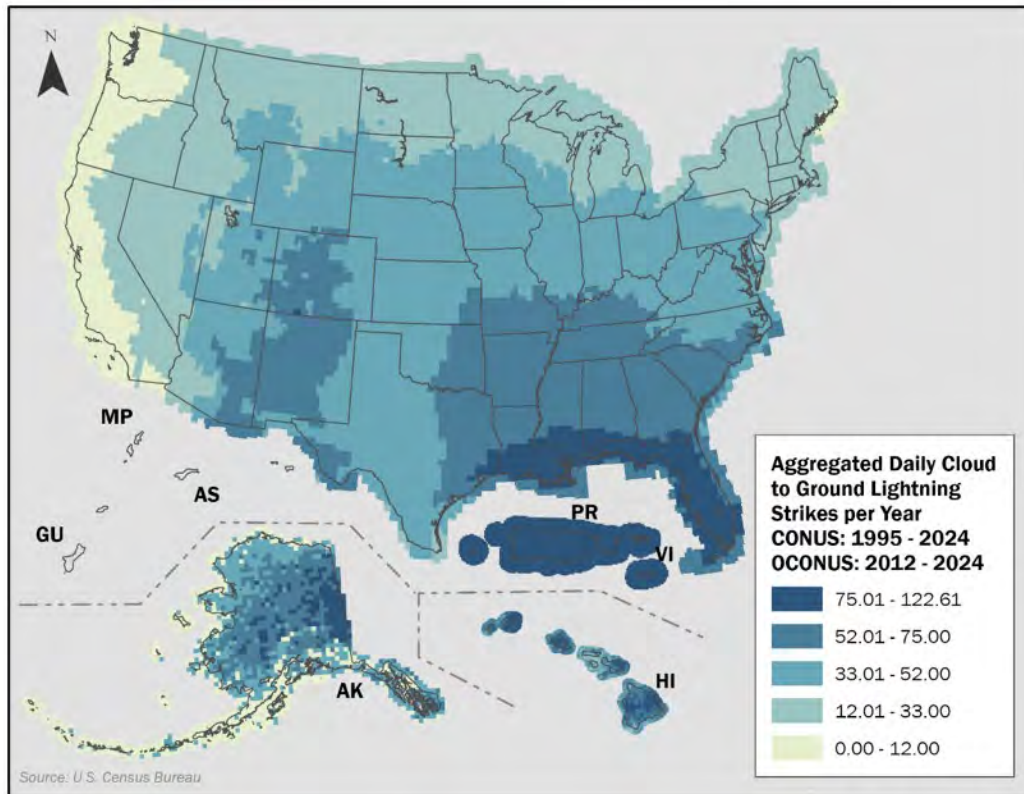
### 17.1. Lightning Spatial Source Data

**Historical Occurrence Source:** [National Oceanic and Atmospheric Administration/National Weather Service/Storm Prediction Center and Vaisala NLDN, Daily Lightning Climatology](#)<sup>112</sup>

The Storm Prediction Center currently maintains a dataset providing yearly gridded Lightning climatologies derived using historical records of daily cloud-to-ground strikes over the Continental United States, Alaska, Hawai'i, and Puerto Rico. Spatiotemporal records are in NetCDF (Network Common Data Form) format for 366 days (including Leap Year). Separate datasets were provided for the Continental United States, Alaska, Hawai'i, and Puerto Rico, with slight variations in spatial resolution and climatologies across the three datasets. For the Continental United States and Alaska, each layer is a grid of 40-by-40-km cells in the Albers Equal Area projection (see [Figure 82](#)); for Hawai'i and Puerto Rico, data are presented in a 3-by-3-km grid. The Continental United States annual probabilities represent the percent days per year that an area within that grid cell would expect to experience at least one Lightning strike based on a 29-year period of record (1995-2024). Outside the Continental United States annual climatology is the true annual likelihood of Lightning, derived over a 12-year period of record (2012-2024). These yearly values are used to calculate annualized frequency at the Census block level.

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<sup>112</sup> NCEI, NOAA. (2017). *Cloud-to-ground lightning strikes, Prototype* [online dataset]. Retrieved from <https://www.ncei.noaa.gov/products/lightning-products>.



**Figure 82: Map of Lightning Strikes**

## 17.2. Lightning Spatial Data Processing

The latitude and longitude values and the cloud-to-ground flash threshold of each entry in the NetCDF file containing gridded annual Lightning climatology data were extracted to points. These points were then buffered by 20km for the Continental United States and Alaska. Points in Hawai'i and Puerto Rico were buffered by 1.5km. After the buffer, a minimum bounding geometry (envelope) was created around each point. The result was a gridded polygon dataset containing the mean percent days per year bucketed by the number of cloud-to-ground flashes per hour that were observed within the buffer of a given grid point. These polygons in the processing database could then be intersected with the Census-block polygons to calculate annualized frequency and exposure.

### 17.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Lightning can occur almost anywhere under the right conditions, so all counties were deemed possible for Lightning strike occurrence. While the current data source does not supply information for the Pacific Territories (Guam, Northern Mariana Islands, American Samoa) or the U.S. Virgin Islands, these areas are still included as possible for Lightning occurrence. No Scores are available for Guam, Mariana Islands, American Samoa, or the U.S. Virgin Islands as the data are insufficient.

### 17.3. Lightning Non-Spatial Data Processing

#### 17.3.1. PERIOD OF RECORD

The Storm Prediction Center Daily Lightning Climatology data incorporate Lightning strikes observed between Jan. 01, 1995 and Dec. 31, 2023 for the Continental United States and between Jan. 01, 2012 and Dec. 31, 2024 for Outside the Continental United States, so the period of record for which Lightning data are utilized is 29 years and 12 years, respectively.

### 17.4. Lightning Exposure

Because Lightning strikes can occur anywhere, the entire building and population value of a Census block, Census tract, and county are considered exposed to Lightning. Population equivalence, which is used in select Expected Annual Loss calculations, is calculated by multiplying population by the Value of Statistical Life (\$13.7 million per person).

### 17.5. Lightning Annualized Frequency

The annualized frequency value represents the estimated number of days with a recorded Lightning strike each year for a specific area. The annualized frequency is calculated at the Census block-level value as an area-weighted aggregation of the annualized frequencies of its intersecting polygons obtained from the source data. The Census block value is used in the Expected Annual Loss calculations.

To calculate annualized frequency at the Census block level, the gridded Lightning polygons obtained from the source data are first intersected with the Census block polygons within the processing database. The resulting table contains the polygon's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 64](#)).

**Table 64: Sample Data from the Lightning Fishnet Census Block Intersection table**

| <i>LightningFishnetID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|---------------------------|--------------------|---------------------------|
| 815373                    | 481130020001002    | 0.0732602925796509        |
| 815373                    | 481130020001003    | 0.0534260125160217        |
| 815373                    | 481130020001004    | 0.048496762966156         |

An area-weighted annualized frequency value is then calculated at the Census block level using the intersection between the Lightning polygons and the Census block as in [Equation 103](#).

**Equation 103: Census Block Area-Weighted Fishnet Lightning Annualized Frequency**

$$Freq_{LTNG_{CB}} = \frac{\sum_{Poly}^{CB} (Freq_{LTNG_{Poly}} \times IntsctArea_{LTNG_{FishCB}})}{Area_{CB}}$$

where:

|                                 |                                                                                                                                          |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $Freq_{LTNG_{CB}}$              | is the area-weighted annualized frequency of Lightning strike event-days determined for the specific Census block (event-days per year). |
| $Freq_{LTNG_{Poly}}$            | is the annualized frequency of Lightning strike event-days for the specific Lightning polygon (event-days per year).                     |
| $IntsctArea_{LTNG_{Fish_{CB}}}$ | is the intersected area of the Lightning polygons with the Census block (in square kilometers).                                          |
| $\sum_{Poly}^{CB}$              | is the sum for all Lightning polygons that intersect the Census block                                                                    |
| $Area_{CB}$                     | is the total area of the Census block (in square kilometers).                                                                            |

### 17.5.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county levels. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 104](#).

#### Equation 104: Census Tract and County Area-Weighted Lightning Annualized Frequency

$$Freq_{LTNG_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{LTNG_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{LTNG_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{LTNG_{CB}} \times Area_{CB})}{Area_{Co}}$$

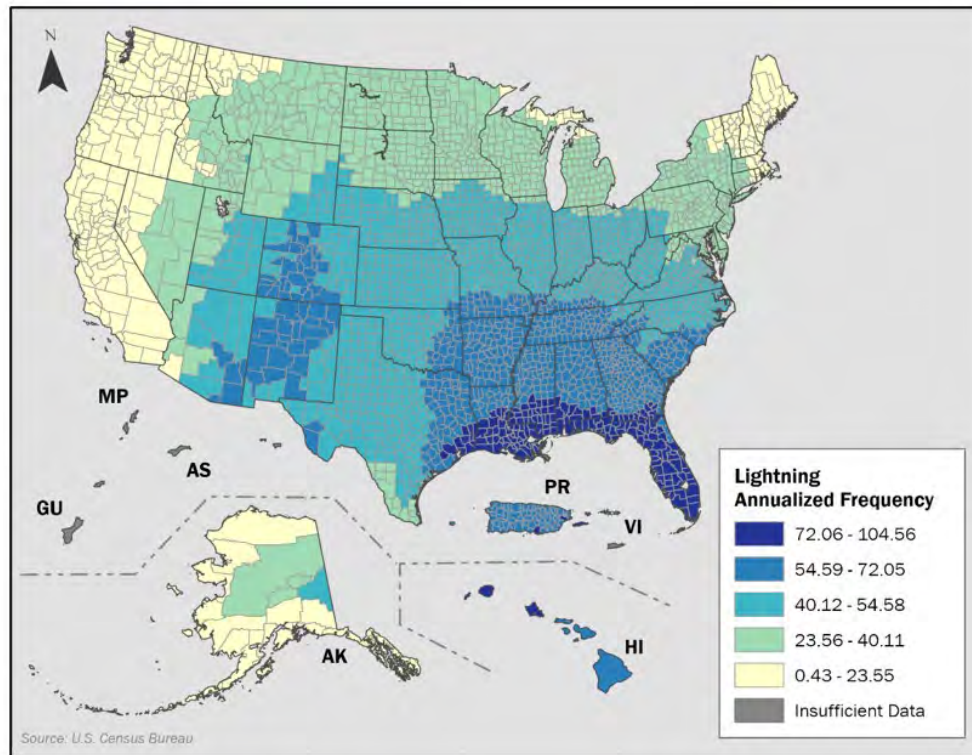
where:

|                    |                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| $Freq_{LTNG_{CT}}$ | is the area-weighted Lightning annualized frequency for a specific Census tract.                                           |
| $Freq_{LTNG_{CB}}$ | is the area-weighted annualized frequency of Lightning strikes determined for the specific Census block (events per year). |
| $Area_{CB}$        | is the total area of the Census block (in square kilometers).                                                              |
| $\sum_{CB}^{CT}$   | is the sum for all Census blocks in the Census tract.                                                                      |
| $Area_{CT}$        | is the total area of the Census tract (in square kilometers).                                                              |
| $Freq_{LTNG_{Co}}$ | is the area-weighted Lightning annualized frequency for a specific county.                                                 |

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 83](#) displays Lightning annualized frequency at the county level.



**Figure 83: Lightning Annualized Frequency by County**

## 17.6. Lightning Historic Loss Ratio

The Lightning Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Lightning strike event day, or the average rate of loss associated with a Lightning strike event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4.4 Historic Loss Ratio](#) Calculation Methodology. The Historic Loss Ratio parameters described below are specific to the Lightning hazard type.

Loss data are provided by SHELUDS<sup>113</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDS events from 1996 to 2023 are included in the Historic Loss Ratio calculation. Two peril types are mapped to the hazard Lightning (see [Table 65](#)). Native records of

<sup>113</sup> For Lightning loss information, SHELUDS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

Lightning events that caused loss over more than one day have a daily average loss value assigned to each day, and all records are aggregated on a single event-day basis (see [Section 17.2](#)).

**Table 65: Lightning Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELUS</i> | <i>Total SHELUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-----------------------------|----------------------------------|--------------------------------------|
| Fire-St Elmo's              | 0                                | 0                                    |
| Lightning                   | 15,416                           | 14,191                               |

The Historic Loss Ratio exposure value used in the formula below is a county-level value that represents the dollar value of the total building value or the entire population of a county as recorded in Hazus 6.0. The Loss Ratio per Basis for each SHELUS-documented event-day and each consequence type (building and population) is calculated using [Equation 105](#).

**Equation 105: Loss Ratio per Basis Calculation for a Single Lightning Strike Event**

$$LRB_{LTNG Co CnsqType} = \frac{LOSS_{LTNG Co CnsqType}}{HLRExposure_{Co CnsqType}}$$

where:

|                             |                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $LRB_{LTNG Co CnsqType}$    | is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the specific Lightning strike event-day. Calculation is performed for each consequence type (building and population). |
| $LOSS_{LTNG Co CnsqType}$   | is the loss (by consequence type) experienced from the Lightning strike event-day documented to have occurred in the county (in dollars or impacted people).                                                                          |
| $HLRExposure_{Co CnsqType}$ | is the total value (by consequence type) of the county estimated to have been exposed to the Lightning strike event-day (in dollars or people).                                                                                       |

Lightning strikes can occur with a high frequency in areas but often result in no recorded loss to buildings or population. SHELUS does not record events in which no loss occurred, so a number of zero-loss event-days are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELUS period of record (1996 to 2023). For Lightning, the historical event-day count is extracted using the intersection between the Lightning polygons and the Census block polygons used to calculate annualized frequency (see [Table 64](#)). The area-weighted count of all Lightning-Census block polygon intersections within the county for each record year is used as the historical event-day count. An annual rate is calculated as the



number of event-days per year and this rate is multiplied by the SHELDDUS period of record of 27 years to estimate a historical event-day count for the appropriate time range.

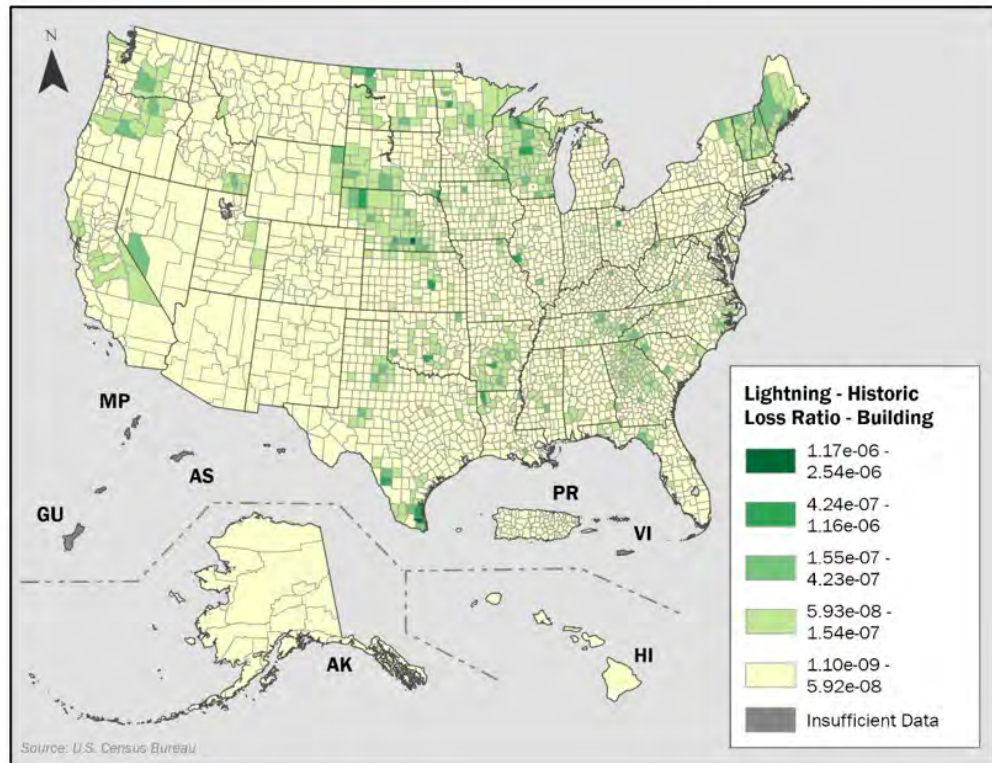
If the number of loss-causing Lightning event-day records from SHELDDUS is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for all Lightning events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Lightning.

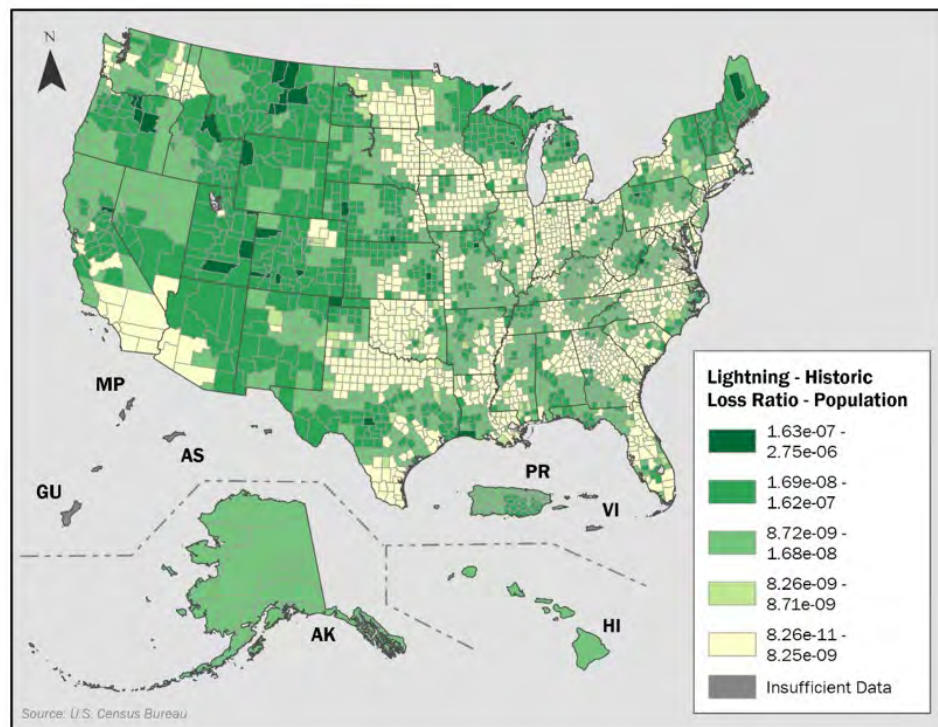
**Table 66: Lightning Historic Loss Ratio Bayesian Weighting Level Sets**

| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>             | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-----------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Population              | P1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P4            | National                          | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | N/A                                              |
| Building                | B1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | B2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B4            | National                          | Large Central Metropolitan Areas (LCMAS), Non-LCMAs | N/A                                              |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Lightning (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 84](#) and [Figure 85](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Lightning. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 84: Lightning Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 85: Lightning Bayesian-Adjusted Historic Loss Ratio – Population**

## 17.7. Lightning Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 106](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 106: Census Block Expected Annual Loss to Lightning

$$EAL_{LTNG_{CB_{Bldg}}} = Exposure_{LTNG_{CB_{Bldg}}} \times Freq_{LTNG_{CB}} \times HLR_{LTNG_{CB_{Bldg}}}$$

$$EAL_{LTNG_{CB_{Pop}}} = Exposure_{LTNG_{CB_{Pop}}} \times Freq_{LTNG_{CB}} \times HLR_{LTNG_{CB_{Pop}}}$$

where:

$EAL_{LTNG_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Lightning occurrences for a specific Census block (in dollars).

$Exposure_{LTNG_{CB_{Bldg}}}$  is the building value exposed to Lightning occurrences in the Census block (in dollars).

$Freq_{LTNG_{CB}}$  is the Lightning annualized frequency for the Census block (event-days per year).

$HLR_{LTNG_{CB_{Bldg}}}$  is the Bayesian-adjusted building Historic Loss Ratio for Lightning for the Census block.

$EAL_{LTNG_{CB_{Pop}}}$  is the population equivalence Expected Annual Loss due to Lightning occurrences for a specific Census block (in dollars).

$Exposure_{LTNG_{CB_{Pop}}}$  is the population equivalence value exposed to Lightning occurrences in the Census block (in dollars).

$HLR_{LTNG_{CB_{Pop}}}$  is the Bayesian-adjusted population Historic Loss Ratio for Lightning for the Census block.

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building and population equivalence Expected Annual Loss values at the Census block level as in [Equation 107](#).

### Equation 107: Census Tract and County Expected Annual Loss to Lightning

$$EAL_{LTNG_{CT}} = \sum_{CB}^{CT} EAL_{LTNG_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{LTNG_{CB_{Pop}}}$$

$$EAL_{LTNG_{Co}} = \sum_{CB}^{Co} EAL_{LTNG_{CB_{Bldg}}} + \sum_{CB}^{Co} EAL_{LTNG_{CB_{Pop}}}$$

where:

$EAL_{LTNG_{CT}}$  is the total Expected Annual Loss due to Lightning occurrences for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{LTNG_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Lightning occurrences for all Census blocks in the Census tract (in dollars).

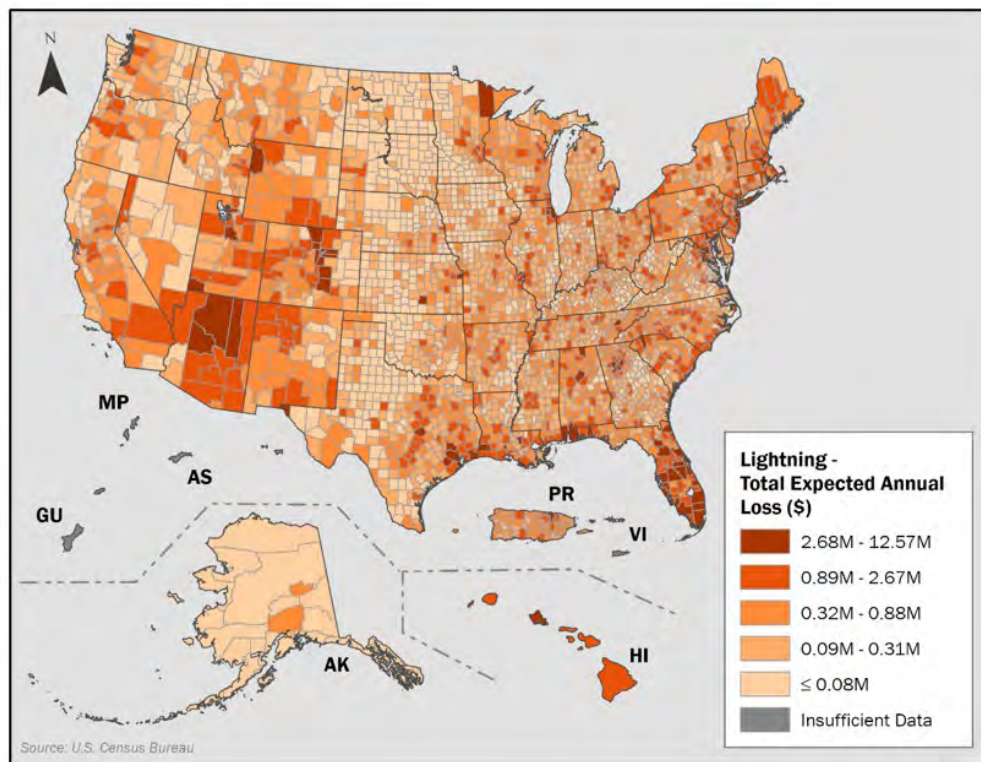
$\sum_{CB}^{CT} EAL_{LTNG_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Lightning occurrences for all Census blocks in the Census tract (in dollars).

$EAL_{LTNG_{Co}}$  is the total Expected Annual Loss due to Lightning occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{LTNG_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Lightning occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{LTNG_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Lightning occurrences for all Census blocks in the county (in dollars).

[Figure 86](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Lightning occurrences.



**Figure 86: Total Expected Annual Loss by County to Lightning**

With the Lightning total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Lightning Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Lightning Expected Annual Loss for population by the total population of the community.

## 17.8. Lightning Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- Reliable data sources for Lightning strikes with a sufficient history are difficult to come by for some United States Territories. As such, the source hazard data for the Lightning hazard are still limited to the Continental United States, Alaska, Hawai'i, and Puerto Rico, leaving American Samoa, Guam, Northern Mariana Islands, and the U.S. Virgin Islands out of the risk calculations for this hazard.



## 18. Strong Wind

Strong Wind consists of damaging winds, often originating from thunderstorms, which are classified as exceeding 58 mph.

Losses due to Strong Wind are modeled using Severe Weather Database Files data from the Storm Prediction Center and SHELDUS. The Severe Weather Database Files provide historical records of Strong Wind events, which are used for estimating event extents, calculating the historical occurrence, and calculating the frequency and exposure components. SHELDUS is used for historical losses, which are used to build the Historic Loss Ratio. The Strong Wind hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 18.1. Strong Wind Spatial Source Data

**Historical Occurrence Source:** [National Weather Service, Storm Prediction Center, Severe Weather Database Files](#)<sup>114</sup>

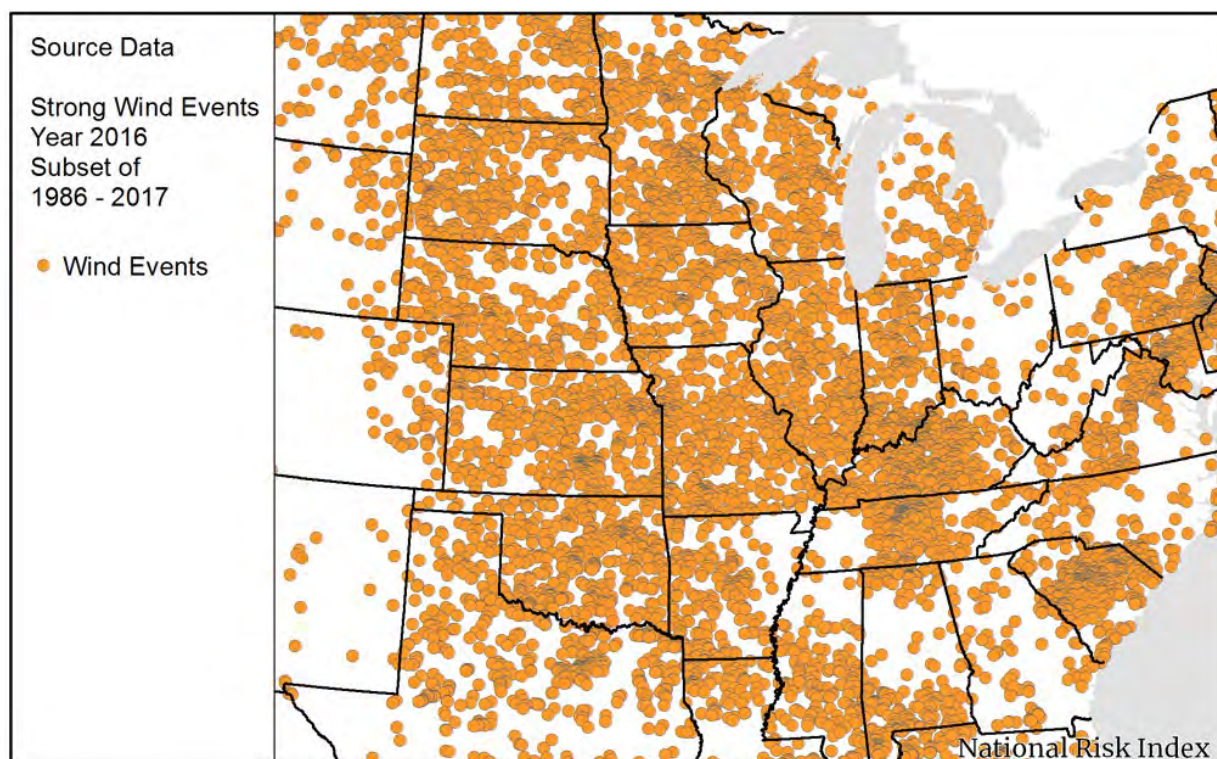
The Storm Prediction Center compiles all records of damaging Wind from the National Weather Service's monthly Storm Data publication and makes them available in CSV format on the Severe Weather Maps, Graphics, and Data Page website. These files record spatiotemporal information (start and end coordinates, date, time) as well as economic loss, wind speed in knots, and, from 2006 on, whether the wind speed was measured or estimated and whether the speed denotes a gust wind speed or a sustained wind speed (see [Table 67](#) and [Figure 87](#)). Many fields are empty for older records, especially those before 1985.

**Table 67: Sample Strong Wind Data from the Storm Prediction Center**

| <i>om</i><br>(Wind ID) | <i>Date</i>           | <i>State</i> | <i>mag</i><br>(Wind Speed [kt]) | <i>slon</i> (Start Longitude) | <i>slat</i> (Start Latitude) | <i>elon</i> (End Longitude) | <i>elat</i> (End Latitude) |
|------------------------|-----------------------|--------------|---------------------------------|-------------------------------|------------------------------|-----------------------------|----------------------------|
| 400                    | Oct. 23, 1955 7:00 PM | Montana      | 0                               | -84.58                        | 43.28                        | 0                           | 0                          |
| 553                    | Feb. 06 1999 10:37 PM | Arkansas     | 52                              | -93.92                        | 33.93                        | 0                           | 0                          |
| 636896                 | June 09, 2017 1:59 AM | Michigan     | 100                             | -111.86                       | 48.85                        | -111.86                     | 48.85                      |

<sup>114</sup> NWS – SPC, NOAA. (2024). Severe Weather Database files, Damaging Wind, 1955-2023 [online dataset]. Retrieved from <https://www.spc.noaa.gov/wcm/>.



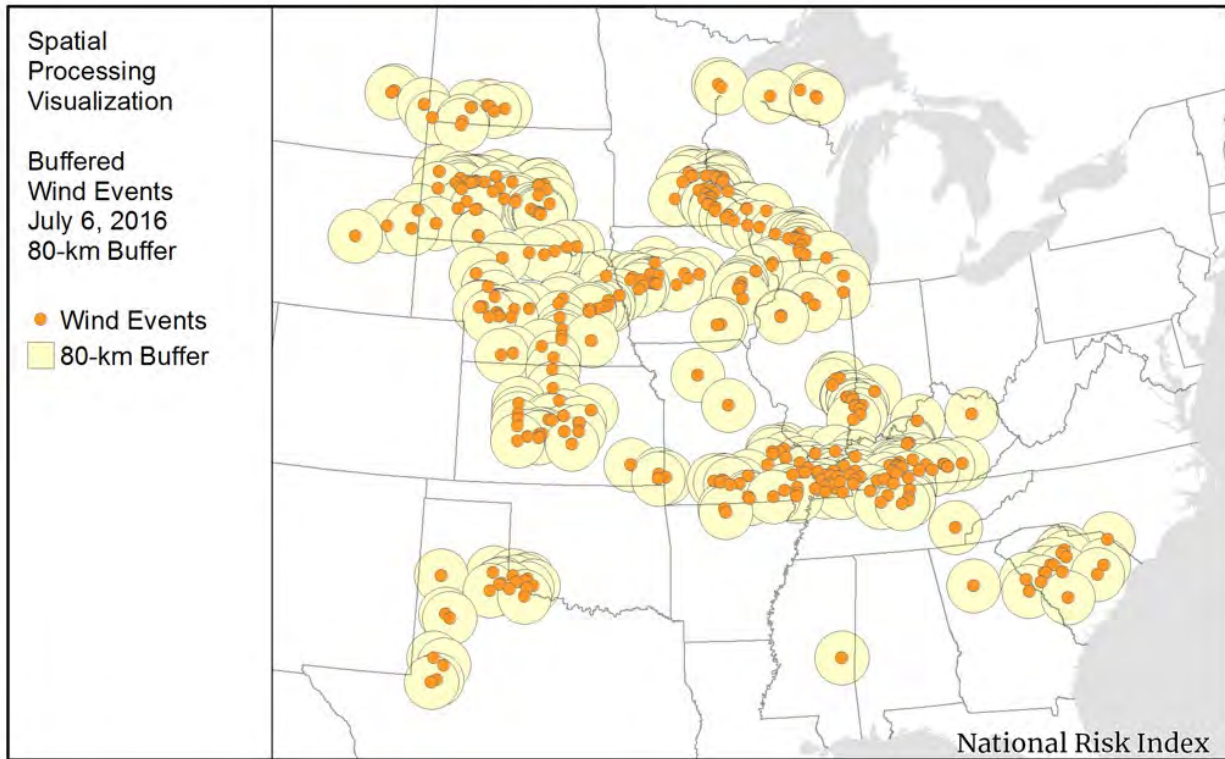


**Figure 87: Map of Wind Points**

## 18.2. Strong Wind Spatial Data Processing

The source data includes fields for two sets of coordinates, a start, and an end. This is mainly because the data share their format with the data for tornadoes. Most Wind events only have start coordinates (or the end coordinates match the start coordinates), so the points are projected from these coordinates. Any events outside of the period of record or with wind speeds of less than 50.4 knots (58 mph)<sup>115</sup> were filtered out. An 80-km buffer was created from the remaining points to produce a layer of Strong Wind event polygons (see [Figure 88](#)). The 80-km buffer is not an attempt to represent the area of impact by a Strong Wind event, but rather an effort to estimate the area where Strong Winds may have been present. The Strong Wind event polygons could then be used to estimate annualized frequency at the Census block level.

<sup>115</sup> This threshold is used by National Oceanic and Atmospheric Administration and the National Weather Service as the minimum wind gust criterion for a Severe Thunderstorm Watch.



**Figure 88: Map of Buffered Wind Points**

### 18.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Strong Winds can occur almost anywhere under the right conditions, so all counties were deemed possible for Strong Wind occurrence.

## 18.3. Strong Wind Non-Spatial Data Processing

### 18.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Strong Wind, in events, is initially computed as the number of distinct Strong Wind event polygons that intersect a 49-by-49-km fishnet grid cell. Buffering the Strong Wind points and using the fishnet grid to count historical Strong Wind events serves to spatially spread the influence of past Strong Wind events to nearby areas that may also be susceptible to Strong Wind but have not experienced Strong Wind as frequently. However, using these methods can overestimate Strong Wind frequency. To adjust for this, a national scaling factor is calculated (see [Equation 108](#)).

#### Equation 108: National Scaling Factor for Strong Wind Event Count

$$NatlScalingFactor_{SWND} = \frac{EventCount_{SWND_{Ntl}}}{\sum FishnetIntsctCount_{SWND_{Ntl}}}$$

where:

$NatlScalingFactor_{SWND}$  is the Strong Wind scaling factor to be applied to the fishnet grid cell event count.

$EventCount_{SWND_{Ntl}}$  is the count of distinct Strong Wind events which have occurred in the United States

$\sum FishnetIntsctCount_{SWND_{Ntl}}$  is the summed total of all Strong Wind event polygon-fishnet grid cell intersections in the United States

The scaling factor is then applied to the fishnet grid Strong Wind event count (see [Equation 109](#)).

#### Equation 109: Scaled Strong Wind Event Fishnet Count

$$ScaledEventCount_{SWND_{Fish}} = EventCount_{SWND_{Fish}} \times NatlScalingFactor_{SWND}$$

where:

$ScaledEventCount_{SWND_{Fish}}$  is the scaled count of Strong Wind events within a fishnet grid cell (in events per year).

$EventCount_{SWND_{Fish}}$  is the count of Strong Wind event polygons that intersect a 49-by-49 km fishnet grid cell.

$NatlScalingFactor_{SWND}$  is the Strong Wind scaling factor to be applied to the fishnet grid cell event count.

The Census block Strong Wind event count is then computed as the scaled event count of the fishnet grid cell that encompasses the Census block, or, if the Census block intersects multiple fishnet grid cells, an area-weighted count of the cells that intersect the Census block (see [Appendix D – Fishnet Occurrence Count](#)). This scaled count is used to compute Strong Wind event annualized frequency.

Historical event counts are also supplied at the Census tract and county levels as the scaled, area-weighted count of Strong Wind events intersecting fishnet grid cells that intersect the Census tract and county, respectively.

### 18.3.2. PERIOD OF RECORD

Wind speed data between Jan. 01, 1996 and Dec. 31, 2023 are analyzed, so the period of record for which Strong Wind data are utilized is 38.0 years.

## 18.4. Strong Wind Exposure

Because Strong Wind can occur anywhere, the entire building, population, and agriculture value of a Census block, Census tract, and county is considered exposed to Strong Wind. Agriculture value is

included as a consequence type for Strong Wind because more than 1% of economic loss due to Strong Wind recorded in SHELDS impacted agriculture. Population equivalence, which is used in select Expected Annual Loss calculations, is calculated by multiplying population by the Value of Statistical Life (\$13.7 million per person).

## 18.5. Strong Wind Annualized Frequency

The number of recorded Strong Wind occurrences, in events, each year over the period of record (38.0 years) is used to estimate the annualized frequency of Strong Wind events in an area. This annualized frequency is calculated at the Census block level, and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the Strong Wind event polygons created from the source data (as described in [Section 18.2 Strong Wind Spatial Data Processing](#)), as well as their corresponding computed duration days from the pre-processing of the data. The Census block Strong Wind event count computed using the scaled event counts of the fishnet grid cells intersecting the Census block is divided by the period of record to compute frequency as in [Equation 110](#).

### Equation 110: Census Block Strong Wind Annualized Frequency

$$Freq_{SWND_{CB}} = \frac{ScaledEventCount_{SWND_{Fish}}}{PeriodRecord_{SWND}}$$

where:

$Freq_{SWND_{CB}}$  is the annualized frequency of Strong Wind events determined for a specific Census block (events per year).

$ScaledEventCount_{SWND_{Fish}}$  is the scaled count of Strong Wind events calculated for the Census block.

$PeriodRecord_{SWND}$  is the period of record for Strong Wind (38.0 years).

### 18.5.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Strong Wind events at the Census tract and county level by the period of record, as the event count for annualized frequency is a fishnet area-weighted event count including Strong Wind events that may have impacted the surrounding area but not the county or Census tract itself. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 111](#).

**Equation 111: Census Tract and County Area-Weighted Strong Wind Annualized Frequency Aggregation**

$$Freq_{SWND_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{SWND_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{SWND_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{SWND_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{SWND_{CT}}$  is the area-weighted Strong Wind annualized frequency for a specific Census tract (events per year).

$Freq_{SWND_{CB}}$  is the annualized frequency of Strong Wind events determined for a specific Census block (events per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

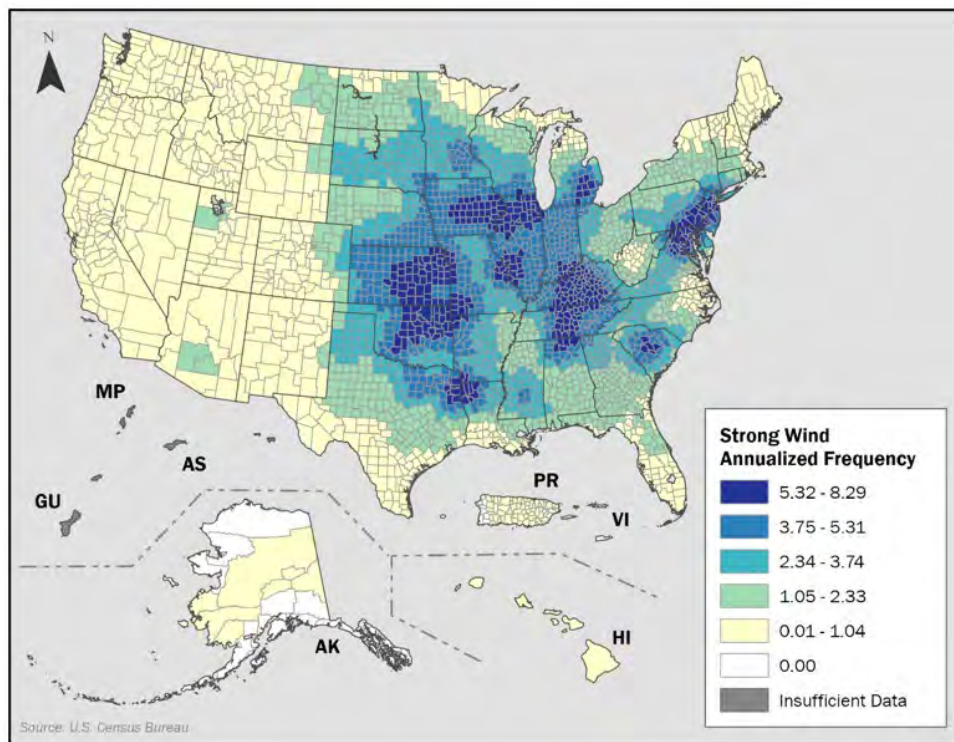
$Freq_{SWND_{Co}}$  is the area-weighted Strong Wind annualized frequency for a specific county (events per year).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 89](#) displays Strong Wind annualized frequency at the county level.





**Figure 89: Strong Wind Annualized Frequency by County**

## 18.6. Strong Wind Historic Loss Ratio

The Strong Wind Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Strong Wind event, or the average rate of loss associated with the occurrence of a Strong Wind event. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Strong Wind hazard type.

Loss data are provided by SHELUDS<sup>116</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELUDS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. Three peril types are mapped to the hazard Strong Wind (see [Table 68](#)). Native records of Strong Wind events that caused loss over more than one day have their loss assigned to the first day, and all records are aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 68: Strong Wind Peril Types and Recorded Events from 1996-2023**

| Peril Type in SHELUDS | Total SHELUDS Loss Records | Total Records per Event Basis |
|-----------------------|----------------------------|-------------------------------|
| Derecho               | 1                          | 1                             |

<sup>116</sup> For Strong Wind loss information, SHELUDS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.



| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Wind                          | 237,093                            | 170,449                              |
| Wind-Straight Line            | 0                                  | 0                                    |

The Historic Loss Ratio exposure value for Strong Wind is a county-level value that represents the dollar value of the total building value or the entire population of a county as recorded in Hazus 6.0, or the total crop and livestock value of the county as estimated in the United States Department of Agriculture 2017 Census of Agriculture data. The Loss Ratio per Basis for each SHEL DUS-documented event and each consequence type (building, population, and agriculture) is calculated using [Equation 112](#).

**Equation 112: Loss Ratio per Basis Calculation for a Single Strong Wind Event**

$$LRB_{SWND\ Co\ CnsqType} = \frac{Loss_{SWND\ Co\ CnsqType}}{HLRExposure_{Co\ CnsqType}}$$

where:

|                              |                                                                                                                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $LRB_{SWND\ Co\ CnsqType}$   | is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Strong Wind event. Calculation is performed for each consequence type (building, population, and agriculture). |
| $Loss_{SWND\ Co\ CnsqType}$  | is the loss (by consequence type) experienced from the Strong Wind event documented to have occurred in the county (in dollars or impacted people).                                                                                                        |
| $HLRExposure_{Co\ CnsqType}$ | is the total value (by consequence type) of the county estimated to have been exposed to the Strong Wind event (in dollars or people).                                                                                                                     |

Strong Wind events can occur with a high frequency in areas, but often result in no recorded loss to buildings, population, or agriculture. SHEL DUS does not record events in which no loss occurred, so a number of zero-loss events are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHEL DUS period of record (1996 through 2023). For Strong Wind, the historical event count is extracted using an intersection between the Census blocks and Strong Wind event polygons for the years 1986-2023. An annual rate is calculated as the event count divided by the period of record of 38.0 years, and this rate is multiplied by the SHEL DUS period of record of 28 years to estimate a historical event count for the appropriate time range.

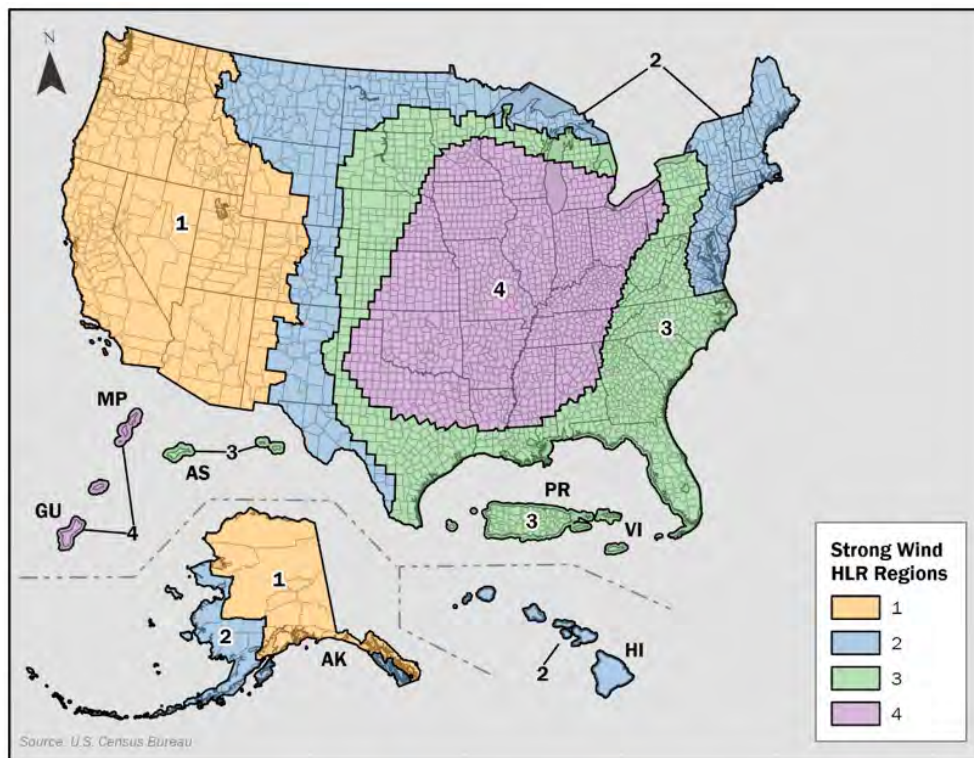
If the number of loss-causing Strong Wind event records from SHEL DUS is less than the scaled event count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

After the Loss Ratios per Basis are calculated for all Strong Wind events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Strong Wind.

**Table 69: Strong Wind Historic Loss Ratio Bayesian Weighting Level Sets**

| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>             | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|-----------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Agriculture             | A1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A2            | Neighbor Counties (Touch & 115km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A3            | Neighbor Counties (Touch & 208km) | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | A4            | Strong Wind Regions               | N/A                                                 | N/A                                              |
|                         | A5            | National                          | N/A                                                 | N/A                                              |
| Population              | P1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | P2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | P4            | Strong Wind Regions               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | P5            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
| Building                | B1            | County                            | N/A                                                 | 25 Loss Ratio per Basis records                  |
|                         | B2            | Neighbor Counties (Touch & 115km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B3            | Neighbor Counties (Touch & 208km) | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                  |
|                         | B4            | Strong Wind Regions               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |
|                         | B5            | National                          | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                              |

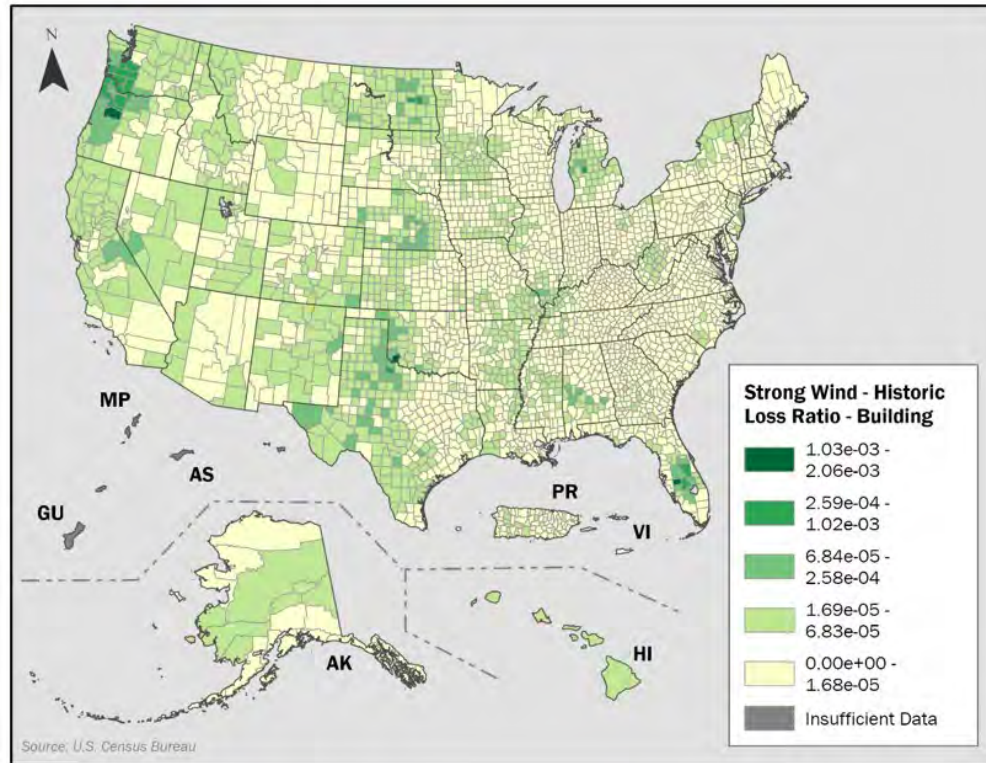
The regional definition for Strong Wind is derived using safe room design tornado speed zones as defined in the Federal Emergency Management Agency P-361 study<sup>117</sup> (see [Figure 90](#)).



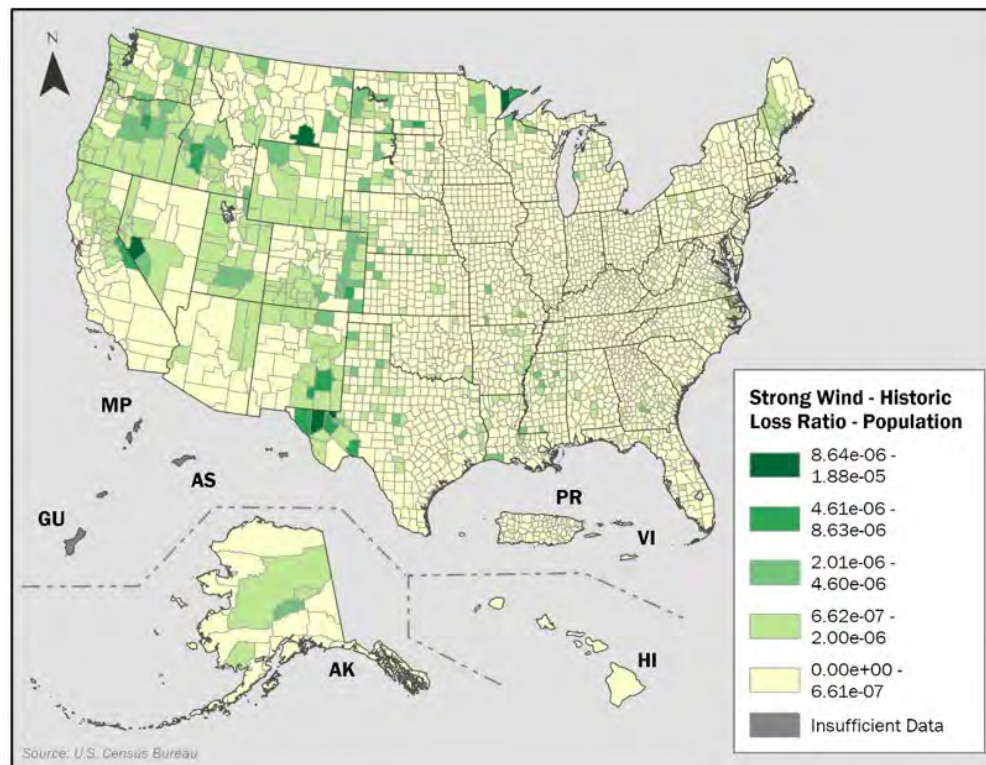
**Figure 90: Strong Wind-specific Historic Loss Ratio Regions**

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Strong Wind (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 91](#), [Figure 92](#), and [Figure 93](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Strong Wind. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.

<sup>117</sup> [FEMA P-361](#), Safe Rooms for Tornadoes and Hurricanes: Guidance for Community and Residential Safe Rooms

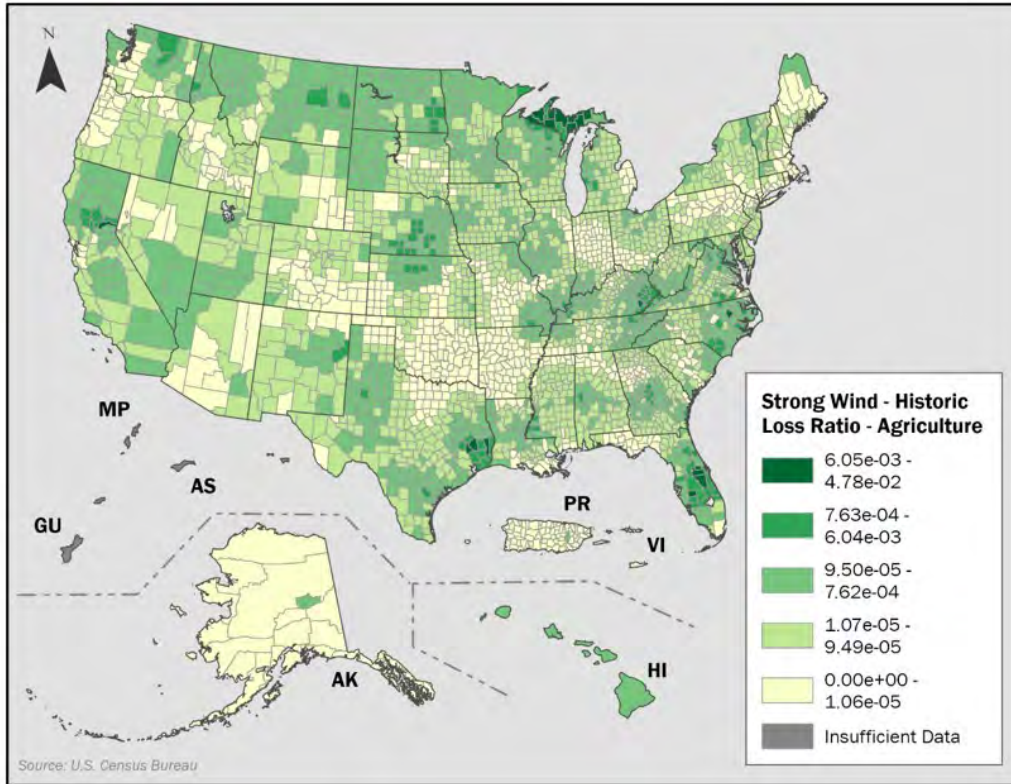


**Figure 91: Strong Wind Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 92: Strong Wind Bayesian-Adjusted Historic Loss Ratio – Population**





**Figure 93: Strong Wind Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

## 18.7. Strong Wind Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 113](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 113: Census Block Expected Annual Loss to Strong Wind

$$EAL_{SWND_{CB_{Bldg}}} = Exposure_{SWND_{CB_{Bldg}}} \times Freq_{SWND_{CB}} \times HLR_{SWND_{CB_{Bldg}}}$$

$$EAL_{SWND_{CB_{Pop}}} = Exposure_{SWND_{CB_{Pop}}} \times Freq_{SWND_{CB}} \times HLR_{SWND_{CB_{Pop}}}$$

$$EAL_{SWND_{CB_{Ag}}} = Exposure_{SWND_{CB_{Ag}}} \times Freq_{SWND_{CB}} \times HLR_{SWND_{CB_{Ag}}}$$

where:

$EAL_{SWND_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Strong Wind occurrences for a specific Census block (in dollars).

$Exposure_{SWND_{CB_{Bldg}}}$  is the building value exposed to Strong Wind occurrences in the Census block (in dollars).

|                              |                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $Freq_{SWND_{CB}}$           | is the Strong Wind annualized frequency for the Census block (events per year).                                             |
| $HLR_{SWND_{CB_{Bldg}}}$     | is the Bayesian-adjusted building Historic Loss Ratio for Strong Wind for the Census block.                                 |
| $EAL_{SWND_{CB_{Pop}}}$      | is the population equivalence Expected Annual Loss due to Strong Wind occurrences for a specific Census block (in dollars). |
| $Exposure_{SWND_{CB_{Pop}}}$ | is the population equivalence value exposed to Strong Wind occurrences in the Census block (in dollars).                    |
| $HLR_{SWND_{CB_{Pop}}}$      | is the Bayesian-adjusted population Historic Loss Ratio for Strong Wind for the Census block.                               |
| $EAL_{SWND_{CB_{Ag}}}$       | is the agriculture Expected Annual Loss due to Strong Wind occurrences for a specific Census block (in dollars).            |
| $Exposure_{SWND_{CB_{Ag}}}$  | is the agriculture value exposed to Strong Wind occurrences in the Census block (in dollars).                               |
| $HLR_{SWND_{CB_{Ag}}}$       | is the Bayesian-adjusted agriculture Historic Loss Ratio for Strong Wind for the Census block.                              |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 114](#).

**Equation 114: Census Tract and County Expected Annual Loss to Strong Wind**

$$EAL_{SWND_{CT}} = \sum_{CB}^{CT} EAL_{SWND_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{SWND_{CB_{Pop}}} + \sum_{CB}^{CT} EAL_{SWND_{CB_{Ag}}}$$

$$EAL_{SWND_{Co}} = \sum_{CB}^{Co} EAL_{SWND_{CB_{Bldg}}} + \sum_{CB}^{Co} EAL_{SWND_{CB_{Pop}}} + \sum_{CB}^{Co} EAL_{SWND_{CB_{Ag}}}$$

where:

|                                         |                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{SWND_{CT}}$                       | is the total Expected Annual Loss due to Strong Wind occurrences for a specific Census tract (in dollars).                           |
| $\sum_{CB}^{CT} EAL_{SWND_{CB_{Bldg}}}$ | is the summed building Expected Annual Loss due to Strong Wind for all Census blocks in the Census tract (in dollars).               |
| $\sum_{CB}^{CT} EAL_{SWND_{CB_{Pop}}}$  | is the summed population equivalence Expected Annual Loss due to Strong Wind for all Census blocks in the Census tract (in dollars). |



$\sum_{CB}^{CT} EAL_{SWND_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Strong Wind for all Census blocks in the Census tract (in dollars).

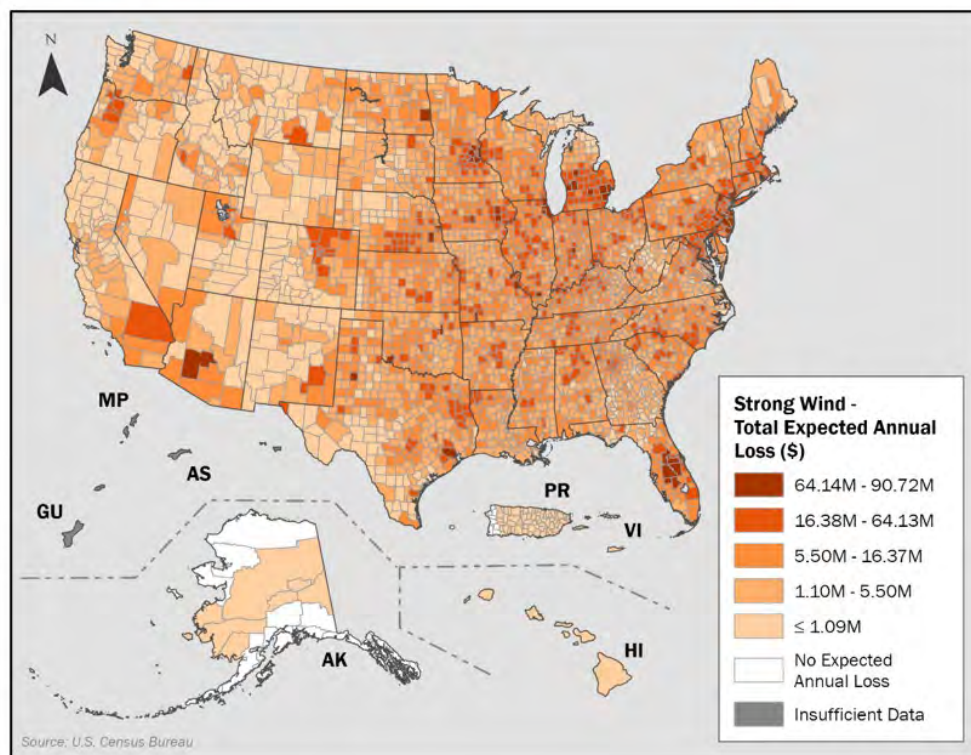
$EAL_{SWND_{Co}}$  is the total Expected Annual Loss due to Strong Wind occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{SWND_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Strong Wind for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{SWND_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Strong Wind for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{SWND_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Strong Wind for all Census blocks in the county (in dollars).

[Figure 94](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Strong Wind occurrences.



**Figure 94: Total Expected Annual Loss by County to Strong Wind**

With the Strong Wind total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Strong Wind Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Strong Wind Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Strong Wind Expected Annual Loss for agriculture by the total agriculture value of the community.

## **18.8. Strong Wind Methodology Discussion**

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- Due to the inexact differentiation of Strong Wind, Hurricane Wind, and Tornado-caused damage for historical wind events, the data development team recognizes that the underlying data may combine losses from different strong wind impacts. Additionally, many hazards are exacerbated by the impacts of strong winds, including wildfire and drought, among others.

## 19. Tornado

A Tornado is a narrow, violently rotating column of air that extends from the base of a thunderstorm to the ground and is visible only if it forms a condensation funnel made up of water droplets, dust, and debris.

The Tornado hazard is a complex model that involves subtype distinctions that are processed separately before being combined into the overall Tornado hazard. These subtypes are based on the severity of the Tornado event. The model uses the Severe Weather Database Files data from the Storm Prediction Center and SHELDUS as its data sources. The Severe Weather Database Files provide historical records of Tornado events, which are used for establishing the three subtypes, calculating the historical occurrence of each subtype, and calculating the frequency and exposure components. SHELDUS is used for historical losses, which are used to build the Historic Loss Ratio. The Tornado hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 19.1. Tornado Spatial Source Data

**Historical Occurrence Source:** [National Weather Service, Storm Prediction Center, Severe Weather Database Files](#)<sup>118</sup>

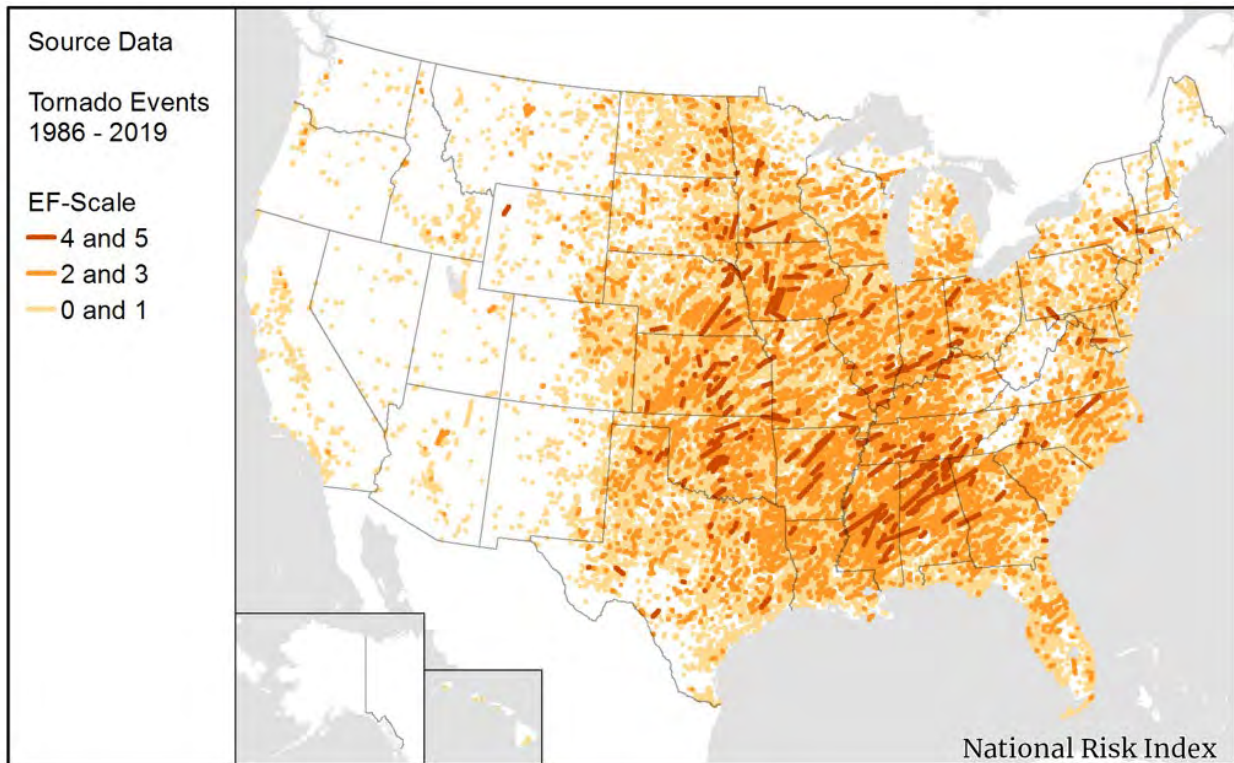
The Storm Prediction Center compiles all records of Tornadoes from the National Weather Service's monthly Storm Data publication and makes them available in CSV and shapefile format on the Severe Weather Maps, Graphics, and Data Page website. Shapefiles representing Tornadoes as both points (initial touchdown points) and lines (paths) were downloaded. These files record spatiotemporal information (start and end coordinates, date, time) as well as economic loss, injuries, fatalities, and, depending on the date of the Tornado, Fujita (F-) or EF- scale category (see [Table 70](#) and [Figure 95](#)). Economic loss information is recorded as either a predefined category of loss (1950-1995), a value representing loss in millions of dollars (1996-2015), or the loss in dollars (2016-present). Tornado records with two distinct sets of start and end coordinates represent a Tornado path. A record with identical start and end coordinates or with no end coordinates represents a Tornado touchdown.

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<sup>118</sup> SPC, NWS. (2024). Severe Weather Database files, Tornado, 1950-2023 [online dataset]. Retrieved from <http://www.spc.noaa.gov/wcm/>.

**Table 70: Sample Tornado Data from the Storm Prediction Center**

| om<br>(Tornado<br>Number<br>[before<br>2007]) | Date                      | State    | Mag<br>(F/EF<br>Scale) | Injuries | Fatalities | loss (Loss<br>Category or \$) | len (Path<br>Length<br>in Miles) | wid (Path<br>Width in<br>Yards) |
|-----------------------------------------------|---------------------------|----------|------------------------|----------|------------|-------------------------------|----------------------------------|---------------------------------|
| 1                                             | Jan. 03, 1950<br>11:00 AM | Missouri | 3                      | 3        | 0          | 6                             | 9.5                              | 150                             |
| 241                                           | May 15, 1989<br>3:35 PM   | Texas    | 1                      | 3        | 0          | 5                             | 5.5                              | 80                              |
| 0                                             | Dec. 20, 2017<br>12:15 PM | Georgia  | 0                      | 0        | 0          | 30000                         | 3.17                             | 125                             |

**Figure 95: Map of Tornado Source Data**

## 19.2. Tornado Spatial Data Processing

Tornado records in the path shapefile provided by the Storm Prediction Center may have empty geometries if the path information is incomplete. To form a complete set of geometries for all Tornadoes within the period of record, path records with empty geometries were replaced by the

record with the same unique ID in the point shapefile. Any Tornadoes outside the period of record or that have an F- or EF-scale of -9 to signify insufficient data were filtered out.

With the intended spatial processing goal of intersecting Tornado events to determine the Census block (and parent county) that the Tornado traversed, Tornado path lines and touchdown points were buffered to create tornado event path polygons. To conservatively estimate the largest area for each Tornado polygon, even those without a complete path geometry from the source data, three methods were used to calculate a possible buffer radius. Whichever method yielded the largest radius was used to buffer the given line or point. Options for buffer radii were:

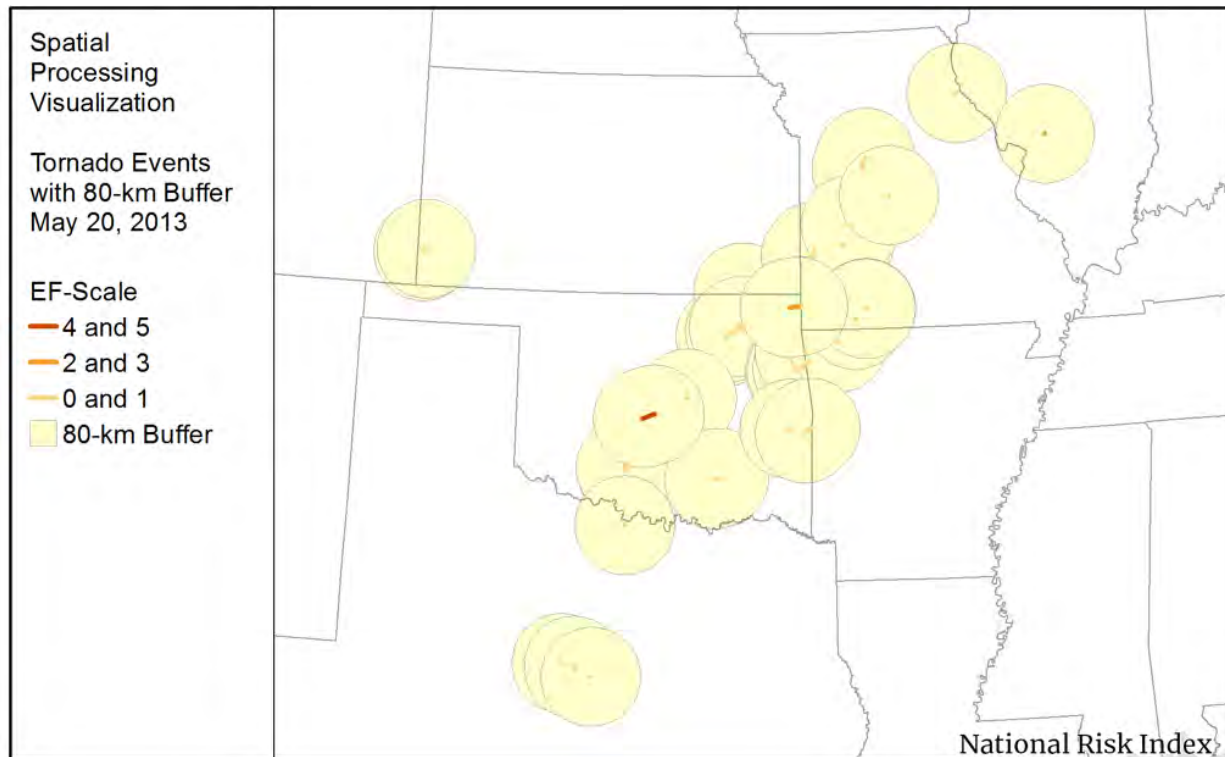
- Half of the Tornado width as specified in [Table 70](#) (converted from yards to meters);
- The calculated radius of the Tornado as extrapolated from its length and width (converted to meters) as provided in the source data; or
- The average radius of impact for a storm of that magnitude based on F- or EF-scale category according to [Table 71](#).

**Table 71: Tornado Categories**

| <i>F-Scale Category</i> | <i>Tornado Touchdown Point Buffer (meter)</i> | <i>Tornado Path Line Buffer (meter)</i> |
|-------------------------|-----------------------------------------------|-----------------------------------------|
| 0                       | 27                                            | 48                                      |
| 1                       | 54                                            | 134                                     |
| 2                       | 110                                           | 269                                     |
| 3                       | 172                                           | 535                                     |
| 4                       | 249                                           | 776                                     |
| 5                       | 249                                           | 1,233                                   |

The resulting category-buffered Tornado event path polygons are intersected with the Census blocks to determine the counties that might have experienced loss from each Tornado event. This relationship is used in the Historic Loss Ratio calculation, as well as for determining historical Tornado event counts at the Census tract and county level.

Because Tornado occurrences are recorded at distinct locations and multiple Tornadoes are often reported on the same day in near proximity, it was necessary for annualized frequency estimation to spread the influence of the reported historical event. Thus, an additional 80-km buffer was created from these category-buffered polygons (see [Figure 96](#)). This 80-km buffer radius is not an attempt to represent the Tornado's impact area. Rather, it is better to represent the area where the event could possibly have occurred. The 80-km buffered Tornado event path polygons are intersected with the 49-by-49-km fishnet grid and used to estimate annualized frequency at the Census tract level.



**Figure 96: Map of Buffered Tornadoes**

### 19.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Tornadoes are capable of occurring almost anywhere under the right conditions, so all counties were deemed possible for Tornado occurrence.

## 19.3. Tornado Non-Spatial Data Processing

### 19.3.1. TORNADO HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Tornado, in events, is computed as the number of distinct Tornado event path polygons that intersect a Census block. Historical event counts are supplied at the Census tract and county levels as the number of distinct Tornado event path polygons that intersect the Census tract and county, respectively. Note that this historical event count is not used to calculate annualized frequency.

### 19.3.2. PERIOD OF RECORD

Tornado data between Jan. 01, 1950 and Dec. 31, 2023 are analyzed, so the period of record for which Tornado data are utilized is 74.0 years. The entire 74-year period of record is utilized for tornadoes in EF-scale categories 2, 3, 4, and 5 while tornado data between Jan. 01, 1986 and Dec. 31, 2023 (38.0 years) are utilized for tornadoes in EF-scale categories 0 and 1.



## 19.4. Tornado Exposure

The size of the damage area caused by a Tornado can vary greatly depending on its magnitude or EF-scale. For this reason, exposure is calculated for three sub-types: 1) EF-scale 0 and 1; 2) EF-scale 2 and 3; and 3) EF-scale 4 and 5. An average historical Tornado damage area is assigned to each Tornado sub-type (see [Table 72](#)). These average damage area values were calculated from the historical set of Tornado event path polygons generated according to one of the three methods described in [Section 19.2 Tornado Spatial Data Processing](#) and not from the subsequent 80-km buffer applied to the Tornado event path polygon used for frequency estimation.

**Table 72: Average Historical Damage Area by Tornado Sub-Type**

| <i>Tornado Sub-Type</i> | <i>Average Historical Damage Area</i> |
|-------------------------|---------------------------------------|
| EF-Scale 0 and 1        | 0.7 km <sup>2</sup>                   |
| EF-Scale 2 and 3        | 9.6 km <sup>2</sup>                   |
| EF-Scale 4 and 5        | 80.7 km <sup>2</sup>                  |

Because a Tornado could occur anywhere in the Census tract, the Census tract average density (the Census tract's total building value, population equivalence, or agriculture value divided by the total area of the Census tract) is applied. Therefore, the exposure area of a Census tract is calculated using [Equation 115](#) and the exposure area of a county is calculated using [Equation 116](#).

**Equation 115: Census Tract Tornado Sub-Type Exposure**

$$Exposure_{TRNDSubCTBldg} = DamageArea_{TRNDSub} \times AvgDen_{CTBldg}$$

$$Exposure_{TRNDSubCTPop} = (DamageArea_{TRNDSub} \times AvgDen_{CTPop}) \times VSL$$

$$Exposure_{TRNDSubCTAg} = DamageArea_{TRNDSub} \times AvgDen_{CTAg}$$

where:

$Exposure_{TRNDSubCTBldg}$  is the building value exposed to Tornadoes of a sub-type in a specific Census tract (in dollars).

$DamageArea_{TRNDSub}$  is the average damage area of a Tornado sub-type (in square kilometers).

$AvgDen_{CTBldg}$  is the average building value density of the Census tract (in dollars per square kilometer).

|                           |                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| $Exposure_{TRNDSubCTPop}$ | is the population equivalence exposed to Tornadoes of a sub-type in a specific Census tract (in dollars). |
| $AvgDen_{CTPop}$          | is the average population density of the Census tract (in people per square kilometer).                   |
| $VSL$                     | is the Value of Statistical Life (\$13.7 million per person).                                             |
| $Exposure_{TRNDSubCTAg}$  | is the agriculture value exposed to Tornadoes of a sub-type in a specific Census tract (in dollars).      |
| $AvgDen_{CTAg}$           | is the average agriculture value density of the Census tract (in dollars per square kilometer).           |

### Equation 116: County Tornado Sub-Type Exposure

$$Exposure_{TRNDSubCoBldg} = DamageArea_{TRNDSub} \times AvgDen_{CoBldg}$$

$$Exposure_{TRNDSubCoPop} = (DamageArea_{TRNDSub} \times AvgDen_{CoPop}) \times VSL$$

$$Exposure_{TRNDSubCoAg} = DamageArea_{TRNDSub} \times AvgDen_{CoAg}$$

where:

|                            |                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------|
| $Exposure_{TRNDSubCoBldg}$ | is the building value exposed to Tornadoes of a sub-type in a specific county (in dollars).               |
| $DamageArea_{TRNDSub}$     | is the average damage area of a Tornado sub-type (in square kilometers).                                  |
| $AvgDen_{CoBldg}$          | is the average building value density of the county (in dollars per square kilometer).                    |
| $Exposure_{TRNDSubCoPop}$  | is the population equivalence value exposed to Tornadoes of a sub-type in a specific county (in dollars). |
| $AvgDen_{CoPop}$           | is the average population density of the county (in people per square kilometer).                         |
| $VSL$                      | is the Value of Statistical Life (\$13.7 million per person).                                             |
| $Exposure_{TRNDSubCoAg}$   | is the agriculture value exposed to Tornadoes of a sub-type in a specific county (in dollars).            |
| $AvgDen_{CoAg}$            | is the average agriculture value density of the county (in dollars per square kilometer).                 |

Note that exposure values for each sub-type are multiplied by their respective sub-type annualized frequency and Historic Loss Ratio to calculate the sub-type Expected Annual Loss. Exposure values provided in the data downloads are surrogates representing the entire building value, population, and agriculture value of the Census tract or county.

## 19.5. Tornado Annualized Frequency

The annualized frequency value represents the estimated number of recorded Tornado occurrences, in events, each year for a specific area. Annualized frequency is calculated for each Tornado sub-type at the county and Census tract levels, which are used in the Expected Annual Loss calculations.

Annualized frequency calculations are determined by intersecting the 80-km buffered Tornado event path polygons generated in [Section 19.2 Tornado Spatial Data Processing](#) with a 49-by-49-km fishnet grid. The count of distinct Tornado event path polygons of each sub-type intersecting each grid cell is recorded (see [Appendix D – Fishnet Occurrence Count](#)).

Buffering the Tornado paths and using the 49-by-49-km fishnet grid to count historical Tornado events serves to spatially spread the influence of past events to nearby areas that may also be susceptible to Tornadoes but have not experienced as many. However, using these methods can overestimate Tornado annualized frequency. To adjust for this, a national scaling factor is calculated for each Tornado sub-type (see [Equation 117](#)).

### Equation 117: National Scaling Factor by Tornado Sub-Type

$$NatlScalingFactor_{TRNDSub} = \frac{EventCount_{TRNDSubNtl}}{FishnetIntsctCount_{TRNDSubNtl}}$$

where:

|                                   |                                                                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $NatlScalingFactor_{TRNDSub}$     | is the Tornado sub-type scaling factor to be applied to fishnet grid cell frequency.                                               |
| $EventCount_{TRNDSubNtl}$         | is the count of distinct Tornado events of a sub-type that have occurred in the United States                                      |
| $FishnetIntsctCount_{TRNDSubNtl}$ | is the summed total count of all 80-km buffered Tornado-fishnet grid cell intersections of a Tornado sub-type in the United States |

A minimum scaling factor is also calculated for fishnet grid cells that do not intersect a historical Tornado path. This scaling factor is set to 1/932 or one divided by the total count of all 49-by-49-km fishnet grid cells that do not intersect a historical Tornado path but intersect one or more United States counties (see [Table 73](#)).

**Table 73: National Scaling Factor by Tornado Sub-Type**

| <i>Tornado Sub-Type</i>      | <i>National Tornado Event Count</i> | <i>National Fishnet Intersect Count</i> | <i>National Scaling Factor</i> |
|------------------------------|-------------------------------------|-----------------------------------------|--------------------------------|
| EF-Scale 0 and 1             | 38,524                              | 622,232                                 | 0.06191                        |
| EF-Scale 2 and 3             | 12,352                              | 213015                                  | 0.05799                        |
| EF-Scale 4 and 5             | 646                                 | 13189                                   | 0.04898                        |
| No historical Tornado events | 1                                   | 932                                     | 0.00107                        |

This national scaling factor is applied to each grid cell sub-type event count and divided by the period of record to calculate a grid cell sub-type annualized frequency. This is then divided by the representative area associated with the grid cell to produce a sub-type annualized frequency rate per square kilometer (see [Equation 118](#)). By definition, the 49-by-49-km fishnet grid cells are equally sized at 2,401 square kilometers; however, the land area a grid cell covers varies by location. Grid cells along the coasts can have significant portions covering water and grid cells along the United States border can have significant portions covering foreign lands. Both cases would result in overestimating frequency for the counties and Census tracts associated with these grid cells. To balance these situations, the area associated with the fishnet grid cell is defined as the average of the fishnet grid size and the area covering United States land. This assists with the apportionment of the fishnet frequency rate per square kilometer.

**Equation 118: Scaled Tornado Sub-Type Fishnet Annualized Frequency and Annualized Frequency Rate**

$$Freq_{TRNDSubFish} = \frac{EventCount_{TRNDSubFish} \times NatlScalingFactor_{TRNDSub}}{PeriodRecord_{TRND}}$$

$$FreqRate_{TRNDSubFish} = \frac{Freq_{TRNDSubFish}}{RepresentativeArea_{Fish}}$$

where:

$Freq_{TRNDSubFish}$  is the annualized frequency of Tornado sub-type events within a fishnet grid cell (in events per year).

$EventCount_{TRNDSubFish}$  is the count of distinct Tornado path polygons that intersect a 49-by-49 km fishnet grid cell.

$NatlScalingFactor_{TRNDSub}$  is the Tornado sub-type scaling factor to be applied to fishnet grid cell frequency.

$PeriodRecord_{TRND}$  is the period of record for Tornado (38.0 years for F0/1 and 74.0 years for F-2/3/4/5).

$FreqRate_{TRNDSubFish}$  is the Tornado sub-type annualized frequency rate of the fishnet grid cell (in annualized frequency per square kilometer).

$RepresentativeArea_{Fish}$  is the average of the fishnet grid cell size (2,401 square kilometers) and the United States land area within the fishnet grid cell (in square kilometers).

Area-weighted Tornado sub-type annualized frequency rates are then calculated at the Census tract and county levels (see [Equation 119](#)).

**Equation 119: Census Tract and County Tornado Sub-Type Area-Weighted Annualized Frequency Rates**

$$FreqRate_{TRNDSubCT} = \sum_{CB}^{CT} \frac{(FreqRate_{TRNDSubFish} \times IntsctArea_{CBFish})}{Area_{CT}}$$

$$FreqRate_{TRNDSubCo} = \sum_{CB}^{Co} \frac{(FreqRate_{TRNDSubFish} \times IntsctArea_{CBFish})}{Area_{Co}}$$

where:

$FreqRate_{TRNDSubCT}$  is the Tornado sub-type annualized frequency rate of the Census tract (in annualized frequency per square kilometer).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

$FreqRate_{TRNDSubFish}$  is the Tornado sub-type annualized frequency rate of the fishnet grid cell (in annualized frequency per square kilometer).

$IntsctArea_{CBFish}$  is the intersected area of the Census block with a specific fishnet grid cell (in square kilometers).

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$FreqRate_{TRNDSubCo}$  is the Tornado sub-type annualized frequency rate of the county (in annualized frequency per square kilometer).

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

In rare cases, a Census tract or county may have experienced more high-magnitude Tornado events than low-magnitude events. To treat these statistical anomalies, lower magnitude sub-type annualized frequency rates are scaled up to at least match the next highest magnitude sub-type annualized frequency rate. For example, if an EF-scale 2 and 3 annualized frequency rate for a county is  $1 \times 10^{-6}$  and its EF-scale 4 and 5 has a higher annualized frequency rate of  $5 \times 10^{-6}$ , the EF-

scale 2 and 3 annualized frequency rate is set to  $5 \times 10^{-6}$ . Additionally, if there is a gap in sub-types, then the missing sub-type is inserted and assigned the annualized frequency rate of the higher magnitude sub-type. These actions ensure that, as the sub-type magnitude increases, the annualized frequency rate is less than or equal to the previous sub-type rate.

The Census tract and county annualized frequencies are then calculated as the Tornado sub-type annualized frequency rate multiplied by the area for the Census tract or county as in [Equation 120](#). Note that if the Census tract or county area is less than the average damage area for the Tornado sub-type, then the annualized frequency is calculated as the annualized frequency rate multiplied by the average damage area for the Tornado sub-type. This is done to match the assumptions made for exposure to annualized frequency.

**Equation 120: Census Tract and County Tornado Sub-Type Annualized Frequency**

$$Freq_{TRNDSub_{CT}} = FreqRate_{TRNDSub_{CT}} \times Area_{CT}$$

$$Freq_{TRNDSub_{Co}} = FreqRate_{TRNDSub_{Co}} \times Area_{Co}$$

where:

$Freq_{TRNDSub_{CT}}$  is the scaled annualized frequency of Tornado sub-type events within a Census tract (in events per year).

$FreqRate_{TRNDSub_{CT}}$  is the Tornado sub-type annualized frequency rate of the Census tract (in frequency per square kilometer).

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{TRNDSub_{Co}}$  is the scaled annualized frequency of Tornado sub-type events within a county (in events per year).

$FreqRate_{TRNDSub_{Co}}$  is the Tornado sub-type annualized frequency rate of the county (in frequency per square kilometer).

$Area_{Co}$  is the total area of the county (in square kilometers).

### 19.5.1. MINIMUM ANNUAL FREQUENCY

A Minimum Annual Frequency is calculated for Census tracts and counties that do not intersect a fishnet grid cell that has experienced a past Tornado event. This is calculated in the same way as other sub-type frequencies with the minimum historical Tornado event count set to 1 and uses the smallest average damage area (0.7 km<sup>2</sup> for EF-scale 0 and 1) to multiply by the annualized frequency rate as in [Equation 121](#).



**Equation 121: Census Tract and County Tornado Minimum Annual Frequency**

$$Freq_{TMAF} = \frac{1 \times NatlScalingFactor_{TMAF}}{PeriodRecord_{TRND}}$$

$$FreqRate_{TMAF_{CT}} = \frac{Freq_{TMAF}}{Area_{CT}}$$

$$FreqRate_{TMAF_{Co}} = \frac{Freq_{TMAF}}{Area_{Co}}$$

$$Freq_{TMAF_{CT}} = FreqRate_{TMAF_{CT}} \times DamageArea_{EF0\&1}$$

$$Freq_{TMAF_{Co}} = FreqRate_{TMAF_{Co}} \times DamageArea_{EF0\&1}$$

where:

$Freq_{TMAF}$  is the scaled Minimum Annual Frequency of Tornado events (in events per year).

$NatlScalingFactor_{TMAF}$  is the minimum scaling factor to be applied to fishnet grid cell Tornado frequency.

$PeriodRecord_{TRND}$  is the period of record for Tornado (38.0 years).

$FreqRate_{TMAF_{CT}}$  is the minimum Tornado annualized frequency rate of the Census tract (in frequency per square kilometer).

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$FreqRate_{TMAF_{Co}}$  is the minimum Tornado frequency rate of the county (in frequency per square kilometer).

$Area_{Co}$  is the total area of the county (in square kilometers).

$Freq_{TMAF_{CT}}$  is the minimum annualized frequency of Tornado events within a Census tract (in events per year).

$DamageArea_{EF0\&1}$  is the average damage area for EF-scale 0 and 1 Tornadoes (0.7 km<sup>2</sup>).

$Freq_{TMAF_{Co}}$  is the minimum annualized frequency of Tornado events within a county (in events per year).

**19.5.2. ANNUALIZED FREQUENCY AGGREGATION**

Annualized frequency values for each Tornado sub-type are multiplied by their respective sub-type exposure and Historic Loss Ratio to calculate the sub-type Expected Annual Loss. Annualized frequency values provided in the dataset are a surrogate value calculated as the sum of all sub-type

frequencies at the Census tract and county levels as in [Equation 122](#) Census tracts and counties with a frequency of 0 for all sub-types use the Minimum Annual Frequency.

**Equation 122: Census Tract and County Tornado Sub-Type Annualized Frequency Aggregation**

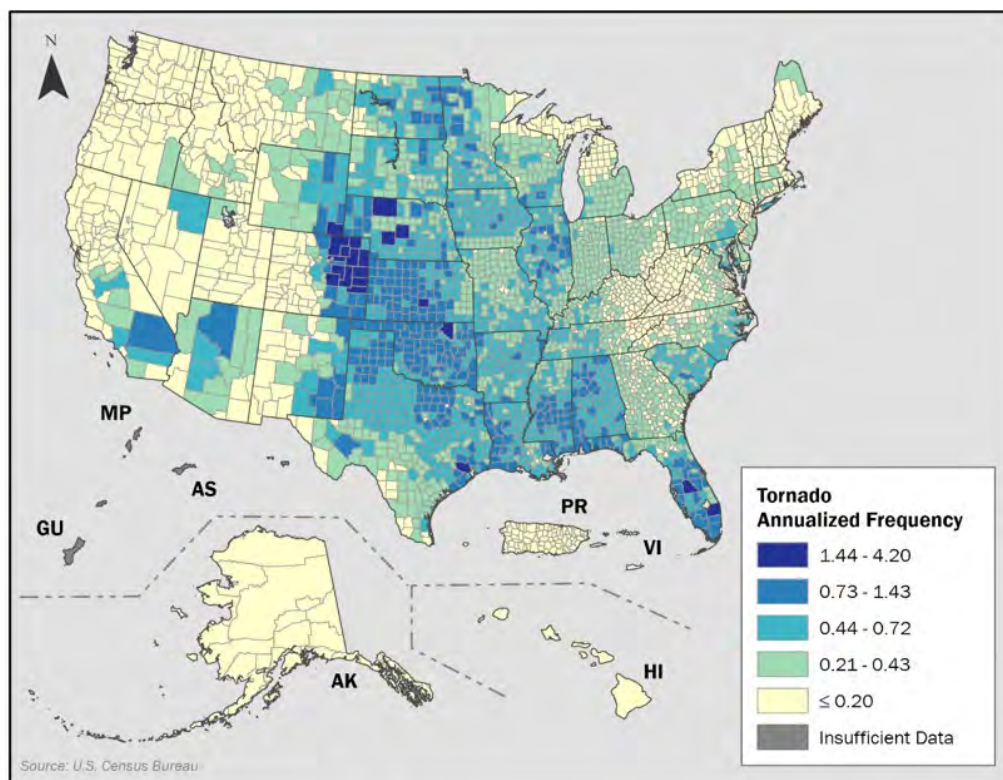
$$Freq_{TRND_{CT}} = Freq_{EF0\&1_{CT}} + Freq_{EF2\&3_{CT}} + Freq_{EF4\&5_{CT}}$$

$$Freq_{TRND_{Co}} = Freq_{EF0\&1_{Co}} + Freq_{EF2\&3_{Co}} + Freq_{EF4\&5_{Co}}$$

where:

|                      |                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| $Freq_{TRND_{CT}}$   | is the Tornado annualized frequency calculated for a specific Census tract (events per year).                  |
| $Freq_{EF0\&1_{CT}}$ | is the EF-scale 0 and 1 Tornado annualized frequency calculated for a specific Census tract (events per year). |
| $Freq_{EF2\&3_{CT}}$ | is the EF-scale 2 and 3 Tornado annualized frequency calculated for a specific Census tract (events per year). |
| $Freq_{EF4\&5_{CT}}$ | is the EF-scale 4 and 5 Tornado annualized frequency calculated for a specific Census tract (events per year). |
| $Freq_{TRND_{Co}}$   | is the Tornado annualized frequency calculated for a specific county (events per year).                        |
| $Freq_{EF0\&1_{Co}}$ | is the EF-scale 0 and 1 Tornado annualized frequency calculated for a specific county (events per year).       |
| $Freq_{EF2\&3_{Co}}$ | is the EF-scale 2 and 3 Tornado annualized frequency calculated for a specific county (events per year).       |
| $Freq_{EF4\&5_{Co}}$ | is the EF-scale 4 and 5 Tornado annualized frequency calculated for a specific county (events per year).       |

[Figure 97](#) displays Tornado annualized frequency at the county level.



**Figure 97: Tornado Annualized Frequency by County**

## 19.6. Tornado Historic Loss Ratio

The Tornado Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Tornado occurrence, or the average rate of loss associated with a Tornado occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Tornado hazard type. Unique to the Tornado hazard, individual Historic Loss Ratios were calculated for each severity sub-type within each consequence type to reflect the diverse levels of destruction that can occur across the EF scale. Unlike other hazards in this analysis, the overall consequence-specific Historic Loss Ratios are determined through a back-calculation from the final Expected Annual Loss value, as discussed later in this section.

Loss data are provided by SHELUDS<sup>119</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated for each of the individual Tornado sub-types. SHELUDS events from 1996 through 2019 are included in the Historic Loss Ratio calculation. Five peril types are mapped to the hazard Tornado (see [Table 74](#)). Native records of Tornadoes that caused loss over more than one day (such as those that occurred overnight) have their loss assigned to the first day (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

<sup>119</sup> For Tornado loss information, SHELUDS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

**Table 74: Tornado Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Fire-Tornado                  | 0                                  | 2                                    |
| Tornado                       | 20,408                             | 48,421                               |
| Waterspout                    | 10                                 | 534                                  |
| Wind-Tornadic                 | 0                                  | 46                                   |
| Wind-Vortex                   | 2                                  | 16                                   |

Tornado Expected Annual Loss calculations require sub-type specific Historic Loss Ratio values. To accomplish this, native SHEL DUS loss records are matched to specific Tornadoes, so that each Loss Ratio per Basis is calculated using loss attributed to a specific Tornado and its exposure within a county as determined by the tornado event-path polygon and Census block intersects. In most cases, a single Tornado event occurs on a single day in a single county. However, multiple Tornadoes can occur in the same county on the same day, and a single Tornado may cause damage in multiple counties. To make this matching as precise as possible, several strategies were implemented.

SHEL DUS Tornado records from 2000 forward typically include a unique identifier to link them to records in the National Centers for Environmental Information Storm Events Database. The National Centers for Environmental Information records have additional information that is not in the SHEL DUS data but is often present in the Tornado spatial source data. This includes event timestamps, EF-scale, length and width of the Tornado path, begin and end coordinates of the path, and the full Tornado event loss data. A fuzzy logic approach was used to map as many historical Tornado paths as possible to their county-specific loss in SHEL DUS using these fields held in common between the spatial source data and National Centers for Environmental Information data. For example, it could be assumed that if a Tornado path record occurred on the same day with the same timestamp and had the same recorded EF-scale, physical dimensions, and geographic coordinates as a National Centers for Environmental Information record, both sources were describing the same Tornado event. This approach was iterated multiple times, each time using less stringent requirements for matching. By matching a National Centers for Environmental Information record to a Tornado path, the exposure of a specific Tornado within a county could be matched to its SHEL DUS county loss data.

Tornadoes with an EF-scale of 4 or 5 are rare and have much larger damage areas. They often occur during particularly bad storms, which can spawn multiple Tornadoes and can have long paths that impact multiple counties. Some of these Tornadoes may not be matched in the source data using the fuzzy logic approach, which could reduce their impact in the Historic Loss Ratio and give an inaccurate estimate of Expected Annual Loss. A decision was made to manually inspect Tornado records of this sub-type and ensure that the loss of each of these Tornado events was appropriately matched to the correct path.

The Tornado match results were reviewed and analyzed. The aggregated total loss (from the multiple county SHELUS loss records attributed to the same Tornado event) was compared to the loss reported for the tornado path in the source data (from the Storm Prediction Center). When averaging the ratio of SHELUS loss to Tornado path loss (from the Storm Prediction Center) among the matched records, a strong correlation is seen (see [Table 75](#)).

**Table 75: Ratio of SHELUS Loss to Tornado Path Loss Among Matched Tornado Records**

| <i>EF-Scale</i> | <i>Average Fatality Ratio</i> | <i>Average Injury Ratio</i> | <i>Average Building Damage Ratio</i> | <i>Average Agriculture Damage Ratio</i> |
|-----------------|-------------------------------|-----------------------------|--------------------------------------|-----------------------------------------|
| 0               | 0.875                         | 0.988                       | 1.179                                | 0.994                                   |
| 1               | 0.985                         | 0.993                       | 1.164                                | 0.983                                   |
| 2               | 0.99                          | 0.993                       | 1.0394                               | 1.054                                   |
| 3               | 0.962                         | 0.991                       | 1.204                                | 0.973                                   |
| 4               | 0.988                         | 0.987                       | 1.094                                | 0.99                                    |
| 5               | 1.0                           | 0.998                       | 0.959                                | 1.0                                     |

Loss Ratios per Basis are calculated for each matched Tornado event occurring in a county. The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is calculated for each Tornado as the average value density of the consequence of the exposed Census blocks multiplied by the area of intersection and then summed to the county level. The SHELUS-recorded loss is divided by the consequence value exposed to the Tornado path to calculate the Loss Ratio per Basis as in [Equation 123](#).

**Equation 123: Loss Ratio per Basis Calculation for a Single Tornado Event**

$$LRB_{TRNDCoCnsqType} = \frac{LOSS_{TRNDCoCnsqType}}{HLRExposure_{TRNDSubCoCnsqType}}$$

where:

$LRB_{TRNDCoCnsqType}$

is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Tornado event. Calculation is performed for each consequence type (building, population, and agriculture).

$LOSS_{TRND Co CnsqType}$  is the loss (by consequence type) experienced from the specific Tornado event documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{TRNDSub Co CnsqType}$  is the value (by consequence type) of the area estimated to have been exposed to the Tornado sub-type event based on the path of the historical Tornado (in dollars or people).

Tornado events may occur in areas without resulting in recorded loss to buildings, population, or agriculture. SHELDUS does not record events in which no loss occurred, so a number of zero-loss events are inserted into the data to align the event count in the Historic Loss Ratio calculation to the historical event count experienced within the SHELDUS period of record (1996 through 2019). For Tornado, the historical event count by sub-type is extracted using an intersection between the Tornado event path polygons (buffered by F-scale, but not the additional 80-km buffer; see [Section 19.2 Tornado Spatial Data Processing](#)) and the Census blocks. Using the path loss data, the percentage of past Tornadoes that caused no loss is calculated by sub-type. This percentage is multiplied by the sub-type count of Tornadoes that were matched to SHELDUS loss records in the county. For each sub-type (except EF-scale 4 and 5), a number of zero-loss records equal to the resulting product are inserted into the Loss Ratio per Basis table with zero values consequence ratios.

After the Loss Ratios per Basis are calculated for all Tornado events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Tornado.

**Table 76: Tornado Historic Loss Ratio Bayesian Weighting Level Sets**

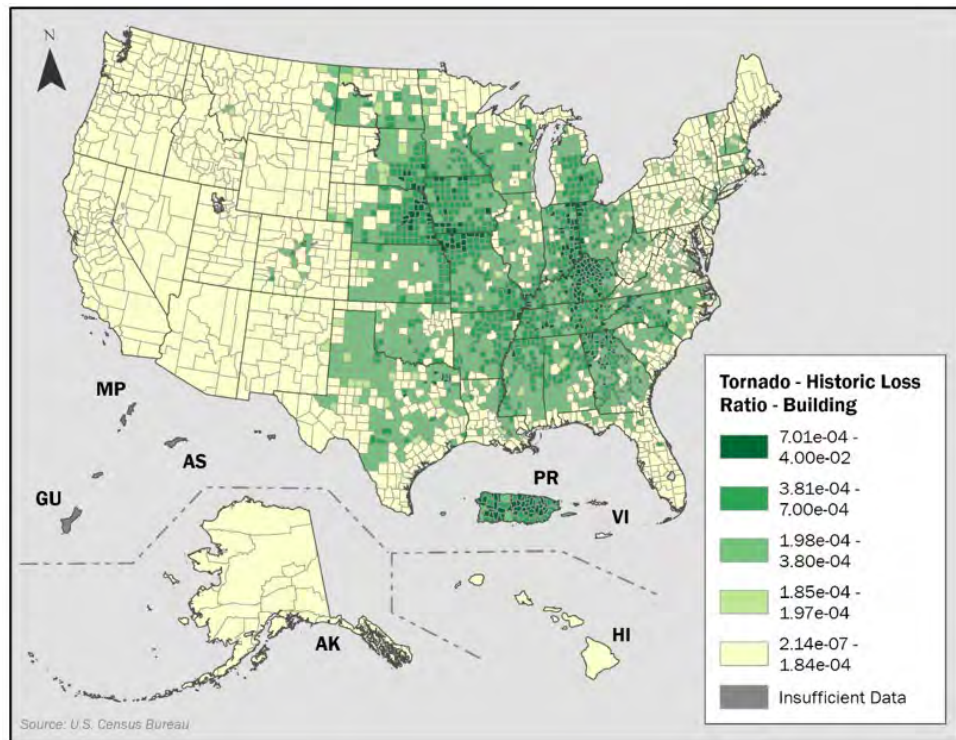
| <b>Tornado Subtype</b> | <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>                           | <b>Set Refinement Filter</b> | <b>Set Utilization Threshold (Set Must Have)</b> |
|------------------------|-------------------------|---------------|-------------------------------------------------|------------------------------|--------------------------------------------------|
| A: EF 0-1              | Agriculture             | A1            | County                                          | N/A                          | N/A                                              |
|                        |                         | A2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | A3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | A4            | National                                        | N/A                          | N/A                                              |



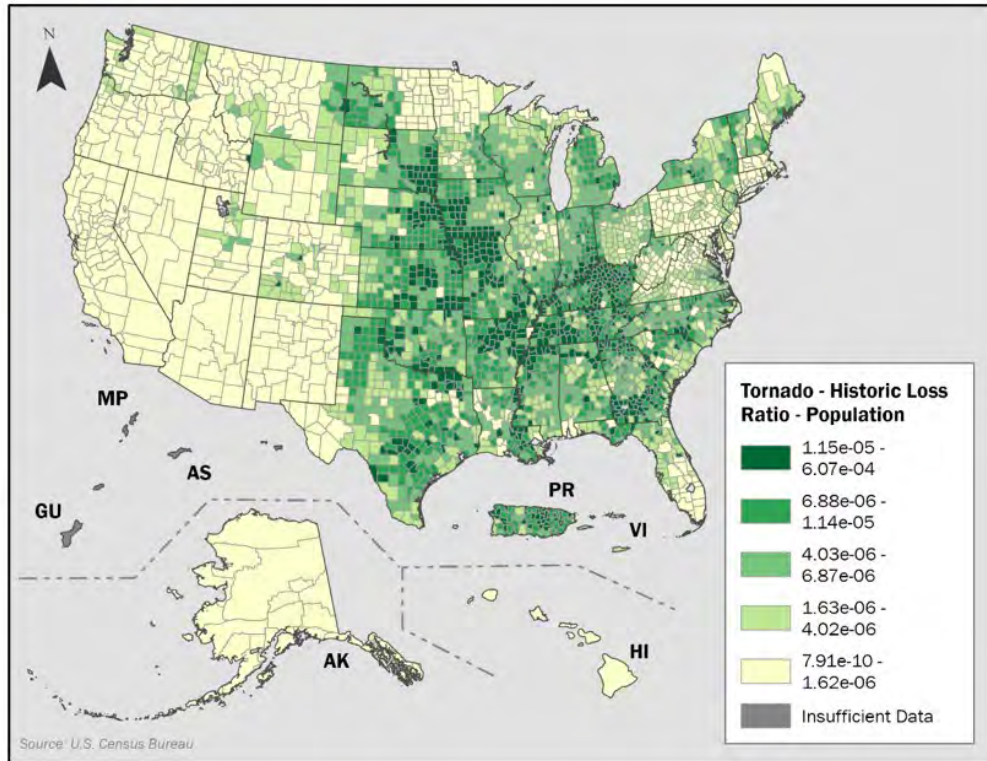
| <b>Tornado Subtype</b> | <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>                           | <b>Set Refinement Filter</b> | <b>Set Utilization Threshold (Set Must Have)</b> |
|------------------------|-------------------------|---------------|-------------------------------------------------|------------------------------|--------------------------------------------------|
| A: EF 0-1              | Population              | P1            | County                                          | N/A                          | N/A                                              |
|                        |                         | P2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | P3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | P4            | National                                        | N/A                          | N/A                                              |
|                        |                         | B1            | County                                          | N/A                          | N/A                                              |
|                        |                         | B2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | B3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | B4            | National                                        | N/A                          | N/A                                              |
| B: EF 2-3              | Agriculture             | A1            | County                                          | N/A                          | N/A                                              |
|                        |                         | A2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | A3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | A4            | National                                        | N/A                          | N/A                                              |
|                        | Population              | P1            | County                                          | N/A                          | N/A                                              |
|                        |                         | P2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | P3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | P4            | National                                        | N/A                          | N/A                                              |

| <b>Tornado Subtype</b> | <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>                           | <b>Set Refinement Filter</b> | <b>Set Utilization Threshold (Set Must Have)</b> |
|------------------------|-------------------------|---------------|-------------------------------------------------|------------------------------|--------------------------------------------------|
| B: EF 2-3              | Building                | B1            | County                                          | N/A                          | N/A                                              |
|                        |                         | B2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | B3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | B4            | National                                        | N/A                          | N/A                                              |
| C: EF 4-5              | Agriculture             | A1            | County                                          | N/A                          | N/A                                              |
|                        |                         | A2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | A3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | A4            | National                                        | N/A                          | N/A                                              |
|                        | Population              | P1            | County                                          | N/A                          | N/A                                              |
|                        |                         | P2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | P3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | P4            | National                                        | N/A                          | N/A                                              |
|                        | Building                | B1            | County                                          | N/A                          | N/A                                              |
|                        |                         | B2            | 196KM Fishnet                                   | N/A                          | N/A                                              |
|                        |                         | B3            | National Risk Index Historic Loss Ratio Regions | N/A                          | N/A                                              |
|                        |                         | B4            | National                                        | N/A                          | N/A                                              |

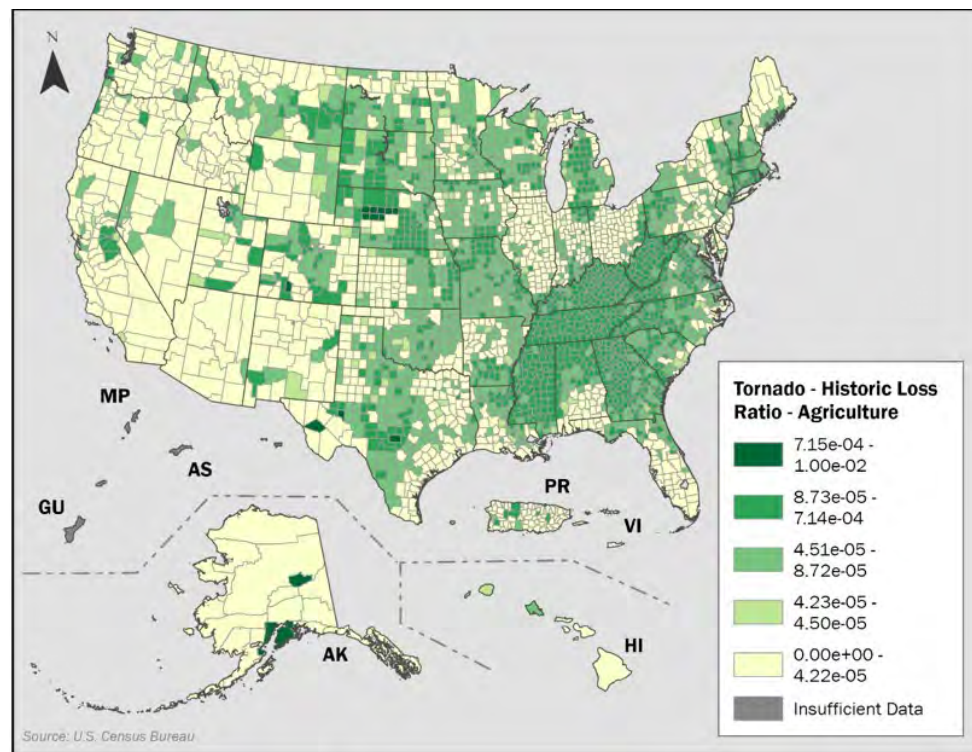
Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values for each sub-type (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). Historic Loss Ratio values for each Tornado sub-type are multiplied by their respective sub-type annualized frequency and exposure to calculate the sub-type Expected Annual Loss. Historic Loss Ratio values provided in the dataset are a surrogate value calculated as the county Expected Annual Loss for a specific consequence type for all sub-types divided by the product of the summed Tornado sub-type annualized frequencies and the surrogate exposure, which is the total building value, population equivalence, or agriculture value of the county. [Figure 98](#), [Figure 99](#), and [Figure 100](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Tornado.



**Figure 98: Tornado Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 99: Tornado Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 100: Tornado Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

The resulting Bayesian-adjusted Historic Loss Ratio by Tornado sub-type and consequence type is then inherited by the Census tracts within the parent county.

## 19.7. Tornado Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated for each Tornado sub-type, the Expected Annual Loss is computed at the Census tract and county level as in [Equation 124](#).

### Equation 124: Census Tract and County Expected Annual Loss to Tornado Sub-Type

$$EAL_{TRNDSubCTBldg} = Exposure_{TRNDSubCTBldg} \times Freq_{TRNDSubCT} \times HLR_{TRNDSubCTBldg}$$

$$EAL_{TRNDSubCTPop} = Exposure_{TRNDSubCTPop} \times Freq_{TRNDSubCT} \times HLR_{TRNDSubCTPop}$$

$$EAL_{TRNDSubCTAg} = Exposure_{TRNDSubCTAg} \times Freq_{TRNDSubCT} \times HLR_{TRNDSubCTAg}$$

$$EAL_{TRNDSubCoBldg} = Exposure_{TRNDSubCoBldg} \times Freq_{TRNDSubCo} \times HLR_{TRNDSubCoBldg}$$

$$EAL_{TRNDSubCoPop} = Exposure_{TRNDSubCoPop} \times Freq_{TRNDSubCo} \times HLR_{TRNDSubCoPop}$$

$$EAL_{TRNDSubCoAg} = Exposure_{TRNDSubCoAg} \times Freq_{TRNDSubCo} \times HLR_{TRNDSubCoAg}$$

where:

$EAL_{TRNDSubCTBldg}$  is the building Expected Annual Loss due to Tornado sub-type occurrences for a specific Census tract (in dollars).

$Exposure_{TRNDSubCTBldg}$  is the building value exposed to Tornado sub-type occurrences in the Census tract (in dollars).

$Freq_{TRNDSubCT}$  is the Tornado sub-type annualized frequency calculated for the Census tract (occurrences per year).

$HLR_{TRNDSubCTBldg}$  is the Bayesian-adjusted building Historic Loss Ratio for Tornado sub-type for the Census tract.

$EAL_{TRNDSubCTPop}$  is the population equivalence Expected Annual Loss due to Tornado sub-type occurrences for a specific Census tract (in dollars).

$Exposure_{TRNDSubCTPop}$  is the population equivalence value exposed to Tornado sub-type occurrences in the Census tract (in dollars).

|                            |                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| $HLR_{TRNDSubCTPop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Tornado sub-type for the Census tract.                         |
| $EAL_{TRNDSubCTAg}$        | is the agriculture Expected Annual Loss due to Tornado sub-type occurrences for a specific Census tract (in dollars).      |
| $Exposure_{TRNDSubCTAg}$   | is the agriculture value exposed to Tornado sub-type occurrences in the Census tract (in dollars).                         |
| $HLR_{TRNDSubCTAg}$        | is the Bayesian-adjusted agriculture Historic Loss Ratio for Tornado sub-type for the Census tract.                        |
| $EAL_{TRNDSubCoBldg}$      | is the building Expected Annual Loss due to Tornado sub-type occurrences for a specific county (in dollars).               |
| $Exposure_{TRNDSubCoBldg}$ | is the building value exposed to Tornado sub-type occurrences in the county (in dollars).                                  |
| $Freq_{TRNDSubCo}$         | is the Tornado sub-type annualized frequency calculated for the county (occurrences per year).                             |
| $HLR_{TRNDSubCoBldg}$      | is the Bayesian-adjusted building Historic Loss Ratio for Tornado sub-type for the county.                                 |
| $EAL_{TRNDSubCoPop}$       | is the population equivalence Expected Annual Loss due to Tornado sub-type occurrences for a specific county (in dollars). |
| $Exposure_{TRNDSubCoPop}$  | is the population equivalence value exposed to Tornado sub-type events in the county (in dollars).                         |
| $HLR_{TRNDSubCoPop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Tornado sub-type for the county.                               |
| $EAL_{TRNDSubCoAg}$        | is the agriculture Expected Annual Loss due to Tornado sub-type occurrences for a specific county (in dollars).            |
| $Exposure_{TRNDSubCoAg}$   | is the agriculture value exposed to Tornado sub-type occurrences in the county (in dollars).                               |
| $HLR_{TRNDSubCoAg}$        | is the Bayesian-adjusted agriculture Historic Loss Ratio for Tornado sub-type for the county.                              |

The total Expected Annual Loss values at the Census tract and county levels are the sums of the Expected Annual Loss values for each Tornado sub-type and consequence type as in [Equation 125](#).



**Equation 125: Census Tract and County Expected Annual Loss to Tornado**

$$EAL_{TRNDCT\ CnsqType} = EAL_{EF0\&1CT\ CnsqType} + EAL_{EF2\&3CT\ CnsqType} + EAL_{EF4\&5CT\ CnsqType}$$

$$EAL_{TRNDCo\ CnsqType} = EAL_{EF0\&1Co\ CnsqType} + EAL_{EF2\&3Co\ CnsqType} + EAL_{EF4\&5Co\ CnsqType}$$

$$EAL_{TRNDCT} = EAL_{TRNDCT\ Bldg} + EAL_{TRNDCT\ Pop} + EAL_{TRNDCT\ Ag}$$

$$EAL_{TRNDCo} = EAL_{TRNDCo\ Bldg} + EAL_{TRNDCo\ Pop} + EAL_{TRNDCo\ Ag}$$

where:

|                            |                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{TRNDCT\ CnsqType}$   | is the total Expected Annual Loss due to Tornado occurrences for a specific Census tract and consequence type (in dollars).  |
| $EAL_{EF0\&1CT\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 0 and 1 Tornadoes for a specific Census tract and consequence type (in dollars). |
| $EAL_{EF2\&3CT\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 2 and 3 Tornadoes for a specific Census tract and consequence type (in dollars). |
| $EAL_{EF4\&5CT\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 4 and 5 Tornadoes for a specific Census tract and consequence type (in dollars). |
| $EAL_{TRNDCo\ CnsqType}$   | is the total Expected Annual Loss due to Tornado occurrences for a specific county and consequence type (in dollars).        |
| $EAL_{EF0\&1Co\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 0 and 1 Tornadoes for a specific county and consequence type (in dollars).       |
| $EAL_{EF2\&3Co\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 2 and 3 Tornadoes for a specific county and consequence type (in dollars).       |
| $EAL_{EF4\&5Co\ CnsqType}$ | is the Expected Annual Loss due to EF-scale 4 and 5 Tornadoes for a specific county and consequence type (in dollars).       |
| $EAL_{TRNDCT}$             | is the total Expected Annual Loss due to Tornado occurrences for a specific Census tract (in dollars).                       |
| $EAL_{TRNDCT\ Bldg}$       | is the building Expected Annual Loss due to Tornado occurrences for a specific Census tract (in dollars).                    |
| $EAL_{TRNDCT\ Pop}$        | is the population equivalence Expected Annual Loss due to Tornado occurrences for a specific Census tract (in dollars).      |

$EAL_{TRNDCTAg}$  is the agriculture Expected Annual Loss due to Tornado occurrences for specific Census tract (in dollars).

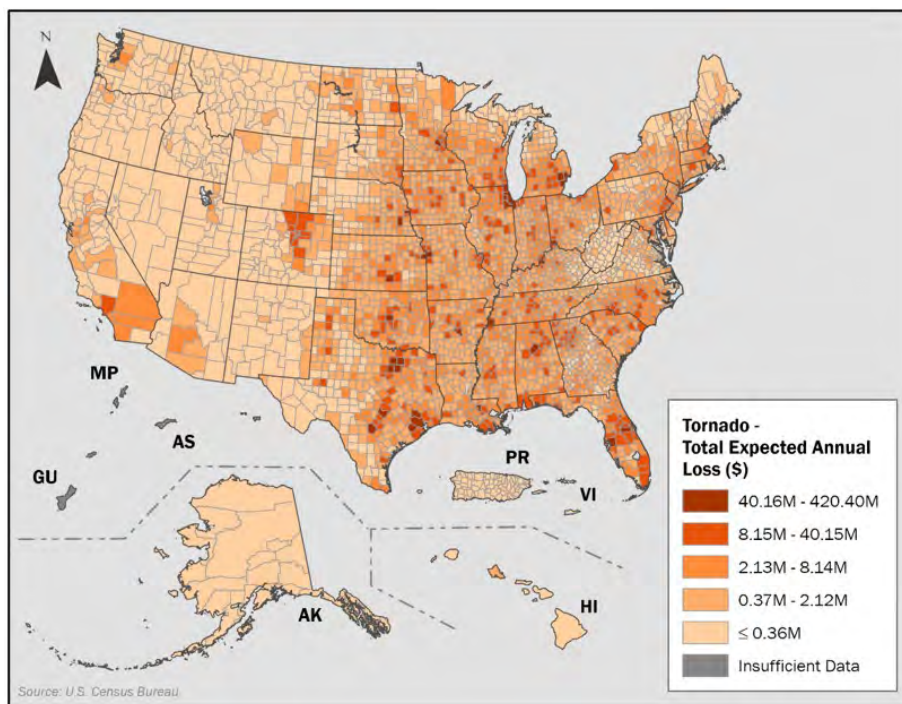
$EAL_{TRNDCo}$  is the total Expected Annual Loss due to Tornado occurrences for a specific county (in dollars).

$EAL_{TRNDCoBldg}$  is the building Expected Annual Loss due to Tornado occurrences for a specific county (in dollars).

$EAL_{TRNDCoPop}$  is the population equivalence Expected Annual Loss due to Tornado occurrences for a specific count (in dollars).

$EAL_{TRNDCoAg}$  is the agriculture Expected Annual Loss due to Tornado occurrences for a specific county (in dollars).

[Figure 101](#) shows the total Expected Annual Loss (building value, population equivalence, and agriculture value combined) to Tornado occurrences.



**Figure 101: Total Expected Annual Loss by County to Tornado**

With the Tornado total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Tornado Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Tornado Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Tornado Expected Annual Loss for agriculture by the total agriculture value of the community.

## **19.8. Tornado Methodology Discussion**

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- The Tornado hazard is a complex model that involves subtype distinctions that are processed separately before being combined into the overall Tornado hazard. These subtypes are based on the severity of the Tornado event. As a result of this sub-hazard approach, the published frequency and Historic Loss Ratio values are surrogates back calculated from the final Expected Annual Loss values. As such, this is one of the few hazards in this analysis that has Census tract-level Historic Loss Ratios.

## 20. Tsunami

A Tsunami is a wave or series of waves generated by gravitational displacements of bodies of water. These displacements can be caused by earthquakes, landslides, volcanic eruptions, or even large meteors hitting the ocean and causing a rise or mounding of water at the ocean surface. A Tsunami can travel across the open ocean at about 500 mph and slow down to about 30 mph as it approaches land, causing it to grow significantly in height.

The data development team collaborated with the National Tsunami Hazard Mitigation Program within the National Oceanic and Atmospheric Administration and U.S. Geological Survey to replace Expected Annual Loss data for Tsunami with results based on the best available hazard data from high-risk tsunami states and territories (Alaska, American Samoa, California, Guam, Hawai'i, Northern Mariana Islands, Oregon, Puerto Rico, Washington and U.S. Virgin Islands). These new data are standardized and processed according to guidance from the National Tsunami Hazard Mitigation Program and U.S. Geological Survey to best estimate the risk to buildings and population. The Tsunami hazard calculates risk for only the Building and Population consequence types. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 20.1. Tsunami Spatial Source Data

**Historical Occurrence Source:** [National Geophysical Data Center, National Oceanic and Atmospheric Administration, National Centers for Environmental Information/World Data Service Global Historical Tsunami Database](#)

The National Oceanic and Atmospheric Administration maintains a database of historical Tsunami runup points with records of Tsunami events dating back to 1800. These records supply spatiotemporal information, including geographic coordinates, observation date, and tsunami event validity, and occasionally some information on magnitude (like water height) or damage, such as deaths, injuries, and destruction to property. Each runup point has a unique identifier and each Tsunami originating event also has a unique identifier. Each Tsunami event typically causes multiple runup events. Runup points are available for download in CSV format (see [Table 77](#)). One known limitation of this dataset is that there are fewer gauges in Puerto Rico and U.S. Virgin Islands; therefore, historical runup events in these locations may be underrepresented.

**Table 77: Sample Data from the Global Historical Tsunami Database**

| <i>Tsunami Runup ID</i> | <i>Tsunami Event ID</i> | <i>Year</i> | <i>Month</i> | <i>Day</i> | <i>Country</i> | <i>State</i> | <i>Location Name</i> | <i>Latitude</i> | <i>Longitude</i> |
|-------------------------|-------------------------|-------------|--------------|------------|----------------|--------------|----------------------|-----------------|------------------|
| 6291                    | 2249                    | 1995        | 7            | 30         | USA            | ALASKA       | KODIAK               | 57.730          | -152.513         |
| 6632                    | 2373                    | 2001        | 6            | 23         | USA            | CALIFORNIA   | LOS ANGELES          | 33.719          | -118.272         |

| <i>Tsunami Runup ID</i> | <i>Tsunami Event ID</i> | <i>Year</i> | <i>Month</i> | <i>Day</i> | <i>Country</i> | <i>State</i> | <i>Location Name</i> | <i>Latitude</i> | <i>Longitude</i> |
|-------------------------|-------------------------|-------------|--------------|------------|----------------|--------------|----------------------|-----------------|------------------|
| 6636                    | 2373                    | 2001        | 6            | 23         | USA            | CALIFORNIA   | SAN DIEGO            | 32.715          | -117.174         |

Tsunami runup point data ranges from Jan. 01, 1800 to Dec. 31, 2021, which is used as the basis for calculating a period of record for historical losses: 222.15 years. This is not to be confused with the period of record of tsunami occurrence used for probabilistic modeling of tsunami events (ex: “10,000-year event”).

#### **Loss Quantification Source: The Federal Emergency Management Agency’s Estimated Average Annualized Tsunami Losses for the United States (P-2426)**

Starting in 2023, the Federal Emergency Management Agency collaborated with the National Tsunami Hazard Mitigation Program, the National Oceanic and Atmospheric Administration and the U.S. Geological Survey to provide a comprehensive national assessment of tsunami risk. It includes earthquake-induced tsunami risk but does not include impacts from tsunamis generated by landslides, volcanic eruptions, glacial calving, near-earth objects, weather, or other events. The results of the study provide economic loss data to buildings (structural, non-structural, contents, and inventory replacement values) and residential populations from modeled earthquake-generated tsunami scenarios. This study is based on the best available hazard data from the United States West Coast (California, Oregon, and Washington), Alaska, Hawai’i, United States Pacific Territories (American Samoa, Guam, and the Northern Mariana Islands) and Caribbean Territories (Puerto Rico and U.S. Virgin Islands). Tsunami risk associated with states along the East Coast, Gulf Coast, and Great Lakes is not included in this study, as Hazus 6.1 software does not currently include the ability to analyze tsunami risk in those states.<sup>120</sup> For more information on the methods used for calculating Tsunami losses, please refer to the original study’s documentation.

## **20.2. Tsunami Spatial Data Processing**

Because this dataset uses the Average Annualized Loss values provided in the Federal Emergency Management Agency Estimated Average Annualized Tsunami Losses study for its Expected Annual Loss values and these calculations were performed by the Federal Emergency Management Agency and its collaborators outside of the Expected Annual Loss database, the only spatial processing required for Tsunami was to the National Oceanic and Atmospheric Administration Global Historical Tsunami data to help calculate the surrogate frequency, Exposure, and Historic Loss Ratio values.

The Tsunami runup points from the National Oceanic and Atmospheric Administration Global Historical Tsunami database were buffered by 500 meters to create polygons (see [Figure 102](#)). Polygons were used instead of the original points to provide a smoother distribution of tsunami

<sup>120</sup> Once modeling capabilities and tsunami hazard data become available for additional states, the Federal Emergency Management Agency intends to incorporate these data into future editions of this study.

events and account for a reasonable likelihood of event occurrence over the geographic area. These buffered polygons were intersected with the Census block geometries to create a per-block list of historical Tsunami events.



**Figure 102: Tsunami Runup Buffer Map**

### 20.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

This dataset used P-2426's results as the basis to determine the areas where Tsunami occurrence is possible, where applicable. The dataset's criteria for inclusion were if a Census tract is within 300 feet of the coast, or if it is further than 300 feet from the coast but has an average annualized loss value. The development team reviewed this initial set of Census tracts with Subject Matter Experts to determine if any modifications should be made to the hazard occurrence areas. The results of this discussion yielded that all Census blocks using this method would be included except for those upriver of St. Helens, Oregon.

In addition to the inclusion criteria above, regional guidance was used to determine hazard possibility for the East, Gulf, and Great Lakes coasts which were outside the bounds of P-2426.



For the East Coast, any Census block in a county that was included in the U.S. Geological Survey's Estimates of population exposure and evacuation potential for earthquake-related tsunami hazard zones in the United States report was deemed as hazard possible.<sup>121</sup>

In the Gulf Coast (Texas, Louisiana, Mississippi, Alabama, and western Florida), any Census block in a county was designated as hazard possible when the county directly touched the waters of the Gulf, as these areas are threatened by submarine landslide sources in the Gulf. In Louisiana, the parishes of Tangipahoa, St. John the Baptist, and St. Charles were also marked as hazard possible, as they border Lake Pontchartrain, where a seiche event was observed in 2003.<sup>122</sup>

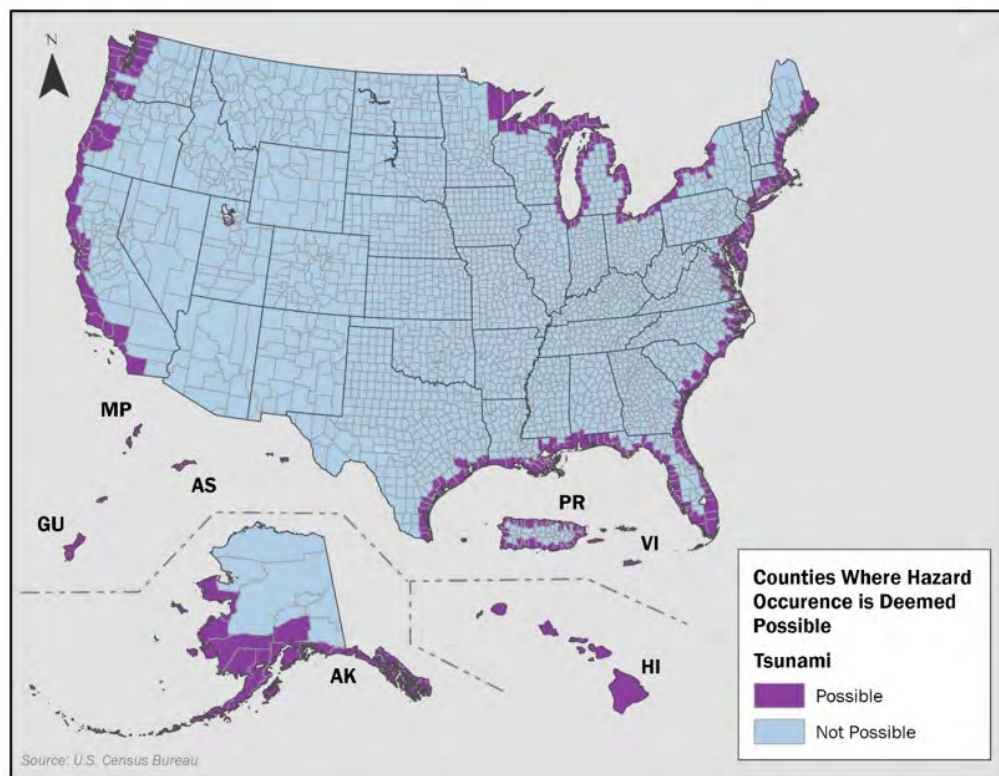
Along the Great Lakes coast, any Census block located in a Census tract bordering a Great Lake, or that was deemed possible for Coastal Flooding (see [Section 7.2](#)), is included as one in which tsunami-like events are considered possible.

In further consultations with Subject Matter Experts, it was determined that additional refinement of these counts was required to incorporate the state-to-state or regional variability of tsunami assessments and conceptualization. Therefore, the final list of counties deemed possible for hazard occurrence were grouped into several Tsunami Regions (see [Figure 103](#)) that share similar source data collection or geographic considerations, and these Tsunami Regions were then used in subsequent calculations.

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<sup>121</sup> Wood, N.J., Peters, J., and Jones, J.L. (2025). Estimates of population exposure and evacuation potential for earthquake-related tsunami hazard zones in the United States: U.S. Geological Survey data release, <https://doi.org/10.5066/P13GXERU>.

<sup>122</sup> <https://www.ngdc.noaa.gov/hazel/view/hazards/tsunami/event-more-info/2402>



**Figure 103: Map of Counties Deemed Possible for Tsunami Occurrence**

There are seventeen Tsunami Regions:

- Alaska - Aleutian Islands
- Alaska - Bering Sea
- Alaska - Gulf of Alaska
- American Samoa
- California - Northern
- California - San Francisco Bay
- California - Southern/Central
- Columbia River
- East and Gulf Coast
- Guam and Northern Mariana Islands
- Great Lakes
- Hawai'i
- Oregon
- Puerto Rico
- U.S. Virgin Islands
- Washington - Inner Coast
- Washington - Outer Coast

### 20.2.2. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Tsunami, in events, is computed as the number of distinct Tsunami events that have caused runup events (from the Global Historical Tsunami Runup Data, see [Section 20.1 Tsunami Spatial Source Data](#)) within a given Tsunami Region. Only runup events assigned a tsunami validity code of 2 (questionable tsunami), 3 (probable tsunami), or 4 (definite tsunami), and whose buffered points intersect with a Census block, are included in this count. The seventeen Tsunami Regions outlined in the previous section serve as the aggregation level, meaning that the count of all the distinct events in the per-block list of historical Tsunami events (see [Section](#)

[20.2 Tsunami Spatial Data Processing](#)) are summed to yield a unique regional event count. For example, the tsunami counts for all Census blocks within the California – Northern Tsunami Region were summed to create its regional event count.

### 20.3. Tsunami Annualized Frequency

The Tsunami annualized frequency value is composed of two values: one frequency based on historical counts of events; another frequency derived from the probabilistic tsunami return periods used by P-2426 for its impact calculations. The annualized frequency value provided to the user is a summation of these two frequencies strictly as a surrogate value meant to represent the annualized frequency of tsunamis in a way that incorporates both the historical and probabilistic components of the tsunami hazard. This surrogate frequency is the figure presented in the dataset.

The first value is based on the regional Tsunami runup-event count value data discussed in [Section 20.2.2 Historical Occurrence Count](#). This Regional event count was then divided by the historical event database’s period of record (222.15 years) to create the representative annualized frequency of historical events within that Region. This value was then assigned to all counties, Census tracts, and Census blocks within that Region, meaning all Census blocks within a Tsunami Region contain the same historical event annualized frequency, but this value is unique for each of the 17 Tsunami Regions. This value is not used in any calculations or results within the dataset and is merely an indicator for the user’s reference.

#### Equation 126: Tsunami Region Historical Annualized Frequency

$$Freq_{TsunRegion} = \frac{EventCount_{TsunRegion}}{PeriodRecord_{Tsun}}$$

where:

$Freq_{TsunRegion}$  is the annualized frequency of Tsunami events determined for a specific Region (events per year).

$EventCount_{TsunRegion}$  is the number of Tsunami runup-event polygons (with distinct originating events) that intersect the Region.

$PeriodRecord_{Tsun}$  is the period of record for Tsunami (222.15 years).

The geological annualized frequency is unique to each Census block and is not aggregated or inherited to other geographic extents (e.g., county, Region).

The Census block inherits its historical component annualized frequency value from the Region that contains it, as shown in [Equation 127](#).

**Equation 127: Census Block Tsunami Annualized Frequency Inheritance**

$$Freq_{TSUN_{Region}} = Freq_{TSUN_{CB}}$$

where:

$Freq_{TSUN_{Region}}$  is the annualized frequency of Tsunami runup events determined for a specific Region (events per year).

$Freq_{TSUN_{CB}}$  is the Tsunami annualized frequency determined for a specific Census block (events per year).

The second value is a probabilistic annualized frequency and represents the annualized frequency of tsunami events that are known to occur based on geological science but have not occurred within the historical period of record (since 1800). These are calculated as the differential probability of events over their respective time periods, specifically using the event data in P-2426 where each Census block was attributed an Average Annualized Loss value resulting from a number of tsunami events, their associated time periods of occurrence and a differential probability value derived from those time periods of occurrence. The probabilistic annualized frequency is calculated at the Census block level by summing the differential probability value for time periods of occurrence where the Differential Average Annualized Loss value for that time period was greater than zero.

**Equation 128: Tsunami Probabilistic Annualized Frequency**

$$PFreq_{TSUN_{CB}} = \sum \Delta Probability_t \text{ where } \Delta AAL_T > 0$$

where:

$PFreq_{TSUN_{CB}}$  is the probabilistic annualized frequency of Tsunami events determined for a specific Census block

$\Delta Probability_t$  is the differential probability value for a specific time period of occurrence

$\Delta AAL_T$  is the Differential Average Annualized Loss value for a specific time period

The annualized frequency value presented to the user in the National Risk Index dataset is the sum of the historical count annualized frequency and the probabilistic annualized frequency values described above. This is calculated at the Census block, Census tract, and county levels. This presented value is not used for any further calculations or results and is strictly for reference.

**Equation 129: Census Block Tsunami Annualized Frequency**

$$Freq_{TSUN_{CB}} = Freq_{TSUN_{Region}} + PFreq_{TSUN_{CB}}$$

where:

$Freq_{TSUN_{CB}}$  is the annualized frequency of Tsunami events determined for a specific Census block

$Freq_{TSUN_{Region}}$  is the historical annualized frequency of Tsunami events determined for the Region a specific Census block is in

$PFreq_{TSUN_{CB}}$  is the Tsunami Probabilistic Annualized Frequency for a specific Census block

**20.3.1. ANNUALIZED FREQUENCY AGGREGATION**

To calculate annualized frequency at the county level, the annualized frequency values at the Census block level are aggregated to the county level using area-weighted functions as in [Equation 130](#). Only Census blocks with a non-zero annualized frequency were included in the aggregation so that landlocked areas did not overly influence the annualized frequency values of the county. Each county inherits the maximum annualized frequency value of its residing Census blocks.

**Equation 130: County Tsunami Annualized Frequency**

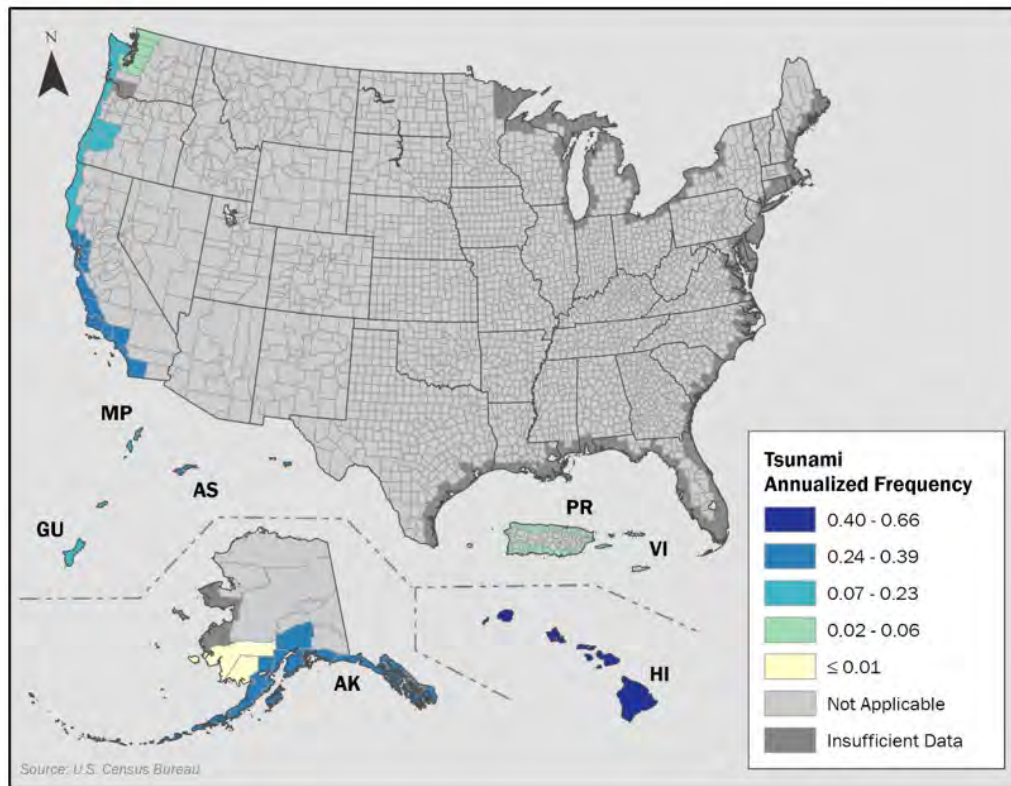
$$Freq_{TSUN_{Co}} = Max(Freq_{TSUN_{CB}})$$

where:

$Freq_{TSUN_{Co}}$  is the Tsunami annualized frequency calculated for a specific county.

$Max(Freq_{TSUN_{CB}})$  is the maximum annualized frequency calculated for a specific Census block within the specific county.

[Figure 104](#) displays Tsunami annualized frequency at the county level, which consists of the historical and geological frequencies added together.



**Figure 104: Tsunami Annualized Frequency by County**

## 20.4. Tsunami Exposure

Exposure values for building and population were sourced from P-2426, provided at the Census block level. The building values were Consumer Price Index-adjusted to December 2024 dollars, and the population equivalents were multiplied by the Value of Statistical Life to convert the provided value into dollars.

### 20.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 131](#)).



**Equation 131: Census Tract and County Tsunami Exposure Aggregation**

$$Exposure_{TSUNCTBldg} = \sum_{CB}^{CT} Exposure_{TSUNCBBldg}$$

$$Exposure_{TSUNCoBldg} = \sum_{CB}^{Co} Exposure_{TSUNCBBldg}$$

$$Exposure_{TSUNCTPop} = \sum_{CB}^{CT} Exposure_{TSUNCBPop}$$

$$Exposure_{TSUNCoPop} = \sum_{CB}^{Co} Exposure_{TSUNCBPop}$$

where:

$Exposure_{TSUNCTBldg}$  is the building value exposed to Tsunami in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{TSUNCBBldg}$  is the summed value of all buildings exposed to Tsunami for each Census block within the Census tract (in dollars).

$Exposure_{TSUNCoBldg}$  is the building value exposed to Tsunami in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{TSUNCBBldg}$  is the summed value of all buildings exposed to Tsunami for each Census block within the county (in dollars).

$Exposure_{TSUNCTPop}$  is the population equivalence value exposed to Tsunami event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{TSUNCBPop}$  is the summed value of all population equivalence exposed to Tsunami for each Census block within the Census tract (in dollars).

$Exposure_{TSUNCoPop}$  is the population equivalence value exposed to Tsunami in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{TSUNCBPop}$  is the summed value of all population equivalence exposed to Tsunami for each Census block within the county (in dollars).

## 20.5. Tsunami Historic Loss Ratio

The Tsunami Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Tsunami event, or the average rate of loss associated with the occurrence of a Tsunami event. Historic Loss Ratio values displayed are surrogate values as the final Expected Annual Loss values at the Census tract and county levels are extracted from the P-2426 study as shown in [Equation 132](#).

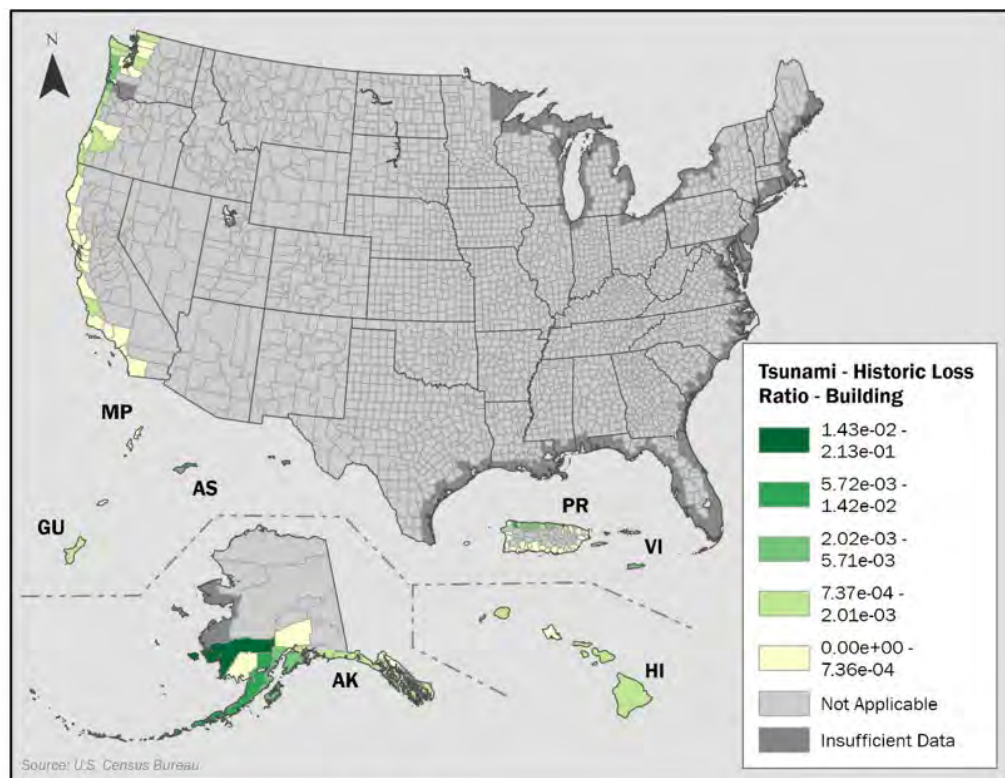
**Equation 132: Historic Loss Ratio Calculation for Tsunami**

$$HLR_{TSUN Co CnsqType} = \frac{EAL_{TSUN Co CnsqType}}{AFREQ_{TSUN Co} \times EXP_{TSUN Co CnsqType}}$$

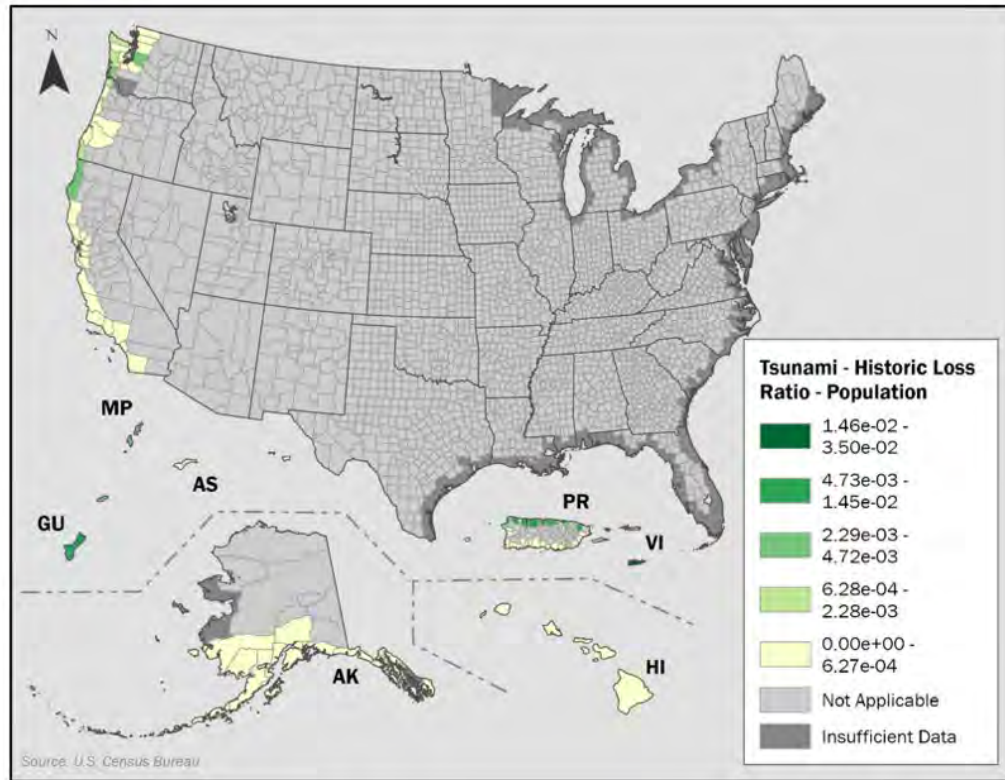
where:

|                          |                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $HLR_{TSUN Co CnsqType}$ | is the Historic Loss Ratio (by consequence type) of a county for Tsunami.                                                                                         |
| $EAL_{TSUN Co CnsqType}$ | is the Expected Annual Loss (by consequence type) of a county for Tsunami.                                                                                        |
| $AFREQ_{TSUN Co}$        | is the annualized frequency of a county for Tsunami.                                                                                                              |
| $EXP_{TSUN Co CnsqType}$ | is the value (by consequence type) of the sum of Census block exposure values obtained from the Federal Emergency Management Agency study (in dollars or people). |

[Figure 105](#) and [Figure 106](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Tsunami.



**Figure 105: Tsunami Historic Loss Ratio – Building Value**



**Figure 106: Tsunami Historic Loss Ratio – Population**

The resulting surrogate Historic Loss Ratio by consequence type is then inherited by the Census tracts within the parent county.

## 20.6. Tsunami Expected Annual Loss

The dataset extracted Average Annualized Loss values as provided by the P-2426 study data at the Census block level to use for its Expected Annual Loss Values. The exposure, annualized frequency, and Historic Loss Ratio discussed in the previous sections are surrogate values and are not used to compute the Expected Annual Loss Values for this hazard.

The total Expected Annual Loss Values at the Census tract and county levels are the sums of the aggregated building and population equivalence Expected Annual Loss Values of their Census block Values as in [Equation 133](#).

### Equation 133: Census Tract and County Expected Annual Loss to Tsunami

$$EAL_{TSUN_{CT}} = \sum_{CB}^{CT} EAL_{TSUN_{CB}Bldg} + \sum_{CB}^{CT} EAL_{TSUN_{CB}Pop}$$

$$EAL_{TSUN_{Co}} = \sum_{CB}^{Co} EAL_{TSUN_{CB}Bldg} + \sum_{CB}^{Co} EAL_{TSUN_{CB}Pop}$$

where:

$EAL_{TSUN_{CT}}$  is the total Expected Annual Loss due to Tsunami for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{TSUN_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Tsunami for all Census blocks in the Census tract (in dollars).

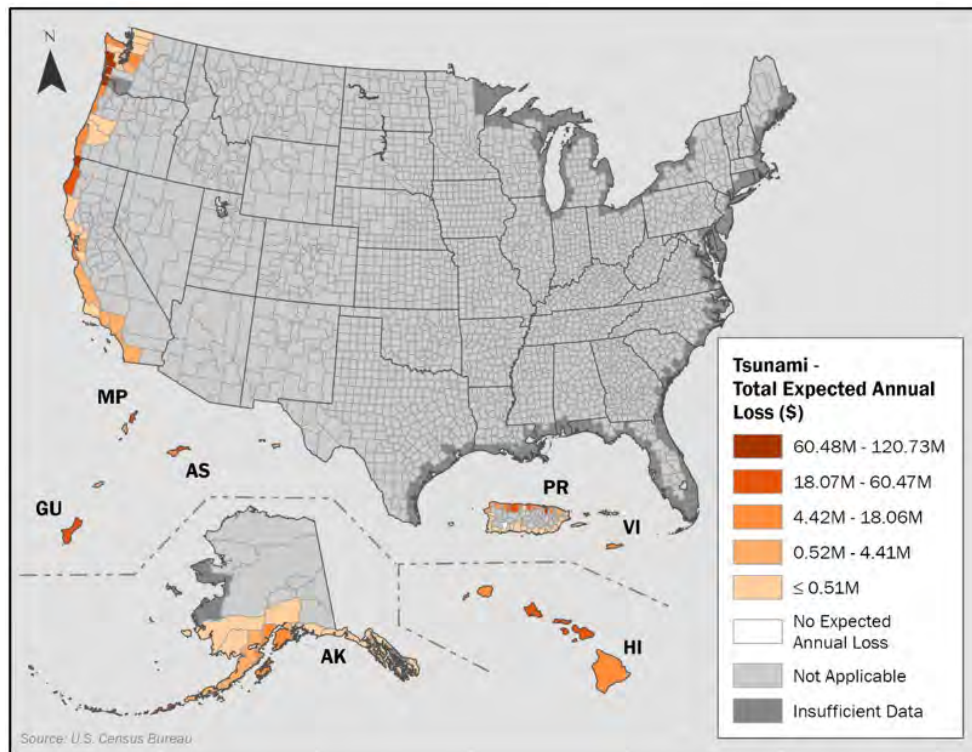
$\sum_{CB}^{CT} EAL_{TSUN_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Tsunami for all Census blocks in the Census tract (in dollars).

$EAL_{TSUN_{Co}}$  is the total Expected Annual Loss due to Tsunami for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{TSUN_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Tsunami for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{TSUN_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Tsunami for all Census blocks in the county (in dollars).

[Figure 107](#) shows the total Expected Annual Loss (building value and population equivalence combined) to Tsunami occurrences.



**Figure 107: Total Expected Annual Loss by County to Tsunami**

With the Tsunami total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)).

The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Tsunami Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Tsunami Expected Annual Loss for population by the total population of the community.

## 20.7. Tsunami Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations and Section 4 of the P-2426 report for the study limitations of the Tsunami Expected Annual Loss source data.

- Due to limitations in data availability and modeling capabilities, this study could not estimate losses for temporary and non-residential populations, harbor facilities, non-seismic tsunami sources, or tsunami risk in the Great Lakes, East Coast, and Gulf Coast regions.
- States and territories included in the average annualized loss study did not possess the same quantity or quality of data, therefore not all sources or return periods for all states and territories were included.
- Expected Annual Loss data are based on modeled tsunami scenarios only.

## 21. Volcanic Activity

Volcanic Activity occurs via vents that act as conduits between the Earth's surface and inner layers, producing the eruption of materials such as gas, molten rock, and volcanic ash when gas pressure and buoyancy drive molten rock upward and through zones of weakness in the Earth's crust.

The Volcanic Activity hazard considers direct impacts from explosive eruptions, pyroclastic density currents, lava flows, debris avalanches, lahars, and ashfall. The data development team engaged in extensive collaboration with subject matter experts to develop susceptibility and impact area layers for volcanoes. Susceptibility zones were determined based on two sub-hazard categories, Near-Volcano (including lahars), and Ashfall. Near-Volcano susceptibility was determined based on the U.S. Geological Survey's National Volcanic Threat Layer and 2018 Update to the U.S. Geological Survey National Volcanic Threat Assessment datasets. For the Ashfall sub-hazard a susceptibility layer was developed with input for U.S. Geological Survey subject matter experts. The Volcanic Activity hazard calculates risk only for the Building and Population consequence types. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 21.1. Volcanic Activity Spatial Source Data

**Susceptible Area Source:** [U.S. Geological Survey National Volcanic Threat Layer](#)<sup>123</sup>

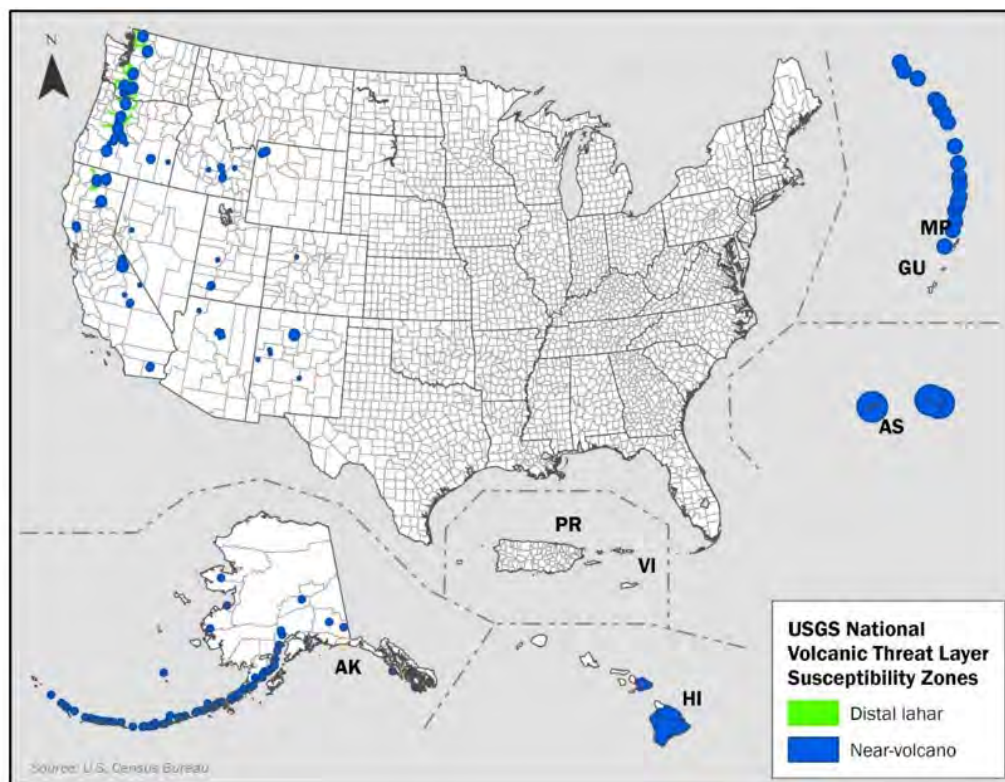
The U.S. Geological Survey's National Volcanic Threat Layer identifies 161 active or potentially active volcanos within the United States and its territories. The layer depicts areas near the volcanic vents that could be affected by proximal volcanic hazards including airborne debris from explosive eruptions (otherwise known as ballistics), pyroclastic density currents, lava flows, debris avalanches, lahars, and heavy ashfall. Where applicable, it also shows areas downstream that could be affected by distal lahar flows that have the potential to extend beyond this proximal hazard boundary. Each threat area is labeled with its associated qualitative threat categorization as determined by the 2018 National Volcanic Threat Assessment.<sup>124</sup>

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<sup>123</sup> Ramsey, D.W., Bard, J.A., and Johnson, E.R., 2024, National Volcanic Threat Layer: U.S. Geological Survey data release, <https://doi.org/10.5066/P13RUCUG>

<sup>124</sup> Ewert, J.W., Diefenbach, A.K., and Ramsey, D.W., 2018, 2018 update to the U.S. Geological Survey national volcanic threat assessment: U.S. Geological Survey Scientific Investigations Report 2018–5140, 40 p., <https://doi.org/10.3133/sir20185140>.





**Figure 108: U.S. Geological Survey's National Volcanic Threat Layer**

**Susceptible Area Source:** [U.S. Geological Survey Plots of Wind Patterns of the World's Volcanoes](#)<sup>125</sup>

This dataset contains the U.S. Geological Survey's wind rose diagrams and tables of wind patterns at volcanoes listed in the Smithsonian Institution's Volcanoes of the World Database. The speed and direction of wind above each volcano were read twice daily from January 1, 1990, to December 28, 2009, and plotted as 20 wind rose diagrams, divided into four seasons and five elevation intervals (0-5, 5-11, 11-16, 16-24, and 24-40 km above sea level), and as one summary plot for year-round winds at 0-5 km elevation above sea level.

**Frequency Source:** [2018 Update to the U.S. Geological Survey National Volcanic Threat Assessment](#)<sup>126</sup>

The 2018 Update to the U.S. Geological Survey National Volcanic Threat Assessment assesses the relative threats posed by each of the 161 volcanoes found within the National Volcanic Threat Layer. The Threat Assessment categorizes the volcanoes on a threat scale from Very Low to Very High based on a 24-factor hazard and exposure matrix. One of those factors is "Eruption recurrence,"

<sup>125</sup> Mastin, L.G. (2017). Plots of Wind Patterns of the World's Volcanoes: U.S. Geological Survey data release, <https://doi.org/10.5066/F7SQ8XKT>

<sup>126</sup> Ewert, J.W., Diefenbach, A.K., and Ramsey, D.W., (2018). 2018 update to the U.S. Geological Survey national volcanic threat assessment: *U.S. Geological Survey Scientific Investigations Report 2018-5140*, 40 p., <https://doi.org/10.3133/sir20185140>

which ranks the volcanoes on a scale of 0 to 4, with increasing factor values associated with decreasing recurrence intervals.

**Historical Loss Data Source:** Various sources were used to provide historical loss data for specific volcanic activity events (see [Table 78](#)).

**Table 78: Historical Loss Data Sources for Specific Volcanic Events**

| <i>Data Source</i>                                                                                                                                    | <i>Description</i>                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">U.S. Geological Survey 2024 Map showing subaerial extents of lava flows and explosive deposits from Kīlauea, 1790–2023</a> <sup>127</sup> | Map showing the subaerial extents of lava flows and explosive deposits from Kīlauea, covering the years 1790 through 2023. Lava flow hazard zones and districts of the County of Hawai‘i are also depicted. |
| National Oceanic and Atmospheric Administration, OCM, <a href="#">Natural Hazard Management in Coastal Areas, November 1976</a> <sup>128</sup>        | A summary of natural hazards within coastal states; includes damage estimates caused by the 1960 Kapoho lava flow in Hawai‘i from the eruption of Kīlauea.                                                  |
| State of Hawai‘i <a href="#">1992 Department of Defense Annual Report</a> <sup>129</sup>                                                              | Hawai‘i Department of Defense 1992 Annual Report that includes damage estimates caused by the 1991 destruction of Kalapana Gardens by the ongoing eruption of Kīlauea that began in 1983.                   |

<sup>127</sup> Mulliken, K.M., Kauahikaua, J.P., Swanson, D.A., and Zoeller, M.H. (2024). Chronology of recent volcanic activity on the Island of Hawai‘i, Hawai‘i: U.S. Geological Survey data release, <https://doi.org/10.5066/P9V3NQYB>

<sup>128</sup> White, G.F., Baker, E.J., Baumann, D.D. and Chow, W.T. (1976). Natural hazard management in coastal areas.

<sup>129</sup> Hawai‘i. Department of Defense, *Annual Report* (1992). <https://dod.hawaii.gov/wp-content/uploads/2024/05/1992-DOD-Annual-Report.pdf>

| <b>Data Source</b>                                                                                                                                                                              | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">National Oceanic and Atmospheric Administration National Centers for Environmental Information/World Data Service Global Significant Volcanic Eruptions Database</a> <sup>130</sup> | The Significant Volcanic Eruption Database is a global listing of over 800 significant eruptions which includes information on the latitude, longitude, elevation, type of volcano, and last known eruption. A significant eruption is classified as one that meets at least one of the following criteria: caused fatalities, caused moderate damage (approximately \$1 million or more), with a Volcanic Explosivity Index of 6 or larger, caused a tsunami, or was associated with a major earthquake. It is the source data for the volcanic hazard damage in SHELDUS. |
| <a href="#">Oregon State University Cost of Volcanic Eruptions</a> <sup>131</sup>                                                                                                               | Loss estimates from the May 18, 1980 eruption of Mount St. Helens and the 1955 eruption of Kīlauea broken down by category. Data is sourced from the Washington State Department of Commerce and Economic development Research Division.                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">Rapid remote volcanic ashfall impact assessment for the 2022 eruption of Hunga volcano, Tonga</a> <sup>132</sup>                                                                    | A methodology for conducting rapid remote volcanic ashfall impact assessments, conducted over a 10-day period following the eruption of Hunga Tonga Hunga Ha'apai volcano in Tonga. Three different hazard models for ashfall thickness were used across the main island of Tongatapu and available asset information and vulnerability functions for buildings, agriculture, electricity networks, water supply, and roads, to provide initial estimates of losses due to ashfall from the 15 January 2022 eruption.                                                      |

<sup>130</sup> National Geophysical Data Center / World Data Service (NGDC/WDS): NCEI/WDS Global Significant Volcanic Eruptions Database. NOAA National Centers for Environmental Information. doi:10.7289/V5JW8BSH

<sup>131</sup> Oregon State University. "Cost of Volcanic Eruptions". Accessed March 10, 2025.

<sup>132</sup> Weir, A.M., Williams, J.H., Wilson, T.M. et al. Rapid remote volcanic ashfall impact assessment for the 2022 eruption of Hunga volcano, Tonga: a bespoke approach and lessons identified. Bull Volcano 86, 88 (2024). <https://doi.org/10.1007/s00445-024-01776-y>

| Data Source                                                                                                                                                  | Description                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">2022 Global Rapid Post Disaster Damage Estimation (GRADE) Report on the Hunga Tonga-Hunga Ha'apai Eruption and Tsunami, Tonga</a> <sup>133</sup> | A rapid post-disaster damage assessment to estimate the potential direct damage caused by the tephra fall (henceforth referred to as ashfall) and tsunami resulting from the 15 January 2022 volcanic eruption of the Hunga Tonga Hunga Ha'apai volcano in Tonga. |

## 21.2. Volcanic Activity Spatial Data Processing

Volcanic Activity was broken down into two sub-hazards during processing due to the significantly different aerial extents and historic loss ratios of the two sub-hazards. These sub-hazards are termed “Near-Volcano” and “Ashfall.”

### 21.2.1. NEAR-VOLCANO SUB-HAZARD EXPOSURE DEFINITION

The Near-Volcano hazard is sourced from the U.S. Geological Survey National Volcanic Threat Layer, which contains two separate hazard identifications, near-volcano (which includes ballistics, pyroclastic density currents, lava flows, debris avalanches, and heavy ashfall) and lahar. Both were treated the same within the Near-Volcano sub-hazard. The U.S. Geological Survey National Volcanic Threat Layer shapefiles were then used to identify Census blocks, Census tracts, and counties susceptible to Near-Volcanic hazards.

### 21.2.2. ASHFALL EXPOSURE DETERMINATION

Though the National Volcanic Threat Layer captures most of the significant impacts to buildings and people due to volcanic activity, it currently does not account for Ashfall beyond the boundaries of the National Volcanic Threat Layer in its hazard extents. Ash from a volcanic eruption can travel for hundreds, sometimes thousands of kilometers from the eruption source, potentially damaging buildings, disrupting transportation corridors, contaminating water supplies, and impacting air quality. According to estimates, structural building damage like what is measured in this analysis requires approximately 50 to 100 mm of Ashfall.<sup>134</sup> While some existing data on Ashfall extents do exist, these datasets do not reflect Ashfall in terms that allow for direct building and population losses to be measured. To account for this data gap, a simplified Ashfall model was created for this analysis. The generalized model presented here is a first step in representing the potential Ashfall impacts in counties outside the Near-Volcano sub-hazard area as defined by the National Volcano Threat Layer.

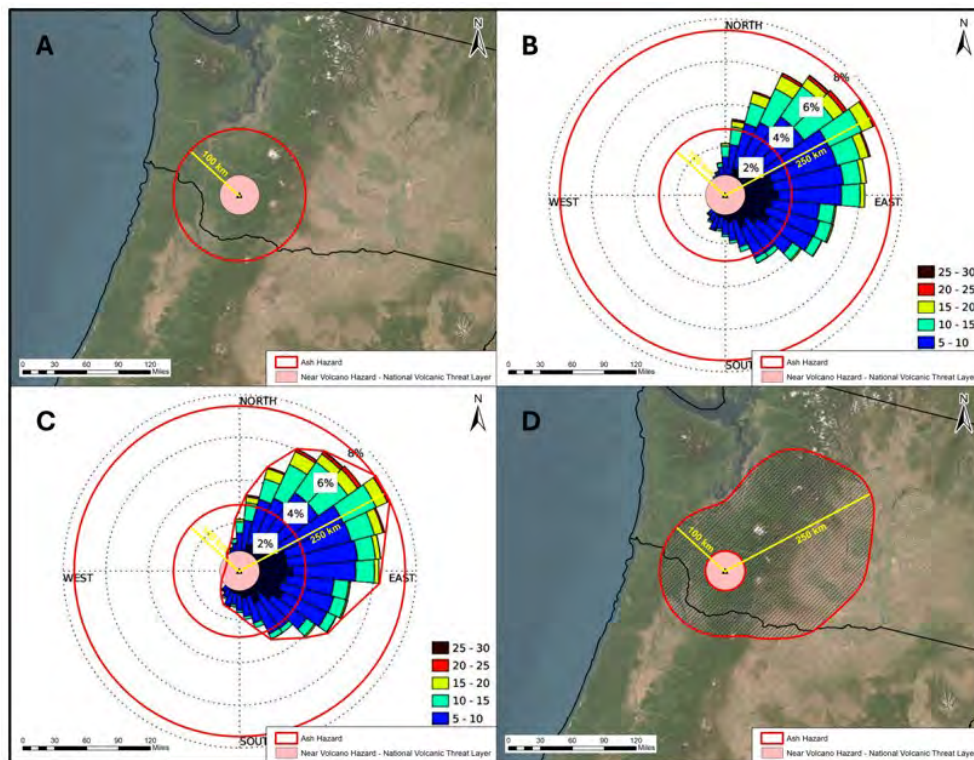
<sup>133</sup> New Zealand Defence Force (NZDF). (2022). The January 15, 2022 Hunga Tonga-Hunga Ha'apai Eruption and Tsunami, Tonga Global Rapid Post Disaster Damage Estimation (Grade) Report.

<sup>134</sup> Jenkins, S.F., Wilson, T.M., Magill, C., Miller, V., Stewart, C., Blong, R., Marzocchi, W., Boulton, M., Bonadonna, C. and Costa, A., 2015. Volcanic ash fall hazard and risk. *Global volcanic hazards and risk*, pp.173-222.

After discussions with subject matter experts, it was decided that the Ashfall sub-hazard would employ a buffer approach based on observed historical events and prevailing wind direction. Wind rose data at 0 to 5 km elevation above each volcano were used to determine generalized Ashfall sub-hazard extents. A minimum buffer was set at 100km based on expert recommendation for a likely event area and to account for variability in wind direction and speed throughout a given year. Each volcano's wind rose informed the general shape of the Ashfall buffer, with a maximum radius of 250km applied along the axis of prevailing wind direction to capture most of the counties that have historically experienced Ashfall losses. The resulting polygons were merged to create a single ashfall susceptibility area, and the outer extent was smoothed using a 75km tolerance. The National Volcanic Threat Layer polygons were then clipped out to avoid duplication as heavy ashfall is already accounted for in that layer.

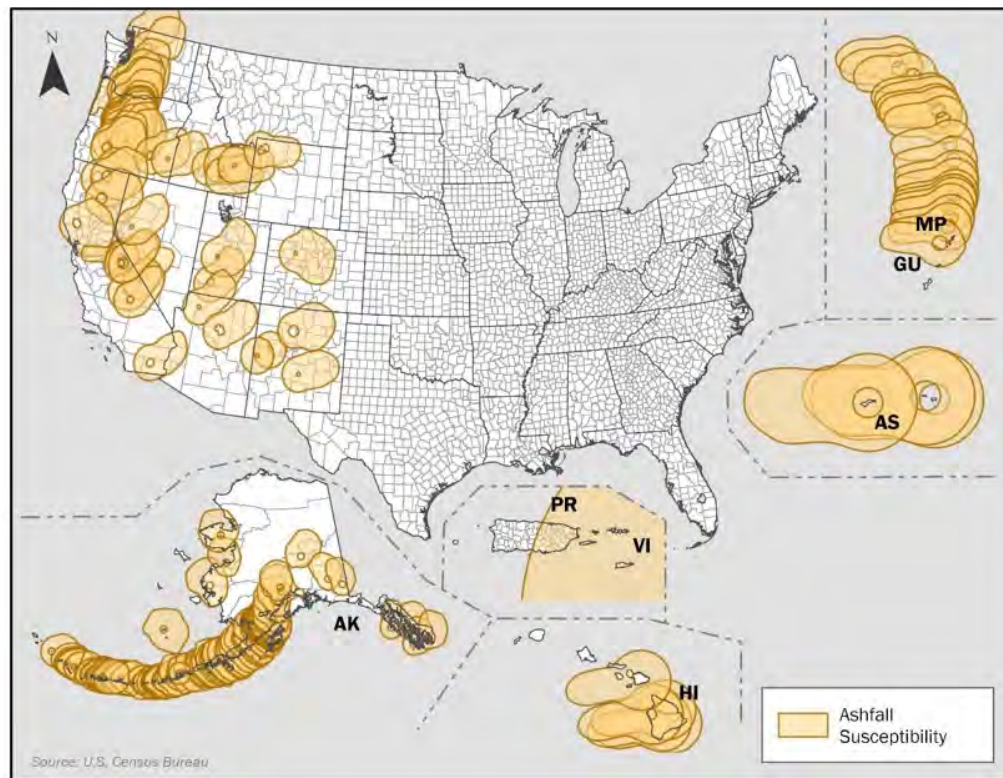
If counties with historical ashfall losses within the SHELUS database were not included within the generated Ashfall susceptibility cloud, the individual volcano's Ashfall "cloud" was extended to include those counties. For example, in 2001, Puerto Rico experienced historical SHELUS ashfall losses from the Soufrière Hills volcano in Montserrat, approximately 475km away from the furthest listed county, so a 475km buffer was added to the volcano and all counties within it were identified as susceptible to Ashfall.

See [Figure 109](#) for a visualization of this process and for the resulting Ashfall susceptibility for all volcanoes.



**Figure 109: Ashfall Susceptibility Development**



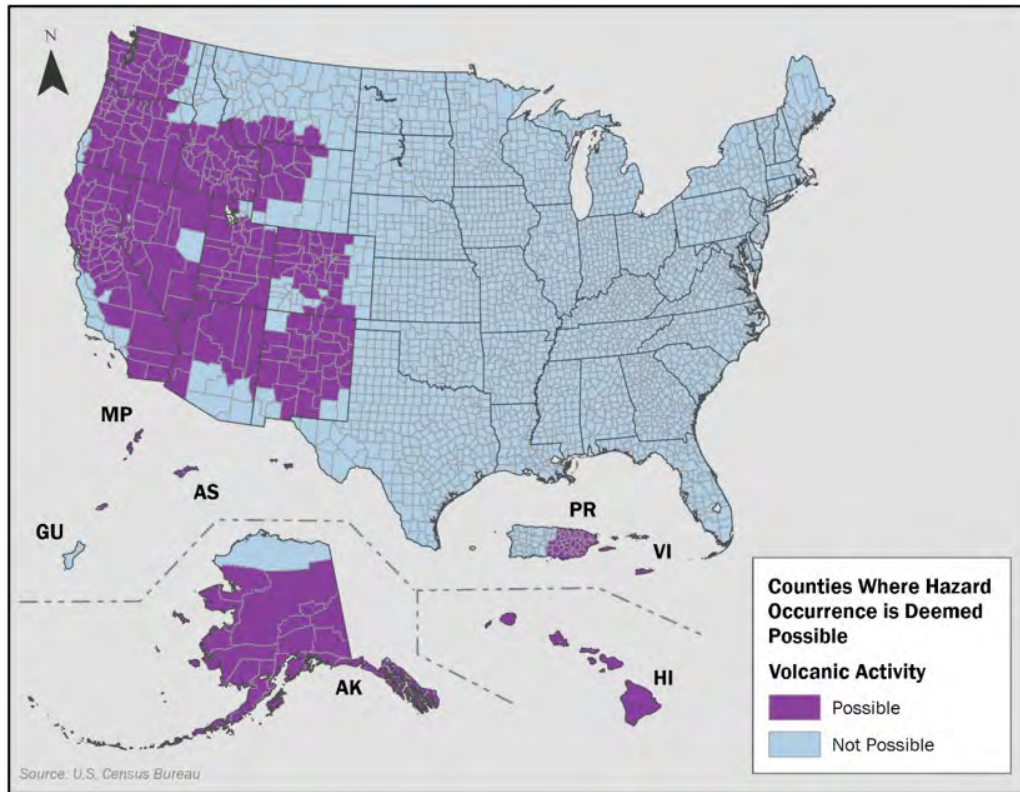


**Figure 110: Ashfall Susceptibility per Volcano**

### 21.2.3. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

The spatial areas with the potential to be impacted by volcanic events were identified using the U.S. Geological Survey National Volcanic Threat Layer and the generated Ashfall sub-hazard areas. To distinguish between areas where no Volcanic Activity has occurred and those where such events are not deemed possible, a control table was generated to designate which counties have some probability of Volcanic Activity. Any county with a Census block that intersected one or more of the identified volcanic threat areas is included as one in which Volcanic Activity is possible (see [Figure 111](#)).





**Figure 111: Map of Counties Deemed Possible for Volcanic Activity Occurrence**

### 21.3. Volcanic Activity Exposure

To identify areas of exposure, the Near-Volcano and Ashfall polygons are intersected with the Census block polygons within the processing database. The resulting table contains the volcano's unique identifier, Census block number, and the intersected area in square kilometers (see [Table 79](#)).

**Table 79: Sample Data from the Volcano Census Block Intersection Table**

| <i>VolcanoID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|------------------|--------------------|---------------------------|
| 332040           | 150010215131003    | 1.0601549430              |
| 332010           | 150010210141001    | 1.6149009619              |
| 332020           | 150010208023001    | 1.1707411354              |

To determine exposure value, the average coverage is found by summing the intersected areas for volcano polygons that intersect the Census block and dividing this sum by the number of intersecting volcano polygons. This is multiplied by the developed area building value density and the developed area population density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 134](#)). These Census block densities have been calculated by dividing the total exposure values (as recorded in Hazus 6.0) by the developed land

area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

#### Equation 134: Census Block Volcano Exposure

$$Exposure_{VLCN_{CB}Bldg} = \frac{\sum IntsctArea_{VLCN_{CB}}}{VolcanoCount_{CB}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{VLCN_{CB}Pop} = \left( \frac{\sum IntsctArea_{VLCN_{CB}}}{VolcanoCount_{CB}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

where:

$Exposure_{VLCN_{CB}Bldg}$  is the building value exposed to Volcanic Activity for a specific Census block (in dollars).

$\sum IntsctArea_{VLCN_{CB}}$  is the sum of the intersected areas of volcano polygons with the Census block (in square kilometers).

$VolcanoCount_{CB}$  is the total number of volcano polygons (each associated with a specific volcano) that intersect the Census block.

$DevAreaDen_{CB_{Bldg}}$  is the developed area building value density of the Census block (in dollars per square kilometer).

$Exposure_{VLCN_{CB}Pop}$  is the population equivalence value exposed to Volcanic Activity at the Census block level (in dollars).

$DevAreaDen_{CB_{Pop}}$  is the developed area population density of the Census block (in people).

$VSL$  is the Value of Statistical Life (\$13.7 million per person).

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

#### 21.3.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 135](#)).

**Equation 135: Census Tract and County Volcano Exposure Aggregation**

$$\begin{aligned}
Exposure_{VLCN\ CT\ Bldg} &= \sum_{CB}^{CT} Exposure_{VLCN\ CB\ Bldg} \\
Exposure_{VLCN\ Co\ Bldg} &= \sum_{CB}^{Co} Exposure_{VLCN\ CB\ Bldg} \\
Exposure_{VLCN\ CT\ Pop} &= \sum_{CB}^{CT} Exposure_{VLCN\ CB\ Pop} \\
Exposure_{VLCN\ Co\ Pop} &= \sum_{CB}^{Co} Exposure_{VLCN\ CB\ Pop}
\end{aligned}$$

where:

$Exposure_{VLCN\ CT\ Bldg}$  is the building value exposed to Volcanic Activity in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{VLCN\ CB\ Bldg}$  is the summed value of all buildings exposed to Volcanic Activity for each Census block within the Census tract (in dollars).

$Exposure_{VLCN\ Co\ Bldg}$  is the building value exposed to Volcanic Activity in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{VLCN\ CB\ Bldg}$  is the summed value of all buildings exposed to Volcanic Activity for each Census block within the county (in dollars).

$Exposure_{VLCN\ CT\ Pop}$  is the population equivalence value exposed to Volcanic Activity in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{VLCN\ CB\ Pop}$  is the summed value of all population equivalence exposed to Volcanic Activity for each Census block within the Census tract (in dollars).

$Exposure_{VLCN\ Co\ Pop}$  is the population equivalence value exposed to Volcanic Activity in a specific county (in dollars).

$\sum_{CB}^{Co} Exposure_{VLCN\ CB\ Pop}$  is the summed value of all population equivalence exposed to Volcanic Activity for each Census block within the county (in dollars).

## 21.4. Volcanic Activity Annualized Frequency

The annualized frequency for Volcanic Activity is exceptionally low when compared to other hazard types. Although the known and dated eruption histories for some volcanoes go back millions of years, and sometimes hundreds of millions of years, the details of volcanic eruption events within

the last few thousands of years are lacking for most volcanoes within the National Volcanic Threat Layer. However, one of the factors within the 24-factor hazard and exposure matrix of the 2018 update to the U.S. Geological Survey National Volcanic Threat Assessment is “Eruption recurrence”, which ranks the volcanoes on a scale of 0 to 4, with increasing factor values associated with decreasing recurrence intervals ([Table 80](#)).

**Table 80: National Volcanic Threat Assessment Eruption Recurrence Interval**

| <i>Hazard Score</i> | <i>USGS Eruption Recurrence Interval Hazard Score Meaning</i>                                                                                                                     | <i>Determined Recurrence Interval Based on Hazard Score</i> | <i>Calculated Annual Frequency</i> |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------|
| 4                   | The eruption interval is estimated to be 1-99 years                                                                                                                               | 99 years                                                    | 0.0101                             |
| 3                   | The eruption interval is estimated to be 100-1,000 years                                                                                                                          | 1,000 years                                                 | 0.001                              |
| 2                   | The eruption interval is estimated to be 1,000 to several thousand years                                                                                                          | 5,000 years                                                 | 0.0002                             |
| 1                   | The eruption interval is estimated to be 5,000-10,000 years, or if no Holocene eruptions but is a large-volume restless silicic system that has erupted in the last 100,000 years | 10,000 years                                                | 0.0001                             |
| 0                   | The volcano has no known Holocene eruption                                                                                                                                        | Not enough data to determine                                | Not enough data to determine       |

Within the matrix, the eruption recurrence factor was determined as a time range due to limited data, where few volcanoes have enough dated historical eruptions events to provide an accurate recurrence interval, and the bimodal nature of volcanic eruptions produces clusters of frequent eruptions that are separated by longer periods of quiescence.<sup>135</sup> From discussions with the U.S. Geological Survey, it was determined that the high end of each estimation range would be ideal for use in frequency estimates for each volcanic hazard. For instance, if a volcanic hazard was labeled with a recurrence interval Hazard Score of 4, it is estimated to have another volcanic eruption within the next 1 to 99 years. For these volcanoes, a recurrence interval of 99 years is used, calculated as an annual estimated volcanic hazard risk of  $1/99 = 0.0101$ . This is based on an analysis of the Holocene (the last ~11,000 years) eruption history from the Smithsonian Institute’s Global Volcanism Program<sup>136</sup> to determine average eruption intervals within recent geological history. Volcanoes with higher threat rankings typically had calculated recurrence intervals that were longer

<sup>135</sup> Ewert, John W. (2007). System for ranking relative threats of US volcanoes. *Natural Hazards Review* 8.4: 112-124., [https://doi.org/10.1061/\(ASCE\)1527-6988\(2007\)8:4\(112\)](https://doi.org/10.1061/(ASCE)1527-6988(2007)8:4(112))

<sup>136</sup> Global Volcanism Program. (2025). [Database] Volcanoes of the World (v. 5.2.6; 5 Feb 2025). Distributed by Smithsonian Institution, compiled by Venzke, E. <https://doi.org/10.5479/si.GVP.VOTW5-2024.5.2>

than the eruption recurrence factor was estimated, indicating that using values on the later end of the scale was a valid decision.

The only addition to these annual frequencies is the inclusion of Montserrat's Soufrière Hills volcano that caused ashfall losses in the U.S. Virgin Islands and Puerto Rico. Since Soufrière Hills is not in the National Volcanic Threat Layer, there is no eruption recurrence interval assigned to it. Therefore, based on the activity of the volcano, discussions with subject matter experts, and comparisons to volcanoes within the National Volcanic Threat Layer, it was determined to use an annualized frequency that matched that of the most active volcanoes currently in the National Volcanic Threat Layer (0.0101).

An annualized frequency value is assigned to each volcano polygon based on the estimated recurrence interval from the 2018 Update to the U.S. Geological Survey National Volcanic Threat Assessment and outlined in [Table 80](#). Annualized frequency is only computed for volcanoes that are designated with a threat hazard category of 1 or greater.

#### Equation 136: Volcanic Activity Annualized Frequency

$$Freq_{VLCN} = \frac{1}{Recurrence_{VLCN}}$$

where:

$Freq_{VLCN}$  is the annualized frequency of Volcanic Activity for the volcano (events per year).

$Recurrence_{VLCN}$  Is the determined recurrence interval based on the threat hazard score

The annualized frequency value at the Census block level represents the estimated number of Volcanic eruptions each year for a specific area, or the probability that a volcano in the area will erupt in a given year. The volcanic activity annualized frequency as calculated in Equation 133 is applied to the Census Block level as is. When multiple Volcanic hazards impact the same Census block, the volcanic activity annualized frequency of these hazards are added together using the mutually exclusive probability equation as described in [Equation 137](#).

#### Equation 137: Multiple Volcanic Hazard Annualized Frequency Calculation

$$Freq_{VLCN}(Volcano A \text{ or } Volcano B) = Freq_{VLCN}(Volcano A) + Freq_{VLCN}(Volcano B)$$

Where:

$Freq_{VLCN}$  is the annualized frequency of Volcanic Activity for the volcano (events per year).

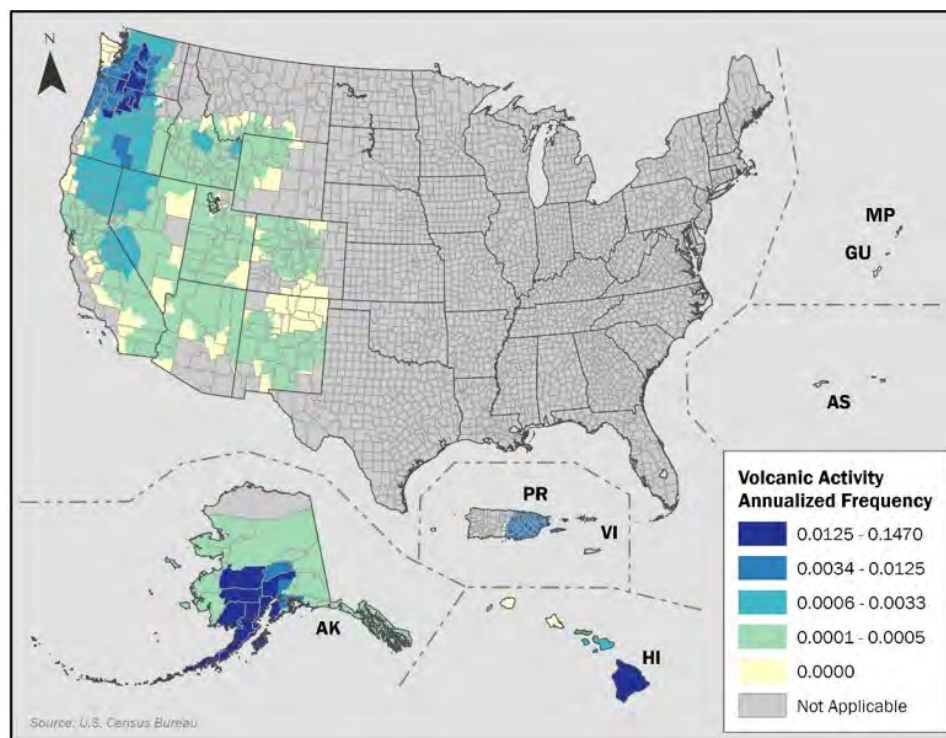
$Volcano A/B$  indicates the probability, or annualized frequency, of a specific volcano erupting in a particular year.

For instance, if there are two volcanoes, one with a Recurrence Interval of 10,000 years (annualized frequency = 0.0001) and one with a Recurrence Interval of 99 years (annualized frequency = 0.0101), then the combined annualized frequency would be  $\text{Freq}_{\text{VLGN}}(\text{A or B}) = 0.0102$ . Annualized frequency calculations use the same intersection between volcano polygons and Census block polygons that were used to calculate exposure.

#### 21.4.1. ANNUALIZED FREQUENCY AGGREGATION

The dataset presents annualized frequency values at both the Census tract and county level. The annualized frequency values at the Census block level are aggregated to the Census tract and county levels by carrying the volcanic activity annualized frequency upwards from Census block to Census tract, to county. This means that whatever the volcanic activity frequency of the Census block is, that same frequency is used for the Census tract and the county. When multiple volcanic hazards impact Census tracts and counties, the annualized frequencies for each individual impacting volcano are added together, as described in [Equation 137](#).

[Figure 112](#) displays Volcanic Activity annualized frequency at the county level.



**Figure 112: Volcanic Activity Annualized Frequency by County**

### 21.5. Volcanic Activity Historic Loss Ratio

The Volcanic Activity Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a volcanic occurrence, or the average rate of loss associated with the volcanic occurrence. For a detailed description of the Historic Loss Ratio calculation



process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Volcanic Activity hazard type.

Most databases of volcanic losses contained either outdated data or were missing key events, so a comprehensive search for all volcanic losses attributed to volcanoes within the United States was conducted. Historic Loss Ratio estimates were broken down into three separate regional categories: Hawai'i, Cascadia, and the Pacific Territories (see [Figure 113](#)).

All volcanic loss events from 1960 to 2023 are included in the Historic Loss Ratio calculation. SHELDUS was used as a baseline for loss estimates and research was done to expand or adjust any listed losses.

### 21.5.1. NEAR-VOLCANO SUB-HAZARD HISTORIC LOSS RATIO

A comprehensive evaluation of the eruption losses identified within the National Oceanic and Atmospheric Administration National Centers for Environmental Information/World Data Service Global Significant Volcanic Eruptions Database was performed to review source data and loss estimates. Further research was performed to identify impactful eruptions not included within the National Centers for Environmental Information/World Data Service database. Subject matter experts were consulted to help clarify inconsistencies among records. Once compiled and verified, the losses were then divided by exposure area.

**Hawai'i:** All volcanic activity losses for Hawai'i occurred from eruptions of Kīlauea, located on the southeastern shore of Hawai'i Island. Building exposure was defined using a map of Kīlauea's historic 1790-2023 lava flows in order to exclude buildings outside the likely flow path. After consultation with the U.S. Geological Survey, a 1-km buffer was added to these lava flows and the building exposure within that area was used for Historic Loss Ratio calculations.

**Cascadia:** The only historical eruption in the United States to cause non-ashfall losses outside of Hawai'i was the May 18, 1980 eruption of Mount St. Helens. Initial estimates placed total losses from the eruption at \$2B, though over time that value has been amended to approximately \$1.1B. Data from Oregon State University citing the Washington State Department of Commerce and Economic Development Research Division divides out the losses from Mount St. Helens by category.<sup>137</sup> In it, the total structural losses are listed as \$106.9 million, about 11% of the total losses. This value was divided by the building exposure within the National Volcanic Threat Layer's identified hazard area for Mount St. Helens. Two subsequent eruptions (May 25, 1980 and June 12, 1980) caused Ashfall losses in northwestern Oregon.

**Pacific Territories:** As there are currently no loss events within the Pacific Territories due to volcanic activity, the nearby eruption of Hunga volcano in Tonga was suggested by subject matter experts as a proxy for territories of the Commonwealth of the Northern Mariana Islands and American Samoa

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<sup>137</sup> <https://volcano.oregonstate.edu/cost-volcanic-eruptions>

(Guam does not have any active volcanoes). A previous study by Weir et al. (2024)<sup>138</sup> identified a range of Tongan loss ratios depending on particulate matter and modeling used. This data was then compared to loss estimates for the eruption divided by the total building exposure of the island. This loss ratio was within the range of the Weir et al (2024) estimates and was therefore used for the Pacific Territories' Historic Loss Ratios.

### 21.5.2. ASHFALL SUB-HAZARD HISTORIC LOSS RATIO

For the Ashfall sub-hazard, a uniform, minimal Historic Loss Ratio was applied across the Ashfall susceptibility layer based on historic losses recorded in SHELDS for the two Ashfall events from Mount St. Helens (see [Table 81](#)).

### 21.5.3. VOLCANIC ACTIVITY HAZARD HISTORIC LOSS RATIO

The Historic Loss Ratio exposure value for Volcanic Activity is a county-level value that represents the dollar value of the total building value, or the entire population of a county as recorded in Hazus 6.0. The Loss Ratio per Basis for each volcanic event and each consequence type (building and population) is calculated using [Equation 138](#).

#### Equation 138: Loss Ratio per Basis Calculation for a Single Volcanic Event

$$LRB_{VLCN Co CnsqType} = \frac{LOSS_{VLCN Co CnsqType}}{HLRExposure_{VLCN Co CnsqType}}$$

where:

|                                  |                                                                                                                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $LRB_{VLCN Co CnsqType}$         | is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific volcanic event. Calculation is performed for each consequence type (building and population). |
| $LOSS_{VLCN Co CnsqType}$        | is the loss (by consequence type) experienced from the specific volcanic event documented to have occurred in the county (in dollars or impacted people).                                                                                 |
| $HLRExposure_{VLCN Co CnsqType}$ | is the value (by consequence type) of the susceptible area estimated to have been exposed to the volcanic event (in dollars or people).                                                                                                   |

Due to the long frequency ranges for volcanic eruptions, no zero-loss occurrences were inserted into the Loss Ratio table. Historic Loss Ratio values are calculated across Volcanic Physiographic Regions

<sup>138</sup> Weir, A.M., Williams, J.H., Wilson, T.M., Hayes, J.L., Stewart, C., Leonard, G.S., Magill, C., Jenkins, S.F., Williams, S., Craig, H.M. and Kula, T. (2024). Rapid remote volcanic ashfall impact assessment for the 2022 eruption of Hunga volcano, Tonga: a bespoke approach and lessons identified. *Bulletin of Volcanology*, 86(11), pp.1-20.

(Figure 113). The regions are defined by the types and frequencies of volcanic eruptions that are anticipated across them.



**Figure 113: Volcanic Physiographic Regions**

Historic Loss Ratio estimates were calculated by dividing total losses for each region by the susceptible area for each volcanic eruption. These Historic Loss Ratio values (Table 81) were then applied across each of the Volcanic Physiographic Regions.

**Table 81: Volcano Historic Loss Ratio Calculations**

| <i>Region</i>          | <i>HLRB –<br/>Near Volcano</i> | <i>HLRP –<br/>Near Volcano</i> | <i>HLRB –<br/>Ashfall</i> | <i>HLRP –<br/>Ashfall</i> |
|------------------------|--------------------------------|--------------------------------|---------------------------|---------------------------|
| ALASKA*                | 0.1129915                      | 0.0037688                      | 0.000000036               | 0.000000173               |
| CASCADIA               | 0.1129915                      | 0.0037688                      | 0.000000036               | 0.000000173               |
| HAWAI'I                | 0.1041740                      | 0.0001489                      | 0.000000036               | 0.000000173               |
| PACIFIC<br>TERRITORIES | 0.0259324                      | 0.0001000                      | 0.000000036               | 0.000000173               |
| PR-VI**                | 0.0051000                      | 0.0001000                      | 0.000000036               | 0.000000173               |
| ROCKIES**              | 0.0051000                      | 0.0001000                      | 0.000000036               | 0.000000173               |

*\*Alaska has no historical eruption losses within the susceptible zones, however since the type of volcanoes within Alaska produce similar eruptions to the Cascadia volcanoes, the Cascadian Historic Loss Ratio was used in Alaska.*

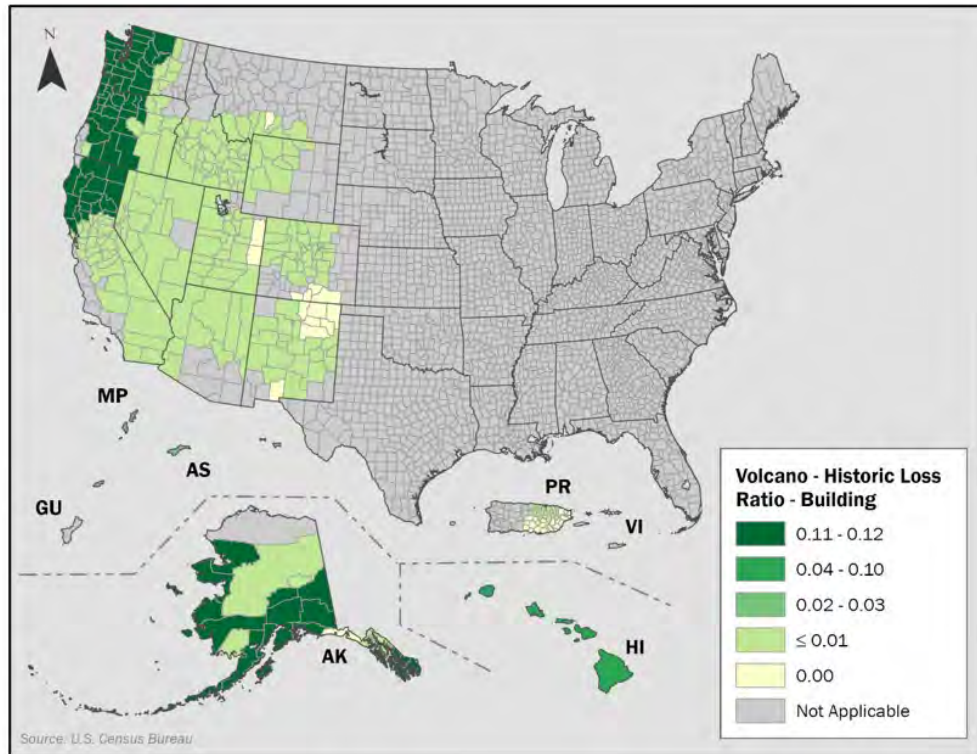
*\*\*No historical eruptions had occurred in this region within the period of record therefore a background Historic Loss Ratio value was determined using ashfall loss estimates.*

After the Loss Ratios per Basis are calculated for all Volcanic Activity events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Volcanic Activity.

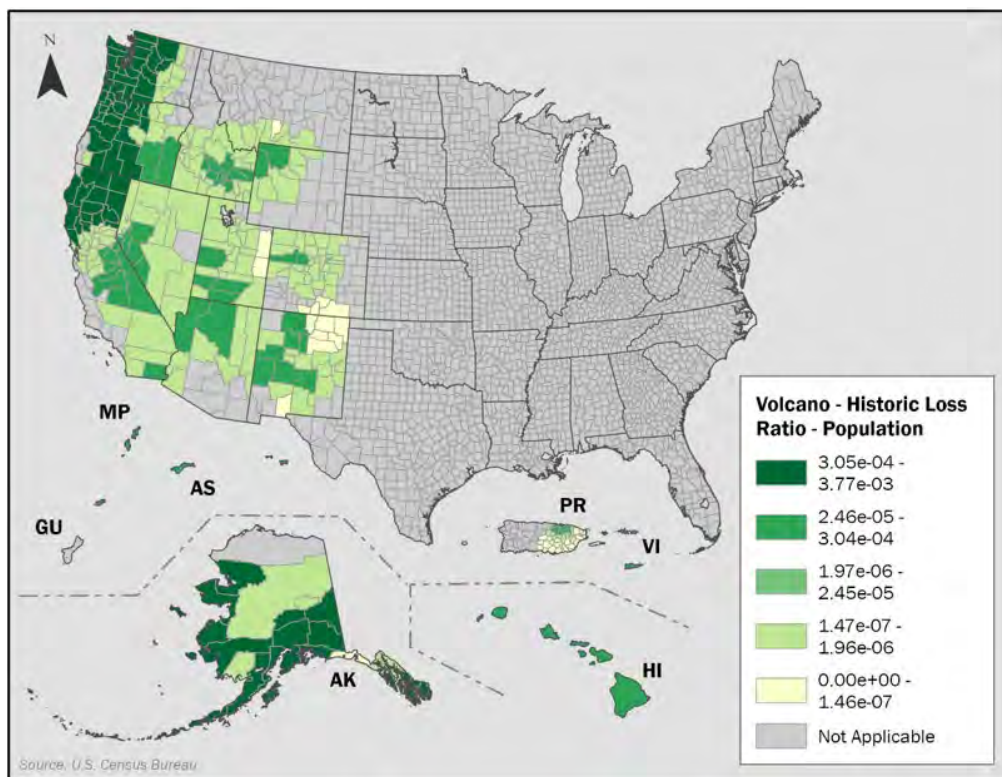
**Table 82: Volcanic Activity Historic Loss Ratio Bayesian Weighting Level Sets**

| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>     | <i>Set Refinement Filter</i> | <i>Set Utilization Threshold (Set Must Have)</i> |
|-------------------------|---------------|---------------------------|------------------------------|--------------------------------------------------|
| Population              | P1            | Volcanic Activity Regions | N/A                          | N/A                                              |
| Building                | B1            | Volcanic Activity Regions | N/A                          | N/A                                              |

Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 114](#) and [Figure 115](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Volcanic Activity. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 114: Volcanic Activity Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 115: Volcanic Activity Bayesian-Adjusted Historic Loss Ratio – Population**



## 21.6. Volcanic Activity Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss for the Near-Volcano and Ashfall sub-hazards is computed at the Census block level as in [Equation 139](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 139: Census Block Expected Annual Loss to Volcanic Activity

$$EAL_{VLCNSubCB Bldg} = Exposure_{VLCNSubCB Bldg} \times Freq_{VLCNSubCB} \times HLR_{VLCNSubCB Bldg}$$

$$EAL_{VLCNSubCB Pop} = Exposure_{VLCNSubCB Pop} \times Freq_{VLCNSubCB} \times HLR_{VLCNSubCB Pop}$$

where:

|                             |                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{VLCNSubCB Bldg}$      | is the building Expected Annual Loss due to Volcanic Activity sub-hazard for a specific Census block (in dollars).               |
| $Exposure_{VLCNSubCB Bldg}$ | is the building value exposed to Volcanic Activity sub-hazard in the Census block (in dollars).                                  |
| $Freq_{VLCNSubCB}$          | is the Volcanic Activity sub-hazard annualized frequency for the Census block (events per year).                                 |
| $HLR_{VLCNSubCB Bldg}$      | is the Bayesian-adjusted building Historic Loss Ratio for Volcanic Activity sub-hazard for the Census block.                     |
| $EAL_{VLCNSubCB Pop}$       | is the population equivalence Expected Annual Loss due to Volcanic Activity sub-hazard for a specific Census block (in dollars). |
| $Exposure_{VLCNSubCB Pop}$  | is the population equivalence value exposed to Volcanic Activity sub-hazard in the Census block (in dollars).                    |
| $HLR_{VLCNSubCB Pop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Volcanic Activity for the Census block.                              |

The total Expected Annual Loss values at the Census tract and county level are the sums of the Expected Annual Loss values for each Volcanic Activity sub-hazard and consequence types at the Census block level as in [Equation 140](#).

### Equation 140: Census Tract and County Expected Annual Loss to Volcanic Activity

$$EAL_{VLCN CB Bldg} = EAL_{VLCNNear CB Bldg} + EAL_{VLCNAsh CB Bldg}$$



$$EAL_{VLCN_{CBPop}} = EAL_{VLCNNear_{CBPop}} + EAL_{VLCNAsh_{CBPop}}$$

$$EAL_{VLCN_{CT}} = \sum_{CB}^{CT} EAL_{VLCN_{CB_{Bldg}}} + \sum_{CB}^{CT} EAL_{VLCN_{CB_{Pop}}}$$

$$EAL_{VLCN_{Co}} = \sum_{CB}^{Co} EAL_{VLCN_{CB_{Bldg}}} + \sum_{CB}^{Co} EAL_{VLCN_{CB_{Pop}}}$$

where:

$EAL_{VLCN_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Volcanic Activity for all sub-hazards for a specific Census block (in dollars).

$EAL_{VLCNNear_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Volcanic Activity for the Near Volcano sub-hazard for a specific Census block (in dollars).

$EAL_{VLCNAsh_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Volcanic Activity for the Ashfall sub-hazard for a specific Census block (in dollars).

$EAL_{VLCN_{CB_{Pop}}}$  is the population equivalence Expected Annual Loss due to Volcanic Activity for all sub-hazards for a specific Census block (in dollars).

$EAL_{VLCNNear_{CB_{Pop}}}$  is the population equivalence Expected Annual Loss due to Volcanic Activity for the Near Volcano sub-hazard for a specific Census block (in dollars).

$EAL_{VLCNAsh_{CB_{Pop}}}$  is the population equivalence Expected Annual Loss due to Volcanic Activity for the Ashfall sub-hazard for a specific Census block (in dollars).

$EAL_{VLCN_{CT}}$  is the total Expected Annual Loss due to Volcanic Activity for all sub-hazards for a specific Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{VLCN_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Volcanic Activity for all sub-hazards for all Census blocks in the Census tract (in dollars).

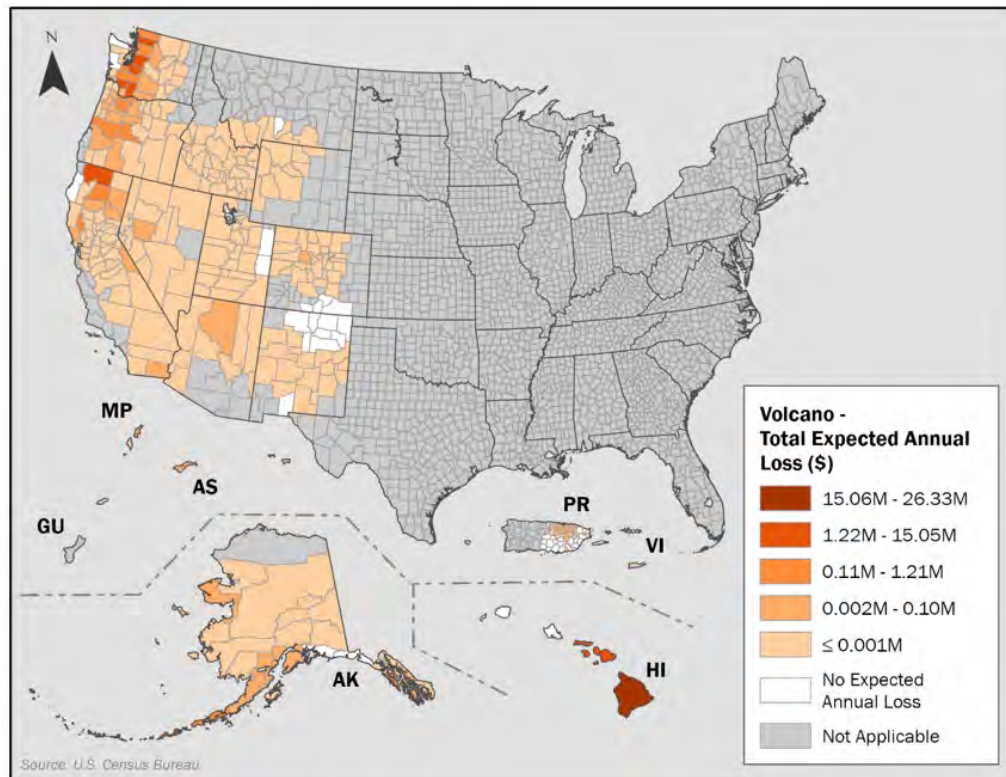
$\sum_{CB}^{CT} EAL_{VLCN_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Volcanic Activity for all sub-hazards for all Census blocks in the Census tract (in dollars).

$EAL_{VLCN_{Co}}$  is the total Expected Annual Loss due to Volcanic Activity for all sub-hazards for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{VLCN_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Volcanic Activity for all sub-hazards for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{VLCN_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Volcanic Activity for all sub-hazards for all Census blocks in the county (in dollars).

[Figure 116](#) shows the total Expected Annual Loss (population equivalence and building value combined) due to Volcanic Activity.



**Figure 116: Total Expected Annual Loss by County to Volcanic Activity**

With the Volcanic Activity total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Volcanic Activity Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Volcanic Activity Expected Annual Loss for population by the total population of the community.

## 21.7. Volcanic Activity Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- This methodology includes ashfall in its assessment of volcanic activity impacts. However, the nature of Ashfall damages – which can be best characterized by indirect damages including supply chain disruptions and long-term health impacts – mostly falls outside the scope of this analysis, which focuses on direct damages to buildings and immediate population losses. The decision to include Ashfall as a sub-hazard came after extensive conversations with subject matter experts and other stakeholders, who felt it was best to acknowledge the potential for Ashfall impacts in order to keep communities better informed of their hazard and risk profiles. The data development team acknowledges that Ashfall losses modeled in this analysis cover only a fraction of the most notable impacts, and those who would use these data for this purpose should not strictly rely on it and will need to seek additional data or resources.
- Ashfall modeling data which would more accurately identify the potential geographic extent of harmful ashfall were not available at the time of ingestion into the dataset. The buffer method used is an overly simplistic approach to an inherently complex task of modelling Ashfall impacts.
- While the U.S. Geological Survey's National Volcanic Threat Layer does provide a breakdown between proximal volcanic hazards and lahars, Ashfall is not included in the volcanic threat potential. This is a major volcanic hazard that has the potential to spread far beyond the volcanic eruption source, wherever that volcanic event may occur.
- Limited eruption history data for nearly all volcanoes within the United States and its territories limits the possibility of calculating a volcano-specific recurrence interval. The nature of volcanic eruptions also limits the accuracy, should a recurrence interval be provided for a subset of the volcano list.

## 22. Wildfire

A Wildfire is an unplanned fire burning in natural or wildland areas, such as forest, shrub lands, grasslands, or prairies.

The Wildfire hazard was developed using probabilistic data from the US Forest Service to calculate frequency of hazard events, susceptible areas, and impacts. As these data are not available for the Territories, the Wildfire hazard does not currently include American Samoa, Guam, Northern Mariana Islands, Puerto Rico, or U.S. Virgin Islands. The Wildfire hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 22.1. Wildfire Spatial Source Data

**Probabilistic Modeling and Susceptible Area Source:** [USDA, Forest Service, Spatial Datasets of Probabilistic Wildfire Risk Components for the United States](#)<sup>139</sup>

The United States Forest Service Missoula Fire Sciences Laboratory generated a series of raster datasets representing burn probability and conditional Flame Length Probability for the conterminous United States, Alaska, and Hawai'i through its geospatial Fire Simulation (FSim) system. FSim is designed to simulate the occurrence and growth of wildfires under tens of thousands of hypothetical contemporary fire seasons in order to estimate the probability of a given area (i.e., pixel) burning under 2024 landscape conditions and fire management practices.

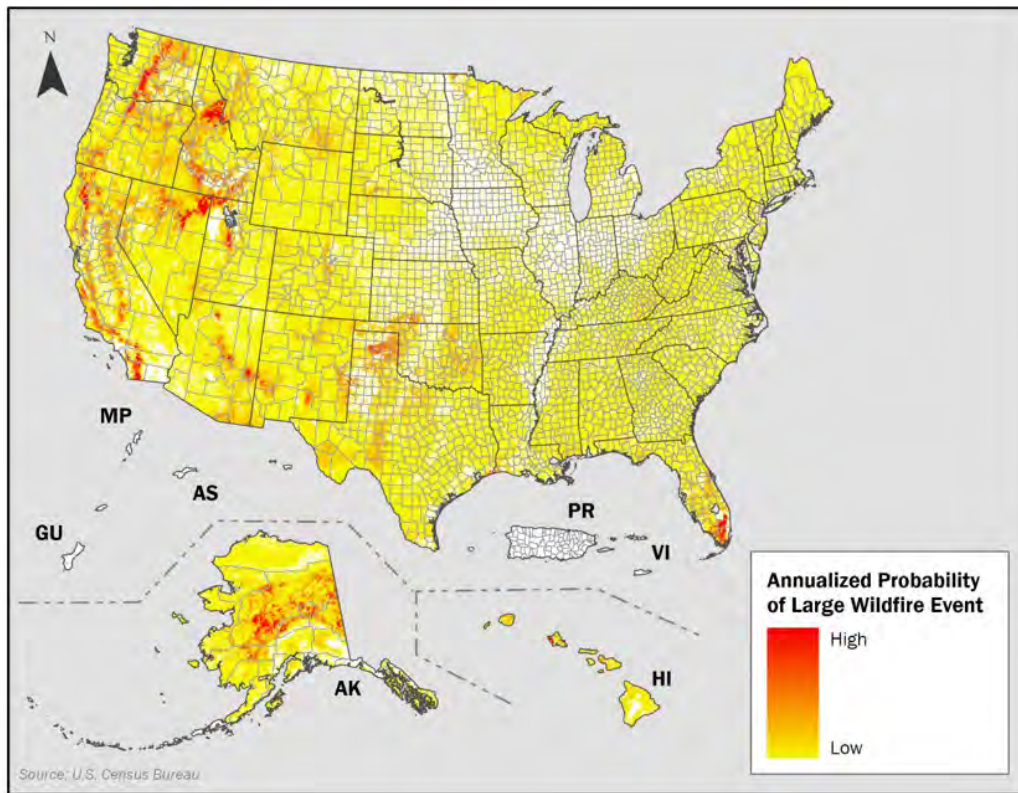
The burn probability raster dataset models the probability of an area being burned by a large fire (i.e., a fire that escapes initial fire suppression and spreads) at a 270-meter grid spatial resolution. The cell value in the raster file contains the mean annual burn probability as a value between 0 and 1 and represents the tendency for the cell area to burn due to a large fire on an annual basis given its landscape, contemporary weather conditions, and probability of containment (see [Figure 117](#)).

Flame-length probability is estimated for six standard Fire Intensity Levels and represents the likelihood of burning at a certain intensity level, given that a fire occurs. The Flame Length Probability (FLP) dataset consists of six raster files (one for each Fire Intensity Level), corresponding to the following flame length classes: FLP1  $\leq 2$  ft.; FLP2 =  $2 < 4$  ft.; FLP3 =  $4 < 6$  ft.; FLP4 =  $6 < 8$  ft.; FLP5 =  $8 < 12$  ft.; and FLP6 =  $12+$  ft. These files are also at a 270-meter grid spatial resolution.

**Note:** Because burn probability and Flame Length Probability data are not available for American Samoa, Guam, the Commonwealth of the Northern Mariana Islands, Puerto Rico, or the U.S. Virgin Islands, exposure, annualized frequency, and, therefore, Expected Annual Loss cannot be computed for these territories.

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<sup>139</sup> Short, Karen C.; Finney, Mark A.; Vogler, Kevin C.; Scott, Joe H.; Gilbertson-Day, Julie W.; Grenfell, Isaac C. (2024). Spatial datasets of probabilistic wildfire risk components for the United States (270m). 3rd Edition. Fort Collins, CO: Forest Service Research Data Archive. <https://doi.org/10.2737/RDS-2016-0034-3>.



**Figure 117: Burn Probability Raster**

## 22.2. Wildfire Spatial Data Processing

To determine the intersections of the raster cells with Census blocks, a custom raster-vector intersect tool was developed that allowed for the direct intersection of the high-resolution burn probability and Flame Length Probability raster layers with the developed area and agriculture area Census blocks. The tool uses the attributes of the raster cell (see [Table 83](#)) to directly calculate area-weighted exposure and frequency for both developed area and agriculture area within each Census block.

**Table 83: Sample Data from the Wildfire Raster Cell Attribute Table**

| Wildfire Raster Cell_ID | Burn Probability Value | Fil1Value | Fil2Value | Fil3Value | Fil4Value | Fil5Value | Fil6Value |
|-------------------------|------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| 6833438                 | 0.0174                 | 0.023     | 0.4483    | 0.454     | 0.0575    | 0.0172    | 0         |
| 6850554                 | 0.0209                 | 0         | 0         | 0         | 0.0048    | 0.2632    | 0.7321    |
| 853511                  | 0                      | 0         | 0         | 0         | 0         | 0         | 0         |

### 22.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Every county covered by the Wildfire probability raster had at least some possibility of Wildfire occurrence, so all counties were deemed possible for Wildfire occurrence. While the current data source does not supply information for American Samoa, Guam, Commonwealth of the Northern Mariana Islands, Puerto Rico or the U.S. Virgin Islands, these United States territories are still included as possible for Wildfire occurrence. No Scores are available for the United States territories as the data are insufficient.

### 22.3. Wildfire Exposure

Areas deemed susceptible to Wildfire are defined for each consequence type as the Census block developed area or agriculture area where the BP is greater than 0 and the modeled possibility of large fires reaching a Fire Intensity Level at or above a given threshold is greater than 0. Agricultural area and a Fire Intensity Level threshold corresponding to FLP4 (flame lengths of 6-8ft) is used for the agriculture consequence type to determine exposure area. To determine exposure area for the population consequence type, the Census block developed area and a FIL threshold corresponding to FLP4 is used. For the building consequence type, exposure area is determined using the Census block developed area and a Fire Intensity Level threshold corresponding to FLP5 (flame lengths of 8-12ft).

To identify areas of exposure, the Wildfire raster cells are intersected with the Census block developed area and agriculture area polygons utilizing the custom raster-vector intersect tool. The tool results provide the raster cell's unique identifier, Census block number, and the intersected area in square meters of either the developed area or agriculture area (see [Table 84](#)).

**Table 84: Sample Data from the Wildfire Raster Cell Census Block Intersection Table**

| <i>WildfireFishnetID</i> | <i>CensusBlock</i> | <i>IntersectedAreaM2</i> |
|--------------------------|--------------------|--------------------------|
| 102645159                | 060510001012069    | 3283.97478103638         |
| 102645160                | 060510001012069    | 9104.0656890869          |
| 102645161                | 060510001012069    | 3616.94129104614         |

To find exposure value, the sum of the intersection areas of the Wildfire raster cells with each Census block developed area or agriculture is multiplied by the developed area building value density, the developed area population density, and the agriculture area value density of the Census block to model exposure within the Census block (see [Equation 141](#)). These developed area densities in the Census block have been calculated by dividing the total Census block values (building value and population as recorded in Hazus 6.0 by the total developed area within a Census block area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.



**Equation 141: Census Block Wildfire Exposure**

$$Exposure_{WFIR_{CB_{Bldg}}} = \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish_{CB}}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{WFIR_{CB_{Pop}}} = \left( \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

$$Exposure_{WFIR_{CB_{Ag}}} = \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish_{CB}}} \times AvgDen_{CB_{Ag}}$$

where:

$Exposure_{WFIR_{CB_{Bldg}}}$  is the building value exposed to Wildfire in a specific Census block (in dollars).

$\sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish_{CB}}}$  is the sum of the intersected areas of Wildfire fishnet polygons within the Census block (in square kilometers) where the burn probability was greater than 0 and the value for the Flame Length Probability is greater than 0 at or above the consequence type specific threshold.

$DevAreaDen_{CB_{Bldg}}$  is the developed area building value density of the Census block (in dollars per square kilometer).

$Exposure_{WFIR_{CB_{Pop}}}$  is the population equivalence value exposed to Wildfire in a specific Census block (in dollars).

$DevAreaDen_{CB_{Pop}}$  is the developed area population value density of the Census block (in people per square kilometer).

$VSL$  is the Value of Statistical Life (\$13.7 million per person).

$Exposure_{WFIR_{CB_{Ag}}}$  is the agriculture area value exposed to Wildfire in a specific Census block (in dollars).

$AvgDen_{CB_{Ag}}$  is the average agriculture value density of the Census block (in dollars per square kilometer).

**22.3.1. EXPOSURE AGGREGATION**

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 142](#)).

**Equation 142: Census Tract and County Wildfire Exposure Aggregation**

$$\begin{aligned}
Exposure_{WFIR_{CB}Bldg} &= \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish}_{CB}} \times AvgDen_{CB_{Bldg}} \\
Exposure_{WFIR_{CB}Pop} &= \left( \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish}_{CB}} \times AvgDen_{CB_{Pop}} \right) \times VSL \\
Exposure_{WFIR_{CB}Ag} &= \sum_{Fish}^{CB} IntsctArea_{WFIR_{Fish}_{CB}} \times AvgDen_{CB_{Ag}} \\
Exposure_{WFIR_{CT}Bldg} &= \sum_{CB}^{CT} Exposure_{WFIR_{CB}Bldg} \\
Exposure_{WFIR_{Co}Bldg} &= \sum_{CB}^{Co} Exposure_{WFIR_{CB}Bldg} \\
Exposure_{WFIR_{CT}Pop} &= \sum_{CB}^{CT} Exposure_{WFIR_{CB}Pop} \\
Exposure_{WFIR_{Co}Pop} &= \sum_{CB}^{Co} Exposure_{WFIR_{CB}Pop} \\
Exposure_{WFIR_{CT}Ag} &= \sum_{CB}^{CT} Exposure_{WFIR_{CB}Ag} \\
Exposure_{WFIR_{Co}Ag} &= \sum_{CB}^{Co} Exposure_{WFIR_{CB}Ag}
\end{aligned}$$

where:

|                                           |                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{WFIR_{CT}Bldg}$                | is the building value exposed to Wildfire in a specific Census tract (in dollars).                                                                                |
| $\sum_{CB}^{CT} Exposure_{WFIR_{CB}Bldg}$ | is the summed value of all buildings where the Fire Intensity Level class of 5 or 6 is greater than 0 for each Census block within the Census tract (in dollars). |
| $Exposure_{WFIR_{Co}Bldg}$                | is the building value exposed to Wildfire in a specific county (in dollars).                                                                                      |
| $\sum_{CB}^{Co} Exposure_{WFIR_{CB}Bldg}$ | is the summed value of all buildings where the Fire Intensity Level class of 5 or 6 is greater than 0 for each Census block within the county (in dollars).       |

|                                       |                                                                                                                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{WFIRCTPop}$                | is the population equivalence value exposed to Wildfire in a specific Census tract (in dollars).                                                                                   |
| $\sum_{CB}^{CT} Exposure_{WFIRCBPop}$ | is the summed value of all population equivalence where the Fire Intensity Level class of 4, 5, or 6 is greater than 0 for each Census block within the Census tract (in dollars). |
| $Exposure_{WFIRCoPop}$                | is the population equivalence value exposed to Wildfire in a specific county (in dollars).                                                                                         |
| $\sum_{CB}^{Co} Exposure_{WFIRCBPop}$ | is the summed value of all population equivalence where the Fire Intensity Level class of 4, 5, or 6 is greater than 0 for each Census block within the county (in dollars).       |
| $Exposure_{WFIRCTAg}$                 | is the agriculture value exposed to Wildfire in a specific Census tract (in dollars).                                                                                              |
| $\sum_{CB}^{CT} Exposure_{WFIRCBAg}$  | is the summed value of all agriculture where the Fire Intensity Level class of 4, 5, or 6 is greater than 0 for each Census block within the Census tract (in dollars).            |
| $Exposure_{WFIRCoAg}$                 | is the agriculture value exposed to Wildfire in a specific county (in dollars).                                                                                                    |
| $\sum_{CB}^{Co} Exposure_{WFIRCBAg}$  | is the summed value of all agriculture where the Fire Intensity Level class of 4, 5, or 6 is greater than 0 for each Census block within the county (in dollars).                  |

## 22.4. Wildfire Annualized Frequency

The annualized frequency value represents the area-weighted burn probability (due to a large fire) of a location in a given year. The annualized frequency is calculated at the Census block level (see [Equation 143](#)), and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations for a Census block were performed within the same custom developed raster-vector tool used to calculate exposure. The tool output provides the area-weighted annualized frequency of Wildfire probability, where the area is defined as the Census block's developed area or agriculture area. Note that, unlike the frequency calculations for other hazards, the Wildfire frequency is consequence-dependent, so agriculture frequency may differ from the building and population frequency.

### Equation 143: Census Block Area-Weighted Wildfire Annualized Frequency

$$Freq_{WFIRCB} = \frac{IntsctArea_{WFIRCB} \times BProb_{WFIRCB}}{Area_{CB}}$$

where:

$Freq_{WFIRCB}$  is the area-weighted annualized frequency of Wildfire probability determined for a specific Census block's developed or agriculture area (probability per year).

$IntsctArea_{WFIRCB}$  is the intersected area of a specific Wildfire raster cell (where the burn probability was greater than 0) with the Census block's developed or agriculture area (in square meters).

$BProb_{WFIRCB}$  is the probability of Wildfire occurrence for the Wildfire raster cell.

$Area_{CB}$  is the total area of the Census block's developed or agriculture area (in square meters).

#### 22.4.1. ANNUALIZED FREQUENCY AGGREGATION

The annualized frequency values at the Census block level are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 144](#).

##### Equation 144: Census Tract and County Area-Weighted Wildfire Annualized Frequency Aggregation

$$Freq_{WFIRCT} = \frac{\sum_{CB}^{CT} (Freq_{WFIRCB} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{WFIRCo} = \frac{\sum_{CB}^{Co} (Freq_{WFIRCB} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{WFIRCT}$  is the area-weighted Wildfire annualized frequency for a specific Census tract.

$Freq_{WFIRCB}$  is the area-weighted annualized frequency of Wildfire probability determined for a specific Census block (probability per year).

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

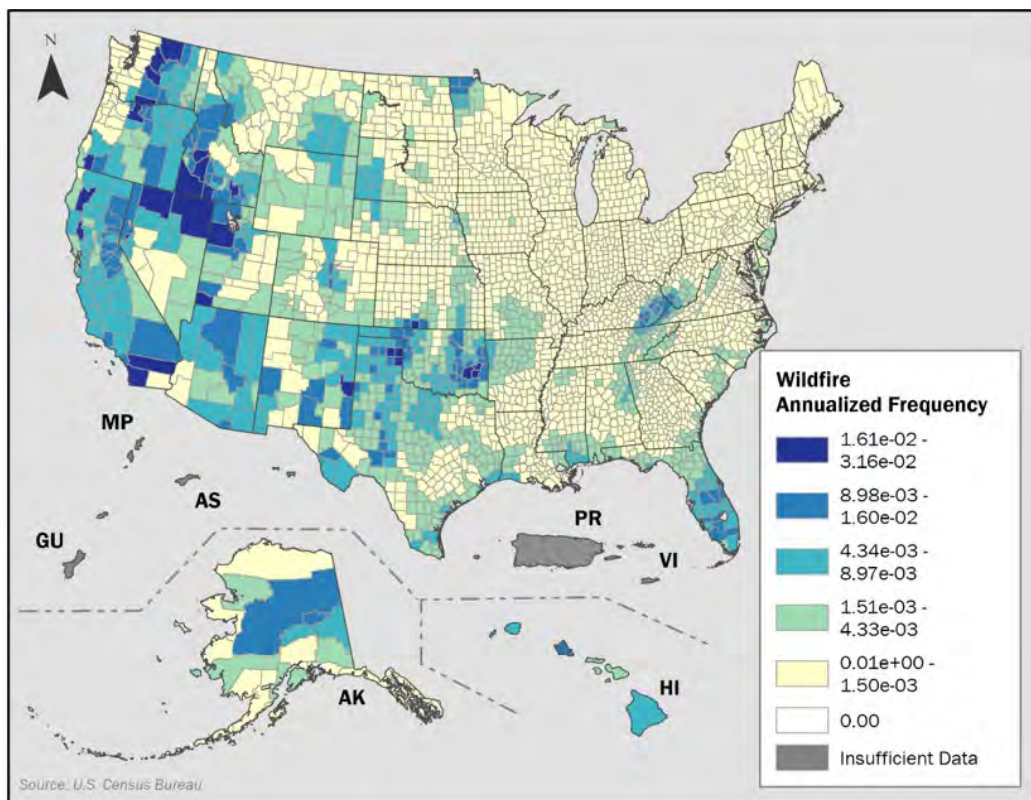
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{WFIRCo}$  is the area-weighted Wildfire annualized frequency for a specific county.

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

Figure 118 displays Wildfire annualized frequency at the county level. The frequency surrogates displayed in this map and provided in the dataset are the maximum of the developed area and agriculture area frequencies when these differ.



**Figure 118: Wildfire Annualized Frequency by County**

## 22.5. Wildfire Historic Loss Ratio

The Wildfire Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Wildfire occurrence, or the average rate of loss associated with a Wildfire occurrence. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Wildfire hazard type.

Loss data are provided by SHELDUS<sup>140</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHELDUS events from 1996 through 2023 are included in the Historic Loss Ratio calculation. Five peril types are mapped to the hazard Wildfire (see [Table 85](#)). These native

<sup>140</sup> For Wildfire loss information, SHELDUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

records are aggregated on a timeframe basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 85: Wildfire Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHELUS</i> | <i>Total SHELUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-----------------------------|----------------------------------|--------------------------------------|
| Fire-Brush                  | 0                                | 0                                    |
| Fire-Bush                   | 0                                | 0                                    |
| Fire-Forest                 | 150                              | 144                                  |
| Fire-Grass                  | 0                                | 0                                    |
| Wildfire                    | 3,708                            | 3,213                                |

For building value Historic Loss Ratio, Wildfire counties that intersect Wildfire fishnet cells for which the FIL reaches class 6 receive a default Historic Loss Ratio value that varies across Wildfire Historic Loss Ratio Regions. Counties in the West Southeast and Northeast Regions receive a value of 0.3, 0.1, and 0.1, respectively. Using these default values resulted in a building Expected Annual Loss to Wildfire that best approximated the average annual building loss reported in SHELUS within each Region.

For population and agriculture, the Historic Loss Ratio exposure values used in the Loss Ratio per Basis calculation are the population and agriculture value of the county's area that is most susceptible to Wildfire. This value is determined by summing the average population density or average agriculture value density exposure values of the Census blocks that intersect Wildfire fishnet cells for which the FIL reaches 6 (average flame length of 12 feet or more). The Loss Ratio per Basis for each SHELUS-documented event is calculated using [Equation 145](#).

**Equation 145: Loss Ratio per Basis Calculation for a Single Wildfire Event**

$$LRB_{WFIRCoCnsqType} = \frac{LOSS_{WFIRCoCnsqType}}{HLRExposure_{CoCnsqType}}$$

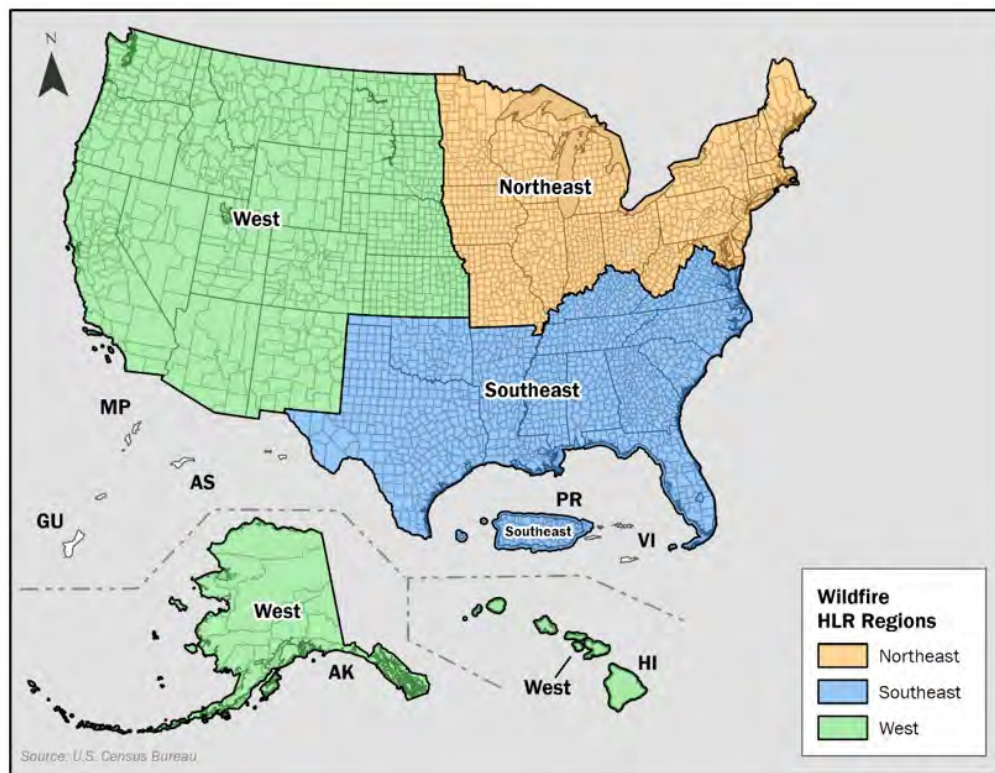
where:

|                         |                                                                                                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $LRB_{WFIRCoCnsqType}$  | is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Wildfire event. Calculation is performed for population and agriculture. |
| $LOSS_{WFIRCoCnsqType}$ | is the loss (by consequence type) experienced from the Wildfire event documented to have occurred in the county (in dollars or impacted people).                                                                     |



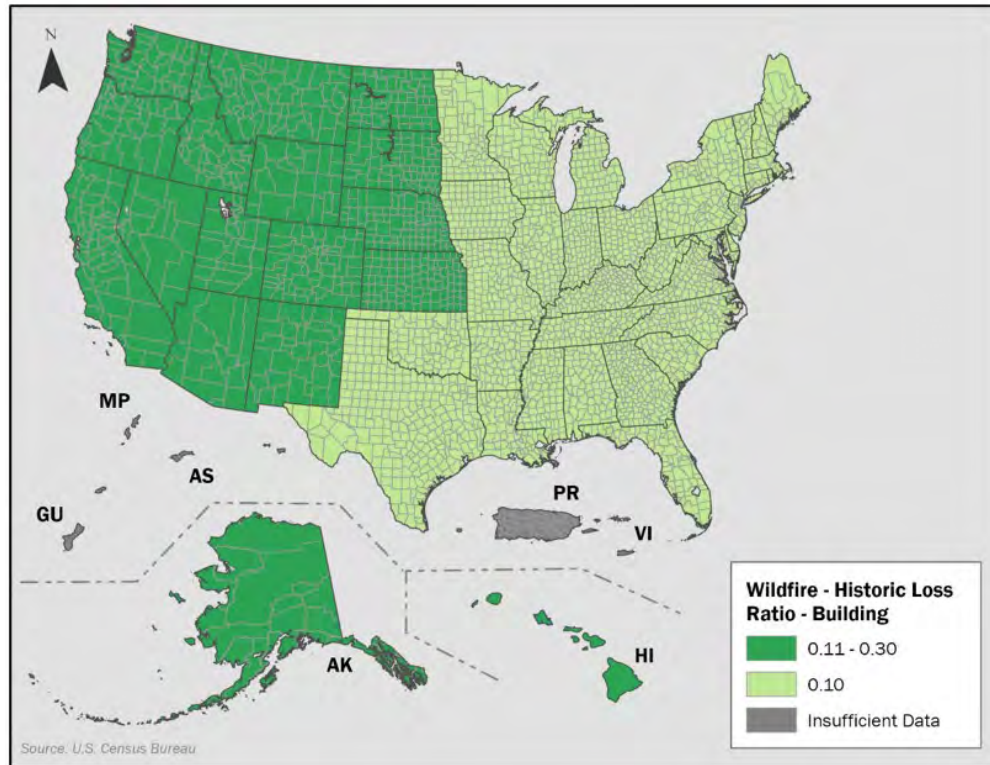
$HLRExposure_{CoCnsqType}$  is the value (by consequence type) of the susceptible area estimated to have been exposed to the Wildfire event (in dollars or people).

Wildfire frequency is based on a probabilistic model, so no zero-loss occurrences are inserted into the Loss Ratio table. After the population and agriculture Loss Ratios per Basis are calculated for each county, average loss ratios are computed at the Region level (for population and agriculture) and assigned to their residing counties.

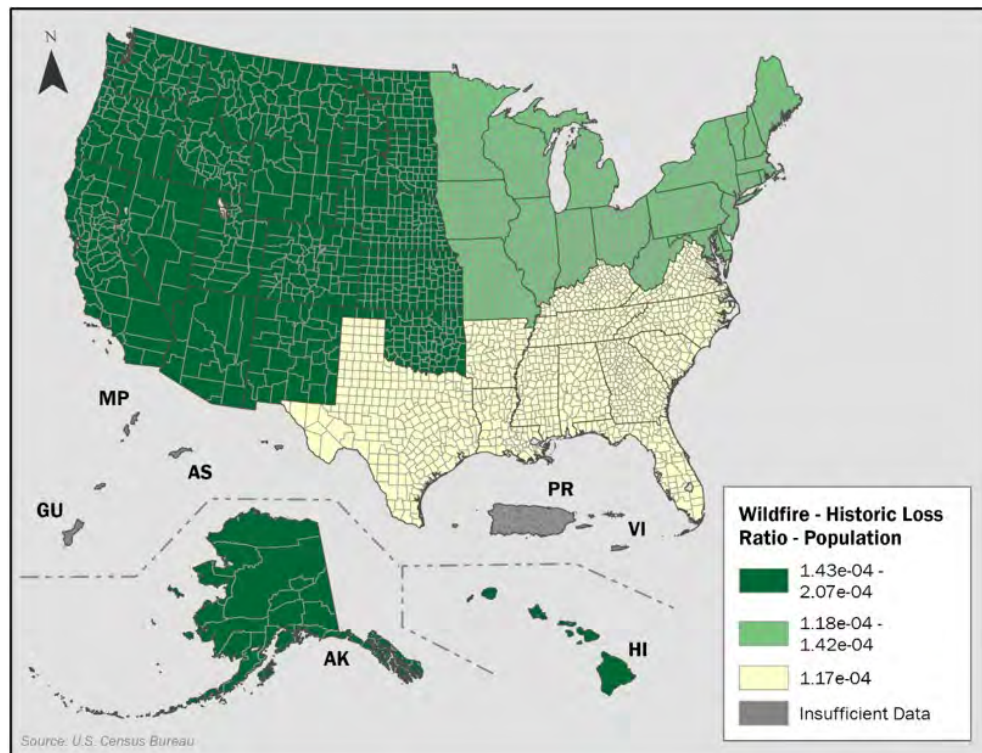


**Figure 119: Wildfire-specific Historic Loss Ratio Regions**

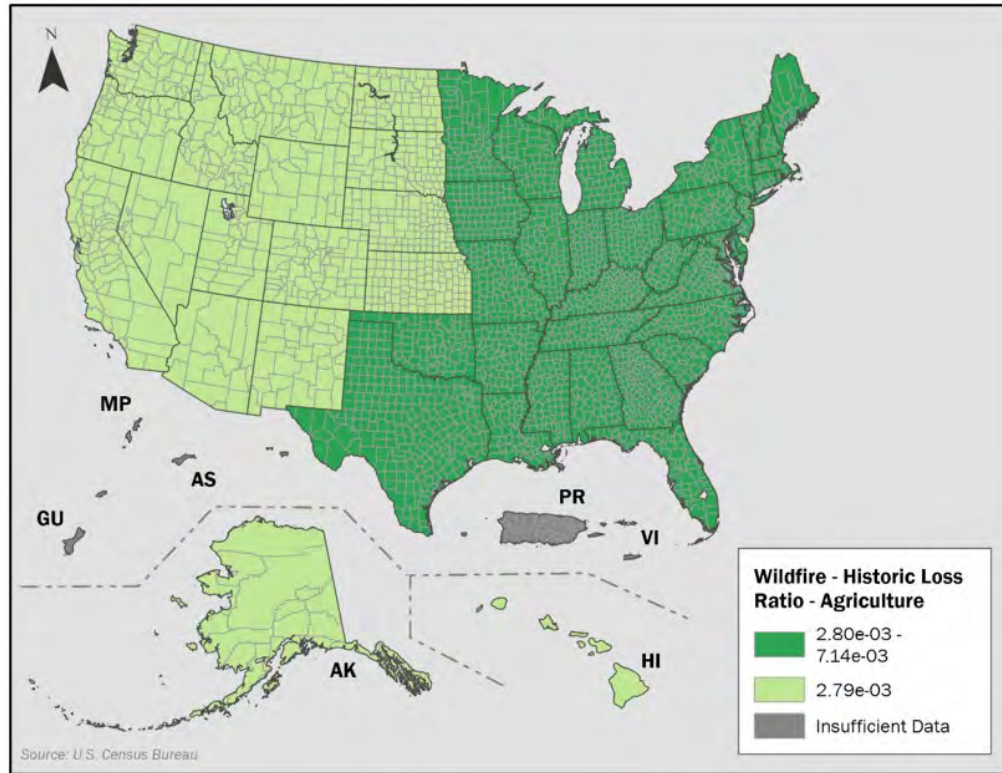
Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing techniques are then implemented to further refine the Historic Loss Ratio values for Wildfire (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 120](#), [Figure 121](#), and [Figure 122](#) represent the final, Bayesian-adjusted county-level population and agriculture Historic Loss Ratio values for Wildfire. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 120: Wildfire Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 121: Wildfire Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 122: Wildfire Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

## 22.6. Wildfire Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 146](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

### Equation 146: Census Block Expected Annual Loss to Wildfire

$$EAL_{WFIRCBldg} = Exposure_{WFIRCBldg} \times Freq_{WFIRCB} \times HLR_{WFIRCBldg}$$

$$EAL_{WFIRCBPop} = Exposure_{WFIRCBPop} \times Freq_{WFIRCB} \times HLR_{WFIRCBPop}$$

$$EAL_{WFIRCB_{Ag}} = Exposure_{WFIRCB_{Ag}} \times Freq_{WFIRCB} \times HLR_{WFIRCB_{Ag}}$$

where:

$EAL_{WFIRCBldg}$  is the building Expected Annual Loss due to Wildfire occurrences for a specific Census block (in dollars).

|                            |                                                                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{WFIR_{CB}Bldg}$ | is the building value where the flame intensity level of 5 or 6 is greater than 0 in the Census block (in dollars).                   |
| $Freq_{WFIR_{CB}}$         | is the Wildfire annualized frequency for the Census block (probability per year).                                                     |
| $HLR_{WFIR_{CB}Bldg}$      | is the Bayesian-adjusted building Historic Loss Ratio for Wildfire for the Census block.                                              |
| $EAL_{WFIR_{CB}Pop}$       | is the population equivalence Expected Annual Loss due to Wildfire occurrences for a specific Census block (in dollars).              |
| $Exposure_{WFIR_{CB}Pop}$  | is the population equivalence value where the flame intensity level of 4, 5, or 6 is greater than 0 in the Census block (in dollars). |
| $HLR_{WFIR_{CB}Pop}$       | is the Bayesian-adjusted population Historic Loss Ratio for Wildfire for the Census block.                                            |
| $EAL_{WFIR_{CB}Ag}$        | is the agriculture Expected Annual Loss due to Wildfire occurrences for a specific Census block (in dollars).                         |
| $Exposure_{WFIR_{CB}Ag}$   | is the agriculture value where the flame intensity level of 4, 5, or 6 is greater than 0 in the Census block (in dollars).            |
| $HLR_{WFIR_{CB}Ag}$        | is the Bayesian-adjusted agriculture Historic Loss Ratio for Wildfire for the Census block.                                           |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 147](#).

**Equation 147: Census Tract and County Expected Annual Loss to Wildfire**

$$EAL_{WFIR_{CT}} = \sum_{CB}^{CT} EAL_{WFIR_{CB}Bldg} + \sum_{CB}^{CT} EAL_{WFIR_{CB}Pop} + \sum_{CB}^{CT} EAL_{WFIR_{CB}Ag}$$

$$EAL_{WFIR_{Co}} = \sum_{CB}^{Co} EAL_{WFIR_{CB}Bldg} + \sum_{CB}^{Co} EAL_{WFIR_{CB}Pop} + \sum_{CB}^{Co} EAL_{WFIR_{CB}Ag}$$

where:

|                                      |                                                                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{WFIR_{CT}}$                    | is the total Expected Annual Loss due to Wildfire occurrences for a specific Census tract (in dollars).                         |
| $\sum_{CB}^{CT} EAL_{WFIR_{CB}Bldg}$ | is the summed building Expected Annual Loss due to Wildfire occurrences for all Census blocks in the Census tract (in dollars). |



$\sum_{CB}^{CT} EAL_{WFIR_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Wildfire occurrences for all Census blocks in the Census tract (in dollars).

$\sum_{CB}^{CT} EAL_{WFIR_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Wildfire occurrences for all Census blocks in the Census tract (in dollars).

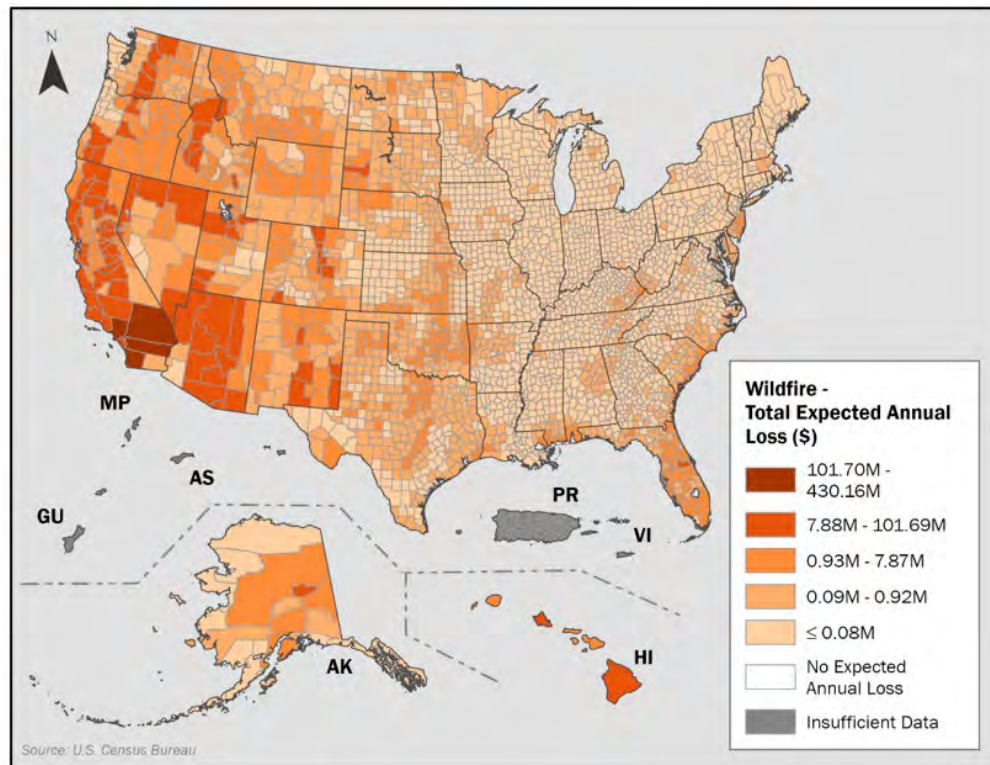
$EAL_{WFIR_{Co}}$  is the total Expected Annual Loss due to Wildfire occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{WFIR_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Wildfire occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{WFIR_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Wildfire occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{WFIR_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Wildfire occurrences for all Census blocks in the county (in dollars).

[Figure 123](#) shows the total Expected Annual Loss (building, population equivalence, and agriculture value combined) to Wildfire occurrences.



**Figure 123: Total Expected Annual Loss by County to Wildfire**

With the Wildfire total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)).

The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.

Building Expected Annual Loss Rate is calculated by dividing the Wildfire Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Wildfire Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by dividing the Wildfire Expected Annual Loss for agriculture by the total agriculture value of the community.

## 22.7. Wildfire Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- Historic Loss Ratio relies on incomplete data regarding extent of damage compared to flame length.
- The Agriculture and Population consequence type calculations use lower flame length thresholds for damage than the Building consequence type based on an assumption that crops do not require 8ft flame lengths to sustain damage.



## 23. Winter Weather

Winter Weather consists of winter storm events in which the main types of precipitation are snow, sleet, or freezing rain.

The Winter Weather hazard modeling uses Mesonet data to provide impact areas for this hazard that are derived from the National Weather Service's warnings and advisories. SHELDUS data were used for historical losses for Winter Weather events and generate the Historic Loss Ratios. These are overlaid with the exposure values provided in Hazus 6.0 for Building, Population, and Agriculture for the final Expected Annual Loss. The Winter Weather hazard calculates risk for all three consequence types: Building, Population, and Agriculture. For a summary of the Hazard Data Characteristics, please see [Appendix B – Hazard Data Characteristics Comparison](#).

### 23.1. Winter Weather Spatial Source Data

**Historical Occurrence Compiling Source:** [Iowa State University, Iowa Environmental Mesonet](#)<sup>141</sup>

The National Weather Service is continuously issuing weather alerts based on current weather conditions.<sup>142</sup> Each alert is coded by type of phenomenon and level of significance, (e.g., Watch, Warning, or Advisory) and conceptually can indicate the potential for weather event activities in a specific area. Archived National Weather Service alerts are aggregated, continuously updated, and made available for download in shapefile format by Iowa State University's Iowa Environmental Mesonet. Data include geometry for each alert's issued area and attributes related to each alert's severity and phenomena type. Weather alerts are also timestamped with the time of issuance and the time of expiration. A table describing this dataset's attributes can be found in [Appendix C – Mesonet-National Weather Service Weather Event Attribute Description](#).

The geometry shape for each alert record represents the geographic area for which the National Weather Service alert applied. However, the Mesonet shapes are simplified versions of the more detailed National Weather Service Public Forecast Zone shape originally associated with the alert record. Because the Mesonet tabular data still retains the reference IDs for the National Weather Service Public Forecast Zones, these IDs are able to be used to relate to the more detailed zone associated with each alert record.

The National Weather Service Public Forecast Zones can be downloaded in shapefile format<sup>143</sup> and represent the codified areas for which weather alerts are issued by National Weather Service. The Public Forecast Zones shape definitions are predominantly derived from county boundaries. While the Public Forecast Zone boundaries are more refined than those substituted into the Mesonet data, they are not at the same resolution as the current county boundaries derived from Census blocks.

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<sup>141</sup> Department of Agronomy, Iowa State University. (2025). *Iowa Environmental Mesonet* [online database]. Retrieved from <https://mesonet.agron.iastate.edu/request/gis/watchwarn.phtml>.

<sup>142</sup> NWS, NOAA. (2018). *Active Alerts* [online dataset]. Retrieved from [Active Alerts](#).

<sup>143</sup> NWS, NOAA. (2022). *NWS Public Forecast Zones* [online dataset]. Retrieved from <https://www.weather.gov/gis/PublicZones>.

Utilizing the Public Forecast Zone shapefile in conjunction with the Public Forecast Zone – County Correlation file,<sup>144</sup> a determination was made as to which Public Forecast Zones have single-county coverage and which are either sub-county zones or made of portions of multiple counties. For perspective, the following approximate distributions of forecast zone composition were found:

- 70% of the zones have single-county coverage.
- 20% are cases where a single county is subdivided into multiple zones.
- 10% are zones that breach parts of multiple contiguous counties.

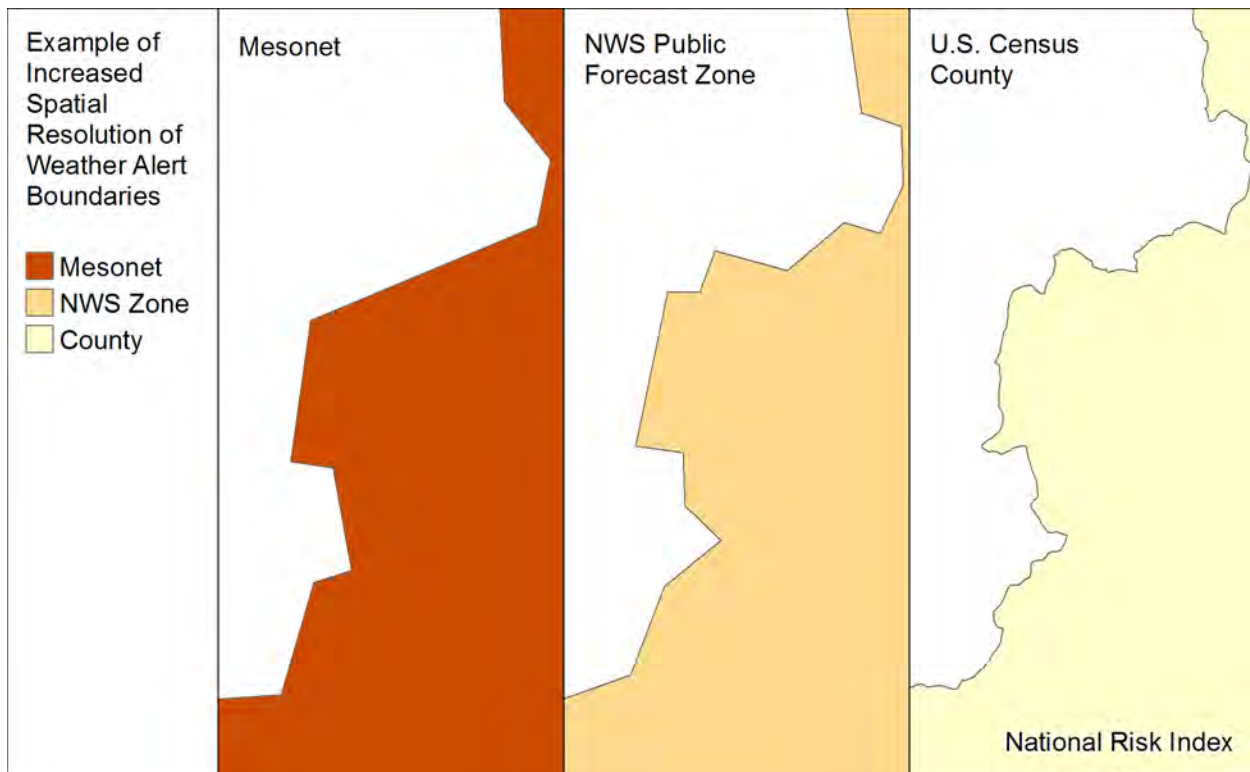
For those Forecast Zones covering a single county, the United States Census Topologically Integrated Geographic Encoding and Referencing 2021 county boundaries (Topologically Integrated Geographic Encoding and Referencing 2024 boundaries for Connecticut) are substituted.

Another aspect of the National Weather Service Public Forecast Zones is that they can and have changed over time. In the Mesonet data (2005 through 2024), there are many distinct Forecast Zones referenced that do not exist in the current National Weather Service Public Forecast Zone shapefile. This occurs when a National Weather Service Public Forecast Zone has been modified in shape, renamed, and/or “retired” from use.

Further research found that the National Weather Service maintains a downloadable Change History log of the various changes in Forecast Zone areas since 1997. This text file does not contain the pre- or post-shape of the altered forecast zone. Archived versions of these changes are likely available via contact with National Weather Service, but the effort to match the National Weather Service issued alert record to the version-controlled shape representation of the forecast zone at the time of alert issue seems to be beyond the scope of the processing effort, though a Mesonet representative was contacted to see if Forecast Zone shapes associated with each year of alert data had been archived. Unfortunately, no such archival information was available. For cases where the more refined National Weather Service Forecast Zone shape is unavailable, the simplified Mesonet boundary version shape is used. See [Figure 124](#) for an example of the differences in the spatial resolution of weather alert boundaries.

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<sup>144</sup> NWS, NOAA. (2022). *Zone-County Correlation File* [online dataset]. Retrieved from <https://www.weather.gov/gis/ZoneCounty>.



**Figure 124: Three Boundary Definitions: Mesonet, Forecast Zone, and United States Census County**

## 23.2. Winter Weather Spatial Data Processing

Winter Weather event alerts were extracted from the Mesonet data based on the Valid Time Extent Code significance code (SIG field) and the phenomena code (PHENOM field) values. Only Warning alerts (SIG = 'W') and Advisory alerts (SIG = 'Y') of any of the Phenomena Types in [Table 86](#) were considered Winter Weather events. (see [Table 87](#)).

**Table 86: Winter Weather Phenomenon Types**

| <i>PHENOM Code</i> | <i>Phenomena Code Description</i> |
|--------------------|-----------------------------------|
| BZ                 | Blizzard                          |
| HS                 | Heavy Snow                        |
| LB                 | Lake Effect Snow and Blowing Snow |
| LE                 | Lake Effect Snow                  |
| SN                 | Snow                              |
| SB                 | Snow and Blowing Snow             |
| WS                 | Winter Storm                      |

| <i>PHENOM Code</i> | <i>Phenomena Code Description</i> |
|--------------------|-----------------------------------|
| WW                 | Winter Weather                    |

**Table 87: Sample Original Mesonet Winter Weather Records**

| <i>WFO</i> | <i>ISSUED</i>             | <i>EXPIRED</i>            | <i>PHENOM</i> | <i>SIG</i> | <i>NWS_UGC</i> | <i>AREA_KM2</i> |
|------------|---------------------------|---------------------------|---------------|------------|----------------|-----------------|
| GJT        | Feb. 27, 2017<br>1:00 PM  | Mar. 01, 2017<br>3:39 AM  | WS            | W          | COZ019         | 9720.85253906   |
| PHI        | Mar. 14, 2017<br>12:00 AM | Mar. 14, 2017<br>6:49 PM  | BZ            | A          | NJZ001         | 1386.35180664   |
| AFG        | Mar. 29, 2008<br>8:00 PM  | Mar. 30, 2008<br>12:17 PM | HS            | W          | AKZ214         | 25092.76593474  |

To remove unintended error in spatial results due to the use of the simplified event area shapes contained in the Mesonet data, event areas with a higher resolution version were substituted. This substitution used the National Weather Service Public Forecast Zone shape associated with the alert record or, in cases where the forecast zone is for a single county, a better resolution version of the county boundary area.

Winter Weather occurrences are measured in event-days as this more accurately represents the variability of Winter Weather event duration. To capture this, each native alert record with a duration greater than a single day was replaced with multiple records, one for each day of the original record's duration.

If a Winter Weather event's duration on any given day was less than 6 hours, then the event was assigned to the day having the greatest duration of the event. This handled cases where the event occurs in the late evening and actually endured for a greater length of time on the next calendar day than on the day the alert was issued.

For cases where the event duration is longer, the following logic was used: If a weather event's duration was greater than 6 hours, assign the event to all days on which 6 or more hours occur. For example, if a 14-hour weather event was issued for 2 AM until 6 PM on January 1, then the event would be assigned to January 1. If the alert was issued from 11 PM on January 1 to 1 PM on January 2, then the event would be assigned to only January 2. If the alert was issued from 7 PM on January 1 to 9 AM on January 2, then the event would be assigned to both January 1 and January 2. To illustrate this concept, the Winter Weather occurrences in [Table 88](#) are expanded to create the Winter Weather event-day records in [Table 89](#).

Additionally, there were some data quality issues with the Mesonet data. For example, some warnings had an expiration date that is prior to the issue date. In these cases, a single record was used and assigned the issue date.

**Table 88: Sample Winter Weather Data After Zone Shape Re-Sourcing**

| <i>Winter StormID</i> | <i>WFO</i> | <i>Issued</i>          | <i>Expired</i>         | <i>PHENOM</i> | <i>SIG</i> | <i>NWS_UGC</i> | <i>AreaKm2</i> | <i>NewShape Source</i> |
|-----------------------|------------|------------------------|------------------------|---------------|------------|----------------|----------------|------------------------|
| 45437                 | GJT        | Feb. 27, 2017 1:00 PM  | Mar. 01, 2017 3:39 AM  | WS            | W          | COZ019         | 9707.610       | Census County          |
| 45253                 | AJK        | Mar. 12, 2017 11:50 PM | Mar. 14, 2017 2:00 AM  | WS            | W          | AKZ022         | 4153.062       | NWS Forecast Zone      |
| 45416                 | CYS        | Feb. 27, 2017 9:00 PM  | Feb. 28, 2017 10:02 AM | WS            | W          | WYZ112         | 2354.592       | NWS Forecast Zone      |

**Table 89: Sample Data from the Winter Weather Date Expansion Table**

| <i>WinterStormDate ExpansionID</i> | <i>WinterStormID</i> | <i>Issued</i>          | <i>Expired</i>         | <i>DateType</i>            | <i>WinterStormHours</i> |
|------------------------------------|----------------------|------------------------|------------------------|----------------------------|-------------------------|
| 35072                              | 45437                | Feb. 27, 2017 1:00 PM  | Feb. 28, 2017 12:00 AM | Expanded Dates - Issued    | 11                      |
| 35073                              | 45437                | Feb. 28, 2017 12:00 AM | Mar. 01, 2017 12:00 AM | Expanded Dates - New Dates | 24                      |
| 35058                              | 45253                | Mar. 13, 2017 12:00 AM | Mar. 14, 2017 12:00 AM | Expanded Dates - New Dates | 24                      |
| 35067                              | 45416                | Feb. 28, 2017 12:00 AM | Feb. 28, 2017 10:02 AM | Expanded Dates - Expired   | 10.033333               |

To avoid overestimating the area of influence a “single” distinct weather event has due to multiple National Weather Service alerts being issued for that same weather event, a process to combine all Winter Weather event areas occurring on the same day (Year, Month, and Day specific) into one representative event shape was performed. This process resulted in an impact area shape for a single event for each day on which a Winter Weather event occurred. These event-day polygons could then be intersected with the Census block polygons to determine Winter Weather exposure and annualized frequency.

### 23.2.1. DETERMINATION OF POSSIBILITY OF HAZARD OCCURRENCE

Winter Weather can occur almost anywhere in the United States as the definition of a Winter Weather occurrence is locally defined by the area’s weather forecast office. For example, a forecast office in Texas may define a Winter Weather occurrence differently than a forecast office in New York. Therefore, all counties were deemed possible for Winter Weather occurrence.

23.2.2. PERIOD OF RECORD

The National Weather Service’s system of recording WWA in the 1990s and early 2000s was too difficult for automated processing. In 2005 the Valid Time Extent Code system was implemented, which allowed for the easy automated parsing of alert data. Therefore, National Weather Service weather events data were downloaded for 2005 through 2024. The date range is Nov. 12, 2005 to Nov. 12, 2024, so the period of record for which Winter Weather data are utilized is 19.0 years.

23.3. Winter Weather Non-Spatial Data Processing

23.3.1. HISTORICAL OCCURRENCE COUNT

The historical occurrence count of Winter Weather, in event-days, is computed as the area-weighted sum of distinct Winter Weather event-day polygons that intersect a Census block. This count uses the same Winter Weather expansion Census block intersection table used to find exposure at the Census block level and are used to compute annualized frequency at the Census block level.

Historical event-day counts are also supplied at the Census tract and county levels as the number of distinct Winter Weather event-day polygons that intersect the Census tract and county, respectively.

23.4. Winter Weather Exposure

To identify areas of exposure, the Winter Weather event-day polygons (also referred to as Winter Storm Date Expansions to acknowledge the spatiotemporal processing described in [Section 23.2 Winter Weather Spatial Data Processing](#)) are intersected with the Census block polygons within the processing database. The resulting table contains the Winter Weather event-day’s unique identifier, Census block number, and the intersected area in square kilometers (see [Table 90](#)).

Table 90: Sample Data from the Winter Weather Census Block Intersection Table

| <i>WinterStormDateExpansionID</i> | <i>CensusBlock</i> | <i>IntersectedAreaKm2</i> |
|-----------------------------------|--------------------|---------------------------|
| 44082                             | 517750105012023    | 0.00380071655273438       |
| 44082                             | 517700023004045    | 0.00177242324829102       |
| 44082                             | 517750102005022    | 0.090136718170166         |

To determine exposure value, the average coverage of a Winter Weather event-day is found by summing the intersected areas for all Winter Weather event-day polygons that intersected the Census block and dividing this sum by the number of intersecting event-day polygons. This is multiplied by the developed area building value density, the developed area population density, and the agriculture area value density of the Census block to model the conservative-case concentration of exposure within the Census block (see [Equation 148](#)). The densities of the Census block were calculated by dividing the total exposure values (building value and population as recorded in Hazus



6.0) by the developed or agriculture land area (in square kilometers). The Value of Statistical Life was used to express population equivalence exposure in terms of dollars.

**Equation 148: Census Block Winter Weather Exposure**

$$Exposure_{WNTW_{CB}Bldg} = \frac{\sum IntsctArea_{WNTW_{CB}}}{EventDayCount_{WNTW_{CB}}} \times DevAreaDen_{CB_{Bldg}}$$

$$Exposure_{WNTW_{CB}Pop} = \left( \frac{\sum IntsctArea_{WNTW_{CB}}}{EventDayCount_{WNTW_{CB}}} \times DevAreaDen_{CB_{Pop}} \right) \times VSL$$

$$Exposure_{WNTW_{CB}Ag} = \frac{\sum IntsctArea_{WNTW_{CB}}}{EventDayCount_{WNTW_{CB}}} \times AgValueDen_{CB}$$

where:

|                               |                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|
| $Exposure_{WNTW_{CB}Bldg}$    | is the building value exposed to Winter Weather event-days in a specific Census block (in dollars).                 |
| $\sum IntsctArea_{WNTW_{CB}}$ | is the sum of the intersected areas of past Winter Weather event-days with the Census block (in square kilometers). |
| $EventDayCount_{WNTW_{CB}}$   | is the total number of Winter Weather event-day polygons that intersect the Census block.                           |
| $DevAreaDen_{CB_{Bldg}}$      | is the developed area building value density of the Census block (in dollars per square kilometer).                 |
| $Exposure_{WNTW_{CB}Pop}$     | is the population equivalence value exposed to Winter Weather event-days in a specific Census block (in dollars).   |
| $DevAreaDen_{CB_{Pop}}$       | is the developed area population density of the Census block (in people per square kilometer).                      |
| $VSL$                         | is the Value of Statistical Life (\$13.7 million per person).                                                       |
| $Exposure_{WNTW_{CB}Ag}$      | is the agriculture value exposed to Winter Weather event-days in a specific Census block (in dollars).              |
| $AgValueDen_{CB}$             | is the agriculture value density of the Census block (in dollars per square kilometer).                             |

It should be noted that, for a Winter Weather event-day polygon's intersection with a Census block to be included, the area of the intersection must cover more than 5% of the Census block. This is a spatial modeling technique to correct for the small intersect "slivers" generated by differing versions of county boundary geometry being used.

Because the exposure model uses a conservative-case concentration of exposure and a developed area density value, it is possible to mathematically generate an exposure value that is greater than the total value of the Census block. The Hazus-recorded population and building value and the Census of Agriculture-reported crop and livestock value for the Census block are considered ceilings on exposure. For example, if the calculated exposed building value exceeds the Hazus-recorded building value, then the Hazus-recorded building value is used as the building exposure value for the Census block.

### 23.4.1. EXPOSURE AGGREGATION

To calculate exposure at the Census tract level, the exposure values for each Census block within the Census tract are summed. To calculate exposure at the county level, the exposure values for each Census block within the county are summed (see [Equation 149](#)).

**Equation 149: Census Tract and County Winter Weather Exposure Aggregation**

$$Exposure_{WNTW_{CT}Bldg} = \sum_{CB}^{CT} Exposure_{WNTW_{CB}Bldg}$$

$$Exposure_{WNTW_{Co}Bldg} = \sum_{CB}^{Co} Exposure_{WNTW_{CB}Bldg}$$

$$Exposure_{WNTW_{CT}Pop} = \sum_{CB}^{CT} Exposure_{WNTW_{CB}Pop}$$

$$Exposure_{WNTW_{Co}Pop} = \sum_{CB}^{Co} Exposure_{WNTW_{CB}Pop}$$

$$Exposure_{WNTW_{CT}Ag} = \sum_{CB}^{CT} Exposure_{WNTW_{CB}Ag}$$

$$Exposure_{WNTW_{Co}Ag} = \sum_{CB}^{Co} Exposure_{WNTW_{CB}Ag}$$

where:

$Exposure_{WNTW_{CT}Bldg}$  is the building value exposed to Winter Weather event-days in a specific Census tract (in dollars).

$\sum_{CB}^{CT} Exposure_{WNTW_{CB}Bldg}$  is the summed value of all buildings exposed to Winter Weather for each Census block within the Census tract (in dollars).

|                                           |                                                                                                                                         |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| $Exposure_{WNTW_{Co}Bldg}$                | is the building value exposed to Winter Weather event-days in a specific county (in dollars).                                           |
| $\sum_{CB}^{Co} Exposure_{WNTW_{CB}Bldg}$ | is the summed value of all buildings exposed to Winter Weather for each Census block within the county (in dollars).                    |
| $Exposure_{WNTW_{CT}Pop}$                 | is the population equivalence value exposed to Winter Weather event-days in a specific Census tract (in dollars).                       |
| $\sum_{CB}^{CT} Exposure_{WNTW_{CB}Pop}$  | is the summed value of all population equivalence exposed to Winter Weather for each Census block within the Census tract (in dollars). |
| $Exposure_{WNTW_{Co}Pop}$                 | is the population equivalence value exposed to Winter Weather event-days in a specific county (in dollars).                             |
| $\sum_{CB}^{Co} Exposure_{WNTW_{CB}Pop}$  | is the summed value of all population equivalence exposed to Winter Weather for each Census block within the county (in dollars).       |
| $Exposure_{WNTW_{CT}Ag}$                  | is the agriculture value exposed to Winter Weather event-days in a specific Census tract (in dollars).                                  |
| $\sum_{CB}^{CT} Exposure_{WNTW_{CB}Ag}$   | is the summed value of all agriculture areas exposed to Winter Weather for each Census block within the Census tract (in dollars).      |
| $Exposure_{WNTW_{Co}Ag}$                  | is the agriculture value exposed to Winter Weather event-days in a specific county (in dollars).                                        |
| $\sum_{CB}^{Co} Exposure_{WNTW_{CB}Ag}$   | is the summed value of all agriculture areas exposed to Winter Weather for each Census block within the county (in dollars).            |

### 23.5. Winter Weather Annualized Frequency

The number of recorded Winter Weather occurrences, in event-days, each year over the period of record (19.0 years) is used to estimate the annualized frequency of Winter Weather events in an area. Because a Winter Weather event can last over several days or a single day, an event-day basis was used to estimate annualized frequency as this method better captures the variability in duration between occurrences. The annualized frequency is calculated at the Census block level using [Equation 150](#), and the Census block-level value is used in the Expected Annual Loss calculations.

Annualized frequency calculations use the same intersection between Winter Weather event-days (or Winter Storm Date Expansion) polygons and Census block polygons that were used to calculate exposure. The count of distinct Winter Weather event-day polygons intersecting each Census block is recorded and used to calculate the annualized frequency of Winter Weather event-days.

**Equation 150: Census Block Winter Weather Annualized Frequency**

$$Freq_{WNTW_{CB}} = \frac{EventDayCount_{WNTW_{CB}}}{PeriodRecord_{WNTW}}$$

where:

$Freq_{WNTW_{CB}}$  is the annualized frequency of Winter Weather event-days determined for a specific Census block (event-days per year).

$EventDayCount_{WNTW_{CB}}$  is the number of Winter Weather event-days that intersect the Census block.

$PeriodRecord_{WNTW}$  is the period of record for Winter Weather (19.0 years).

**23.5.1. ANNUALIZED FREQUENCY AGGREGATION**

The dataset provides area-weighted average annualized frequency values at both the Census tract and county level. These values may not exactly match that of dividing the number of recorded Winter Weather occurrences at the Census tract and county level by the period of record. The annualized frequency values at the Census block are aggregated to the Census tract and county levels using area-weighted functions as in [Equation 151](#).

**Equation 151: Census Tract and County Area Weighted Winter Weather Annualized Frequency Aggregation**

$$Freq_{WNTW_{CT}} = \frac{\sum_{CB}^{CT} (Freq_{WNTW_{CB}} \times Area_{CB})}{Area_{CT}}$$

$$Freq_{WNTW_{Co}} = \frac{\sum_{CB}^{Co} (Freq_{WNTW_{CB}} \times Area_{CB})}{Area_{Co}}$$

where:

$Freq_{WNTW_{CT}}$  is the area-weighted Winter Weather annualized frequency for a specific Census tract.

$Freq_{WNTW_{CB}}$  is the Winter Weather annualized frequency associated with a specific Census block.

$Area_{CB}$  is the total area of the Census block (in square kilometers).

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

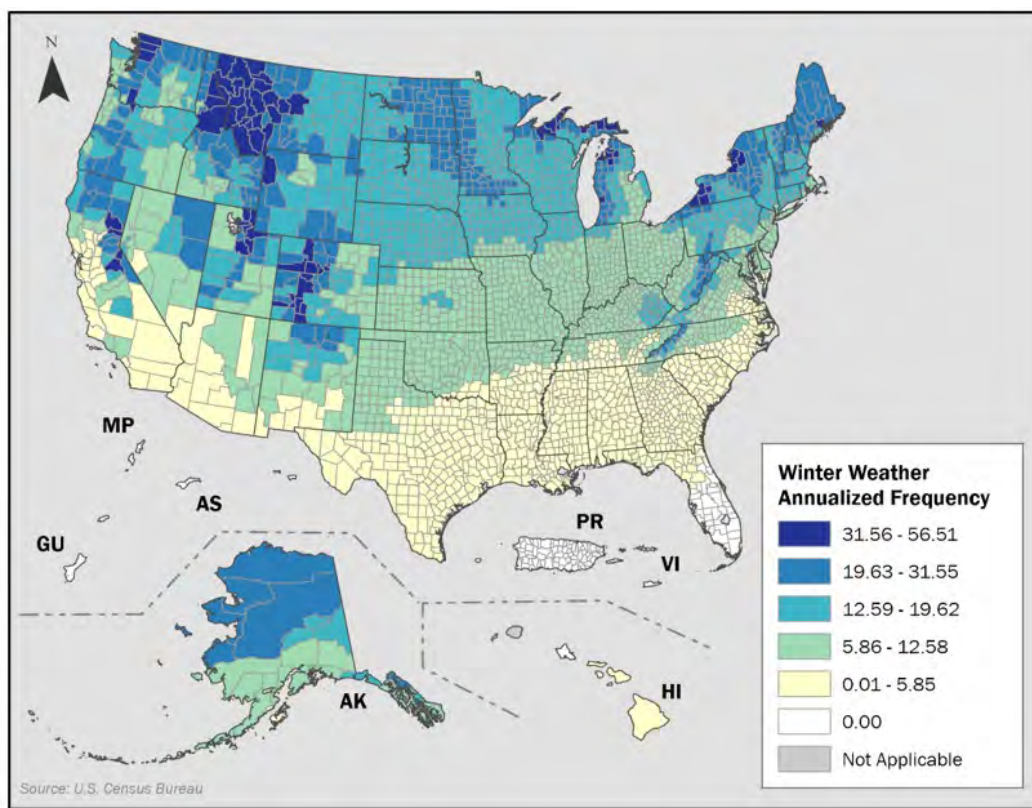
$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$Freq_{WNTW_{Co}}$  is the annualized area-weighted Winter Weather annualized frequency for a specific county.

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

[Figure 125](#) displays Winter Weather annualized frequency at the county level.



**Figure 125: Winter Weather Annualized Frequency by County**

## 23.6. Winter Weather Historic Loss Ratio

The Winter Weather Historic Loss Ratio is the representative percentage of a location's hazard exposure that experiences loss due to a Winter Weather event-day, or the average rate of loss associated with the occurrence of a Winter Weather event-day. For a detailed description of the Historic Loss Ratio calculation process, see [Section 5.4](#). The Historic Loss Ratio parameters described below are specific to the Winter Weather hazard type.

Loss data are provided by SHEL DUS<sup>145</sup> at the county level, so this is the lowest level at which Historic Loss Ratio is calculated. SHEL DUS events from 1996 through 2023 are included in the Historic Loss

<sup>145</sup> For Winter Weather loss information, SHEL DUS compiles data from the monthly Storm Data publication produced by National Oceanic and Atmospheric Administration's National Centers for Environmental Information.

Ratio calculation. Three peril types are mapped to the hazard Winter Weather (see [Table 91](#)). These native records are expanded on an event-day basis (to a maximum of 31 event-days) and aggregated on a single-event-per-day basis (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)).

**Table 91: Winter Weather Peril Types and Recorded Events from 1996-2023**

| <i>Peril Type in SHEL DUS</i> | <i>Total SHEL DUS Loss Records</i> | <i>Total Records per Event Basis</i> |
|-------------------------------|------------------------------------|--------------------------------------|
| Blizzard                      | 1,998                              | 3,797                                |
| Storm-Winter                  | 10,402                             | 18,989                               |
| Winter Weather                | 3,800                              | 4,592                                |

The Historic Loss Ratio exposure value used in the Loss Ratio per Basis calculation is a county-level value that represents the dollar value of the total building value and the entire population of a county as recorded in Hazus 6.1. The Loss Ratio per Basis for each SHEL DUS-documented event-day and each consequence type (building, population, and agriculture) is calculated using [Equation 152](#).

**Equation 152: Loss Ratio per Basis Calculation for a Single Winter Weather Event-Day**

$$LRB_{WNTW Co CnsqType} = \frac{Loss_{WNTW Co CnsqType}}{HLRExposure_{Co CnsqType}}$$

where:

$LRB_{WNTW Co CnsqType}$  is the Loss Ratio per Basis representing the ratio of loss to exposure experienced by a specific county due to the occurrence of a specific Winter Weather event-day. Calculation is performed for each consequence type (building, population, and agriculture).

$Loss_{WNTW Co CnsqType}$  is the loss (by consequence type) experienced from the Winter Weather event-day documented to have occurred in the county (in dollars or impacted people).

$HLRExposure_{Co CnsqType}$  is the total value (by consequence type) of the county estimated to have been exposed to the Winter Weather event-day (in dollars or people).

Winter Weather event-days can occur with a high frequency in areas, but often result in no recorded loss to buildings, population, or agriculture. SHEL DUS does not record events in which no loss occurred, so a number of zero-loss event-day records are inserted into the loss data to align the event-day count in the Historic Loss Ratio calculation to the historical event-day count within the SHEL DUS period of record (1996 through 2023). For Winter Weather, the historical event-day count is extracted using the intersection between the Winter Weather event-day polygons for the years 2005-2024 and the Census block polygons used to calculate exposure and annualized frequency



(see [Table 89](#)). An annual rate is calculated as the event-day count divided by the period of record of 19.0 years, and this rate is multiplied by the SHELDUS period of record of 28 years to estimate a historical event-day count for the appropriate time range.

If the number of loss-causing Winter Weather event-day records from SHELDUS is less than the scaled event-day count for the county, then a number of zero-loss records equal to the difference are inserted into the Loss Ratio per Basis table with zero values for the consequence ratios.

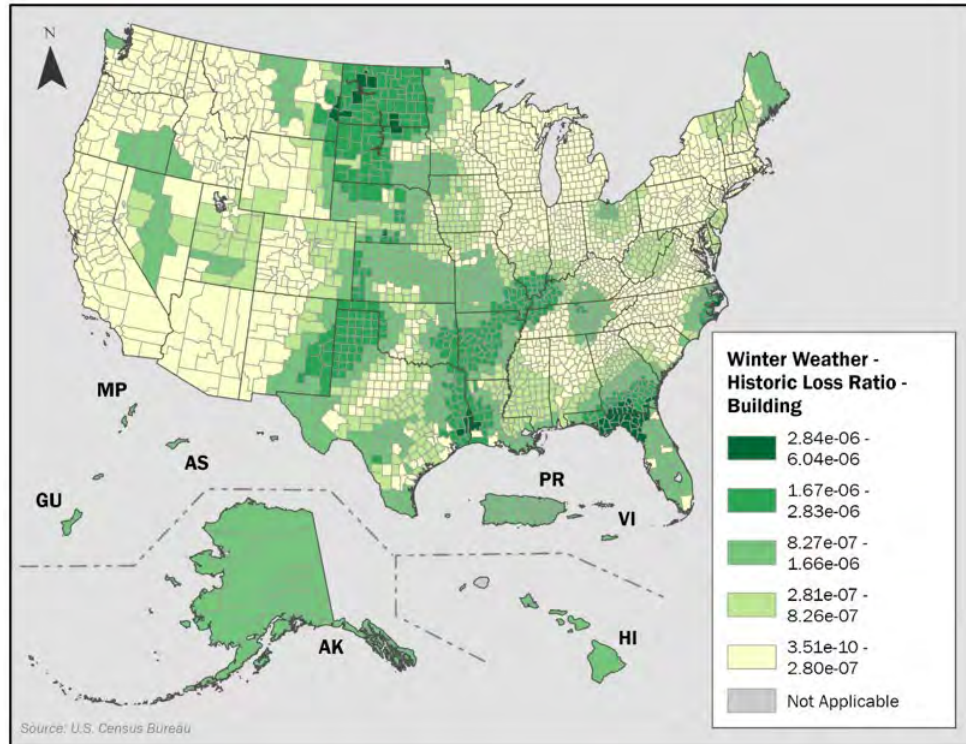
After the Loss Ratios per Basis are calculated for all Winter Weather events, various sets of Loss Ratios per Basis are created for use in the Historic Loss Ratio Bayesian credibility process. Loss Ratio per Basis Sets are defined by consequence type, Bayesian level, the possible use of a National Center for Health Statistics Urban-Rural Code filter and usage threshold settings. Each of these settings can be thought of as a condition for the inclusion of a specific Loss Ratio per Basis record into the set and then the criteria for determining if the set should be used for a specific county. The following table displays the Loss Ratio per Basis Sets used in creating the Historic Loss Ratios for Winter Weather.

**Table 92: Winter Weather Historic Loss Ratio Bayesian Weighting Level Sets**

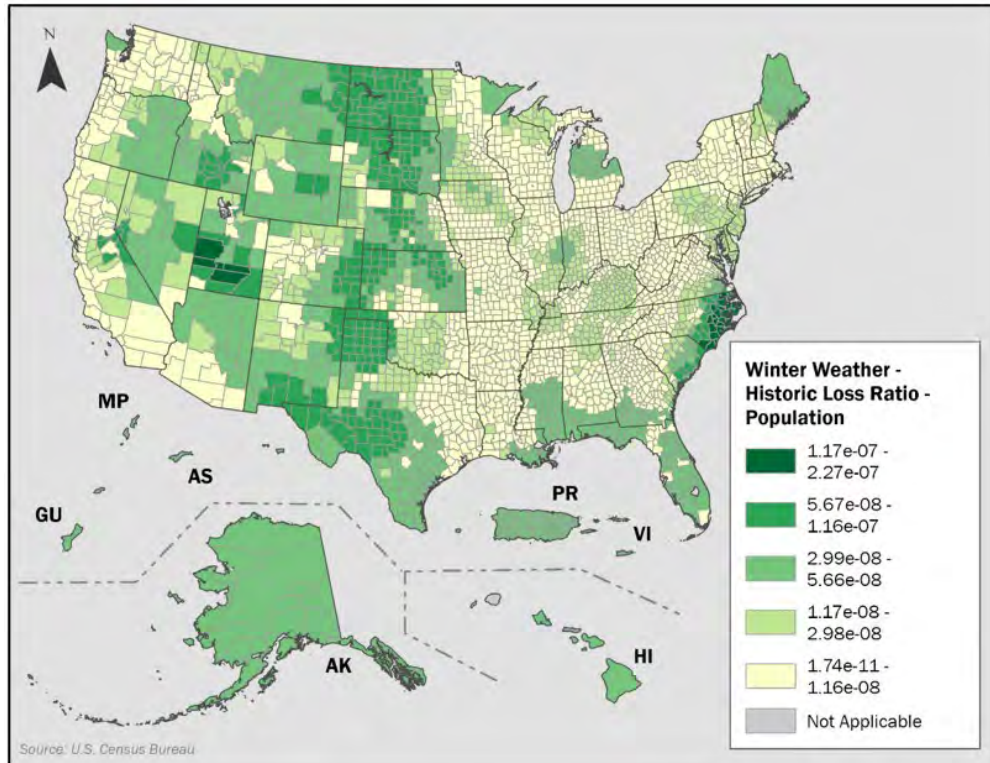
| <i>Consequence Type</i> | <i>Set ID</i> | <i>Bayesian Level</i>                           | <i>Set Refinement Filter</i>                        | <i>Set Utilization Threshold<br/>(Set Must Have)</i> |
|-------------------------|---------------|-------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| Agriculture             | A1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                      |
|                         | A2            | Neighbor Counties (Touch & 115km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                      |
|                         | A3            | Neighbor Counties (Touch & 208km)               | N/A                                                 | 25 Loss Ratio per Basis records                      |
|                         | A4            | National Risk Index Historic Loss Ratio Regions | N/A                                                 | N/A                                                  |
|                         | A5            | National                                        | N/A                                                 | N/A                                                  |
| Population              | P1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                      |
|                         | P2            | Neighbor Counties (Touch & 115km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                      |
|                         | P3            | Neighbor Counties (Touch & 208km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                      |
|                         | P4            | National Risk Index Historic Loss Ratio Regions | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                                  |
|                         | P5            | National                                        | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                                  |

| <b>Consequence Type</b> | <b>Set ID</b> | <b>Bayesian Level</b>                           | <b>Set Refinement Filter</b>                        | <b>Set Utilization Threshold<br/>(Set Must Have)</b> |
|-------------------------|---------------|-------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| Building                | B1            | County                                          | N/A                                                 | 25 Loss Ratio per Basis records                      |
|                         | B2            | Neighbor Counties (Touch & 115km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                      |
|                         | B3            | Neighbor Counties (Touch & 208km)               | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | 25 Loss Ratio per Basis records                      |
|                         | B4            | National Risk Index Historic Loss Ratio Regions | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                                  |
|                         | B5            | National                                        | Large Central Metropolitan Areas (LCMAs), Non-LCMAs | N/A                                                  |

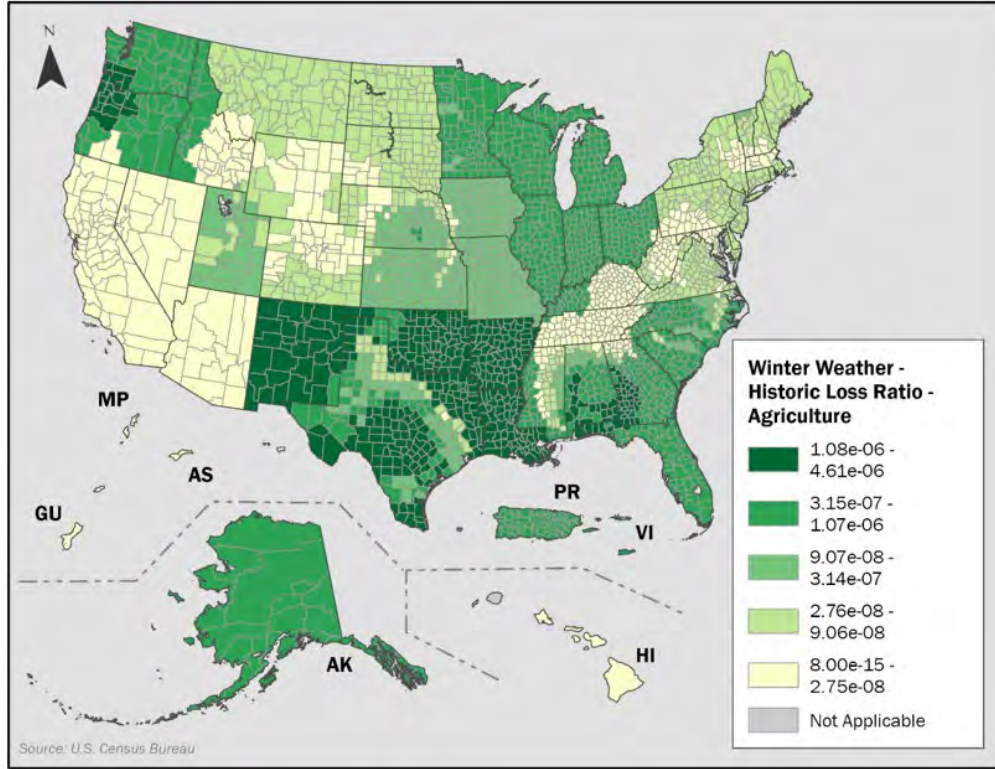
Loss Ratio per Basis Set statistics are then used to develop the Bayesian credibility weighting factors that are used to generate the Bayesian-adjusted county-level Historic Loss Ratio values. Back testing and smoothing techniques are then implemented to further refine the Historic Loss Ratio values for Winter Weather (see [Section 5.4.4 Historic Loss Ratio Calculation Methodology](#)). [Figure 126](#), [Figure 127](#), and [Figure 128](#) represent the final, Bayesian-adjusted county-level Historic Loss Ratio values for Winter Weather. The resulting Bayesian-adjusted Historic Loss Ratio by consequence type is then inherited by the Census blocks and Census tracts within the parent county.



**Figure 126: Winter Weather Bayesian-Adjusted Historic Loss Ratio – Building Value**



**Figure 127: Winter Weather Bayesian-Adjusted Historic Loss Ratio – Population**



**Figure 128: Winter Weather Bayesian-Adjusted Historic Loss Ratio – Agriculture Value**

### 23.7. Winter Weather Expected Annual Loss

Once exposure, annualized frequency, and Historic Loss Ratio have been calculated, the Expected Annual Loss is computed at the Census block level as in [Equation 153](#). Performing the base calculations once at the Census block level and aggregating up allows for the most detailed and accurate estimation of Expected Annual Loss at higher levels.

**Equation 153: Census Block Expected Annual Loss to Winter Weather**

$$EAL_{WNTW_{CB_{Bldg}}} = Exposure_{WNTW_{CB_{Bldg}}} \times Freq_{WNTW_{CB}} \times HLR_{WNTW_{CB_{Bldg}}}$$

$$EAL_{WNTW_{CB_{Pop}}} = Exposure_{WNTW_{CB_{Pop}}} \times Freq_{WNTW_{CB}} \times HLR_{WNTW_{CB_{Pop}}}$$

$$EAL_{WNTW_{CB_{Ag}}} = Exposure_{WNTW_{CB_{Ag}}} \times Freq_{WNTW_{CB}} \times HLR_{WNTW_{CB_{Ag}}}$$

where:

$EAL_{WNTW_{CB_{Bldg}}}$  is the building Expected Annual Loss due to Winter Weather occurrences for a specific Census block (in dollars).

$Exposure_{WNTW_{CB_{Bldg}}}$  is the building value exposed to Winter Weather occurrences in the Census block (in dollars).

|                           |                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $Freq_{WNTW_{CB}}$        | is the Winter Weather annualized frequency for the Census block (event-days per year).                                         |
| $HLR_{WNTW_{CB}Bldg}$     | is the Bayesian-adjusted building Historic Loss Ratio for Winter Weather for the Census block.                                 |
| $EAL_{WNTW_{CB}Pop}$      | is the population equivalence Expected Annual Loss due to Winter Weather occurrences for a specific Census block (in dollars). |
| $Exposure_{WNTW_{CB}Pop}$ | is the population equivalence value exposed to Winter Weather occurrences in the Census block (in dollars).                    |
| $HLR_{WNTW_{CB}Pop}$      | is the Bayesian-adjusted population Historic Loss Ratio for Winter Weather or the Census block.                                |
| $EAL_{WNTW_{CB}Ag}$       | is the agriculture Expected Annual Loss due to Winter Weather occurrences for a specific Census block (in dollars).            |
| $Exposure_{WNTW_{CB}Ag}$  | is the agriculture value exposed to Winter Weather occurrences in the Census block (in dollars).                               |
| $HLR_{WNTW_{CB}Ag}$       | is the Bayesian-adjusted agriculture Historic Loss Ratio for Winter Weather for the Census block.                              |

The total Expected Annual Loss values at the Census tract and county level are the sums of the aggregated building, population equivalence, and agriculture Expected Annual Loss values at the Census block level as in [Equation 154](#).

**Equation 154: Census Tract and County Expected Annual Loss to Winter Weather**

$$EAL_{WNTW_{CT}} = \sum_{CB}^{CT} EAL_{WNTW_{CB}Bldg} + \sum_{CB}^{CT} EAL_{WNTW_{CB}Pop} + \sum_{CB}^{CT} EAL_{WNTW_{CB}Ag}$$

$$EAL_{WNTW_{Co}} = \sum_{CB}^{Co} EAL_{WNTW_{CB}Bldg} + \sum_{CB}^{Co} EAL_{WNTW_{CB}Pop} + \sum_{CB}^{Co} EAL_{WNTW_{CB}Ag}$$

where:

|                                      |                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $EAL_{WNTW_{CT}}$                    | is the total Expected Annual Loss due to Winter Weather occurrences for a specific Census tract (in dollars).                                       |
| $\sum_{CB}^{CT} EAL_{WNTW_{CB}Bldg}$ | is the summed building Expected Annual Loss due to Winter Weather occurrences for all Census blocks in the Census tract (in dollars).               |
| $\sum_{CB}^{CT} EAL_{WNTW_{CB}Pop}$  | is the summed population equivalence Expected Annual Loss due to Winter Weather occurrences for all Census blocks in the Census tract (in dollars). |



$\sum_{CB}^{CT} EAL_{WNTW_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss to Winter Weather occurrence for all Census blocks in the Census tract (in dollars).

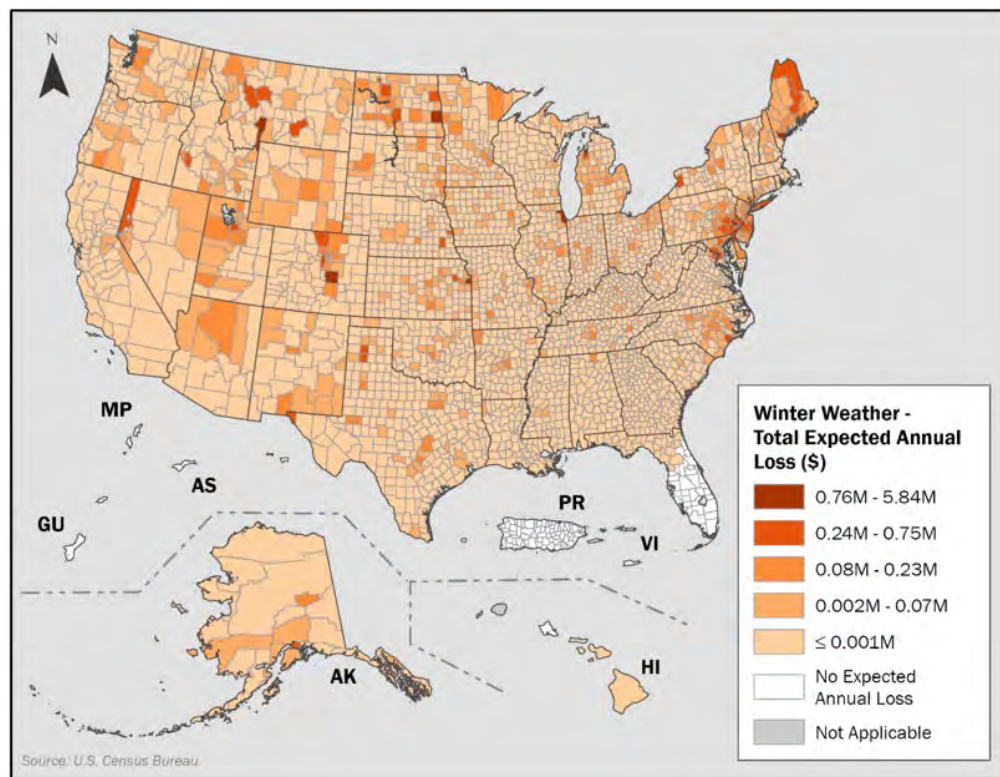
$EAL_{WNTW_{Co}}$  is the total Expected Annual Loss due to Winter Weather occurrences for a specific county (in dollars).

$\sum_{CB}^{Co} EAL_{WNTW_{CB_{Bldg}}}$  is the summed building Expected Annual Loss due to Winter Weather occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{WNTW_{CB_{Pop}}}$  is the summed population equivalence Expected Annual Loss due to Winter Weather occurrences for all Census blocks in the county (in dollars).

$\sum_{CB}^{Co} EAL_{WNTW_{CB_{Ag}}}$  is the summed agriculture Expected Annual Loss due to Winter Weather occurrences for all Census blocks in the county (in dollars).

[Figure 129](#) shows the total Expected Annual Loss (building, population equivalence, and agriculture value combined) to Winter Weather occurrences.



**Figure 129: Total Expected Annual Loss by County to Winter Weather**

With the Winter Weather total Expected Annual Loss Value computed for each Census tract and county, the companion Expected Annual Loss Score is computed (see [Section 3.2 Values, Scores and Ratings](#)). The Expected Annual Loss Score represents a community's national percentile ranking relative to all communities at the same level.



Building Expected Annual Loss Rate is calculated by dividing the Winter Weather Expected Annual Loss for building by the total building value of the community. Population Expected Annual Loss Rate is calculated by dividing the Winter Weather Expected Annual Loss for population by the total population of the community. Agriculture Expected Annual Loss Rate is calculated by Winter Weather the Wildfire Expected Annual Loss for agriculture by the total agriculture value of the community.

## 23.8. Winter Weather Methodology Discussion

This section briefly presents some of the limitations of the current methodology and data, implications of the current methodology and datasets, and considerations for using the data provided to the user. See [Section 3.3 Assumptions and Limitations](#) for non-hazard-specific assumptions and limitations.

- National Weather Service Warnings and Advisories are issued at the discretion of each individual National Weather Service Forecast Office, based on regionally determined criteria. Though this accounts for varying capacities across the country, it results in an inconsistent record of dangerous winter weather events.
- The data development team made several decisions regarding the delineation between what is considered Winter Weather versus other cold-based hazards. These have been outlined in this Technical Document.

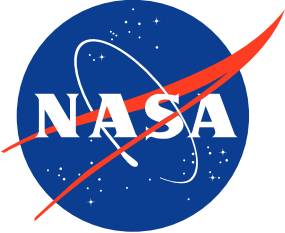
## Appendix A – Contributors

Multiple entities contributed to the development of the data by providing domain expertise and/or data.

| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                            | <i>Expertise / Source Data</i>                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|    | Agency for Toxic Substances and Disease Registry is a federal public health agency of the United States Department of Health and Human Services. Agency for Toxic Substances and Disease Registry protects communities from harmful health effects related to exposure to natural and man-made hazardous substances.                                                                          | Social Vulnerability Source Data                                   |
|    | The mission of the State of Alaska Division of Homeland Security and Emergency Management's Disaster Assistance Section is to coordinate state and federal actions with local jurisdictions to assist affected communities in responding to, recovering from, and mitigating against major disasters or emergencies declared by the Governor of Alaska or the President of the United States. | Tsunami Source Data                                                |
|   | The Geologic Hazards Section collects, analyzes, and compiles geologic data useful for hazard risk-mitigation purposes. Studies of major geologic hazards such as earthquakes, landslides, active faults, and tsunamis result in reports outlining potential hazards in susceptible areas.                                                                                                    | Tsunami Source Data                                                |
|  | Argonne National Laboratory is a multidisciplinary science and engineering research center that seeks to answer the biggest questions facing humanity, from how to obtain affordable clean energy to protect ourselves and our environment.                                                                                                                                                   | Natural Hazards Expertise                                          |
|  | The Center for Emergency Management and Homeland Security at Arizona State University is a university-wide interdisciplinary hub for the research and practice of emergency management and homeland security.                                                                                                                                                                                 | Expected Annual Loss Expertise;<br>Historic Loss Ratio Source Data |
|  | The California Department of Conservation administers a variety of programs vital to California's public safety, environment, and economy. Its services are designed to balance today's needs with tomorrow's obligations by fostering the wise use and conservation of energy, land, and mineral resources.                                                                                  | Tsunami Source Data                                                |

| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                              | <i>Expertise / Source Data</i>                          |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|    | The mission of the California Geological Survey is to provide scientific products and services about the state's geology, seismology, and mineral resources, including their related hazards that affect the health, safety, and business interests of the people of California.                                                                                                                | Tsunami Source Data                                     |
|    | The California Office of Emergency Services takes a proactive approach to addressing the risks, threats, and vulnerabilities of California's people, property, economy, and environment.                                                                                                                                                                                                        | Tsunami Source Data                                     |
|    | The Centers for Disease Control and Prevention is the nation's leading science-based, data-driven, service organization that protects the public's health. For more than 70 years, they have put science into action to help children stay healthy so they can grow and learn; to help families, businesses, and communities fight disease and stay strong; and to protect the public's health. | Social Vulnerability Source Data                        |
|   | City of Augusta, Georgia                                                                                                                                                                                                                                                                                                                                                                        | Natural Hazards Expertise                               |
|  | The Colorado Avalanche Information Center is a part of the Colorado Geological Survey, provides avalanche-safety classes and issues forecasts of avalanche and mountain weather conditions, and maintains an inventory of all avalanche related fatalities for the US.                                                                                                                          | Avalanche Source Data                                   |
|  | The Community and Regional Resilience Institute is an organization that assists communities across the nation with understanding their strengths and vulnerabilities, taking positive collective actions to limit the impact of disruptive crises, and providing guidance to communities recovering from disasters.                                                                             | Community Resilience and Social Vulnerability Expertise |
|  | The Federal Alliance for Safe Homes (FLASH) is a consumer advocate that promotes life safety, property protection, and resiliency by empowering the community with knowledge and resources for strengthening homes and safeguarding families from natural and manmade disasters.                                                                                                                | Natural Hazards Expertise                               |

| <b>Contributor</b>                                                                  | <b>Description</b>                                                                                                                                                                                                                                                                                                                                             | <b>Expertise / Source Data</b>                                                                    |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|    | The Federal Emergency Management Agency is a federal agency responsible for helping people before, during, and after disasters.                                                                                                                                                                                                                                | Natural Hazards Expertise; Coastal Flooding; Earthquake; Exposure and Inland Flooding Source Data |
|    | Georgetown University is a private research university located in Washington, the District of Columbia                                                                                                                                                                                                                                                         | Social Vulnerability Expertise                                                                    |
|    | Harvard University is a private Ivy League research university in Cambridge, Massachusetts.                                                                                                                                                                                                                                                                    | Social Vulnerability Expertise                                                                    |
|   | The Humanitarian Data Exchange, managed by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA)'s Centre for Humanitarian Data, is an open platform for sharing data across crises and organizations. The goal of the Humanitarian Data Exchange is to make humanitarian data easy to find and use for analysis.                      | Volcanic Activity Source Data                                                                     |
|  | Idaho National Laboratory is one of the national laboratories of the United States Department of Energy.                                                                                                                                                                                                                                                       | Community Resilience and Social Vulnerability Expertise                                           |
|  | The Insurance Institute for Business & Home Safety is an independent, nonprofit, scientific research and communications organization of property insurers and reinsurers that conducts objective research to identify and promote the most effective ways to strengthen homes, businesses, and communities against natural disasters and other causes of loss. | Natural Hazards Expertise                                                                         |
|  | The Iowa Environmental Mesonet of Iowa State University collects environmental data from cooperating members with observing networks, and stores and makes the data publicly available.                                                                                                                                                                        | Cold Wave, Heat Wave, Winter Weather Source Data                                                  |
|  | Louisiana State University is a public research university located in Baton Rouge, Louisiana.                                                                                                                                                                                                                                                                  | Natural Hazards Expertise                                                                         |

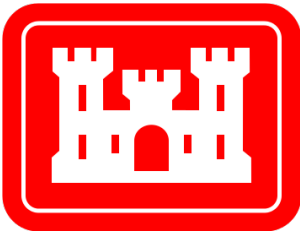
| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Expertise / Source Data</i>           |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|    | The Multi-Resolution Land Characteristics (MRLC) consortium is a group of federal agencies who coordinate and generate consistent and relevant land cover information at the national scale for a wide variety of environmental, land management, and modeling applications.                                                                                                                                                                                                                              | Land Cover Source Data                   |
|    | The National Alliance for Public Safety GIS (NAPSG) Foundation is a 501(c)(3) nonprofit organization that was formed in 2005 to overcome the challenges faced by Federal, tribal, state, and local public safety agencies in the adoption and use of GIS as a tool to protect their citizens.                                                                                                                                                                                                             | Data and Methods Expertise               |
|    | The Jet Propulsion Laboratory is a federally-funded research and development center managed by Caltech with an array of mission areas, including scientific data collection.                                                                                                                                                                                                                                                                                                                              | Natural Hazard Expertise and Source Data |
|  | The National Earthquake Hazards Reduction Program was established by the United States Congress to reduce the risks of life and property from future earthquakes in the United States through the establishment and maintenance of an effective earthquake hazards reduction program. Four primary agencies contribute to the program's mitigation efforts: Federal Emergency Management Agency, National Institute of Standards and Technology, National Science Foundation, and U.S. Geological Survey. | Earthquake Source Data                   |
|  | The National Integrated Drought Information System is a multi-agency partnership that coordinates drought monitoring, forecasting, planning, and information at federal, tribal, state, and local levels across the country.                                                                                                                                                                                                                                                                              | Drought Source Data                      |

| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                 | <i>Expertise / Source Data</i>                             |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|    | <p>The National Integrated Heat Health Information System builds societal understanding of heat risks, develops science-based solutions, improves capacity, communication, and decision-making to reduce heat-related illness and death.</p>                                                                                                       | <p>Expertise on Heat Wave</p>                              |
|    | <p>The National Institute of Standards and Technology is a physical science and measurement standards laboratory with programs in nanoscale science and technology, engineering, information technology, neutron research, material measurement, and physical measurement, and a mission to promote innovation and industrial competitiveness.</p> | <p>Natural Hazards Expertise</p>                           |
|   | <p>The National Centers for Environmental Information is responsible for preserving, monitoring, assessing, and providing public access to the nation's largest archive of climate and historical weather data and information. It provides over 25 petabytes of comprehensive atmospheric, coastal, oceanic, and geophysical data.</p>            | <p>Lightning, Inland Flooding, and Tsunami Source Data</p> |
|  | <p>The mission of the National Hurricane Center is to save lives, mitigate property loss, and improve economic efficiency by issuing the best watches, warnings, forecasts, and analyses of hazardous tropical weather and by increasing understanding of these hazards.</p>                                                                       | <p>Coastal Flooding, Hurricane Source Data</p>             |
|  | <p>The mission of the National Weather Service is to provide weather, water, and climate data, forecasts, and warnings for the protection of life and property and enhancement of the national economy.</p>                                                                                                                                        | <p>Cold Wave, Heat Wave, Winter Weather Source Data</p>    |
|  | <p>The Office for Coastal Management provides access to the science and environmental intelligence communities need to identify the best ways to address storm preparedness, erosion, development, habitat loss, sea level rise, public access, and threats to water quality.</p>                                                                  | <p>Coastal Flooding Source Data</p>                        |

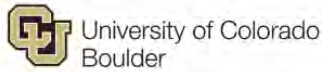
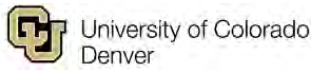


| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Expertise / Source Data</i>              |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|    | <p>The National Oceanic and Atmospheric Administration's Pacific Marine Environmental Laboratory conducts investigations and observations related to oceanic and marine science. These observations are used to improve weather and climate predictions, fisheries management, and coastal resilience. Research areas at Pacific Marine Environmental Laboratory include ocean acidification, tsunami detection and forecasting, air-sea interactions, hydrothermal vent systems, fisheries oceanography, and long-term climate monitoring and analysis.</p> | Tsunami Source Data                         |
|    | <p>The mission of the Storm Prediction Center is to use innovative science and technology to deliver timely and accurate watch and forecast products/information dealing with tornadoes, severe thunderstorms, lightning, wildfires, and winter weather for the United States to protect lives and property.</p>                                                                                                                                                                                                                                             | Hail, Strong Wind, Tornado Source Data      |
|   | <p>A part of the National Weather Service's Tsunami Program, the National Tsunami Hazard Mitigation Program is a coordinated United States national effort to mitigate the impact of tsunamis through public education, community response planning, hazard assessment, and warning coordination.</p>                                                                                                                                                                                                                                                        | Tsunami Source Data                         |
|  | <p>The Natural Hazard Mitigation Association is a professional association that promotes reducing the risk and consequences of natural hazard events with a special emphasis on protecting the most vulnerable populations in our communities.</p>                                                                                                                                                                                                                                                                                                           | Natural Hazards Expertise                   |
|  | <p>The New York State Homeland Security and Emergency Services department provides leadership, coordination, and support for efforts to prevent, protect against, prepare for, respond to, and recover from terrorism and other man-made and natural disasters, threats, fires, and other emergencies.</p>                                                                                                                                                                                                                                                   | How the Data can Help Expertise             |
|  | <p>Old Dominion University is a public research university in Norfolk, Virginia.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Data and Methods; Natural Hazards Expertise |

| <b>Contributor</b>                                                                                                                                                           | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                       | <b>Expertise / Source Data</b>                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|                                                                                             | The Oregon Department of Geology and Mineral Resources seeks to increase understanding of Oregon's geologic resources and hazards through science and stewardship.                                                                                                                                                                                                                       | Tsunami Source Data                                 |
|                                                                                             | The Pacific Disaster Center is an applied science, information, and technology center working to reduce disaster risks and impacts on life, property, and economies worldwide.                                                                                                                                                                                                           | Natural Hazards Expertise                           |
|                                                                                             | The Puerto Rico Seismic Network is the regional authority for monitoring earthquakes and tsunamis in Puerto Rico and the Virgin Islands.                                                                                                                                                                                                                                                 | Earthquake Expertise                                |
|  Smithsonian Institution<br>National Museum of Natural History<br>Global Volcanism Program | The mission of the Smithsonian Institution's Global Volcanism Program is to document, understand, and disseminate information about global volcanic activity.                                                                                                                                                                                                                            | Volcanic Activity Source Data                       |
|                                                                                           | The Southern Climate Impacts Planning Program assists organizations with decision making that builds resilience by collaboratively producing research, tools, and knowledge that reduce weather and climate risks and impacts across the South-Central United States.                                                                                                                    | Natural Hazards Expertise                           |
|                                                                                           | The Hawai'i Emergency Management Agency plans for and responds to both natural and human-caused emergencies. These include emergencies resulting from all hazards, from tsunamis, wildfires, and hurricanes to incidents involving hazardous materials or nuclear power.                                                                                                                 | Tsunami Source Data                                 |
|                                                                                           | The mission of the Albany Visualization and Informatics Lab at State University of New York at Albany is to use the latest technology to solve modern transportation problems. Using a modern, web-based, and extensible visualization platform, AVAIL seeks to explore the interaction of current planning and research procedures through the use of visual analytics and informatics. | Data and Methods;<br>Expected Annual Loss Expertise |

| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Expertise / Source Data</i>                          |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|    | Texas A&M University is a public research university in College Station, Texas.                                                                                                                                                                                                                                                                                                                                                                                                                                  | Social Vulnerability Expertise                          |
|    | The Polis Center at Indiana University-Purdue University Indianapolis is a collaborative, applied research center that specializes in community-based research and analysis, and advanced information technologies to build understanding of community issues from a variety of perspectives.                                                                                                                                                                                                                    | Community Resilience and Social Vulnerability Expertise |
|   | The Cold Regions Research and Engineering Laboratory solves interdisciplinary, strategically important problems for the Corps of Engineers, Army, Department of Defense, and the nation. The Cold Regions Research and Engineering Laboratory discovers, develops, and delivers advanced and applied science and engineering to complex environments, materials, and processes in all seasons and climates.                                                                                                      | Ice Storm Source Data                                   |
|  | The United States Census Bureau is a principal agency of the United States Federal Statistical System, responsible for producing data about the American people and economy.                                                                                                                                                                                                                                                                                                                                     | Data and Methods; Social Vulnerability Expertise        |
|  | The United States Department of Agriculture provides economic opportunity through innovation, helping rural America to thrive; to promote agriculture production; and to preserve our Nation's natural resources. The United States Department of Agriculture's Forest Service has been managing wildland fire on national forests and grasslands for more than 100 years. The agency works alongside state and local partners to protect people, communities, and resources across the entire shared landscape. | Wildfire Expertise                                      |

| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Expertise / Source Data</i>                                             |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|    | <p>The United States Department of Agriculture's National Agricultural Statistics Service conducts hundreds of surveys every year and prepares reports covering virtually every aspect of United States agriculture. Production and supplies of food and fiber, prices paid and received by farmers, farm labor and wages, farm finances, chemical use, and changes in the demographics of United States producers are only a few examples. The National Agricultural Statistics Service is committed to providing timely, accurate, and useful statistics in service to United States agriculture.</p> | <p>Exposure Source Data</p>                                                |
|    | <p>The United States Environmental Protection Agency works to protect human health and the environment, and ensure that Americans have clean air, land, and water in order to reduce environmental risks based on the best available scientific information. To accomplish this mission, the Environmental Protection Agency develops and enforces environmental regulations, gives grants, studies environmental issues, sponsors partnerships, publishes information, and teaches people about the environment.</p>                                                                                   | <p>Social Vulnerability Expertise</p>                                      |
|   | <p>The Fire Modeling Institute's Missoula Fire Sciences Lab has a national charter to conduct fundamental and applied research relating to wildland fire processes, terrestrial and atmospheric effects of fire, and ecological adaptations to fire. It also develops associated tools and applications for scientists and managers.</p>                                                                                                                                                                                                                                                                | <p>Wildfire Source Data</p>                                                |
|  | <p>The National Avalanche Center provides program guidance and support to Forest Service avalanche centers and military artillery programs, as well as field support and the transfer of information and technology.</p>                                                                                                                                                                                                                                                                                                                                                                                | <p>Avalanche Source Data</p>                                               |
|  | <p>The United States Geological Service is a federal agency that provides new scientific methods and tools to enable timely, relevant, and useful information about the Earth and its processes.</p>                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Natural Hazards Expertise;<br/>Landslide and Earthquake Source Data</p> |

| <b>Contributor</b>                                                                  | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Expertise / Source Data</b>                                    |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|    | The Earthquake Hazards Program's role is to provide earth sciences information and products for earthquake loss reduction. The goals of the program are to improve earthquake hazard identification and risk assessment methods and their use, maintain and improve comprehensive earthquake monitoring in the United States, and improve the understanding of earthquake occurrences and their effects and consequences. | Data and Methods and Earthquake Expertise; Earthquake Source Data |
|    | The Landslide Hazards Program has the primary objective of reducing long-term losses from landslide hazards by improving our understanding of the causes of ground failure and suggesting mitigation strategies.                                                                                                                                                                                                          | Earthquake and Landslide Source Data                              |
|    | The United Nations Office for Disaster Risk Reduction is an organizational unit of the United Nations Secretariat that serves as the focal point in the United Nations system for the coordination of disaster reduction and to ensure synergies among disaster reduction activities.                                                                                                                                     | Volcanic Activity Source Data                                     |
|   | The Alaska Earthquake Center at the University of Alaska – Fairbanks is dedicated to reducing the impacts of earthquakes, tsunamis, and volcanic eruptions in Alaska. It provides definitive earthquake information to the public, emergency managers, scientists, and engineers.                                                                                                                                         | Tsunami Source Data                                               |
|  | The University of Central Florida is a public research university located in Orlando, Florida.                                                                                                                                                                                                                                                                                                                            | Expected Annual Loss and Natural Hazards Expertise                |
|  | The University of Colorado – Boulder is a public research university located in Boulder, Colorado.                                                                                                                                                                                                                                                                                                                        | Community Resilience and Social Vulnerability Expertise           |
|  | The University of Colorado – Denver is a public research university located in Denver, Colorado.                                                                                                                                                                                                                                                                                                                          | Natural Hazards Expertise                                         |
|  | University of Hawai'i at Manoa Pacific Islands Ocean Observing System empowers ocean users and stakeholders in the Pacific Islands by providing accurate and reliable coastal and ocean information, tools, and services that are easy to access and use.                                                                                                                                                                 | Tsunami Source Data                                               |



| <i>Contributor</i>                                                                                                                        | <i>Description</i>                                                                                                                                                                                                                                                                                                                      | <i>Expertise / Source Data</i>                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
|                                                          | The University of Idaho is a public research university located in Moscow, Idaho.                                                                                                                                                                                                                                                       | Natural Hazards Expertise                                               |
|                                                          | The Wind Engineering Laboratory at the University of Illinois at Urbana-Champaign is a research laboratory that focuses on developing techniques and physical resources necessary to extend current understanding of windstorm hazards and their impacts on structures.                                                                 | Natural Hazards Expertise                                               |
|                                                          | The University of Iowa is a public research university in Iowa City, Iowa.                                                                                                                                                                                                                                                              | Social Vulnerability Expertise                                          |
|                                                          | The University of Michigan is a public research university located in Ann Arbor, Michigan.                                                                                                                                                                                                                                              | Natural Hazards Expertise                                               |
|                                                         | The University of Missouri is a public research university located in Columbia, Missouri.                                                                                                                                                                                                                                               | Community Resilience Expertise; Social Vulnerability Expertise          |
|                                                        | The National Drought Mitigation Center at the University of Nebraska-Lincoln helps people and institutions develop and implement measures to reduce societal vulnerability to drought, stressing preparedness and risk management rather than crisis management.                                                                        | Drought Source Data                                                     |
|  THE UNIVERSITY<br>of NORTH CAROLINA<br>at CHAPEL HILL | The University of North Carolina at Chapel Hill is a public research university located in Chapel Hill, North Carolina.                                                                                                                                                                                                                 | Community Resilience and Social Vulnerability Expertise                 |
|                                                        | The Hazards Vulnerability & Resilience Institute at the University of South Carolina is an interdisciplinary research and graduate/undergraduate training center focused on the development of theory, data, metrics, methods, applications, and spatial analytical models for understanding the field of hazard vulnerability science. | Community Resilience and Social Vulnerability Expertise and Source Data |



| <i>Contributor</i>                                                                  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                      | <i>Expertise / Source Data</i>                          |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|    | The Tsunami Research Center of the University of Southern California is actively involved with all aspects of tsunami research, including field surveys, numerical and analytical modeling, and hazard assessment, mitigation, and planning.                                                                                                                                                            | Tsunami Source Data                                     |
|    | The University of Texas Arlington is a public research university in Arlington, Texas.                                                                                                                                                                                                                                                                                                                  | Social Vulnerability Expertise                          |
|    | The Space Science and Engineering Center at the University of Wisconsin-Madison develops and utilizes space-, aircraft-, and ground-based instrumentation to collect and analyze observations of the Earth's atmosphere, oceans, land surface, and other planetary atmospheres to improve our understanding of weather, climate, and atmospheric processes.                                             | Natural Hazards Expertise                               |
|   | The Urban Institute is a nonprofit research organization that provides unbiased, authoritative insights to inform consequential choices about the well-being of people and places in the United States. Their experts diagnose current challenges and look ahead to identify opportunities for change and help stakeholders develop solutions and strategies to address concerns and remove roadblocks. | Community Resilience and Social Vulnerability Expertise |
|  | Virgin Islands Territorial Emergency Management Agency is the territorial agency with the primary responsibility for ensuring the territory's resilience to disasters including tsunamis.                                                                                                                                                                                                               | Tsunami Source Data                                     |
|  | Virginia Tech University is a public research university with a main campus in Blacksburg, Virginia.                                                                                                                                                                                                                                                                                                    | Social Vulnerability Expertise                          |
|  | The mission of the Washington State Department of Natural Resources is to manage, sustain, and protect the health and productivity of Washington's lands and waters to meet the needs of present and future generations.                                                                                                                                                                                | Tsunami Source Data                                     |

| <b>Contributor</b>                                                                | <b>Description</b>                                                                                                                                                                                                                                                  | <b>Expertise / Source Data</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
|  | During state emergencies, Washington State's Emergency Management Division manages the State Emergency Operations Center located on Camp Murray, near Tacoma, and coordinates the response to ensure help is provided to those who need it quickly and effectively. | Tsunami Source Data            |

This document contains references and links to non-federal resources and organizations. This information is meant solely for informational purposes and is not intended to be an endorsement of any nonfederal entity by the Federal Emergency Management Agency, United States Department of Homeland Security, or the United States government.

## Appendix B – Hazard Data Characteristics Comparison

**Table 93: Avalanche Hazard Characteristics**

|                                                                         |                                                                                                                                                                |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                      | Avalanche                                                                                                                                                      |
| <b>Consequence Types</b>                                                | Population, Building                                                                                                                                           |
| <b>Annualized Frequency Data Source</b>                                 | Colorado Avalanche Information Center, SHELDUS, National Oceanic and Atmospheric Administration Injury Data                                                    |
| <b>Annualized Frequency Period of Record</b>                            | 1994-2023                                                                                                                                                      |
| <b>Hazard Occurrence Basis</b>                                          | Distinct events                                                                                                                                                |
| <b>Exposure Extent Data Source</b>                                      | Default exposure values                                                                                                                                        |
| <b>Exposure Type</b>                                                    | Default exposure values                                                                                                                                        |
| <b>Historic Loss Ratio Data Source</b>                                  | SHELDUS                                                                                                                                                        |
| <b>Historic Loss Ratio Period of Record</b>                             | <ul style="list-style-type: none"> <li>Population: 1994-2023</li> <li>Building: 1996-2023</li> </ul>                                                           |
| <b>Loss Allocation Date Expansion</b>                                   | No                                                                                                                                                             |
| <b>Loss Aggregation</b>                                                 | Timeframe                                                                                                                                                      |
| <b>Zero-Loss Record Padding</b>                                         | No                                                                                                                                                             |
| <b>Bayesian Settings - Population - County</b>                          | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Population - Avalanche Physiographic Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - County</b>                            | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - Avalanche Physiographic Regions</b>   | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |

**Table 94: Coastal Flooding Hazard Characteristics**

|                                                                                       |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                    | Coastal Flooding                                                                                                                                                                                                                                                       |
| <b>Consequence Types</b>                                                              | Population, Building                                                                                                                                                                                                                                                   |
| <b>Annualized Frequency Data Source</b>                                               | Federal Emergency Management Agency, Geospatial Flood Risk Assessment; NFIP NFHL (SFHA 100- & 500-Year); National Oceanic and Atmospheric Administration, OCM HTF Probability                                                                                          |
| <b>Annualized Frequency Period of Record</b>                                          | Not applicable; Annualized frequency modeled on each sub-type layer's annualized frequency and exposure area                                                                                                                                                           |
| <b>Hazard Occurrence Basis</b>                                                        | Distinct events                                                                                                                                                                                                                                                        |
| <b>Exposure Extent Data Source</b>                                                    | Federal Emergency Management Agency, Geospatial Flood Risk Assessment; NFIP NFHL (SFHA 100- & 500-Year); National Oceanic and Atmospheric Administration, OCM HTF Probability; National Oceanic and Atmospheric Administration, NHC SLOSH Model Data                   |
| <b>Exposure Type</b>                                                                  | <ul style="list-style-type: none"> <li>Developed area and agriculture density applied to the floodplain area</li> <li>Building in Federal Emergency Management Agency GFRA Great Lakes designated areas: Exposure derived from the GFRA building loss rates</li> </ul> |
| <b>Historic Loss Ratio Data Source</b>                                                | SHELDUS                                                                                                                                                                                                                                                                |
| <b>Historic Loss Ratio Period of Record</b>                                           | 1996-2023                                                                                                                                                                                                                                                              |
| <b>Loss Allocation Date Expansion</b>                                                 | No                                                                                                                                                                                                                                                                     |
| <b>Loss Aggregation</b>                                                               | Consecutive day                                                                                                                                                                                                                                                        |
| <b>Zero-Loss Record Padding</b>                                                       | No                                                                                                                                                                                                                                                                     |
| <b>Bayesian Settings - Building - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                            |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                              |
| <b>Bayesian Settings - Building - National Risk Index Historic Loss Ratio Regions</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 2 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                               |
| <b>Bayesian Settings - Building - National</b>                                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                                                            |
| <b>Bayesian Settings - Population - County</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                            |

|                                                                                         |                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Bayesian Settings - Population - Surrounding Area - 208km</i></b>                 | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Population - National Risk Index Historic Loss Ratio Regions</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 2 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No  |
| <b>Bayesian Settings - Population - National</b>                                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                               |

**Table 95: Cold Wave Hazard Characteristics**

|                                                                   |                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                | Cold Wave                                                                                                                                                                                                                                    |
| <b>Consequence Types</b>                                          | Population, Building, Agriculture                                                                                                                                                                                                            |
| <b>Annualized Frequency Data Source</b>                           | Iowa Environmental Mesonet                                                                                                                                                                                                                   |
| <b>Annualized Frequency Period of Record</b>                      | 2005-2024                                                                                                                                                                                                                                    |
| <b>Hazard Occurrence Basis</b>                                    | Event-days                                                                                                                                                                                                                                   |
| <b>Exposure Extent Data Source</b>                                | Iowa Environmental Mesonet                                                                                                                                                                                                                   |
| <b>Exposure Type</b>                                              | Developed area density applied to area of the average hazard event                                                                                                                                                                           |
| <b>Historic Loss Ratio Data Source</b>                            | NCEI Storm Events Database                                                                                                                                                                                                                   |
| <b>Historic Loss Ratio Period of Record</b>                       | 1996-2023                                                                                                                                                                                                                                    |
| <b>Loss Allocation Date Expansion</b>                             | Yes (maximum of 31 days)                                                                                                                                                                                                                     |
| <b>Loss Aggregation</b>                                           | Single day                                                                                                                                                                                                                                   |
| <b>Zero-Loss Record Padding</b>                                   | Yes                                                                                                                                                                                                                                          |
| <b>Zero-Loss Record Padding Period of Record</b>                  | 2005-2024                                                                                                                                                                                                                                    |
| <b>Bayesian Settings - Agriculture - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 115km</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Agriculture - Census Regions</b>           | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Agriculture - National</b>                 | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential                                                               |
| <b>Bayesian Settings - Building - County</b>                      | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |



|                                                                  |                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Census Regions</b>             | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Building - National</b>                   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |
| <b>Bayesian Settings - Population - Surrounding Area - 115km</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Surrounding Area - 208km</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Census Regions</b>           | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential                                                               |

**Table 96: Drought Hazard Characteristics**

|                                                                                          |                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                       | Drought                                                                                                                                                                                                                                     |
| <b>Consequence Types</b>                                                                 | Agriculture                                                                                                                                                                                                                                 |
| <b>Annualized Frequency Data Source</b>                                                  | United States Drought Monitor                                                                                                                                                                                                               |
| <b>Annualized Frequency Period of Record</b>                                             | 2000-2025                                                                                                                                                                                                                                   |
| <b>Hazard Occurrence Basis</b>                                                           | Event-days                                                                                                                                                                                                                                  |
| <b>Exposure Extent Data Source</b>                                                       | United States Drought Monitor                                                                                                                                                                                                               |
| <b>Exposure Type</b>                                                                     | Agriculture value density applied to area of the average hazard event                                                                                                                                                                       |
| <b>Historic Loss Ratio Data Source</b>                                                   | SHELDUS                                                                                                                                                                                                                                     |
| <b>Historic Loss Ratio Period of Record</b>                                              | 1996-2023                                                                                                                                                                                                                                   |
| <b>Loss Allocation Date Expansion</b>                                                    | Yes (maximum of 365 days)                                                                                                                                                                                                                   |
| <b>Loss Aggregation</b>                                                                  | Single day                                                                                                                                                                                                                                  |
| <b>Zero-Loss Record Padding</b>                                                          | Yes                                                                                                                                                                                                                                         |
| <b>Zero-Loss Record Padding Period of Record</b>                                         | 2000-2025                                                                                                                                                                                                                                   |
| <b>Bayesian Settings - Agriculture - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b>                        | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - National Risk Index Historic Loss Ratio Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Agriculture - National</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |

**Table 97: Earthquake Hazard Characteristics**

|                                                     |                                                                                                           |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b><i>Hazard Type</i></b>                           | Earthquake                                                                                                |
| <b><i>Consequence Types</i></b>                     | Population, Building                                                                                      |
| <b><i>Annualized Frequency Data Source</i></b>      | USGS 2023 National Seismic Hazard Model                                                                   |
| <b><i>Annualized Frequency Period of Record</i></b> | Annualized probability                                                                                    |
| <b><i>Hazard Occurrence Basis</i></b>               | Distinct events                                                                                           |
| <b><i>Exposure Extent Data Source</i></b>           | Federal Emergency Management Agency Hazus Earthquake Analysis                                             |
| <b><i>Exposure Type</i></b>                         | Building value and population exposure from Federal Emergency Management Agency Hazus Earthquake Analysis |
| <b><i>Historic Loss Ratio Data Source</i></b>       | N/A – reverse solve                                                                                       |

**Table 98: Hail Hazard Characteristics**

|                                                                   |                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                | Hail                                                                                                                                                                                                                                        |
| <b>Consequence Types</b>                                          | Population, Building, Agriculture                                                                                                                                                                                                           |
| <b>Annualized Frequency Data Source</b>                           | NOAA, SPC Severe Weather Database                                                                                                                                                                                                           |
| <b>Annualized Frequency Period of Record</b>                      | 1986-2023                                                                                                                                                                                                                                   |
| <b>Hazard Occurrence Basis</b>                                    | Distinct events                                                                                                                                                                                                                             |
| <b>Exposure Extent Data Source</b>                                | Total county/census tract value                                                                                                                                                                                                             |
| <b>Exposure Type</b>                                              | Total building value, population, and agriculture value                                                                                                                                                                                     |
| <b>Historic Loss Ratio Data Source</b>                            | SHELDUS                                                                                                                                                                                                                                     |
| <b>Historic Loss Ratio Period of Record</b>                       | 1996-2023                                                                                                                                                                                                                                   |
| <b>Loss Allocation Date Expansion</b>                             | Yes (maximum of 1 day)                                                                                                                                                                                                                      |
| <b>Loss Aggregation</b>                                           | Single day                                                                                                                                                                                                                                  |
| <b>Zero-Loss Record Padding</b>                                   | Yes                                                                                                                                                                                                                                         |
| <b>Zero-Loss Record Padding Period of Record</b>                  | 1986-2023                                                                                                                                                                                                                                   |
| <b>Bayesian Settings - Agriculture - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 115km</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Hail Regions</b>             | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Agriculture - National</b>                 | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Building - County</b>                      | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |

|                                                                  |                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>   | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>   | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Hail Regions</b>               | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Building - National</b>                   | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                 |
| <b>Bayesian Settings - Population - Surrounding Area - 115km</b> | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Surrounding Area - 208km</b> | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Hail Regions</b>             | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - National</b>                 | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHEL DUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |

**Table 99: Heat Wave Hazard Characteristics**

|                                                                   |                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                | Heat Wave                                                                                                                                                                                                                                                   |
| <b>Consequence Types</b>                                          | Population, Building, Agriculture                                                                                                                                                                                                                           |
| <b>Annualized Frequency Data Source</b>                           | Iowa Environmental Mesonet                                                                                                                                                                                                                                  |
| <b>Annualized Frequency Period of Record</b>                      | 2005-2024                                                                                                                                                                                                                                                   |
| <b>Hazard Occurrence Basis</b>                                    | Event-days                                                                                                                                                                                                                                                  |
| <b>Exposure Extent Data Source</b>                                | Iowa Environmental Mesonet                                                                                                                                                                                                                                  |
| <b>Exposure Type</b>                                              | Developed area and agriculture density applied to area of the average hazard event                                                                                                                                                                          |
| <b>Historic Loss Ratio Data Source</b>                            | SHELDUS                                                                                                                                                                                                                                                     |
| <b>Historic Loss Ratio Period of Record</b>                       | 1996-2023                                                                                                                                                                                                                                                   |
| <b>Loss Allocation Date Expansion</b>                             | Yes (maximum of 31 days)                                                                                                                                                                                                                                    |
| <b>Loss Aggregation</b>                                           | Single day                                                                                                                                                                                                                                                  |
| <b>Zero-Loss Record Padding</b>                                   | Yes                                                                                                                                                                                                                                                         |
| <b>Zero-Loss Record Padding Period of Record</b>                  | 2005-2024                                                                                                                                                                                                                                                   |
| <b>Bayesian Settings - Agriculture - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                 |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                 |
| <b>Bayesian Settings - Agriculture - Census Regions</b>           | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                                               |
| <b>Bayesian Settings - Agriculture - National</b>                 | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                                               |
| <b>Bayesian Settings - Building - County</b>                      | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                 |
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>    | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>    | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building -</b>                             | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/>                                                                                                                                                            |



|                                                                  |                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Census Regions</b>                                            | SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                                                                                                                                |
| <b>Bayesian Settings - Building - National</b>                   | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                 |
| <b>Bayesian Settings - Population - Surrounding Area - 208km</b> | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Census Regions</b>           | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - National</b>                 | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |

**Table 100: Hurricane Hazard Characteristics**

|                                                  |                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------|
| <b>Hazard Type</b>                               | Hurricane                                                          |
| <b>Consequence Types</b>                         | Population, Building, Agriculture                                  |
| <b>Annualized Frequency Data Source</b>          | NOAA, NHC HURDAT2 Best Track Data                                  |
| <b>Annualized Frequency Period of Record</b>     | Atlantic: 1851-2024; Pacific: 1949-2024                            |
| <b>Hazard Occurrence Basis</b>                   | Distinct events                                                    |
| <b>Exposure Extent Data Source</b>               | NOAA HURDAT2 Best Track Data                                       |
| <b>Exposure Type</b>                             | Developed area density applied to area of the average hazard event |
| <b>Historic Loss Ratio Data Source</b>           | SHELDUS                                                            |
| <b>Historic Loss Ratio Period of Record</b>      | 1996-2023                                                          |
| <b>Loss Allocation Date Expansion</b>            | No                                                                 |
| <b>Loss Aggregation</b>                          | Consecutive day                                                    |
| <b>Zero-Loss Record Padding</b>                  | Yes                                                                |
| <b>Zero-Loss Record Padding Period of Record</b> | Atlantic 1851-2024; Pacific 1949-2024                              |

**Table 101: Ice Storm Hazard Characteristics**

|                                                                                       |                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                    | Ice Storm                                                                                                                                                                                                                                    |
| <b>Consequence Types</b>                                                              | Population, Building                                                                                                                                                                                                                         |
| <b>Annualized Frequency Data Source</b>                                               | USACE, CRREL Damaging Ice Storm GIS                                                                                                                                                                                                          |
| <b>Annualized Frequency Period of Record</b>                                          | 1946-2014                                                                                                                                                                                                                                    |
| <b>Hazard Occurrence Basis</b>                                                        | Event-days                                                                                                                                                                                                                                   |
| <b>Exposure Extent Data Source</b>                                                    | USACE CRREL Damaging Ice Storm GIS                                                                                                                                                                                                           |
| <b>Exposure Type</b>                                                                  | Developed area density applied to area of the average hazard event                                                                                                                                                                           |
| <b>Historic Loss Ratio Data Source</b>                                                | SHELDUS                                                                                                                                                                                                                                      |
| <b>Historic Loss Ratio Period of Record</b>                                           | 1996-2023                                                                                                                                                                                                                                    |
| <b>Loss Allocation Date Expansion</b>                                                 | Yes (maximum of 31 days)                                                                                                                                                                                                                     |
| <b>Loss Aggregation</b>                                                               | Single day                                                                                                                                                                                                                                   |
| <b>Zero-Loss Record Padding</b>                                                       | Yes                                                                                                                                                                                                                                          |
| <b>Zero-Loss Record Padding Period of Record</b>                                      | 1946-2014                                                                                                                                                                                                                                    |
| <b>Bayesian Settings - Building - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                 |
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - National Risk Index Historic Loss Ratio Regions</b> | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |
| <b>Bayesian Settings - Building - National</b>                                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |
| <b>Bayesian Settings - Population - County</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                 |

|                                                                                                |                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Bayesian Settings - Population - Surrounding Area - 115km</i></b>                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b><i>Bayesian Settings - Population - Surrounding Area - 208km</i></b>                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b><i>Bayesian Settings - Population - National Risk Index Historic Loss Ratio Regions</i></b> | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |
| <b><i>Bayesian Settings - Population - National</i></b>                                        | Filter: LCMA's & Non-LCMA's <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |

**Table 102: Landslide Hazard Characteristics**

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                      | Landslide                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Consequence Types</b>                                                | Population, Building                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Annualized Frequency Data Source</b>                                 | <ul style="list-style-type: none"> <li>CONUS, Alaska &amp; Hawai'i: Luna et al., 2025, Constraining landslide frequency across the United States to inform county-level risk reduction</li> <li>Puerto Rico: Derived from "Map of slope-failure locations in Puerto Rico after Hurricane Maria" and "Landslides Triggered by Hurricane Maria: Assessment of an Extreme Event in Puerto Rico"</li> </ul> |
| <b>Annualized Frequency Period of Record</b>                            | Annualized probability                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Hazard Occurrence Basis</b>                                          | Distinct events                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Exposure Extent Data Source</b>                                      | USGS Landslide Hazard Map                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Exposure Type</b>                                                    | Developed area density applied to Landslide susceptible areas                                                                                                                                                                                                                                                                                                                                           |
| <b>Historic Loss Ratio Data Source</b>                                  | SHELDUS                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Historic Loss Ratio Period of Record</b>                             | 1996-2023                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Loss Allocation Date Expansion</b>                                   | No                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Loss Aggregation</b>                                                 | No                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Zero-Loss Record Padding</b>                                         | Yes - default loss values are inserted for susceptible counties with no past hazard events                                                                                                                                                                                                                                                                                                              |
| <b>Zero-Loss Record Padding Period of Record</b>                        | 1996-2023                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Bayesian Settings - Building - County</b>                            | Filter: N/A <input type="radio"/> Set Usage Threshold: 1 Loss Ratio per Basis record in Non-Zero Pad Record Count modeN/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                                                                                                           |
| <b>Bayesian Settings - Building - Landslide Physiographic Regions</b>   | Filter: N/A <input type="radio"/> Set Usage Threshold: 1 Loss Ratio per Basis record in Non-Zero Pad Record Count modeN/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                                                                                                           |
| <b>Bayesian Settings - Population - County</b>                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 1 Loss Ratio per Basis record in Non-Zero Pad Record Count modeN/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                                                                                                           |
| <b>Bayesian Settings - Population - Landslide Physiographic Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: 2 Loss Ratio per Basis records in Non-Zero Pad Record Count modeN/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                                                                                                          |

**Table 103: Lightning Hazard Characteristics**

|                                                                  |                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                               | Lightning                                                                                                                                                                                                                                                 |
| <b>Consequence Types</b>                                         | Population, Building                                                                                                                                                                                                                                      |
| <b>Annualized Frequency Data Source</b>                          | NOAA/Vaisala NLDN Annual Lightning Climatology Data                                                                                                                                                                                                       |
| <b>Annualized Frequency Period of Record</b>                     | Annualized probability                                                                                                                                                                                                                                    |
| <b>Hazard Occurrence Basis</b>                                   | Distinct events                                                                                                                                                                                                                                           |
| <b>Exposure Extent Data Source</b>                               | Total county/census tract value                                                                                                                                                                                                                           |
| <b>Exposure Type</b>                                             | Total building value and population                                                                                                                                                                                                                       |
| <b>Historic Loss Ratio Data Source</b>                           | SHELDUS                                                                                                                                                                                                                                                   |
| <b>Historic Loss Ratio Period of Record</b>                      | 1996-2023                                                                                                                                                                                                                                                 |
| <b>Loss Allocation Date Expansion</b>                            | Yes (maximum of 1 day)                                                                                                                                                                                                                                    |
| <b>Loss Aggregation</b>                                          | Single day                                                                                                                                                                                                                                                |
| <b>Zero-Loss Record Padding</b>                                  | Yes                                                                                                                                                                                                                                                       |
| <b>Zero-Loss Record Padding Period of Record</b>                 | 1996-2024                                                                                                                                                                                                                                                 |
| <b>Bayesian Settings - Building - County</b>                     | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - National</b>                   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |
| <b>Bayesian Settings - Population - Surrounding Area - 115km</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |



|                                                                         |                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Bayesian Settings - Population - Surrounding Area - 208km</i></b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b><i>Bayesian Settings - Population - National</i></b>                 | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |

**Table 104: Inland Flooding Hazard Characteristics**

|                                                                                          |                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                       | Inland Flooding                                                                                                                                                                                                                                         |
| <b>Consequence Types</b>                                                                 | Population, Building, Agriculture                                                                                                                                                                                                                       |
| <b>Annualized Frequency Data Source</b>                                                  | NCEI Storm Events Database                                                                                                                                                                                                                              |
| <b>Annualized Frequency Period of Record</b>                                             | 1996-2023                                                                                                                                                                                                                                               |
| <b>Hazard Occurrence Basis</b>                                                           | Event-days                                                                                                                                                                                                                                              |
| <b>Exposure Extent Data Source</b>                                                       | Federal Emergency Management Agency, Geospatial Flood Risk Assessment; NFIP NFHL (SFHA 100-Year Floodplain); CoreLogic SFHA Layer                                                                                                                       |
| <b>Exposure Type</b>                                                                     | <ul style="list-style-type: none"> <li>Population &amp; Agriculture: Developed area and agriculture density applied to the floodplain area</li> <li>Building: Exposure derived from the Geospatial Flood Risk Assessment building loss rates</li> </ul> |
| <b>Historic Loss Ratio Data Source</b>                                                   | SHELDUS                                                                                                                                                                                                                                                 |
| <b>Historic Loss Ratio Period of Record</b>                                              | 1996-2023                                                                                                                                                                                                                                               |
| <b>Loss Allocation Date Expansion</b>                                                    | Yes (maximum of 31 days)                                                                                                                                                                                                                                |
| <b>Loss Aggregation</b>                                                                  | Single day                                                                                                                                                                                                                                              |
| <b>Zero-Loss Record Padding</b>                                                          | Yes                                                                                                                                                                                                                                                     |
| <b>Zero-Loss Record Padding Period of Record</b>                                         | 1996-2023                                                                                                                                                                                                                                               |
| <b>Bayesian Settings - Agriculture - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                            |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b>                        | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                            |
| <b>Bayesian Settings - Agriculture - National Risk Index Historic Loss Ratio Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                          |
| <b>Bayesian Settings - Agriculture - National</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                                                          |
| <b>Bayesian Settings - Building - County</b>                                             | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                             |

|                                                                                                |                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Bayesian Settings - Building - Surrounding Area - 208km</i></b>                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                |
| <b><i>Bayesian Settings - Building - National Risk Index Historic Loss Ratio Regions</i></b>   | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                                              |
| <b><i>Bayesian Settings - Building - National</i></b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                                              |
| <b><i>Bayesian Settings - Population - County</i></b>                                          | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b><i>Bayesian Settings - Population - Surrounding Area - 208km</i></b>                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No |
| <b><i>Bayesian Settings - Population - National Risk Index Historic Loss Ratio Regions</i></b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |
| <b><i>Bayesian Settings - Population - National</i></b>                                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: No                                                               |

**Table 105: Strong Wind Hazard Characteristics**

|                                                                   |                                                                                                                                                                                 |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                | Strong Wind                                                                                                                                                                     |
| <b>Consequence Types</b>                                          | Population, Building, Agriculture                                                                                                                                               |
| <b>Annualized Frequency Data Source</b>                           | NOAA, SPC Severe Weather Database                                                                                                                                               |
| <b>Annualized Frequency Period of Record</b>                      | 1986-2023                                                                                                                                                                       |
| <b>Hazard Occurrence Basis</b>                                    | Distinct events                                                                                                                                                                 |
| <b>Exposure Extent Data Source</b>                                | Total county/census tract value                                                                                                                                                 |
| <b>Exposure Type</b>                                              | Total building value, population, and agriculture value                                                                                                                         |
| <b>Historic Loss Ratio Data Source</b>                            | SHELDUS                                                                                                                                                                         |
| <b>Historic Loss Ratio Period of Record</b>                       | 1996-2023                                                                                                                                                                       |
| <b>Loss Allocation Date Expansion</b>                             | Yes (maximum of 1 day)                                                                                                                                                          |
| <b>Loss Aggregation</b>                                           | Single day                                                                                                                                                                      |
| <b>Zero-Loss Record Padding</b>                                   | Yes                                                                                                                                                                             |
| <b>Zero-Loss Record Padding Period of Record</b>                  | 1986-2023                                                                                                                                                                       |
| <b>Bayesian Settings - Agriculture - County</b>                   | Filter: N/A ○ Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 115km</b> | Filter: N/A ○ Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b> | Filter: N/A ○ Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Agriculture - Strong Wind Regions</b>      | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes – exponential                                                               |
| <b>Bayesian Settings - Agriculture - National</b>                 | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes – exponential                                                               |
| <b>Bayesian Settings - Building - County</b>                      | Filter: N/A ○ Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode ○ SHELDUS back testing applied: Yes ○ Smoothing applied: Yes – exponential |

|                                                                  |                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Building - Strong Wind Regions</b>        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential                                                               |
| <b>Bayesian Settings - Building - National</b>                   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                   | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential               |
| <b>Bayesian Settings - Population - Surrounding Area - 115km</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Surrounding Area - 208km</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Strong Wind Regions</b>      | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - National</b>                 | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |

**Table 106: Tornado Hazard Characteristics**

|                                                                                          |                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                       | Tornado                                                                                                                                                                                                                                            |
| <b>Consequence Types</b>                                                                 | Population, Building, Agriculture                                                                                                                                                                                                                  |
| <b>Annualized Frequency Data Source</b>                                                  | NOAA, SPC Severe Weather Database                                                                                                                                                                                                                  |
| <b>Annualized Frequency Period of Record</b>                                             | 1986-2023                                                                                                                                                                                                                                          |
| <b>Hazard Occurrence Basis</b>                                                           | Distinct events                                                                                                                                                                                                                                    |
| <b>Exposure Extent Data Source</b>                                                       | <p>Average impact area per magnitude sub-type:</p> <ul style="list-style-type: none"> <li>• 0.8 km<sup>2</sup> for EF-Scale 0 and 1</li> <li>• 13 km<sup>2</sup> for EF-Scale 2 and 3</li> <li>• 79 km<sup>2</sup> for EF-Scale 4 and 5</li> </ul> |
| <b>Exposure Type</b>                                                                     | Average density applied to the average area of hazard impact per magnitude sub-type                                                                                                                                                                |
| <b>Historic Loss Ratio Data Source</b>                                                   | SHELDUS                                                                                                                                                                                                                                            |
| <b>Historic Loss Ratio Period of Record</b>                                              | 1996-2019                                                                                                                                                                                                                                          |
| <b>Loss Allocation Date Expansion</b>                                                    | Yes (maximum of 1 day)                                                                                                                                                                                                                             |
| <b>Loss Aggregation</b>                                                                  | No                                                                                                                                                                                                                                                 |
| <b>Zero-Loss Record Padding</b>                                                          | <ul style="list-style-type: none"> <li>• Yes (for EF-Scale 0 and 1)</li> <li>• Yes (for EF-Scale 2 and 3)</li> <li>• No (for EF-Scale 4 and 5)</li> </ul>                                                                                          |
| <b>Zero-Loss Record Padding Period of Record</b>                                         | 1986-2023                                                                                                                                                                                                                                          |
| <b>Bayesian Settings - Agriculture - County</b>                                          | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: No ○ Smoothing applied: No                                                                                                                                                  |
| <b>Bayesian Settings - Agriculture - 196KM Fishnet</b>                                   | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: No ○ Smoothing applied: No                                                                                                                                                  |
| <b>Bayesian Settings - Agriculture - National Risk Index Historic Loss Ratio Regions</b> | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: No ○ Smoothing applied: No                                                                                                                                                  |
| <b>Bayesian Settings - Agriculture - National</b>                                        | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: No ○ Smoothing applied: No                                                                                                                                                  |
| <b>Bayesian Settings - Building - County</b>                                             | Filter: N/A ○ Set Usage Threshold: N/A ○ SHELDUS back testing applied: No ○ Smoothing applied: No                                                                                                                                                  |



|                                                                                       |                                                                                                                                                               |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bayesian Settings - Building - 196KM Fishnet</b>                                   | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - National Risk Index Historic Loss Ratio Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Building - National</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Population - County</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |
| <b>Bayesian Settings - Population - 196KM Fishnet</b>                                 | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No |

**Table 107: Tsunami Hazard Characteristics**

|                                                     |                                                                                                                                                                                                                 |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Hazard Type</i></b>                           | Tsunami                                                                                                                                                                                                         |
| <b><i>Consequence Types</i></b>                     | Population, Building                                                                                                                                                                                            |
| <b><i>Annualized Frequency Data Source</i></b>      | Federal Emergency Management Agency's Estimated Average Annualized Tsunami Losses for the United States (P-2426) and National Oceanic and Atmospheric Administration, NCEI Global Historical Tsunami Runup Data |
| <b><i>Annualized Frequency Period of Record</i></b> | Annualized probability                                                                                                                                                                                          |
| <b><i>Hazard Occurrence Basis</i></b>               | Distinct events                                                                                                                                                                                                 |
| <b><i>Exposure Extent Data Source</i></b>           | Federal Emergency Management Agency's Estimated Average Annualized Tsunami Losses for the United States (P-2426)                                                                                                |
| <b><i>Exposure Type</i></b>                         | Building and population values sourced from P-2426                                                                                                                                                              |
| <b><i>Historic Loss Ratio Data Source</i></b>       | N/A – Reverse Solve                                                                                                                                                                                             |

**Table 108: Volcanic Activity Hazard Characteristics**

|                                                                   |                                                                                                                                                                                                                      |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                | Volcanic Activity                                                                                                                                                                                                    |
| <b>Consequence Types</b>                                          | Population, Building                                                                                                                                                                                                 |
| <b>Annualized Frequency Data Source</b>                           | 2018 Update to the US Geological Survey National Volcanic Threat Assessment                                                                                                                                          |
| <b>Annualized Frequency Period of Record</b>                      | Annualized probability                                                                                                                                                                                               |
| <b>Hazard Occurrence Basis</b>                                    | Distinct events                                                                                                                                                                                                      |
| <b>Exposure Extent Data Source</b>                                | Exposure extent data source per sub-hazard: <ul style="list-style-type: none"> <li>Near Volcano: USGS National Volcanic Threat Layer</li> <li>Ashfall: Susceptibility extents based on wind-rose patterns</li> </ul> |
| <b>Exposure Type</b>                                              | Developed area density applied to areas exposed to possible Volcanic eruption                                                                                                                                        |
| <b>Historic Loss Ratio Data Source</b>                            | SHELDUS                                                                                                                                                                                                              |
| <b>Historic Loss Ratio Period of Record</b>                       | 1960-2023                                                                                                                                                                                                            |
| <b>Loss Allocation Date Expansion</b>                             | No                                                                                                                                                                                                                   |
| <b>Loss Aggregation</b>                                           | No                                                                                                                                                                                                                   |
| <b>Zero-Loss Record Padding</b>                                   | No                                                                                                                                                                                                                   |
| <b>Bayesian Settings - Building - Volcanic Activity Regions</b>   | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                        |
| <b>Bayesian Settings - Population - Volcanic Activity Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: No <input type="radio"/> Smoothing applied: No                                                        |

**Table 109: Wildfire Hazard Characteristics**

|                                                     |                                                                          |
|-----------------------------------------------------|--------------------------------------------------------------------------|
| <b><i>Hazard Type</i></b>                           | Wildfire                                                                 |
| <b><i>Consequence Types</i></b>                     | Population, Building, Agriculture                                        |
| <b><i>Annualized Frequency Data Source</i></b>      | USDA, Forest Service FSim Burn Probability and Fire Intensity Level Data |
| <b><i>Annualized Frequency Period of Record</i></b> | Annualized probability                                                   |
| <b><i>Hazard Occurrence Basis</i></b>               | Distinct events                                                          |
| <b><i>Exposure Extent Data Source</i></b>           | USDA, Forest Service FSim Burn Probability and Fire Intensity Level Data |
| <b><i>Exposure Type</i></b>                         | Average density applied to Wildfire susceptible areas                    |
| <b><i>Historic Loss Ratio Data Source</i></b>       | SHELDUS                                                                  |
| <b><i>Historic Loss Ratio Period of Record</i></b>  | 1996-2023                                                                |
| <b><i>Loss Allocation Date Expansion</i></b>        | No                                                                       |
| <b><i>Loss Aggregation</i></b>                      | Timeframe                                                                |
| <b><i>Zero-Loss Record Padding</i></b>              | No                                                                       |

**Table 110: Winter Weather Hazard Characteristics**

|                                                                                          |                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Type</b>                                                                       | Winter Weather                                                                                                                                                                                                                                            |
| <b>Consequence Types</b>                                                                 | Population, Building, Agriculture                                                                                                                                                                                                                         |
| <b>Annualized Frequency Data Source</b>                                                  | Iowa Environmental Mesonet                                                                                                                                                                                                                                |
| <b>Annualized Frequency Period of Record</b>                                             | 2005-2024                                                                                                                                                                                                                                                 |
| <b>Hazard Occurrence Basis</b>                                                           | Event-days                                                                                                                                                                                                                                                |
| <b>Exposure Extent Data Source</b>                                                       | Iowa Environmental Mesonet                                                                                                                                                                                                                                |
| <b>Exposure Type</b>                                                                     | Developed area and agriculture density applied to area of the average hazard event                                                                                                                                                                        |
| <b>Historic Loss Ratio Data Source</b>                                                   | SHELDUS                                                                                                                                                                                                                                                   |
| <b>Historic Loss Ratio Period of Record</b>                                              | 1996-2023                                                                                                                                                                                                                                                 |
| <b>Loss Allocation Date Expansion</b>                                                    | Yes (maximum of 31 days)                                                                                                                                                                                                                                  |
| <b>Loss Aggregation</b>                                                                  | Single day                                                                                                                                                                                                                                                |
| <b>Zero-Loss Record Padding</b>                                                          | Yes                                                                                                                                                                                                                                                       |
| <b>Zero-Loss Record Padding Period of Record</b>                                         | 2005-2024                                                                                                                                                                                                                                                 |
| <b>Bayesian Settings - Agriculture - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 115km</b>                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential |
| <b>Bayesian Settings - Agriculture - Surrounding Area - 208km</b>                        | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |
| <b>Bayesian Settings - Agriculture - National Risk Index Historic Loss Ratio Regions</b> | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential                                                                             |
| <b>Bayesian Settings - Agriculture - National</b>                                        | Filter: N/A <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential                                                                             |
| <b>Bayesian Settings - Building - County</b>                                             | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes – exponential               |

|                                                                                         |                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bayesian Settings - Building - Surrounding Area - 115km</b>                          | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - Surrounding Area - 208km</b>                          | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Building - National Risk Index Historic Loss Ratio Regions</b>   | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Building - National</b>                                          | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - County</b>                                          | Filter: N/A <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential               |
| <b>Bayesian Settings - Population - Surrounding Area - 115km</b>                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - Surrounding Area - 208km</b>                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: 25 Loss Ratio per Basis records in Non-Zero Pad Record Count mode <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential |
| <b>Bayesian Settings - Population - National Risk Index Historic Loss Ratio Regions</b> | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |
| <b>Bayesian Settings - Population - National</b>                                        | Filter: LCMAs & Non-LCMAs <input type="radio"/> Set Usage Threshold: N/A <input type="radio"/> SHELDUS back testing applied: Yes <input type="radio"/> Smoothing applied: Yes - exponential                                                               |



## Appendix C – Mesonet-National Weather Service Weather Event Attribute Description

**Table 111: Mesonet-National Weather Service Weather Event Attribute Descriptions**

| <i>Attribute</i> | <i>Description</i>                                                                                                                                                                                                                                               |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WFO              | Three letter ID for issuing Weather Forecast Office                                                                                                                                                                                                              |
| ISSUED           | String representing product initial valid UTC timestamp YYYYMMDDHHMM                                                                                                                                                                                             |
| EXPIRED          | String representing product expiration, this is not the original product expiration, but the actual time the product was no longer valid. For example, the product could have been extended in time or cancelled.                                                |
| INIT_ISS         | String representing the issuance time of the product UTC timestamp YYYYMMDDHHMM                                                                                                                                                                                  |
| INIT_EXP         | String representing the initial time of the product expiration UTC timestamp YYYYMMDDHHMM                                                                                                                                                                        |
| PHENOM           | VTEC phenomena code. Ex) SV == Severe Thunderstorm, TO == Tornado                                                                                                                                                                                                |
| GTYPE            | Geographical type of polygon. C == County, P == Polygon                                                                                                                                                                                                          |
| SIG              | VTEC significance. Ex) A == Watch, W == Warning, Y == Advisories                                                                                                                                                                                                 |
| ETN              | VTEC Event Tracking Number                                                                                                                                                                                                                                       |
| STATUS           | The three-character code for the VTEC status field. (i.e., EXP, CAN, NEW. For the case of polygons of GTYPE='P'(Storm Based Warnings), the STATUS code is always NEW. For all other cases, this STATUS is the last status parsed for the associated WWA product. |
| NWS_UGC          | NWS code used for a zone of a county                                                                                                                                                                                                                             |
| AREA_KM2         | Area of the geometry in sq. kilometers (Projection: EPSG 2163)                                                                                                                                                                                                   |

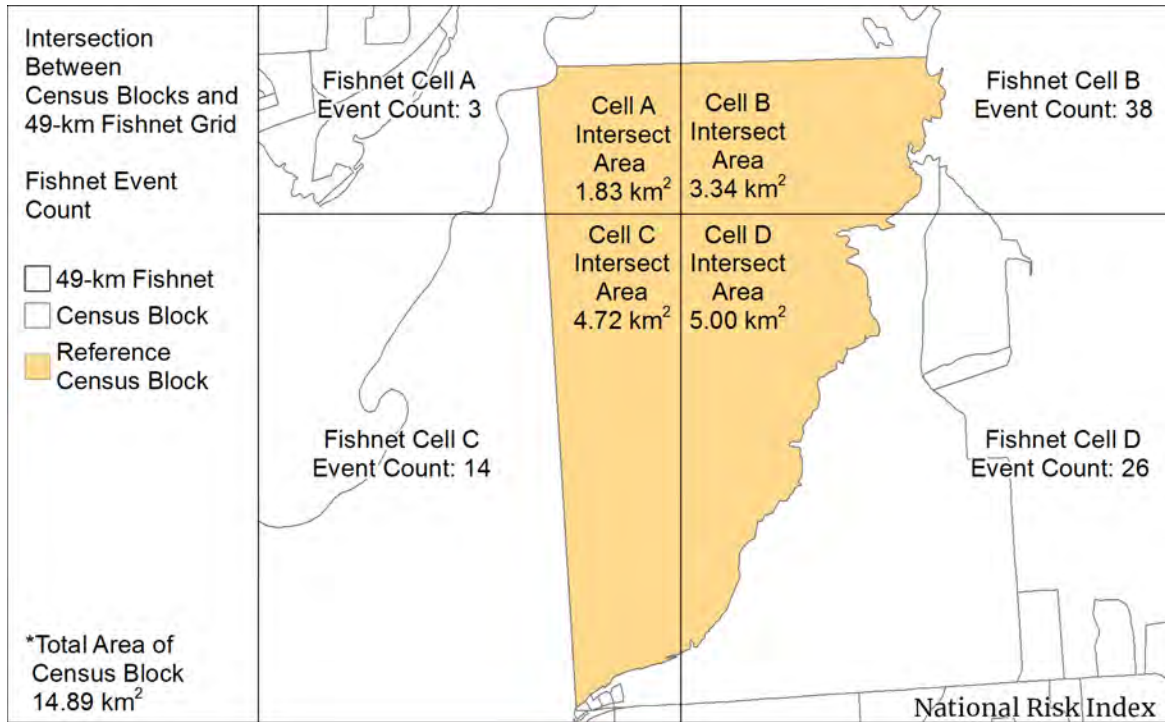
## Appendix D – Fishnet Occurrence Count

**Table 112: Sample Historic Fishnet Hazard Event/Event-Day Count Data**

| <i>Fishnet49kmID</i> | <i>NumberHail<br/>Events</i> | <i>NumberHurricane<br/>Events</i> | <i>NumberIceStorm<br/>EventDays</i> | <i>NumberTornado<br/>Events:<br/>EF0&amp;1<sup>146</sup></i> | <i>NumberWind<br/>Events</i> |
|----------------------|------------------------------|-----------------------------------|-------------------------------------|--------------------------------------------------------------|------------------------------|
| 170                  | 1                            | 39                                | 0                                   | 8                                                            | 5                            |
| 171                  | 1                            | 41                                | 0                                   | 13                                                           | 8                            |
| 172                  | 1                            | 36                                | 0                                   | 15                                                           | 8                            |

For widespread hazards that can occur anywhere within a county, a historic occurrence count (event or event-day) is performed at the level of a 49-by-49-km fishnet grid cell (see [Table 112](#)) which is then intersected with the Census block, Census tract, or county to estimate annualized frequency. If a Census block, Census tract, or county intersects multiple fishnet grid cells (see [Figure 130](#)), an area-weighted average count is calculated. For example, the reference Census block below is intersected by four fishnet cells, each of which intersects a different count of occurrence event (or event-day) polygons. The hazard occurrence count for this Census block would be calculated according to [Equation 155](#) and aggregated to the Census tract and county levels according to [Equation 156](#).

<sup>146</sup> Tornado event counts are actually performed for each EF-scale and follow a different methodology using the fishnet grid cell counts. See the Tornado frequency documentation for more information.



**Figure 130: Intersection Between Census Blocks and 49-by-49 km Fishnet Grid**

**Equation 155: Census Block Area-Weighted Fishnet Event Count**

$$EventCount_{Hazard_{CB}} = \frac{\sum_{Fish}^{CB} (EventCount_{Hazard_{Fish}} \times IntsctArea_{CB_{Fish}})}{Area_{CB}}$$

where:

$EventCount_{Hazard_{CB}}$  is the number of hazard occurrences (event or event-day) calculated for a specific Census block.

$EventCount_{Hazard_{Fish}}$  is the number of hazard occurrences (event or event-day) calculated for the fishnet grid cell.

$IntsctArea_{CB_{Fish}}$  is the intersected area of the Census block with a specific fishnet grid cell (in square kilometers).

$\sum_{Fish}^{CB}$  is the sum for all fishnet grid cells that intersect the Census block.

$Area_{CB}$  is the total area of the Census block (in square kilometers).

**Equation 156: Census Tract and County Area-Weighted Fishnet Event Count**

$$EventCount_{Hazard_{CT}} = \frac{\sum_{CB}^{CT} (IntsctArea_{CB_{Fish}} \times EventCount_{Hazard_{Fish}})}{Area_{CT}}$$

$$EventCount_{Hazard_{Co}} = \frac{\sum_{CB}^{Co} (IntsctArea_{CB_{Fish}} \times EventCount_{Hazard_{Fish}})}{Area_{Co}}$$

where:

$EventCount_{Hazard_{CT}}$  is the count of hazard occurrences (event or event-day) calculated for a specific Census tract.

$IntsctArea_{CB_{Fish}}$  is the intersected area of a specific fishnet grid cell with a specific Census block (in square kilometers).

$EventCount_{Hazard_{Fish}}$  is the count of hazard occurrences (event or event-day) calculated for the fishnet grid cell.

$\sum_{CB}^{CT}$  is the sum for all Census blocks in the Census tract.

$Area_{CT}$  is the total area of the Census tract (in square kilometers).

$EventCount_{Hazard_{Co}}$  is the count of hazard occurrences (event or event-day) calculated for a specific county.

$\sum_{CB}^{Co}$  is the sum for all Census blocks in the county.

$Area_{Co}$  is the total area of the county (in square kilometers).

## Appendix E – Social Vulnerability Index

Technical Documentation for processing the Social Vulnerability data obtained from the Census Community Resilience Estimates team. This document describes how the Social Vulnerability data obtained from Census is used to calculate the Social Vulnerability scores used in the National Risk Index risk calculation.

### Overview

This document describes how community-level inputs provided by the U.S. Census are transformed into:

- a dominance count using a one-standard deviation (~68%) interval, and
- a corresponding percentile score based on that dominance count.

The resulting percentile score is the Social Vulnerability value used in the FEMA National Risk Index risk calculation and provided in the data downloads. The method incorporates the Census-provided uncertainty around each raw percentile by converting the published 90% confidence interval to an approximate 1 standard deviation interval via a normal approximation, as recommended by Census for this use case.

### Why use 1 standard deviation instead of 90% Confidence Interval?

Census advised that a 1 standard deviation interval better differentiates communities (the 90% interval produces too many ties), and agreed the normal approximation used to obtain the standard error is reasonable for this application.

### Input Data

For each community, we use the following Census-provided values:

#### Raw percentile score

The base Social Vulnerability percentile for the community (0–100). Higher values indicate higher relative vulnerability.

#### 90% confidence interval lower bound

The lower bound of the 90% confidence interval for the raw percentile score.

#### 90% confidence interval upper bound

The upper bound of the 90% confidence interval for the raw percentile score.

Note: The 90% interval bounds are produced by Census using a bootstrap with underlying survey data.

## Method Summary

- Derive an estimated standard error from the 90% interval using a normal approximation, then form a 1 standard deviation interval around the raw percentile.
- Compute a dominance count for each community: the number of other communities that are clearly less vulnerable using the 1 standard deviation intervals.
- Convert dominance counts to a percentile scale to produce the final Social Vulnerability percentile used by the National Risk Index.

### STEP 1 — DERIVE THE 1 STANDARD DEVIATION INTERVAL FROM THE 90% INTERVAL

Under a normal approximation, a two-sided 90% confidence interval can be expressed as: (point estimate)  $\pm 1.645 \times$  (standard error).

Therefore, we estimate the standard error as:

$$\text{standard error} = (\text{upper 90\% bound} - \text{lower 90\% bound}) / (2 \times 1.645)$$

We then form a one-standard deviation interval (approximately 68% coverage) around the raw percentile:

$$\begin{aligned} \text{1 standard deviation lower bound} &= \text{raw percentile} - \text{standard error} \\ \text{1 standard deviation upper bound} &= \text{raw percentile} + \text{standard error} \end{aligned}$$

### STEP 2 — DOMINANCE COUNTING (USING 1 STANDARD DEVIATION INTERVALS)

For each community A, we compare its 1 standard deviation lower bound to the 1 standard deviation upper bounds of all other communities B:

A is considered more vulnerable than B (with ~68% confidence) if:  
(A's 1 standard deviation lower bound)  $\geq$  (B's 1 standard deviation upper bound)

- The dominance count for a community is the total number of other communities for which this condition holds.
- Self-comparisons are excluded.
- A dominance proportion can be computed as the dominance count divided by (total communities – 1).

Intuition: If the lower edge of A's plausible range is at least as high as the upper edge of B's plausible range, then A's vulnerability is clearly higher than B's at the 1 standard deviation level.



### STEP 3 — CONVERT DOMINANCE COUNTS TO PERCENTILE SCORES

To produce the final Social Vulnerability percentile score used by the National Risk Index, we rank communities by their 1 standard deviation dominance counts and convert those ranks to percentiles. For a given community, this is interpreted as the percentage of communities with an equal or lower 1 standard deviation dominance count.

This transformation places the result on a familiar 0–100 scale and helps align the output with the broader National Risk Index percentile framework.

#### Output

- 1 standard deviation dominance count — number of other communities that are less vulnerable than a community at the ~68% (1 standard deviation) level.
- Social Vulnerability percentile (based on 1 standard deviation dominance) — the percentile rank of the 1 standard deviation dominance count among all communities; this is the score used in National Risk Index risk calculations and provided in data downloads.

#### Interpretation

- A higher 1 standard deviation dominance count indicates a community clearly exceeds more peers in vulnerability when accounting for uncertainty using the 1 standard deviation interval.
- The final percentile reflects a community's position relative to all others on this “clear vulnerability” dimension; for example, a value of 85 means the community's 1 standard deviation dominance count is greater than or equal to 85% of communities.

#### Reasoning

- Use of 1 standard deviation intervals: Recommended by Census to reduce ties and increase differentiation among communities.
- Normal approximation for standard error: Recommended by Census for converting the published 90% interval into a standard error for the 1 standard deviation interval.
- Dominance definition: Comparing the lower bound of one community to the upper bound of another ensures a “clear separation” in vulnerability given uncertainty.

#### Assumptions and Limitations

- The conversion from the 90% interval to a standard error and then to a 1 standard deviation interval assumes approximate normality of the raw percentile estimates provided by Census.

- The 90% interval bounds originate from a bootstrap performed by Census on underlying survey responses that are not publicly available. We treat those intervals as valid summaries of uncertainty.

## Appendix F – Community Resilience Variable Gathering: Technical Documentation

### Overview

This document provides comprehensive technical documentation for the 12 community resilience variables derived from U.S. Census American Community Survey (ACS) data and 3 variables derived from the U.S. Census Community Business Patterns data. Each variable represents an input into the Social or Economic Pillars of the HVRI BRIC. Each pillar must be recalculated to align with the new direction of FEMA and National Risk Index Data. These data were used, along with data from the LEAD tool and County Health Rankings, to recalculate the Baseline Resilience Indicators for Communities index with variables pertaining to food security, racial/ethnic income inequality, and gender income inequality removed.

**Data Sources:** American Community Survey (ACS) 5-Year Estimates, primarily 2020 (2015 for one variable); Community Business Patterns (CBP) 2020

### Variable Documentation

#### 1. EDUCATIONAL ATTAINMENT

**Definition:** Percentage of population over 25 with college education or more

**Census Table:** B15003 - Educational Attainment for the Population 25 Years and Over

**Variable Codes:**

- B15003\_001E: Total population 25 years and over
- B15003\_022E: Bachelor's degree
- B15003\_023E: Master's degree
- B15003\_024E: Professional school degree
- B15003\_025E: Doctorate degree

**Calculation Formula:**

Educational Attainment = (Bachelor's + Master's + Professional + Doctorate) / Total Population 25+  
× 100

**Data Quality:** Excellent

**Caveats:**

None

## 2. PRE-RETIREMENT AGE (WORKING AGE POPULATION)

**Definition:** Percentage of population between 15 to 64 years of age

**Census Table:** B01001 - Sex by Age

**Variable Codes:**

- B01001\_001E: Total population
- B01001\_006E to B01001\_019E: Male population, ages 15–64 (5-year brackets)
- B01001\_030E to B01001\_043E: Female population, ages 15–64 (5-year brackets)

**Calculation Formula:**

Pre-retirement Age (%) = Sum of population ages 15-64 / Total population × 100

**Data Quality:** Excellent

**Caveats:**

None

---

## 3. TRANSPORTATION ACCESS

**Definition:** Percentage of households with at least one vehicle

**Census Table:** B25044 - Tenure by Vehicles Available

**Variable Codes:**

- B25044\_001E: Total households
- B25044\_004E: Owner occupied, 1 vehicle
- B25044\_005E: Owner occupied, 2 vehicles
- B25044\_006E: Owner occupied, 3+ vehicles
- B25044\_011E: Renter occupied, 1 vehicle
- B25044\_012E: Renter occupied, 2 vehicles
- B25044\_013E: Renter occupied, 3+ vehicles

**Calculation Formula:**

Transportation Access (%) = (Sum of all households with 1+ vehicles / Total households) × 100

**Data Quality:** Excellent

**Caveats:**

- None

---

#### 4. COMMUNICATION CAPACITY

**Definition:** Percentage of households with telephone service available

**Census Table:** DP04 – Selected Housing Characteristic

**Variable Codes:**

- DP04\_0075PE: of occupied units with NO telephone service

**Calculation Formula:**

Households with phone service (%) = 100 - % without phone service

**Data Quality:** Good, but vague description from source

**Caveats:**

- Cell phones are included.
- 

#### 5. ENGLISH LANGUAGE COMPETENCY

**Definition:** Percentage of population that are proficient English speakers

**Census Table:** B16001 - Language Spoken at Home by Ability to Speak English

**Variable Codes (2015 data):**

- B16001\_001E: Total population 5 years and over
- B16001\_002E: Speak only English
- B16001\_005E: Spanish at home, speak English “very well”
- B16001\_008E: Other Indo-European languages, “very well”
- B16001\_011E: Asian and Pacific Islander languages, “very well”
- B16001\_014E: Other languages, “very well”

**Calculation Formula:**

Language Competency (%) = (Only English + All "very well" English speakers) / Total population 5+ × 100

**Data Quality:** Good but dated

**Caveats:**

- Uses 2015 ACS due to 2020 missingness (Consistent with notes from HVRI)
  - Includes only “very well” or only-English; excludes “well,” “not well,” “not at all”
-

## 6. NON-SPECIAL NEEDS (DISABILITY STATUS)

**Definition:** Percentage of population without sensory, physical, or mental disability

**Census Table:** B18101 - Sex by Age by Disability Status

**Variable Codes** (subset):

- B01003\_001E: Total population
- B18101\_004E: Male under 5 with disability
- B18101\_007E: Male 5–17 with disability
- B18101\_010E: Male 18–34 with disability
- B18101\_013E: Male 35–64 with disability
- B18101\_016E: Male 65–74 with disability
- B18101\_019E: Male 75+ with disability
- B18101\_023E: Female under 5 with disability
- B18101\_026E: Female 5–17 with disability
- B18101\_029E: Female 18–34 with disability
- B18101\_032E: Female 35–64 with disability
- B18101\_035E: Female 65–74 with disability
- B18101\_038E: Female 75+ with disability

**Calculation Formula:**

Non-special Needs (%) = (Total population - Sum with disabilities) / Total population × 100

**Data Quality:** Excellent

**Caveats:**

None

---

## 7. HEALTH INSURANCE COVERAGE

**Definition:** Percentage of population under age 65 with health insurance

**Census Table:** B27001 - Health Insurance Coverage Status by Sex by Age

**Variable Codes** (extensive under-65 set):

- Multiple variables for total and insured by sex/age: Under 6, 6–18, 19–25, 26–34, 35–44, 45–54, 55–64; separate male/female counts

**Calculation Formula:**

Health Coverage (%) = Sum insured population under 65 / Sum total population under 65 × 100

**Data Quality:** Excellent



**Caveats:**

None

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## 8. HEALTH ACCESS (PHYSICIAN ACCESS)

**Definition:** Health practitioners per 10,000 population (Note: Not physician-specific)

**Census Tables:**

- C24010 - Sex by Occupation for the Civilian Employed Population 16+ (health diagnosing and treating practitioners)
- B01003 - Total Population

**Variable Codes:**

- C24010\_016E: Male health diagnosing and treating practitioners
- C24010\_052E: Female health diagnosing and treating practitioners
- B01003\_001E: Total population

**Calculation Formula:**

Health Access (per 10,000) = (Male health practitioners + Female health practitioners) / Total population × 10,000

**Data Quality:** Poor

**Caveats:**

- Not physician-specific (includes physicians, dentists, nurses, therapists, etc.)
  - ACS standard tables lack physician-specific occupation detail
  - Use Physician Rate from *County Health Rankings* instead!
- 

## 9. HOMEOWNERSHIP

**Definition:** Percentage of owner-occupied housing units

**Census Table:** B25003 - Tenure

**Variable Codes:**

- B25003\_001E: Total occupied housing units
- B25003\_002E: Owner-occupied housing units

**Calculation Formula:**

Homeownership (%) = Owner-occupied units / Total occupied units × 100

**Data Quality:** Excellent

**Caveats:**

- Denominator excludes vacant units
  - Standard HUD/federal definition
- 

## 10. EMPLOYMENT RATE

**Definition:** Percentage of labor force employed

**Census Table:** B23025 - Employment Status for the Population 16+ Years

**Variable Codes:**

- B23025\_002E: Total labor force
- B23025\_004E: Civilian employed population

**Calculation Formula:**

Employment Rate (%) = Civilian employed / Total labor force × 100

**Data Quality:** Excellent

**Caveats:**

- Labor force = employed + unemployed actively seeking; excludes not-in-labor-force
  - Civilian focus (excludes armed forces)
- 

## 11. FEDERAL EMPLOYMENT

**Definition:** Percentage of labor force employed by federal government

**Census Tables:**

- B23025 - Employment Status for the Population 16+ Years (denominator)
- B24080 - Sex by Class of Worker for the Civilian Employed Population 16+ Years

**Variable Codes:**

- B23025\_002E: Total labor force (denominator)
- B24080\_009E: Male federal government workers
- B24080\_019E: Female federal government workers

**Calculation Formula:**

Employment (%) = (Male federal workers + Female federal workers) / Total labor force × 100

**Data Quality:** Excellent

**Caveats:**

- Civilian focus (excludes military)
  - Federal only (excludes state/local)
- 

## **12. NON-DEPENDENCE ON PRIMARY/TOURISM SECTORS**

**Definition:** Percentage of employees not in farming, fishing, forestry, extractive industry, or tourism

**Census Tables:**

- C24070 - Industry by Class of Worker for the Civilian Employed Population 16 Years and Over

**Variable Codes:**

- C24070\_001E: Total civilian employed population 16+
- C24070\_002E: Agriculture Forestry Fishing Hunting Mining Male
- C24070\_003E: Agriculture Forestry Fishing Hunting Mining Female
- C24070\_004E: Construction Male
- C24070\_005E: Construction Female
- C24070\_006E: Manufacturing Male
- C24070\_007E: Manufacturing Female
- C24070\_008E: Wholesale Trade Male
- C24070\_009E: Wholesale Trade Female
- C24070\_010E: Retail Trade Male
- C24070\_011E: Retail Trade Female
- C24070\_012E: Transportation Warehousing Utilities Male
- C24070\_013E: Transportation Warehousing Utilities Female
- C24070\_014E: Information Male
- C24070\_015E: Information Female
- C24070\_016E: Finance Insurance Real Estate Male

-C24070\_017E: Finance Insurance Real Estate Female

-C24070\_018E: Professional Scientific Management Male

-C24070\_019E: Professional Scientific Management Female

-C24070\_020E: Educational Health Social Services Male

-C24070\_021E: Educational Health Social Services Female

-C24070\_022E: Arts Entertainment Recreation Accommodation Food Male

-C24070\_023E: Arts Entertainment Recreation Accommodation Food Female

-C24070\_024E: Other Services Male

-C24070\_025E: Other Services Female

-C24070\_026E: Public Administration Male

-C24070\_027E: Public Administration Female

**Calculation Formula:**

Non-dependence (%) =  $(1 - (\text{Total dependence workers} / \text{Total employed})) \times 100$

**Data Quality:** Excellent

**Caveats:**

None

---

### 13. BUSINESS RATIO I

**Definition:** Ratio of large to small businesses

**Census Table:** County Business Patterns (CBP) 2020

**Variable Codes:**

-ESTAB: Total establishments (all employment sizes)

-EMPSZES 220: Establishments with 100-249 employees

-EMPSZES 230: Establishments with 250-499 employees

-EMPSZES 240: Establishments with 500-999 employees

-EMPSZES 254: Establishments with 1000+ employees

**Calculation Formula:**

Ratio I = (Total establishments – Establishments with 100+ Employees) / Large establishments

**Data Quality:** Good, but limited data and unclear definition from source

**Caveats:**

- Notes from HVRI mention small businesses as being less than 10 employees
- Establishment counts with fewer than 10 employees are suppressed
- Incomplete coverage for employee categories less than 100

---

## 14. BUSINESS RATIO II

**Definition:** Average business size (employees per establishment)

**Census Table:** County Business Patterns (CBP) 2020

**Variable Codes:**

- ESTAB: Total number of establishments
- EMP: Total Employment

**Calculation Formula:**

Ratio II = Total employment / Total establishments

**Data Quality:** Excellent

**Caveats:**

None

---

## 15. LARGE RETAIL-REGIONAL/NATIONAL GEOGRAPHIC DISTRIBUTION

**Definition:** Large retail stores per 10,000 population

**Census Table:**

- County Business Patterns (CBP) 2020
- B01003: Total Population

**Variable Codes:**

- ESTAB (NAICS 452): Establishments in General Merchandise Stores
- B01003\_001E: Total population

**Calculation Formula:**

Large retail stores rate = (NAICS 452 establishments / Total population) × 10,000

**Data Quality:** Good, but vague definition from source data

**Caveats:**

- Notes from HVRI reference NAICS 452 establishments, which capture general merchandise stores including department stores and warehouse clubs/supercenters (e.g., Walmart, Target, Costco, Meijer)
- Does not include supermarkets (e.g., Kroger, Albertsons, Publix, Aldi)
- BRIC definition is vague, but mentions large *regional* retail as well (e.g., Menards, Vons/Pavilions, Wegmans, Harris Teeter)

## Variable Quality Summary

| <i>Variable</i>                           | <i>Data Quality</i> | <i>Issues/Limitations</i> |
|-------------------------------------------|---------------------|---------------------------|
| Educational Attainment                    | Excellent           | None                      |
| Pre-retirement Age                        | Excellent           | None                      |
| Transportation Access                     | Excellent           | None                      |
| Communication Capacity                    | Excellent           | Includes cell phones      |
| Language Competency                       | Good                | Uses 2015 data            |
| Non-special Needs                         | Excellent           | None                      |
| Health Coverage                           | Excellent           | None                      |
| Health Access                             | Limited             | Not physician-specific    |
| Homeownership                             | Excellent           | None                      |
| Employment Rate                           | Excellent           | None                      |
| Federal Employment                        | Excellent           | None                      |
| Non-dependence on primary/tourism sectors | Excellent           | None                      |
| Business Ratio I                          | Good                | Source definition unclear |



| <b><i>Variable</i></b>                                 | <b><i>Data Quality</i></b> | <b><i>Issues/Limitations</i></b> |
|--------------------------------------------------------|----------------------------|----------------------------------|
| Business Ratio II                                      | Excellent                  | None                             |
| Large retail-regional/national geographic distribution | Good                       | Source definition unclear        |